

Publications by Model: ISSM

Total Publications: 556 (sorted newest to oldest)

1. Predicting the Onset of Subglacial Drainage Channels

Authors: K. L. P. Warburton (University of Cambridge, Dartmouth College); C. R. Meyer (Dartmouth College); A. N. Sommers (Dartmouth College)

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2024jf007758

URL: <https://doi.org/10.1029/2024jf007758>

Publication Date: 2024-12-24

Volume/Issue: Volume 129, Issue 12

References: 53

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: subglacial channels, subglacial hydrology, glacier categorization, stability analysis

Funding: National Science Foundation, National Aeronautics and Space Administration, Army Research Office (and 1 more)

License: CC BY 4.0

Open Access: N/A

Abstract: The sliding speed of glaciers depends strongly on the water pressure at the ice-sediment interface, which is controlled by the efficiency of water transport through a subglacial hydrological system. The least efficient component of the system consists of "distributed" flow everywhere beneath the ice, whereas the "channelized" drainage through large, thermally eroded conduits is more efficient. To understand the conditions under which the subglacial network channelizes, we perform a linear stability analysis of distributed flow, considering competition between thermal erosion and viscous ice collapse. The calculated growth rate gives a stability criterion, describing the minimum subglacial meltwater flux needed for channels to form, but also indicates the tendency to generate infinitely narrow channels in existing models. We demonstrate the need to include lateral heat diffusion when modeling melt incision to resolve channel widths, which allows continuum models to recover Röthlisberger channel behavior. We also show that low numerical resolution can suppress channel formation and lead to overestimates of water pressure. Our derived channelization criterion can be used to predict the character of subglacial hydrological systems without recourse to numerical simulations, with practical implications for understanding changes in ice velocity due to changes in surface melt runoff.

2. A fast and simplified subglacial hydrological model for the Antarctic Ice Sheet and outlet glaciers

Authors: Elise Kazmierczak; Thomas Gregov; Violaine Coulon; Frank Pattyn

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-18-5887-2024

URL: <https://doi.org/10.5194/tc-18-5887-2024>

Publication Date: 2024-12-16

Volume/Issue: Volume 18, Issue 12

References: 110

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Horizon 2020

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Open Access: N/A

Abstract: . We present a novel and computationally efficient subglacial hydrological model that represents in a simplified way both hard- and soft-bed rheologies, as well as an automatic switch between efficient and inefficient subglacial discharge, designed for the Antarctic Ice Sheet. The subglacial model is dynamically linked to a regularized Coulomb friction law, allowing for a coupled evolution of the ice sheet on decadal to centennial timescales. It does not explicitly simulate the details of water conduits at the local scale and assumes that subglacial hydrology is in quasi-static equilibrium with the ice sheet, which makes the computations very fast. The hydrological model is tested on an idealized marine ice sheet and subsequently applied to the drainage basin of Thwaites Glacier, West Antarctica, that is composed of a heterogeneous (hard/soft) bed. We find that accounting for subglacial hydrology in the sliding law accelerates the grounding-line retreat of Thwaites Glacier under present-day climatic conditions. Highest retreat rates are obtained for hard-bed configurations and/or inefficient drainage systems. We show that the sensitivity is particularly driven by large gradients in effective pressure, more so than the value of effective pressure itself, in the vicinity of the grounding line. Therefore, we advocate for a better understanding of the subglacial system with respect to both the spatial and temporal variability in effective pressure and the rheological conditions/properties of the bed.

3. Projections of precipitation and temperatures in Greenland and the impact of spatially uniform anomalies on the evolution of the ice sheet

Authors: Nils Bochow; Anna Poltronieri; Niklas Boers

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-18-5825-2024

URL: <https://doi.org/10.5194/tc-18-5825-2024>

Publication Date: 2024-12-12

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License: CC BY 4.0

Open Access: N/A

Abstract: . Simulations of the Greenland ice sheet (GrIS) at millennial timescales and beyond often assume spatially and temporally uniform temperature anomalies and precipitation sensitivities over these timescales or rely on simple parameterisation schemes for the precipitation rates. However, there is no a priori reason to expect spatially and temporally uniform sensitivities across the whole GrIS. Precipitation is frequently modelled to increase with the standard thermodynamic scaling of $\sim 7\% \text{ K}^{-1}$ derived from the Clausius-Clapeyron relation and often based on older model generations. Here, we update the commonly used parameters for long-term modelling of the GrIS, based on the output of the latest generation of coupled Earth system models (CMIP6), using the historical time period and four different future emission scenarios. We show that the precipitation sensitivities in Greenland have a strong spatial dependence, with values ranging from $-3\% \text{ K}^{-1}$ in southern Greenland to $13\% \text{ K}^{-1}$ in northeastern Greenland relative to the local annual mean near-surface temperature in the CMIP6 ensemble mean. Additionally, we show that the annual mean temperatures in Greenland increase between 1.29 and 1.53 times faster than the global mean temperature (GMT), with northern Greenland warming up to 2 times faster than southern Greenland in all emission scenarios. However, we also show that there is a considerable spread in the model responses that can, at least partially, be attributed to differences in the Atlantic meridional overturning circulation (AMOC) response across models. Finally, using the Parallel Ice Sheet Model (PISM), we show that assuming uniform temperature and precipitation anomalies and sensitivities leads to overestimation of near-surface temperatures and underestimation of precipitation in key regions of the GrIS, such as southwestern Greenland. This, in turn, can result in substantial overestimation of ice loss in the long-term evolution of the GrIS.

4. Exploring grid sensitivity in an ice sheet model: A case study of the Amery Ice Shelf

Authors: Qian-Xi Wang; Teng Li; Xiao Cheng; Chen Zhao; Lei Zheng; Qi Liang

Journal: Advances in Climate Change Research

Publisher: Elsevier BV

DOI: 10.1016/j.accre.2024.12.001

URL: <https://doi.org/10.1016/j.accre.2024.12.001>

Publication Date: 2024-12-08

Volume/Issue: Volume 15, Issue 6

References: 56

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

5. On the Identification of the Basal Drag Parameter in Ice Sheet Models Using L-Curves

Authors: Lea-Sophie Höyns (University of Bremen, Alfred Wegener Institute); Thomas Kleiner (Alfred Wegener Institute); Felix Kranz (University of Bremen); Tim Meyer (University of Bremen); Andreas Rademacher (University of Bremen); Michael Wolovick (Alfred Wegener Institute); Angelika Humbert (University of Bremen, Alfred Wegener Institute)

Journal: PAMM

Publisher: Wiley

DOI: 10.1002/pamm.202400076

URL: <https://doi.org/10.1002/pamm.202400076>

Publication Date: 2024-12-02

Volume/Issue: Volume 24, Issue 4

References: 27

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Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: The unknown and not observable basal drag distribution underneath the glaciers strongly impacts ice flow speeds and, with that, the ongoing mass loss of the Antarctic Ice Sheet. Therefore, basal drag is required for precise ice sheet modeling and accurate projections of future sea-level rise. This can be achieved by applying an inverse method based on observed ice surface velocity data. The forward model equations, including boundary conditions, represent the ice dynamics in an approximated way. The ice-base boundary condition is the main focus here, as it describes a non-linear friction law. This law depends on the unknown basal drag parameter determined by the inversion and utilizes an effective pressure from a subglacial hydrology model. The inverse method minimizes a cost function consisting of the sum of a term quantifying the misfit between simulated and observed surface velocities and a regularization term to penalize unrealistic oscillations in the basal drag parameter. An L-curve analysis determines the optimal weighting of both cost function terms. Here, we perform inversions for three different domains of Antarctica, comprising about 9.4 Mio , to compare the variability of the resulting basal drag and the L-curves. The results present a low basal drag, as well as a low variability, predominating over large parts of the interior of EAIS and WAIS. In contrast, some fast-flowing glaciers reveal a patchy pattern of alternating high and low basal drag. In addition, parts of the grounding line exhibit a high basal drag, which potentially affects the future retreat behavior of the ice sheets.

6. The future of Upernavik Isstrøm through the ISMIP6 framework: sensitivity analysis and Bayesian calibration of ensemble prediction

Authors: Eliot Jager; Fabien Gillet-Chaulet; Nicolas Champollion; Romain Millan; Heiko Goelzer; Jérémie Mouginot

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-18-5519-2024

URL: <https://doi.org/10.5194/tc-18-5519-2024>

Publication Date: 2024-11-28

Volume/Issue: Volume 18, Issue 11

References: 78

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Language: en

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Keywords: N/A

Funding: Research Council of Finland

License: CC BY 4.0

Open Access: N/A

Abstract: . This study investigates the uncertain future contribution to sea-level rise in response to global warming of Upernavik Isstrøm, a tidewater glacier in Greenland. We analyse multiple sources of uncertainty, including Shared Socioeconomic Pathways (SSPs), climate models (global and regional), ice-ocean interactions, and ice sheet model (ISM) parameters. We use weighting methods based on spatio-temporal velocity and elevation data to reduce ice flow model uncertainty and evaluate their ability to prevent overconfidence. Our developed initialization method demonstrates the capability of Elmer/Ice to accurately replicate the hindcast mass loss of Upernavik Isstrøm. Future mass loss predictions in 2100 range from a contribution to sea-level rise from 1.5 to 7.2 mm, with an already committed sea-level contribution projection from 0.6 to 1.3 mm. At the end of the century, SSP-related uncertainty constitutes the predominant component of total uncertainty, accounting for 40 %, while uncertainty linked to the ISM represents 15 % of the overall uncertainty. We find that calibration does not reduce uncertainty in the future mass loss between today and 2100 (+2 %) but significantly reduces uncertainty in the hindcast mass loss between 1985 and 2015 (-32 % to -61 % depending on the weighting method). Combining calibration of the ice sheet model with SSP weighting yields uncertainty reductions in future mass loss in 2050 (-1.5 %) and in 2100 (-32 %).

7. Graph convolutional network as a fast statistical emulator for numerical ice sheet modeling

Authors: Younghyun Koo; Maryam Rahnemoonfar

Journal: Journal of Glaciology

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/jog.2024.93

URL: <https://doi.org/10.1017/jog.2024.93>

Publication Date: 2024-11-20

Volume/Issue: Volume 71

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Language: en

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Keywords: N/A

Funding: National Science Foundation, National Science Foundation

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Open Access: N/A

Abstract: The Ice-sheet and Sea-level System Model (ISSM) provides numerical solutions for ice sheet dynamics using finite element and fine mesh adaption. However, considering ISSM is compatible only with central processing units (CPUs), it has limitations in economizing computational time to explore the linkage between climate forcings and ice dynamics. Although several deep learning emulators using graphic processing units (GPUs) have been proposed to accelerate ice sheet modeling, most of them rely on convolutional neural networks (CNNs) designed for regular grids. Since they are not appropriate for the irregular meshes of ISSM, we use a graph convolutional network (GCN) to replicate the adapted mesh structures of the ISSM. When applied to transient simulations of the Pine Island Glacier (PIG), Antarctica, the GCN successfully reproduces ice thickness and velocity with a correlation coefficient of approximately 0.997, outperforming non-graph models, including fully convolutional network (FCN) and multi-layer perceptron (MLP). Compared to the fixed-resolution approach of the FCN, the flexible-resolution structure of the GCN accurately captures detailed ice dynamics in fast-ice regions. By leveraging 60-100 times faster computational time of the GPU-based GCN emulator, we efficiently examine the impacts of basal melting rates on the ice sheet dynamics in the PIG.

8. Numerical simulation of glacier terminus evolution using the dual action principle for momentum balance

Authors: Daniel R. Shapero; Gonzalo Gonzalez de Diego

Journal: Journal of Glaciology

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/jog.2024.92

URL: <https://doi.org/10.1017/jog.2024.92>

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Language: en

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Keywords: N/A

Funding: Earth Sciences Division, Earth Sciences Division, Office of Advanced Cyberinfrastructure

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Open Access: N/A

Abstract: The momentum conservation equation for glacier flow can be described through minimization of an action functional. Several software packages for glacier flow modeling use this action principle in the design of numerical solution procedures. We derive here an equivalent dual action principle for the shallow stream approximation and implement this model using the finite element method. The key feature of the dual action is that the flow law and friction law are both inverted, which changes the character of the non-linearities. This altered character makes it possible to implement numerical solvers for the dual form that work even when the ice thickness or strain rate are exactly equal to zero. Solvers for the primal form typically fail on such input data and require regularization of the problem. This robustness makes it possible to implement iceberg calving in a simple way: the modeler sets the ice thickness to zero in the desired area. We provide several demonstrations and a reference implementation.

9. Recent and future variability of the ice-sheet catchment of Sermeq Kujalleq (Jakobshavn Isbræ), Greenland

Authors: Anja Lørkkegaard; William Colgan; Andy Aschwanden; Shfaqat Abbas Khan

Journal: Journal of Glaciology

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/jog.2024.73

URL: <https://doi.org/10.1017/jog.2024.73>

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References: 63

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: Knowledge of ice-sheet catchments is critical for mass-balance assessments, especially glacier-scale input-output budgets. This study explores variations in the catchment of Sermeq Kujalleq, or Jakobshavn Isbrø, Greenland. Six observation-based catchment delineations are evaluated along with a 16-member catchment ensemble calculated from ice-sheet models within the Ice Sheet Model Intercomparison Project for CMIP6 (ISMIP6). The 'present-day' ISMIP6 ensemble mean area was found to be $\sim 6.3\%$ larger than the mean of the observed catchments. Ensemble spreads were comparable in size, $\pm 12.3\%$ and $\pm 15.4\%$, suggesting models are able to delineate the present-day catchment with the same degree of uncertainty as observational methods. The mean catchment area of a 13-member ISMIP6 ensemble shows temporal variation, increasing $\sim [2.7, 5.7, 9.1]\%$ under three ocean forcing scenarios and a RCP8.5 projection based on one GCM from 2015 to 2100, primarily as the southern catchment boundary migrates southward. This is interpreted as Sermeq Kujalleq exhibiting dynamic piracy, re-directing ice away from adjacent land terminating glaciers. For mass-balance assessments, present-day catchment delineation is more important than capturing the temporal evolution of individual catchments. However, the modeled temporal changes in catchment area are potentially underestimated, as the models exhibit insufficient acceleration of inland ice flow.

10. Sensitivity of the future evolution of the Wilkes Subglacial Basin ice sheet to grounding-line melt parameterizations

Authors: Yu Wang; Chen Zhao; Rupert Gladstone; Thomas Zwinger; Benjamin K. Galton-Fenzi; Poul Christoffersen

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-18-5117-2024

URL: <https://doi.org/10.5194/tc-18-5117-2024>

Publication Date: 2024-11-13

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Language: en

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Funding: Australian Government, Research Council of Finland

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Open Access: N/A

Abstract: . Projections of Antarctic Ice Sheet mass loss and therefore global sea level rise are hugely uncertain, partly due to how mass loss of the ice sheet occurs at the grounding line. The Wilkes Subglacial Basin (WSB), a vast region of the East Antarctic Ice Sheet, is thought to be particularly vulnerable to deglaciation under future climate warming scenarios. However, future projections of ice loss, driven by grounding-line migration, are known to be sensitive to the parameterization of ocean-induced basal melt of the floating ice shelves and, specifically, to the adjacent grounding line - termed grounding-line melt parameterizations (GLMPs). This study investigates future ice sheet dynamics in the WSB with respect to four GLMPs under both the upper and lower bounds of climate warming scenarios from the present to 2500, with different model resolutions, ice shelf melt parameterizations (ISMPs) and choices of sliding relationships. The variation in these GLMPs determines the distribution and the amount of melt applied in the finite-element assembly procedure on partially grounded elements (i.e. elements containing the grounding line). Our findings indicate that the GLMPs significantly affect both the trigger timings of tipping points and the overall magnitude of ice mass loss. We conclude that applying full melting to the partially grounded elements, which causes melting on the grounded side of the grounding line, should be avoided under all circumstances due to its poor numerical convergence and substantial overestimation of ice mass loss. We recommend preferring options that depend on the specific model context, by either (1) not applying any melt immediately adjacent to the grounding line or (2) employing a sub-element parameterization.

11. Perspective on Regional Sea-level Change and Coastal Impacts

Authors: Kathleen L. McInnes; Robert J. Nicholls; Roderik van de Wal; David Behar; Ivan D. Haigh; Benjamin D. Hamlington; Jochen Hinkel; Daniella Hirschfeld; Benjamin P. Horton; Angelique Melet; Matthew D. Palmer; Alexander A. Robel; Detlef Stammer; Abby Sullivan

Journal: Cambridge Prisms: Coastal Futures

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/cft.2024.15

URL: <https://doi.org/10.1017/cft.2024.15>

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Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: We synthesize sea-level science developments, priorities and practitioner needs at the end of the 10-year World Climate Research Program Grand Challenge 'Regional Sea-Level Change and Coastal Impacts'. Sea-level science and associated climate services have progressed but are unevenly distributed. There remains deep uncertainty concerning high-end and long-term sea-level projections due to indeterminate emissions, the ice sheet response and other climate tipping points. These are priorities for sea-level science. At the same time practitioners need climate services that provide localized information including median and curated high-end sea-level projections for long-term planning, together with information to address near-term pressures, including extreme sea level-related hazards and land subsidence, which can greatly exceed current rates of climate-induced sea-level rise in some populous coastal settlements. To maximise the impact of scientific knowledge, ongoing co-production between science and practitioner communities is essential. Here we report on recent progress and ways forward for the next decade.

12. Precipitation drives western Patagonian glacier variability and may curb future ice mass loss

Authors: Matthias Troch; Henning Åkesson; Joshua K. Cuzzone; Sebastien Bertrand

Journal: Scientific Reports

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41598-024-77486-4

URL: <https://doi.org/10.1038/s41598-024-77486-4>

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Subjects: N/A

Keywords: N/A

Funding: Universiteit Gent, Belgian American Educational Foundation, Research Council of Norway (and 2 more)

License: Creative Commons

Open Access: N/A

Abstract: N/A

13. The organization of subglacial drainage during the demise of the Finnish Lake District Ice Lobe

Authors: Adam J. Hepburn; Christine F. Dow; Antti Ojala; Joni Mäkinen; Elina Ahokangas; Jussi Hovikoski; Jukka-Pekka Palmu; Kari Kajuutti

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-18-4873-2024

URL: <https://doi.org/10.5194/tc-18-4873-2024>

Publication Date: 2024-10-29

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References: 132

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Research Council of Finland, Canada Excellence Research Chairs, Government of Canada, European Space Agency (and 1 more)

License: CC BY 4.0

Open Access: N/A

Abstract: . Unknown basal characteristics limit our ability to simulate the subglacial hydrology of rapidly melting contemporary ice sheets. Sediment-based landforms generated beneath Late Pleistocene ice sheets, together with detailed digital elevation models, offer a valuable means of testing basal hydrology models, which describe the flow and dynamics of water in the subglacial system. However, to date no work has evaluated how well process-based subglacial hydrology models represent the hypothesized conditions associated with glaciofluvial landform formation in the palaeo setting. Previous work comparing model output to geomorphological evidence has typically done so using models that do not resolve subglacial processes and instead express likely subglacial water pathways. Here, we explore the ability of the Glacier Drainage System model (GlaDS), a process-based subglacial hydrology model, to represent the genesis conditions associated with a specific glaciofluvial landform termed "murtoos". Distinctive triangular landforms found throughout Finland and Sweden, murtoos are hypothesized to form 40-60 km from the former Fennoscandian Ice Sheet margin within a "semi-distributed" system at the onset of channelized drainage in small cavities where water pressure is equal to or exceeds ice overburden pressure. Concentrating within a specific ice lobe of the former Fennoscandian Ice Sheet and using digital elevation models with a simulated former ice surface geometry, we forced GlaDS with transient surface melt and explored the sensitivity of our model outcomes to parameter decisions such as the system conductivity and bed topography. Our model outputs closely match the general spacing, direction, and complexity of eskers and mapped assemblages of features related to subglacial drainage in "meltwater routes". Many of the predictions for murtoo formation are produced by the model, including the location of water pressure equal to ice overburden, the onset of channelized drainage, the transition in drainage modes, and importantly the seasonal sequence of drainage conditions inferred from murtoo sedimentology. These conclusions are largely robust to a range of parameter decisions, and we explore seasonal and inter-annual drainage behaviour associated with murtoo zones and meltwater pathways. Our results demonstrate that

examining palaeo basal topography alongside subglacial hydrology model outputs holds promise for the mutually beneficial analyses of palaeo and contemporary ice sheets to assess the controls of hydrology on ice dynamics and subglacial landform evolution.

14. Characteristics of dynamic thickness change across diverse outlet glacier geometries and basal conditions

Authors: Donglai Yang; Kristin Poinar; Sophie Nowicki; Beata Csatho

Journal: Journal of Glaciology

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/jog.2024.50

URL: <https://doi.org/10.1017/jog.2024.50>

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References: 85

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Language: en

Subjects: N/A

Keywords: N/A

Funding: NASA Headquarters

License: CC BY 4.0

Open Access: N/A

Abstract: Outlet glaciers in Greenland are undergoing retreat and diffusive thinning in response to external forcings, but the rates and magnitudes of these responses differ from glacier to glacier for unclear reasons. We test how changes in ice overburden pressure and basal lubrication affect diffusive thinning rates and their spatial patterns by conducting numerical experiments over various idealized Greenland-like glacier domains. We find that ~10 km frontal retreat over a decade can produce sustained thinning rates as large as 16 m a⁻¹ due to ice overburden pressure changes, at outlet glaciers with high basal drag (>60 kPa) and lateral resistive stress (>70 kPa). Localized basal lubrication perturbations induce upstream thinning and downstream thickening up to 12 m a⁻¹; the duration of the lubrication forcing generally has a greater effect than its intensity on induced thickness changes. Lastly, episodic grounding line retreats over a rough bed produce a stepped time series of thinning broadly consistent with observations of dynamic elevation change on multiple Greenland glaciers. Our findings highlight the critical role of the total grounding zone - not ice front position - through the resistive stress change in relation to total glacier thinning.

15. The influence of present-day regional surface mass balance uncertainties on the future evolution of the Antarctic Ice Sheet

Authors: Christian Wirths; Thomas F. Stocker; Johannes C. R. Sutter

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-18-4435-2024

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Keywords: N/A

Funding: Schweizerischer Nationalfonds zur Förderung der Wissenschaftlichen Forschung, Horizon 2020, Deutsche Forschungsgemeinschaft

License: CC BY 4.0

Open Access: N/A

Abstract: . Rising global sea levels are one of many impacts associated with current anthropogenic global warming. The Antarctic Ice Sheet (AIS) has the potential to contribute several meters of sea level rise over the next few centuries. To predict future sea level rise contributions from ice sheets, both global and regional climate model (RCM) outputs are used as forcing in ice sheet model simulations. While the impact of different global models on future projections is well-studied, the effect of different regional models on the evolution of the AIS is mostly unknown. In our study, we present the impact of the choice of present-day reference RCM forcing on the evolution of the AIS. We used the Parallel Ice Sheet Model (PISM) to study the AIS in a quasi-equilibrium state and under future projections, combining present-day RCM output with global climate model projections. Our study suggests differences in projected Antarctic sea level contributions due to the choice of different present-day surface mass balance (SMB) and temperature baseline forcings of 10.6 mm in the year 2100 and 70.0 mm in 2300 under the RCP8.5 scenario. Those uncertainties are an order of magnitude smaller than what is estimated from uncertainties related to ice sheet and climate models. However, we observe an increase in RCM-induced uncertainties over time and for higher-emission scenarios. Additionally, our study shows that the complex relationship between the selected RCM baseline climatology and its impact on future sea level rise is closely related to the stability of West Antarctic Ice Sheet (WAIS), particularly the dynamic response of Thwaites and Pine Island glaciers. On millennial timescale, the choice of the RCM reference leads to ice volume differences up to 2.3 m and can result in the long-term collapse of the West Antarctic Ice Sheet.

16. How well can satellite altimetry and firn models resolve Antarctic firn thickness variations?

Authors: Maria T. Kappelsberger; Martin Horwath; Eric Buchta; Matthias O. Willen; Ludwig Schröder; Sanne B. M. Veldhuijsen; Peter Kuipers Munneke; Michiel R. van den Broeke

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-18-4355-2024

URL: <https://doi.org/10.5194/tc-18-4355-2024>

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Subjects: N/A

Keywords: N/A

Funding: Deutsche Forschungsgemeinschaft, Nederlandse Organisatie voor Wetenschappelijk Onderzoek

License: CC BY 4.0

Open Access: N/A

Abstract: . Elevation changes of the Antarctic Ice Sheet (AIS) related to surface mass balance and firn processes vary strongly in space and time. Their subdecadal natural variability is large and hampers the detection of long-term climate trends. Firn models or satellite altimetry observations are typically used to investigate such firn thickness changes. However, there is a large spread among firn models. Further, they do not fully explain observed firn thickness changes, especially on smaller spatial scales. Reconciled firn thickness variations will facilitate the detection of long-term trends from satellite altimetry; the resolution of the spatial patterns of such trends; and, hence, their attribution to the underlying mechanisms. This study has two objectives. First, we quantify interannual Antarctic firn thickness variations on a 10 km grid scale. Second, we characterise errors in both the altimetry products and firn models. To achieve this, we jointly analyse satellite altimetry and firn modelling results in time and space. We use the timing of firn thickness variations from firn models and the satellite-observed amplitude of these variations to generate a combined product ("adjusted firn thickness variations") over the AIS for 1992-2017. The combined product characterises spatially resolved variations better than either firn models alone or altimetry alone. It provides a higher resolution and a more precise spatial distribution of the variations compared to model-only solutions and eliminates most of the altimetry errors compared to altimetry-only solutions. Relative uncertainties in basin-mean time series of the adjusted firn thickness variations range from 20 % to 108 %. At the grid cell level, relative uncertainties are higher, with median values per basin in the range of 54 % to 186 %. This is due to the uncertainties in the large and very dry areas of central East Antarctica, especially over large megadune fields, where the low signal-to-noise ratio poses a challenge for both models and altimetry to resolve firn thickness variations. A large part of the variance in the altimetric time series is not explained by the adjusted firn thickness variations. Analysis of the altimetric residuals indicate that they contain firn model errors, such as firn signals not captured by the models, and altimetry errors, such as time-variable radar penetration effects and errors in intermission calibration. This highlights the

need for improvements in firn modelling and altimetry analysis.

17. Global convergence of Newton's method for the regularized p -Stokes equations

Authors: Niko Schmidt

Journal: Numerical Algorithms

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s11075-024-01941-6

URL: <https://doi.org/10.1007/s11075-024-01941-6>

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Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Christian-Albrechts-Universität zu Kiel

License: CC BY 4.0

Open Access: N/A

Abstract: The motion of glaciers can be simulated with the $\vec{p}$ -Stokes equations. Up to now, Newton's method to solve these equations has been analyzed in finite-dimensional settings only. We analyze the problem in infinite dimensions to gain a new viewpoint. We do that by proving global convergence of the infinite-dimensional Newton's method with Armijo step sizes to the solution of these equations. We only have to add an arbitrarily small diffusion term for this convergence result. We prove that the additional diffusion term only causes minor differences in the solution compared to the original $\vec{p}$ -Stokes equations under the assumption of some regularity. Finally, we test our algorithms on two experiments: A reformulation of the experiment ISMIP-HOM $\vec{B}$ without sliding and a block with sliding. For the former, the approximation of exact step sizes for the Picard iteration and exact step sizes and Armijo step sizes for Newton's method are superior in the experiment compared to the Picard iteration. For the latter experiment, Newton's method with Armijo step sizes needs many iterations until it converges fast to the solution. Thus, Newton's method with approximately exact step sizes is better than Armijo step sizes in this experiment.

18. Implications of high-resolution velocity and strain rate observations for modelling of Greenlandic tidewater glaciers

Authors: Dominik Fahrner; Pablo J. González; Douglas W. F. Mair; James M. Lea

Journal: Journal of Glaciology

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/jog.2024.63

URL: <https://doi.org/10.1017/jog.2024.63>

Publication Date: 2024-09-16

Volume/Issue: Volume 70

References: 70

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Language: en

Subjects: N/A

Keywords: N/A

Funding: Natural Environment Research Council, Agencia Estatal de Investigación, Agencia Estatal de Investigación

License: CC BY 4.0

Open Access: N/A

Abstract: Synthetic Aperture Radar (SAR) has been used extensively to determine the surface ice flow velocity of tidewater glaciers and investigate changes in seasonal or annual ice dynamics at medium spatial resolution (■100 m). However, assessing tidewater glacier behaviour at these resolutions risks missing key details of glacier dynamics, which is particularly important for determination of strain rates that relate to crevasse formation, depth, and ice damage. Here we present surface ice velocity and strain maps with a 16 m posting derived from high-resolution (1 m) PAZ Ciencia spotlight mode SAR imagery for Narsap Sermia, SW Greenland, for October 2019 to February 2021. Results reveal fine details in strain rate, including an area of compression proximal to the terminus, with an upstream shift of strains through time. The velocity evolution of Narsap Sermia shows distinct seasonal changes starting in summer 2020, which are largely modulated by the subglacial drainage system. Comparison of our results with medium-resolution velocity products shows that while these can capture general strain and velocity patterns, our high-resolution data reveals considerably larger ranges of strain values. This is likely to have implications for tuning strain rate dependent calving and ice damage parameterisations within numerical models.

19. Inland Summer Speedup at Zachariae Isstrøm, Northeast Greenland, Driven by Subglacial Hydrology

Authors: Shfaqat A. Khan (Technical University of Denmark); Mathieu Morlighem (Dartmouth College); Shivani Ehrenfeucht (Department of Earth System Science University of California-Irvine Irvine CA USA, University of Waterloo); Helene Seroussi (Dartmouth College); Youngmin Choi (University of Maryland); Eric Rignot (Department of Earth System Science University of California-Irvine Irvine CA USA, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Angelika Humbert (University of Bremen, Alfred Wegener Institute); Derek Pickell (Dartmouth College); Javed Hassan (Technical University of Denmark)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2024gl110691

URL: <https://doi.org/10.1029/2024gl110691>

Publication Date: 2024-09-15

Volume/Issue: Volume 51, Issue 18

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Language: en

Subjects: N/A

Keywords: ice mass change, numerical ice flow modeling, Greenland glaciers, glacier dynamics, subglacial hydrology (and 1 more)

Funding: Carlsbergfondet

License: Creative Commons

Open Access: N/A

Abstract: The Northeast Greenland Ice Stream (NEGIS) has experienced substantial dynamic thinning in recent years. Here, we examine the evolving behavior of NEGIS, with focus on summer speedup at Zachariae Isstrøm, one of the NEGIS outlet glaciers, which has exhibited rapid retreat and acceleration, indicative of its vulnerability to changing climate conditions. Through a combination of Sentinel-1 data, in-situ GPS observations, and numerical ice flow modeling from 2007, we investigate the mechanisms driving short-term changes. Our analysis reveals a summer speedup in ice flow both near the terminus and inland, with satellite data detecting changes up to 60 km inland, while GPS data capture changes up to 190 km inland along the glacier center line. We attribute this summer speedup to variations in subglacial hydrology, where surface meltwater runoff influences basal friction over the melt season. Incorporating subglacial hydrology into numerical models makes it possible to replicate observed ice velocity patterns.

20. New perspectives on ice forcing in continental arc magma plumbing systems

Authors: Brad S. Singer; Pablo Moreno-Yaeger; Meredith Townsend; Christian Huber; Joshua Cuzzone; Benjamin R. Edwards; Matias Romero; Yasmeen Orellana-Salazar; Shaun A. Marcott; Rachel E. Breunig; Ken L. Ferrier; Kathryn Scholz; Allie N. Coonin; Brent V. Alloway; Marissa M. Tremblay; Sally Stevens; Ivo Fustos-Toribio; Patricio I. Moreno; Franco Vera; Álvaro Amigo

Journal: Journal of Volcanology and Geothermal Research

Publisher: Elsevier BV

DOI: 10.1016/j.jvolgeores.2024.108187

URL: <https://doi.org/10.1016/j.jvolgeores.2024.108187>

Publication Date: 2024-09-11

Volume/Issue: Volume 455

References: 196

Cited by: 2 publications

Language: en

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Keywords: N/A

Funding: National Science Foundation, College of Engineering, University of Wisconsin-Madison, University of California, Los Angeles

License: N/A

Open Access: N/A

Abstract: N/A

21. Ice viscosity governs hydraulic fracture that causes rapid drainage of supraglacial lakes

Authors: Tim Hageman; Jessica Mejía; Ravindra Duddu; Emilio Martínez-Pañeda

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-18-3991-2024

URL: <https://doi.org/10.5194/tc-18-3991-2024>

Publication Date: 2024-09-05

Volume/Issue: Volume 18, Issue 9

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Keywords: N/A

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Open Access: N/A

Abstract: . Full-thickness crevasses can transport water from the glacier surface to the bedrock where high water pressures can open kilometre-long cracks along the basal interface, which can accelerate glacier flow. We present a first computational modelling study that describes time-dependent fracture propagation in an idealised glacier causing rapid supraglacial lake drainage. A novel two-scale numerical method is developed to capture the elastic and viscoelastic deformations of ice along with crevasse propagation. The fluid-conserving thermo-hydro-mechanical model incorporates turbulent fluid flow and accounts for melting and refreezing in fractures. Applying this model to observational data from a 2008 rapid-lake-drainage event indicates that viscous deformation exerts a much stronger control on hydrofracture propagation compared to thermal effects. This finding contradicts the conventional assumption that elastic deformation is adequate to describe fracture propagation in glaciers over short timescales (minutes to several hours) and instead demonstrates that viscous deformation must be considered to reproduce observations of lake drainage rates and local ice surface elevation changes. As supraglacial lakes continue expanding inland and as Greenland Ice Sheet temperatures become warmer than -8 degreesC, our results suggest rapid lake drainage events are likely to occur without refreezing, which has implications for the rate of sea level rise.

22. Evolution of the Antarctic Ice Sheet Over the Next Three Centuries From an ISMIP6 Model Ensemble

Authors: Hélène Seroussi (Dartmouth College); Tyler Pelle (Scripps Institution of Oceanography / University of California San Diego); William H. Lipscomb (National Center for Atmospheric Research); Ayako Abe-Ouchi (University of Tokyo); Torsten Albrecht (Potsdam Institute for Climate Impact Research (PIK) Leibniz Association Potsdam Germany); Jorge Alvarez-Solas (Instituto de Geociencias Consejo Superior de Investigaciones Científicas Universidad Complutense de Madrid Madrid Spain, Departamento de Física de la Tierra y Astrofísica Facultad de Ciencias Físicas Universidad Complutense de Madrid Madrid Spain); Xylar Asay-Davis (Los Alamos National Laboratory); Jean-Baptiste Barre (University of California Irvine); Constantijn J. Berends (Utrecht University); Jorge Bernaldes (Utrecht University); Javier Blasco (Laboratoire de Glaciologie Université libre de Bruxelles Brussels Belgium, Instituto de Geociencias Consejo Superior de Investigaciones Científicas Universidad Complutense de Madrid Madrid Spain, Departamento de Física de la Tierra y Astrofísica Facultad de Ciencias Físicas Universidad Complutense de Madrid Madrid Spain); Justine Caillet (Institut de Recherche pour le Développement (IRD)); David M. Chandler (NORCE Norwegian Research Centre, Bjerknes Centre for Climate Research); Violaine Coulon (Laboratoire de Glaciologie Université libre de Bruxelles Brussels Belgium); Richard Cullather (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Christophe Dumas (Laboratoire des Sciences du Climat et de l'environnement LSCE-IPSL CEA-CNRS-UVSQ Université Paris-Saclay Paris France); Benjamin K. Galton-Fenzi (University of Tasmania, Australian Antarctic Division Kingston TAS Australia); Julius Garbe (Institute of Physics and Astronomy University of Potsdam Potsdam Germany, Potsdam Institute for Climate Impact Research (PIK) Leibniz Association Potsdam Germany); Fabien Gillet-Chaulet (Institut de Recherche pour le Développement (IRD)); Rupert Gladstone (University of Lapland); Heiko Goelzer (NORCE Norwegian Research Centre, Bjerknes Centre for Climate Research); Nicholas Golledge (Victoria University of Wellington); Ralf Greve (Hokkaido University); G. Hilmar Gudmundsson (Northumbria University Newcastle UK); Holly Kyeore Han (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory, Los Alamos National Laboratory); Trevor R. Hillebrand (Los Alamos National Laboratory); Matthew J. Hoffman (Los Alamos National Laboratory); Philippe Huybrechts (Vrije Universiteit Brussel); Nicolas C. Jourdain (Institut de Recherche pour le Développement (IRD)); Ann Kristin Klose (Institute of Physics and Astronomy University of Potsdam Potsdam Germany, Potsdam Institute for Climate Impact Research (PIK) Leibniz Association Potsdam Germany); Petra M. Langebroek (NORCE Norwegian Research Centre, Department of Geosciences iC3: Centre for Ice, Cryosphere, Carbon and Climate UiT The Arctic University of Norway Tromsø Norway, Bjerknes Centre for Climate Research); Gunter R. Leguy (National Center for Atmospheric Research); Daniel P. Lowry (GNS Science Lower Hutt New Zealand); Pierre Mathiot (Institut de Recherche pour le Développement (IRD)); Marisa Montoya (Instituto de Geociencias Consejo Superior de Investigaciones Científicas Universidad Complutense de Madrid Madrid Spain, Departamento de Física de la Tierra y Astrofísica Facultad de Ciencias Físicas Universidad Complutense de Madrid Madrid Spain); Mathieu Morlighem (Dartmouth College); Sophie Nowicki (Department of Geology University at Buffalo Buffalo NY USA); Frank Pattyn (Laboratoire de Glaciologie Université libre de Bruxelles Brussels Belgium); Antony J. Payne (University of Liverpool); Aurélien Quiquet (Laboratoire des Sciences du Climat et de l'environnement LSCE-IPSL CEA-CNRS-UVSQ Université Paris-Saclay Paris France); Ronja Reese (Potsdam Institute for Climate Impact Research (PIK) Leibniz Association Potsdam Germany, Northumbria University Newcastle UK); Alexander Robinson (Alfred Wegener Institute, Departamento de Física de la Tierra y Astrofísica Facultad de Ciencias Físicas Universidad Complutense de Madrid Madrid Spain); Leopekka Saraste (CSC-IT Center for Science Espoo Finland); Erika G. Simon (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Sainan Sun (Northumbria University Newcastle UK); Jake P. Twarog (Dartmouth College); Luke D. Trusel (Pennsylvania State University); Benoit Urruty (Institut de Recherche pour le Développement (IRD)); Jonas Van Breedam (Vrije Universiteit Brussel); Roderik S. W. van de Wal (Utrecht University); Yu Wang (University of Tasmania); Chen Zhao (University of Tasmania); Thomas Zwinger (CSC-IT Center for Science Espoo Finland)

Journal: Earth's Future

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2024ef004561

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Publication Date: 2024-09-04

Volume/Issue: Volume 12, Issue 9

References: 200

Cited by: 10 publications

Language: en

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Keywords: N/A

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Abstract: The Ice Sheet Model Intercomparison Project for CMIP6 (ISMIP6) is the primary effort of CMIP6 (Coupled Model Intercomparison Project-Phase 6) focusing on ice sheets, designed to provide an ensemble of process-based projections of the ice-sheet contribution to sea-level rise over the twenty-first century. However, the behavior of the Antarctic Ice Sheet beyond 2100 remains largely unknown: several instability mechanisms can develop on longer time scales, potentially destabilizing large parts of Antarctica. Projections of Antarctic Ice Sheet evolution until 2300 are presented here, using an ensemble of 16 ice-flow models and forcing from global climate models. Under high-emission scenarios, the Antarctic sea-level contribution is limited to less than 30 cm sea-level equivalent (SLE) by 2100, but increases rapidly thereafter to reach up to 4.4 m SLE by 2300. Simulations including ice-shelf collapse lead to an additional 1.1 m SLE on average by 2300, and can reach 6.9 m SLE. Widespread retreat is observed on that timescale in most West Antarctic basins, leading to a collapse of large sectors of West Antarctica by 2300 in 30%-40% of the ensemble. While the onset date of retreat varies among ice models, the rate of upstream propagation is highly consistent once retreat begins. Calculations of sea-level contribution including water density corrections lead to an additional ~10% sea level and up to 50% for contributions accounting for bedrock uplift in response to ice loading. Overall, these results highlight large sea-level contributions from Antarctica and suggest that the choice of ice sheet model remains the leading source of uncertainty in multi-century projections.

23. Numerical stabilization methods for level-set-based ice front migration

Authors: Gong Cheng; Mathieu Morlighem; G. Hilmar Gudmundsson

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-17-6227-2024

URL: <https://doi.org/10.5194/gmd-17-6227-2024>

Publication Date: 2024-08-22

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References: 39

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Heising-Simons Foundation, National Science Foundation, Natural Environment Research Council

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Open Access: N/A

Abstract: . Numerical modeling of ice sheet dynamics is a critical tool for projecting future sea level rise. Among all the processes responsible for the loss of mass of the ice sheets, enhanced ice discharge triggered by the retreat of marine-terminating glaciers is one of the key drivers. Numerical models of ice sheet flow are therefore required to include ice front migration in order to reproduce today's mass loss and to be able to predict their future. However, the discontinuous nature of calving poses a significant numerical challenge for accurately capturing the motion of the ice front. In this study, we explore different stabilization techniques combined with varying reinitialization strategies to enhance the numerical stability and accuracy of solving the level-set function, which tracks the position of the ice front. Through rigorous testing on an idealized domain with a semicircular and a straight-line ice front, including scenarios with diverse front velocities, we assess the performance of these techniques. The findings contribute to advancing our ability to model ice sheet dynamics, specifically calving processes, and provide valuable insights into the most effective strategies for simulating and tracking the motion of the ice front.

24. The West Antarctic Ice Sheet may not be vulnerable to marine ice cliff instability during the 21st century

Authors: Mathieu Morlighem (Dartmouth College); Daniel Goldberg (University of Edinburgh); Jowan M. Barnes (University of Edinburgh, School of Geography and Environmental Sciences, Northumbria University, Newcastle upon Tyne, UK.); Jeremy N. Bassis (University of Michigan); Douglas I. Benn (University of St Andrews); Anna J. Crawford (Division of Biological and Environmental Sciences, University of Stirling, Stirling FK9 4LA, UK.); G. Hilmar Gudmundsson (School of Geography and Environmental Sciences, Northumbria University, Newcastle upon Tyne, UK.); H  l  ne Seroussi (Dartmouth College)

Journal: Science Advances

Publisher: American Association for the Advancement of Science (AAAS)

DOI: 10.1126/sciadv.ado7794

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Publication Date: 2024-08-21

Volume/Issue: Volume 10, Issue 34

References: 30

Cited by: 10 publications

Language: en

Subjects: N/A

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Abstract: The collapse of ice shelves could expose tall ice cliffs at ice sheet margins. The marine ice cliff instability (MICI) is a hypothesis that predicts that, if these cliffs are tall enough, ice may fail structurally leading to self-sustained retreat. To date, projections that include MICI have been performed with a single model based on a simple parameterization. Here, we implement a physically motivated parameterization in three ice sheet models and simulate the response of the Amundsen Sea Embayment after a hypothetical collapse of floating ice. All models show that Thwaites Glacier would not retreat further in the 21st century. In another set of simulations, we force the grounding line to retreat into Thwaites' deeper basin to expose a taller cliff. In these simulations, rapid thinning and velocity increase reduce the calving rate, stabilizing the cliff. These experiments show that Thwaites may be less vulnerable to MICI than previously thought, and model projections that include this process should be re-evaluated.

25. Open Radar Code Architecture (ORCA): A Platform for Software-Defined Coherent Chirped Radar Systems

Authors: Thomas O. Teisberg (Department of Electrical Engineering, Stanford University, Stanford, CA, USA); Anna L. Broome (Department of Electrical Engineering, Stanford University, Stanford, CA, USA); Dustin M. Schroeder (Department of Electrical Engineering, Department of Geophysics, Stanford University, Stanford, CA, USA)

Journal: IEEE Transactions on Geoscience and Remote Sensing

Publisher: Institute of Electrical and Electronics Engineers (IEEE)

DOI: 10.1109/tgrs.2024.3446368

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Language: N/A

Subjects: N/A

Keywords: Ice-penetrating radar, radar remote sensing, radar sounding, software-defined radio (SDR)

Funding: National Science Foundation, National Aeronautics and Space Administration, NASA Future Investigators in NASA Earth and Space Science and Technology (and 2 more)

License: Creative Commons

Open Access: N/A

Abstract: N/A

26. Holocene warmth explains the Little Ice Age advance of Sermeq Kujalleq

Authors: Karita Kajanto; Camilla Andresen; Helene Seroussi; Therese Rieckh; Jason P. Briner; Basile de Fleurian; Andreas Born; Kerim Nisancioglu

Journal: Quaternary Science Reviews

Publisher: Elsevier BV

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Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Norwegian Research Centre for Hydropower Technology, Kansas NSF EPSCoR, Health Research Fund of Central Denmark Region (and 7 more)

License: N/A

Open Access: N/A

Abstract: N/A

27. Velocity of Greenland's Helheim Glacier Controlled Both by Terminus Effects and Subglacial Hydrology With Distinct Realms of Influence

Authors: A. N. Sommers (Dartmouth College); C. R. Meyer (Dartmouth College); K. Poinar (University at Buffalo Buffalo NY USA); J. Mejia (University at Buffalo Buffalo NY USA); M. Morlighem (Dartmouth College); H. Rajaram (Johns Hopkins University); K. L. P. Warburton (Dartmouth College); W. Chu (Georgia Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2024gl109168

URL: <https://doi.org/10.1029/2024gl109168>

Publication Date: 2024-08-07

Volume/Issue: Volume 51, Issue 15

References: 50

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: coupled modeling, Helheim glacier, SHAKTI-ISSM, Greenland, ice dynamics (and 1 more)

Funding: Heising-Simons Foundation

License: CC BY 4.0

Open Access: N/A

Abstract: Two outstanding questions for the future of the Greenland Ice Sheet are (a) how enhanced meltwater draining beneath the ice will impact the behavior of large tidewater glaciers, and (b) to what extent tidewater glacier velocity is driven by changes at the terminus versus changes in sliding velocity due to meltwater. We present a two-way coupled framework to simulate the nonlinear feedbacks of evolving subglacial hydrology and ice dynamics using the Subglacial Hydrology And Kinetic, Transient Interactions (SHAKTI) model within the Ice-sheet and Sea-level System Model (ISSM). Through coupled simulations of Helheim Glacier, we find that terminus effects dominate the seasonal velocity pattern up to 15 km from the terminus, while hydrology drives the velocity response upstream. With increased melt, the hydrology influence yields seasonal acceleration of several hundred meters per year in the interior, suggesting that hydrology will play an important role in future mass balance of tidewater glaciers.

28. Investigating the dynamics and interactions of surface features on Pine Island Glacier using remote sensing and deep learning

Authors: Qi Zhu; Hua-Dong Guo; Lu Zhang; Dong Liang; Zhe-Rong Wu; Zhuo-Ran Lyu; Xiao-Bing Du

Journal: Advances in Climate Change Research

Publisher: Elsevier BV

DOI: 10.1016/j.accre.2024.07.011

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Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Key Project of Research and Development Plan of Hunan Province, National Aerospace Science Foundation of China, International Foodservice Editorial Council (and 1 more)

License: N/A

Open Access: N/A

Abstract: N/A

29. Holocene thinning in central Greenland controlled by the Northeast Greenland Ice Stream

Authors: Ilaria Tabone; Alexander Robinson; Marisa Montoya; Jorge Alvarez-Solas

Journal: Nature Communications

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41467-024-50772-5

URL: <https://doi.org/10.1038/s41467-024-50772-5>

Publication Date: 2024-07-31

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References: 68

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: N/A

30. Forward and Inverse Modeling of Ice Sheet Flow Using Physics-Informed Neural Networks: Application to Helheim Glacier, Greenland

Authors: Gong Cheng (Dartmouth College); Mathieu Morlighem (Dartmouth College); Sade Francis (Dartmouth College)

Journal: Journal of Geophysical Research: Machine Learning and Computation

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2024jh000169

URL: <https://doi.org/10.1029/2024jh000169>

Publication Date: 2024-07-31

Volume/Issue: Volume 1, Issue 3

References: 60

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Heising-Simons Foundation, National Science Foundation

License: CC BY 4.0

Open Access: N/A

Abstract: Predicting the future contribution of the ice sheets to sea level rise over the next decades presents several challenges due to a poor understanding of critical boundary conditions, such as basal sliding. Traditional numerical models often rely on data assimilation methods to infer spatially variable friction coefficients by solving an inverse problem, given an empirical friction law. However, these approaches are not versatile, as they sometimes demand extensive code development efforts when integrating new physics into the model. Furthermore, this approach makes it difficult to handle sparse data effectively. To tackle these challenges, we use the Physics-Informed Neural Networks (PINNs) to seamlessly integrate observational data and governing equations of ice flow into a unified loss function, facilitating the solution of both forward and inverse problems within the same framework. We illustrate the versatility of this approach by applying the framework to two-dimensional problems on the Helheim Glacier in southeast Greenland. By systematically concealing one variable (e.g., ice speed, ice thickness, etc.), we demonstrate the ability of PINNs to accurately reconstruct hidden information. Furthermore, we extend this application to address a challenging mixed inversion problem. We show how PINNs are capable of inferring the basal friction coefficient while simultaneously filling gaps in the sparsely observed ice thickness. This unified framework offers a promising avenue to enhance the predictive capabilities of ice sheet models, reducing uncertainties, and advancing our understanding of poorly constrained physical processes.

31. Calving front monitoring at a subseasonal resolution: a deep learning application for Greenland glaciers

Authors: Erik Loebel; Mirko Scheinert; Martin Horwath; Angelika Humbert; Julia Sohn; Konrad Heidler; Charlotte Liebezeit; Xiao Xiang Zhu

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-18-3315-2024

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Volume/Issue: Volume 18, Issue 7

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Cited by: 1 publications

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Keywords: N/A

Funding: Helmholtz Association, Bundesministerium für Bildung und Forschung, Bundesministerium für Wirtschaft und Klimaschutz

License: CC BY 4.0

Open Access: N/A

Abstract: . The mass balance of the Greenland Ice Sheet is strongly influenced by the dynamics of its outlet glaciers. Therefore, it is of paramount importance to accurately and continuously monitor these glaciers, especially the variation in their frontal positions. A temporally comprehensive parameterization of glacier calving is essential for understanding dynamic changes and constraining ice sheet modeling. However, many current calving front records are limited in terms of temporal resolution as they rely on manual delineation, which is laborious and not appropriate considering the increasing amount of satellite imagery available. In this contribution, we address this problem by applying an automated method to extract calving fronts from optical satellite imagery. The core of this workflow builds on recent advances in the field of deep learning while taking full advantage of multispectral input information. The performance of the method is evaluated using three independent test datasets. For the three datasets, we calculate mean delineation errors of 61.2, 73.7, and 73.5 m, respectively. Eventually, we apply the technique to Landsat-8 imagery. We generate 9243 calving front positions across 23 outlet glaciers in Greenland for the period 2013-2021. Resulting time series not only resolve long-term and seasonal signals but also resolve subseasonal patterns. We discuss the implications for glaciological studies and present a first application for analyzing the effect of bedrock topography on calving front variations. Our method and derived results represent an important step towards the development of intelligent processing strategies for glacier monitoring, opening up new possibilities for studying and modeling the dynamics of Greenland's outlet glaciers.

32. The increasingly dominant role of climate change on length of day variations

Authors: Mostafa Kiani Shahvandi (ETH Zurich); Surendra Adhikari (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Mathieu Dumberry (University of Alberta); Siddhartha Mishra (ETH Zurich); Benedikt Soja (ETH Zurich)

Journal: Proceedings of the National Academy of Sciences

Publisher: Proceedings of the National Academy of Sciences

DOI: 10.1073/pnas.2406930121

URL: <https://doi.org/10.1073/pnas.2406930121>

Publication Date: 2024-07-15

Volume/Issue: Volume 121, Issue 30

References: 67

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NASA Sea-Level Change Team (N-SLCT), Earth Surface and Interior (ESI) Focus Area, and the Cryospheric Science Program, Discovery Grant from NSERC/CRSNG of Canada

License: Creative Commons

Open Access: N/A

Abstract: The melting of ice sheets and global glaciers results in sea-level rise, a pole-to-equator mass transport increasing Earth's oblateness and resulting in an increase in the length of day (LOD). Here, we use observations and reconstructions of mass variations at the Earth's surface since 1900 to show that the climate-induced LOD trend hovered between 0.3 and 1.0 ms/cy in the 20th century, but has accelerated to 1.33 ± 0.03 ms/cy since 2000. We further show that surface mass transport fully explains the accelerating trend in the Earth oblateness observed in the past three decades. We derive an independent measure of the decreasing LOD trend induced by Glacial Isostatic Adjustment (GIA) of -0.80 ± 0.10 ms/cy, which provides a constraint for the mantle viscosity. The sum of this GIA rate and lunar tidal friction fully explains the secular LOD trend that is inferred from the eclipse record in the past three millennia prior to the onset of contemporary climate change. Projections of future climate warming under high emission scenarios suggest that the climate-induced LOD rate may reach 2.62 ± 0.79 ms/cy by 2100, overtaking lunar tidal friction as the single most important contributor to the long-term LOD variations.

33. Contributions of core, mantle and climatological processes to Earth's polar motion

Authors: Mostafa Kiani Shahvandi; Surendra Adhikari; Mathieu Dumbery; Sadegh Modiri; Robert Heinkelmann; Harald Schuh; Siddhartha Mishra; Benedikt Soja

Journal: Nature Geoscience

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41561-024-01478-2

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Open Access: N/A

Abstract: Earth's spin axis slowly moves relative to the crust over time. A 120-year-long record of this polar motion from astronomical and more modern geodetic measurements displays interannual and multidecadal fluctuations of 20 to 40 milliarcseconds superimposed on a secular trend of about 3 milliarcseconds per year. Earth's polar motion is thought to be driven by various surface and interior processes, but how these processes operate and interact to produce the observed signal remains enigmatic. Here we show that predictions made by an ensemble of physics-informed neural networks trained on measurements to capture geophysical processes can explain the main features of the observed polar motion. We find that glacial isostatic adjustment and mantle convection primarily account for the secular trend. Mass redistribution on the Earth's surface--for example, ice melting and global changes in water storage--yields a relatively weak trend but explains about 90% of the interannual and multidecadal variations. We also find that core processes contribute to both the secular trend and fluctuations in polar motion, either due to variations in torque at the core-mantle boundary or dynamical feedback of the core in response to surface mass changes. Our findings provide constraints on core-mantle interactions for which observations are rare and global ice mass balance over the past century and suggest feedback operating between climate-related surface processes and core dynamics.

34. FastIsostasy v1.0 - a regional, accelerated 2D glacial isostatic adjustment (GIA) model accounting for the lateral variability of the solid Earth

Authors: Jan Swierczek-Jereczek; Marisa Montoya; Konstantin Latychev; Alexander Robinson; Jorge Alvarez-Solas; Jerry Mitrovica

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-17-5263-2024

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Keywords: N/A

Funding: H2020 European Research Council, Ministerio de Ciencia e Innovación, HORIZON EUROPE European Research Council

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Open Access: N/A

Abstract: . The vast majority of ice-sheet modelling studies rely on simplified representations of the glacial isostatic adjustment (GIA), which, among other limitations, do not account for lateral variations in the lithospheric thickness and upper-mantle viscosity. In studies of the last glacial cycle using 3D GIA models, this has however been shown to have major impacts on the dynamics of marine-based sectors of Antarctica, which are likely to be the greatest contributors to sea-level rise in the coming centuries. This gap in comprehensiveness is explained by the fact that 3D GIA models are computationally expensive, rarely open-source and require a complex coupling scheme. To close this gap between "best" and "tractable" GIA models, we propose FastIsostasy here, a regional GIA model capturing lateral variations in the lithospheric thickness and mantle viscosity. By means of fast Fourier transforms and a hybrid collocation scheme to solve its underlying partial differential equation, FastIsostasy can simulate 100 000 years of high-resolution bedrock displacement in only minutes of single-CPU computation, including the changes in sea-surface height due to mass redistribution. Despite its 2D grid, FastIsostasy parameterises the depth-dependent viscosity and therefore represents the depth dimension to a certain extent. FastIsostasy is benchmarked here against analytical, as well as 1D and 3D numerical solutions, and shows good agreement with them. For a simulation of the last glacial cycle, its mean and maximal error over time and space respectively yield less than 5 % and 16 % compared to a 3D GIA model over the regional solution domain. FastIsostasy is open-source, is documented with many examples and provides a straightforward interface for coupling to an ice-sheet model. The model is benchmarked here based on its implementation in Julia, while a Fortran version is also provided to allow for compatibility with most existing ice-sheet models. The Julia version provides additional features, including a vast library of adaptive time-stepping methods and GPU support.

35. Subglacial Discharge Accelerates Dynamic Retreat of Aurora Subglacial Basin Outlet Glaciers, East Antarctica, Over the 21st Century

Authors: T. Pelle (Scripps Institution of Oceanography / University of California San Diego); J. S. Greenbaum (Scripps Institution of Oceanography / University of California San Diego); S. Ehrenfeucht (University of Waterloo); C. F. Dow (University of Waterloo); F. S. McCormack (Monash University)

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2023jf007513

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Subjects: N/A

Keywords: Totten glacier, ice-ocean interactions, Antarctica, ice sheet modeling, sea level rise (and 1 more)

Funding: Earth Sciences Division, National Science Foundation, Natural Sciences and Engineering Research Council of Canada

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Open Access: N/A

Abstract: Recent studies have revealed the presence of a complex freshwater system underlying the Aurora Subglacial Basin (ASB), a region of East Antarctica that contains ~7 m of global sea level potential in ice mainly grounded below sea level. However, the impact that subglacial freshwater has on driving the evolution of the dynamic outlet glaciers that drain this basin has yet to be tested in a coupled ice sheet-subglacial hydrology numerical modeling framework. Here, we project the evolution of the primary outlet glaciers draining the ASB (Moscow University Ice Shelf, Totten, Vanderford, and Adams Glaciers) in response to an evolving subglacial hydrology system and to ocean forcing through 2100, following low and high CMIP6 emission scenarios. By 2100, ice-hydrology feedbacks enhance the ASB's 2100 sea level contribution by ~30% (7.50-9.80 mm) in high emission scenarios and accelerate the retreat of Totten Glacier's main ice stream by 25 years. Ice-hydrology feedbacks are particularly influential in the retreat of the Vanderford and Adams Glaciers, driving an additional 10 km of retreat in fully coupled simulations relative to uncoupled simulations. Hydrology-driven ice shelf melt enhancements are the primary cause of domain-wide mass loss in low emission scenarios, but are secondary to ice sheet frictional feedbacks under high emission scenarios. The results presented here demonstrate that ice-subglacial hydrology interactions can significantly accelerate retreat of dynamic Antarctic glaciers and that future Antarctic sea level assessments that do not take these interactions into account might be severely underestimating Antarctic Ice Sheet mass loss.

36. Assessing the benefits of approximately exact step sizes for Picard and Newton solver in simulating ice flow (FEniCS-full-Stokes v.1.3.2)

Authors: Niko Schmidt; Angelika Humbert; Thomas Slawig

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-17-4943-2024

URL: <https://doi.org/10.5194/gmd-17-4943-2024>

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Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: . Solving the momentum balance is the computationally expensive part of simulating the evolution of ice sheets. The momentum balance is described by the nonlinear full-Stokes equations, which are solved iteratively. We use the Picard iteration and Newton's method combined with Armijo step sizes and approximately exact step sizes, respectively, to solve these equations. The Picard iteration uses either no step size control or the approximately exact step sizes. We compare the variants of Newton's method and the Picard iteration in benchmark experiments, called ISMIP-HOM experiments A, B, E1, and E2. The ISMIP-HOM experiments consist of a more realistic domain and are designed to test the quality of ice models. For an even more realistic test case, we simulate the experiments E1 and E2 with a time-dependent surface. We obtain that approximately exact step sizes greatly reduce the necessary number of iterations for the Picard iteration and Newton's method with nearly no increase in the computation time for each iteration.

37. Seawater Intrusion in the Observed Grounding Zone of Petermann Glacier Causes Extensive Retreat

Authors: Shivani Ehrenfeucht (Department of Earth System Science University of California - Irvine Irvine CA USA); Eric Rignot (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, Department of Earth System Science University of California - Irvine Irvine CA USA); Mathieu Morlighem (Dartmouth College)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Subjects: N/A

Keywords: glacier dynamics, basal melt, ice-ocean interactions, seawater intrusion, ice sheet modeling (and 1 more)

Funding: Earth Sciences Division, University of California, Irvine, National Aeronautics and Space Administration

License: Creative Commons

Open Access: N/A

Abstract: Understanding grounding line dynamics is critical for projecting glacier evolution and sea level rise. Observations from satellite radar interferometry reveal rapid grounding line migration forced by oceanic tides that are several kilometers larger than predicted by hydrostatic equilibrium, indicating the transition from grounded to floating ice is more complex than previously thought. Recent studies suggest seawater intrusion beneath grounded ice may play a role in driving rapid ice loss. Here, we investigate its impact on the evolution of Petermann Glacier, Greenland, using an ice sheet model. We compare model results with observed changes in grounding line position, velocity, and ice elevation between 2010 and 2022. We match the observed retreat, speed up, and thinning using 3-km-long seawater intrusion that drive peak ice melt rates of 50 m/yr; but we cannot obtain the same agreement without seawater intrusion. Including seawater intrusion in glacier modeling will increase the sensitivity to ocean warming.

38. Biases in ice sheet models from missing noise-induced drift

Authors: Alexander A. Robel; Vincent Verjans; Aminat A. Ambelorun

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-18-2613-2024

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Keywords: N/A

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Open Access: N/A

Abstract: . Most climatic and glaciological processes exhibit internal variability, which is omitted from many ice sheet model simulations. Prior studies have found that climatic variability can change ice sheet sensitivity to the long-term mean and trend in climate forcing. In this study, we use an ensemble of simulations with a stochastic large-scale ice sheet model to demonstrate that variability in frontal ablation of marine-terminating glaciers changes the mean state of the Greenland Ice Sheet through noise-induced drift. Conversely, stochastic variability in surface mass balance does not appear to cause noise-induced drift in these ensembles. We describe three potential causes for noise-induced drift identified in prior statistical physics literature: noise-induced bifurcations, multiplicative noise, and nonlinearities in noisy processes. Idealized simulations and Reynolds decomposition theory show that for marine ice sheets in particular, noise-induced bifurcations and nonlinearities in variable ice sheet processes are likely the cause of the noise-induced drift. We argue that the omnipresence of variability in climate and ice sheet systems means that the state of real-world ice sheets includes this tendency to drift. Thus, the lack of representation of such noise-induced drift in spin-up and transient ice sheet simulations is a potentially ubiquitous source of bias in ice sheet models.

39. Combining "Deep Learning" and Physically Constrained Neural Networks to Derive Complex Glaciological Change Processes from Modern High-Resolution Satellite Imagery: Application of the GEOCLASS-Image System to Create VarioCNN for Glacier Surges

Authors: Ute C. Herzfeld (University of Colorado Boulder); Lawrence J. Hessburg (University of Colorado Boulder); Thomas M. Trantow (University of Colorado Boulder); Adam N. Hayes (University of Colorado Boulder)

Journal: Remote Sensing

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Subjects: N/A

Keywords: image classification, satellite remote sensing, glaciology, connectionist-geostatistical classification, neural networks (and 4 more)

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License: CC BY 4.0

Open Access: N/A

Abstract: The objectives of this paper are to investigate the trade-offs between a physically constrained neural network and a deep, convolutional neural network and to design a combined ML approach ("VarioCNN"). Our solution is provided in the framework of a cyberinfrastructure that includes a newly designed ML software, GEOCLASS-image (v1.0), modern high-resolution satellite image data sets (Maxar WorldView data), and instructions/descriptions that may facilitate solving similar spatial classification problems. Combining the advantages of the physically-driven connectionist-geostatistical classification method with those of an efficient CNN, VarioCNN provides a means for rapid and efficient extraction of complex geophysical information from submeter resolution satellite imagery. A retraining loop overcomes the difficulties of creating a labeled training data set. Computational analyses and developments are centered on a specific, but generalizable, geophysical problem: The classification of crevasse types that form during the surge of a glacier system. A surge is a glacial catastrophe, an acceleration of a glacier to typically 100-200 times its normal velocity. GEOCLASS-image is applied to study the current (2016-2024) surge in the Negribreen Glacier System, Svalbard. The geophysical result is a description of the structural evolution and expansion of the surge, based on crevasse types that capture ice deformation in six simplified classes.

40. Widespread seawater intrusions beneath the grounded ice of Thwaites Glacier, West Antarctica

Authors: Eric Rignot (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Enrico Ciraci (University of California Irvine); Bernd Scheuchl (University of California Irvine); Valentyn Tolpekin (ICEYE Oy); Michael Wollersheim (ICEYE Oy); Christine Dow (University of Waterloo)

Journal: Proceedings of the National Academy of Sciences

Publisher: Proceedings of the National Academy of Sciences

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Open Access: N/A

Abstract: Warm water from the Southern Ocean has a dominant impact on the evolution of Antarctic glaciers and in turn on their contribution to sea level rise. Using a continuous time series of daily-repeat satellite synthetic-aperture radar interferometry data from the ICEYE constellation collected in March-June 2023, we document an ice grounding zone, or region of tidally controlled migration of the transition boundary between grounded ice and ice afloat in the ocean, at the main trunk of Thwaites Glacier, West Antarctica, a strong contributor to sea level rise with an ice volume equivalent to a 0.6-m global sea level rise. The ice grounding zone is 6 km wide in the central part of Thwaites with shallow bed slopes, and 2 km wide along its flanks with steep basal slopes. We additionally detect irregular seawater intrusions, 5 to 10 cm in thickness, extending another 6 km upstream, at high tide, in a bed depression located beyond a bedrock ridge that impedes the glacier retreat. Seawater intrusions align well with regions predicted by the GlaDS subglacial water model to host a high-pressure distributed subglacial hydrology system in between lower-pressure subglacial channels. Pressurized seawater intrusions will induce vigorous melt of grounded ice over kilometers, making the glacier more vulnerable to ocean warming, and increasing the projections of ice mass loss. Kilometer-wide, widespread seawater intrusion beneath grounded ice may be the missing link between the rapid, past, and present changes in ice sheet mass and the slower changes replicated by ice sheet models.

41. Modeling the timing of Patagonian Ice Sheet retreat in the Chilean Lake District from 22-10 ka

Authors: Joshua Cuzzone; Matias Romero; Shaun A. Marcott

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-18-1381-2024

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Open Access: N/A

Abstract: . Studying the retreat of the Patagonian Ice Sheet (PIS) during the last deglaciation represents an important opportunity to understand how ice sheets outside the polar regions have responded to deglacial changes in temperature and large-scale atmospheric circulation. At the northernmost extension of the PIS during the Last Glacial Maximum (LGM), the Chilean Lake District (CLD) was influenced by the southern westerly winds (SWW), which strongly modulated the hydrologic and heat budgets of the region. Despite progress in constraining the nature and timing of deglacial ice retreat across this area, considerable uncertainty in the glacial history still exists due to a lack of geologic constraints on past ice margin change. Where the glacial chronology is lacking, ice sheet models can provide important insight into our understanding of the characteristics and drivers of deglacial ice retreat. Here we apply the Ice Sheet and Sea-level System Model (ISSM) to simulate the LGM and last deglacial ice history of the PIS across the CLD at high spatial resolution (450 m). We present a transient simulation of ice margin change across the last deglaciation using climate inputs from the National Center for Atmospheric Research Community Climate System Model (CCSM3) Trace-21ka experiment. At the LGM, the simulated ice extent across the CLD agrees well with the most comprehensive reconstruction of PIS ice history (PATICE). Coincident with deglacial warming, ice retreat ensues after 19 ka, with large-scale ice retreat occurring across the CLD between 18 and 16.5 ka. By 17 ka, the northern portion of the CLD becomes ice free, and by 15 ka, ice only persists at high elevations as mountain glaciers and small ice caps. Our simulated ice history agrees well with PATICE for early deglacial ice retreat but diverges at and after 15 ka, where the geologic reconstruction suggests the persistence of an ice cap across the southern CLD until 10 ka. However, given the high uncertainty in the geologic reconstruction of the PIS across the CLD during the later deglaciation, this work emphasizes a need for improved geologic constraints on past ice margin change. While deglacial warming drove the ice retreat across this region, sensitivity tests reveal that modest variations in wintertime precipitation (~10 %) can modulate the pacing of ice retreat by up to 2 ka, which has implications when comparing simulated outputs of ice margin change to geologic reconstructions. While we find that TraCE-21ka simulates large-scale changes in the SWW across the CLD that are consistent

with regional paleoclimate reconstructions, the magnitude of the simulated precipitation changes is smaller than what is found in proxy records. From our sensitivity analysis, we can deduce that larger anomalies in precipitation, as found in paleoclimate proxies, may have had a large impact on modulating the magnitude and timing of deglacial ice retreat. This fact highlights an additional need for better constraints on the deglacial change in strength, position, and extent of the SWW as it relates to understanding the drivers of deglacial PIS behavior.

42. WAVI.jl: Ice Sheet Modelling in Julia

Authors: Alexander T. Bradley; Robert J. Arthern; David T. Bett; C. Rosie Williams; James Byrne

Journal: Journal of Open Source Software

Publisher: The Open Journal

DOI: 10.21105/joss.05584

URL: <https://doi.org/10.21105/joss.05584>

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Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

43. Impact of boundary conditions on the modeled thermal regime of the Antarctic ice sheet

Authors: In-Woo Park; Emilia Kyung Jin; Mathieu Morlighem; Kang-Kun Lee

Journal: The Cryosphere

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DOI: 10.5194/tc-18-1139-2024

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Keywords: N/A

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Open Access: N/A

Abstract: . A realistic initialization of ice flow models is critical for predicting future changes in ice sheet mass balance and their associated contribution to sea level rise. The initial thermal state of an ice sheet is particularly important, as it controls ice viscosity and basal conditions, thereby influencing the overall ice velocity. Englacial and subglacial conditions, however, remain poorly understood due to insufficient direct measurements, which complicate the initialization and validation of thermal models. Here, we investigate the impact of using different geothermal heat flux (GHF) datasets and vertical velocity profiles on the thermal state of the Antarctic ice sheet and compare our modeled temperatures to in situ measurements from 15 boreholes. We find that the temperature profile is more sensitive to vertical velocity than to GHF. The basal temperature of grounded ice and the amount of basal melting are influenced by both selection of GHF and vertical velocity. More importantly, we find that the standard approach, which consists of combining basal sliding speed and incompressibility to derive vertical velocities, provides reasonably good results in fast-flow regions (ice velocity >50 m yr⁻¹) but performs poorly in slow-flow regions (ice velocity <50 m yr⁻¹). Furthermore, the modeled temperature profiles in ice streams, where bed geometry is generated using a mass conservation approach, show better agreement with observed borehole temperatures compared to kriging-based bed geometry.

44. Modeling a Century of Change: Kangerlussuaq Glacier's Mass Loss From 1933 to 2021

Authors: E. Y. H. Lippert (Technical University of Denmark); M. Morlighem (Dartmouth College); G. Cheng (Dartmouth College); S. A. Khan (Technical University of Denmark)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2023gl106286

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References: 51

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Open Access: N/A

Abstract: Kangerlussuaq Glacier (KG) is a major contributor to central-eastern Greenland mass loss, but existing estimates of its mass balance over the last century are inconsistent, and specific drivers of change remain poorly understood. We present a novel approach that combines numerical modeling and a 1933-2021 climate forcing to reconstruct its mass balance over the past century. The model's final state aligns remarkably well with present-day observations. The model reveals a total ice mass loss of 285 Gt over the past century, equivalent to 0.68 mm global sea level rise, 51% of which occurred since 2003 alone. Dynamic thinning from ice front retreat is responsible for 88% of mass change since 1933, with short-term ice front variations having minimal impact on centennial mass loss. Compared to earlier studies, our findings suggest that KG lost 59% (or 301 Gt) less mass over the century than previously thought.

45. Holocene southwest Greenland ice sheet behavior constrained by sea-level modeling

Authors: Raf Antwerpen; Jacqueline Austermann; Nicolás Young; David Porter; Lauren Lewright; Konstantin Latychev

Journal: Quaternary Science Reviews

Publisher: Elsevier BV

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Keywords: N/A

Funding: National Science Foundation

License: N/A

Open Access: N/A

Abstract: N/A

46. Mesh size effect on finite source inversion with 3-D finite-element modelling

Authors: Minsu Kim (Kangwon National University); Byung-Dal So (Kangwon National University); SatByul Kim (Kangwon National University); Taehwan Jo (Korea Meteorological Administration); Sung-Joon Chang (Kangwon National University)

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

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URL: <https://doi.org/10.1093/gji/ggae060>

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Open Access: N/A

Abstract: SUMMARY Three-dimensional finite-element models, which can handle the stress perturbations caused by subsurface mechanical heterogeneities and fault interactions, have been combined with the finite source inversion to estimate the coseismic slip distribution over the fault plane. However, the mesh grid for discretizing the governing equations in the finite-element model significantly affects the numerical accuracy. In this study, we performed kinematic finite source inversion with idealized (regular observation point array; M1A-M1D) and regional (GEONET, GPS Earthquake Observation Network System stations in Japan; M2A-M2H) models with different mesh sizes to quantitatively analyse the effect of the mesh grid size around the fault plane on the inverted fault slip distribution. Synthetic observation data vectors obtained from the finest models (M1A and M2A) are compared with those from the coarser models (M1B-M1D and M2B-M2H), which were adopted to construct Green's function matrix. We found that the coarser mesh models derived a smaller surface displacement, leading to a decrease in the norm of Green's function matrix, which in turn influences the fault slip magnitude from the finite source inversion. Finally, we performed the source inversion for the fault slip distribution of the 2011 Mw 9.0 Tohoku-Oki earthquake using the coseismic surface displacements recorded at the GEONET and seafloor stations and finite-element modelling. By reducing the mesh size on the fault, we confirmed that the estimated magnitude of fault slip converged to approximately 80 m, which is consistent with the range of fault slip amounts from previous studies based on the Okada model. At least 0.88 million total domain elements and a 6.7 km² mesh size on the fault plane with an area of 240 x 720 km² are required for the convergence of the fault slip. Furthermore, we found that the location of the maximum fault slip is less sensitive to the mesh size, implying that source inversion based on a coarse mesh model (i.e. less than 0.5 million elements and > ~60 km² mesh size) can quickly provide the rough fault slip distribution.

47. Southern Ocean High-Resolution (SOhi) Modeling Along the Antarctic Ice Sheet Periphery

Authors: Andy Dinh (University of California Irvine); Eric Rignot (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Matthew Mazloff (Scripps Institution of Oceanography / University of California San Diego); Ian Fenty (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Subjects: N/A

Keywords: Southern Ocean, ocean-ice interactions, ocean modeling, Antarctica

Funding: N/A

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Open Access: N/A

Abstract: The Southern Ocean plays a major role in controlling the evolution of Antarctic glaciers and in turn their impact on sea level rise. We present the Southern Ocean high-resolution (SOhi) simulation of the MITgcm ocean model to reproduce ice-ocean interaction at 1/24 degrees around Antarctica, including all ice shelf cavities and oceanic tides. We evaluate the model accuracy on the continental shelf using Marine Mammals Exploring the Oceans Pole to Pole data and compare the results with three other MITgcm ocean models (ECCO4, SOSE, and LLC4320) and the ISMIP6 temperature reconstruction. Below 400 m, all the models exhibit a warm bias on the continental shelf, but the bias is reduced in the high-resolution simulations. We hypothesize some of the bias is due to an overestimation of sea ice cover, which reduces heat loss to the atmosphere. Both high-resolution and accurate bathymetry are required to improve model accuracy around Antarctica.

48. A stochastic parameterization of ice sheet surface mass balance for the Stochastic Ice-Sheet and Sea-Level System Model (StISSM v1.0)

Authors: Lizz Ultee; Alexander A. Robel; Stefano Castruccio

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-17-1041-2024

URL: <https://doi.org/10.5194/gmd-17-1041-2024>

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Keywords: N/A

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Open Access: N/A

Abstract: . Many scientific and societal questions that draw on ice sheet modeling necessitate sampling a wide range of potential climatic changes and realizations of internal climate variability. For example, coastal planning literature demonstrates a demand for probabilistic sea level projections with quantified uncertainty. Further, robust attribution of past and future ice sheet change to specific processes or forcings requires a full understanding of the space of possible ice sheet behaviors. The wide sampling required to address such questions is computationally infeasible with sophisticated numerical climate models at the resolution required to accurately force ice sheet models. Stochastic generation of climate forcing of ice sheets offers a complementary alternative. Here, we describe a method to construct a stochastic generator for ice sheet surface mass balance varying in time and space. We demonstrate the method with an application to Greenland Ice Sheet surface mass balance for 1980-2012. We account for spatial correlations among glacier catchments using sparse covariance techniques, and we apply an elevation-dependent downscaling to recover gridded surface mass balance fields suitable for forcing an ice sheet model while including feedback from changing ice sheet surface elevation. The efficiency gained in the stochastic method supports large-ensemble simulations of ice sheet change in a new stochastic ice sheet model. We provide open source Python workflows to support use of our stochastic approach for a broad range of applications.

49. Validating ensemble historical simulations of Upernavik Isstrøm (1985-2019) using observations of surface velocity and elevation

Authors: Eliot Jager; Fabien Gillet-Chaulet; Jérémie Mouginot; Romain Millan

Journal: Journal of Glaciology

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/jog.2024.10

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License: CC BY 4.0

Open Access: N/A

Abstract: The future of tidewater glaciers in response to climate warming is one of the largest sources of uncertainty in the contribution of the Greenland ice sheet to global sea-level rise. In this study, we investigate the ability of an ice-sheet model to reproduce the past evolution of the velocity and surface elevation of a tidewater glacier, Upernavik Isstrøm, by prescribing front positions. To achieve this, we run two ensembles of simulations with a Weertman and a regularised-Coulomb friction law. We show that the ice-flow model has to include a reduction in friction in the first 15 km upstream of the ice front in fast-flowing regions to capture the trends observed during the 1985-2019 period. Without this process, the ensemble model overestimates the ice flow before the retreat of the front in 2005 and does not fully reproduce its acceleration during the retreat. This results in an overestimation of the total mass loss between 1985 and 2019 of 50% (300 vs 200 Gt). Using a variance-based sensitivity analysis, we show that uncertainties in the friction law and the ice-flow law have a greater impact on the model results than surface mass balance and initial surface elevation.

50. Graphics-processing-unit-accelerated ice flow solver for unstructured meshes using the Shallow-Shelf Approximation (FastIceFlo v1.0.1)

Authors: Anjali Sandip; Ludovic Räss; Mathieu Morlighem

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-17-899-2024

URL: <https://doi.org/10.5194/gmd-17-899-2024>

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References: 31

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: Solver|Graphics processing unit

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: . Ice-sheet flow models capable of accurately projecting their future mass balance constitute tools to improve flood risk assessment and assist sea-level rise mitigation associated with enhanced ice discharge. Some processes that need to be captured, such as grounding-line migration, require high spatial resolution (under the kilometer scale). Conventional ice flow models mainly execute on central processing units (CPUs), which feature limited parallel processing capabilities and peak memory bandwidth. This may hinder model scalability and result in long run times, requiring significant computational resources. As an alternative, graphics processing units (GPUs) are ideally suited for high spatial resolution, as the calculations can be performed concurrently by thousands of threads, processing most of the computational domain simultaneously. In this study, we combine a GPU-based approach with the pseudo-transient (PT) method, an accelerated iterative and matrix-free solution strategy, and investigate its performance for finite elements and unstructured meshes with application to two-dimensional (2-D) models of real glaciers at a regional scale. For both the Jakobshavn and Pine Island glacier models, the number of nonlinear PT iterations required to converge a given number of vertices (N) scales in the order of $O(N^{1.2})$ or better. We further compare the performance of the PT CUDA C implementation with a standard finite-element CPU-based implementation using the price-to-performance metric. The price of a single Tesla V100 GPU is 1.5 times that of two Intel Xeon Gold 6140 CPUs. We expect a minimum speedup of at least 1.5 times to justify the Tesla V100 GPU price to performance. Our developments result in a GPU-based implementation that achieves this goal with a speedup beyond 1.5 times. This study represents a first step toward leveraging GPU processing power, enabling more accurate polar ice discharge predictions. The insights gained will benefit efforts to diminish spatial resolution constraints at higher computing performance. The higher computing performance will allow for ensembles of ice-sheet flow simulations to be run at the continental scale and higher resolution, a previously challenging task. The advances will further enable the quantification of model sensitivity to changes in upcoming climate forcings. These findings will significantly benefit process-oriented sea-level-projection studies over the coming decades.

51. Monitoring Earth's climate variables with satellite laser altimetry

Authors: Lori A. Magruder; Sinead L. Farrell; Amy Neuenschwander; Laura Duncanson; Beata Csatho; Sahra Kacimi; Helen A. Fricker

Journal: Nature Reviews Earth & Environment

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s43017-023-00508-8

URL: <https://doi.org/10.1038/s43017-023-00508-8>

Publication Date: 2024-01-30

Volume/Issue: Volume 5, Issue 2

References: 220

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

52. Accelerating Subglacial Hydrology for Ice Sheet Models With Deep Learning Methods

Authors: Vincent Verjans (Center for Climate Physics, Institute for Basic Science, Georgia Institute of Technology); Alexander Robel (Georgia Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2023gl105281

URL: <https://doi.org/10.1029/2023gl105281>

Publication Date: 2024-01-22

Volume/Issue: Volume 51, Issue 2

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Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: ice sheet modeling, deep learning, subglacial hydrology

Funding: Heising-Simons Foundation

License: Creative Commons

Open Access: N/A

Abstract: Subglacial drainage networks regulate the response of ice sheet flow to surface meltwater input to the subglacial environment. Simulating subglacial hydrology evolution is critical to projecting ice sheet sensitivity to climate, and contribution to sea-level change. However, current numerical subglacial hydrology models are computationally expensive, and, consequently, evolving subglacial hydrology is neglected in large-scale ice sheet simulations. We present a deep learning emulator of a state-of-the-art subglacial hydrology model, trained at multiple Greenland glaciers. Our emulator performs strongly in both temporal ($R^2 > 0.99$) and spatial ($R^2 > 0.95$) generalization, offers high computational savings, and can be used to force numerical ice sheet models. This will enable century- and large-scale ice sheet model simulations, including interactions between ice flow and increased meltwater input to the subglacial environment. Generally, our work demonstrates that machine learning can further improve ice sheet models, reduce computational bottlenecks, and exploit information from high-fidelity models and novel observational platforms.

53. Heterogeneous Basal Thermal Conditions Underpinning the Adélie-George V Coast, East Antarctica

Authors: Eliza J. Dawson (Department of Geophysics Stanford University Stanford CA USA); Dustin M. Schroeder (Department of Geophysics Stanford University Stanford CA USA, Department of Electrical Engineering Stanford University Stanford CA USA); Winnie Chu (Georgia Institute of Technology); Elisa Mantelli (Department of Earth and Environmental Sciences Ludwig-Maximilians-Universitaet Munich Germany, Alfred Wegener Institute); Helene Seroussi (Dartmouth College)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2023gl105450

URL: <https://doi.org/10.1029/2023gl105450>

Publication Date: 2024-01-19

Volume/Issue: Volume 51, Issue 2

References: 68

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY-NC 4.0

Open Access: N/A

Abstract: Adélie-George V Land in East Antarctica, encompassing the vast Wilkes Subglacial Basin, has a configuration that could be prone to ice sheet instability: the basin's retrograde bed slope could make its marine terminating glaciers vulnerable to warm seawater intrusion and irreversible retreat under predicted climate forcing. However, future projections are uncertain, due in part to limited subglacial observations near the grounding zone. Here, we develop a novel statistical approach to characterize subglacial conditions from radar sounding observations. Our method reveals intermixed frozen and thawed bed within 100 km of the grounding-zone near the Wilkes Subglacial Basin outflow, and enables comparisons to ice sheet model-inferred thermal states. The signs of intermixed or near thawed conditions raises the possibility that changes in basal thermal state could impact the stability of Adélie-George V Land, adding to the region's potentially vulnerable topographic configuration and sensitivity to ocean forcing driven grounding line retreat.

54. Ubiquitous acceleration in Greenland Ice Sheet calving from 1985 to 2022

Authors: Chad A. Greene; Alex S. Gardner; Michael Wood; Joshua K. Cuzzone

Journal: Nature

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41586-023-06863-2

URL: <https://doi.org/10.1038/s41586-023-06863-2>

Publication Date: 2024-01-17

Volume/Issue: Volume 625, Issue 7995

References: 84

Cited by: 18 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

55. What can radar-based measures of subglacial hydrology tell us about basal shear stress? A case study at Thwaites Glacier, West Antarctica

Authors: Rohaiz Haris; Winnie Chu; Alexander Robel

Journal: Journal of Glaciology

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/jog.2024.3

URL: <https://doi.org/10.1017/jog.2024.3>

Publication Date: 2024-01-15

Volume/Issue: Volume 70

References: 59

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: Ice sheet models use observations to infer basal shear stress, but the variety of methods and datasets available has resulted in a wide range of estimates. Radar-based metrics such as reflectivity and specularity content have been used to characterize subglacial hydrologic conditions that are linked to spatial variations in basal shear stress. We explore whether radar metrics can be used to inform models about basal shear stress. At Thwaites Glacier, West Antarctica, we sample basal shear stress inversions across a wide range of ice sheet models to see how the basal shear stress distribution changes in regions of varying relative reflectivity and specularity content. Our results reveal three key findings: (1) Regions of high specularity content exhibit lower mean basal shear stresses (2) Wet and bumpy regions, as characterized by high relative reflectivity and low specularity content, exhibit higher mean basal shear stresses (3) Models disagree about what basal shear stress should be at the onset of rapid ice flow and high basal melt where relative reflectivity and specularity content are low.

56. Using specularity content to evaluate eight geothermal heat flow maps of Totten Glacier

Authors: Yan Huang; Liyun Zhao; Michael Wolovick; Yiliang Ma; John C. Moore

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-18-103-2024

URL: <https://doi.org/10.5194/tc-18-103-2024>

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Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, National Key Research and Development Program of China, State Key Laboratory of Earth Surface Processes and Resource Ecology (and 1 more)

License: CC BY 4.0

Open Access: N/A

Abstract: . Geothermal heat flow (GHF) is the dominant factor affecting the basal thermal regime of ice sheet dynamics. But it is poorly defined for the Antarctic ice sheet. We compare the basal thermal state of the Totten Glacier catchment as simulated by eight different GHF datasets. We use a basal energy and water flow model coupled with a 3D full-Stokes ice dynamics model to estimate the basal temperature, basal friction heat and basal melting rate. In addition to the location of subglacial lakes, we use specularity content of the airborne radar returns as a two-sided constraint to discriminate between local wet or dry basal conditions and compare the returns with the basal state simulations with different GHFs. Two medium magnitude GHF distribution maps derived from seismic modelling rank well at simulating both cold- and warm-bed regions, the GHFs from Shen et al. (2020) and Shapiro and Ritzwoller (2004). The best-fit simulated result shows that most of the inland bed area is frozen. Only the central inland subglacial canyon, co-located with high specularity content, reaches the pressure melting point consistently in all the eight GHFs. Modelled basal melting rates in the slow-flowing region are generally 0-5 mm yr⁻¹ but with local maxima of 10 mm yr⁻¹ at the central inland subglacial canyon. The fast-flowing grounded glaciers close to the Totten ice shelf are lubricating their bases with meltwater at rates of 10-400 mm yr⁻¹.

57. A Variational LSTM Emulator of Sea Level Contribution From the Antarctic Ice Sheet

Authors: Peter Van Katwyk (Brown University); Baylor Fox-Kemper (Brown University); H  l  ne Seroussi (Dartmouth College); Sophie Nowicki (Department of Geology University at Buffalo Buffalo NY USA); Karianne J. Bergen (Brown University)

Journal: Journal of Advances in Modeling Earth Systems

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2023ms003899

URL: <https://doj.org/10.1029/2023ms003899>

Publication Date: 2023-12-23

Volume/Issue: Volume 15, Issue 12

References: 66

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: Antarctic ice sheet|Divergence (linguistics)

Funding: National Science Foundation, National Aeronautics and Space Administration

License: CC BY 4.0

Open Access: N/A

Abstract: The Antarctic ice sheet (AIS) will be a dominant contributor to global mean sea level rise in the 21st century but remains a major source of uncertainty. The Ice Sheet Model Intercomparison for CMIP6 (ISMIP6) is an ensemble of continental-scale models for studying the evolution of the AIS and projecting its future contribution to sea level. Due to their complexity and computational cost, ISMIP6 simulations are sparse and generated infrequently. Emulators are smaller-scale models that approximate ISMs and enable experimentation and exploration into the drivers of sea level change. We introduce a neural network (NN) emulator to approximate the ISMIP6 ensemble, using a variational Long Short-Term Memory (LSTM) with Monte Carlo dropout to quantify single-projection uncertainty. The proposed NN emulator is compared to a Gaussian Process (GP) emulator on four criteria: accuracy of point estimates and predictive distributions of individual model projections, approximation of the ensemble projections, and model training time. The NN predicts more accurately on single projections, with a mean absolute error of 0.46 mm Sea Level Equivalent (SLE) versus 0.73 mm SLE for the GP, and has more accurate uncertainty estimates. The NN emulator also better approximates the ensemble distribution of ISMIP6 model projections, with a Kullback-Leibler divergence of 18.26 versus 199.14 for GP at the projection year 2100. The NN enables more accurate experimentation with a reduced runtime, offering a new tool for understanding the important role of regional precipitation, ice sheet drainage systems, and interannual and longer timescale dynamics.

58. Impact of time-dependent data assimilation on ice flow model initialization and projections: a case study of Kjer Glacier, Greenland

Authors: Youngmin Choi; Helene Seroussi; Mathieu Morlighem; Nicole-Jeanne Schlegel; Alex Gardner

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-17-5499-2023

URL: <https://doi.org/10.5194/tc-17-5499-2023>

Publication Date: 2023-12-22

Volume/Issue: Volume 17, Issue 12

References: 71

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation

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Open Access: N/A

Abstract: . Ice sheet models are often initialized with data assimilation of present-day conditions, in which unknown model parameters are estimated using the inverse method. While assimilation of snapshot observations has been widely used for regional- and large-scale ice sheet simulations, data assimilation based on time-dependent data has recently started to emerge to constrain model parameters while capturing the transient evolution of the system. However, this method has been applied only to a few glaciers with fixed ice front positions, using spatially and temporally limited observations, and has not been applied to marine-terminating glaciers of the Greenland Ice Sheet that have been retreating over the last 30 years. In this study, we assimilate time series of surface velocity into a model of Kjer Glacier in West Greenland to better capture the observed acceleration over the past 3 decades. We compare snapshot and transient inverse methods and investigate the impact of initialization procedures on the parameters inferred, as well as model projections. We find that transient-calibrated simulations better capture past trends and better reproduce changes after the calibration period, even when a short period of observations is used. The results show the feasibility and clear benefits of a time-dependent data assimilation for initializing ice sheet models. This approach is now possible with the development of longer observational records, though it remains computationally challenging.

59. The Framework for Assessing Changes To Sea-level (FACTS) v1.0: a platform for characterizing parametric and structural uncertainty in future global, relative, and extreme sea-level change

Authors: Robert E. Kopp; Gregory G. Garner; Tim H. J. Hermans; Shantenu Jha; Praveen Kumar; Alexander Reedy; Aimée B. A. Slangen; Matteo Turilli; Tamsin L. Edwards; Jonathan M. Gregory; George Koubbe; Anders Levermann; Andre Merzky; Sophie Nowicki; Matthew D. Palmer; Chris Smith

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-16-7461-2023

URL: <https://doi.org/10.5194/gmd-16-7461-2023>

Publication Date: 2023-12-21

Volume/Issue: Volume 16, Issue 24

References: 108

Cited by: 28 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, National Aeronautics and Space Administration, Natural Environment Research Council (and 2 more)

License: CC BY 4.0

Open Access: N/A

Abstract: . Future sea-level rise projections are characterized by both quantifiable uncertainty and unquantifiable structural uncertainty. Thorough scientific assessment of sea-level rise projections requires analysis of both dimensions of uncertainty. Probabilistic sea-level rise projections evaluate the quantifiable dimension of uncertainty; comparison of alternative probabilistic methods provides an indication of structural uncertainty. Here we describe the Framework for Assessing Changes To Sea-level (FACTS), a modular platform for characterizing different probability distributions for the drivers of sea-level change and their consequences for global mean, regional, and extreme sea-level change. We demonstrate its application by generating seven alternative probability distributions under multiple emissions scenarios for both future global mean sea-level change and future relative and extreme sea-level change at New York City. These distributions, closely aligned with those presented in the Intergovernmental Panel on Climate Change Sixth Assessment Report, emphasize the role of the Antarctic and Greenland ice sheets as drivers of structural uncertainty in sea-level change projections.

60. The role of near-terminus conditions in the ice-flow speed of Upernavik Isstrøm in northwest Greenland

Authors: Kelsey M. Voss; Karen E. Alley; David A. Lilien; Dorthe Dahl-Jensen

Journal: Annals of Glaciology

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/aog.2023.76

URL: <https://doi.org/10.1017/aog.2023.76>

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References: 72

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Language: en

Subjects: N/A

Keywords: N/A

Funding: Natural Sciences and Engineering Research Council of Canada, Natural Sciences and Engineering Research Council of Canada

License: CC BY 4.0

Open Access: N/A

Abstract: Upernavik Isstrøm, the largest contributor to sea-level rise in northwest Greenland, has experienced complex and contrasting ice-flow-speed changes across its five outlets over the last two decades. In this study, we present a detailed remote-sensing analysis of the ice dynamics at Upernavik's outlets from 2000 to 2021 to evaluate the details of these changes. Previous research suggested that the presence or absence of floating ice tongues strongly influences Upernavik's ice dynamics. We use several lines of evidence to document the presence of floating ice tongues, and find that, while several outlets experienced ice-tongue formation and/or loss during the study period, these changes do not explain observed fluctuations in ice-flow velocity. Further exploration of ice-dynamic forcings using a flowline model suggests that changes in basal slipperiness near the terminus have a strong impact on upstream ice dynamics and can explain the velocity variations. Our results suggest that speed fluctuations at Upernavik's outlets may be seasonally and interannually controlled by bed conditions near the terminus, and highlight the need for further research on the influence of basal conditions on complex tidewater glacier dynamics.

61. Insights into the vulnerability of Antarctic glaciers from the ISMIP6 ice sheet model ensemble and associated uncertainty

Authors: Hélène Seroussi; Vincent Verjans; Sophie Nowicki; Antony J. Payne; Heiko Goelzer; William H. Lipscomb; Ayako Abe-Ouchi; Cécile Agosta; Torsten Albrecht; Xylar Asay-Davis; Alice Barthel; Reinhard Calov; Richard Cullather; Christophe Dumas; Benjamin K. Galton-Fenzi; Rupert Gladstone; Nicholas R. Golledge; Jonathan M. Gregory; Ralf Greve; Tore Hattermann; Matthew J. Hoffman; Angelika Humbert; Philippe Huybrechts; Nicolas C. Jourdain; Thomas Kleiner; Eric Larour; Gunter R. Leguy; Daniel P. Lowry; Christopher M. Little; Mathieu Morlighem; Frank Pattyn; Tyler Pelle; Stephen F. Price; Aurélien Quiquet; Ronja Reese; Nicole-Jeanne Schlegel; Andrew Shepherd; Erika Simon; Robin S. Smith; Fiammetta Straneo; Sainan Sun; Luke D. Trusel; Jonas Van Breedam; Peter Van Katwyk; Roderik S. W. van de Wal; Ricarda Winkelmann; Chen Zhao; Tong Zhang; Thomas Zwinger

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-17-5197-2023

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References: 60

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Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration, Office of Science, Research Council of Finland (and 10 more)

License: CC BY 4.0

Open Access: N/A

Abstract: . The Antarctic Ice Sheet represents the largest source of uncertainty in future sea level rise projections, with a contribution to sea level by 2100 ranging from -5 to 43 cm of sea level equivalent under high carbon emission scenarios estimated by the recent Ice Sheet Model Intercomparison for CMIP6 (ISMIP6). ISMIP6 highlighted the different behaviors of the East and West Antarctic ice sheets, as well as the possible role of increased surface mass balance in offsetting the dynamic ice loss in response to changing oceanic conditions in ice shelf cavities. However, the detailed contribution of individual glaciers, as well as the partitioning of uncertainty associated with this ensemble, have not yet been investigated. Here, we analyze the ISMIP6 results for high carbon emission scenarios, focusing on key glaciers around the Antarctic Ice Sheet, and we quantify their projected dynamic mass loss, defined here as mass loss through increased ice discharge into the ocean in response to changing oceanic conditions. We highlight glaciers contributing the most to sea level rise, as well as their vulnerability to changes in oceanic conditions. We then investigate the different sources of uncertainty and their relative role in projections, for the entire continent and for key individual glaciers. We show that, in addition to Thwaites and Pine Island glaciers in West Antarctica, Totten and Moscow University glaciers in East Antarctica present comparable future dynamic mass loss and high sensitivity to ice shelf basal melt. The overall uncertainty in additional dynamic mass loss in response to changing oceanic conditions, compared to a scenario with constant oceanic conditions, is dominated by the

choice of ice sheet model, accounting for 52 % of the total uncertainty of the Antarctic dynamic mass loss in 2100. Its relative role for the most dynamic glaciers varies between 14 % for MacAyeal and Whillans ice streams and 56 % for Pine Island Glacier at the end of the century. The uncertainty associated with the choice of climate model increases over time and reaches 13 % of the uncertainty by 2100 for the Antarctic Ice Sheet but varies between 4 % for Thwaites Glacier and 53 % for Whillans Ice Stream. The uncertainty associated with the ice-climate interaction, which captures different treatments of oceanic forcings such as the choice of melt parameterization, its calibration, and simulated ice shelf geometries, accounts for 22 % of the uncertainty at the ice sheet scale but reaches 36 % and 39 % for Institute Ice Stream and Thwaites Glacier, respectively, by 2100. Overall, this study helps inform future research by highlighting the sectors of the ice sheet most vulnerable to oceanic warming over the 21st century and by quantifying the main sources of uncertainty.

62. Controls on calving at a large Greenland tidewater glacier: stress regime, self-organised criticality and the crevasse-depth calving law

Authors: Douglas I. Benn; Joe Todd; Adrian Luckman; Suzanne Bevan; Thomas R. Chudley; Jan Åström; Thomas Zwinger; Samuel Cook; Poul Christoffersen

Journal: Journal of Glaciology

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/jog.2023.81

URL: <https://doi.org/10.1017/jog.2023.81>

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Volume/Issue: Volume 69, Issue 278

References: 47

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NERC Environmental Bioinformatics Centre, Natural Environment Research Council, Horizon 2020 Framework Programme

License: CC BY 4.0

Open Access: N/A

Abstract: We investigate the physical basis of the crevasse-depth (CD) calving law by analysing relationships between glaciological stresses and calving behaviour at Sermeq Kujalleq (Store Glacier), Greenland. Our observations and model simulations show that the glacier has a stable position defined by a compressive arch between lateral pinning points. Ice advance beyond the arch results in calving back to the stable position; conversely, if melt-undercutting forces the ice front behind the stable position, it readvances because ice velocities exceed subaqueous melt rates. This behaviour is typical of self-organising criticality, in which the stable ice-front position acts as an attractor between unstable super-critical and sub-critical regimes. This perspective provides strong support for a 'position-law' approach to modelling calving at Sermeq Kujalleq, because any calving 'rate' is simply a by-product of how quickly ice is delivered to the critical point. The CD calving law predicts ice-front position from the penetration of surface and basal crevasse fields, and accurately simulates super-critical calving back to the compressive arch and melt-driven calving into the sub-critical zone. The CD calving law reflects the glaciological controls on calving at Sermeq Kujalleq and exhibits considerable skill in simulating its mean position and seasonal fluctuations.

63. Recent irreversible retreat phase of Pine Island Glacier

Authors: Brad Reed; J. A. Mattias Green; Adrian Jenkins; G. Hilmar Gudmundsson

Journal: Nature Climate Change

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41558-023-01887-y

URL: <https://doi.org/10.1038/s41558-023-01887-y>

Publication Date: 2023-12-04

Volume/Issue: Volume 14, Issue 1

References: 74

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: RCUK | Natural Environment Research Council, EC | Horizon 2020 Framework Programme

License: CC BY 4.0

Open Access: N/A

Abstract: Pine Island Glacier (PIG), a part of the West Antarctic marine ice sheet, has recently undergone substantial changes including speed up, retreat and thinning. Theoretical arguments and modelling work suggest that marine ice sheets can become unstable and undergo irreversible retreat. Here, we use an ice-flow model validated by observational data to show that a rapid PIG retreat in the 1970s from a subglacial ridge to an upstream ice plain was self-enhancing and irreversible. The results suggest that by the early 1970s, the retreat of PIG had reached a point beyond which its original position at the ridge could not be recovered, even during subsequent periods of cooler ocean conditions. The irreversible phase ended by the early 1990s after almost 40 km of retreat and 0.34 mm added to global mean sea level, making PIG the main contributor from the Antarctic ice sheet in this period.

64. Regularization and L-curves in ice sheet inverse models: a case study in the Filchner-Ronne catchment

Authors: Michael Wolovick; Angelika Humbert; Thomas Kleiner; Martin Rückamp

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-17-5027-2023

URL: <https://doi.org/10.5194/tc-17-5027-2023>

Publication Date: 2023-11-29

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Cited by: 1 publications

Language: en

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Keywords: N/A

Funding: Alfred-Wegener-Institut, Helmholtz-Zentrum für Polar- und Meeresforschung

License: CC BY 4.0

Open Access: N/A

Abstract: . Over the past 3 decades, inversions for ice sheet basal drag have become commonplace in glaciological modeling. Such inversions require regularization to prevent over-fitting and ensure that the structure they recover is a robust inference from the observations, confidence which is required if they are to be used to draw conclusions about processes and properties of the ice base. While L-curve analysis can be used to select the optimal regularization level, the treatment of L-curve analysis in glaciological inverse modeling has been highly variable. Building on the history of glaciological inverse modeling, we demonstrate general best practices for regularizing glaciological inverse problems, using a domain in the Filchner-Ronne catchment of Antarctica as our test bed. We show a step-by-step approach to cost function normalization and L-curve analysis. We explore the spatial and spectral characteristics of the solution as a function of regularization, and we test the sensitivity of L-curve analysis and regularization to model resolution, effective pressure, sliding nonlinearity, and the flow equation. We find that the optimal regularization level converges towards a finite non-zero limit in the continuous problem, associated with a best knowable basal drag field. Nonlinear sliding laws outperform linear sliding in our analysis, with both a lower total variance and a more sharply cornered L-curve. By contrast, geometry-based approximations for effective pressure degrade inversion performance when added to a sliding law, but an actual hydrology model may marginally improve performance in some cases. Our results with 3D inversions suggest that the additional model complexity may not be justified by the 2D nature of the surface velocity data. We conclude with recommendations for best practices in future glaciological inversions.

65. Properties of water, carbon dioxide, and nitrogen ices in planetary surface environments

Authors: K.B. Fard; I.B. Smith

Journal: Icarus

Publisher: Elsevier BV

DOI: 10.1016/j.icarus.2023.115895

URL: <https://doi.org/10.1016/j.icarus.2023.115895>

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Language: en

Subjects: N/A

Keywords: N/A

Funding: Canada Research Chairs, Natural Sciences and Engineering Research Council of Canada

License: N/A

Open Access: N/A

Abstract: N/A

66. Evaluation of four calving laws for Antarctic ice shelves

Authors: Joel A. Wilner; Mathieu Morlighem; Gong Cheng

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-17-4889-2023

URL: <https://doi.org/10.5194/tc-17-4889-2023>

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Language: en

Subjects: N/A

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License: CC BY 4.0

Open Access: N/A

Abstract: . Many floating ice shelves in Antarctica buttress the ice streams feeding them, thereby reducing the discharge of icebergs into the ocean. The rate at which ice shelves calve icebergs and how fast they flow determine whether they advance, retreat, or remain stable, exerting a first-order control on ice discharge. To parameterize calving within ice sheet models, several empirical and physical calving "laws" have been proposed in the past few decades. Such laws emphasize dissimilar features, including along- and across-flow strain rates (the eigencalving law), a fracture yield criterion (the von Mises law), longitudinal stretching (the crevasse depth law), and a simple ice thickness threshold (the minimum thickness law), among others. Despite the multitude of established calving laws, these laws remain largely unvalidated for the Antarctic Ice Sheet, rendering it difficult to assess the broad applicability of any given law in Antarctica. We address this shortcoming through a set of numerical experiments that evaluate existing calving laws for 10 ice shelves around the Antarctic Ice Sheet. We utilize the Ice-sheet and Sea-level System Model (ISSM) and implement four calving laws under constant external forcing, calibrating the free parameter of each of these calving laws for each ice shelf by assuming that the current position of the ice front is in steady state and finding the set of parameters that best achieves this position over a simulation of 200 years. We find that, in general, the eigencalving and von Mises laws best reproduce observed calving front positions under the steady-state position assumption. These results will streamline future modeling efforts of Antarctic ice shelves by better informing the relevant physics of Antarctic-style calving on a shelf-by-shelf basis.

67. Basal conditions of Denman Glacier from glacier hydrology and ice dynamics modeling

Authors: Koi McArthur; Felicity S. McCormack; Christine F. Dow

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-17-4705-2023

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Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Australian Research Council, Canada Research Chairs, Natural Sciences and Engineering Research Council of Canada

License: CC BY 4.0

Open Access: N/A

Abstract: . Basal sliding in Antarctic glaciers is often modeled using a friction law that relates basal shear stresses to the effective pressure. As few ice sheet models are dynamically coupled to subglacial hydrology models, variability in subglacial hydrology associated with the effective pressure is often implicitly captured in the basal friction coefficient - an unknown parameter in the basal friction law. We investigate the impact of using effective pressures calculated from the Glacier Drainage System (GlaDS) model on basal friction coefficients calculated using inverse methods in the Ice-sheet and Sea-level System Model (ISSM) at Denman Glacier, East Antarctica, for the Schoof and Budd friction laws. For the Schoof friction law, a positive correlation emerges between the GlaDS effective pressure and basal friction coefficient in regions of fast ice flow. Using GlaDS effective pressures generally leads to smoother basal friction coefficients and basal shear stresses, and larger differences between the simulated and observed ice surface velocities compared with using an effective pressure equal to the ice overburden pressure plus the gravitational potential energy of the water. Compared with the Budd friction law, the Schoof friction law offers improved capabilities in capturing the spatial variations associated with known physics of the subglacial hydrology. Our results indicate that ice sheet model representation of basal sliding is more realistic when using direct outputs from a subglacial hydrology model, demonstrating the importance of coupling between ice sheet and subglacial hydrological systems. However, using our outputs we have also developed an empirical parameterization of effective pressure that improves the application of the Schoof friction law without requiring explicit hydrological modeling.

68. Reconstructing subglacial lake activity with an altimetry-based inverse method

Authors: Aaron G. Stubblefield; Colin R. Meyer; Matthew R. Siegfried; Wilson Sauthoff; Marc Spiegelman

Journal: Journal of Glaciology

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/jog.2023.90

URL: <https://doi.org/10.1017/jog.2023.90>

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Volume/Issue: Volume 69, Issue 278

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Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration, National Aeronautics and Space Administration, Office of Polar Programs (and 1 more)

License: CC BY 4.0

Open Access: N/A

Abstract: Subglacial lake water-volume changes produce ice-elevation anomalies that provide clues about water flow beneath glaciers and ice sheets. Significant challenges remain in the quantitative interpretation of these elevation-change anomalies because the surface expression of subglacial lake activity depends on basal conditions, rate of water-volume change, and ice rheology. To address these challenges, we introduce an inverse method that reconstructs subglacial lake activity from altimetry data while accounting for the effects of viscous ice flow. We use a linearized approximation of a Stokes ice-flow model under the assumption that subglacial lake activity only induces small perturbations relative to a reference ice-flow state. We validate this assumption by accurately reconstructing lake activity from synthetic data that are produced with a fully nonlinear model. We then apply the method to estimate the water-volume changes of several active subglacial lakes in Antarctica by inverting data from NASA's Ice, Cloud, and land Elevation Satellite 2 (ICESat-2) laser altimetry mission. The results show that there can be substantial discrepancies (20% or more) between the inversion and traditional estimation methods due to the effects of viscous ice flow. The inverse method will help refine estimates of subglacial water transport and further constrain the role of subglacial hydrology in ice-sheet evolution.

69. A Review of Bayesian Modelling in Glaciology

Authors: Giri Gopalan; Andrew Zammit-Mangion; Felicity McCormack

Journal: Statistical Modeling Using Bayesian Latent Gaussian Models

Publisher: Springer International Publishing

DOI: 10.1007/978-3-031-39791-2_2

URL: https://doi.org/10.1007/978-3-031-39791-2_2

Publication Date: 2023-11-08

Volume/Issue: N/A

References: 75

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

70. Choice of observation type affects Bayesian calibration of Greenland Ice Sheet model simulations

Authors: Denis Felikson; Sophie Nowicki; Isabel Nias; Beata Csatho; Anton Schenk; Michael J. Croteau; Bryant Loomis

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-17-4661-2023

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Publication Date: 2023-11-07

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Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration

License: CC BY 4.0

Open Access: N/A

Abstract: . Determining reliable probability distributions for ice sheet mass change over the coming century is critical to refining uncertainties in sea-level rise projections. Bayesian calibration, a method for constraining projection uncertainty using observations, has been previously applied to ice sheet projections but the impact of the chosen observation type on the calibrated posterior probability distributions has not been quantified. Here, we perform three separate Bayesian calibrations to constrain uncertainty in Greenland Ice Sheet (GrIS) simulations of the committed mass loss in 2100 under the current climate, using observations of velocity change, dynamic ice thickness change, and mass change. Comparing the posterior probability distributions shows that the median ice sheet mass change can differ by 119 % for the particular model ensemble that we used, depending on the observation type used in the calibration. More importantly for risk-averse sea-level planning, posterior probabilities of high-end mass change scenarios are highly sensitive to the observation selected for calibration. Furthermore, we show that using mass change observations alone may result in model simulations that overestimate flow acceleration and underestimate dynamic thinning around the margin of the ice sheet. Finally, we look ahead and present ideas for ways to improve Bayesian calibration of ice sheet projections.

71. Assessing the potential for ice flow piracy between the Totten and Vanderford glaciers, East Antarctica

Authors: Felicity S. McCormack; Jason L. Roberts; Bernd Kulesa; Alan Aitken; Christine F. Dow; Lawrence Bird; Benjamin K. Galton-Fenzi; Katharina Hochmuth; Richard S. Jones; Andrew N. Mackintosh; Koi McArthur

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-17-4549-2023

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Cited by: 7 publications

Language: en

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Keywords: N/A

Funding: Australian Research Council, Natural Sciences and Engineering Research Council of Canada, Canada Research Chairs

License: CC BY 4.0

Open Access: N/A

Abstract: . The largest regional drivers of current surface elevation increases in the Antarctic Ice Sheet are associated with ice flow reconfiguration in previously active ice streams, highlighting the important role of ice dynamics in mass balance calculations. Here, we investigate controls on the evolution of the flow configuration of the Vanderford and Totten glaciers - key outlet glaciers of the Aurora Subglacial Basin (ASB) - the most rapidly thinning region of the East Antarctic Ice Sheet (EAIS). We synthesise factors that influence the ice flow in this region and use an ice sheet model to investigate the sensitivity of the catchment divide location to changes in surface elevation due to thinning at the Vanderford Glacier (VG) associated with ongoing retreat and thickening at the Totten Glacier (TG) associated with an intensification of the east-west snowfall gradient. The present-day catchment divide between the Totten and Vanderford glaciers is not constrained by the geology or topography but is determined by the large-scale ice sheet geometry and its long-term evolution in response to climate forcing. Furthermore, the catchment divide migrates under relatively small changes in surface elevation, leading to ice flow and basal water piracy from the Totten to the Vanderford Glacier. Our findings show that ice flow reconfigurations occur not only in regions of West Antarctica like the Siple Coast but also in the east, motivating further investigations of past, and the potential for future, ice flow reconfigurations around the whole Antarctic coastline. Modelling of ice flow and basal water piracy may require coupled ice sheet thermomechanical and subglacial hydrology models constrained by field observations of subglacial conditions. Our results have implications for ice sheet mass budget studies that integrate over catchments and the validity of the zero flow assumption when selecting sites for ice core records of past climate.

72. Subglacial discharge accelerates future retreat of Denman and Scott Glaciers, East Antarctica

Authors: Tyler Pelle (Scripps Institution of Oceanography / University of California San Diego); Jamin S. Greenbaum (Scripps Institution of Oceanography / University of California San Diego); Christine F. Dow (University of Waterloo); Adrian Jenkins (Department of Geography and Environmental Sciences, Northumbria University, Newcastle upon Tyne, NE1 8ST, UK.); Mathieu Morlighem (Dartmouth College)

Journal: Science Advances

Publisher: American Association for the Advancement of Science (AAAS)

DOI: 10.1126/sciadv.adi9014

URL: <https://doi.org/10.1126/sciadv.adi9014>

Publication Date: 2023-10-27

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Cited by: 12 publications

Language: en

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Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: Ice shelf basal melting is the primary mechanism driving mass loss from the Antarctic Ice Sheet, yet it is unknown how the localized melt enhancement from subglacial discharge will affect future Antarctic glacial retreat. We develop a parameterization of ice shelf basal melt that accounts for both ocean and subglacial discharge forcing and apply it in future projections of Denman and Scott Glaciers, East Antarctica, through 2300. In forward simulations, subglacial discharge accelerates the onset of retreat of these systems into the deepest continental trench on Earth by 25 years. During this retreat, Denman Glacier alone contributes 0.33 millimeters per year to global sea level rise, comparable to half of the contemporary sea level contribution of the entire Antarctic Ice Sheet. Our results stress the importance of resolving complex interactions between the ice, ocean, and subglacial environments in future Antarctic Ice Sheet projections.

73. Fractures in glaciers--Crack tips and their stress fields by observation and modeling

Authors: Angelika Humbert (University of Bremen, Alfred Wegener Institute); Dietmar Gross (Technical University of Darmstadt); Rabea Sondershaus (Technical University of Darmstadt); Ralf Müller (Technical University of Darmstadt); Holger Steeb (University of Stuttgart); Matthias Braun (Institut für Geographie Friedrich-Alexander-Universität Erlangen-Nürnberg Erlangen Germany); Jörg Brauchle (German Aerospace Center Institute of Optical Sensor Systems Berlin-Adlershof Germany); Karsten Stebner (German Aerospace Center Institute of Optical Sensor Systems Berlin-Adlershof Germany); Martin Rückamp (Bavarian Academy of Science and Humanities)

Journal: PAMM

Publisher: Wiley

DOI: 10.1002/pamm.202300260

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References: 10

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Language: en

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Keywords: N/A

Funding: Deutsche Forschungsgemeinschaft

License: Creative Commons

Open Access: N/A

Abstract: High-resolution optical camera systems are opening new opportunities to study fractures in ice. Here, we present data obtained from the Modular Aerial Camera System camera system operated onboard of Alfred Wegener Institute Helmholtz Centre for Polar and Marine Research (AWI) polar aircraft in northeast Greenland in 2022. In addition, we are using optical and radar satellite imagery. The study area is the 79 degreesN Glacier (Nioghalvfjærdsbræ, 79NG), an outlet glacier of the Northeast Greenland Ice Stream. We found that crack tips are exhibiting additional isolated cracks ahead of the main crack. Subsequent crack propagation is starting from those isolated cracks, leading to an advance of the crack, with bridges between crack faces. The bridges provide information of the episodic crack propagation. Fractures have typically a length scale of kilometers and the distance of crack faces is in the order of meters to tenths of meters. Fracture modes will be inferred from stress fields computed by an inverse modeling approach using the Ice Sheet and Sea Level System Model. To this end, a surface velocity field derived from satellite remote sensing is used for the optimal control method that constrains model parameters, for example, basal friction coefficient or rheology.

74. Sensitivity analysis of a King George Island outlet glacier, South Shetlands, Antarctica

Authors: THIAGO DIAS DOS SANTOS (Universidade de São Paulo, Brazil; Universidade Federal do Rio Grande do Sul, Brazil; University of California, USA); MATHIEU MORLIGHEM (Dartmouth College); JEFFERSON CARDIA SIMÕES (Universidade Federal do Rio Grande do Sul, Brazil); PHILIPPE REMY BERNARD DEVLOO (Universidade Estadual de Campinas, Brazil)

Journal: Anais da Academia Brasileira de Ciências

Publisher: FapUNIFESP (SciELO)

DOI: 10.1590/0001-3765202320210560

URL: <https://doi.org/10.1590/0001-3765202320210560>

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References: 73

Cited by: N/A publications

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Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: N/A

75. Development and Benchmarking of the Shallow Shelf Approximation Ice Sheet Dynamics Module

Authors: Yi-Jeong Baek; Su-Jeong Lim; Byung-Dal So

Journal: Ocean Science Journal

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s12601-023-00120-3

URL: <https://doi.org/10.1007/s12601-023-00120-3>

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Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Research Foundation of Korea, National Research Foundation of Korea, Korea Institute of Ocean & Technology (KIOST) research program

License: N/A

Open Access: N/A

Abstract: N/A

76. A parallel implementation of the confined-unconfined aquifer system model for subglacial hydrology: design, verification, and performance analysis (CUAS-MPI v0.1.0)

Authors: Yannic Fischler; Thomas Kleiner; Christian Bischof; Jeremie Schmiedel; Roiy Sayag; Raban Emunds; Lennart Frederik Oestreich; Angelika Humbert

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-16-5305-2023

URL: <https://doi.org/10.5194/gmd-16-5305-2023>

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Volume/Issue: Volume 16, Issue 18

References: 65

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: German-Israeli Foundation for Scientific Research and Development

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Open Access: N/A

Abstract: . The subglacial hydrological system affects (i) the motion of ice sheets through sliding, (ii) the location of lakes at the ice margin, and (iii) the ocean circulation by freshwater discharge directly at the grounding line or (iv) via rivers flowing over land. For modeling this hydrology system, a previously developed porous-media concept called the confined-unconfined aquifer system (CUAS) is used. To allow for realistic simulations at the ice sheet scale, we developed CUAS-MPI, an MPI-parallel C/C++ implementation of CUAS (MPI: Message Passing Interface), which employs the Portable, Extensible Toolkit for Scientific Computation (PETSc) infrastructure for handling grids and equation systems. We validate the accuracy of the numerical results by comparing them with a set of analytical solutions to the model equations, which involve two types of boundary conditions. We then investigate the scaling behavior of CUAS-MPI and show that CUAS-MPI scales up to 3840 MPI processes running a realistic Greenland setup on the Lichtenberg HPC system. Our measurements also show that CUAS-MPI reaches a throughput comparable to that of ice sheet simulations, e.g., the Ice-sheet and Sea-level System Model (ISSM). Lastly, we discuss opportunities for ice sheet modeling, explore future coupling possibilities of CUAS-MPI with other simulations, and consider throughput bottlenecks and limits of further scaling.

77. A thicker Antarctic ice stream during the mid-Pliocene warm period

Authors: Martim Mas e Braga; Richard S. Jones; Jorge Bernales; Jane Lund Andersen; Ola Fredin; Mathieu Morlighem; Alexandria J. Koester; Nathaniel A. Lifton; Jonathan M. Harbor; Yusuke Suganuma; Neil F. Glasser; Irina Rogozhina; Arjen P. Stroeven

Journal: Communications Earth & Environment

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s43247-023-00983-3

URL: <https://doi.org/10.1038/s43247-023-00983-3>

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References: 84

Cited by: 3 publications

Language: en

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Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: Ice streams regulate most ice mass loss in Antarctica. Determining ice stream response to warmer conditions during the Pliocene could provide insights into their future behaviour, but this is hindered by a poor representation of subglacial topography in ice-sheet models. We address this limitation using a high-resolution model for Dronning Maud Land (East Antarctica). We show that contrary to dynamic thinning of the region's ice streams following ice-shelf collapse, the largest ice stream, Jutulstraumen, thickens by 700 m despite lying on a retrograde bed slope. We attribute this counterintuitive thickening to a shallower Pliocene subglacial topography and inherent high lateral stresses at its flux gate. These conditions constrict ice drainage and, combined with increased snowfall, allow ice accumulation upstream. Similar stress balances and increased precipitation projections occur across 27% of present-day East Antarctica, and understanding how lateral stresses regulate ice-stream discharge is necessary for accurately assessing Antarctica's future sea-level rise contribution.

78. Statistically parameterizing and evaluating a positive degree-day model to estimate surface melt in Antarctica from 1979 to 2022

Authors: Yaowen Zheng; Nicholas R. Golledge; Alexandra Gossart; Ghislain Picard; Marion Leduc-Leballeur

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-17-3667-2023

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References: 60

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Royal Society Te Apārangi, Ministry for Business Innovation and Employment

License: CC BY 4.0

Open Access: N/A

Abstract: . Surface melting is one of the primary drivers of ice shelf collapse in Antarctica and is expected to increase in the future as the global climate continues to warm because there is a statistically significant positive relationship between air temperature and melting. Enhanced surface melt will impact the mass balance of the Antarctic Ice Sheet (AIS) and, through dynamic feedbacks, induce changes in global mean sea level (GMSL). However, the current understanding of surface melt in Antarctica remains limited in terms of the uncertainties in quantifying surface melt and understanding the driving processes of surface melt in past, present and future contexts. Here, we construct a novel grid-cell-level spatially distributed positive degree-day (PDD) model, forced with 2 m air temperature reanalysis data and spatially parameterized by minimizing the error with respect to satellite estimates and surface energy balance (SEB) model outputs on each computing cell over the period 1979 to 2022. We evaluate the PDD model by performing a goodness-of-fit test and cross-validation. We assess the accuracy of our parameterization method, based on the performance of the PDD model when considering all computing cells as a whole, independently of the time window chosen for parameterization. We conduct a sensitivity experiment by adding $\pm 10\%$ to the training data (satellite estimates and SEB model outputs) used for PDD parameterization and a sensitivity experiment by adding constant temperature perturbations (+1, +2, +3, +4 and +5 $^{\circ}\text{C}$) to the 2 m air temperature field to force the PDD model. We find that the PDD melt extent and amounts change analogously to the variations in the training data with steady statistically significant correlations and that the PDD melt amounts increase nonlinearly with the temperature perturbations, demonstrating the consistency of our parameterization and the applicability of the PDD model to warmer climate scenarios. Within the limitations discussed, we suggest that an appropriately parameterized PDD model can be a valuable tool for exploring Antarctic surface melt beyond the satellite era.

79. Inferring the Basal Friction Law From Long Term Changes of Glacier Length, Thickness and Velocity on an Alpine Glacier

Authors: A. Gilbert (Institut de Recherche pour le Développement (IRD)); F. Gimbert (Institut de Recherche pour le Développement (IRD)); O. Gagliardini (Institut de Recherche pour le Développement (IRD)); C. Vincent (Institut de Recherche pour le Développement (IRD))

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2023gl104503

URL: <https://doi.org/10.1029/2023gl104503>

Publication Date: 2023-08-29

Volume/Issue: Volume 50, Issue 16

References: 56

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: ice dynamics, mountain glaciers, glacier sliding

Funding: Agence Nationale de la Recherche

License: CC BY 4.0

Open Access: N/A

Abstract: Basal sliding of glaciers and ice sheets remains a source of uncertainty in simulating the long-term evolution of ice masses. In particular, the response of ice flow to changes in driving stress depends strongly on the value of the exponent m in nonlinear friction laws (e.g., Weertman's law), which is poorly constrained by observations. Here we constrain the friction law at a natural scale on Argentière Glacier (French Alps, hard-bed), taking advantage of well-resolved observations of glacier mass balance, geometry and basal sliding over time spans that include large changes in driving stress. By combining three different independent methods based on (a) surface velocity inversion, (b) transient length change modeling, and (c) direct local sliding measurements, we consistently find a value of $m = 3.1 \pm 0.3$. We suggest that Weertman's law is suitable for modeling the long-term evolution of hard-bedded glaciers and ice sheets.

80. Widespread slowdown in thinning rates of West Antarctic ice shelves

Authors: Fernando S. Paolo; Alex S. Gardner; Chad A. Greene; Johan Nilsson; Michael P. Schodlok; Nicole-Jeanne Schlegel; Helen A. Fricker

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-17-3409-2023

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References: 97

Cited by: 3 publications

Language: en

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Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: . Antarctica's floating ice shelves modulate discharge of grounded ice into the ocean by providing a backstress. Ice shelf thinning and grounding line retreat have reduced this backstress, driving rapid drawdown of key unstable areas of the Antarctic Ice Sheet, leading to sea-level rise. If ice shelf loss continues, it may initiate irreversible glacier retreat through the marine ice sheet instability. Identification of areas undergoing significant change requires knowledge of spatial and temporal patterns in recent ice shelf loss. We used 26 years (1992-2017) of satellite-derived Antarctic ice shelf thickness, flow, and basal melt rates to construct a time-dependent dataset of ice shelf thickness and basal melt on a 3 km grid every 3 months. We used a novel data fusion approach, state-of-the-art satellite-derived velocities, and a new surface mass balance model. Our data revealed an overall pattern of thinning all around Antarctica, with a thinning slowdown starting around 2008 widespread across the Amundsen, Bellingshausen, and Wilkes sectors. We attribute this slowdown partly to modulation in external ocean forcing, altered in West Antarctica by negative feedbacks between ice shelf thinning rates and grounded ice flow, and sub-ice-shelf cavity geometry and basal melting. In agreement with earlier studies, the highest rates of ice shelf thinning are found for those ice shelves located in the Amundsen and Bellingshausen sectors. Our study reveals that over the 1992-2017 observational period the Amundsen and Bellingshausen ice shelves experienced a slight reduction in rates of basal melting, suggesting that high rates of thinning are largely a response to changes in ocean conditions that predate our satellite altimetry record, with shorter-term variability only resulting in small deviations from the long-term trend. Our work demonstrates that causal inference drawn from ice shelf thinning and basal melt rates must take into account complex feedbacks between thinning and ice advection and between ice shelf draft and basal melt rates.

81. A hybrid deep neural operator/finite element method for ice-sheet modeling

Authors: QiZhi He; Mauro Perego; Amanda A. Howard; George Em Karniadakis; Panos Stinis

Journal: Journal of Computational Physics

Publisher: Elsevier BV

DOI: 10.1016/j.jcp.2023.112428

URL: <https://doi.org/10.1016/j.jcp.2023.112428>

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Volume/Issue: Volume 492

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Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

82. Amundsen Sea Embayment ice-sheet mass-loss predictions to 2050 calibrated using observations of velocity and elevation change

Authors: Suzanne Bevan; Stephen Cornford; Lin Gilbert; Inés Otosaka; Daniel Martin; Trystan Surawy-Stepney

Journal: Journal of Glaciology

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/jog.2023.57

URL: <https://doi.org/10.1017/jog.2023.57>

Publication Date: 2023-08-14

Volume/Issue: Volume 69, Issue 278

References: 47

Cited by: 4 publications

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Keywords: N/A

Funding: Horizon 2020 Framework Programme

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Open Access: N/A

Abstract: Mass loss from the Amundsen Sea Embayment of the West Antarctic Ice Sheet is a major contributor to global sea-level rise (SLR) and has been increasing over recent decades. Predictions of future SLR are increasingly modelled using ensembles of simulations within which model parameters and external forcings are varied within credible ranges. Accurately reporting the uncertainty associated with these predictions is crucial in enabling effective planning for, and construction of defences against, rising sea levels. Calibrating model simulations against current observations of ice-sheet behaviour enables the uncertainty to be reduced. Here we calibrate an ensemble of BISICLES ice-sheet model simulations of ice loss from the Amundsen Sea Embayment using remotely sensed observations of surface elevation and ice speed. Each calibration type is shown to be capable of reducing the 90% credibility bounds of predicted contributions to SLR by 34 and 43% respectively.

83. Missing sea level rise in southeastern Greenland during and since the Little Ice Age

Authors: Sarah A. Woodroffe; Leanne M. Wake; Kristian K. Kjeldsen; Natasha L. M. Barlow; Antony J. Long; Kurt H. Kjær

Journal: Climate of the Past

Publisher: Copernicus GmbH

DOI: 10.5194/cp-19-1585-2023

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License: CC BY 4.0

Open Access: N/A

Abstract: . The Greenland Ice Sheet has been losing mass at an accelerating rate over the past 2 decades. Understanding ice mass and glacier changes during the preceding several hundred years prior to geodetic measurements is more difficult because evidence of past ice extent in many places was later overridden. Salt marshes provide the only continuous records of relative sea level (RSL) from close to the Greenland Ice Sheet that span the period of time during and since the Little Ice Age (LIA) and can be used to reconstruct ice mass gain and loss over recent centuries. Salt marsh sediments collected at the mouth of Dronning Marie Dal, close to the Greenland Ice Sheet margin in southeastern Greenland, record RSL changes over the past ca. 300 years through changing sediment and diatom stratigraphy. These RSL changes record a combination of processes that are dominated by local and regional changes in Greenland Ice Sheet mass balance during this critical period that spans the maximum of the LIA and 20th-century warming. In the early part of the record (1725-1762 CE) the rate of RSL rise is higher than reconstructed from the closest isolation basin at Timmiarmiut, but between 1762 and 1880 CE the RSL rate is within the error range of the rate of RSL change recorded in the isolation basin. RSL begins to slowly fall around 1880 CE, with a total amount of RSL fall of 0.09 $\pm$ 0.1 m in the last 140 years. Modelled RSL, which takes into account contributions from post-LIA Greenland Ice Sheet glacio-isostatic adjustment (GIA), ongoing deglacial GIA, the global non-ice sheet glacial melt fingerprint, contributions from thermohaline effects, the Antarctic mass loss sea level fingerprint and terrestrial water storage, overpredicts the amount of RSL fall since the end of the LIA by at least 0.5 m. The GIA signal caused by post-LIA Greenland Ice Sheet mass loss is by far the largest contributor to this modelled RSL, and error in its calculation has a large impact on RSL predictions at Dronning Marie Dal. We cannot reconcile the modelled RSL and the salt marsh observations, even when moving the termination of the LIA to 1700 CE and reducing the post-LIA Greenland mass loss signal by 30 %, and a "budget residual" of $\sim$ 3 mm yr⁻¹ since the end of the LIA remains unexplained. This new RSL record backs up other studies that suggest that there are significant regional differences in the timing and magnitude of the response of the Greenland Ice Sheet to the climate shift from the LIA into the 20th

century.

84. Melting ice and rising seas - connecting projected change in Antarctica's ice sheets to communities in Aotearoa New Zealand

Authors: Richard Levy (GNS Science - Te Pu Ao, Lower Hutt, New Zealand, Victoria University of Wellington); Timothy Naish (Victoria University of Wellington); Daniel Lowry (GNS Science - Te Pu Ao, Lower Hutt, New Zealand); Rebecca Priestley (Te Pūnaha Matatini, Centre of Research Excellence for Complex Systems and Networks, Auckland, New Zealand, Victoria University of Wellington); Rachelle Winefield (Te Pūnaha Matatini, Te Whenua Land Information New Zealand, Wellington, New Zealand); Alanna Alevropolous-Borrill (Victoria University of Wellington); Emory Beck (Te Pūnaha Matatini, Te Whenua Land Information New Zealand, Wellington, New Zealand); Rob Bell (Environmental Planning Programme, School of Social Sciences, University of Waikato, Hamilton, New Zealand, Bell Adapt Ltd, Hamilton, New Zealand); Graeme Blick (Te Pūnaha Matatini, Te Whenua Land Information New Zealand, Wellington, New Zealand); Ruzica Dadic (WSL Institute of Snow and Avalanche Research SLF, Davos, Switzerland, Victoria University of Wellington); Tasman Gillies (Takiwa Data Analytics, Auckland, New Zealand); Nicholas Golledge (Victoria University of Wellington); Zoe Heine (Victoria University of Wellington); Stefan Jendersie (Victoria University of Wellington); Judy Lawrence (Victoria University of Wellington); Katherine O'Leary (GNS Science - Te Pu Ao, Lower Hutt, New Zealand); Ryan Paulik (National Institute of Water and Atmosphere, Wellington, New Zealand); Ceridwyn Roberts (Ceridwyn Roberts Science Communication, Wellington, New Zealand); Mike Taitoko (Takiwa Data Analytics, Auckland, New Zealand); Natalie Trayling (GNS Science - Te Pu Ao, Lower Hutt, New Zealand)

Journal: Journal of the Royal Society of New Zealand

Publisher: Informa UK Limited

DOI: 10.1080/03036758.2023.2232743

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Volume/Issue: Volume 54, Issue 4

References: 121

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: New Zealand Ministry of Business, Innovation and Employment, Our Changing Coast Programme, New Zealand Antarctic Science Platform

License: Creative Commons

Open Access: N/A

Abstract: N/A

85. The Glacier-Climate Interaction Over the Tibetan Plateau and Its Surroundings During the Last Glacial Maximum

Authors: Qiang Wei (Peking University); Yonggang Liu (Peking University); Qing Yan (Nanjing University, Chinese Academy of Sciences); Tandong Yao (Chinese Academy of Sciences); Miao Wang (China Meteorological Administration); Han Huang (McGill University); Yongyun Hu (Peking University)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2023gl103538

URL: <https://doi.org/10.1029/2023gl103538>

Publication Date: 2023-07-20

Volume/Issue: Volume 50, Issue 14

References: 68

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: paleoclimate, glacier, Last Glacial Maximum, Tibetan Plateau, coupled simulation

Funding: National Natural Science Foundation of China

License: Creative Commons

Open Access: N/A

Abstract: Glacier growth affects the local climate, and in turn, can either promote or prohibit its own growth. Such feedback has not been considered in modeling the glaciers of the Tibetan Plateau and its surroundings (TPS) during the Last Glacial Maximum (LGM; ~28-23 ka). We find that the volume/area of the glaciers simulated by a coupled glacier-climate model is 20%/10% less than that by a standalone glacier model forced with fixed climate fields; glaciers advance toward their western rims and yet decrease in the interior of TPS. Such changes in spatial patterns improve model-data comparison. Moreover, the expansion of glaciers warms the winter surface temperature of the eastern TPS and decreases precipitation almost everywhere. These effects are primarily due to the added surface elevation, which blocks the water vapor brought by westerlies and south-westerlies, reducing precipitation and increasing surface temperatures to the east and northeast of the newly grown glaciers.

86. The internal structure of a debris-covered glacier on Mars revealed by gully incision

Authors: F.E.G. Butcher; N.S. Arnold; S.J. Conway; D.C. Berman; J.M. Davis; M.R. Balme

Journal: Icarus

Publisher: Elsevier BV

DOI: 10.1016/j.icarus.2023.115717

URL: <https://doi.org/10.1016/j.icarus.2023.115717>

Publication Date: 2023-07-20

Volume/Issue: Volume 419

References: 66

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

87. Precursor of disintegration of Greenland's largest floating ice tongue

Authors: Angelika Humbert; Veit Helm; Niklas Neckel; Ole Zeising; Martin Rückamp; Shfaqat Abbas Khan; Erik Loebel; Jörg Brauchle; Karsten Stebner; Dietmar Gross; Rabea Sondershaus; Ralf Müller

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-17-2851-2023

URL: <https://doi.org/10.5194/tc-17-2851-2023>

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Volume/Issue: Volume 17, Issue 7

References: 56

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Bundesministerium für Bildung und Forschung, Deutsche Forschungsgemeinschaft, Helmholtz Association

License: CC BY 4.0

Open Access: N/A

Abstract: . The largest floating tongue of Greenland's ice sheet, Nioghalvfjærdsbræ, has been relatively stable with respect to areal retreat until 2022. Draining more than 6 % of the ice sheet, a disintegration of Nioghalvfjærdsbræ's floating tongue and subsequent acceleration due to loss in buttressing are likely to lead to sea level rise. Therefore, the stability of the floating tongue is a focus of this study. We employed a suite of observational methods to detect recent changes at the calving front. We found that the calving style has changed since 2016 at the southern part of the eastern calving front, from tongue-type calving to a crack evolution initiated at frontal ice rises reaching 5-7 km and progressing further upstream compared to 2010. The calving front area is further weakened by an area upstream of the main calving front that consists of open water and an ice mélange that has substantially expanded, leading to the formation of a narrow ice bridge. These geometric and mechanical changes may be a precursor of instability of the floating tongue. We complement our study by numerical ice flow simulations to estimate the impact of future ice-front retreat and complete ice shelf disintegration on the discharge of grounded ice. These idealized scenarios reveal that a loss of the south-eastern area of the ice shelf would lead to a 0.2 % increase in ice discharge at the grounding line, while a sudden collapse of the frontal area (46 % of the floating tongue area) will enhance the ice discharge by 5.1 % due to loss in buttressing. Eventually, a full collapse of the floating tongue increases the grounding line flux by 166 %.

88. Brief communication: Is vertical shear in an ice shelf (still) negligible?

Authors: Chris Miele; Timothy C. Bartholomaus; Ellyn M. Enderlin

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-17-2701-2023

URL: <https://doi.org/10.5194/tc-17-2701-2023>

Publication Date: 2023-07-12

Volume/Issue: Volume 17, Issue 7

References: 20

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, National Aeronautics and Space Administration

License: CC BY 4.0

Open Access: N/A

Abstract: . Vertical shear is recognized today as a key component of the stress balance of ice shelves. However, the first ice shelf models were built on the neglect of vertical shear. Partly due to its historical treatment, it remains common to discuss vertical shear as though it were still considered negligible in ice shelf models. Here, we offer a historical perspective on the changing treatment of vertical shear over time, and we emphasize the term's non-negligibility in current ice shelf modeling. We illustrate our discussion in the simplest context of an analytic, isothermal, shallow-ice-shelf model.

89. Satellite record reveals 1960s acceleration of Totten Ice Shelf in East Antarctica

Authors: Rongxing Li; Yuan Cheng; Tian Chang; David E. Gwyther; Martin Forbes; Lu An; Menglian Xia; Xiaohan Yuan; Gang Qiao; Xiaohua Tong; Wenkai Ye

Journal: Nature Communications

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41467-023-39588-x

URL: <https://doi.org/10.1038/s41467-023-39588-x>

Publication Date: 2023-07-10

Volume/Issue: Volume 14, Issue 1

References: 70

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China

License: CC BY 4.0

Open Access: N/A

Abstract: Wilkes Land and Totten Glacier (TG) in East Antarctica (EA) have been losing ice mass significantly since 1989. There is a lack of knowledge of long-term mass balance in the region which hinders the estimation of its contribution to global sea level rise. Here we show that this acceleration trend in TG has occurred since the 1960s. We reconstruct ice flow velocity fields of 1963-1989 in TG from the first-generation satellite images of ARGON and Landsat-1&4, and build a five decade-long record of ice dynamics. We find a persistent long-term ice discharge rate of 68 ± 1 Gt/y and an acceleration of 0.17 ± 0.02 Gt/y² from 1963 to 2018, making TG the greatest contributor to global sea level rise in EA. We attribute the long-term acceleration near grounding line from 1963 to 2018 to basal melting likely induced by warm modified Circumpolar Deep Water. The speed up in shelf front during 1973-1989 was caused by a large calving front retreat. As the current trend continues, intensified monitoring in the TG region is recommended in the next decades.

90. Review "Evaluation of four calving laws for Antarctic ice shelves"

Authors: Javier Blasco

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/tc-2023-86-rc2

URL: <https://doi.org/10.5194/tc-2023-86-rc2>

Publication Date: 2023-07-05

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Crevasse|Iceberg|Buttress

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

91. Performance portable ice-sheet modeling with MALI

Authors: Jerry Watkins (Sandia National Laboratories, Livermore, CA, USA); Max Carlson (Sandia National Laboratories, Livermore, CA, USA); Kyle Shan (Micron Technology, Boise, ID, USA); Irina Tezaur (Sandia National Laboratories, Livermore, CA, USA); Mauro Perego (Sandia National Laboratories, Albuquerque, NM, USA); Luca Bertagna (Sandia National Laboratories, Albuquerque, NM, USA); Carolyn Kao (TSMC, Hsinchu, Taiwan); Matthew J Hoffman (Los Alamos National Laboratory); Stephen F Price (Los Alamos National Laboratory)

Journal: The International Journal of High Performance Computing Applications

Publisher: SAGE Publications

DOI: 10.1177/10943420231183688

URL: <https://doi.org/10.1177/10943420231183688>

Publication Date: 2023-06-27

Volume/Issue: Volume 37, Issue 5

References: 70

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: ice-sheet modeling, Kokkos, Trilinos, changepoint detection, performance (and 5 more)

Funding: Office of Science

License: N/A

Open Access: N/A

Abstract: High-resolution simulations of polar ice sheets play a crucial role in the ongoing effort to develop more accurate and reliable Earth system models for probabilistic sea-level projections. These simulations often require a massive amount of memory and computation from large supercomputing clusters to provide sufficient accuracy and resolution; therefore, it has become essential to ensure performance on these platforms. Many of today's supercomputers contain a diverse set of computing architectures and require specific programming interfaces in order to obtain optimal efficiency. In an effort to avoid architecture-specific programming and maintain productivity across platforms, the ice-sheet modeling code known as MPAS-Albany Land Ice (MALI) uses high-level abstractions to integrate Trilinos libraries and the Kokkos programming model for performance portable code across a variety of different architectures. In this article, we analyze the performance portable features of MALI via a performance analysis on current CPU-based and GPU-based supercomputers. The analysis highlights not only the performance portable improvements made in finite element assembly and multigrid preconditioning within MALI with speedups between 1.26 and 1.82x across CPU and GPU architectures but also identifies the need to further improve performance in software coupling and preconditioning on GPUs. We perform a weak scalability study and show that simulations on GPU-based machines perform 1.24-1.92x faster when utilizing the GPUs. The best performance is found in finite element assembly, which achieved a speedup of up to 8.65x and a weak scaling efficiency of 82.6% with GPUs. We additionally describe an automated performance testing framework developed for this code base using a changepoint detection method. The framework is used to make actionable decisions about performance within MALI. We provide several concrete examples of scenarios in which the framework has identified performance regressions, improvements, and algorithm differences over the course of 2 years of development.

92. Subglacial hydrology modeling predicts high winter water pressure and spatially variable transmissivity at Helheim Glacier, Greenland

Authors: Aleah Sommers; Colin Meyer; Mathieu Morlighem; Harihar Rajaram; Kristin Poinar; Winnie Chu; Jessica Mejia

Journal: Journal of Glaciology

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/jog.2023.39

URL: <https://doi.org/10.1017/jog.2023.39>

Publication Date: 2023-06-21

Volume/Issue: Volume 69, Issue 278

References: 78

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Heising-Simons Foundation, Heising-Simons Foundation, Heising-Simons Foundation (and 1 more)

License: CC BY 4.0

Open Access: N/A

Abstract: Water pressure beneath glaciers influences ice velocity. Subglacial hydrology models are helpful for gaining insight into basal conditions, but models depend on unconstrained parameters, and a current challenge is reproducing elevated water pressures in winter. We eliminate terms related to englacial storage, opening by sliding, and melt due to changes in the pressure-melting-point temperature, to create a minimalist version of the Subglacial Hydrology And Kinetic, Transient Interactions (SHAKTI) model, and apply this model to Helheim Glacier in east Greenland to explore the winter base state of the subglacial drainage system. Our results suggest that meltwater produced at the bed alone supports active winter drainage with large areas of elevated water pressure and preferential drainage pathways, using a continuum approach that allows for transitions between flow regimes. Transmissivity varies spatially over several orders of magnitude from 10^{-4} to $10^3 \text{ m}^2 \text{ s}^{-1}$, with regions of weak transmissivity representing poorly connected regions of the system. Bed topography controls the location of primary drainage pathways, and high basal melt rates occur along the steep valley walls. Frictional heat from sliding is a dominant source of basal melt; different approaches for calculating basal shear stress produce significantly different basal melt rates and subglacial discharge.

93. Communicating future sea-level rise uncertainty and ambiguity to assessment users

Authors: Robert E. Kopp; Michael Oppenheimer; Jessica L. O'Reilly; Sybren S. Drijfhout; Tamsin L. Edwards; Baylor Fox-Kemper; Gregory G. Garner; Nicholas R. Golledge; Tim H. J. Hermans; Helene T. Hewitt; Benjamin P. Horton; Gerhard Krinner; Dirk Notz; Sophie Nowicki; Matthew D. Palmer; Aimée B. A. Slangen; Cunde Xiao

Journal: Nature Climate Change

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41558-023-01691-8

URL: <https://doi.org/10.1038/s41558-023-01691-8>

Publication Date: 2023-06-19

Volume/Issue: Volume 13, Issue 7

References: 100

Cited by: 16 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

94. Highly variable friction and slip observed at Antarctic ice stream bed

Authors: T. S. Hudson; S. K. Kufner; A. M. Brisbourne; J. M. Kendall; A. M. Smith; R. B. Alley; R. J. Arthern; T. Murray

Journal: Nature Geoscience

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41561-023-01204-4

URL: <https://doi.org/10.1038/s41561-023-01204-4>

Publication Date: 2023-06-15

Volume/Issue: Volume 16, Issue 7

References: 84

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: The slip of glaciers over the underlying bed is the dominant mechanism governing the migration of ice from land into the oceans, with accelerating slip contributing to sea-level rise. Yet glacier slip remains poorly understood, and observational constraints are sparse. Here we use passive seismic observations to measure both frictional shear stress and slip at the bed of the Rutford Ice Stream in Antarctica using 100,000 repetitive stick-slip icequakes. We find that basal shear stresses and slip rates vary from 104 to 107 Pa and 0.2 to 1.5 m per day, respectively. Friction and slip vary temporally over the order of hours, and spatially over 10s of metres, due to corresponding variations in effective normal stress and ice-bed interface material. Our findings suggest that the bed is substantially more complex than currently assumed in ice stream models and that basal effective normal stresses may be significantly higher than previously thought. Our observations can provide constraints on the basal boundary conditions for ice-dynamics models. This is critical for constraining the primary contribution of ice mass loss in Antarctica and hence for reducing uncertainty in sea-level rise projections.

95. Limited Impact of Thwaites Ice Shelf on Future Ice Loss From Antarctica

Authors: G. H. Gudmundsson (Department of Geography and Environmental Sciences Northumbria University Newcastle Upon Tyne UK); J. M. Barnes (Department of Geography and Environmental Sciences Northumbria University Newcastle Upon Tyne UK); D. N. Goldberg (University of Edinburgh); M. Morlighem (Department of Earth Sciences Dartmouth NH USA)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2023gl102880

URL: <https://doi.org/10.1029/2023gl102880>

Publication Date: 2023-05-31

Volume/Issue: Volume 50, Issue 11

References: 32

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, Natural Environment Research Council, Office of Nuclear Energy

License: CC BY 4.0

Open Access: N/A

Abstract: Thwaites Ice Shelf (TWIS), the floating extension of Thwaites Glacier, West Antarctica, is changing rapidly and may completely disintegrate in the near future. Any buttressing that the ice shelf provides to the upstream grounded Thwaites glacier will then be lost. Previously, it has been argued that this could lead to onset of dynamical instability and the rapid demise of the entire glacier. Here we provide the first systematic quantitative assessment of how strongly the upstream ice is buttressed by TWIS and how its collapse affects future projections. By modeling the stresses acting along the current grounding line, we show that they deviate insignificantly from the stresses after ice shelf collapse. Using three ice-flow models, we furthermore model the transient evolution of Thwaites Glacier and find that a complete disintegration of the ice shelf will not substantially impact future mass loss over the next 50 years.

96. In the Quest of a Parametric Relation Between Ice Sheet Model Inferred Weertman's Sliding-Law Parameter and Airborne Radar-Derived Basal Reflectivity Underneath Thwaites Glacier, Antarctica

Authors: Indrani Das (Columbia University); Mathieu Morlighem (Dartmouth College); Jowan Barnes (Northumbria University Newcastle UK); G. Hilmar Gudmundsson (Northumbria University Newcastle UK); Daniel Goldberg (University of Edinburgh); Thiago Dias dos Santos (University of São Paulo)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2022gl098910

URL: <https://doi.org/10.1029/2022gl098910>

Publication Date: 2023-05-26

Volume/Issue: Volume 50, Issue 10

References: 44

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: Weertman's sliding law, Thwaites Glacier, statistical correlation, basal reflectivity

Funding: N/A

License: CC BY-NC 4.0

Open Access: N/A

Abstract: Numerical ice sheet models use sliding laws to connect basal shear stress and ice velocity to simulate ice sliding. A sliding-law parameter β_2 is used to control Weertman's sliding law in numerical ice sheet models. Basal reflectivity derived from ice-penetrating radar also provides information about frozen or thawed conditions underneath glaciers. To assess whether basal reflectivity can be used to constrain β_2 , we carry out statistical experiments between two recently published datasets: β_2 inferred from three numerical ice sheet models (ISSM, Úa and STREAMICE) and airborne radar-derived relative basal reflectivity from the AGASEA-BBAS mission over Thwaites Glacier (TG). Our results show no robust correlation between the β_2 -relative reflectivity pair. Pearson's correlation coefficient, a test of linearity, ranges from -0.26 to -0.38. Spearman's correlation coefficient, which does not require a linear assumption, is also modest (~ -0.35). We conclude that β_2 and relative basal reflectivity underneath TG do not infer similar basal conditions.

97. Hierarchical off-diagonal low-rank approximation of Hessians in inverse problems, with application to ice sheet model initialization

Authors: Tucker Hartland; Georg Stadler; Mauro Perego; Kim Liegeois; Noémi Petra

Journal: Inverse Problems

Publisher: IOP Publishing

DOI: 10.1088/1361-6420/acd719

URL: <https://doi.org/10.1088/1361-6420/acd719>

Publication Date: 2023-05-19

Volume/Issue: Volume 39, Issue 8

References: 54

Cited by: 4 publications

Language: N/A

Subjects: N/A

Keywords: Hessians, hierarchical matrices, ice-sheet models, inverse problems, HODLR matrices (and 1 more)

Funding: Office of Science, National Energy Research Scientific Computing Center, National Science Foundation

License: N/A

Open Access: N/A

Abstract: Obtaining lightweight and accurate approximations of discretized objective functional Hessians in inverse problems governed by partial differential equations (PDEs) is essential to make both deterministic and Bayesian statistical large-scale inverse problems computationally tractable. The cubic computational complexity of dense linear algebraic tasks, such as Cholesky factorization, that provide a means to sample Gaussian distributions and determine solutions of Newton linear systems is a computational bottleneck at large-scale. These tasks can be reduced to log-linear complexity by utilizing hierarchical off-diagonal low-rank (HODLR) matrix approximations. In this work, we show that a class of Hessians that arise from inverse problems governed by PDEs are well approximated by the HODLR matrix format. In particular, we study inverse problems governed by PDEs that model the instantaneous viscous flow of ice sheets. In these problems, we seek a spatially distributed basal sliding parameter field such that the flow predicted by the ice sheet model is consistent with ice sheet surface velocity observations. We demonstrate the use of HODLR Hessian approximation to efficiently sample the Laplace approximation of the posterior distribution with covariance further approximated by HODLR matrix compression. Computational studies are performed which illustrate ice sheet problem regimes for which the Gauss-Newton data-misfit Hessian is more efficiently approximated by the HODLR matrix format than the low-rank (LR) format. We then demonstrate that HODLR approximations can be favorable, when compared to global LR approximations, for large-scale problems by studying the data-misfit Hessian associated with inverse problems governed by the first-order Stokes flow model on the Humboldt glacier and Greenland ice sheet.

98. Robust reconstruction of glacier beds using transient 2D assimilation with Stokes

Authors: Samuel Cook; Fabien Gillet-Chaulet; Johannes Fürst

Journal: Journal of Glaciology

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/jog.2023.26

URL: <https://doi.org/10.1017/jog.2023.26>

Publication Date: 2023-05-12

Volume/Issue: Volume 69, Issue 277

References: 25

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Agence Nationale de la Recherche, H2020 European Research Council

License: CC BY 4.0

Open Access: N/A

Abstract: Initialising model glaciers such that they match well with their real counterparts and are thus able to make more accurate predictions is an ongoing challenge in glacier modelling. We set out a data-assimilation approach using an ensemble Kalman filter in a 2D flowline example that provides one possible solution to this problem. We show that our approach is valid across a range of parameters and scenarios, including deliberately data-deficient or inaccurate ones, and leads to robust retrieval of the glacier bed. We also provide some suggestions for how best to use data assimilation within a mountain-glacier context.

99. Model insights into bed control on retreat of Thwaites Glacier, West Antarctica

Authors: Emily Schwans; Byron R. Parizek; Richard B. Alley; Sridhar Anandakrishnan; Mathieu M. Morlighem

Journal: Journal of Glaciology

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/jog.2023.13

URL: <https://doi.org/10.1017/jog.2023.13>

Publication Date: 2023-05-09

Volume/Issue: Volume 69, Issue 277

References: 81

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, National Science Foundation, National Science Foundation

License: CC BY 4.0

Open Access: N/A

Abstract: Thwaites Glacier (TG) plays an important role in future sea-level rise (SLR) contribution from the West Antarctic Ice Sheet. Recent observations show that TG is losing mass, and its grounding zone is retreating. Previous modeling has produced a wide range of results concerning whether, when, and how rapidly further retreat will occur under continued warming. These differences arise at least in part from ill-constrained processes, including friction from the bed, and future atmosphere and ocean forcing affecting ice-shelf and grounding-zone buttressing. Here, we apply the Ice Sheet and Sea-level System Model (ISSM) with a range of specifications of basal sliding behavior in response to varying ocean forcing. We find that basin-wide bed character strongly affects TG's response to sub-shelf melt by modulating how changes in driving stress are balanced by the bed as the glacier responds to external forcing. Resulting differences in dynamic thinning patterns alter modeled grounding-line retreat across Thwaites' catchment, affecting both modeled rates and magnitudes of SLR contribution from this critical sector of the ice sheet. Bed character introduces large uncertainties in projections of TG under equal external forcing, pointing to this as a crucial constraint needed in predictive models of West Antarctica.

100. Glacier Energy and Mass Balance (GEMB): a model of firn processes for cryosphere research

Authors: Alex S. Gardner; Nicole-Jeanne Schlegel; Eric Larour

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-16-2277-2023

URL: <https://doi.org/10.5194/gmd-16-2277-2023>

Publication Date: 2023-04-27

Volume/Issue: Volume 16, Issue 8

References: 120

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NASA Headquarters

License: CC BY 4.0

Open Access: N/A

Abstract: . This paper provides the first description of the open-source Glacier Energy and Mass Balance model. GEMB models the ice sheet and glacier surface-atmospheric energy and mass exchange, as well as the firn state. It is a column model (no horizontal communication) of intermediate complexity that includes those processes deemed most relevant to glacier studies. GEMB prioritizes computational efficiency to accommodate the very long (thousands of years) spin-ups necessary for initializing deep firn columns and sensitivity experiments needed to characterize model uncertainty on continental scales. The model is one-way coupled with the atmosphere, which allows the model to be run offline with a diversity of climate forcing but neglects feedback to the atmosphere. GEMB provides numerous parameterization choices for various key processes (e.g., albedo, subsurface shortwave absorption, and compaction), making it well suited for uncertainty quantification and model exploration. The model is evaluated against the current state of the art and in situ observations and is shown to perform well.

101. Bias Correction and Statistical Modeling of Variable Oceanic Forcing of Greenland Outlet Glaciers

Authors: Vincent Verjans (Georgia Institute of Technology); Alexander Robel (Georgia Institute of Technology); Andrew F. Thompson (California Institute of Technology); Helene Seroussi (Dartmouth College)

Journal: Journal of Advances in Modeling Earth Systems

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2023ms003610

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Keywords: statistical modeling, ocean forcing, ice sheet

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Abstract: Variability in oceanic conditions directly impacts ice loss from marine outlet glaciers in Greenland, influencing the ice sheet mass balance. Oceanic conditions are available from Atmosphere-Ocean Global Climate Model (AOGCM) output, but these models require extensive computational resources and lack the fine resolution needed to simulate ocean dynamics on the Greenland continental shelf and close to glacier marine termini. Here, we develop a statistical approach to generate ocean forcing for ice sheet model simulations, which incorporates natural spatiotemporal variability and anthropogenic changes. Starting from raw AOGCM ocean heat content, we apply: (a) a bias-correction using ocean reanalysis, (b) an extrapolation accounting for on-shelf ocean dynamics, and (c) stochastic time series models to generate realizations of natural variability. The bias-correction reduces model errors by ~25% when compared to independent in-situ measurements. The bias-corrected time series are subsequently extrapolated to fjord mouth locations using relations constrained from available high-resolution regional ocean model results. The stochastic time series models reproduce the spatial correlation, characteristic timescales, and the amplitude of natural variability of bias-corrected AOGCMs, but at negligible computational expense. We demonstrate the efficiency of this method by generating >6,000 time series of ocean forcing for >200 Greenland marine-terminating glacier locations until 2100. As our method is computationally efficient and adaptable to any ocean model output and reanalysis product, it provides flexibility in exploring sensitivity to ocean conditions in Greenland ice sheet model simulations. We provide the output and workflow in an open-source repository, and discuss advantages and future developments for our method.

102. Compensating errors in inversions for subglacial bed roughness: same steady state, different dynamic response

Authors: Constantijn J. Berends; Roderik S. W. van de Wal; Tim van den Akker; William H. Lipscomb

Journal: The Cryosphere

Publisher: Copernicus GmbH

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URL: <https://doi.org/10.5194/tc-17-1585-2023>

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Keywords: N/A

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Open Access: N/A

Abstract: . Subglacial bed roughness is one of the main factors controlling the rate of future Antarctic ice-sheet retreat and also one of the most uncertain. A common technique to constrain the bed roughness using ice-sheet models is basal inversion, tuning the roughness to reproduce the observed present-day ice-sheet geometry and/or surface velocity. However, many other factors affecting ice-sheet evolution, such as the englacial temperature and viscosity, the surface and basal mass balance, and the subglacial topography, also contain substantial uncertainties. Using a basal inversion technique intrinsically causes any errors in these other quantities to lead to compensating errors in the inverted bed roughness. Using a set of idealised-geometry experiments, we quantify these compensating errors and investigate their effect on the dynamic response of the ice sheet to a prescribed forcing. We find that relatively small errors in ice viscosity and subglacial topography require substantial compensating errors in the bed roughness in order to produce the same steady-state ice sheet, obscuring the realistic spatial variability in the bed roughness. When subjected to a retreat-inducing forcing, we find that these different parameter combinations, which per definition of the inversion procedure result in the same steady-state geometry, lead to a rate of ice volume loss that can differ by as much as a factor of 2. This implies that ice-sheet models that use basal inversion to initialise their model state can still display a substantial model bias despite having an initial state which is close to the observations.

103. Exploring ice sheet model sensitivity to ocean thermal forcing and basal sliding using the Community Ice Sheet Model (CISM)

Authors: Mira Berdahl; Gunter Leguy; William H. Lipscomb; Nathan M. Urban; Matthew J. Hoffman

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-17-1513-2023

URL: <https://doi.org/10.5194/tc-17-1513-2023>

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Keywords: N/A

Funding: National Center for Atmospheric Research, Los Alamos National Laboratory, National Science Foundation

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Abstract: . Multi-meter sea level rise (SLR) is thought to be possible within the next few centuries, with most of the uncertainty originating from the Antarctic land ice contribution. One source of uncertainty relates to the ice sheet model initialization. Since ice sheets have a long response time (compared to other Earth system components such as the atmosphere), ice sheet model initialization methods can have significant impacts on how the ice sheet responds to future forcings. To assess this, we generated 25 different ice sheet spin-ups, using the Community Ice Sheet Model (CISM) at a 4 km resolution. During each spin-up, we varied two key parameters known to impact the sensitivity of the ice sheet to future forcing: one related to the sensitivity of the ice shelf melt rate to ocean thermal forcing (TF) and the other related to the basal friction. The spin-ups all nudge toward observed thickness and enforce a no-advance calving criterion, such that all final spin-up states resemble observations but differ in their melt and friction parameter settings. Each spin-up was then forced with future ocean thermal forcings from 13 different CMIP6 models under the Shared Socioeconomic Pathway (SSP)5-8.5 emissions scenario and modern climatological surface mass balance data. Our results show that the effects of the ice sheet and ocean parameter settings used during the spin-up are capable of impacting multi-century future SLR predictions by as much as 2 m. By the end of this century, the effects of these choices are more modest, but still significant, with differences of up to 0.2 m of SLR. We have identified a combined ocean and ice parameter space that leads to widespread mass loss within 500 years (low friction and high melt rate sensitivity). To explore temperature thresholds, we also ran a synthetically forced CISM ensemble that is focused on the Amundsen region only. Given certain ocean and ice parameter choices, Amundsen mass loss can be triggered with thermal forcing anomalies between 1.5 and 2 °C relative to the spin-up. Our results emphasize the critical importance of considering ice sheet and ocean parameter choices during spin-up for SLR predictions and suggest the importance of including glacial isostatic adjustment in ice sheet simulations.

104. Ocean-Ice Sheet Coupling in the Totten Glacier Area, East Antarctica: Analysis of the Feedbacks and Their Response to a Sudden Ocean Warming

Authors: Guillian Van Achter (Earth and Life Institute, UCLouvain, 1348 Louvain-la-Neuve, Belgium); Thierry Fichefet (Earth and Life Institute, UCLouvain, 1348 Louvain-la-Neuve, Belgium); Hugues Goosse (Earth and Life Institute, UCLouvain, 1348 Louvain-la-Neuve, Belgium); Charles Pelletier (European Centre for Medium-Range Weather Forecasts (ECMWF)); Konstanze Haubner (University of Bergen, Laboratoire de Glaciologie, Université libre de Bruxelles, 1050 Brussels, Belgium, Bjerknes Centre for Climate Research); Frank Pattyn (Laboratoire de Glaciologie, Université libre de Bruxelles, 1050 Brussels, Belgium)

Journal: Geosciences

Publisher: MDPI AG

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References: 80

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Fonds De La Recherche Scientifique—FNRS

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Open Access: N/A

Abstract: We coupled together high-resolution versions of the ocean-sea ice model NEMO and the ice sheet model BISICLES configured to the Totten Glacier area and ran a series of simulations over the recent past (1995-2014) and under warming conditions (2081-2100; SSP4-4.5) with NEMO in stand-alone mode and with the coupled model to assess the effects of the coupling. During the recent past, the ocean-ice sheet coupling has increased the time-averaged value of the basal melt rate in both the Totten and Moscow University ice shelf cavities by 6.7% and 14.2%, respectively. The relationship between the changes in ice shelf thickness and ice shelf basal melt rate suggests that the effect of the coupling is not a linear response to the melt rate but rather a more complex response, driven partly by the dynamical component of the ice sheet model. The response of the ice sheet-ocean coupling due to the ocean warming is a 10% and 3% basal melt rate decrease in the Totten and Moscow University ice shelf cavities, respectively. This indicates that the ocean-ice sheet coupling under climate warming conditions dampens the basal melt rates. Our study highlights the importance of incorporating ocean-ice sheet coupling in climate simulations, even over short time periods.

105. Variational inference of ice shelf rheology with physics-informed machine learning

Authors: Bryan Riel; Brent Minchew

Journal: Journal of Glaciology

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/jog.2023.8

URL: <https://doi.org/10.1017/jog.2023.8>

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References: 66

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Keywords: N/A

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Open Access: N/A

Abstract: Floating ice shelves that fringe the coast of Antarctica resist the flow of grounded ice into the ocean. One of the key factors governing the amount of flow resistance an ice shelf provides is the rigidity (related to viscosity) of the ice that constitutes it. Ice rigidity is highly heterogeneous and must be calibrated from spatially continuous surface observations assimilated into an ice-flow model. Realistic uncertainties in calibrated rigidity values are needed to quantify uncertainties in ice sheet and sea-level forecasts. Here, we present a physics-informed machine learning framework for inferring the full probability distribution of rigidity values for a given ice shelf, conditioned on ice surface velocity and thickness fields derived from remote-sensing data. We employ variational inference to jointly train neural networks and a variational Gaussian Process to reconstruct surface observations, rigidity values and uncertainties. Applying the framework to synthetic and large ice shelves in Antarctica demonstrates that rigidity is well-constrained where ice deformation is measurable within the noise level of the observations. Further reduction in uncertainties can be achieved by complementing variational inference with conventional inversion methods. Our results demonstrate a path forward for continuously updated calibrations of ice flow parameters from remote-sensing observations.

106. Differences in food web structure and composition between new and nearby older lakes in West Greenland suggest succession trajectories driven by glacier retreat

Authors: Erik Jeppesen; Thomas A. Davidson; Mariana Meerhoff; Luc De Meester; Iván González-Bergonzoni; Nicolás Vidal; Hartmut Arndt; Klaus Jürgens; Ruben Sommaruga; Korhan Özkan; Torben L. Lauridsen; Sh Tserenpil

Journal: Hydrobiologia

Publisher: Springer Science and Business Media LLC

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URL: <https://doi.org/10.1007/s10750-023-05189-4>

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Cited by: 4 publications

Language: en

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Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

107. On the Multiscale Oceanic Heat Transports Toward the Bases of the Antarctic Ice Shelves

Authors: Zhaomin Wang (Southern Marine Science and Engineering Guangdong Laboratory); Chengyan Liu (Southern Marine Science and Engineering Guangdong Laboratory); Chen Cheng (Southern Marine Science and Engineering Guangdong Laboratory); Qing Qin (Hohai University, Ministry of Natural Resources); Liangjun Yan (Hohai University, Ministry of Natural Resources); Jiangchao Qian (Hohai University, Ministry of Natural Resources); Chong Sun (Hohai University, Ministry of Natural Resources); Li Zhang (Sun Yat-Sen University)

Journal: Ocean-Land-Atmosphere Research

Publisher: American Association for the Advancement of Science (AAAS)

DOI: 10.34133/olar.0010

URL: <https://doi.org/10.34133/olar.0010>

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References: 229

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The mass balance of the Antarctic Ice Sheet (AIS) is important to global sea-level change. The AIS loses mass mainly through basal melting and subsequent calving of the Antarctic ice shelves. However, the simulated basal melting rates are very uncertain in ice sheet models, partially resulting from the poor understanding of oceanic heat transports. In this article, we review the recent progress in understanding and simulating such heat transports. Regulated by major circulation features, Circumpolar Deep Water (CDW) is much closer to the Bellingshausen-Amundsen Seas and the Cooperation Sea (60 degreesE to 90 degreesE) and the sector further east to 160 degreesE. The ice shelves within these sectors are experiencing enhanced basal melting resulting from tropical forcing and intensified westerlies. Around West Antarctica, the isopycnal structure favors the delivery of CDW across slopes and shelves, while around East Antarctica, the persistent and strong westward Antarctic Slope Current (Front) acts to prevent warm-water intrusion. Both eddies and troughs favor heat transport to the fronts of the ice shelves and even into the cavities. The sharp contrast between the water column thicknesses on both sides of ice shelf fronts blocks the barotropic inflows and can excite topographic Rossby waves. Inside the cavities, the heat fluxes to the bases of the ice shelves are controlled by the cavity geometry, the circulations in the cavities, and the properties of the water masses beneath the ice shelves. Limited direct observations of cavities have promoted the development of various models. To improve basal melting simulations, meltwater plume models have been developed to study meltwater-laden mixed layer dynamics by increasing the vertical resolution, with recent advanced studies considering the vertical structures of frazil ice concentration and velocity. To reduce the uncertainties in the simulated and projected basal mass loss of the Antarctic ice shelves, future efforts should be devoted to improving the bathymetry and cavity geometry, investigating

small-scale processes and parameterizing these processes in coupled climate-ice sheet models, and quantifying the feedback from the mass loss of the AIS.

108. The role of subglacial hydrology in Antarctic ice sheet dynamics and stability: a modelling perspective

Authors: Christine F. Dow

Journal: Annals of Glaciology

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/aog.2023.9

URL: <https://doi.org/10.1017/aog.2023.9>

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Language: en

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Keywords: N/A

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Open Access: N/A

Abstract: Subglacial hydrology is an important component of the ice dynamic system in Antarctica but is challenging to investigate due to the large spatial scales of the catchment systems, the ice thickness, and remote location. Here I discuss key discoveries about Antarctic subglacial hydrology from the Glacier Drainage System (GlaDS) model, including the presence of long, often high-pressure, subglacial channels. These channels pump tens of cubic metres per second of freshwater into ice-shelf cavities and directly affect melt rates at the critical grounding zone regions. Future ice dynamics and ice-shelf cavity models should take subglacial hydrology into account if they are to accurately predict future behaviour of the Antarctic Ice Sheet.

109. Grounding Zone of Amery Ice Shelf, Antarctica, From Differential Synthetic-Aperture Radar Interferometry

Authors: Hanning Chen (University of California Irvine); Eric Rignot (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Bernd Scheuchl (University of California Irvine); Shivani Ehrenfeucht (University of California Irvine)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Cited by: 8 publications

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Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: We employ a time series of Sentinel-1 differential radar interferometry data from 2018 to detect the variability in grounding line position of the Fisher, Mellor, and Lambert glaciers, which drain about 47 billion tons of ice per year from East Antarctica. We observe kilometer-scale tidal migration, two orders of magnitude larger than expected for ice flowing over a hard bed. The migration is not in phase with changes in oceanic tide. In two estuaries underlain by subglacial channels, we observe two states of migration that switch on and off over time scales of several weeks. The range of vertical motion reveals a water column thickness of 2-20 cm. Such intrusions of seawater over wide grounding zones are not accounted for in physical models. Including them will add vigorous melting of grounded ice that will enhance the sensitivity of glaciers to ocean warming and increase projections of mass loss.

110. Assessing controls on ice dynamics at Crane Glacier, Antarctic Peninsula, using a numerical ice flow model

Authors: Rainey Aberle; Ellyn M. Enderlin; Hans-Peter Marshall; Michal Kopera; Tate G. Meehan

Journal: Journal of Glaciology

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/jog.2023.2

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Publication Date: 2023-03-15

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Keywords: N/A

Funding: National Science Foundation

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Open Access: N/A

Abstract: The Antarctic Peninsula's widespread glacier retreat and ice shelf collapse have been attributed to atmospheric and oceanic warming. Following the initial post-collapse period of retreat, several former tributary glaciers of the Larsen A and B ice shelves have been slowly re-advancing for more than a decade. Here, we use a flowline model of Crane Glacier to gauge the sensitivity of former tributary glaciers to future climate change following this period of long-term dynamic adjustment. The glacier's long-term geometry and speed changes are similar to those of other former Larsen A and B tributaries, suggesting that Crane Glacier is a reasonable representation of regional dynamics. For the unperturbed climate simulations, discharge remains nearly unchanged in 2018-2100, indicating that dynamic readjustment to shelf collapse took ~15 years. Despite large uncertainties in Crane Glacier's past and future climate forcing, a wide range of future climate scenarios leads to a relatively modest range in grounding line discharge (0.8-1.5 Gt a⁻¹) by 2100. Based on the model results for Crane, we infer that although former ice shelf tributaries may readvance following collapse, similar to the tidewater glacier cycle, their dynamic response to future climate perturbations should be less than their response to ice shelf collapse.

111. Multidecadal pre- and post-collapse dynamics of the northern Larsen Ice Shelf

Authors: Shujie Wang; Hongxing Liu; Richard B. Alley; Kenneth Jezek; Patrick Alexander; Karen E. Alley; Zhengrui Huang; Lei Wang

Journal: Earth and Planetary Science Letters

Publisher: Elsevier BV

DOI: 10.1016/j.epsl.2023.118077

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Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

112. Migration of the Shear Margins at Thwaites Glacier: Dependence on Basal Conditions and Testability Against Field Data

Authors: Paul T. Summers (Department of Geophysics Stanford University Stanford CA USA); Cooper W. Elsworth (Watershed San Francisco CA USA); Christine F. Dow (University of Waterloo); Jenny Suckale (Department of Geophysics Stanford University Stanford CA USA, Institute for Computational and Mathematical Engineering Stanford University Stanford CA USA)

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2022jf006958

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Publication Date: 2023-03-07

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Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

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Abstract: Projections of global sea level depend sensitively on whether Thwaites Glacier, Antarctica, will continue to lose ice rapidly. Prior studies have focused primarily on understanding the evolution of ice velocity and whether the reverse-sloping bed at Thwaites Glacier could drive irreversible retreat. However, the overall ice flux to the ocean and the possibility of irreversible retreat depend not only on the ice speed but also on the width of the main ice trunk. Here, we complement prior work by focusing specifically on understanding whether the lateral boundaries of the main ice trunk, termed shear margins, might migrate over time. We hypothesize that the shear margins at Thwaites Glacier will migrate on a decadal timescale in response to continued ice thinning and surface steepening. We test this hypothesis by developing a depth-averaged, thermomechanical free-boundary model that captures the complex topography underneath the glacier and solves for both the ice velocity and for the position of the shear margins. We find that both shear margins are prone to migration in response to ice thinning with basal strength and surface slope steepening determining their relative motion. We construct four end-member cases of basal strength that represent different physical properties governing friction at the glacier bed and present two cases of ice thinning to contrast the effects of surface steepening and ice thinning. We test our model by hindcasting historic data and discuss how data from ongoing field campaigns could further be used to test our model.

113. SICOPOLIS-AD v2: tangent linear and adjoint modeling framework for ice sheet modeling enabled by automatic differentiation tool Tapenade

Authors: Shreyas Sunil Gaikwad; Laurent Hascoet; Sri Hari Krishna Narayanan; Liz Curry-Logan; Ralf Greve; Patrick Heimbach

Journal: Journal of Open Source Software

Publisher: The Open Journal

DOI: 10.21105/joss.04679

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References: 22

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Automatic differentiation

Funding: N/A

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Open Access: N/A

Abstract: N/A

114. Modeling the Greenland Ice Sheet's Committed Contribution to Sea Level During the 21st Century

Authors: Isabel J. Nias (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, University of Maryland, University of Liverpool); Sophie Nowicki (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, Department of Geology University at Buffalo Buffalo NY USA); Denis Felikson (Goddard Earth Sciences Technology and Research Studies and Investigations II Morgan State University Baltimore MD USA, National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Bryant Loomis (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center)

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2022jf006914

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Language: en

Subjects: N/A

Keywords: ice sheet modeling, Bayesian calibration, sea level rise, Greenland Ice Sheet

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Open Access: N/A

Abstract: Mass loss from the Greenland Ice Sheet can be partitioned between surface mass balance and discharge due to ice dynamics through its marine-terminating outlet glaciers. A perturbation to a glacier terminus (e.g., a calving event) results in both an instantaneous response in velocity and mass loss and a diffusive response due to the evolution of ice thickness over time. This diffusive response means the total impact of a retreat event can take decades to be fully realized. Here we model the committed response of the Greenland Ice Sheet by applying perturbations to the marine-terminating glacier termini that represent recent observed changes, and simulating the response over the 21st century, while holding the climate forcing constant. The sensitivity of the ice sheet response to model parameter uncertainty is explored within an ensemble framework, and Gravity Recovery and Climate Experiment data is used to constrain the results using a Bayesian calibration approach. We find that the Greenland Ice Sheet's committed contribution to 21st century sea level rise is at least 33.5 [17.5 52.4] mm (25th and 75th percentiles in brackets), with at least 6 mm being attributable directly to terminus retreat that occurred between 2007 and 2015. The spread in our projections is driven by uncertainty in the basal friction coefficient. Our results complement the ISMIP6 Greenland projections, which report the ice sheet response to future forcing, excluding the background response. In this way, we can obtain estimates of Greenland's total contribution to sea level rise in 2100.

115. The impact of spatially varying ice sheet basal conditions on sliding at glacial time scales

Authors: Evan J. Gowan; Sebastian Hinck; Lu Niu; Caroline Clason; Gerrit Lohmann

Journal: Journal of Glaciology

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/jog.2022.125

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References: 70

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: Spatially variable basal conditions are thought to govern how ice sheets behave at glacial time scales (>1000 years) and responsible for changes in dynamics between the core and peripheral regions of the Laurentide and Fennoscandian ice sheets. Basal motion is accomplished via the deformation of unconsolidated sediments, or via sliding of the ice over an undeformable bed. We present an ice sheet sliding module for the Parallel Ice Sheet Model (PISM) that takes into account changes in sediment cover and incorporates surface meltwater. This model routes meltwater, produced at the surface and base of the ice sheet, toward the margin of the ice sheet. Basal sliding is accomplished through the deformation of water saturated sediments, or sliding at the ice-bed interface. In areas with continuous, water saturated sediments, sliding is almost always accomplished through sediment deformation. In areas with incomplete cover, sliding has a stronger dependence on the supply of water. We find that the addition of surface meltwater to the base is a more important factor for ice sheet evolution than the style of sliding. In a glacial cycle simulation, our model causes a more rapid buildup of the Laurentide Ice Sheet.

116. Incorporating Horizontal Density Variations Into Large-Scale Modeling of Ice Masses

Authors: Camilla A. O. Schelpe (Department of Geography and Environmental Sciences Northumbria University Newcastle upon Tyne UK); G. Hilmar Gudmundsson (Department of Geography and Environmental Sciences Northumbria University Newcastle upon Tyne UK)

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

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Open Access: N/A

Abstract: Gravity-driven flow of large ice masses such as the Antarctic Ice Sheet (AIS) depends on both the geometry and the mass density of the ice sheet. The vertical density profile can be approximated as pure ice overlain by a firn layer of varying thickness, and for the AIS the firn thickness is not uncommonly 10%-20% of the total thickness, leading to not insignificant variation in horizontal density. Nevertheless, in most vertically integrated ice-flow models today, the density is assumed constant, sometimes with an adjustment in thickness to compensate. In this study, we explore the treatment of horizontal density variations (HDVs) within vertically integrated ice-sheet models. We assess the relative merits and shortcomings of previously proposed approaches and provide new formulations for including HDVs. We use perturbation analysis to derive analytical solutions that describe the impact of density variations on ice flow for both grounded ice and floating ice shelves, which reveal significant qualitative differences between each of the proposed density formulations. We recommend always including the horizontal gradient of the density in the body-force term of the momentum equation, and in the mass-conservation equation. Furthermore, by modeling the transient evolution of a large sector of the West Antarctic Ice Sheet (WAIS), we find that doing so leads to about a 10% correction in the estimated change in volume above flotation over 40 years. We conclude that including HDVs in flow modeling of the AIS is important for accurate predictions of mass loss.

117. SUHMO: an adaptive mesh refinement SUBglacial Hydrology MOdel v1.0

Authors: Anne M. Felden; Daniel F. Martin; Esmond G. Ng

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-16-407-2023

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References: 52

Cited by: 4 publications

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Subjects: N/A

Keywords: N/A

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Open Access: N/A

Abstract: . Water flowing under ice sheets and glaciers can have a strong influence on ice dynamics, particularly through pressure changes, suggesting that a comprehensive ice sheet model should include the effect of basal hydrology. Modeling subglacial hydrology remains a challenge, however, mainly due to the range of spatial and temporal scales involved - from subglacial channels to vast subglacial lakes. Additionally, networks of subglacial drainage channels dynamically evolve over time. To address some of these challenges, we have developed an adaptive mesh refinement (AMR) model based on the Chombo software framework. We extend the model proposed by Sommers et al. (2018) with a small but significant change to accommodate the transition from unresolved to resolved flow features. We handle the strong nonlinearities present in the equations by resorting to an efficient nonlinear full approximation scheme multigrid (FAS-MG) algorithm. We outline the details of the algorithm and present convergence analysis results demonstrating its good performance. Additionally, we present results validating our approach, using test cases from the Subglacial Hydrology Model Intercomparison Project (SHMIP) (de Fleurian et al., 2018). We finish by presenting a more complex, 100 km-by-100 km synthetic test case with peaks and valleys that we use to investigate the effective pressure distribution as the number of AMR levels increases. These preliminary results suggest that a minimum spatial resolution is needed to properly capture channel features, but additional work is required to precisely quantify this and its impact on accurately modeling the coupled ice sheet-hydrology system. The efficiency of our approach, relying on localized refinement, is also demonstrated. Future work will include coupling the SUBglacial Hydrology MOdel (SUHMO) with the BISICLES AMR ice sheet model (Cornford et al., 2013), both built on the same numerical framework.

118. Automatic Detection of Subglacial Water Bodies in the AGAP Region, East Antarctica, Based on Short-Time Fourier Transform

Authors: Tong Hao (Tongji University); Liwen Jing (Tongji University); Jiashu Liu (Peking University); Dailiang Wang (Tongji University); Tiantian Feng (Tongji University); Aiguo Zhao (Tongji University); Rongxing Li (Tongji University)

Journal: Remote Sensing

Publisher: MDPI AG

DOI: 10.3390/rs15020363

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Cited by: 9 publications

Language: en

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Keywords: N/A

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Open Access: N/A

Abstract: Subglacial water bodies are critical components in analyzing the instability of the Antarctic ice sheet. Their detection and identification normally rely on geophysical and remote sensing methods such as airborne radar echo sounding (RES), ground seismic, and satellite/airborne altimetry and gravity surveys. In particular, RES surveys are able to detect basal terrain with a relatively high accuracy that can assist with the mapping of subglacial hydrology systems. Traditional RES processing methods for the identification of subglacial water bodies mostly rely on their brightness in radargrams and hydraulic flatness. In this study, we propose an automatic method with the main objective to differentiate the basal materials by quantitatively evaluating the shape of the A-scope waveform near the basal interface in RES radargrams, which has been seldom investigated. We develop an automatic algorithm mainly based on the traditional short-time Fourier transform (STFT) to quantify the shape of the A-scope waveform in radargrams. Specifically, with an appropriate window width applied on the main peak of each A-scope waveform in the RES radargram, STFT shows distinct and contrasting frequency responses at the ice-water interface and ice-rock interface, which is largely dependent upon their different reflection characteristics at the basal interface. We apply this method on 882 RES radargrams collected in the Antarctic's Gamburtsev Province (AGAP) in East Antarctica. There are 8822 identified A-scopes with the calculated detection value larger than the set threshold, out of the overall 1,515,065 valid A-scopes in these 882 RES radargrams. Although these identified A-scopes only takes 0.58% of the overall A-scope population, they show exceptionally continuous distribution to represent the subglacial water bodies. Through a comprehensive comparison with existing inventories of subglacial lakes, we successfully verify the validity and advantages of our method in identifying subglacial water bodies using the detection probability for other basal materials of theoretically the highest along-track resolution. The frequency signature obtained by the proposed joint time-frequency analysis provides a new corridor of investigation that can be further expanded to multivariable deep learning approaches for subglacial and englacial material characterization, as well as subglacial

hydrology mapping.

119. Seasonal Acceleration of Petermann Glacier, Greenland, From Changes in Subglacial Hydrology

Authors: Shivani Ehrenfeucht (Department of Earth System Science University of California- Irvine Irvine CA USA); Mathieu Morlighem (Department of Earth System Science University of California- Irvine Irvine CA USA, Dartmouth College); Eric Rignot (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, Department of Earth System Science University of California- Irvine Irvine CA USA); Christine F. Dow (University of Waterloo); Jeremie Mouginot (Institut de Recherche pour le Développement (IRD), Department of Earth System Science University of California- Irvine Irvine CA USA)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2022gl098009

URL: <https://doi.org/10.1029/2022gl098009>

Publication Date: 2022-12-24

Volume/Issue: Volume 50, Issue 1

References: 86

Cited by: 19 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: Petermann Glacier is a major outlet glacier of northern Greenland that drains a marine-based basin vulnerable to destabilization from enhanced oceanic and atmospheric forcings. Using satellite radar interferometry data from the Sentinel-1a/b missions, we observe a seasonal glacier acceleration of 15% in the summer, from 1,250 to 1,500 m/yr near the grounding line, but the physical drivers of this seasonality have not been elucidated. Here, we use a subglacial hydrology model coupled one-way to an ice sheet model to evaluate the role of subglacial hydrology as a physical mechanism explaining the seasonal acceleration. We find excellent agreement between the observed and predicted velocity in terms of timing and magnitude with the addition of an applied lower limit on effective pressure of 6% of ice overburden pressure. We conclude that seasonal changes in subglacial hydrology are sufficient to explain the observed seasonal speed up of Petermann Glacier.

120. Probabilistic Sea Level Rise Hazard Analysis Based on the Current Generation of Data and Protocols

Authors: Xiao Luo (Graduate Student, Dept. of Civil, Environmental, and Construction Engineering, Texas Tech Univ., Lubbock, TX 79409. ORCID: .); Ting Lin (Assistant Professor, Dept. of Civil, Environmental, and Construction Engineering, Texas Tech Univ., Lubbock, TX 79409 (corresponding author). ORCID: .)

Journal: Journal of Structural Engineering

Publisher: American Society of Civil Engineers (ASCE)

DOI: 10.1061/jsendh.steng-11413

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Publication Date: 2022-12-20

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References: 73

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

121. NASA's Earth Information System: Sea-Level Change

Authors: Denis Felikson (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Ian Fenty (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Ben Hamlington (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Alexey Shiklomanov (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Carmen Blackwood (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Mark Carroll (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Michael Croteau (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Cedric David (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Kyla Drushka (University of Washington); Daniel Duffy (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); J. Thomas Farrar (Woods Hole Oceanographic Institution); Gael Forget (Massachusetts Institute of Technology); Severine Fournier (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Thomas Frederikse (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Ichiro Fukumori (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Alexandre Guillaume (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Patrick Heimbach (University of Texas at Austin); Thomas Huang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Eric Larour (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Kyo Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Jian Li (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Brooke Medley (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Sophie Nowicki (University at Buffalo, Dept. of Geology, Buffalo, USA); Justin Quinn (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Ramon Ramirez-Linan (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Richard Ray (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Mike Rilee (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center)

Journal: OCEANS 2022, Hampton Roads

Publisher: IEEE

DOI: 10.1109/oceans47191.2022.9977250

URL: <https://doi.org/10.1109/oceans47191.2022.9977250>

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Language: N/A

Subjects: N/A

Keywords: Earth observation|Earth system science

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

122. Performance analysis of high-resolution ice-sheet simulations

Authors: Ed Bueler

Journal: Journal of Glaciology

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/jog.2022.113

URL: <https://doi.org/10.1017/jog.2022.113>

Publication Date: 2022-12-14

Volume/Issue: Volume 69, Issue 276

References: 34

Cited by: 1 publications

Language: en

Subjects: N/A

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Open Access: N/A

Abstract: Numerical glacier and ice-sheet models compute evolving ice geometry and velocity fields using various stress-balance approximations and boundary conditions. At high spatial resolution, with horizontal mesh/grid resolutions of a few kilometers or smaller, these models usually require time steps shorter than climate-coupling time scales because they update ice thickness after each velocity solution. High-resolution performance is degraded by the stability restrictions of such explicit time-stepping. This short note, which considers the shallow ice approximation and Stokes models as stress-balance end members, clarifies the scaling of numerical model performance by quantifying simulation cost per model year in terms of mesh resolution and the number of degrees of freedom. The performance of current-generation explicit time-stepping models is assessed, and then compared to the prospective performance of implicit schemes. The main results highlight the key roles played by the algorithmic scaling of stress-balance solvers and coupled, implicit-step solvers.

123. The Stochastic Ice-Sheet and Sea-Level System Model v1.0 (StlSSM v1.0)

Authors: Vincent Verjans; Alexander A. Robel; Helene Seroussi; Lizz Ultee; Andrew F. Thompson

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-15-8269-2022

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Publication Date: 2022-11-18

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References: 116

Cited by: 4 publications

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Subjects: N/A

Keywords: N/A

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Open Access: N/A

Abstract: . We introduce the first version of the Stochastic Ice-sheet and Sea-level System Model (StlSSM v1.0), which adds stochastic parameterizations within a state-of-the-art large-scale ice sheet model. In StlSSM v1.0, stochastic parameterizations target climatic fields with internal variability, as well as glaciological processes exhibiting variability that cannot be resolved at the spatiotemporal resolution of ice sheet models: calving and subglacial hydrology. Because both climate and unresolved glaciological processes include internal variability, stochastic parameterizations allow StlSSM v1.0 to account for the impacts of their high-frequency variability on ice dynamics and on the long-term evolution of modeled glaciers and ice sheets. StlSSM v1.0 additionally includes statistical models to represent surface mass balance and oceanic forcing as autoregressive processes. Such models, once appropriately calibrated, allow users to sample irreducible uncertainty in climate prediction without the need for computationally expensive ensembles from climate models. When combined together, these novel features of StlSSM v1.0 enable quantification of irreducible uncertainty in ice sheet model simulations and of ice sheet sensitivity to noisy forcings. We detail the implementation strategy of StlSSM v1.0, evaluate its capabilities in idealized model experiments, demonstrate its applicability at the scale of a Greenland ice sheet simulation, and highlight priorities for future developments. Results from our test experiments demonstrate the complexity of ice sheet response to variability, such as asymmetric and/or non-zero mean responses to symmetric, zero-mean imposed variability. They also show differing levels of projection uncertainty for stochastic variability in different processes. These features are in line with results from stochastic experiments in climate and ocean models, as well as with the theoretical expected behavior of noise-forced non-linear systems.

124. Reconstruction of Near-Surface Air Temperature over the Greenland Ice Sheet Based on MODIS Data and Machine Learning Approaches

Authors: Jiahang Che (College of Geography and Environment, Shandong Normal University, Jinan 250358, China); Minghu Ding (State Key Laboratory of Severe Weather, China Academy of Meteorological Sciences, Beijing 100081, China); Qinglin Zhang (College of Geography and Environment, Shandong Normal University, Jinan 250358, China); Yetang Wang (College of Geography and Environment, Shandong Normal University, Jinan 250358, China); Weijun Sun (College of Geography and Environment, Shandong Normal University, Jinan 250358, China); Yuzhe Wang (College of Geography and Environment, Shandong Normal University, Jinan 250358, China); Lei Wang (College of Geography and Environment, Shandong Normal University, Jinan 250358, China); Baojuan Huai (College of Geography and Environment, Shandong Normal University, Jinan 250358, China)

Journal: Remote Sensing

Publisher: MDPI AG

DOI: 10.3390/rs14225775

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Volume/Issue: Volume 14, Issue 22

References: 96

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: temperature, machine learning methods, MODIS, GrIS

Funding: Natural Science Foundation of China, Key Laboratory of Oceanic Atmospheric Chemistry and Global Change, Ministry of Natural Resources

License: CC BY 4.0

Open Access: N/A

Abstract: High spatial and temporal resolution products of near-surface air temperature (T2m) over the Greenland Ice Sheet (GrIS) are required as baseline information in a variety of research disciplines. Due to the sparse network of in situ data on the GrIS, remote sensing data and machine learning methods provide great advantages, due to their capacity and accessibility. The Land Surface Temperature (LST) at 780 m resolution from the Moderate Resolution Imaging Spectroradiometer (MODIS) and T2m observation from 25 Automatic Weather Stations (AWSs) are used to establish a relationship over the GrIS by comparing multiple machine learning approaches. Four machine learning methods--neural network (NN), gaussian process regression (GPR), support vector machine (SVM), and random forest (RF)--are used to reconstruct the T2m at daily and monthly scales. We develop a reliable T2m reconstruction model based on key meteorological parameters, such as albedo, wind speed, and specific humidity. The reconstructions daily and monthly products are generated on a 780 m x 780 m spatial grid spanning from 2007 to 2019. When compared with in situ observations, the NN method presents the highest accuracy, with R of 0.96, RMSE of 2.67 degreesC, and BIAS of -0.36 degreesC. Similar to the regional climate model (RACMO2.3p2), the reconstructed T2m can better reflect the spatial pattern in term of latitude, longitude, and altitude effects.

125. Extensive inland thinning and speed-up of Northeast Greenland Ice Stream

Authors: Shfaqat A. Khan; Youngmin Choi; Mathieu Morlighem; Eric Rignot; Veit Helm; Angelika Humbert; Jérémie Mouginot; Romain Millan; Kurt H. Kjær; Anders A. Bjørk

Journal: Nature

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41586-022-05301-z

URL: <https://doi.org/10.1038/s41586-022-05301-z>

Publication Date: 2022-11-09

Volume/Issue: Volume 611, Issue 7937

References: 77

Cited by: 32 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: Over the past two decades, ice loss from the Greenland ice sheet (GrIS) has increased owing to enhanced surface melting and ice discharge to the ocean¹⁻⁵. Whether continuing increased ice loss will accelerate further, and by how much, remains contentious⁶⁻⁹. A main contributor to future ice loss is the Northeast Greenland Ice Stream (NEGIS), Greenland's largest basin and a prominent feature of fast-flowing ice that reaches the interior of the GrIS¹⁰⁻¹². Owing to its topographic setting, this sector is vulnerable to rapid retreat, leading to unstable conditions similar to those in the marine-based setting of ice streams in Antarctica¹³⁻²⁰. Here we show that extensive speed-up and thinning triggered by frontal changes in 2012 have already propagated more than 200 km inland. We use unique global navigation satellite system (GNSS) observations, combined with surface elevation changes and surface speeds obtained from satellite data, to select the correct basal conditions to be used in ice flow numerical models, which we then use for future simulations. Our model results indicate that this marine-based sector alone will contribute 13.5-15.5 mm sea-level rise by 2100 (equivalent to the contribution of the entire ice sheet over the past 50 years) and will cause precipitous changes in the coming century. This study shows that measurements of subtle changes in the ice speed and elevation inland help to constrain numerical models of the future mass balance and higher-end projections show better agreement with observations.

126. Antarctic basal environment shaped by high-pressure flow through a subglacial river system

Authors: C. F. Dow; N. Ross; H. Jeofry; K. Siu; M. J. Siegert

Journal: Nature Geoscience

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41561-022-01059-1

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Cited by: 27 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Gouvernement du Canada | Natural Sciences and Engineering Research Council of Canada, Canada Research Chairs, Newcastle University (and 2 more)

License: N/A

Open Access: N/A

Abstract: N/A

127. Antarctic geothermal heat flow and its implications for tectonics and ice sheets

Authors: Anya M. Reading; Tobias Stål; Jacqueline A. Halpin; Mareen Lösing; Jörg Ebbing; Weisen Shen; Felicity S. McCormack; Christine S. Siddoway; Derrick Hasterok

Journal: Nature Reviews Earth & Environment

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s43017-022-00348-y

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References: 180

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

128. On the evolution of an ice shelf melt channel at the base of Filchner Ice Shelf, from observations and viscoelastic modeling

Authors: Angelika Humbert; Julia Christmann; Hugh F. J. Corr; Veit Helm; Lea-Sophie Höyngs; Coen Hofstede; Ralf Müller; Niklas Neckel; Keith W. Nicholls; Timm Schultz; Daniel Steinhage; Michael Wolovick; Ole Zeising

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-16-4107-2022

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Volume/Issue: Volume 16, Issue 10

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Language: en

Subjects: N/A

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Funding: Alfred Wegener Institute Helmholtz Centre for Polar and Marine Research, Helmholtz-Gemeinschaft, British Antarctic Survey

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Open Access: N/A

Abstract: . Ice shelves play a key role in the stability of the Antarctic Ice Sheet due to their buttressing effect. A loss of buttressing as a result of increased basal melting or ice shelf disintegration will lead to increased ice discharge. Some ice shelves exhibit channels at the base that are not yet fully understood. In this study, we present in situ melt rates of a channel which is up to 330 m high and located in the southern Filchner Ice Shelf. Maximum observed melt rates are 2 m yr⁻¹. Melt rates inside the channel decrease in the direction of ice flow and turn to freezing ~55 km downstream of the grounding line. While closer to the grounding line melt rates are higher within the channel than outside, this relationship reverses further downstream. Comparing the modeled evolution of this channel under present-day climate conditions over 250 years with its present geometry reveals a mismatch. Melt rates twice as large as the present-day values are required to fit the observed geometry. In contrast, forcing the model with present-day melt rates results in a closure of the channel, which contradicts observations. The ice shelf experiences strong tidal variability in vertical strain rates at the measured site, and discrete pulses of increased melting occurred throughout the measurement period. The type of melt channel in this study diminishes in height with distance from the grounding line and is hence not a destabilizing factor for ice shelves.

129. Uncovering Basal Friction in Northwest Greenland Using an Ice Flow Model and Observations of the Past Decade

Authors: Youngmin Choi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Helene Seroussi (Dartmouth College); Alex Gardner (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Nicole-Jeanne Schlegel (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2022jf006710

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Publication Date: 2022-10-08

Volume/Issue: Volume 127, Issue 10

References: 72

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: sea level rise, Greenland ice sheet, friction, modeling

Funding: Jet Propulsion Laboratory

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Open Access: N/A

Abstract: Glaciers move under the pull of gravity due to a combination of internal deformation (creep) and basal sliding. Basal sliding dominates ice motion in fast flowing areas, and an accurate description of sliding in ice flow models is therefore critical to adequately capture ice dynamics and its temporal evolution. However, there is currently no consensus on the exact form of basal friction law, as direct observations of basal properties remain limited. Using an ice flow model and observations of the past decade, we evaluate a variety of friction laws and determine their ability to reproduce the observed acceleration and mass loss of marine-terminating glaciers in northwest Greenland. We find that friction laws that include an explicit dependence on the effective pressure better reproduce the observed acceleration and mass loss overall, with semi-plastic basal laws describing some glaciers well. Our glacier-scale model evaluation suggests heterogeneous bed conditions under northwest Greenland, which cannot be detected from the results of regional changes. Accounting for heterogeneous bed conditions, this region might contribute 15 mm to sea level rise by 2100, twice as much as predicted from models based on commonly used uniform bed conditions.

130. A comparison between three-dimensional, transient, thermomechanically coupled first-order and Stokes ice flow models

Authors: Zhan Yan; Wei Leng; Yuzhe Wang; Cunde Xiao; Tong Zhang

Journal: Journal of Glaciology

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/jog.2022.77

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Volume/Issue: Volume 69, Issue 275

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Language: en

Subjects: N/A

Keywords: N/A

Funding: State Key Laboratory of Earth Surface Processes and Resource Ecology, Fundamental Research Funds for the Central Universities, National Key Research and Development Program of China

License: CC BY 4.0

Open Access: N/A

Abstract: In this study, we investigate the differences between two transient, three-dimensional, thermomechanically coupled ice-sheet models, namely, a first-order approximation model (FOM) and a 'full' Stokes ice-sheet model (FSM) under the same numerical framework. For all numerical experiments, we take the FSM outputs as the reference values and calculate the mean relative errors in the velocity and temperature fields for the FOM over 100 years. Four different boundary conditions (ice slope, geothermal heat flux, basal topography and basal sliding) are tested, and by changing these parameters, we verify the thermomechanical behavior of the FOM and discover that the velocity and temperature biases of the FOM generally increase with increases in the ice slope, geothermal heat flux, undulation amplitude of the ice base, and with the existence of basal sliding. In addition, the model difference between the FOM and FSM may accumulate over time, and the spatial distribution patterns of the relative velocity and temperature errors are in good agreement.

131. Surface topographic impact of subglacial water beneath the south polar ice cap of Mars

Authors: N. S. Arnold; F. E. G. Butcher; S. J. Conway; C. Gallagher; M. R. Balme

Journal: Nature Astronomy

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41550-022-01782-0

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Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

132. Detecting the Greenland Ice Sheet Strong Surface Melt During Summer 2021 using SMAP L-Band Microwave Radiometry

Authors: Mohammad Mousavi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Andreas Colliander (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Nicole-Jeanne Schlegel (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Julie Z. Miller (University of Colorado, Earth Science and Observation Center, Boulder, CO, USA, 80309); John S. Kimball (University of Montana)

Journal: IGARSS 2022 - 2022 IEEE International Geoscience and Remote Sensing Symposium

Publisher: IEEE

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Language: N/A

Subjects: N/A

Keywords: Greenland ice sheet|Radiometry|Ice-sheet model

Funding: Jet Propulsion Laboratory

License: N/A

Open Access: N/A

Abstract: N/A

133. Trends and uncertainties of mass-driven sea-level change in the satellite altimetry era

Authors: Carolina M. L. Camargo; Riccardo E. M. Riva; Tim H. J. Hermans; Aimée B. A. Slangen

Journal: Earth System Dynamics

Publisher: Copernicus GmbH

DOI: 10.5194/esd-13-1351-2022

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Publication Date: 2022-09-27

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References: 91

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Netherlands Space Office

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Open Access: N/A

Abstract: . Ocean mass change is one of the main drivers of present-day sea-level change (SLC). Also known as barystatic SLC, ocean mass change is caused by the exchange of freshwater between the land and the ocean, such as melting of continental ice from glaciers and ice sheets, and variations in land water storage. While many studies have quantified the present-day barystatic contribution to global mean SLC, fewer works have looked into regional changes. This study provides an analysis of regional patterns of contemporary mass redistribution associated with barystatic SLC since 1993 (the satellite altimetry era), with a focus on the uncertainty budget. We consider three types of uncertainties: intrinsic (the uncertainty from the data/model itself), temporal (related to the temporal variability in the time series) and spatial-structural (related to the spatial distribution of the mass change sources). Regional patterns (fingerprints) of barystatic SLC are computed from a range of estimates of the individual freshwater sources and used to analyze the different types of uncertainty. Combining all contributions, we find that regional sea-level trends range from -0.4 to 3.3 mm yr⁻¹ for 2003-2016 and from -0.3 to 2.6 mm yr⁻¹ for 1993-2016, considering the 5-95th percentile range across all grid points and depending on the choice of dataset. When all types of uncertainties from all contributions are combined, the total barystatic uncertainties regionally range from 0.6 to 1.3 mm yr⁻¹ for 2003-2016 and from 0.4 to 0.8 mm yr⁻¹ for 1993-2016, also depending on the dataset choice. We find that the temporal uncertainty dominates the budget, responsible on average for 65 % of the total uncertainty, followed by the spatial-structural and intrinsic uncertainties, which contribute on average 16 % and 18 %, respectively. The main source of uncertainty is the temporal uncertainty from the land water storage contribution, which is responsible for 35 %-60 % of the total uncertainty, depending on the region of interest. Another important contribution comes from the spatial-structural uncertainty from Antarctica and land water storage, which shows that different locations of mass change can lead to trend deviations larger than 20 %. As the barystatic SLC contribution and its uncertainty vary significantly from region to region, better insights into regional SLC are important for local management and adaptation planning.

134. Declining Basal Motion Dominates the Long-Term Slowing of Athabasca Glacier, Canada

Authors: William H. Armstrong (Appalachian State University); David Polashenski (University of Alaska Fairbanks); Martin Truffer (University of Alaska Fairbanks); Greg Horne (Jasper National Park Jasper AB Canada); Jacob B. Hanson (Appalachian State University); Robert L. Hawley (Dartmouth College); Anthony M. Hengst (Appalachian State University); Lily Vowels (Appalachian State University); Brian Menounos (University of Northern British Columbia); Wesley Van Wychen (University of Waterloo)

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Publisher: American Geophysical Union (AGU)

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References: 115

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: basal motion, glacier change, climate change, glacier dynamics, subglacial hydrology (and 1 more)

Funding: National Science Foundation

License: Creative Commons

Open Access: N/A

Abstract: Globally, glaciers are shrinking in response to climate change, with implications for global sea level rise as well as downstream ecosystems and water resources. Sliding at the ice-bed interface (basal motion) provides a mechanism for glaciers to respond rapidly to climate change. While the short-term dynamics of glacier basal motion (<10 years) have received substantial attention, little is known about how basal motion and its sensitivity to subglacial hydrology changes over long (>50 year) timescales--this knowledge is required for accurate prediction of future glacier change. We compare historical data with modern estimates from field and satellite data at Athabasca Glacier and show that the glacier thinned by 60 m (-21%) over 1961-2020. However, a concurrent increase in surface slope results in minimal change in the average driving stress (-6 kPa and -4%). These geometric changes coincide with relatively uniform slowing (-15 m a⁻¹ and -45%). Simplified ice modeling suggests that declining basal motion accounts for most of this slow down (91% on average and 46% at minimum). A decline in basal motion can be explained by increasing basal friction resulting from geometric change in addition to increasing meltwater flux through a more efficient subglacial hydrologic system. These results highlight the need to include time-varying dynamics of basal motion in glacier models and analyses. If these findings are generalizable, they suggest that declining basal motion reduces the flux of ice to lower elevations, helping to mitigate glacier mass loss in a warming climate.

135. Sensitivity of the Ross Ice Shelf to environmental and glaciological controls

Authors: Francesca Baldacchino; Mathieu Morlighem; Nicholas R. Golledge; Huw Horgan; Alena Malyarenko

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-16-3723-2022

URL: <https://doi.org/10.5194/tc-16-3723-2022>

Publication Date: 2022-09-20

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Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Ministry of Science and Innovation, New Zealand, Royal Society Te Apārangi

License: CC BY 4.0

Open Access: N/A

Abstract: . The Ross Ice Shelf (RIS) is currently stable but recent observations have indicated that basal melt rates beneath the ice shelf are expected to increase. It is important to know which areas of the RIS are more sensitive to enhanced basal melting as well as other external forcings or internal material properties of the ice to understand how climate change will influence RIS mass balance. In this paper, we use automatic differentiation and the Ice Sheet and Sea-level System Model to quantify the sensitivity of the RIS to changes in basal friction, ice rigidity, surface mass balance, and basal melting. Using volume above flotation (VAF) as our quantity of interest, we find that the RIS is most sensitive to changes in basal friction and ice rigidity close to grounding lines and along shear margins of the Siple Coast Ice Streams and Transantarctic Mountains Outlet Glaciers. The RIS sensitivity to surface mass balance is uniform over grounded ice, while the sensitivity to basal melting is more spatially variable. Changes in basal melting close to the grounding lines of the Siple Coast Ice Streams and Transantarctic Mountains outlet glaciers have a larger impact on the final VAF compared to elsewhere. Additionally, the pinning points and ice shelf shear margins are highly sensitive to changes in basal melt. Our sensitivity maps allow areas of greatest future vulnerability to be identified.

136. Ice mass loss sensitivity to the Antarctic ice sheet basal thermal state

Authors: Eliza J. Dawson; Dustin M. Schroeder; Winnie Chu; Elisa Mantelli; Hélène Seroussi

Journal: Nature Communications

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41467-022-32632-2

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Volume/Issue: Volume 13, Issue 1

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Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, I am a National Science Foundation Graduate Research Fellow

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Open Access: N/A

Abstract: Sea-level rise projections rely on accurate predictions of ice mass loss from Antarctica. Climate change promotes greater mass loss by destabilizing ice shelves and accelerating the discharge of upstream grounded ice. Mass loss is further exacerbated by mechanisms such as the Marine Ice Sheet Instability and the Marine Ice Cliff Instability. However, the effect of basal thermal state changes of grounded ice remains largely unexplored. Here, we use numerical ice sheet modeling to investigate how warmer basal temperatures could affect the Antarctic ice sheet mass balance. We find increased mass loss in response to idealized basal thawing experiments run over 100 years. Most notably, frozen-bed patches could be tenuously sustaining the current ice configuration in parts of George V, Adélie, Enderby, and Kemp Land regions of East Antarctica. With less than 5 degrees of basal warming, these frozen patches may begin to thaw, producing new loci of mass loss.

137. Greenland ice sheet climate disequilibrium and committed sea-level rise

Authors: Jason E. Box; Alun Hubbard; David B. Bahr; William T. Colgan; Xavier Fettweis; Kenneth D. Mankoff; Adrien Wehrlé; Brice Noël; Michiel R. van den Broeke; Bert Wouters; Anders A. Bjørk; Robert S. Fausto

Journal: Nature Climate Change

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41558-022-01441-2

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Publication Date: 2022-08-29

Volume/Issue: Volume 12, Issue 9

References: 66

Cited by: 64 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: Ice loss from the Greenland ice sheet is one of the largest sources of contemporary sea-level rise (SLR). While process-based models place timescales on Greenland's deglaciation, their confidence is obscured by model shortcomings including imprecise atmospheric and oceanic couplings. Here, we present a complementary approach resolving ice sheet disequilibrium with climate constrained by satellite-derived bare-ice extent, tidewater sector ice flow discharge and surface mass balance data. We find that Greenland ice imbalance with the recent (2000-2019) climate commits at least 274 +/- 68 mm SLR from 59 +/- 15 x 10³ km² ice retreat, equivalent to 3.3 +/- 0.9% volume loss, regardless of twenty-first-century climate pathways. This is a result of increasing mass turnover from precipitation, ice flow discharge and meltwater run-off. The high-melt year of 2012 applied in perpetuity yields an ice loss commitment of 782 +/- 135 mm SLR, serving as an ominous prognosis for Greenland's trajectory through a twenty-first century of warming.

138. Increasing stable time-step sizes of the free-surface problem arising in ice-sheet simulations

Authors: André Löfgren; Josefin Ahlkrona; Christian Helanow

Journal: Journal of Computational Physics: X

Publisher: Elsevier BV

DOI: 10.1016/j.jcp.2022.100114

URL: <https://doi.org/10.1016/j.jcp.2022.100114>

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Volume/Issue: Volume 16

References: 35

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Vetenskapsrådet, Swedish e-Science Research Centre

License: N/A

Open Access: N/A

Abstract: N/A

139. Investigating the Effects of Climate Change on Structural Actions

Authors: André Orcesi (Cerema, Research team ENDSUM, DTecITM/DTOA/GITEX, Champs-sur-Marne, France; Department MAST-EMGCU, Université Gustave Eiffel, IFSTTAR, Marne-la-Vallée, France); Alan O'Connor (Trinity College Dublin); Dimitris Diamantidis (OTH Regensburg, Germany); Miroslav Sykora (Czech Technical University in Prague, Czech Republic); Teng Wu (University at Buffalo, USA); Mitsuyoshi Akiyama (Waseda University, Japan); Abdul Kadir Alhamid (Waseda University, Japan); Franziska Schmidt (Department MAST-EMGCU, Université Gustave Eiffel, IFSTTAR, Marne-la-Vallée, France); Maria Pregnolato (University of Bristol); Yue Li (Case Western Reserve University, USA); Babak Salarieh (University of Alabama); Abdullahi M. Salman (University of Alabama); Emilio Bastidas-Arteaga (LaSIE UMR CNRS 7356, La Rochelle University, La Rochelle, France); Olga Markogiannaki (University of Western Macedonia, Greece); Franck Schoefs (GeM /IUML, CNRS UMR 6183/FR 3473, Nantes Université, Ecole Centrale de Nantes, F44322 Nantes, France.)

Journal: Structural Engineering International

Publisher: Informa UK Limited

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Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: hurricanes, extreme weather events, sea-level rise, scour, flooding (and 2 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

140. Modelling the influence of marine ice on the dynamics of an idealised ice shelf

Authors: Lisa Crow; Felicity S. McCormack; Sue Cook; Jason Roberts; Adam Treverrow

Journal: Journal of Glaciology

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/jog.2022.66

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Publication Date: 2022-08-15

Volume/Issue: Volume 69, Issue 274

References: 70

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: Understanding the dynamic behaviour of ice shelves, specifically the controls on their ability to buttress the flow of ice into the ocean, is critical for predicting future ice-sheet contributions to sea level rise. Many large ice shelves, which are predominantly composed of meteoric ice, have a basal layer of marine ice (formed from accumulated platelets at the ice-ocean interface), comprising up to 40% of their thickness locally. Differences in temperature, chemistry and microstructure between marine and meteoric ice mean the rheological properties of the ice vary throughout the ice shelf. These differences are not explicitly accounted for in ice-sheet modelling applications, and may have an important influence on ice shelf dynamics. We tested the sensitivity of a model of an idealised ice shelf to variations in temperature distribution and flow enhancement, and found that incorporating a realistic thermal profile (where the marine ice layer is isothermal) had an order of magnitude greater effect on ice mass flux and thinning than incorporating the mechanical properties of the marine ice. The presence of marine ice at the ice shelf base has the potential to significantly increase deviatoric stresses at the surface and ice mass flux across the front of an ice shelf.

141. Antarctic calving loss rivals ice-shelf thinning

Authors: Chad A. Greene; Alex S. Gardner; Nicole-Jeanne Schlegel; Alexander D. Fraser

Journal: Nature

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41586-022-05037-w

URL: <https://doi.org/10.1038/s41586-022-05037-w>

Publication Date: 2022-08-10

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References: 74

Cited by: 61 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

142. Inferring time-dependent calving dynamics at Helheim Glacier

Authors: Jacob Downs; Douglas Brinkerhoff; Mathieu Morlighem

Journal: Journal of Glaciology

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/jog.2022.68

URL: <https://doi.org/10.1017/jog.2022.68>

Publication Date: 2022-08-10

Volume/Issue: Volume 69, Issue 274

References: 54

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Heising-Simons Foundation, Heising-Simons Foundation, Heising-Simons Foundation

License: CC BY 4.0

Open Access: N/A

Abstract: We perform Bayesian inference of the parameters of a time-dependent model of ice flow and calving at Helheim Glacier, East Greenland. We find that, while a time-independent calving parameterization can recover the mean observed terminus position, such a model is unable to recover sub-annual variability, even when forced with seasonally varying climate. To address this, we develop a simple stochastic model relating surface runoff rates and calving threshold. Again inferring model parameters from observations, we find that this parameterization is able to reproduce observations with respect to both mean position and characteristic temporal variability. This result demonstrates the importance of considering potential sub-annual controls on calving rates in numerical models, which may include variable undercutting rates or surface runoff-dependent surface crevasse propagation.

143. Stochastic renewal process model of time-variant tsunami hazard assessment under nonstationary effects of sea-level rise due to climate change

Authors: Abdul Kadir Alhamid; Mitsuyoshi Akiyama; Koki Aoki; Shunichi Koshimura; Dan M. Frangopol

Journal: Structural Safety

Publisher: Elsevier BV

DOI: 10.1016/j.strusafe.2022.102263

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Publication Date: 2022-08-08

Volume/Issue: Volume 99

References: 74

Cited by: 20 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

144. Inferring forms of glacier slip laws from estimates of ice-bed separation during glacier slip

Authors: Jacob B. Woodard; Lucas K. Zoet; Neal R. Iverson; Christian Helanow

Journal: Journal of Glaciology

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/jog.2022.63

URL: <https://doi.org/10.1017/jog.2022.63>

Publication Date: 2022-08-05

Volume/Issue: Volume 69, Issue 274

References: 42

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: United States National Science Foundation

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Open Access: N/A

Abstract: Sea-level projections depend sensitively on the parameterization used for basal slip in glacier flow models. During slip over rock-beds, ice-bed separation increases with slip velocity and basal water pressure. We present a method for using these variables and measured bed topography to estimate the average bed slope in contact with ice, $\bar{m}$. Three-dimensional numerical modeling of slip over small areas of former beds has shown that changes in $\bar{m}$ with increasing slip velocity and water pressure mimic changes in basal drag. Computed values of $\bar{m}$ can thus provide the form of the slip law that relates drag to velocity and water pressure, avoiding computationally expensive numerical modeling. The method is applied to 618 sections from four former glacier beds. Results generally show an increase in $\bar{m}$, and hence inferred basal drag, with slip velocity up to a limiting value, consistent with a regularized Coulomb slip law.

145. Fine-Scale Geothermal Heat Flow in Antarctica Can Increase Simulated Subglacial Melt Estimates

Authors: Felicity S. McCormack (Monash University); Jason L. Roberts (Australian Antarctic Division Kingston TAS Australia); Christine F. Dow (University of Waterloo); Tobias Stål (University of Tasmania); Jacqueline A. Halpin (University of Tasmania); Anya M. Reading (University of Tasmania); Martin J. Siegert (Imperial College London)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: basal melt, geothermal heat flow, ice sheet model, Antarctica, Aurora Subglacial Basin

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: Antarctic geothermal heat flow (GHF) affects the thermal regime of ice sheets and simulations of ice and subglacial meltwater discharge to the ocean, but remains poorly constrained. We use an ice sheet model to investigate the impact of GHF anomalies on subglacial meltwater production in the Aurora Subglacial Basin, East Antarctica. We find that spatially-variable GHF fields produce more meltwater than a constant GHF with the same background mean, and meltwater production increases as the resolution of GHF anomalies increases. Our results suggest that model simulations of this region systematically underestimate meltwater production using current GHF models. We determine the minimum basal heating required to bring the basal ice temperature to the pressure melting point, which should be taken together with the scale-length of likely local variability in targeting in-situ GHF field campaigns.

146. Benchmarking the vertically integrated ice-sheet model IMAU-ICE (version 2.0)

Authors: Constantijn J. Berends; Heiko Goelzer; Thomas J. Reerink; Lennert B. Stap; Roderik S. W. van de Wal

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-15-5667-2022

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Language: en

Subjects: N/A

Keywords: N/A

Funding: Horizon 2020, Aard- en Levenswetenschappen, Nederlandse Organisatie voor Wetenschappelijk Onderzoek

License: CC BY 4.0

Open Access: N/A

Abstract: . Ice-dynamical processes constitute a large uncertainty in future projections of sea-level rise caused by anthropogenic climate change. Improving our understanding of these processes requires ice-sheet models that perform well at simulating both past and future ice-sheet evolution. Here, we present version 2.0 of the ice-sheet model IMAU-ICE, which uses the depth-integrated viscosity approximation (DIVA) to solve the stress balance. We evaluate its performance in a range of benchmark experiments, including simple analytical solutions and both schematic and realistic model intercomparison exercises. IMAU-ICE has adopted recent developments in the numerical treatment of englacial stress and sub-shelf melt near the grounding line, which result in good performance in experiments concerning grounding-line migration (MISMIP, MISMIP+) and buttressing (ABUMIP). This makes it a model that is robust, versatile, and user-friendly, which will provide a firm basis for (palaeo-)glaciological research in the coming years.

147. Simulating the Holocene deglaciation across a marine-terminating portion of southwestern Greenland in response to marine and atmospheric forcings

Authors: Joshua K. Cuzzzone; Nicolás E. Young; Mathieu Morlighem; Jason P. Briner; Nicole-Jeanne Schlegel

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-16-2355-2022

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Publication Date: 2022-06-17

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References: 70

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Division of Arctic Sciences

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Open Access: N/A

Abstract: . Numerical simulations of the Greenland Ice Sheet (GrIS) over geologic timescales can greatly improve our knowledge of the critical factors driving GrIS demise during climatically warm periods, which has clear relevance for better predicting GrIS behavior over the upcoming centuries. To assess the fidelity of these modeling efforts, however, observational constraints of past ice sheet change are needed. Across southwestern Greenland, geologic records detail Holocene ice retreat across both terrestrial-based and marine-terminating environments, providing an ideal opportunity to rigorously benchmark model simulations against geologic reconstructions of ice sheet change. Here, we present regional ice sheet modeling results using the Ice-sheet and Sea-level System Model (ISSM) of Holocene ice sheet history across an extensive fjord region in southwestern Greenland covering the landscape around the Kangerlussuaq Nunaata Sermia (KNS) glacier and extending outward along the 200 km Nuup Kangerula (Godthåbsfjord). Our simulations, forced by reconstructions of Holocene climate and recently implemented calving laws, assess the sensitivity of ice retreat across the KNS region to atmospheric and oceanic forcing. Our simulations reveal that the geologically reconstructed ice retreat across the terrestrial landscape in the study area was likely driven by fluctuations in surface mass balance in response to Early Holocene warming - and was likely not influenced significantly by the response of adjacent outlet glaciers to calving and ocean-induced melting. The impact of ice calving within fjords, however, plays a significant role by enhancing ice discharge at the terminus, leading to interior thinning up to the ice divide that is consistent with reconstructed magnitudes of Early Holocene ice thinning. Our results, benchmarked against geologic constraints of past ice-margin change, suggest that while calving did not strongly influence Holocene ice-margin migration across terrestrial portions of the KNS forefield, it strongly impacted regional mass loss. While these results imply that the implementation and resolution of ice calving in paleo-ice-flow models is important towards making more robust estimations of past ice mass change, they also illustrate the importance these processes have on contemporary and future long-term ice mass change across similar fjord-dominated regions of the GrIS.

148. Impact of runoff temporal distribution on ice dynamics

Authors: Basile de Fleurian; Richard Davy; Petra M. Langebroek

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-16-2265-2022

URL: <https://doi.org/10.5194/tc-16-2265-2022>

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References: 60

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Language: en

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Keywords: N/A

Funding: Norges Forskningsråd

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Open Access: N/A

Abstract: . Record highs of meltwater production at the surface of the Greenland ice sheet have been recorded with a high recurrence over the last decades. Those melt seasons with longer durations, larger intensities, or with both increased length and melt intensity have a direct impact on the surface mass balance of the ice sheet and on its contribution to sea level rise. Moreover, the surface melt also affects the ice dynamics through the meltwater lubrication feedback. It is still not clear how the meltwater lubrication feedback impacts the long-term ice velocities on the Greenland ice sheet. Here we take a modeling approach with simplified ice sheet geometry and climate forcings to investigate in more detail the impacts of the changing characteristics of the melt season on ice dynamics. We model the ice dynamics through the coupling of the Double Continuum (DoCo) subglacial hydrology model with a shallow shelf approximation for the ice dynamics in the Ice-sheet and Sea-level System Model (ISSM). The climate forcing is generated from the ERA5 dataset to allow the length and intensity of the melt season to be varied in a comparable range of values. Our simulations present different behaviors between the lower and higher part of the glacier, but overall, a longer melt season will yield a faster glacier for a given runoff value. However, an increase in the intensity of the melt season, even under increasing runoff, tends to reduce glacier velocities. Those results emphasize the complexity of the meltwater lubrication feedback and urge us to use subglacial drainage models with both inefficient and efficient drainage components to give an accurate assessment of its impact on the overall dynamics of the Greenland ice sheet.

149. Resolving glacial isostatic adjustment (GIA) in response to modern and future ice loss at marine grounding lines in West Antarctica

Authors: Jeannette Xiu Wen Wan; Natalya Gomez; Konstantin Latychev; Holly Kyeore Han

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-16-2203-2022

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References: 71

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Natural Sciences and Engineering Research Council of Canada, Canada Research Chairs, Canada Foundation for Innovation

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Open Access: N/A

Abstract: . Accurate glacial isostatic adjustment (GIA) modelling in the cryosphere is required for interpreting satellite, geophysical and geological records and for assessing the feedbacks of Earth deformation and sea-level change on marine ice-sheet grounding lines. GIA modelling in areas of active ice loss in West Antarctica is particularly challenging because the ice is underlain by laterally varying mantle viscosities that are up to several orders of magnitude lower than the global average, leading to a faster and more localised response of the solid Earth to ongoing and future ice-sheet retreat and necessitating GIA models that incorporate 3-D viscoelastic Earth structure. Improvements to GIA models allow for computation of the viscoelastic response of the Earth to surface ice loading at sub-kilometre resolution, and ice-sheet models and observational products now provide the inputs to GIA models at comparably unprecedented detail. However, the resolution required to accurately capture GIA in models remains poorly understood, and high-resolution calculations come at heavy computational expense. We adopt a 3-D GIA model with a range of Earth structure models based on recent seismic tomography and geodetic data to perform a comprehensive analysis of the influence of grid resolution on predictions of GIA in the Amundsen Sea Embayment (ASE) in West Antarctica. Through idealised sensitivity testing down to sub-kilometre resolution with spatially isolated ice loading changes, we find that a grid resolution of ~ 13 of the radius of the load or higher is required to accurately capture the elastic response of the Earth. However, when we consider more realistic, spatially coherent ice loss scenarios based on modern observational records and future ice-sheet model projections and adopt a viscoelastic Earth, we find that predicted deformation and sea-level change along the grounding line converge to within 5 % with grid resolutions of 7.5 km or higher, and to within 2 % for grid resolutions of 3.75 km and higher, even when the input ice model is on a 1 km grid. Furthermore, we show that low mantle viscosities beneath the ASE lead to viscous deformation that contributes to the instrumental record on decadal timescales and equals or dominates over elastic effects by the end of the 21st century. Our findings suggest that for the range of resolutions of 1.9–15 km that we considered, the error due to adopting a coarser grid in this region is negligible compared to the

effect of neglecting viscous effects and the uncertainty in the adopted mantle viscosity structure.

150. A Semi-Empirical Framework for ice sheet response analysis under Oceanic forcing in Antarctica and Greenland

Authors: Xiao Luo; Ting Lin

Journal: Climate Dynamics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s00382-022-06317-x

URL: <https://doi.org/10.1007/s00382-022-06317-x>

Publication Date: 2022-05-20

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Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Texas Tech University

License: N/A

Open Access: N/A

Abstract: N/A

151. A scalability study of the Ice-sheet and Sea-level System Model (ISSM, version 4.18)

Authors: Yannic Fischler; Martin Rückamp; Christian Bischof; Vadym Aizinger; Mathieu Morlighem; Angelika Humbert

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-15-3753-2022

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References: 57

Cited by: 4 publications

Language: en

Subjects: N/A

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Funding: Deutsche Forschungsgemeinschaft, LOEWE Zentrum AdRIA

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Open Access: N/A

Abstract: . Accurately modelling the contribution of Greenland and Antarctica to sea level rise requires solving partial differential equations at a high spatial resolution. In this paper, we discuss the scaling of the Ice-sheet and Sea-level System Model (ISSM) applied to the Greenland Ice Sheet with horizontal grid resolutions varying between 10 and 0.25 km. The model setup used as benchmark problem comprises a variety of modules with different levels of complexity and computational demands. The core builds the so-called stress balance module, which uses the higher-order approximation (or Blatter-Pattyn) of the Stokes equations, including free surface and ice-front evolution as well as thermodynamics in form of an enthalpy balance, and a mesh of linear prismatic finite elements, to compute the ice flow. We develop a detailed user-oriented, yet low-overhead, performance instrumentation tailored to the requirements of Earth system models and run scaling tests up to 6144 Message Passing Interface (MPI) processes. The results show that the computation of the Greenland model scales overall well up to 3072 MPI processes but is eventually slowed down by matrix assembly, the output handling and lower-dimensional problems that employ lower numbers of unknowns per MPI process. We also discuss improvements of the scaling and identify further improvements needed for climate research. The instrumented version of ISSM thus not only identifies potential performance bottlenecks that were not present at lower core counts but also provides the capability to continually monitor the performance of ISSM code basis. This is of long-term significance as the overall performance of ISSM model depends on the subtle interplay between algorithms, their implementation, underlying libraries, compilers, runtime systems and hardware characteristics, all of which are in a constant state of flux. We believe that future large-scale high-performance computing (HPC) systems will continue to employ the MPI-based programming paradigm on the road to exascale. Our scaling study pertains to a particular modelling setup available within ISSM and does not address accelerator techniques such as the use of vector units or GPUs. However, with 6144 MPI processes, we identified issues that need to be addressed in order to improve the ability of the ISSM code base to take advantage of upcoming systems that will require scaling to even higher numbers of MPI processes.

152. GREB-ISM v1.0: A coupled ice sheet model for the Globally Resolved Energy Balance model for global simulations on timescales of 100 kyr

Authors: Zhiang Xie; Dietmar Dommenget; Felicity S. McCormack; Andrew N. Mackintosh

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-15-3691-2022

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License: CC BY 4.0

Open Access: N/A

Abstract: . We introduce a newly developed global ice sheet model coupled to the Globally Resolved Energy Balance (GREB) climate model for the simulation of global ice sheet evolution on timescales of 100 kyr or longer (GREB-ISM v1.0). Ice sheets and ice shelves are simulated on a global grid, fully interacting with the climate simulation of surface temperature, precipitation, albedo, land-sea mask, topography and sea level. Thus, it is a fully coupled atmosphere, ocean, land and ice sheet model. We test the model in ice sheet stand-alone and fully coupled simulations. The ice sheet model dynamics behave similarly to other hybrid SIA (shallow ice approximation) and SSA (shallow shelf approximation) models, but the West Antarctic ice Sheet accumulates too much ice using present-day boundary conditions. The coupled model simulations produce global equilibrium ice sheet volumes and calving rates like those observed for present-day boundary conditions. We designed a series of idealized experiments driven by oscillating solar radiation forcing on periods of 20, 50 and 100 kyr in the Northern Hemisphere. These simulations show clear interactions between the climate system and ice sheets, resulting in slow buildup and fast decay of ice-covered areas and global ice volume. The results also show that Northern Hemisphere ice sheets respond more strongly to timescales longer than 100 kyr. The coupling to the atmosphere and sea level leads to climate interactions between the Northern and Southern Hemispheres. The model can run global simulations of 100 kyr d-1 on a desktop computer, allowing the simulation of the whole Quaternary period (2.6 Myr) within 1 month.

153. Petermann ice shelf may not recover after a future breakup

Authors: Henning Åkesson; Mathieu Morlighem; Johan Nilsson; Christian Stranne; Martin Jakobsson

Journal: Nature Communications

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41467-022-29529-5

URL: <https://doi.org/10.1038/s41467-022-29529-5>

Publication Date: 2022-05-09

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References: 83

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Abstract: Floating ice shelves buttress inland ice and curtail grounded-ice discharge. Climate warming causes melting and ultimately breakup of ice shelves, which could escalate ocean-bound ice discharge and thereby sea-level rise. Should ice shelves collapse, it is unclear whether they could recover, even if we meet the goals of the Paris Agreement. Here, we use a numerical ice-sheet model to determine if Petermann Ice Shelf in northwest Greenland can recover from a future breakup. Our experiments suggest that post-breakup recovery of confined ice shelves like Petermann's is unlikely, unless iceberg calving is greatly reduced. Ice discharge from Petermann Glacier also remains up to 40% higher than today, even if the ocean cools below present-day temperatures. If this behaviour is not unique for Petermann, continued near-future ocean warming may push the ice shelves protecting Earth's polar ice sheets into a new retreated high-discharge state which may be exceedingly difficult to recover from.

154. Constraining the Earth's Dynamical Ellipticity From Ice Age Dynamics

Authors: Mohammad Farhat (IMCCE CNRS Observatoire de Paris PSL University Sorbonne Université Paris France); Jacques Laskar (IMCCE CNRS Observatoire de Paris PSL University Sorbonne Université Paris France); Gwenaél Boué (IMCCE CNRS Observatoire de Paris PSL University Sorbonne Université Paris France)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

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Language: en

Subjects: N/A

Keywords: Earth evolution, dynamical ellipticity, glacial isostatic adjustment, ice ages

Funding: Agence Nationale de la Recherche

License: N/A

Open Access: N/A

Abstract: The dynamical ellipticity of a planet expresses the departure of its mass distribution from spherical symmetry. It enters as a parameter in the description of a planet's precession and nutation, as well as other rotational normal modes. In the case of the Earth, uncertainties in this quantity's history produce an uncertainty in the solutions for the past evolution of the Earth-Moon system. Constraining this history has been a target of interdisciplinary efforts as it represents an astro-geodetic parameter whose variation is shaped by geophysical processes, and whose imprints can be found in the geological signal. We revisit the classical problem of its variation during ice ages, where glacial cycles exerted a varying surface loading that had altered the shape of the geoid. In the framework of glacial isostatic adjustment, and with the help of a recent paleoclimatic proxy of ice volume, we present the evolution of the dynamical ellipticity over the Cenozoic ice ages. We map out the problem in full generality identifying major sensitivities to surface loading and internal variations in parameter space. This constrained evolution is aimed to be used in future astronomical computations of the orbital and insolation quantities of the Earth.

155. Comparison of ice dynamics using full-Stokes and Blatter-Pattyn approximation: application to the Northeast Greenland Ice Stream

Authors: Martin Rückamp; Thomas Kleiner; Angelika Humbert

Journal: The Cryosphere

Publisher: Copernicus GmbH

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URL: <https://doi.org/10.5194/tc-16-1675-2022>

Publication Date: 2022-05-05

Volume/Issue: Volume 16, Issue 5

References: 61

Cited by: 5 publications

Language: en

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Funding: N/A

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Open Access: N/A

Abstract: . Full-Stokes (FS) ice sheet models provide the most sophisticated formulation of ice sheet flow. However, their applicability is often limited due to the high computational demand and numerical challenges. To balance computational demand and accuracy, the so-called Blatter-Pattyn (BP) stress regime is frequently used. Here, we explore the dynamic consequences of using simplified approaches by solving FS and the BP stress regime applied to the Northeast Greenland Ice Stream. To ensure a consistent comparison, we use one single ice sheet model to run the simulations under identical numerical conditions. A sensitivity study to the horizontal grid resolution (from 12.8 to a resolution of 0.1 km) reveals that velocity differences between the FS and BP solution emerge below ~ 1 km horizontal resolution and continuously increase with resolution. Over the majority of the modelling domain both models reveal similar surface velocity patterns. At the grounding line of the 79° North Glacier the simulations show considerable differences whereby the BP model overestimates ice discharge of up to 50 % compared to FS. A sensitivity study to the friction type reveals that differences are stronger for a power-law friction than a linear friction law. Model differences are attributed to topographic variability and the basal drag, in which neglected stress terms in BP become important.

156. Carbon Dioxide Ice Glaciers at the South Pole of Mars

Authors: I. B. Smith (Planetary Science Institute Lakewood CO USA, York University Toronto ON Canada); N.-J. Schlegel (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); E. Larour (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); I. Isola (York University Toronto ON Canada); P. B. Buhler (Planetary Science Institute Lakewood CO USA); N. E. Putzig (Planetary Science Institute Lakewood CO USA); R. Greve (Hokkaido University)

Journal: Journal of Geophysical Research: Planets

Publisher: American Geophysical Union (AGU)

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Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: ice, Mars, climate, carbon dioxide, glaciers

Funding: Canada Research Chairs

License: N/A

Open Access: N/A

Abstract: Massive, kilometer thick deposits of carbon dioxide (CO₂) ice have been detected at the south polar cap of Mars by radar investigations. These deposits are divided into several units that are separated by thin water ice bounding layers. Recent studies investigated the accumulation history of CO₂ ice and found that the deposits most likely formed during several episodes in the past, when Martian obliquity was much lower than now. Those studies, while able to predict total volumes of CO₂ ice consistent with those observed, did not attempt to explain the anomalous three-dimensional distribution (thickness or extent) of CO₂ ice or the ice's offset from the topographic high of the polar cap. In this paper we use a combination of feature analysis and numerical modeling to demonstrate that the CO₂ deposits flow as glaciers and that glacial flow distributes the ice into its current position. Further, this distribution allows the ice to survive during high obliquity excursions.

157. Atypical landslide induces speedup, advance, and long-term slowdown of a tidewater glacier

Authors: Maximillian Van Wyk de Vries (University of Minnesota); Andrew D. Wickert (University of Minnesota); Kelly R. MacGregor (Department of Geology, Macalester College, Saint Paul, Minnesota 55105, USA); Camilo Rada (Centro de Investigación GAlA Antártica, Universidad de Magallanes, 6210427 Punta Arenas, Chile); Michael J. Willis (University of Colorado Boulder)

Journal: Geology

Publisher: Geological Society of America

DOI: 10.1130/g49854.1

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Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Atmospheric and oceanic warming over the past century have driven rapid glacier thinning and retreat, destabilizing hillslopes and increasing the frequency of landslides. The impact of these landslides on glacier dynamics and resultant secondary landslide hazards are not fully understood. We investigated how a $262 \pm 77 \times 106 \text{ m}^3$ landslide affected the flow of Amalia Glacier, Chilean Patagonia. Despite being one of the largest recorded landslides in a glaciated region, it emplaced little debris onto the glacier surface. Instead, it left a series of landslide-perpendicular ridges, landslide-parallel fractures, and an apron of ice debris--with blocks as much as 25 m across. Our observations suggest that a deep-seated failure of the mountainside impacted the glacier flank, propagating brittle deformation through the ice and emplacing the bulk of the rock mass below the glacier. The landslide triggered a brief downglacier acceleration of Amalia Glacier followed by a slowdown of as much as 60% of the pre-landslide speed and increased suspended-sediment concentrations in the fjord. These results highlight that landslides may induce widespread and long-lasting disruptions to glacier dynamics.

158. Assessment of MODIS Surface Temperature Products of Greenland Ice Sheet Using In-Situ Measurements

Authors: Xiaoge Yu (College of Geography and Environment, Shandong Normal University, Jinan 250061, China); Tingting Wang (College of Geography and Environment, Shandong Normal University, Jinan 250061, China); Minghu Ding (Chinese Academy of Meteorological Sciences); Yetang Wang (College of Geography and Environment, Shandong Normal University, Jinan 250061, China); Weijun Sun (College of Geography and Environment, Shandong Normal University, Jinan 250061, China); Qinglin Zhang (College of Geography and Environment, Shandong Normal University, Jinan 250061, China); Baojuan Huai (College of Geography and Environment, Shandong Normal University, Jinan 250061, China)

Journal: Land

Publisher: MDPI AG

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URL: <https://doi.org/10.3390/land11050593>

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References: 60

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Keywords: N/A

Funding: National Natural Science Foundation of China

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Open Access: N/A

Abstract: Satellite-based data have promoted the research progress in polar regions under global climate change, meanwhile the uncertainties and limitations of satellite-derived surface temperatures are widely discussed over Greenland. This study validated the accuracy of ice surface temperature (IST) from the moderate-resolution imaging spectroradiometer (MODIS) over the Greenland ice sheet (GrIS). Daily MODIS IST was validated against the observational surface temperature from 24 automatic weather stations (AWSs) using the mean bias (MB), the root mean square (RMSE), and the correlation coefficient (R). The temporal and spatial variability over the GrIS spanning from March 2000 to December 2019 and the IST melt threshold (-1 degreesC) were analyzed. Generally, the MODIS IST was underestimated by an average of -2.68 degreesC compared to AWSs, with cold bias mainly occurring in winter. Spatially, the R and RMSE performed the better accuracy of MODIS IST on the northwest, northeast, and central part of the GrIS. Furthermore, the mean IST is mainly concentrated between -20 degreesC and -10 degreesC in summer while between -50 degreesC and -30 degreesC in winter. The largest positive IST anomalies (exceeds 3 degreesC) occurred in southwestern GrIS during 2010. IST shows the positive trends mainly in spring and summer and negative in autumn and winter.

159. Quantifying the Impact of Bedrock Topography Uncertainty in Pine Island Glacier Projections for This Century

Authors: Andreas Wernecke (School of Environment Earth and Ecosystem Sciences The Open University Milton Keynes UK, University of Hamburg, Max-Planck-Institute for Meteorology Hamburg Germany); Tamsin L. Edwards (Kings College London London UK); Philip B. Holden (School of Environment Earth and Ecosystem Sciences The Open University Milton Keynes UK); Neil R. Edwards (School of Environment Earth and Ecosystem Sciences The Open University Milton Keynes UK); Stephen L. Cornford (Faculty of Science and Engineering Swansea University Swansea UK)

Journal: Geophysical Research Letters

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Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: Pine Island Glacier, bedrock topography, uncertainty quantification, ice sheet modeling, BISICLES (and 1 more)

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Open Access: N/A

Abstract: The predicted Antarctic contribution to global-mean sea-level rise is one of the most uncertain among all major sources. Partly this is because of instability mechanisms of the ice flow over deep basins. Errors in bedrock topography can substantially impact the projected resilience of glaciers against such instabilities. Here we analyze the Pine Island Glacier topography to derive a statistical model representation. Our model allows for inhomogeneous and spatially dependent uncertainties and avoids unnecessary smoothing from spatial averaging or interpolation. A set of topography realizations is generated representing our best estimate of the topographic uncertainty in ice sheet model simulations. The bedrock uncertainty alone creates a 5%-25% uncertainty in the predicted sea level rise contribution at year 2100, depending on friction law and climate forcing. Pine Island Glacier simulations on this new set are consistent with simulations on the BedMachine reference topography but diverge from Bedmap2 simulations.

160. Arctic observations and sustainable development goals - Contributions and examples from ERA-PLANET iCUPE data

Authors: Steffen M. Noe; Ksenia Tabakova; Alexander Mahura; Hanna K. Lappalainen; Miriam Kosmale; Jyri Heilimo; Roberto Salzano; Mattia Santoro; Rosamaria Salvatori; Andrea Spolaor; Warren Cairns; Carlo Barbante; Fidel Pankratov; Angelika Humbert; Jeroen E. Sonke; Kathy S. Law; Tatsuo Onishi; Jean-Daniel Paris; Henrik Skov; Andreas Massling; Aurélien Dommergue; Mikhail Arshinov; Denis Davydov; Boris Belan; Tuukka Petäjä

Journal: Environmental Science & Policy

Publisher: Elsevier BV

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References: 77

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

161. Ice viscosity is more sensitive to stress than commonly assumed

Authors: Joanna D. Millstein; Brent M. Minchew; Samuel S. Pegler

Journal: Communications Earth & Environment

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s43247-022-00385-x

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Cited by: 29 publications

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Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: Accurate representation of the viscous flow of ice is fundamental to understanding glacier dynamics and projecting sea-level rise. Ice viscosity is often described by a simple but largely untested and uncalibrated constitutive relation, Glen's Flow Law, wherein the rate of deformation is proportional to stress raised to the power n . The value $n = 3$ is commonly prescribed in ice-flow models, though observations and experiments support a range of values across stresses and temperatures found on Earth. Here, we leverage recent remotely-sensed observations of Antarctic ice shelves to show that Glen's Flow Law approximates the viscous flow of ice with $n = 4.1 \pm 0.4$ in fast-flowing areas. The viscosity and flow rate of ice are therefore more sensitive to changes in stress than most ice-flow models allow. By calibrating the governing equation of ice deformation, our result is a pathway towards improving projections of future glacier change.

162. A rapidly retreating, marine-terminating glacier's modeled response to perturbations in basal traction

Authors: Jacob Downs; Jesse V. Johnson

Journal: Journal of Glaciology

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/jog.2022.5

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Publication Date: 2022-03-08

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References: 56

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

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Open Access: N/A

Abstract: Upernavik Isstrøm, a marine glacier undergoing rapid retreat, is simulated by forcing a numerical model with ocean-driven melt. A review of processes driving retreat led us to hypothesize that a glacier undergoing rapid retreat may be less sensitive to perturbations in the balance of forces than a glacier that is undergoing moderate changes or a glacier in steady state. Numerical experiments suggest this is not the case, and that a system in rapid retreat is as sensitive to basal traction perturbations as a system that is near to steady state. This result is important when considering other glacier systems experiencing marine-forced retreat. While the ice-ocean interface is of primary importance, additional perturbations from meltwater-forced decoupling of the glacier from its bed continue to feature in glacier dynamics.

163. Derivation of bedrock topography measurement requirements for the reduction of uncertainty in ice-sheet model projections of Thwaites Glacier

Authors: Blake A. Castleman; Nicole-Jeanne Schlegel; Lambert Caron; Eric Larour; Ala Khazendar

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-16-761-2022

URL: <https://doi.org/10.5194/tc-16-761-2022>

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References: 71

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Jet Propulsion Laboratory, National Aeronautics and Space Administration

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Open Access: N/A

Abstract: . Determining the future evolution of the Antarctic IceSheet is critical for understanding and narrowing the large existing uncertainties in century-scale global mean sea-level-rise (SLR) projections. One of the most significant glaciers and ice streams in Antarctica, Thwaites Glacier, is at risk of destabilization and, if destabilized, has the potential to be the largest regional-scale contributor of SLR on Earth. This is because Thwaites Glacier is vulnerable to the marine ice-sheet instability as its grounding line is significantly influenced by ocean-driven basal melting rates, and its bedrock topography retrogrades into kilometer-deep troughs. In this study, we investigate how bedrock topography features influence the grounding line migration beneath Thwaites Glacier when extreme ocean-driven basal melt rates are applied. Specifically, we design experiments using the Ice-sheet and Sea-level System Model (ISSM) to quantify the SLR projection uncertainty due to reported errors in the current bedrock topography maps that are often used by ice-sheet models. We find that spread in model estimates of sea-level-rise contribution from Thwaites Glacier due to the reported bedrock topography error could be as large as 21.9 cm after 200 years of extreme ocean warming. Next, we perturb the bedrock topography beneath Thwaites Glacier using wavelet decomposition techniques to introduce realistic noise (within error). We explore the model space with multiple realizations of noise to quantify what spatial and vertical resolutions in bedrock topography are required to minimize the uncertainty in our 200-year experiment. We conclude that at least a 2 km spatial and 8 m vertical resolution would independently constrain possible SLR to ± 2 cm over 200 years, fulfilling requirements outlined by the 2017 Decadal Survey for Earth Science. Lastly, we perform an ensemble of simulations to determine in which regions our model of Thwaites Glacier is most sensitive to perturbations in bedrock topography. Our results suggest that the retreat of the grounding line is most sensitive to bedrock topography in proximity to the grounding line's initial position. Additionally, we find that the location and amplitude of the bedrock perturbation is more significant than its sharpness and shape. Overall, these findings inform and benchmark observational requirements for future missions that will measure ice-sheet bedrock topography, not only in the case of Thwaites Glacier but for Antarctica on the continental scale.

164. Simulating ice-shelf extent using damage mechanics

Authors: Samuel B. Kachuck; Morgan Whitcomb; Jeremy N. Bassis; Daniel F. Martin; Stephen F. Price

Journal: Journal of Glaciology

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/jog.2022.12

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Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, U.S. Department of Energy, U.S. Department of Energy

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Open Access: N/A

Abstract: Inaccurate representations of iceberg calving from ice shelves are a large source of uncertainty in mass-loss projections from the Antarctic ice sheet. Here, we address this limitation by implementing and testing a continuum damage-mechanics model in a continental scale ice-sheet model. The damage-mechanics formulation, based on a linear stability analysis and subsequent long-wavelength approximation of crevasses that evolve in a viscous medium, links damage evolution to climate forcing and the large-scale stresses within an ice shelf. We incorporate this model into the BISICLES ice-sheet model and test it by applying it to idealized (1) ice tongues, for which we present analytical solutions and (2) buttressed ice-shelf geometries. Our simulations show that the model reproduces the large disparity in lengths of ice shelves with geometries and melt rates broadly similar to those of four Antarctic ice shelves: Erebus Glacier Tongue (length ~ 13 km), the unembayed portion of Drygalski Ice Tongue (~ 65 km), the Amery Ice Shelf (~ 350 km) and the Ross Ice Shelf (~ 500 km). These results demonstrate that our simple continuum model holds promise for constraining realistic ice-shelf extents in large-scale ice-sheet models in a computationally tractable manner.

165. Helheim Glacier's Terminus Position Controls Its Seasonal and Inter-Annual Ice Flow Variability

Authors: Gong Cheng (University of California Irvine, Dartmouth College); Mathieu Morlighem (University of California Irvine, Dartmouth College); Jérémie Mouginot (University of California Irvine, Institut de Géosciences de l'Environnement Université Grenoble Alpes Grenoble France); Daniel Cheng (University of California Irvine)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: ice flow variability, Helheim glacier, ice sheet modeling, calving, ISSM

Funding: Heising-Simons Foundation, Agence Nationale de la Recherche

License: N/A

Open Access: N/A

Abstract: Over the past decade, one of the largest contributors to total ice discharge across the Greenland ice sheet, Helheim Glacier, has experienced large fluctuations in ice velocity. In this study, we simulate the dynamics of Helheim, from 2007 to 2020, using the Ice-sheet and Sea-level System Model to identify the drivers of these large changes in ice discharge. By quantifying the impact of individual external forcing and model parameters on Helheim's modeled velocity, we find that the position of the calving front alone explains the dynamic variability of the glacier, as it has a direct and large impact on Helheim's ice velocity. The seasonal to inter-annual variability of Helheim Glacier is, however, relatively insensitive to the choice of friction law or ice rheology factor. This study shows that more research on calving dynamics and ice-ocean interactions is required to project the future of this sector of Greenland.

166. Modeling the Deformation Regime of Thwaites Glacier, West Antarctica, Using a Simple Flow Relation for Ice Anisotropy (ESTAR)

Authors: F. S. McCormack (University of Tasmania, Monash University); R. C. Warner (University of Tasmania); H. Seroussi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, Dartmouth College); C. F. Dow (University of Waterloo); J. L. Roberts (Australian Antarctic Division Kingston TAS Australia); A. Treverrow (Australian Antarctic Division Kingston TAS Australia)

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021jf006332

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References: 98

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

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Open Access: N/A

Abstract: Ice deformation dominates the evolution of ice shelf flow and the slow-moving regions in the interior of ice sheets. However, deformation may be poorly represented in large-scale ice sheet models that use the Glen flow relation, due to its questionable applicability to the steady-state flow of anisotropic ice that prevails in ice sheets, having been derived from secondary creep rates of isotropic ice. We assess the deformation regimes of Thwaites Glacier, West Antarctica, using the Glen and "Empirical Scalar Tertiary Anisotropy Regime", (ESTAR) flow relations, the latter being derived from steady-state deformation rates of anisotropic ice. For grounded ice, the character of the flow relation determines the contribution of deformation to overall flow, with ESTAR producing greater bed-parallel shear deformation than the standard Glen flow relation. The ESTAR experiments show larger basal shear stress maxima than the standard Glen experiment because ESTAR treats the responses to simple shear stresses and compression stresses differently, reducing the role of lateral and longitudinal stresses in momentum balance. On the Thwaites Glacier Tongue, ESTAR provides the best match to observed speeds by accounting for the differing effects of stresses on ice flow. Our results highlight the importance of the numerical description of anisotropy, particularly: In regions of transition from deformation-dominated to sliding-dominated flow; in the approach to the grounding line, and across ice shelves. Given the importance of these locations in determining mass flux into the ocean, our results have implications for projections of sea level change from Antarctic ice loss.

167. Controls on Larsen C Ice Shelf Retreat From a 60-Year Satellite Data Record

Authors: Shujie Wang (Pennsylvania State University); Hongxing Liu (University of Alabama, Key Laboratory of Geographic Information Science Ministry of Education East China Normal University Shanghai China); Kenneth Jezek (Ohio State University); Richard B. Alley (Pennsylvania State University); Lei Wang (Louisiana State University); Patrick Alexander (Columbia University); Yan Huang (Key Laboratory of Geographic Information Science Ministry of Education East China Normal University Shanghai China)

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

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References: 90

Cited by: 20 publications

Language: en

Subjects: N/A

Keywords: flow velocity, ice shelf, Antarctica, retreat, Larsen C (and 1 more)

Funding: National Science Foundation, Heising-Simons Foundation

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Open Access: N/A

Abstract: Rapid retreat of the Larsen A and B ice shelves has provided important clues about the ice shelf destabilization processes. The Larsen C Ice Shelf, the largest remaining ice shelf on the Antarctic Peninsula, may also be vulnerable to future collapse in a warming climate. Here, we utilize multisource satellite images collected over 1963-2020 to derive multidecadal time series of ice front, flow velocities, and critical rift features over Larsen C, with the aim of understanding the controls on its retreat. We complement these observations with modeling experiments using the Ice-sheet and Sea-level System Model to examine how front geometry conditions and mechanical weakening due to rifts affect ice shelf dynamics. Over the past six decades, Larsen C lost over 20% of its area, dominated by rift-induced tabular iceberg calving. The Bawden Ice Rise and Gipps Ice Rise are critical areas for rift formation, through their impact on the longitudinal deviatoric stress field. Mechanical weakening around Gipps Ice Rise is found to be an important control on localized flow acceleration and the propagation of two rifts that caused a major calving event in 2017. Capturing the time-varying effects of rifts on ice rigidity in ice shelf models is essential for making realistic predictions of ice shelf flow dynamics and instability. In the context of the Larsen A and Larsen B collapses, we infer a chronology of destabilization processes for embayment-confined ice shelves, which provides a useful framework for understanding the historical and future destabilization of Antarctic ice shelves.

168. Mapping high-resolution basal topography of West Antarctica from radar data using non-stationary multiple-point geostatistics (MPS-BedMappingV1)

Authors: Zhen Yin; Chen Zuo; Emma J. MacKie; Jef Caers

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-15-1477-2022

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References: 94

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License: CC BY 4.0

Open Access: N/A

Abstract: . The subglacial bed topography is critical for modelling the evolution of Thwaites Glacier in the Amundsen Sea Embayment (ASE), where rapid ice loss threatens the stability of the West Antarctic Ice Sheet. However, mapping of subglacial topography is subject to uncertainties of up to hundreds of metres, primarily due to large gaps of up to tens of kilometres in airborne ice-penetrating radar flight lines. Deterministic interpolation approaches do not reflect such spatial uncertainty. While traditional geostatistical simulations can model such uncertainty, they become difficult to apply because of the significant non-stationary spatial variation of topography over such large surface area. In this study, we develop a non-stationary multiple-point geostatistical (MPS) approach to interpolate large areas with irregular geophysical data and apply it to model the spatial uncertainty of entire ASE basal topography. We collect 166 high-quality topographic training images (TIs) of resolution 500 m to train the gap-filling of radar data gaps, thereby simulating realistic topographic maps. The TIs are extensively sampled from deglaciated regions in the Arctic as well as Antarctica. To address the non-stationarity in topographic modelling, we introduce a Bayesian framework that models the posterior distribution of non-stationary TIs assigned to the local line data. Sampling from this distribution then provides candidate training images for local topographic modelling with uncertainty, constrained to radar flightline data. Compared to traditional MPS approaches that do not consider uncertain TI sampling, our approach results in a significant improvement in the topographic modelling quality and efficiency of the simulation algorithm. Finally, we simulate multiple realizations of high-resolution ASE topographic maps. We use the multiple realizations to investigate the impact of basal topography uncertainty on subglacial hydrological flow patterns.

169. Geometric controls of tidewater glacier dynamics

Authors: Thomas Frank; Henning Åkesson; Basile de Fleurian; Mathieu Morlighem; Kerim H. Nisancioglu

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-16-581-2022

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Keywords: N/A

Funding: Vetenskapsrådet, Norges Forskningsråd

License: CC BY 4.0

Open Access: N/A

Abstract: . Retreat of marine outlet glaciers often initiates depletion of inland ice through dynamic adjustments of the upstream glacier. The local topography of a fjord may promote or inhibit such retreat, and therefore fjord geometry constitutes a critical control on ice sheet mass balance. To quantify the processes of ice-topography interactions and enhance the understanding of the dynamics involved, we analyze a multitude of topographic fjord settings and scenarios using the Ice-sheet and Sea-level System Model (ISSM). We systematically study glacier retreat through a variety of artificial fjord geometries and quantify the modeled dynamics directly in relation to topographic features. We find that retreat in an upstream-widening or upstream-deepening fjord does not necessarily promote retreat, as suggested by previous studies. Conversely, it may stabilize a glacier because converging ice flow towards a constriction enhances lateral and basal shear stress gradients. An upstream-narrowing or upstream-shoaling fjord, in turn, may promote retreat since fjord walls or bed provide little stability to the glacier where ice flow diverges. Furthermore, we identify distinct quantitative relationships directly linking grounding line discharge and retreat rate to fjord topography and transfer these results to a long-term study of the retreat of Jakobshavn Isbræ. These findings offer new perspectives on ice-topography interactions and give guidance to an ad hoc assessment of future topographically induced ice loss based on knowledge of the upstream fjord geometry.

170. Ice Sheet Surface and Subsurface Melt Water Discrimination Using Multi-Frequency Microwave Radiometry

Authors: Andreas Colliander (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Mohammad Mousavi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Shawn Marshall (University of Calgary); Samira Samimi (University of Calgary); John S. Kimball (University of Montana); Julie Z. Miller (Cooperative Institute for Research in Environmental Sciences (CIRES) Boulder CO USA); Joel Johnson (Ohio State University); Mariko Burgin (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Funding: N/A

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Open Access: N/A

Abstract: For understanding englacial hydrology and its impact on ice sheet mass balance, observations of the liquid water content (LWC) within the ice sheets are needed. We combined 1.4-10.7 GHz passive microwave measurements with traditional 18.7-36.5 GHz measurements to detect subsurface LWC. In situ measurements from the DYE-2 experiment site in Greenland and a modeled LWC at this site were used to calibrate and validate the method. Our analysis showed sensitivity of the lower microwave frequencies to LWC in surface and subsurface layers down to at least 2 m, enabling detection of seasonal subsurface LWC and its refreezing. A simplified retrieval detected a delayed refreezing of subsurface LWC following surface freezing, while also capturing total seasonal meltwater production. These advancements open the door to detection of subsurface meltwater and refreezing twice a day at pan-Greenland scale, thereby enabling improved estimates of ice sheet contributions to global sea level rise.

171. Implementation of a Gaussian Markov random field sampler for forward uncertainty quantification in the Ice-sheet and Sea-level System Model v4.19

Authors: Kevin Bulthuis; Eric Larour

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-15-1195-2022

URL: <https://doi.org/10.5194/gmd-15-1195-2022>

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References: 62

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

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Open Access: N/A

Abstract: . Assessing the impact of uncertainties in ice-sheet models is a major and challenging issue that needs to be faced by the ice-sheet community to provide more robust and reliable model-based projections of ice-sheet mass balance. In recent years, uncertainty quantification (UQ) has been increasingly used to characterize and explore uncertainty in ice-sheet models and improve the robustness of their projections. A typical UQ analysis first involves the (probabilistic) characterization of the sources of uncertainty, followed by the propagation and sensitivity analysis of these sources of uncertainty. Previous studies concerned with UQ in ice-sheet models have generally focused on the last two steps but have paid relatively little attention to the preliminary and critical step of the characterization of uncertainty. Sources of uncertainty in ice-sheet models, like uncertainties in ice-sheet geometry or surface mass balance, typically vary in space and potentially in time. For that reason, they are more adequately described as spatio-(temporal) random fields, which account naturally for spatial (and temporal) correlation. As a means of improving the characterization of the sources of uncertainties for forward UQ analysis within the Ice-sheet and Sea-level System Model (ISSM), we present in this paper a stochastic sampler for Gaussian random fields with Matérn covariance function. The class of Matérn covariance functions provides a flexible model able to capture statistical dependence between locations with different degrees of spatial correlation or smoothness properties. The implementation of this stochastic sampler is based on a notable explicit link between Gaussian random fields with Matérn covariance function and a certain stochastic partial differential equation. Discretization of this stochastic partial differential equation by the finite-element method results in a sparse, scalable and computationally efficient representation known as a Gaussian Markov random field. In addition, spatio-temporal samples can be generated by combining an autoregressive temporal model and the Matérn field. The implementation is tested on a set of synthetic experiments to verify that it captures the desired spatial and temporal correlations well. Finally, we illustrate the interest of this stochastic sampler for forward UQ analysis in an application concerned with assessing the impact of various sources of uncertainties on the Pine Island Glacier, West Antarctica. We find that larger spatial and temporal correlations lengths will both likely result in increased uncertainty in the projections.

172. Layered seawater intrusion and melt under grounded ice

Authors: Alexander A. Robel; Earle Wilson; Helene Seroussi

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-16-451-2022

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Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: . Increasing melt of ice sheets at their floating or vertical interfaces with the ocean is a major driver of marine ice sheet retreat and sea level rise. However, the extent to which warm, salty seawater may drive melting under the grounded portions of ice sheets is still not well understood. Previous work has explored the possibility that dense seawater intrudes beneath relatively light subglacial freshwater discharge, similar to the "salt wedge" observed in many estuarine systems. In this study, we develop a generalized theory of layered seawater intrusion under grounded ice, including where subglacial hydrology occurs as a macroporous water sheet over impermeable beds or as microporous Darcy flow through permeable till. Using predictions from this theory, we show that seawater intrusion over flat or reverse-sloping impermeable beds may feasibly occur up to tens of kilometers upstream of a glacier terminus or grounding line. On the other hand, seawater is unlikely to intrude more than tens of meters through permeable till. Simulations using the Ice-sheet and Sea-level System Model (ISSM) show that even just a few hundred meters of basal melt caused by seawater intrusion upstream of marine ice sheet grounding lines can cause projections of marine ice sheet volume loss to be 10 %-50 % higher. Kilometers of intrusion-induced basal melt can cause projected ice sheet volume loss to more than double. These results suggest that further observational, experimental and numerical investigations are needed to determine the conditions under which seawater intrusion occurs and whether it will indeed drive rapid marine ice sheet retreat and sea level rise in the future.

173. Weakening of the pinning point buttressing Thwaites Glacier, West Antarctica

Authors: Christian T. Wild; Karen E. Alley; Atsuhiko Muto; Martin Truffer; Ted A. Scambos; Erin C. Pettit

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-16-397-2022

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Cited by: 28 publications

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Funding: National Science Foundation, Natural Environment Research Council

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Open Access: N/A

Abstract: . The Thwaites Eastern Ice Shelf buttresses a significant portion of Thwaites Glacier through contact with a pinning point 40 km offshore of the present grounding line. Predicting future rates of Thwaites Glacier's contribution to sea-level rise depends on the evolution of this pinning point and the resultant change in the ice-shelf stress field since the breakup of the Thwaites Western Glacier Tongue in 2009. Here we use Landsat-8 feature tracking of ice velocity in combination with ice-sheet model perturbation experiments to show how past changes in flow velocity have been governed in large part by changes in lateral shear and pinning point interactions with the Thwaites Western Glacier Tongue. We then use recent satellite altimetry data from ICESat-2 to show that Thwaites Glacier's grounding line has continued to retreat rapidly; in particular, the grounded area of the pinning point is greatly reduced from earlier mappings in 2014, and grounded ice elevations are continuing to decrease. This loss has created two pinned areas with ice flow now funneled between them. If current rates of surface lowering persist, the Thwaites Eastern Ice Shelf will unpin from the seafloor in less than a decade, despite our finding from airborne radar data that the seafloor underneath the pinning point is about 200 m shallower than previously reported. Advection of relatively thin and mechanically damaged ice onto the remaining portions of the pinning point and feedback mechanisms involving basal melting may further accelerate the unpinning. As a result, ice discharge will likely increase up to 10 % along a 45 km stretch of the grounding line that is currently buttressed by the Thwaites Eastern Ice Shelf.

174. Seasonal Tidewater Glacier Terminus Oscillations Bias Multi-Decadal Projections of Ice Mass Change

Authors: D. Felikson (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, Now at Goddard Earth Sciences Technology and Research Studies and Investigations II Morgan State University Baltimore MD USA, Universities Space Research Association); S. Nowicki (Department of Geology University at Buffalo Buffalo NY USA); I. Nias (University of Liverpool); M. Morlighem (University of California Irvine, Dartmouth College); H. Seroussi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, Dartmouth College)

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

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References: 63

Cited by: 8 publications

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Keywords: N/A

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Open Access: N/A

Abstract: Numerical, process-based simulations of tidewater glacier evolution are necessary to project future sea-level change under various climate scenarios. Previous work has shown that nonlinearities in tidewater glacier and ice stream dynamics can lead to biases in simulated ice mass change in the presence of noisy forcings. Ice sheet modeling projections that are used in the IPCC Assessment Report 6 (AR6) utilize atmospheric and oceanic forcings at annual temporal resolution, omitting any higher frequency forcings. Here, we quantify the effect of seasonal (<1 year) tidewater glacier terminus oscillations on decadal-scale (30 years) mass change. We use an idealized geometry to mimic realistic tidewater glacier geometries and investigate the impact of the magnitude of seasonal oscillations, bed slope at the glacier terminus, and basal friction law. We find that omitting seasonal terminus motion results in biased mass change projections, with up to an 39% underestimate or 25% overestimate of mass loss when seasonality is neglected, depending on bed slope and magnitude of oscillations. The bias is most sensitive to the magnitude of the seasonal terminus oscillations and exhibits very little sensitivity to choice of friction law. Although omitting seasonal terminus motion may bias mass change projections of individual glaciers, further work is needed to investigate the impact on ice sheet mass change projections. In order to achieve this, seasonality in atmospheric and oceanic forcings must be adequately represented and observations of seasonal terminus positions and tidewater glacier thickness changes must be acquired to evaluate numerical models.

175. Authors' Response to Reviewer Comments

Authors: Blake Castleman

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/tc-2021-274-ac1

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Language: N/A

Subjects: N/A

Keywords: Bedrock|Glacier morphology|Future sea level

Funding: N/A

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Open Access: N/A

Abstract: N/A

176. Impact of paleoclimate on present and future evolution of the Greenland Ice Sheet

Authors: Hu Yang; Uta Krebs-Kanzow; Thomas Kleiner; Dmitry Sidorenko; Christian Bernd Rodehacke; Xiaoxu Shi; Paul Gierz; Lu Niu; Evan J. Gowan; Sebastian Hinck; Xingxing Liu; Lennert B. Stap; Gerrit Lohmann

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Publisher: Public Library of Science (PLoS)

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Keywords: N/A

Funding: Deutsche Forschungsgemeinschaft (DFG) as part of the Special Priority Program, German Federal Ministry of Education, Research (Bundesministerium für Bildung und Forschung: BMBF) project ZUWEISS (and 4 more)

License: CC BY 4.0

Open Access: N/A

Abstract: Using transient climate forcing based on simulations from the Alfred Wegener Institute Earth System Model (AWI-ESM), we simulate the evolution of the Greenland Ice Sheet (GrIS) from the last interglacial (125 ka, kiloyear before present) to 2100 AD with the Parallel Ice Sheet Model (PISM). The impact of paleoclimate, especially Holocene climate, on the present and future evolution of the GrIS is explored. Our simulations of the past show close agreement with reconstructions with respect to the recent timing of the peaks in ice volume and the climate of Greenland. The maximum and minimum ice volume at around 18-17 ka and 6-5 ka lag the respective extremes in climate by several thousand years, implying that the ice volume response of the GrIS strongly lags climatic changes. Given that Greenland's climate was getting colder from the Holocene Thermal Maximum (i.e., 8 ka) to the Pre-Industrial era, our simulation implies that the GrIS experienced growth from the mid-Holocene to the industrial era. Due to this background trend, the GrIS still gains mass until the second half of the 20th century, even though anthropogenic warming begins around 1850 AD. This is also in agreement with observational evidence showing mass loss of the GrIS does not begin earlier than the late 20th century. Our results highlight that the present evolution of the GrIS is not only controlled by the recent climate changes, but is also affected by paleoclimate, especially the relatively warm Holocene climate. We propose that the GrIS was not in equilibrium throughout the entire Holocene and that the slow response to Holocene climate needs to be represented in ice sheet simulations in order to predict ice mass loss, and therefore sea level rise, accurately.

177. A new vertically integrated MOno-Layer Higher-Order (MOLHO) ice flow model

Authors: Thiago Dias dos Santos; Mathieu Morlighem; Douglas Brinkerhoff

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-16-179-2022

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References: 36

Cited by: 9 publications

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Keywords: N/A

Funding: Heising-Simons Foundation

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Open Access: N/A

Abstract: . Numerical simulations of ice sheets rely on the momentum balance to determine how ice velocities change as the geometry of the system evolves. Ice is generally assumed to follow a Stokes flow with a nonlinear viscosity. Several approximations have been proposed in order to lower the computational cost of a full-Stokes stress balance. A popular option is the Blatter-Pattyn or higher-order model (HO), which consists of a three-dimensional set of equations that solves the horizontal velocities only. However, it still remains computationally expensive for long transient simulations. Here we present a depth-integrated formulation of the HO model, which can be solved on a two-dimensional mesh in the horizontal plane. We employ a specific polynomial function to describe the vertical variation in the velocity, which allows us to integrate the vertical dimension using a semi-analytic integration. We assess the performance of this MOno-Layer Higher-Order (MOLHO) model to compute ice velocities and simulate grounding line dynamics on standard benchmarks (ISMIP-HOM and MISIP3D). We compare MOLHO results to the ones obtained with the original three-dimensional HO model. We also compare the time performance of both models in time-dependent runs. Our results show that the ice velocities and grounding line positions obtained with MOLHO are in very good agreement with the ones from HO. In terms of computing time, MOLHO requires less than 10 % of the computational time of a typical HO model, for the same simulations. These results suggest that the MOno-Layer Higher-Order formulation provides improved computational time performance and a comparable accuracy compared to the HO formulation, which opens the door to higher-order paleo simulations.

178. Millennial-scale migration of the frozen/melted basal boundary, western Greenland ice sheet

Authors: Aidan Stansberry; Joel Harper; Jesse V. Johnson; Toby Meierbachtol

Journal: Journal of Glaciology

Publisher: Cambridge University Press (CUP)

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Cited by: 1 publications

Language: en

Subjects: N/A

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Open Access: N/A

Abstract: The geometry and thermal structure of western Greenland ice sheet are known to have undergone relatively substantial change over the Holocene. Evolution of the frozen and melted fractions of the bed associated with the ice-sheet retreat over this time frame remains unclear. We address this question using a thermo-mechanically coupled flowline model to simulate a 11 ka period of ice-sheet retreat in west central Greenland. Results indicate an episode of ~100 km of terminus retreat corresponded to ~16 km of upstream frozen/melted basal boundary migration. The majority of migration of the frozen area is associated with the enhancement of the frictional and strain heating fields, which are accentuated toward the retreating ice margin. The thermally active bedrock layer acts as a heat sink, tending to slow contraction of frozen-bed conditions. Since the bedrock heat flux in our region is relatively low compared to other regions of the ice sheet, the frozen region is relatively greater and therefore more susceptible to marginward changes in the frictional and strain heating fields. Migration of melted regions thus depends on both geometric changes and the antecedent thermal state of the bedrock and ice, both of which vary considerably around the ice sheet.

179. Impact of the melt-albedo feedback on the future evolution of the Greenland Ice Sheet with PISM-dEBM-simple

Authors: Maria Zeitz; Ronja Reese; Johanna Beckmann; Uta Krebs-Kanzow; Ricarda Winkelmann

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-15-5739-2021

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Keywords: N/A

Funding: Deutsche Forschungsgemeinschaft, Leibniz-Gemeinschaft

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Open Access: N/A

Abstract: . Surface melting of the Greenland Ice Sheet contributes a large amount to current and future sea level rise. Increased surface melt may lower the reflectivity of the ice sheet surface and thereby increase melt rates: the so-called melt-albedo feedback describes this self-sustaining increase in surface melting. In order to test the effect of the melt-albedo feedback in a prognostic ice sheet model, we implement dEBM-simple, a simplified version of the diurnal Energy Balance Model dEBM, in the Parallel Ice Sheet Model (PISM). The implementation includes a simple representation of the melt-albedo feedback and can thereby replace the positive-degree-day melt scheme. Using PISM-dEBM-simple, we find that this feedback increases ice loss through surface warming by 60 % until 2300 for the high-emission scenario RCP8.5 when compared to a scenario in which the albedo remains constant at its present-day values. With an increase of 90 % compared to a fixed-albedo scenario, the effect is more pronounced for lower surface warming under RCP2.6. Furthermore, assuming an immediate darkening of the ice surface over all summer months, we estimate an upper bound for this effect to be 70 % in the RCP8.5 scenario and a more than 4-fold increase under RCP2.6. With dEBM-simple implemented in PISM, we find that the melt-albedo feedback is an essential contributor to mass loss in dynamic simulations of the Greenland Ice Sheet under future warming.

180. Brief communication: A roadmap towards credible projections of ice sheet contribution to sea level

Authors: Andy Aschwanden; Timothy C. Bartholomaeus; Douglas J. Brinkerhoff; Martin Truffer

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-15-5705-2021

URL: <https://doi.org/10.5194/tc-15-5705-2021>

Publication Date: 2021-12-17

Volume/Issue: Volume 15, Issue 12

References: 58

Cited by: 41 publications

Language: en

Subjects: N/A

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Open Access: N/A

Abstract: . Accurately projecting mass loss from ice sheets is of critical societal importance. However, despite recent improvements in ice sheet models, our analysis of a recent effort to project ice sheet contribution to future sea level suggests that few models reproduce historical mass loss accurately and that they appear much too confident in the spread of predicted outcomes. The inability of models to reproduce historical observations raises concerns about the models' skill at projecting mass loss. Here we suggest that uncertainties in the future sea level contribution from Greenland and Antarctica may well be significantly higher than reported in that study. We propose a roadmap to enable a more realistic accounting of uncertainties associated with such forecasts and a formal process by which observations of mass change should be used to refine projections of mass change. Finally, we note that tremendous government investment and planning affecting tens to hundreds of millions of people is founded on the work of just a few tens of scientists. To achieve the goal of credible projections of ice sheet contribution to sea level, we strongly believe that investment in research must be commensurate with the scale of the challenge.

181. Seasonal evolution of basal environment conditions of Russell sector, West Greenland, inverted from satellite observation of surface flow

Authors: Anna Derkacheva; Fabien Gillet-Chaulet; Jeremie Mouginot; Eliot Jager; Nathan Maier; Samuel Cook

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-15-5675-2021

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Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Agence Nationale de la Recherche

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Open Access: N/A

Abstract: . Due to increasing surface melting on the Greenland ice sheet, better constraints on seasonally evolving basal water pressure and sliding speed are required by models. Here we assess the potential of using inverse methods on a dense time series of surface speeds to recover the seasonal evolution of the basal conditions in a well-documented region in southwest Greenland. Using data compiled from multiple satellite missions, we document seasonally evolving surface velocities with a temporal resolution of 2 weeks between 2015 and 2019. We then apply the inverse control method using the ice flow model Elmer/Ice to infer the basal sliding and friction corresponding to each of the 24 surface velocity data sets. Near the margin where the uncertainty in the velocity and bed topography are small, we obtain clear seasonal variations that can be mostly interpreted in terms of an effective-pressure-based hard-bed friction law. We find for valley bottoms or "troughs" in the bed topography that the changes in modelled basal conditions directly respond to local modelled water pressure variations, while the link is more complex for subglacial "ridges" which are often non-locally forced. At the catchment scale, in-phase variations in the water pressure, surface velocities, and surface runoff variations are found. Our results show that time series inversions of observed surface velocities can be used to understand the evolution of basal conditions over different timescales and could therefore serve as an intermediate validation for subglacial hydrology models to achieve better coupling with ice flow models.

182. Modelling alpine glacier geometry and subglacial erosion patterns in response to contrasting climatic forcing

Authors: Fabio Magrani (University of Bern); Pierre G. Valla (University of Bern, Institut de Recherche pour le Développement (IRD)); David Egholm (Aarhus University)

Journal: Earth Surface Processes and Landforms

Publisher: Wiley

DOI: 10.1002/esp.5302

URL: <https://doi.org/10.1002/esp.5302>

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Keywords: N/A

Funding: Agence Nationale de la Recherche, Schweizerischer Nationalfonds zur Förderung der Wissenschaftlichen Forschung

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Open Access: N/A

Abstract: Climate exerts a primary control on glacier mass balance, driving changes in ice flux, which affects basal sliding and subglacial erosion. Although past glacier reconstructions have been widely used as paleo-climate proxies, extracting quantitative temperature/precipitation data from paleo-glaciers is still challenging. The iSOSIA ice model was used over a synthetic landscape to quantitatively investigate the non-linear dependence of glacier geometry and ice dynamics on climate forcing, as well as to analyse how spatial patterns of subglacial erosion and erosion rates vary between the accumulation and ablation zones for different climatic settings. We performed 10 calibrated climatic scenarios with different combinations of temperature and precipitation rate in two separate sets of simulations [maintaining either same equilibrium line altitude (ELA) or same ice extent; SA and ST, respectively]. Our results reinforce the role of climate and the importance of ice flux in conditioning glacier geometry. In SA simulations, contrasting climatic scenarios showed a major influence on glacier length and thickness. In ST simulations, calibrated ELAs presented a variability of over 300 m in order to maintain similar ice extent, but showed reduced changes in glacier thickness. These results highlight the importance of ice flux to predict glacier geometry, when compared to the overall mass balance from an ELA estimate. Subglacial abrasion and quarrying showed notable differences between the accumulation and ablation zones for varying climatic forcing, with multimodal erosion distributions for increased precipitation/temperature, shifting erosion towards higher-elevation tributaries and connecting erosion patches in the ablation zone. Such trends highlight the role of subglacial hydrology and ice-bed contact (cavitation) in dictating patterns of subglacial erosion, while the depth of erosion relates closely to ice flux and climate inputs. Our results have potential implications for paleo-climate reconstructions based on glacier geometry and provide insights on how subglacial erosion can help estimate paleo-climatic conditions from paleo-glacial records.

183. Centennial- and Orbital-Scale Erosion Beneath the Greenland Ice Sheet Near Jakobshavn Isbræ

Authors: A. Balter-Kennedy (Columbia University); N. E. Young (Columbia University); J. P. Briner (Department of Geology University at Buffalo Buffalo NY USA); B. L. Graham (Department of Geology University at Buffalo Buffalo NY USA); J. M. Schaefer (Columbia University)

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

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Subjects: N/A

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License: CC BY 4.0

Open Access: N/A

Abstract: Erosion beneath glaciers and ice sheets is a fundamental Earth-surface process dictating landscape development, which in turn influences ice-flow dynamics and the climate sensitivity of ice masses. The rate at which subglacial erosion takes place, however, is notoriously difficult to observe because it occurs beneath modern glaciers in a largely inaccessible environment. Here, we present (a) cosmogenic-nuclide measurements from bedrock surfaces with well constrained exposure and burial histories in front of Jakobshavn Isbræ in western Greenland to quantify centennial-scale erosion rates since ~1850 CE, and (b) a new method combining cosmogenic-nuclide measurements in a shallow bedrock core with cosmogenic-nuclide modeling to determine orbital-scale erosion rates across the same landscape. Twenty-seven ^{10}Be measurements in surficial bedrock constrain the erosion rate during historical times to 0.4-0.8 mm yr⁻¹. Seventeen ^{10}Be measurements in a 4-m-long bedrock core yield a centennial-scale erosion rate of 0.3-0.6 mm yr⁻¹, corroborating the results from our surface samples, and reveal that ^{10}Be concentrations below ~2 m depth are greater than what is predicted by an idealized production-rate depth profile. We utilize this excess ^{10}Be at depth to constrain orbital-scale erosion rates at Jakobshavn Isbræ to 0.1-0.3 mm yr⁻¹. The broad similarity between centennial- and orbital-scale erosion rates suggests that subglacial erosion rates have remained relatively uniform throughout the Pleistocene adjacent to Jakobshavn Isbræ.

184. Mapping the Sensitivity of the Amundsen Sea Embayment to Changes in External Forcings Using Automatic Differentiation

Authors: Mathieu Morlighem (University of California Irvine, Dartmouth College); Daniel Goldberg (University of Edinburgh); Thiago Dias dos Santos (University of California Irvine, University of São Paulo, Centro Polar e Climático Universidade Federal do Rio Grande do Sul Porto Alegre Brazil); Jane Lee (University of California Irvine); Max Sagebaum (TU Kaiserslautern Kaiserslautern Germany)

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Keywords: N/A

Funding: National Science Foundation, Natural Environment Research Council

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Open Access: N/A

Abstract: Thwaites and Pine Island Glaciers as well as other ice streams in West Antarctica have been changing dramatically over the past decades. Although changes in ocean conditions are likely the primary driver of these changes, it remains unclear where other processes could cause more mass loss. By employing Automatic Differentiation and two independent ice-sheet models, we construct maps of the sensitivity of the volume above floatation to changes in ocean-induced melt rates, ice rigidity, basal friction, and surface mass balance. We find that changes in basal melt close to the grounding lines and along shear margins have a larger impact on the glaciers' final volume. The glaciers are sensitive to changes in basal friction on regions close to the grounding lines, while changes in ice rigidity has a larger impact along the shear margins of Pine Island. The sensitivity to surface mass balance is uniform over grounded ice.

185. Supraglacial lake bathymetry automatically derived from ICESat-2 constraining lake depth estimates from multi-source satellite imagery

Authors: Rajashree Tri Datta; Bert Wouters

Journal: The Cryosphere

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Abstract: . We introduce an algorithm (Watta) which automatically calculates supraglacial lake bathymetry and detects potential ice layers along tracks of the ICESat-2 (Ice, Cloud, and Land Elevation Satellite) laser altimeter. Watta uses photon heights estimated by the ICESat-2 ATL03 product and extracts supraglacial lake surface, bottom, and depth corrected for refraction and (sub-)surface ice cover in addition to producing surface heights at the native resolution of the ATL03 photon cloud. These measurements are used to constrain empirical estimates of lake depth from satellite imagery, which were thus far dependent on sparse sets of in situ measurements for calibration. Imagery sources include Landsat 8 Operational Land Imager (OLI), Sentinel-2, and high-resolution Planet Labs PlanetScope and SkySat data, used here for the first time to calculate supraglacial lake depths. The Watta algorithm was developed and tested using a set of 46 lakes near Sermeq Kujalleq (Jakobshavn) glacier in western Greenland, and we use multiple imagery sources (available for 45 of these lakes) to assess the use of the red vs. green band to extrapolate depths along a profile to full lake volumes. We use Watta-derived estimates in conjunction with high-resolution imagery from both satellite-based sources (tasked over the season) and nearly simultaneous Operation IceBridge CAMBOT (Continuous Airborne Mapping By Optical Translator) imagery (on a single airborne flight) for a focused study of the drainage of a single lake over the 2019 melt season. Our results suggest that the use of multiple imagery sources (both publicly available and commercial), in combination with altimetry-based depths, can move towards capturing the evolution of supraglacial hydrology at improved spatial and temporal scales.

186. Elastic deformation plays a non-negligible role in Greenland's outlet glacier flow

Authors: Julia Christmann; Veit Helm; Shfaqat Abbas Khan; Thomas Kleiner; Ralf Müller; Mathieu Morlighem; Niklas Neckel; Martin Rückamp; Daniel Steinhage; Ole Zeising; Angelika Humbert

Journal: Communications Earth & Environment

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s43247-021-00296-3

URL: <https://doi.org/10.1038/s43247-021-00296-3>

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Language: en

Subjects: N/A

Keywords: N/A

Funding: Bundesministerium für Bildung und Forschung, European Union's Horizon 2020 Research and Innovation Programme

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Open Access: N/A

Abstract: Future projections of global mean sea level change are uncertain, partly because of our limited understanding of the dynamics of Greenland's outlet glaciers. Here we study Nioghalvfjerdsbræ, an outlet glacier of the Northeast Greenland Ice Stream that holds 1.1 m sea-level equivalent of ice. We use GPS observations and numerical modelling to investigate the role of tides as well as the elastic contribution to glacier flow. We find that ocean tides alter the basal lubrication of the glacier up to 10 km inland of the grounding line, and that their influence is best described by a viscoelastic rather than a viscous model. Further inland, sliding is the dominant mechanism of fast glacier motion, and the ice flow induces persistent elastic strain. We conclude that elastic deformation plays a role in glacier flow, particularly in areas of steep topographic changes and fast ice velocities.

187. Notes on a compressible extended Burgers model of rheology

Authors: Erik R Ivins (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Lambert Caron (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Surendra Adhikari (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Eric Larour (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Journal International

Publisher: Oxford University Press (OUP)

DOI: 10.1093/gji/ggab452

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Language: en

Subjects: N/A

Keywords: N/A

Funding: Jet Propulsion Laboratory, California Institute of Technology, National Aeronautics and Space Administration

License: CC BY 4.0

Open Access: N/A

Abstract: SUMMARY Constitutive theory for viscoelasticity has broad application to solid mantle or ice deformations driven by tides, surface mass variations, and post-seismic flow. Geophysical models using higher order viscoelasticity can better accommodate geodetic observations than lower-order theory, typically provided by tensor versions of Maxwell, 4-parameter Burgers or standard linear (Zener) rheology. We derive, for the first time, a mathematical description of a compressible version of the extended Burgers material (EBM) model paradigm which has a distribution function of relaxation spectra. The latter model is often used for parametrizing high temperature background transient responses in the rock physics and mechanics laboratory setting and have demonstrated application to low frequency seismic wave attenuation. A new generalization of this practical anelastic model is presented and applied to the glacial isostatic adjustment momentum equations, thus providing useful guidance for generating initial-value boundary problem-solving software for quite general coding strategies. The solutions for the vertical motion response to a suddenly imposed surface load reveal a short-term transience of substantial amplitude.

188. Nunataks as barriers to ice flow: implications for palaeo ice sheet reconstructions

Authors: Martim Mas e Braga; Richard Selwyn Jones; Jennifer C. H. Newall; Irina Rogozhina; Jane L. Andersen; Nathaniel A. Lifton; Arjen P. Stroeven

Journal: The Cryosphere

Publisher: Copernicus GmbH

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License: CC BY 4.0

Open Access: N/A

Abstract: . Numerical models predict that discharge from the polar ice sheets will become the largest contributor to sea-level rise over the coming centuries. However, the predicted amount of ice discharge and associated thinning depends on how well ice sheet models reproduce glaciological processes, such as ice flow in regions of large topographic relief, where ice flows around bedrock summits (i.e. nunataks) and through outlet glaciers. The ability of ice sheet models to capture long-term ice loss is best tested by comparing model simulations against geological data. A benchmark for such models is ice surface elevation change, which has been constrained empirically at nunataks and along margins of outlet glaciers using cosmogenic exposure dating. However, the usefulness of this approach in quantifying ice sheet thinning relies on how well such records represent changes in regional ice surface elevation. Here we examine how ice surface elevations respond to the presence of strong topographic relief that acts as an obstacle by modelling ice flow around and between idealised nunataks during periods of imposed ice sheet thinning. We find that, for realistic Antarctic conditions, a single nunatak can exert an impact on ice thickness over 20 km away from its summit, with its most prominent effect being a local increase (decrease) of the ice surface elevation of hundreds of metres upstream (downstream) of the obstacle. A direct consequence of this differential surface response for cosmogenic exposure dating is a delay in the time of bedrock exposure upstream relative to downstream of a nunatak summit. A nunatak elongated transversely to ice flow is able to increase ice retention and therefore impose steeper ice surface gradients, while efficient ice drainage through outlet glaciers produces gentler gradients. Such differences, however, are not typically captured by continent-wide ice sheet models due to their coarse grid resolutions. Their inability to capture site-specific surface elevation changes appears to be a key reason for the observed mismatches between the timing of ice-free conditions from cosmogenic exposure dating and model simulations. We conclude that a model grid refinement over complex topography and information about sample position relative to ice flow near the nunatak are necessary to improve data-model comparisons of ice surface

elevation and therefore the ability of models to simulate ice discharge in regions of large topographic relief.

189. Framework for probabilistic tsunami hazard assessment considering the effects of sea-level rise due to climate change

Authors: Abdul Kadir Alhamid; Mitsuyoshi Akiyama; Hiroki Ishibashi; Koki Aoki; Shunichi Koshimura; Dan M. Frangopol

Journal: Structural Safety

Publisher: Elsevier BV

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Language: en

Subjects: N/A

Keywords: N/A

Funding: Japan Society for the Promotion of Science, JSPS

License: N/A

Open Access: N/A

Abstract: N/A

190. Evaluation of the Local Sea-Level Budget at Tide Gauges Since 1958

Authors: Jinping Wang (Ocean University of China, Centre for Southern Hemisphere Oceans Research (CSHOR) CSIRO Oceans and Atmosphere Hobart TAS Australia); John A. Church (University of New South Wales); Xuebin Zhang (Centre for Southern Hemisphere Oceans Research (CSHOR) CSIRO Oceans and Atmosphere Hobart TAS Australia); Jonathan M. Gregory (Met Office Hadley Centre, Met Office, University of Reading); Laure Zanna (Courant Institute of Mathematical Sciences New York University New York NY USA); Xian Yao Chen (Ocean University of China)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Cited by: 19 publications

Language: en

Subjects: N/A

Keywords: sterodynamic sea level, tide gauge, sea-level budget, sea-level fingerprint, vertical land motion (and 1 more)

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Open Access: N/A

Abstract: Although global mean sea-level rise since 1900 and regional mean sea-level change since the 1960s have been accounted for in terms of the sum of contributions, the same budget closure has not been achieved for local relative sea-level change from a global network of tide gauges. To address this, we combine new estimates of sterodynamic sea-level change (SDSL; including ocean dynamics), glacial isostatic adjustment (GIA), change in land ice mass and terrestrial water storage, and other local vertical land motion. We find that the observed trends over 1958-2015 at all 272 tide gauges distributed worldwide agree with the sum of contributions (within 90% confidence estimates), with similar mean trend (1.1 mm yr⁻¹) and comparable spatial variability (standard deviation of 2.0 and 1.9 mm yr⁻¹ respectively). SDSL is the dominant contribution to both local observed mean trend and spatial variability, except at locations close to former ice-sheets, where GIA dominates.

191. The influence of emissions scenarios on future Antarctic ice loss is unlikely to emerge this century

Authors: Daniel P. Lowry; Mario Krapp; Nicholas R. Golledge; Alanna Alevropoulos-Borrill

Journal: Communications Earth & Environment

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s43247-021-00289-2

URL: <https://doi.org/10.1038/s43247-021-00289-2>

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Cited by: 22 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Ministry of Business, Innovation and Employment

License: CC BY 4.0

Open Access: N/A

Abstract: Of all the components of the global sea-level budget, the future contribution of the Antarctic Ice Sheet is the most uncertain in sea-level rise projections. Dynamic ice sheet model simulations show considerable overlap in the projected Antarctic Ice Sheet sea-level contribution under various greenhouse gas emissions scenarios and the timescale at which scenario dependence will emerge is unclear. With historically constrained ice sheet simulations and a statistical emulator, we demonstrate that a high-emissions signature of the Antarctic Ice Sheet sea-level contribution will not unambiguously emerge from the wide potential range of low-emission sea-level projections for over 100 years due to current limitations in our understanding in ice flow and sliding. However, the results also indicate that the total global warming that occurs over the 21st century controls the resulting long-term Antarctic Ice Sheet sea-level commitment, with multi-meter differences between the highest and lowest emissions scenarios in subsequent centuries.

192. Quantifying the potential future contribution to global mean sea level from the Filchner-Ronne basin, Antarctica

Authors: Emily A. Hill; Sebastian H. R. Rosier; G. Hilmar Gudmundsson; Matthew Collins

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-15-4675-2021

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Cited by: 15 publications

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Keywords: N/A

Funding: Natural Environment Research Council, Horizon 2020

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Open Access: N/A

Abstract: . The future of the Antarctic Ice Sheet in response to climate warming is one of the largest sources of uncertainty in estimates of future changes in global mean sea level (ΔGMSL). Mass loss is currently concentrated in regions of warm circumpolar deep water, but it is unclear how ice shelves currently surrounded by relatively cold ocean waters will respond to climatic changes in the future. Studies suggest that warm water could flush the Filchner-Ronne (FR) ice shelf cavity during the 21st century, but the inland ice sheet response to a drastic increase in ice shelf melt rates is poorly known. Here, we use an ice flow model and uncertainty quantification approach to project the GMSL contribution of the FR basin under RCP emissions scenarios, and we assess the forward propagation and proportional contribution of uncertainties in model parameters (related to ice dynamics and atmospheric/oceanic forcing) on these projections. Our probabilistic projections, derived from an extensive sample of the parameter space using a surrogate model, reveal that the FR basin is unlikely to contribute positively to sea level rise by the 23rd century. This is primarily due to the mitigating effect of increased accumulation with warming, which is capable of suppressing ice loss associated with ocean-driven increases in sub-shelf melt. Mass gain (negative ΔGMSL) from the FR basin increases with warming, but uncertainties in these projections also become larger. In the highest emission scenario RCP8.5, ΔGMSL is likely to range from -103 to 26 mm, and this large spread can be apportioned predominantly to uncertainties in parameters driving increases in precipitation (30 %) and sub-shelf melting (44 %). There is potential, within the bounds of our input parameter space, for major collapse and retreat of ice streams feeding the FR ice shelf, and a substantial positive contribution to GMSL (up to approx. 300 mm), but we consider such a scenario to be very unlikely. Adopting uncertainty quantification techniques in future studies will help to provide robust estimates of potential sea level rise and further identify target areas for constraining projections.

193. Derivation of bedrock topography measurement requirements for the reduction of uncertainty in ice sheet model projections of Thwaites Glacier

Authors: Blake A. Castleman; Nicole-Jeanne Schlegel; Lambert Caron; Eric Larour; Ala Khazendar

Journal: Unknown Journal

Publisher: Copernicus GmbH

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Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Bedrock|Glacier morphology|Antarctic ice sheet|Future sea level

Funding: Jet Propulsion Laboratory, National Aeronautics and Space Administration

License: CC BY 4.0

Open Access: N/A

Abstract: . Determining the future evolution of the Antarctic Ice Sheet is critical for understanding and narrowing the large existing uncertainties in century-scale global mean sea level rise (SLR) projections. One of the most significant glaciers/ice streams in Antarctica, Thwaites Glacier, is at risk of destabilization and, if destabilized, has the potential to be the largest regional-scale contributor of SLR on Earth. This is because Thwaites Glacier is vulnerable to the marine ice sheet instability, as its grounding line is significantly influenced by ocean-driven basal melting rates, and its bedrock topography retrogrades into kilometer deep troughs. In this study, we investigate how bedrock topography features influence the grounding line migration beneath Thwaites Glacier when extreme ocean-driven basal melt rates are applied. Specifically, we design experiments using the Ice-Sheet and Sea-level System Model (ISSM) to quantify the SLR projection uncertainty due to reported errors in the current bedrock topography maps that are often used by ice sheet models. We find that spread in model estimates of sea level rise contribution from Thwaites glacier due to the reported bedrock topography error could be as large as 21.9 cm after 200 years of extreme ocean warming. Next, we perturb the bedrock topography beneath Thwaites Glacier using wavelet decomposition techniques to introduce realistic noise (within error). We explore the model space with multiple realizations of noise to quantify what spatial and vertical resolutions in bedrock topography are required to minimize the uncertainty in our 200-year experiment. We conclude that at least a 2 km spatial and 8 m vertical resolution would independently constrain possible SLR to ± 2 cm over 200 years, fulfilling requirements outlined by the 2017 Decadal Survey for Earth Science. Lastly, we perform an ensemble of simulations to determine in which regions our model of Thwaites Glacier is most sensitive to perturbations in bedrock topography. Our results suggest that the retreat of the grounding line is most sensitive to bedrock topography in proximity to the grounding line's initial position. Additionally, we find that the location and amplitude of the bedrock perturbation is more significant than its sharpness and shape. Overall, these findings inform and benchmark observational requirements for future missions that will measure ice sheet bedrock topography, not only in the case of Thwaites Glacier but for Antarctica on the continental scale.

194. Drivers of Change of Thwaites Glacier, West Antarctica, Between 1995 and 2015

Authors: Thiago Dias dos Santos (University of California Irvine, Centro Polar e Climático Universidade Federal do Rio Grande do Sul Porto Alegre RS Brazil, University of São Paulo); Jowan M. Barnes (Department of Geography and Environmental Sciences Northumbria University Newcastle upon Tyne UK); Daniel N. Goldberg (University of Edinburgh); G. Hilmar Gudmundsson (Department of Geography and Environmental Sciences Northumbria University Newcastle upon Tyne UK); Mathieu Morlighem (University of California Irvine, Dartmouth College)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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References: 61

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: ice sheet modeling, Thwaites Glacier, drivers of change

Funding: National Science Foundation, Natural Environment Research Council

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Open Access: N/A

Abstract: Using three independent ice-flow models and several satellite-based datasets, we assess the importance of correctly capturing ice-shelf breakup, shelf thinning, and reduction in basal traction from ungrounding in reproducing observed speed-up and thinning of Thwaites Glacier between 1995 and 2015. We run several transient numerical simulations applying these three perturbations individually. Our results show that ocean-induced ice-shelf thinning generates most of the observed grounding line retreat, inland speed-up, and mass loss, in agreement with previous work. We improve the agreement with observed inland speed-up and thinning by prescribing changes in ice-shelf geometry and a reduction in basal traction over areas that became ungrounded since 1995, suggesting that shelf breakups and thinning-induced reduction in basal traction play a critical role on Thwaites's dynamics, as pointed out by previous studies. These findings suggest that modeling Thwaites's future requires reliable ocean-induced melt estimates in models that respond accurately to downstream perturbations.

195. Layered seawater intrusion and melt under grounded ice

Authors: Alexander A. Robel; Earle Wilson; Helene Seroussi

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/tc-2021-262

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Language: N/A

Subjects: N/A

Keywords: Lead (geology)|Antarctic ice sheet

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Open Access: N/A

Abstract: . Increasing melt of ice sheets at their floating or vertical interface with the ocean is a major driver of marine ice sheet retreat and sea level rise. However, the extent to which warm, salty seawater may drive melting under the grounded portions of ice sheets is still not well understood. Previous work has explored the possibility that dense seawater intrudes beneath relatively light subglacial freshwater discharge, similar to the salt wedge observed in many estuarine systems. In this study, we develop a generalized theory of layered seawater intrusion under grounded ice, including where subglacial hydrology occurs as a macroporous water sheet over impermeable beds or as microporous Darcy flow through permeable till. Using predictions from this theory, we show that seawater intrusion over hard beds may feasibly occur up to tens of kilometers upstream of a glacier terminus or grounding line. On the other hand, seawater is unlikely to intrude more than tens of meters through subglacial till. High-resolution simulations using the Ice-Sheet and Sea-Level System Model (ISSM) show that even just a few hundred meters of basal melt caused by seawater intrusion upstream of marine ice sheet grounding lines can cause projections of marine ice sheet volume loss to be 10-50 % higher or 100 % higher for kilometers of intrusion-induced basal melt. These results suggest that further observational, experimental and numerical investigations are needed to determine whether the conditions under which extensive seawater intrusion occurs and whether it will indeed drive rapid marine ice sheet retreat and sea level rise in the future.

196. A new sampling capability for uncertainty quantification in the Ice-sheet and Sea-level System Model v4.19 using Gaussian Markov random fields

Authors: Kevin Bulthuis; Eric Y. Larour

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-2021-321

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Cited by: 1 publications

Language: N/A

Subjects: N/A

Keywords: Uncertainty Quantification|Greenland ice sheet|Propagation of uncertainty

Funding: Jet Propulsion Laboratory

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Open Access: N/A

Abstract: . Assessing the impact of uncertainties in ice-sheet models is a major and challenging issue that needs to be faced by the ice-sheet community to provide more robust and reliable model-based projections of ice-sheet mass balance. In recent years, uncertainty quantification (UQ) has been increasingly used to characterize and explore uncertainty in ice-sheet models and improve the robustness of their projections. A typical UQ analysis involves first the (probabilistic) characterization of the sources of uncertainty followed by the propagation and sensitivity analysis of these sources of uncertainty. Previous studies concerned with UQ in ice-sheet models have generally focused on the last two steps but paid relatively little attention to the preliminary and critical step of the characterization of uncertainty. Sources of uncertainty in ice-sheet models, like uncertainties in ice-sheet geometry or surface mass balance, typically vary in space and potentially in time. For that reason, they are more adequately described as spatio(-temporal) random fields, which account naturally for spatial (and temporal) correlation. As a means of improving the characterization of the sources of uncertainties in ice-sheet models, we propose in this paper to represent them as Gaussian random fields with Matérn covariance function. The class of Matérn covariance functions provides a flexible model able to capture statistical dependence between locations with different degrees of spatial correlation or smoothness properties. Samples from a Gaussian random field with Matérn covariance function can be generated efficiently by solving a certain stochastic partial differential equation. Discretization of this stochastic partial differential equation by the finite element method results in a sparse approximation known as a Gaussian Markov random field. We solve this equation efficiently using the finite element method within the Ice-sheet and Sea-level System Model (ISSM). In addition, spatio-temporal samples can be generated by combining an autoregressive temporal model and the Matérn field. The implementation is tested on a set of synthetic experiments to verify that it captures well the desired spatial and temporal correlations. Finally, we demonstrate the interest of this sampling capability in an illustration concerned with assessing the impact of various sources of uncertainties on the Pine Island Glacier, West Antarctica. We find that both larger spatial and temporal correlations lengths will likely result in increased uncertainty in the projections.

197. Physical processes controlling the rifting of Larsen C Ice Shelf, Antarctica, prior to the calving of iceberg A68

Authors: E. Larour (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); E. Rignot (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); M. Poinelli (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, Delft University of Technology); B. Scheuchl (University of California Irvine)

Journal: Proceedings of the National Academy of Sciences

Publisher: Proceedings of the National Academy of Sciences

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URL: <https://doi.org/10.1073/pnas.2105080118>

Publication Date: 2021-09-27

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References: 83

Cited by: 23 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration

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Open Access: N/A

Abstract: Significance The stability of Antarctica and its contribution to sea-level rise are determined by the evolution of its ice shelves, which are vast expanses of floating ice that buttress the continent. Ice shelves have been undergoing major changes in recent decades, many of them collapsing. The presumption is that these events are caused by hydrofracturing and unusual wave forcing. We find that a main control on fracturing is the thickness of the ice mélange encased in and around preexisting rifts that penetrate the entire ice shelf thickness. If the ice mélange thins beyond a threshold value, the rifts reactivate and trigger iceberg calving. This process linking climate forcing and ice shelf retreat is missing from models and does not require hydrofracture.

198. fenics_ice 1.0: a framework for quantifying initialization uncertainty for time-dependent ice sheet models

Authors: Conrad P. Koziol; Joe A. Todd; Daniel N. Goldberg; James R. Maddison

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

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References: 57

Cited by: 2 publications

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Open Access: N/A

Abstract: . Mass loss due to dynamic changes in ice sheets is a significant contributor to sea level rise, and this contribution is expected to increase in the future. Numerical codes simulating the evolution of ice sheets can potentially quantify this future contribution. However, the uncertainty inherent in these models propagates into projections of sea level rise and is hence crucial to understand. Key variables of ice sheet models, such as basal drag or ice stiffness, are typically initialized using inversion methodologies to ensure that models match present observations. Such inversions often involve tens or hundreds of thousands of parameters, with unknown uncertainties and dependencies. The computationally intensive nature of inversions along with their high number of parameters mean traditional methods such as Monte Carlo are expensive for uncertainty quantification. Here we develop a framework to estimate the posterior uncertainty of inversions and project them onto sea level change projections over the decadal timescale. The framework treats parametric uncertainty as multivariate Gaussian and exploits the equivalence between the Hessian of the model and the inverse covariance of the parameter set. The former is computed efficiently via algorithmic differentiation, and the posterior covariance is propagated in time using a time-dependent model adjoint to produce projection error bars. This work represents an important step in quantifying the internal uncertainty of projections of ice sheet models.

199. Decadal to Centennial Timescale Mantle Viscosity Inferred From Modern Crustal Uplift Rates in Greenland

Authors: S. Adhikari (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); G. A. Milne (Department of Earth and Environmental Sciences University of Ottawa Ottawa ON Canada); L. Caron (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); S. A. Khan (Technical University of Denmark); K. K. Kjeldsen (Geological Survey of Denmark and Greenland); J. Nilsson (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); E. Larour (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); E. R. Ivins (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021gl094040

URL: <https://doi.org/10.1029/2021gl094040>

Publication Date: 2021-09-24

Volume/Issue: Volume 48, Issue 19

References: 73

Cited by: 30 publications

Language: en

Subjects: N/A

Keywords: mantle rheology, Little Ice Age, crustal motion, glacial isostatic adjustment, Greenland Ice Sheet

Funding: Science Mission Directorate, Jet Propulsion Laboratory

License: CC BY-NC 4.0

Open Access: N/A

Abstract: The observed crustal uplift rates in Greenland are caused by the combined response of the solid Earth to both ongoing and past surface mass changes. Existing elastic Earth models and Maxwell linear viscoelastic GIA (glacial isostatic adjustment) models together underpredict the observed uplift rates. These models do not capture the ongoing mantle deformation induced by significant ice melting since the Little Ice Age. Using a simple Earth model within a Bayesian framework, we show that this recent mass loss can explain the data-model misfits but only when a reduced mantle strength is considered. The inferred viscosity for sub-centennial timescale mantle deformation is roughly one order of magnitude smaller than the upper mantle viscosity inferred from GIA analysis of geological data. Reconciliation of geological sea-level and modern crustal motion data may require that the model effective viscosity be treated with greater sophistication than in the simple Maxwell rheological paradigm.

200. Data-Driven Inference of the Mechanics of Slip Along Glacier Beds Using Physics-Informed Neural Networks: Case Study on Rutford Ice Stream, Antarctica

Authors: B. Riel (Massachusetts Institute of Technology); B. Minchew (Massachusetts Institute of Technology); T. Bischoff (California Institute of Technology)

Journal: Journal of Advances in Modeling Earth Systems

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021ms002621

URL: <https://doi.org/10.1029/2021ms002621>

Publication Date: 2021-09-23

Volume/Issue: Volume 13, Issue 11

References: 94

Cited by: 15 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: Reliable projections of sea-level rise depend on accurate representations of how fast-flowing glaciers slip along their beds. The mechanics of slip are often parameterized as a constitutive relation (or "sliding law") whose proper form remains uncertain. Here, we present a novel deep learning-based framework for learning the time evolution of drag at glacier beds from time-dependent ice velocity and elevation observations. We use a feedforward neural network, informed by the governing equations of ice flow, to infer spatially and temporally varying basal drag and associated uncertainties from data. We test the framework on 1D and 2D ice flow simulation outputs and demonstrate the recovery of the underlying basal mechanics under various levels of observational and modeling uncertainties. We apply this framework to time-dependent velocity data for Rutford Ice Stream, Antarctica, and present evidence that ocean-tide-driven changes in subglacial water pressure drive changes in ice flow over the tidal cycle.

201. Satellite and ground based estimates for ice surface velocities in the part of central Dronning Maud Land, East Antarctica: Implications for ice flux calculations

Authors: Praveen K. Thakur; Ashit K. Swain; Pankaj R. Dhote; Pradeep Kumar; Suvrat Kaushik; Deepak Gajbhiye; Anirudha Mahagaonkar; Vishal Sharma; Amit Dharwadkar; S.P. Aggarwal; Javed Beg; Prakash Chauhan; A. Senthil Kumar

Journal: Polar Science

Publisher: Elsevier BV

DOI: 10.1016/j.polar.2021.100737

URL: <https://doi.org/10.1016/j.polar.2021.100737>

Publication Date: 2021-09-08

Volume/Issue: Volume 30

References: 110

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Centre for Polar and Ocean Research, Ministry of Earth Sciences, Indian Space Research Organisation, Ministry of Earth Sciences (and 1 more)

License: N/A

Open Access: N/A

Abstract: N/A

202. A Scalability Study of the Ice-sheet and Sea-level System Model (ISSM, Version 4.18)

Authors: Yannic Fischler; Martin Rückamp; Christian Bischof; Vadym Aizinger; Mathieu Morlighem; Angelika Humbert

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-2021-265

URL: <https://doi.org/10.5194/gmd-2021-265>

Publication Date: 2021-08-26

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Benchmark (surveying)

Funding: Deutsche Forschungsgemeinschaft, LOEWE Zentrum AdRIA

License: CC BY 4.0

Open Access: N/A

Abstract: . Accurately modeling the contribution of Greenland and Antarctica to sea level rise requires to solve partial differential equations at a high spatial resolution. It is important to test the scalability of existing ice sheet models in order to assess whether they are ready to take advantage of new cluster architectures. In this paper, we discuss the overall scaling of the Ice-sheet and Sea-level System Model (ISSM) applied to the Greenland ice sheet. The model setup used as benchmark problem comprises a variety of modules with different levels of complexity and computational demands. The core builds the so-called stress balance module, which uses the higher-order approximation (or Blatter-Pattyn) of the Stokes equations and a mesh of linear prismatic finite elements to compute the ice flow. We develop a detailed user-oriented, yet low-overhead performance instrumentation tailored to the requirements of earth system models and run scaling tests up to 6 144 MPI processes. The results show that the computation of the Greenland model scales overall well up to 3 072 MPI processes, but is eventually slowed down by matrix assembly, the output handling, and lower-dimensional problems that employ lower numbers of unknowns per MPI process. We also discuss improvements of the scaling and identify further improvements needed for climate research. The instrumented version of ISSM, thus, not only identifies potential performance bottlenecks that were not present at lower core counts but also provides the capability to continually monitor the performance of ISSM code basis. This is of long-term significance as the overall performance of ISSM model depends on the subtle interplay between algorithms, their implementation, underlying libraries, compilers, run-time systems and hardware characteristics, all of which are in a constant state of flux.

203. An iterative process for efficient optimisation of parameters in geoscientific models: a demonstration using the Parallel Ice Sheet Model (PISM) version 0.7.3

Authors: Steven J. Phipps; Jason L. Roberts; Matt A. King

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-14-5107-2021

URL: <https://doi.org/10.5194/gmd-14-5107-2021>

Publication Date: 2021-08-17

Volume/Issue: Volume 14, Issue 8

References: 69

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Australian Research Council

License: CC BY 4.0

Open Access: N/A

Abstract: . Physical processes within geoscientific models are sometimes described by simplified schemes known as parameterisations. The values of the parameters within these schemes can be poorly constrained by theory or observation. Uncertainty in the parameter values translates into uncertainty in the outputs of the models. Proper quantification of the uncertainty in model predictions therefore requires a systematic approach for sampling parameter space. In this study, we develop a simple and efficient approach to identify regions of multi-dimensional parameter space that are consistent with observations. Using the Parallel Ice Sheet Model to simulate the present-day state of the Antarctic Ice Sheet, we find that co-dependencies between parameters preclude any simple identification of a single optimal set of parameter values. Approaches such as large ensemble modelling are therefore required in order to generate model predictions that incorporate proper quantification of the uncertainty arising from the parameterisation of physical processes.

204. Widespread Grounding Line Retreat of Totten Glacier, East Antarctica, Over the 21st Century

Authors: Tyler Pelle (University of California Irvine); Mathieu Morlighem (University of California Irvine, Dartmouth College); Yoshihiro Nakayama (Hokkaido University); Hélène Seroussi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, Dartmouth College)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021gl093213

URL: <https://doi.org/10.1029/2021ql093213>

Publication Date: 2021-08-16

Volume/Issue: Volume 48, Issue 17

References: 47

Cited by: 26 publications

Language: en

Subjects: N/A

Keywords: ice-ocean interactions, model coupling, Totten Glacier, ice sheet modeling, climate change (and 1 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Totten Glacier (TG), the primary ice discharger of East Antarctica, contains 3.85 m sea level rise equivalent (SLRe) ice mass and has displayed ocean-driven dynamic change since at least the early 2000s. We project TG's evolution through 2100 in an asynchronously coupled ice-ocean model, forced at the ocean boundaries with anomalies in CMIP6 projected temperature, salinity, and velocity. Consistent with previous studies, the Antarctic Slope Current continues to modulate warm water inflow toward TG in future simulations. Warm water (-0.5 - $^{\circ}\text{C}$) accesses TG's sub-ice shelf cavity through depressions along the eastern ice front, driving sustained retreat of TG's eastern grounding zone that cannot be captured in uncoupled models. In high emission scenarios, warm water overcomes topographic barriers and dislodges TG's southern grounding zone around 2070, increasing the rate of grounded ice loss 3.5-fold (10-35 Gt/yr) and resulting in a total 4.20 mm SLRe loss by 2100.

205. Reanalysis Surface Mass Balance of the Greenland Ice Sheet Along K-Transect (2000-2014)

Authors: Mahdi Navari (University of Maryland, National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Steven A. Margulis (University of California Los Angeles); Marco Tedesco (National Aeronautics and Space Administration (NASA) Goddard Institute for Space Studies, National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, Columbia University); Xavier Fettweis (University of Liège); Roderik S. W. van de Wal (Utrecht University)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021gl094602

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Publication Date: 2021-08-16

Volume/Issue: Volume 48, Issue 17

References: 36

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: data assimilation, surface mass balance, Greenland ice sheet, K-transect stations, particle batch smoother

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Accurate estimates of surface mass balance over the Greenland ice sheet (GrIS) would contribute to understanding the cause of recent changes and would help to better estimate the future contribution of the GrIS to sea-level rise. Given the limitations of in-situ measurements, modeling, and remote sensing, it is critical to explore the opportunity to merge the available data to better characterize the spatial and temporal variation of the GrIS surface mass balance (SMB). This work utilizes a particle batch smoother data assimilation technique that yields SMB estimates that benefit from the snow model Crocus and a 16-day albedo product derived from satellite remote sensing data. Comparison of the results against in-situ SMB measurements shows that the assimilation of the albedo product reduces the root mean square error of the posterior estimates of SMB by 51% and reduces bias by 95%.

206. Inferred basal friction and mass flux affected by crystal-orientation fabrics

Authors: Nicholas M. Rathmann; David A. Lilien

Journal: Journal of Glaciology

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/jog.2021.88

URL: <https://doi.org/10.1017/jog.2021.88>

Publication Date: 2021-08-09

Volume/Issue: Volume 68, Issue 268

References: 82

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Villum Fonden, Seventh Framework Programme

License: CC BY 4.0

Open Access: N/A

Abstract: We investigate the errors caused by neglecting the crystal-orientation fabric when inferring the basal friction coefficient field, and whether such errors can be alleviated by inferring an isotropic enhancement factor field to compensate for missing fabric information. We calculate the steady states that arise from ice flowing over a sticky spot and a bedrock bump using a vertical-slab numerical ice-flow model, consisting of a Weertman sliding law and the anisotropic Johnson flow law, coupled to a spectral fabric model of lattice rotation and dynamic recrystallisation. Given the steady or transient states as input for a canonical adjoint-based inversion, we find that Glen's isotropic flow law cannot necessarily be used to infer the true basal drag or friction coefficient field, which are obscured by the orientation fabric, thus potentially affecting vertically integrated mass fluxes. By inverting for an equivalent isotropic enhancement factor, a more accurate mass flux can be recovered, suggesting that joint inversions for basal friction and the isotropic flow-rate factor may be able to compensate for mechanical anisotropies caused by the fabric. Thus, in addition to other sources of rheological uncertainty, fabric might complicate attempts to relate subglacial conditions to basal properties inferred from an inversion relying on Glen's law.

207. icepack: a new glacier flow modeling package in Python, version 1.0

Authors: Daniel R. Shapero; Jessica A. Badgeley; Andrew O. Hoffman; Ian R. Joughin

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-14-4593-2021

URL: <https://doi.org/10.5194/gmd-14-4593-2021>

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References: 91

Cited by: 15 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration, National Science Foundation

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Open Access: N/A

Abstract: . We introduce a new software package called "icepack" for modeling the flow of glaciers and ice sheets. The icepack package is built on the finite element modeling library Firedrake, which uses the Unified Form Language (UFL), a domain-specific language embedded into Python for describing weak forms of partial differential equations. The diagnostic models in icepack are formulated through action principles that are specified in UFL. The components of each action functional can be substituted for different forms of the user's choosing, which makes it easy to experiment with the model physics. The action functional itself can be used to define a solver convergence criterion that is independent of the mesh and requires little tuning on the part of the user. The icepack package includes the 2D shallow ice and shallow stream models. We have also defined a 3D hybrid model based on spectral semi-discretization of the Blatter-Pattyn equations. Finally, icepack includes a Gauss-Newton solver for inverse problems that runs substantially faster than the Broyden-Fletcher-Goldfarb-Shanno (BFGS) method often used in the glaciological literature. The overall design philosophy of icepack is to be as usable as possible for a wide swath of the glaciological community, including both experts and novices in computational science.

208. Marine ice sheet experiments with the Community Ice Sheet Model

Authors: Gunter R. Leguy; William H. Lipscomb; Xylar S. Asay-Davis

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-15-3229-2021

URL: <https://doi.org/10.5194/tc-15-3229-2021>

Publication Date: 2021-07-14

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References: 68

Cited by: 15 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation

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Open Access: N/A

Abstract: . Ice sheet models differ in their numerical treatment of dynamical processes. Simulations of marine-based ice are sensitive to the choice of Stokes flow approximation and basal friction law and to the treatment of stresses and melt rates near the grounding line. We study the effects of these numerical choices on marine ice sheet dynamics in the Community Ice Sheet Model (CISM). In the framework of the Marine Ice Sheet Model Intercomparison Project 3d (MISMIP3d), we show that a depth-integrated, higher-order solver gives results similar to a 3D (Blatter-Pattyn) solver. We confirm that using a grounding line parameterization to approximate stresses in the grounding zone leads to accurate representation of ice sheet flow with a resolution of ~ 2 km, as opposed to ~ 0.5 km without the parameterization. In the MISMIP+ experimental framework, we compare different treatments of sub-shelf melting near the grounding line. In contrast to recent studies arguing that melting should not be applied in partly grounded cells, it is usually beneficial in CISM simulations to apply some melting in these cells. This suggests that the optimal treatment of melting near the grounding line can depend on ice sheet geometry, forcing, or model numerics. In both experimental frameworks, ice flow is sensitive to the choice of basal friction law. To study this sensitivity, we evaluate friction laws that vary the connectivity between the basal hydrological system and the ocean near the grounding line. CISM yields accurate results in steady-state and perturbation experiments at a resolution of ~ 2 km (arguably 4 km) when the connectivity is low or moderate and ~ 1 km (arguably 2 km) when the connectivity is strong.

209. Glaciological Monitoring Using the Sun as a Radio Source for Echo Detection

Authors: S. T. Peters (Department of Electrical Engineering Stanford University Stanford CA USA); D. M. Schroeder (Department of Geophysics Stanford University Stanford CA USA, Department of Electrical Engineering Stanford University Stanford CA USA); W. Chu (Georgia Institute of Technology); D. Castelletti (Department of Geophysics Stanford University Stanford CA USA); M. S. Haynes (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); P. Christoffersen (University of Cambridge); A. Romero-Wolf (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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URL: <https://doi.org/10.1029/2021gl092450>

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References: 68

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: passive sounding, remote sensing, radioglaciology, cryosphere, radio-echo sounding

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: Ice-penetrating radar observations are critical for projecting ice-sheet contribution to sea-level rise; however, these prognostic models have significant uncertainties due to an incomplete understanding of glacial subsurface processes. Existing radars that can characterize subsurface conditions are too resource-intensive to simultaneously monitor ice sheets at both the necessary temporal--daily to multiannual--and spatial--tributary to continental--scales. Here, we investigate using an ambient radio source, instead of transmitting a signal, for glaciological monitoring. We demonstrate, for the first time, passive radio sounding using the Sun to accurately measure ice thickness on Store Glacier, Greenland. Passive radar sounding could provide low-resource time-series measurements of the cryosphere, enabling us to observe and understand evolving englacial and subglacial conditions across Greenland and Antarctica with unprecedented coverage and resolution.

210. Indication of high basal melting at the EastGRIP drill site on the Northeast Greenland Ice Stream

Authors: Ole Zeising; Angelika Humbert

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-15-3119-2021

URL: <https://doi.org/10.5194/tc-15-3119-2021>

Publication Date: 2021-07-06

Volume/Issue: Volume 15, Issue 7

References: 31

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: . The accelerated ice flow of ice streams that reach far into the interior of the ice sheets is associated with lubrication of the ice sheet base by basal meltwater. However, the amount of basal melting under the large ice streams - such as the Northeast Greenland Ice Stream (NEGIS) - is largely unknown. In situ measurements of basal melt rates are important from various perspectives as they indicate the heat budget, the hydrological regime and the relative importance of sliding in glacier motion. The few previous estimates of basal melt rates in the NEGIS region were 0.1 m a⁻¹ and more, based on radiostratigraphy methods. These findings raised the question of the heat source, since even an increased geothermal heat flux could not deliver the necessary amount of heat. Here, we present basal melt rates at the recent deep drill site EastGRIP, located in the centre of NEGIS. Within 2 subsequent years, we found basal melt rates of 0.19±0.04 m a⁻¹ that are based on analysis of repeated phase-sensitive radar measurements. In order to quantify the contribution of processes that contribute to melting, we carried out an assessment of the energy balance at the interface and found the subglacial water system to play a key role in facilitating such high melt rates.

211. Retrieve Ice Velocities and Invert Spatial Rigidity of the Larsen C Ice Shelf Based on Sentinel-1 Interferometric Data

Authors: Faming Gong (School of Microelectronics and Communication Engineering, Chongqing University, Chongqing 400044, China); Kui Zhang (School of Microelectronics and Communication Engineering, Chongqing University, Chongqing 400044, China); Shujun Liu (School of Microelectronics and Communication Engineering, Chongqing University, Chongqing 400044, China)

Journal: Remote Sensing

Publisher: MDPI AG

DOI: 10.3390/rs13122361

URL: <https://doi.org/10.3390/rs13122361>

Publication Date: 2021-06-17

Volume/Issue: Volume 13, Issue 12

References: 37

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, Chongqing Science and Technology Bureau

License: CC BY 4.0

Open Access: N/A

Abstract: The Larsen C Ice Shelf (LCIS) is the largest ice shelf in the Antarctica Peninsula, and its state can be considered to be an indicator of local climate change. The goal of this paper is to invert the rigidity of the LCIS based on the interferometric synthetic aperture radar (InSAR) technique using Sentinel-1 images. A targeted processing chain is first used to obtain reliable interferometric phase measurements under the circumstance of rapid ice flow. Unfortunately, only the descending data are available, which disallows the corresponding 2-D velocity field to be directly obtained from such measurements. A new approach is thus proposed to estimate the interferometric phase-based 2-D velocity field with the assistance of speckle tracking offsets. This approach establishes an implicit relationship between range and azimuth displacements based on speckle tracking observations. By taking advantage of such a relationship, the equivalent interferometric signals in the azimuth direction are estimated, thereby recovering the interferometric phase-based 2-D ice velocity field of the LCIS. To further investigate the state of the LCIS, the recovered 2-D velocity field is utilized to invert the ice rigidity. The shallow-shelf approximation (SSA) is the core of the reverse model, which is closely dependent on boundary conditions, including kinematic and dynamic conditions. The experimental results demonstrate that the spatial distribution of the rigidity varies approximately from 70 MPa·s^{1/3} to 300 MPa·s^{1/3}. This rigidity distribution can reproduce a similar ice flow pattern to the observations.

212. Mechanics and dynamics of pinning points on the Shirase Coast, West Antarctica

Authors: Holly Still; Christina Hulbe

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-15-2647-2021

URL: <https://doi.org/10.5194/tc-15-2647-2021>

Publication Date: 2021-06-14

Volume/Issue: Volume 15, Issue 6

References: 74

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: New Zealand Antarctic Research Institute, University of Otago

License: CC BY 4.0

Open Access: N/A

Abstract: . Ice rises and rumples, sites of localised ice-shelf grounding, modify ice-shelf flow by generating lateral and basal shear stresses, upstream compression, and downstream tension. Studies of pinning points typically quantify this role indirectly, through related metrics such as a buttressing number. Here, we quantify the dynamic effects of pinning points directly, by comparing model-simulated stress states in the Ross Ice Shelf (RIS) with and without a specific set of pinning points located downstream of the MacAyeal and Bindshadler ice streams (MacIS and BIS, respectively). Because ice properties are only known indirectly, the experiment is repeated with different realisations of the ice softness. While longitudinal stretching, and thus ice velocity, is smaller with the pinning points, flow resistance generated by other grounded features is also smaller. Conversely, flow resistance generated by other grounded features increases when the pinning points are absent, providing a non-local control on the net effect of the pinning points on ice-shelf flow. We find that an ice stream located directly upstream of the pinning points, MacIS, is less responsive to their removal than the obliquely oriented BIS. This response is due to zones of locally higher basal drag acting on MacIS, which may itself be a consequence of the coupled ice-shelf and ice-stream response to the pinning points. We also find that inversion of present-day flow and thickness for basal friction and ice softness, without feature-specific, a posteriori adjustment, leads to the incorrect representation of ice rumple morphology and an incorrect boundary condition at the ice base. Viewed from the perspective of change detection, we find that, following pinning point removal, the ice shelf undergoes an adjustment to a new steady state that involves an initial increase in ice speeds across the eastern ice shelf, followed by decaying flow speeds, as mass flux reduces thickness gradients in some areas and increases thickness gradients in others. Increases in ice-stream flow speeds persist with no further adjustment, even without sustained grounding-line retreat. Where pinning point effects are important, model tuning that respects their morphology is necessary to represent the system as a whole and inform interpretations of observed change.

213. Surface Mass Balance Models Vs. Stake Observations: A Comparison in the Lake Vostok Region, Central East Antarctica

Authors: Andreas Richter; Alexey A. Ekaykin; Matthias O. Willen; Vladimir Ya. Lipenkov; Andreas Groh; Sergey V. Popov; Mirko Scheinert; Martin Horwath; Reinhard Dietrich

Journal: Frontiers in Earth Science

Publisher: Frontiers Media SA

DOI: 10.3389/feart.2021.669977

URL: <https://doi.org/10.3389/feart.2021.669977>

Publication Date: 2021-05-25

Volume/Issue: Volume 9

References: 39

Cited by: 8 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: Deutsche Forschungsgemeinschaft, Russian Foundation for Basic Research

License: CC BY 4.0

Open Access: N/A

Abstract: The surface mass balance (SMB) is very low over the vast East Antarctic Plateau, for example in the Vostok region, where the mean SMB is on the order of 20-35 kg m⁻² a⁻¹. The observation and modeling of spatio-temporal SMB variations are equally challenging in this environment. Stake measurements carried out in the Vostok region provide SMB observations over half a century (1970-2019). This unique data set is compared with SMB estimations of the regional climate models RACMO2.3p2 (RACMO) and MAR3.11 (MAR). We focus on the SMB variations over time scales from months to decades. The comparison requires a rigorous assessment of the uncertainty in the stake observations and the spatial scale dependence of the temporal SMB variations. Our results show that RACMO estimates of annual and multi-year SMB agree well with the observations. The regression slope between modelled and observed temporal variations is close to 1.0 for this model. SMB simulations by MAR are affected by a positive bias which amounts to 6 kg m⁻² a⁻¹ at Vostok station and 2 kg m⁻² a⁻¹ along two stake profiles between Lake Vostok and Ridge B. None of the models is capable to reproduce the seasonal distributions of SMB and precipitation. Model SMB estimates are used in assessing the ice-mass balance and sea-level contribution of the Antarctic Ice Sheet by the input-output method. Our results provide insights into the uncertainty contribution of the SMB models to such assessments.

214. The impact of glaciers on mountain erosion

Authors: Frédéric Herman; Fien De Doncker; Ian Delaney; Günther Prasicek; Michèle Koppes

Journal: Nature Reviews Earth & Environment

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s43017-021-00165-9

URL: <https://doi.org/10.1038/s43017-021-00165-9>

Publication Date: 2021-05-18

Volume/Issue: Volume 2, Issue 6

References: 211

Cited by: 69 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

215. The diurnal Energy Balance Model (dEBM): a convenient surface mass balance solution for ice sheets in Earth system modeling

Authors: Uta Krebs-Kanzow; Paul Gierz; Christian B. Rodehacke; Shan Xu; Hu Yang; Gerrit Lohmann

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-15-2295-2021

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References: 73

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

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Open Access: N/A

Abstract: . The surface mass balance scheme dEBM (diurnal Energy Balance Model) provides a novel interface between the atmosphere and land ice for Earth system modeling, which is based on the energy balance of glaciated surfaces. In contrast to empirical schemes, dEBM accounts for changes in the Earth's orbit and atmospheric composition. The scheme only requires monthly atmospheric forcing (precipitation, temperature, shortwave and longwave radiation, and cloud cover). It is also computationally inexpensive, which makes it particularly suitable to investigate the ice sheets' response to long-term climate change. After calibration and validation, we analyze the surface mass balance of the Greenland Ice Sheet (GrIS) based on climate simulations representing two warm climate states: a simulation of the mid-Holocene (approximately 6000 years before present) and a climate projection based on an extreme emission scenario which extends to the year 2100. The former period features an intensified summer insolation while the 21st century is characterized by reduced outgoing longwave radiation. Specifically, we investigate whether the temperature-melt relationship, as used in empirical temperature-index methods, remains stable under changing insolation and atmospheric composition. Our results indicate that the temperature-melt relation is sensitive to changes in insolation on orbital timescales but remains mostly invariant under the projected warming climate of the 21st century.

216. Thermal refraction: implications for subglacial heat flux

Authors: Simon Willcocks; Derrick Hasterok; Samuel Jennings

Journal: Journal of Glaciology

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/jog.2021.38

URL: <https://doi.org/10.1017/jog.2021.38>

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Keywords: N/A

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Open Access: N/A

Abstract: In this study, we explore small-scale (~1 to 20 km) thermal-refractive effects on basal geothermal heat flux (BGHF) at subglacial boundaries resulting from lateral thermal conductivity contrasts associated with subglacial topography and geologic contacts. We construct a series of two-dimensional, conductive, steady-state models that exclude many of the complexities of ice sheets in order to demonstrate the effect of thermal refraction. We show that heat can preferentially flow into or around a subglacial valley depending on the thermal conductivity contrast with underlying bedrock, with anomalies of local BGHF at the ice-bedrock interface between 80 and 120% of regional BGHF and temperature anomalies on the order of +/-15% for the typical range of bedrock conductivities. In the absence of bed topography, subglacial contacts can produce significant heat flux and temperature anomalies that are locally extensive (>10 km). Thermal refraction can result in either an increase or decrease in the likelihood of melting and ice-sheet stability depending on the conductivity contrast and bed topography. While our models exclude many of the physical complexities of ice behavior, they illustrate the need to include refractive effects created by realistic geology into future glacial models to improve the prediction of subglacial melting and ice viscosity.

217. Assessment of numerical schemes for transient, finite-element ice flow models using ISSM v4.18

Authors: Thiago Dias dos Santos; Mathieu Morlighem; H  l  ne Seroussi

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-14-2545-2021

URL: <https://doi.org/10.5194/gmd-14-2545-2021>

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Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: Time-dependent simulations of ice sheets require two equations to be solved: the mass transport equation, derived from the conservation of mass, and the stress balance equation, derived from the conservation of momentum. The mass transport equation controls the advection of ice from the interior of the ice sheet towards its periphery, thereby changing its geometry. Because it is based on an advection equation, a stabilization scheme needs to be employed when solved using the finite-element method. Several stabilization schemes exist in the finite-element method framework, but their respective accuracy and robustness have not yet been systematically assessed for glaciological applications. Here, we compare classical schemes used in the context of the finite-element method: (i) artificial diffusion, (ii) streamline upwinding, (iii) streamline upwind Petrov-Galerkin, (iv) discontinuous Galerkin, and (v) flux-corrected transport. We also look at the stress balance equation, which is responsible for computing the ice velocity that "advects" the ice downstream. To improve the velocity computation accuracy, the ice-sheet modeling community employs several sub-element parameterizations of physical processes at the grounding line, the point where the grounded ice starts to float onto the ocean. Here, we introduce a new sub-element parameterization for the driving stress, the force that drives the ice-sheet flow. We analyze the response of each stabilization scheme by running transient simulations forced by ice-shelf basal melt. The simulations are based on an idealized ice-sheet geometry for which there is no influence of bedrock topography. We also perform transient simulations of the Amundsen Sea Embayment, West Antarctica, where real bedrock and surface elevations are employed. In both idealized and real ice-sheet experiments, stabilization schemes based on artificial diffusion lead systematically to a bias towards more mass loss in comparison to the other schemes and therefore should be avoided or employed with a sufficiently high mesh resolution in the vicinity of the grounding line. We also run diagnostic simulations to assess the accuracy of the driving stress parameterization, which, in combination with an adequate parameterization for basal stress, provides improved numerical convergence in ice speed computations and more accurate results.

218. Projected land ice contributions to twenty-first-century sea level rise

Authors: Tamsin L. Edwards; Sophie Nowicki; Ben Marzeion; Regine Hock; Heiko Goelzer; Hélène Seroussi; Nicolas C. Jourdain; Donald A. Slater; Fiona E. Turner; Christopher J. Smith; Christine M. McKenna; Erika Simon; Ayako Abe-Ouchi; Jonathan M. Gregory; Eric Larour; William H. Lipscomb; Antony J. Payne; Andrew Shepherd; Cécile Agosta; Patrick Alexander; Torsten Albrecht; Brian Anderson; Xylar Asay-Davis; Andy Aschwanden; Alice Barthel; Andrew Bliss; Reinhard Calov; Christopher Chambers; Nicolas Champollion; Youngmin Choi; Richard Cullather; Joshua Cuzzzone; Christophe Dumas; Denis Felikson; Xavier Fettweis; Koji Fujita; Benjamin K. Galton-Fenzi; Rupert Gladstone; Nicholas R. Golledge; Ralf Greve; Tore Hattermann; Matthew J. Hoffman; Angelika Humbert; Matthias Huss; Philippe Huybrechts; Walter Immerzeel; Thomas Kleiner; Philip Kraaijenbrink; Sébastien Le clec'h; Victoria Lee; Gunter R. Leguy; Christopher M. Little; Daniel P. Lowry; Jan-Hendrik Malles; Daniel F. Martin; Fabien Maussion; Mathieu Morlighem; James F. O'Neill; Isabel Nias; Frank Pattyn; Tyler Pelle; Stephen F. Price; Aurélien Quiquet; Valentina Radi; Ronja Reese; David R. Rounce; Martin Rückamp; Akiko Sakai; Courtney Shafer; Nicole-Jeanne Schlegel; Sarah Shannon; Robin S. Smith; Fiammetta Straneo; Sainan Sun; Lev Tarasov; Luke D. Trusel; Jonas Van Breedam; Roderik van de Wal; Michiel van den Broeke; Ricarda Winkelmann; Harry Zekollari; Chen Zhao; Tong Zhang; Thomas Zwinger

Journal: Nature

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41586-021-03302-y

URL: <https://doi.org/10.1038/s41586-021-03302-y>

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Language: en

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Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

219. The Utrecht Finite Volume Ice-Sheet Model: UFEMISM (version 1.0)

Authors: Constantijn J. Berends; Heiko Goelzer; Roderik S. W. van de Wal

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-14-2443-2021

URL: <https://doi.org/10.5194/gmd-14-2443-2021>

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Funding: Netherlands Earth System Science Centre

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Open Access: N/A

Abstract: . Improving our confidence in future projections of sea-level rise requires models that can simulate ice-sheet evolution both in the future and in the geological past. A physically accurate treatment of large changes in ice-sheet geometry requires a proper treatment of processes near the margin, like grounding line dynamics, which in turn requires a high spatial resolution in that specific region, so that small-scale topographical features are resolved. This leads to a demand for computationally efficient models, where such a high resolution can be feasibly applied in simulations of 105-107 years in duration. Here, we present and evaluate a new ice-sheet model that solves the hybrid SIA-SSA approximation of the stress balance, including a heuristic rule for the grounding-line flux. This is done on a dynamic adaptive mesh which is adapted to the modelled ice-sheet geometry during a simulation. Mesh resolution can be configured to be fine only at specified areas, such as the calving front or the grounding line, as well as specified point locations such as ice-core drill sites. This strongly reduces the number of grid points where the equations need to be solved, increasing the computational efficiency. A high resolution allows the model to resolve small geometrical features, such as outlet glaciers and sub-shelf pinning points, which can significantly affect large-scale ice-sheet dynamics. We show that the model reproduces the analytical solutions or model intercomparison benchmarks for a number of schematic ice-sheet configurations, indicating that the numerical approach is valid. Because of the unstructured triangular mesh, the number of vertices increases less rapidly with resolution than in a square-grid model, greatly reducing the required computation time for high resolutions. A simulation of all four continental ice sheets during an entire 120 kyr glacial cycle, with a 4 km resolution near the grounding line, is expected to take 100-200 wall clock hours on a 16-core system (1600-3200 core hours), implying that this model can be feasibly used for high-resolution palaeo-ice-sheet simulations.

220. Future Projections of Petermann Glacier Under Ocean Warming Depend Strongly on Friction Law

Authors: Henning Åkesson (Stockholm University); Mathieu Morlighem (University of California Irvine); Matt O'Regan (Stockholm University); Martin Jakobsson (Stockholm University)

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020jf005921

URL: <https://doi.org/10.1029/2020jf005921>

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Cited by: 19 publications

Language: en

Subjects: N/A

Keywords: ice shelf, sea level, ice sheet, Greenland, friction (and 1 more)

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Open Access: N/A

Abstract: Basal friction heavily controls the dynamics of fast-flowing glaciers. However, the best approach to modeling friction is unclear, increasing uncertainties in projections of future mass loss and sea-level rise. Here, we compare six friction laws and evaluate them for Petermann Glacier in northern Greenland, using a higher order three-dimensional ice-sheet model. We model glacier retreat and mass loss under an ocean-only warming until year 2300, while not considering the effects of a future warmer atmosphere. Regardless of the friction law, we find that breakup of Petermann's ice shelf is likely to occur within the next decades. However, future grounding-line retreat differs by 10s of km and estimates of sea-level rise may quadruple, depending on the friction law employed. A bedrock ridge halts the retreat for four of the laws, and Petermann retreats furthest when applying a Budd or a Coulomb-type "till law." Depending on the friction law, sea-level contributions differ by 133% and 282% by 2300 for 2 degreesC and 5 degreesC ocean warming scenarios, respectively.

221. Future Sea Level Change Under Coupled Model Intercomparison Project Phase 5 and Phase 6 Scenarios From the Greenland and Antarctic Ice Sheets

Authors: Antony J. Payne (University of Bristol); Sophie Nowicki (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, RENEW Institute); Ayako Abe-Ouchi (University of Tokyo); Cécile Agosta (Laboratoire des Sciences du Climat et de l'Environnement LSCE-IPSL CEA-CNRS-UVSQ Université Paris-Saclay Gif-sur-Yvette France); Patrick Alexander (National Aeronautics and Space Administration (NASA) Goddard Institute for Space Studies, National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, Columbia University); Torsten Albrecht (Potsdam Institute for Climate Impact Research (PIK) Member of the Leibniz Association Potsdam Germany); Xylar Asay-Davis (Los Alamos National Laboratory); Andy Aschwanden (University of Alaska Fairbanks); Alice Barthel (Los Alamos National Laboratory); Thomas J. Bracegirdle (British Antarctic Survey Cambridge UK); Reinhard Calov (Potsdam Institute for Climate Impact Research (PIK) Member of the Leibniz Association Potsdam Germany); Christopher Chambers (Hokkaido University); Youngmin Choi (University of California Irvine); Richard Cullather (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Joshua Cuzzone (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Christophe Dumas (Laboratoire des Sciences du Climat et de l'Environnement LSCE-IPSL CEA-CNRS-UVSQ Université Paris-Saclay Gif-sur-Yvette France); Tamsin L. Edwards (King's College London); Denis Felikson (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, Universities Space Research Association); Xavier Fettweis (University of Liège); Benjamin K. Galton-Fenzi (University of Tasmania, Australian Antarctic Division Kingston TAS Australia); Heiko Goelzer (NORCE Norwegian Research Centre, Utrecht University, Laboratoire de Glaciologie Université Libre de Bruxelles Brussels Belgium, Bjerknes Centre for Climate Research); Rupert Gladstone (University of Lapland); Nicholas R. Golledge (Victoria University of Wellington); Jonathan M. Gregory (Met Office Hadley Centre, Met Office, University of Reading); Ralf Greve (Hokkaido University, British Antarctic Survey Cambridge UK); Tore Hattermann (Norwegian Polar Institute, Department of Physics and Technology The Arctic University University of Tromsø Norway); Matthew J. Hoffman (Los Alamos National Laboratory); Angelika Humbert (University of Bremen, Alfred Wegener Institute); Philippe Huybrechts (Vrije Universiteit Brussel); Nicolas C. Jourdain (Institut de Recherche pour le Développement (IRD)); Thomas Kleiner (Alfred Wegener Institute); Peter Kuipers Munneke (Utrecht University); Eric Larour (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Sebastien Le clec'h (Vrije Universiteit Brussel); Victoria Lee (University of Bristol); Gunter Leguy (National Center for Atmospheric Research); William H. Lipscomb (National Center for Atmospheric Research); Christopher M. Little (Atmospheric and Environmental Research Inc.); Daniel P. Lowry (GNS Science Lower Hutt New Zealand); Mathieu Morlighem (University of California Irvine); Isabel Nias (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, University of Liverpool); Frank Pattyn (Laboratoire de Glaciologie Université Libre de Bruxelles Brussels Belgium); Tyler Pelle (University of California Irvine); Stephen F. Price (Los Alamos National Laboratory); Aurélien Quiquet (Laboratoire des Sciences du Climat et de l'Environnement LSCE-IPSL CEA-CNRS-UVSQ Université Paris-Saclay Gif-sur-Yvette France); Ronja Reese (Potsdam Institute for Climate Impact Research (PIK) Member of the Leibniz Association Potsdam Germany); Martin Rückamp (Alfred Wegener Institute); Nicole-Jeanne Schlegel (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Hélène Seroussi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Andrew Shepherd (University of Leeds); Erika Simon (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Donald Slater (Scripps Institution of Oceanography / University of California San Diego); Robin S. Smith (University of Reading); Fiammetta Straneo (Scripps Institution of Oceanography / University of California San Diego); Sainan Sun (Laboratoire de Glaciologie Université Libre de Bruxelles Brussels Belgium); Lev Tarasov (Department of Physics and Physical Oceanography Memorial University of Newfoundland Canada); Luke D. Trusel (Pennsylvania State University); Jonas Van Breedam (Vrije Universiteit Brussel); Roderik van de Wal (Utrecht University); Michiel van den Broeke (Utrecht University); Ricarda Winkelmann (Potsdam Institute for Climate Impact Research (PIK) Member of the

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Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Keywords: N/A

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Abstract: Projections of the sea level contribution from the Greenland and Antarctic ice sheets (GrIS and AIS) rely on atmospheric and oceanic drivers obtained from climate models. The Earth System Models participating in the Coupled Model Intercomparison Project phase 6 (CMIP6) generally project greater future warming compared with the previous Coupled Model Intercomparison Project phase 5 (CMIP5) effort. Here we use four CMIP6 models and a selection of CMIP5 models to force multiple ice sheet models as part of the Ice Sheet Model Intercomparison Project for CMIP6 (ISMIP6). We find that the projected sea level contribution at 2100 from the ice sheet model ensemble under the CMIP6 scenarios falls within the CMIP5 range for the Antarctic ice sheet but is significantly increased for Greenland. Warmer atmosphere in CMIP6 models results in higher Greenland mass loss due to surface melt. For Antarctica, CMIP6 forcing is similar to CMIP5 and mass gain from increased snowfall counteracts increased loss due to ocean warming.

222. The Impact of Variable Ocean Temperatures on Totten Glacier Stability and Discharge

Authors: F. S. McCormack (University of Tasmania, Monash University); J. L. Roberts (University of Tasmania, Australian Antarctic Division Kingston TAS Australia); D. E. Gwyther (Centre for Applications in Natural Resource Mathematics School of Mathematics and Physics University of Queensland St Lucia QLD Australia); M. Morlighem (University of California Irvine); T. Pelle (University of California Irvine); B. K. Galton-Fenzi (University of Tasmania, Australian Antarctic Division Kingston TAS Australia)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Publication Date: 2021-05-04

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References: 64

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: ocean variability, ice sheet model, Totten Glacier

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Open Access: N/A

Abstract: A major uncertainty in Antarctica's contribution to future sea-level rise is the ice sheet response timescales to ocean warming. Totten Glacier drains a region containing 3.9 m global sea level equivalent and has been losing mass over recent decades. We use an ice sheet model coupled to an ice-shelf cavity combined ocean box and plume model to investigate Totten's response to variable ocean forcing. Totten's grounding line is stable for a limited range of ocean temperatures near current observations (i.e., -0.95 degreesC to -0.75 degreesC), with topography influencing the discharge periodicity. For increases of ≥ 0.2 degreesC in temperatures beyond this range, grounding line retreat occurs. Variable ocean forcing can reduce retreat relative to constant forcing, and different variability amplitudes can cause centennial-scale delays in retreat through interactions with topography. Our results highlight the need for long-term ocean state observations and to include forcing variability in ice sheet model simulations of future change.

223. Deglacial Ice Sheet Instabilities Induced by Proglacial Lakes

Authors: Aurélien Quiquet (Laboratoire des Sciences du Climat et de l'Environnement, LSCE/IPSL, CEA-CNRS-UVSQ Université Paris-Saclay Gif-sur-Yvette France, Now at NumClimSolutions Palaiseau France); Christophe Dumas (Laboratoire des Sciences du Climat et de l'Environnement, LSCE/IPSL, CEA-CNRS-UVSQ Université Paris-Saclay Gif-sur-Yvette France); Didier Paillard (Laboratoire des Sciences du Climat et de l'Environnement, LSCE/IPSL, CEA-CNRS-UVSQ Université Paris-Saclay Gif-sur-Yvette France); Gilles Ramstein (Laboratoire des Sciences du Climat et de l'Environnement, LSCE/IPSL, CEA-CNRS-UVSQ Université Paris-Saclay Gif-sur-Yvette France); Catherine Ritz (Institut de Recherche pour le Développement (IRD)); Didier M. Roche (Laboratoire des Sciences du Climat et de l'Environnement, LSCE/IPSL, CEA-CNRS-UVSQ Université Paris-Saclay Gif-sur-Yvette France, Vrije Universiteit Amsterdam)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Cited by: 17 publications

Language: en

Subjects: N/A

Keywords: proglacial lake, grounding line instability, last deglaciation, North American ice sheet

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License: Creative Commons

Open Access: N/A

Abstract: During the last deglaciation (21-7 kaBP), the gradual retreat of Northern Hemisphere ice sheet margins produced large proglacial lakes. While the climatic impacts of these lakes have been widely acknowledged, their role on ice sheet grounding line dynamics has received very little attention so far. Here, we show that proglacial lakes had dramatic implications for the North American ice sheet dynamics through a self-sustained mechanical instability which has similarities with the known marine ice sheet instability consequently providing fast retreat of large portions of the ice sheet over the continent. This instability mechanism is likely important in contributing to deglaciation of terrestrial glaciers and ice sheets with proglacial lakes at their margins as it can substantially accelerate the mass loss. Echoing our knowledge of Antarctic ice sheet dynamics, proglacial lakes are another manifestation of the importance of grounding line dynamics for ice sheet evolution.

224. The transferability of adjoint inversion products between different ice flow models

Authors: Jowan M. Barnes; Thiago Dias dos Santos; Daniel Goldberg; G. Hilmar Gudmundsson; Mathieu Morlighem; Jan De Rydt

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-15-1975-2021

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Language: en

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Keywords: N/A

Funding: Natural Environment Research Council, National Science Foundation

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Abstract: . Among the most important challenges faced by ice flow models is how to represent basal and rheological conditions, which are challenging to obtain from direct observations. A common practice is to use numerical inversions to calculate estimates for the unknown properties, but there are many possible methods and not one standardised approach. As such, every ice flow model has a unique initialisation procedure. Here we compare the outputs of inversions from three different ice flow models, each employing a variant of adjoint-based optimisation to calculate basal sliding coefficients and flow rate factors using the same observed surface velocities and ice thickness distribution. The region we focus on is the Amundsen Sea Embayment in West Antarctica, the subject of much investigation due to rapid changes in the area over recent decades. We find that our inversions produce similar distributions of basal sliding across all models, despite using different techniques, implying that the methods used are highly robust and represent the physical equations without much influence by individual model behaviours. Transferring the products of inversions between models results in time-dependent simulations displaying variability on the order of or lower than existing model intercomparisons. Focusing on contributions to sea level, the highest variability we find in simulations run in the same model with different inversion products is 32 %, over a 40-year period, a difference of 3.67 mm. There is potential for this to be improved with further standardisation of modelling processes, and the lowest variability within a single model is 13 %, or 1.82 mm over 40 years. While the successful transfer of inversion outputs from one model to another requires some extra effort and technical knowledge of the particular models involved, it is certainly possible and could indeed be useful for future intercomparison projects.

225. Joint Inversion for Surface Accumulation Rate and Geothermal Heat Flow From Ice-Penetrating Radar Observations at Dome A, East Antarctica. Part I: Model Description, Data Constraints, and Inversion Results

Authors: M. J. Wolovick (Beijing Normal University); J. C. Moore (Beijing Normal University, University of Lapland); L. Zhao (Southern Marine Science and Engineering Guangdong Laboratory, Beijing Normal University)

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

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Language: en

Subjects: N/A

Keywords: East Antarctica, inversion, ice model, Dome A, genetic algorithm (and 1 more)

Funding: Suomalainen Tiedekatemia, National Natural Science Foundation of China

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Open Access: N/A

Abstract: Ice-penetrating radar data contain a wealth of information about the bed and internal structure of the ice sheet. While these data have long been used to diagnose the presence of basal water, infer attenuation rates, or explore the internal stratigraphy of the ice sheet, they have rarely been used jointly in a formal inverse model for the ice sheet temperature structure. Here, we invert a coupled thermomechanical ice sheet and basal hydrology model to infer both geothermal heat flow (GHF) and accumulation rate from multiple classes of radar observations in the area around Dome A, East Antarctica. Our forward model solves for a coupled steady state between the ice sheet flow field, temperature, and basal hydrology, including melt, water transport, and freeze-on. We fit radar observations of basal water, freeze-on, and internal layers, along with a GHF prior based on aeromagnetic observations. We minimize the combined misfit function by first using an evolutionary algorithm followed by localized perturbation tests. In addition to inferring the spatial distribution of GHF and accumulation rate, we are also able to estimate the uncertainty about our best-fit answer, as well as quantify how our result depends on each individual data constraint. Our results demonstrate a new method for combining multiple glaciological constraints into a single inverse model of the ice sheet, and give us a more rigorous picture of the information content provided by each data set. In a companion paper we analyze and interpret the best-fit model.

226. A cut finite element method for non-Newtonian free surface flows in 2D - application to glacier modelling

Authors: Josefin Ahlkrona; Daniel Elfverson

Journal: Journal of Computational Physics: X

Publisher: Elsevier BV

DOI: 10.1016/j.jcp.2021.100090

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Open Access: N/A

Abstract: N/A

227. The Polar Radiant Energy in the Far Infrared Experiment: A New Perspective on Polar Longwave Energy Exchanges

Authors: Tristan S. L'Ecuyer (University of Wisconsin-Madison); Brian J. Drouin (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); James Anheuser (University of Wisconsin-Madison); Meredith Grames (University of Wisconsin-Madison); David S. Henderson (Space Science and Engineering Center, Madison, Wisconsin); Xianglei Huang (University of Michigan); Brian H. Kahn (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Jennifer E. Kay (University of Colorado Boulder); Boon H. Lim (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Marian Mateling (University of Wisconsin-Madison); Aronne Merrelli (Space Science and Engineering Center, Madison, Wisconsin); Nathaniel B. Miller (Space Science and Engineering Center, Madison, Wisconsin); Sharmila Padmanabhan (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Colten Peterson (University of Michigan); Nicole-Jeanne Schlegel (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Mary L. White (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Yan Xie (University of Michigan)

Journal: Bulletin of the American Meteorological Society

Publisher: American Meteorological Society

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References: 99

Cited by: 19 publications

Language: N/A

Subjects: N/A

Keywords: Longwave|Middle latitudes|Radiant energy

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Open Access: N/A

Abstract: Earth's climate is strongly influenced by energy deficits at the poles that emit more thermal energy than they receive from the sun. Energy exchanges between the surface and atmosphere influence the local environment while heat transport from lower latitudes drives midlatitude atmospheric and oceanic circulations. In the Arctic, in particular, local energy imbalances induce strong seasonality in surface-atmosphere heat exchanges and an acute sensitivity to forced climate variations. Despite these important local and global influences, the largest contributions to the polar atmospheric and surface energy budgets have not been fully characterized. The spectral variation of far-infrared radiation that makes up 60% of polar thermal emission has never been systematically measured impeding progress toward consensus in predicted rates of Arctic warming, sea ice decline, and ice sheet melt. Enabled by recent advances in sensor miniaturization and CubeSat technology, the Polar Radiant Energy in the Far Infrared Experiment (PREFIRE) mission will document, for the first time, the spectral, spatial, and temporal variations of polar far-infrared emission. Selected under NASA's Earth

Ventures Instrument (EVI) program, PREFIRE will utilize new lightweight, low-power, ambient temperature detectors capable of measuring at wavelengths up to 50 μm to quantify Earth's far-infrared spectrum. Estimates of spectral surface emissivity, water vapor, cloud properties, and the atmospheric greenhouse effect derived from these measurements offer the potential to advance our understanding of the factors that modulate thermal fluxes in the cold, dry conditions characteristic of the polar regions.

228. Sea-Level Change: Emergent Hazard in a Warming World

Authors: John J. Clague

Journal: Treatise on Geomorphology

Publisher: Elsevier

DOI: 10.1016/b978-0-12-818234-5.00065-1

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References: 211

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Language: en

Subjects: N/A

Keywords: Risk, Inundation, Hazard, Coastline, Shorelines (and 2 more)

Funding: N/A

License: N/A

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Abstract: N/A

229. Geometric Controls of Tidewater Glacier Dynamics

Authors: Thomas Frank; Henning Åkesson; Basile de Fleurian; Mathieu Morlighem; Kerim H. Nisancioglu

Journal: Unknown Journal

Publisher: Copernicus GmbH

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Cited by: 2 publications

Language: N/A

Subjects: N/A

Keywords: Fjord|Tidewater glacier cycle|Shoaling and schooling

Funding: Vetenskapsrådet, Norges Forskningsråd

License: CC BY 4.0

Open Access: N/A

Abstract: . Retreat of marine outlet glaciers often initiates depletion of inland ice through dynamic adjustments of the upstream glacier. The local topography of a fjord may promote or inhibit such retreat, and therefore fjord geometry constitutes a critical control on ice sheet mass balance. To quantify the processes of ice-topography interactions and enhance the understanding of the dynamics involved, we analyze a multitude of topographic fjord settings and scenarios using the Ice-sheet and Sea-level System Model (ISSM). We systematically study glacier retreat through a variety of artificial fjord geometries and quantify the modeled dynamics directly in relation to topographic features. We find that retreat in an upstream widening or deepening fjord does not necessarily promote retreat, but conversely, may stabilize a glacier because converging ice flow towards a constriction enhances lateral shear. An upstream narrowing or shoaling fjord, in turn, may promote retreat since fjord walls or bed provide little stability to the glacier where ice flow diverges. Furthermore, we identify distinct quantitative relationships directly linking grounding line discharge and retreat rate to fjord topography, and transfer these results to a long-term study of the retreat of Jakobshavn Isbræ. These findings offer new perspectives on ice-topography interactions, and give guidance to an ad-hoc assessment of future topographically induced ice loss based on knowledge of the upstream fjord geometry.

230. Challenges in predicting Greenland supraglacial lake drainages at the regional scale

Authors: Kristin Poinar; Lauren C. Andrews

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-15-1455-2021

URL: <https://doi.org/10.5194/tc-15-1455-2021>

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Volume/Issue: Volume 15, Issue 3

References: 124

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Earth Sciences Division

License: CC BY 4.0

Open Access: N/A

Abstract: . A leading hypothesis for the mechanism of fast supraglacial lake drainages is that transient extensional stresses briefly allow crevassing in otherwise compressional ice flow regimes. Lake water can then hydrofracture a crevasse to the base of the ice sheet, and river inputs can maintain this connection as a moulin. If future ice sheet models are to accurately represent moulins, we must understand their formation processes, timescales, and locations. Here, we use remote-sensing velocity products to constrain the relationship between strain rates and lake drainages across ~ 1600 km² in Pâkitsoq, western Greenland, between 2002-2019. We find significantly more extensional background strain rates at moulins associated with fast-draining lakes than at slow-draining or non-draining lake moulins. We test whether moulins in more extensional background settings drain their lakes earlier, but we find insignificant correlation. To investigate the frequency at which strain-rate transients are associated with fast lake drainage, we examined Landsat-derived strain rates over 16 and 32 d periods at moulins associated with 240 fast-lake-drainage events over 18 years. A low signal-to-noise ratio, the presence of water, and the multi-week repeat cycle obscured any resolution of the hypothesized transient strain rates. Our results support the hypothesis that transient strain rates drive fast lake drainages. However, the current generation of ice sheet velocity products, even when stacked across hundreds of fast lake drainages, cannot resolve these transients. Thus, observational progress in understanding lake drainage initiation will rely on field-based tools such as GPS networks and photogrammetry.

231. Retreat of Humboldt Gletscher, North Greenland, Driven by Undercutting From a Warmer Ocean

Authors: Eric Rignot (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Lu An (University of California Irvine); Nolwenn Chauche (Access Arctic SARL); Mathieu Morlighem (University of California Irvine); Seongsu Jeong (University of California Irvine); Michael Wood (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Jeremie Mouginot (University of California Irvine, Institut des Geosciences de l'Environnement Universite Grenoble-Alpes CNRS Grenoble France); Josh K. Willis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ingo Klaucke (GEOMAR Helmholtz Centre for Ocean Research); Wilhelm Weinrebe (GEOMAR Helmholtz Centre for Ocean Research); Andreas Muenchow (University of Delaware)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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References: 39

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration

License: CC BY-NC 4.0

Open Access: N/A

Abstract: Humboldt Gletscher is a 100-km wide, slow-moving glacier in north Greenland which holds a 19-cm global sea level equivalent. Humboldt has been the fourth largest contributor to sea level rise since 1972 but the cause of its mass loss has not been elucidated. Multi-beam echo sounding data collected in 2019 indicate a seabed 200 m deeper than previously known. Conductivity temperature depth data reveal the presence of warm water of Atlantic origin at 0 degreesC at the glacier front and a warming of the ocean waters by 0.9 +/- 0.1 degreesC since 1962. Using an ocean model, we reconstruct grounded ice undercutting by the ocean, combine it with calculated retreat caused by ice thinning to floatation, and are able to fully explain the observed retreat. Two thirds of the retreat are caused by undercutting of grounded ice, which is a physical process not included in most ice sheet models.

232. Recent retreat of Greenland's marine terminating glaciers has a lasting impact on velocity and mass loss during the 21st Century

Authors: Isabel Nias; Sophie Nowicki; Denis Felikson

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/egusphere-egu21-9908

URL: <https://doi.org/10.5194/egusphere-egu21-9908>

Publication Date: 2021-03-04

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References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Greenland ice sheet|Ice-sheet model|Iceberg|Future sea level|Forcing (mathematics)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: <p>Mass loss from the Greenland Ice Sheet (GrIS) can be partitioned between surface mass balance (SMB) and discharge due to ice dynamics through its marine-terminating outlet glaciers. A perturbation to a glacier terminus (e.g. a calving event) results in an instantaneous response in velocity and mass loss, but also a diffusive response due to the evolution of ice thickness over time. This diffusive response means the total impact of a retreat event can take decades to be fully realised. Here we model the committed response of the GrIS to recent observed changes in terminus position, neglecting any future climate perturbations. Our simulations quantify the sea level contribution that is locked in due to the slow dynamic response of the ice. Using the Ice Sheet System Model (ISSM), we run forward simulations starting from an initial state representative of the 2007 ice sheet. We apply perturbations to the marine-terminating glacier termini that represent recent observed changes, and simulate the response over the 21st Century, holding the climate forcing constant. The sensitivity of the ice sheet response to model parameter uncertainty is explored with in an ensemble framework, and GRACE data is used to constrain the results. We find that terminus retreat observed between 2007 and 2015 results in approximately 6 mm of sea level rise by 2100, with retreat having a lasting impact on velocity and mass loss. Our results complement the ISMIP6 projections, which report the ice sheet response to future forcing, excluding the background committed response. In this way, we can obtain estimates of Greenland's total contribution to sea level rise by 2100.</p>

233. The meltwater feedbacks on ice dynamics, influence of melt amplitude, duration and extent.

Authors: Basile de Fleurian; Petra M. Langebroeke; Richard Davy

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/egusphere-egu21-1099

URL: <https://doi.org/10.5194/egusphere-egu21-1099>

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Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Meltwater|Greenland ice sheet|Glacier morphology|Ice-sheet model|Forcing (mathematics)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: <p>In recent years, temperatures over the Greenland ice sheet have been rising, leading to an increase in surface melt. This increase however can not be reduced to a simple number. Throughout the recent years we have seen some extreme melt seasons with melt extending over the whole surface of the ice sheet (2012) or melt seasons of lower amplitudes but with a longer duration (2010). The effect of those variations on the subglacial system and hence on ice dynamic are poorly understood and are still mainly deduced from studies based on mountain glaciers.</p><p>Here we apply the Ice-sheet and Sea-level System Model (ISSM) to a synthetic glacier with a geometry similar to a Greenland ice sheet land terminating glacier. The forcing is designed such that it allows to investigate different characteristics of the melt season: its length, intensity or the spatial extension of the melt. Subglacial hydrology and ice dynamics are coupled within ISSM is coupled to a subglacial hydrology model, allowing to study the response of the system in terms of subglacial water pressure and the final impact on ice dynamics. Of particular interest is the evolution of the distribution of the efficient and inefficient component of the subglacial drainage system which directly impacts the water pressure evolution at the base of the glacier.</p><p>We note that the initiation of the melt season and the intensity of the melt at this period is a crucial parameter when studying the dynamic response of the glacier to different melt season characteristics. From those results, we can infer a more precise evolution of the dynamics of land terminating glaciers that are heavily driven by their subglacial drainage system. We also highlight which changes in the melt season pattern would be the most damageable for glacier stability in the future.</p>

234. Investigating the drivers of Helheim Glacier's variability from 2007 to 2020

Authors: Gong Cheng; Mathieu Morlighem

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/egusphere-egu21-3405

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Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Greenland ice sheet|Tidewater glacier cycle|Glacier mass balance|Ice-sheet model|Glacier ice accumulation|Glacier terminus

Funding: N/A

License: N/A

Open Access: N/A

Abstract: <p>At least half of today's mass loss of the Greenland ice sheet is due to the retreat of tidewater glaciers. For example, Helheim Glacier, in southwest Greenland, has one of the largest ice discharge records in the Greenland ice sheet during the previous decade. While there is broad agreement that the intrusion of warmer current in the Sermilik Fjord triggers the acceleration and retreat of this marine terminating glacier, other processes such as changes in basal conditions, surface mass balance or calving dynamics may have also played important roles in controlling the retreat of these glaciers. However, our understanding of these processes and their contributions to the retreat and acceleration of the glaciers remains still limited. The individual contributions of these processes have not been quantified, which makes it difficult to determine which of these processes should be included in ice sheet models to correctly capture the present and future retreat and associated mass loss of the ice sheet. Here, we simulate the dynamics of Helheim Glacier from 2007 to 2020 using the Ice-sheet and Sea-level System Model (ISSM) to investigate the model response to changes in external forcings and boundary conditions, such as basal friction, surface mass balance, ice rheology, and ice-ocean interactions at the calving front. The relative importance of each mechanism to the model is quantified within a series of perturbation experiments. We evaluate the ability of the model to match surface speed and terminus position from the observations collected during the simulation period. Preliminary results suggest that Helheim's dynamics is relatively insensitive to the choice of friction law or the surface mass balance, but that the position of the calving front and changes in basal sliding conditions are critical to explain the high variability of Helheim's surface speed. This study, as a result, can be used as a guide for model development of similar glaciers.</p>

235. Automated extraction of calving front locations from multi-spectral satellite imagery using deep learning: methodology and application to Greenland outlet glaciers

Authors: Erik Loebel; Mirko Scheinert; Julia Christmann; Konrad Heidler; Martin Horwath; Angelika Humbert

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/egusphere-egu21-4528

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References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract:

The calving of tidewater glaciers has a strong impact on the stresses of outlet glaciers and their discharge. However, it is still underrepresented in current ice-sheet models incorporating the dynamics of marine-terminating glaciers. This has an impact on simulation results when projecting future sea-level contributions of the Greenland ice sheet. The increasing availability and quality of remote sensing imagery enable us to realize a continuous and precise mapping of relevant parameters such as calving front locations. However, the huge amount of data also accentuates the necessity for intelligent analysis strategies.

In this contribution, we apply an automated workflow to extract calving front positions from multi-spectral Landsat-8 imagery utilizing deep learning. The core of the proposed workflow comprises a convolutional neural network (CNN) for image segmentation exploiting the full range of Landsat-8 multi-spectral capabilities, a statistical textural feature analysis performed on the high-resolution panchromatic band as well as topography model data. The proposed method is evaluated by an independent set of diverse test images as well as by comparing with already available ESA-CCI, MEaSUREs and PROMICE data products. With an estimated prediction error of fewer than two pixels (which equals a spatial resolution of 60 m), automatically extracted calving front locations show very small or even non-distinguishable differences to manually delineated locations. The importance of multi-spectral, textural and topographic features used as input for the CNN is estimated by a permute-and-relearn approach emphasizing their benefit, especially in challenging ice-melange, cloud, and illumination conditions. Jointly with the proposed methodology we present an exceedingly dense dataset for 20 of the most important Greenlandic outlet glaciers for the period from 2013 to 2021.

Eventually, the derived calving front positions are incorporated into the

Ice Sheet and Sea-Level System Model (ISSM). For this, we engage a level set method. This method allows deriving a continuous function in time and space from discrete information at satellite acquisition time steps. As the satellite data is mainly available for fast-flowing outlet glaciers, we use simulated front positions for all remaining ice margins. An alpha-shape method seamlessly links the temporal changing calving fronts to the Greenlandic ice sheet.

236. Potential future recovery of Petermann Glacier in northwest Greenland simulated using ISSM

Authors: Henning Åkesson; Mathieu Morlighem; Martin Jakobsson; Johan Nilsson; Christian Stranne

Journal: Unknown Journal

Publisher: Copernicus GmbH

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Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Iceberg|Future sea level

Funding: N/A

License: N/A

Open Access: N/A

Abstract:

Ice shelves in Antarctica and Greenland are thinning and breaking up, and marine outlet glaciers are retreating, where the ocean is known to play a key role. This pattern is projected to continue over the next decades to centuries due to ocean warming induced by global carbon emissions. Given that we halt or even reverse the current warming trend and fulfil the Paris agreement, it is unclear whether ice shelves and glaciers can recover from the preceding breakup and retreat. Here, we use the numerical ice sheet model ISSM to assess whether Petermann Glacier in northwest Greenland can recover after future ice shelf breakup and grounding line retreat. Petermann's ice tongue is one of the few remaining in Greenland, where several major calving events occurred over the last decade.

Our experiments suggest that if Petermann's ice shelf collapses due to future ocean warming, the ice shelf will not regrow even if that warming is reversed. Neither an ocean warming reversal nor a more positive surface mass balance help the ice shelf to regrow once it has collapsed. Future ocean warming may thus push Petermann into a new dynamic state from where recovery is exceedingly difficult. Finally, we investigate whether reduced calving activity allows for future grounding line readvance and ice shelf recovery. We discuss our findings in light of both potential future recovery and ice shelf collapse and regrowth in the past.

237. Review of "Marine ice-sheet experiments with the Community Ice Sheet Model"

Authors: Rupert Gladstone

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/tc-2020-304-rc2

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Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Ice-sheet model

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

238. The GRISLI-LSCE contribution to the Ice Sheet Model Intercomparison Project for phase 6 of the Coupled Model Intercomparison Project (ISMIP6) - Part 2: Projections of the Antarctic ice sheet evolution by the end of the 21st century

Authors: Aurélien Quiquet; Christophe Dumas

Journal: The Cryosphere

Publisher: Copernicus GmbH

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References: 52

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: . The Antarctic ice sheet's contribution to global sea level rise over the 21st century is of primary societal importance and remains largely uncertain as of yet. In particular, in the recent literature, the contribution of the Antarctic ice sheet by 2100 can be negative (sea level fall) by a few centimetres or positive (sea level rise), with some estimates above 1 m. The Ice Sheet Model Intercomparison Project for the Coupled Model Intercomparison Project - phase 6 (ISMIP6) aimed at reducing the uncertainties in the fate of the ice sheets in the future by gathering various ice sheet models in a common framework. Here, we present the GRISLI-LSCE (Grenoble Ice Sheet and Land Ice model of the Laboratoire des Sciences du Climat et de l'Environnement) contribution to ISMIP6-Antarctica. We show that our model is strongly sensitive to the climate forcing used, with a contribution of the Antarctic ice sheet to global sea level rise by 2100 that ranges from -50 to +150 mm sea level equivalent (SLE). Future oceanic warming leads to a decrease in thickness of the ice shelves, resulting in grounding-line retreat, while increased surface mass balance partially mitigates or even overcompensates the dynamic ice sheet contribution to global sea level rise. Most of the ice sheet changes over the next century are dampened under low-greenhouse-gas-emission scenarios. Uncertainties related to sub-ice-shelf melt rates induce large differences in simulated grounding-line retreat, confirming the importance of this process and its representation in ice sheet models for projections of the Antarctic ice sheet's evolution.

239. Ocean-Driven and Topography-Controlled Nonlinear Glacier Retreat During the Holocene: Southwestern Ross Sea, Antarctica

Authors: R. S. Jones (Science Laboratories Department of Geography Durham University Durham UK, Monash University); G. H. Gudmundsson (Geography and Environmental Sciences Northumbria University Newcastle upon Tyne UK); A. N. Mackintosh (Monash University); F. S. McCormack (Monash University); R. J. Whitmore (Monash University)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020gl091454

URL: <https://doi.org/10.1029/2020gl091454>

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References: 73

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: ice sheet modeling, Antarctica, Holocene

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License: N/A

Open Access: N/A

Abstract: Recent ice sheet mass loss in Antarctica has been attributed to an influx of warm ocean waters, which drove grounding-line retreat and ice thinning. Episodic retreat and rapid thinning also occurred in the southwestern Ross Sea during the Holocene, which today accommodates cold ocean waters. We applied finite element ice-flow modeling to investigate the roles of ocean temperature and bed topography in the deglaciation of this region. First, our experiments demonstrate that bed topography controlled the spatial pattern of grounding-line retreat. Topographic pinning points limited the rate of ice loss until retreat progressed beyond a bathymetric threshold. Second, ocean thermal forcing determined the timing of this ice loss. Enhanced ocean-driven melt is required during the Early-to-Mid Holocene to replicate geological records of deglaciation, possibly indicating that warm ocean waters were once present in this region. On multi-centennial timescales, ocean temperature drove, while bed topography controlled, nonlinear rates of ice mass loss.

240. Insolation vs. meltwater control of productivity and sea surface conditions off SW Greenland during the Holocene

Authors: Estelle Allan (Centre de recherche sur la dynamique du système Terre (Geotop) Université du Québec à Montréal C.P. 8888, succursale centre-Ville Montréal QC H3C 3P8 Canada); Anne de Vernal (Centre de recherche sur la dynamique du système Terre (Geotop) Université du Québec à Montréal C.P. 8888, succursale centre-Ville Montréal QC H3C 3P8 Canada); Marit-Solveig Seidenkrantz (Aarhus University); Jason P. Briner (Department of Geology University at Buffalo 411 Cooke Hall Buffalo NY 14260 USA); Claude Hillaire-Marcel (Centre de recherche sur la dynamique du système Terre (Geotop) Université du Québec à Montréal C.P. 8888, succursale centre-Ville Montréal QC H3C 3P8 Canada); Christof Pearce (Aarhus University); Lorenz Meire (Greenland Institute of Natural Resources, Royal Netherlands Institute for Sea Research, Utrecht University); Hans Røy (Aarhus University); Anders Møller Mathiasen (Aarhus University); Mikkel Thy Nielsen (Aarhus University); Jane Lund Plesner (Aarhus University); Kerstin Perner (Leibniz Institute for Baltic Sea Research)

Journal: Boreas

Publisher: Wiley

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Cited by: 10 publications

Language: en

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Keywords: N/A

Funding: Det Frie Forskningsråd, Canadian Network for Research and Innovation in Machining Technology, Natural Sciences and Engineering Research Council of Canada, Fonds de recherche du Québec – Nature et technologies (and 1 more)

License: N/A

Open Access: N/A

Abstract: We address here the specific timing and amplitude of sea-surface conditions and productivity changes off SW Greenland, northern Labrador Sea, in response to the high deglacial meltwater rates, the Early Holocene maximum insolation and Neoglacial cooling. Dinocyst assemblages from sediment cores collected off Nuuk were used to set up quantitative records of sea ice cover, seasonal sea-surface temperature (SST), salinity (SSS), and primary productivity, with a centennial to millennial scale resolution. Until ~10 ka BP, ice-proximal conditions are suggested by the quasi-exclusive dominance of heterotrophic taxa and low dinocyst concentrations. At about 10 ka BP, an increase in species diversity and abundance of phototrophic taxa marks the onset of interglacial conditions at a regional scale, with summer SST reaching up to 10 degreesC between 8 and 5 ka BP, thus in phase with the Holocene Thermal Maximum as recorded in the southern Greenlandic areas/northern Labrador Sea. During this interval, low SSS but high productivity prevailed in response to high meltwater discharge and nutrient inputs from the Greenland Ice Sheet. After ~5 ka BP, a decrease in phototrophic taxa marks a two-step cooling of surface waters. The first started at ~5 ka BP, and the second at ~3 ka

BP, with a shift toward colder conditions and higher SSS suggesting reduced meltwater discharge during the Neoglacial. This second step coincides with the disappearance of the Saqqaq culture. The gap in human occupation in west Greenland, between the Dorset and the Norse settlements from 2000 to 1000 years BP, might be linked to high amplitude and high frequency variability of ocean and climate conditions.

241. In situ cosmogenic ^{10}Be - ^{14}C - ^{26}Al measurements from recently deglaciated bedrock as a new tool to decipher changes in Greenland Ice Sheet size

Authors: Nicolás E. Young; Alia J. Lesnek; Josh K. Cuzzzone; Jason P. Briner; Jessica A. Badgeley; Alexandra Balter-Kennedy; Brandon L. Graham; Allison Cluett; Jennifer L. Lamp; Roseanne Schwartz; Thibaut Tuna; Edouard Bard; Marc W. Caffee; Susan R. H. Zimmerman; Joerg M. Schaefer

Journal: Climate of the Past

Publisher: Copernicus GmbH

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References: 99

Cited by: 17 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation

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Open Access: N/A

Abstract: . Sometime during the middle to late Holocene (8.2 ka to ~ 1850-1900 CE), the Greenland Ice Sheet (GrIS) was smaller than its current configuration. Determining the exact dimensions of the Holocene ice-sheet minimum and the duration that the ice margin rested inboard of its current position remains challenging. Contemporary retreat of the GrIS from its historical maximum extent in southwestern Greenland is exposing a landscape that holds clues regarding the configuration and timing of past ice-sheet minima. To quantify the duration of the time the GrIS margin was near its modern extent we develop a new technique for Greenland that utilizes in situ cosmogenic ^{10}Be - ^{14}C - ^{26}Al in bedrock samples that have become ice-free only in the last few decades due to the retreating ice-sheet margin at Kangiata Nunaata Sermia (n=12 sites, 36 measurements; KNS), southwest Greenland. To maximize the utility of this approach, we refine the deglaciation history of the region with stand-alone ^{10}Be measurements (n=49) and traditional ^{14}C ages from sedimentary deposits contained in proglacial-threshold lakes. We combine our reconstructed ice-margin history in the KNS region with additional geologic records from southwestern Greenland and recent model simulations of GrIS change to constrain the timing of the GrIS minimum in southwest Greenland and the magnitude of Holocene inland GrIS retreat, as well as to explore the regional climate history influencing Holocene ice-sheet behavior. Our ^{10}Be - ^{14}C - ^{26}Al measurements reveal that (1) KNS retreated behind its modern margin just before 10 ka, but it likely stabilized near the present GrIS margin for several thousand years before retreating farther inland, and (2) pre-Holocene ^{10}Be detected in several of our sample sites is most easily explained by several thousand years of surface exposure during the last interglaciation. Moreover, our new results indicate that the minimum extent of the GrIS likely occurred after ~5 ka, and the GrIS margin may have approached its eventual historical maximum extent as early as ~2 ka. Recent simulations of GrIS change are able to match the geologic record of ice-sheet change in regions

dominated by surface mass balance, but they produce a poorer model-data fit in areas influenced by oceanic and dynamic processes. Simulations that achieve the best model-data fit suggest that inland retreat of the ice margin driven by early to middle Holocene warmth may have been mitigated by increased precipitation. Triple ^{10}Be - ^{14}C - ^{26}Al measurements in recently deglaciated bedrock provide a new tool to help decipher the duration of smaller-than-present ice cover multiple timescales. Modern retreat of the GrIS margin in southwest Greenland is revealing a bedrock landscape that was also exposed during the migration of the GrIS margin towards its Holocene minimum extent, but it has yet to tap into a landscape that remained ice-covered throughout the entire Holocene.

242. Hydrological and Kinematic Precursors of the 2017 Calving Event at the Petermann Glacier in Greenland Observed from Multi-Source Remote Sensing Data

Authors: Daan Li (Chinese Academy of Sciences); Liming Jiang (Chinese Academy of Sciences); Ronggang Huang (Chinese Academy of Sciences)

Journal: Remote Sensing

Publisher: MDPI AG

DOI: 10.3390/rs13040591

URL: <https://doi.org/10.3390/rs13040591>

Publication Date: 2021-02-10

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Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

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Abstract: Both a decrease of sea ice and an increase of surface meltwater, which may induce ice-flow speedup and frontal collapse, have a significant impact on the stability of the floating ice shelf in Greenland. However, detailed dynamic precursors and drivers prior to a fast-calving process remain unclear due to sparse remote sensing observations. Here, we present a comprehensive investigation on hydrological and kinematic precursors before the calving event on 26 July 2017 of Petermann Glacier in northern Greenland, by jointly using remote sensing observations at high-temporal resolution and an ice-flow model. Time series of ice-flow velocity fields during July 2017 were retrieved with Sentinel-2 observations with a sub-weekly sampling interval. The ice-flow speed quickly reached 30 m/d on 26 July (the day before the calving), which is roughly 10 times quicker than the mean glacier velocity. Additionally, a significant decrease in the radar backscatter coefficient of Sentinel-1 images suggests a rapid transformation from landfast sea ice into open water, associated with a decrease in sea ice extent. Additionally, the area of melt ponds on the floating ice tongue began to increase in mid-May, quickly reached a peak at the end of June and lasted for nearly one month until the calving occurred. We used the ice sheet system model to model the spatial-temporal damage and stress on the floating ice, thereby finding an abnormal stress distribution in a cracked region. It is inferred that this calving event may relate to a weakening of the sea ice, shearing of the tributary glacier, and meltwater infiltrating crevasses.

243. Ice dynamics will remain a primary driver of Greenland ice sheet mass loss over the next century

Authors: Youngmin Choi; Mathieu Morlighem; Eric Rignot; Michael Wood

Journal: Communications Earth & Environment

Publisher: Springer Science and Business Media LLC

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Funding: National Science Foundation, National Aeronautics and Space Administration, NASA | Jet Propulsion Laboratory

License: CC BY 4.0

Open Access: N/A

Abstract: The mass loss of the Greenland Ice Sheet is nearly equally partitioned between a decrease in surface mass balance from enhanced surface melt and an increase in ice dynamics from the acceleration and retreat of its marine-terminating glaciers. Much uncertainty remains in the future mass loss of the Greenland Ice Sheet due to the challenges of capturing the ice dynamic response to climate change in numerical models. Here, we estimate the sea level contribution of the Greenland Ice Sheet over the 21st century using an ice-sheet wide, high-resolution, ice-ocean numerical model that includes surface mass balance forcing, thermal forcing from the ocean, and iceberg calving dynamics. The model is calibrated with ice front observations from the past eleven years to capture the recent evolution of marine-terminating glaciers. Under a business as usual scenario, we find that northwest and central west Greenland glaciers will contribute more mass loss than other regions due to ice front retreat and ice flow acceleration. By the end of century, ice discharge from marine-terminating glaciers will contribute 50 +/- 20% of the total mass loss, or twice as much as previously estimated although the contribution from the surface mass balance increases towards the end of the century.

244. Exploring the impact of atmospheric forcing and basal drag on the Antarctic Ice Sheet under Last Glacial Maximum conditions

Authors: Javier Blasco; Jorge Alvarez-Solas; Alexander Robinson; Marisa Montoya

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-15-215-2021

URL: <https://doi.org/10.5194/tc-15-215-2021>

Publication Date: 2021-01-18

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References: 103

Cited by: 6 publications

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Keywords: N/A

Funding: European Commission

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Open Access: N/A

Abstract: . Little is known about the distribution of ice in the Antarctic Ice Sheet (AIS) during the Last Glacial Maximum (LGM). Whereas marine and terrestrial geological data indicate that the grounded ice advanced to a position close to the continental-shelf break, the total ice volume is unclear. Glacial boundary conditions are potentially important sources of uncertainty, in particular basal friction and climatic boundary conditions. Basal friction exerts a strong control on the large-scale dynamics of the ice sheet and thus affects its size and is not well constrained. Glacial climatic boundary conditions determine the net accumulation and ice temperature and are also poorly known. Here we explore the effect of the uncertainty in both features on the total simulated ice storage of the AIS at the LGM. For this purpose we use a hybrid ice sheet shelf model that is forced with different basal drag choices and glacial background climatic conditions obtained from the LGM ensemble climate simulations of the third phase of the Paleoclimate Modelling Intercomparison Project (PMIP3). Overall, we find that the spread in the simulated ice volume for the tested basal drag parameterizations is about the same range as for the different general circulation model (GCM) forcings (4 to 6 m sea level equivalent). For a wide range of plausible basal friction configurations, the simulated ice dynamics vary widely but all simulations produce fully extended ice sheets towards the continental-shelf break. More dynamically active ice sheets correspond to lower ice volumes, while they remain consistent with the available constraints on ice extent. Thus, this work points to the possibility of an AIS with very active ice streams during the LGM. In addition, we find that the surface boundary temperature field plays a crucial role in determining the ice extent through its effect on viscosity. For ice sheets of a similar extent and comparable dynamics, we find that the precipitation field determines the total AIS volume. However, precipitation is highly uncertain. Climatic fields simulated by climate models show more precipitation in coastal regions than a spatially uniform anomaly, which can lead to larger ice volumes. Our results strongly support using these paleoclimatic fields to simulate and study the LGM and potentially other time periods like the last interglacial. However, their accuracy must be assessed as well, as differences between climate model forcing lead to a large spread in the simulated ice volume and extension.

245. Constraining subglacial processes from surface velocity observations using surrogate-based Bayesian inference

Authors: Douglas Brinkerhoff; Andy Aschwanden; Mark Fahnestock

Journal: Journal of Glaciology

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/jog.2020.112

URL: <https://doi.org/10.1017/jog.2020.112>

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References: 77

Cited by: 29 publications

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Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: Basal motion is the primary mechanism for ice flux in Greenland, yet a widely applicable model for predicting it remains elusive. This is due to the difficulty in both observing small-scale bed properties and predicting a time-varying water pressure on which basal motion putatively depends. We take a Bayesian approach to these problems by coupling models of ice dynamics and subglacial hydrology and conditioning on observations of surface velocity in southwestern Greenland to infer the posterior probability distributions for eight spatially and temporally constant parameters governing the behavior of both the sliding law and hydrologic model. Because the model is computationally expensive, characterization of these distributions using classical Markov Chain Monte Carlo sampling is intractable. We skirt this issue by training a neural network as a surrogate that approximates the model at a sliver of the computational cost. We find that surface velocity observations establish strong constraints on model parameters relative to a prior distribution and also elucidate correlations, while the model explains 60% of observed variance. However, we also find that several distinct configurations of the hydrologic system and stress regime are consistent with observations, underscoring the need for continued data collection and model development.

246. Ocean forcing drives glacier retreat in Greenland

Authors: Michael Wood (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Eric Rignot (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ian Fenty (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Lu An (University of California Irvine); Anders Bjørk (University of Copenhagen); Michiel van den Broeke (Utrecht University); Cilan Cai (University of California Irvine); Emily Kane (University of California Irvine); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Romain Millan (University of Grenoble Alpes, Institut de Recherche pour le Développement (IRD)); Mathieu Morlighem (University of California Irvine); Jeremie Mouginot (University of Grenoble Alpes, University of California Irvine, Institut de Recherche pour le Développement (IRD)); Brice Noël (Utrecht University); Bernd Scheuchl (University of California Irvine); Isabella Velicogna (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Josh K. Willis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Hong Zhang (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Science Advances

Publisher: American Association for the Advancement of Science (AAAS)

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URL: <https://doi.org/10.1126/sciadv.aba7282>

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Language: en

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Keywords: N/A

Funding: National Science Foundation, Jet Propulsion Laboratory, Netherlands Earth System Science Centre (and 6 more)

License: CC BY 4.0

Open Access: N/A

Abstract: Many Greenland glaciers have been attacked by the ocean from below and, in turn, contributed to sea-level rise from the ice sheet.

247. The evolution of ice fabrics: A continuum modelling approach validated against laboratory experiments

Authors: Daniel H.M. Richards; Samuel S. Pegler; Sandra Piazzolo; Oliver G. Harlen

Journal: Earth and Planetary Science Letters

Publisher: Elsevier BV

DOI: 10.1016/j.epsl.2020.116718

URL: <https://doi.org/10.1016/j.epsl.2020.116718>

Publication Date: 2020-12-29

Volume/Issue: Volume 556

References: 50

Cited by: 16 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Engineering and Physical Sciences Research Council

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Open Access: N/A

Abstract: N/A

248. Past, Present, and Future Pacific Sea-Level Change

Authors: B. D. Hamlington (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); T. Frederikse (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); P. R. Thompson (University of Hawaii at Manoa); J. K. Willis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); R. S. Nerem (Cooperative Institute for Research in Environmental Sciences University of Colorado at Boulder Boulder CO USA); J. T. Fasullo (Cooperative Institute for Research in Environmental Sciences University of Colorado at Boulder Boulder CO USA)

Journal: Earth's Future

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020ef001839

URL: <https://doi.org/10.1029/2020ef001839>

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Volume/Issue: Volume 9, Issue 4

References: 45

Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: trends, sea-level, satellite altimetry

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: Satellite altimeters have provided near-global coverage of the ocean with a continuous record now approaching 3 decades. These observations have led to definitive evidence of an increase in global mean sea-level, while offering a depiction of the spatial variability in sea-level change. As a result of the increasing length of the altimeter record, studies have sought to understand and detect the emergence of a forced or anthropogenic response in sea-level. The central idea behind these studies is that the altimeter record is now long enough that the influence of interannual to decadal variability is reduced and does not obscure underlying regional trends in the way that it did for the early part of the altimeter record. Two locations--the western tropical Pacific and the northeastern Pacific off the west coast of the United States--are noteworthy for their sea-level variations during the altimeter era, exhibiting large departures from the global average albeit in opposite directions. In this study, we examine satellite altimeter-measured sea-level in these two regions. The goals are to understand the evolution of the sea-level trend, and determine what can be inferred about future sea-level change from the satellite altimeter record. We find that in the two locations considered here the altimetry record is still too short to rule out natural variability contributions to the trend on the order of 1 mm/yr. Using longer records from tide-gauge observations and models, an assessment of the potential future changes in both regions is made for both regions.

249. Complex Basal Conditions and Their Influence on Ice Flow at the Onset of the Northeast Greenland Ice Stream

Authors: Steven Franke (Alfred Wegener Institute); Daniela Jansen (Alfred Wegener Institute); Sebastian Beyer (University of Bremen, Alfred Wegener Institute); Niklas Neckel (Alfred Wegener Institute); Tobias Binder (Alfred Wegener Institute); John Paden (University of Kansas); Olaf Eisen (University of Bremen, Alfred Wegener Institute)

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020jf005689

URL: <https://doi.org/10.1029/2020jf005689>

Publication Date: 2020-12-20

Volume/Issue: Volume 126, Issue 3

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Cited by: 19 publications

Language: en

Subjects: N/A

Keywords: Northeast Greenland Ice Stream, bed conditions, basal roughness, radio-echo sounding, Greenland Ice Sheet (and 1 more)

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: The ice stream geometry and large ice surface velocities at the onset region of the Northeast Greenland Ice Stream (NEGIS) are not yet well reproduced by ice sheet models. The quantification of basal sliding and a parametrization of basal conditions remains a major gap. In this study, we assess the basal conditions of the onset region of the NEGIS in a systematic analysis of airborne ultra-wideband radar data. We evaluate basal roughness and basal return echoes in the context of the current ice stream geometry and ice surface velocity. We observe a change from a smooth to a rougher bed where the ice stream widens, and a distinct roughness anisotropy, indicating a preferred orientation of subglacial structures. In the upstream region, the excess ice mass flux through the shear margins is evacuated by ice flow acceleration and along-flow stretching of the ice. At the downstream part, the generally rougher bed topography correlates with a decrease in flow acceleration and lateral variations in ice surface velocity. Together with basal water routing pathways, this hints to two different zones in this part of the NEGIS: the upstream region collecting water, with a reduced basal traction, and downstream, where the ice stream is slowing down and is widening on a rougher bed, with a distribution of basal water toward the shear margins. Our findings support the hypothesis that the NEGIS is strongly interconnected to the subglacial water system in its onset region, but also to the subglacial substrate and morphology.

250. GHub: Building a glaciology gateway to unify a community

Authors: Jeanette M. Sperhac (Center for Computational Research University at Buffalo Buffalo New York USA); Kristin Poinar (Department of Geology University at Buffalo Buffalo New York USA); Renette Jones-Ivey (Institute for Computational and Data Sciences University at Buffalo Buffalo New York USA); Jason Briner (Department of Geology University at Buffalo Buffalo New York USA); Beata Csatho (Department of Geology University at Buffalo Buffalo New York USA); Sophie Nowicki (Department of Geology University at Buffalo Buffalo New York USA); Erika Simon (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Eric Larour (Sea Level and Ice Group NASA Jet Propulsion Lab Pasadena California USA); Justin Quinn (Sea Level and Ice Group NASA Jet Propulsion Lab Pasadena California USA); Abani Patra (Data Intensive Studies Center Tufts University Boston Massachusetts USA)

Journal: Concurrency and Computation: Practice and Experience

Publisher: Wiley

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URL: <https://doi.org/10.1002/cpe.6130>

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References: 74

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation

License: N/A

Open Access: N/A

Abstract: There is no consensus on how quickly the earth's ice sheets are melting due to global warming, nor on the ramifications to sea level rise. Due to its potential effects on coastal populations and global economies, sea level rise is a grave concern, making ice melt rates an important area of study. The ice-sheet science community consists of two groups that perform related but distinct kinds of research: a data community, and a model building community. The data community characterizes past and current states of the ice sheets by assembling data from field and satellite observations. The modeling community forecasts the rate of ice-sheet decline with computational models validated against observations. Although observational data and models depend on one another, these two groups are not well integrated. Better coordination between data collection efforts and modeling efforts is imperative if we are to improve our understanding of ice sheet loss rates. We present a new science gateway, GHub, a collaboration space for ice sheet scientists. This web-accessible gateway will host datasets and modeling workflows, and provide access to codes that enable tool building by the ice sheet science community. Using GHub, we will collect and centralize existing datasets, creating data products that more completely catalog the ice sheets of Greenland and Antarctica. We will build workflows for model validation and uncertainty quantification, extending existing ice sheet models. Finally, we will host existing community codes, enabling scientists to build new tools utilizing them. With this new cyberinfrastructure, ice sheet scientists will gain integrated tools to quantify the rate and extent of sea level rise, benefitting human societies around the globe.

251. Drygalski Ice Tongue stability influenced by rift formation and ice morphology

Authors: Christine Indrigo; Christine F. Dow; Jamin S. Greenbaum; Mathieu Morlighem

Journal: Journal of Glaciology

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/jog.2020.99

URL: <https://doi.org/10.1017/jog.2020.99>

Publication Date: 2020-12-07

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References: 65

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: The Drygalski Ice Tongue in East Antarctica stretches 90 km into the Ross Sea and influences the local ocean circulation, and persistence of the Terra Nova Bay Polynya. We examine the controls on the size of this floating ice body by comparing the propagation of six large fractures on the ice tongue's northern side using 21 years of Landsat imagery with hydrostatic ice thickness maps and strain rate calculations. We also apply a subglacial hydrology model to estimate the location and discharge from subglacial channels over the grounding line and compare these with basal channels identified along the ice tongue using remote sensing and airborne radar data. Our results suggest that large fractures are inhibited from full-width propagation by thicker ice between basal channels. We hypothesize that only once the ice tongue thins towards the terminus, can fractures propagate and cause large calving events. This suggests an important relationship between the melting of floating ice from subglacial and ocean sources and the expansion of fractures that lead to ice tongue calving.

252. Review of Mechanics and dynamics of pinning points on the Shirase Coast, West Antarctica

Authors: No authors listed

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/tc-2020-298-rc2

URL: <https://doi.org/10.5194/tc-2020-298-rc2>

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Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Dynamics|West coast

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License: N/A

Open Access: N/A

Abstract: N/A

253. Crustal Strain Patterns Associated With Normal, Drought, and Heavy Precipitation Years in California

Authors: Jeonghyeop Kim (Department of Geosciences Stony Brook University Stony Brook NY USA); Alireza Bahadori (Department of Geosciences Stony Brook University Stony Brook NY USA); William E. Holt (Department of Geosciences Stony Brook University Stony Brook NY USA)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020jb019560

URL: <https://doi.org/10.1029/2020jb019560>

Publication Date: 2020-12-02

Volume/Issue: Volume 126, Issue 1

References: 73

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration, Southern California Earthquake Center

License: CC BY 4.0

Open Access: N/A

Abstract: We invert continuously operating Global Positioning System (cGPS) data obtained between 2007 and 2019 to quantify non steady-state horizontal strain anomalies in California. Our long-wavelength transient strain model shows seasonal and multiannual variations in horizontal strain anomalies within the plate boundary zone. During the summer, in general, a zone of extensional dilatation develops along the San Andreas Fault zone and Sierra Nevada, whereas contractional dilatation develops along the Eastern California Shear Zone (ECSZ) north of 36.5 degreesN. The patterns of dilatational strain are opposite during the winter. We find that these seasonal strain anomaly patterns vary in magnitude, depending on precipitation intensity in California. Investigating hydrologic loading models and their horizontal elastic responses reveal that water mass loads on the surface from the precipitation in California are the major sources of the observed long-wavelength horizontal transient strains. We show, however, that a heavy damping in the inversion of the cGPS data is required for the long-wavelength horizontal strain solutions to best match with the expected elastic response from hydrologic loading. Appropriate fitting of the horizontal cGPS yields amplified horizontal strain signals in the Sierra Nevada, along regions adjacent to the San Andreas Fault, and within the ECSZ. The larger-than-expected amplitudes may be associated with poroelastic responses or thermoelastic changes that are superimposed on the hydrologic response. We demonstrate that there is a persistent sharp boundary of horizontal dilatational strain domains at the transition between the High Sierra and Basin and Range Province, caused by the sharp gradient in hydrologic loading there.

254. Time-Varying Ice Sheet Mask: Implications on Ice-Sheet Mass Balance and Crustal Uplift

Authors: Kristian K. Kjeldsen (Geological Survey of Denmark and Greenland); Shfaqat A. Khan (Technical University of Denmark); William T. Colgan (Geological Survey of Denmark and Greenland); Joseph A. MacGregor (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Robert S. Fausto (Geological Survey of Denmark and Greenland)

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020jf005775

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Publication Date: 2020-11-20

Volume/Issue: Volume 125, Issue 12

References: 63

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: mass balance, time-varying, elastic uplift, ice mask, Greenland Ice Sheet

Funding: N/A

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Open Access: N/A

Abstract: Mass-balance estimates of the polar ice sheets presently rely on static ice masks to identify the area under investigation. However, the vintage of available ice masks can differ by more than a decade, and the resolution of the imagery used to delineate ice extent can differ by an order of magnitude, despite evidence of ongoing ice-sheet retreat. We show that using a fine-resolution, time-varying ice mask significantly improves mass-balance estimates of the northwestern sector of the Greenland Ice Sheet and affects interpretation of local crustal uplift. Accounting for ice margin retreat using a time-varying ice mask between 2003 and 2015 results in 1.0%-6.8% less ice-sheet mass loss relative to assuming one of several existing static ice masks. This indicates that mass-balance estimates that use a static ice mask will progressively overestimate mass loss where the ice sheet is predominantly retreating, meaning a negative bias on mass-balance. This trend is anticipated to be amplified in a future warming scenario unless updated ice masks are incorporated. For altimetry-derived mass-balance estimates, this ice-mask-induced bias approaches the magnitude of the ice-sheet-wide uncertainty presently associated with spatiotemporal variability in modeled firn compaction. Further, at five Greenland GPS Network (GNET) stations, the ice-mask-induced bias is equivalent to 5%-15% of the modeled uplift due to elastic deformation. Given present and projected negative trends in Greenland Ice Sheet mass-balance and areal extent, we conclude that studies of its mass-balance should begin incorporating dynamic, periodically updated, fine-resolution ice masks. This methodological refinement will reduce bias in estimates of both ice-sheet mass-balance and crustal uplift.

255. Possible impacts of a 1000 km long hypothetical subglacial river valley towards Petermann Glacier in northern Greenland

Authors: Christopher Chambers; Ralf Greve; Bas Altena; Pierre-Marie Lefeuve

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-14-3747-2020

URL: <https://doi.org/10.5194/tc-14-3747-2020>

Publication Date: 2020-11-12

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References: 35

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: . Greenland basal topographic data show a segmented valley extending from Petermann Fjord into the centre of Greenland; however, the locations of radar scan lines, used to create the bedrock topography data, indicate that valley segmentation is due to data interpolation. Therefore, as a thought experiment, simulations where the valley is opened are used to investigate its effects on basal water movement and distribution. The simulations indicate that the opening of this valley can result in an uninterrupted water pathway from the interior to Petermann Fjord. Along its length, the path of the valley progresses gradually down an ice surface slope, causing a lowering of ice overburden pressure that could enable water flow along its path. The fact that the valley base appears to be relatively flat and follows a path near the interior ice divide that roughly intersects the east and west basal hydrological basins is presented as evidence that its present day form may have developed in conjunction with an overlying ice sheet. Experiments where basal melting is increased solely within the deep interior near the known large area of basal melting result in an increase in the flux of water northwards along the entire valley. The results are consistent with a long subglacial river; however, considerable uncertainty remains over aspects such as whether adequate water is available at the bed, whether water escapes from the valley or is refrozen, and over what form a hydrological conduit could take along the valley base.

256. Sensitivity of ice loss to uncertainty in flow law parameters in an idealized one-dimensional geometry

Authors: Maria Zeitz; Anders Levermann; Ricarda Winkelmann

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-14-3537-2020

URL: <https://doi.org/10.5194/tc-14-3537-2020>

Publication Date: 2020-10-27

Volume/Issue: Volume 14, Issue 10

References: 60

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: . Acceleration of the flow of ice drives mass losses in both the Antarctic and the Greenland Ice Sheet. The projections of possible future sea-level rise rely on numerical ice-sheet models, which solve the physics of ice flow, melt, and calving. While major advancements have been made by the ice-sheet modeling community in addressing several of the related uncertainties, the flow law, which is at the center of most process-based ice-sheet models, is not in the focus of the current scientific debate. However, recent studies show that the flow law parameters are highly uncertain and might be different from the widely accepted standard values. Here, we use an idealized flow-line setup to investigate how these uncertainties in the flow law translate into uncertainties in flow-driven mass loss. In order to disentangle the effect of future warming on the ice flow from other effects, we perform a suite of experiments with the Parallel Ice Sheet Model (PISM), deliberately excluding changes in the surface mass balance. We find that changes in the flow parameters within the observed range can lead up to a doubling of the flow-driven mass loss within the first centuries of warming, compared to standard parameters. The spread of ice loss due to the uncertainty in flow parameters is on the same order of magnitude as the increase in mass loss due to surface warming. While this study focuses on an idealized flow-line geometry, it is likely that this uncertainty carries over to realistic three-dimensional simulations of Greenland and Antarctica.

257. ALPS: A Unified Framework for Modeling Time Series of Land Ice Changes

Authors: Prashant Shekhar; Beata Csatho; Tony Schenk; Carolyn Roberts; Abani K. Patra

Journal: IEEE Transactions on Geoscience and Remote Sensing

Publisher: Institute of Electrical and Electronics Engineers (IEEE)

DOI: 10.1109/tgrs.2020.3027190

URL: <https://doi.org/10.1109/tgrs.2020.3027190>

Publication Date: 2020-10-16

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References: 63

Cited by: 5 publications

Language: N/A

Subjects: N/A

Keywords: time series (TS), penalized approximations, robust modeling, Land ice changes

Funding: NASA's Operation IceBridge, Sea Level Change Science Team

License: N/A

Open Access: N/A

Abstract: N/A

258. ISSM-SLPS: geodetically compliant Sea-Level Projection System for the Ice-sheet and Sea-level System Model v4.17

Authors: Eric Larour; Lambert Caron; Mathieu Morlighem; Surendra Adhikari; Thomas Frederikse; Nicole-Jeanne Schlegel; Erik Ivins; Benjamin Hamlington; Robert Kopp; Sophie Nowicki

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-13-4925-2020

URL: <https://doi.org/10.5194/gmd-13-4925-2020>

Publication Date: 2020-10-15

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References: 78

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: . Understanding future impacts of sea-level rise at the local level is important for mitigating its effects. In particular, quantifying the range of sea-level rise outcomes in a probabilistic way enables coastal planners to better adapt strategies, depending on cost, timing and risk tolerance. For a time horizon of 100 years, frameworks have been developed that provide such projections by relying on sea-level fingerprints where contributions from different processes are sampled at each individual time step and summed up to create probability distributions of sea-level rise for each desired location. While advantageous, this method does not readily allow for including new physics developed in forward models of each component. For example, couplings and feedbacks between ice sheets, ocean circulation and solid-Earth uplift cannot easily be represented in such frameworks. Indeed, the main impediment to inclusion of more forward model physics in probabilistic sea-level frameworks is the availability of dynamically computed sea-level fingerprints that can be directly linked to local mass changes. Here, we demonstrate such an approach within the Ice-sheet and Sea-level System Model (ISSM), where we develop a probabilistic framework that can readily be coupled to forward process models such as those for ice sheets, glacial isostatic adjustment, hydrology and ocean circulation, among others. Through large-scale uncertainty quantification, we demonstrate how this approach enables inclusion of incremental improvements in all forward models and provides fidelity to time-correlated processes. The projection system may readily process input and output quantities that are geodetically consistent with space and terrestrial measurement systems. The approach can also account for numerous improvements in our understanding of sea-level processes.

259. Intercomparison of surface meltwater routing models for the Greenland ice sheet and influence on subglacial effective pressures

Authors: Kang Yang; Aleah Sommers; Lauren C. Andrews; Laurence C. Smith; Xin Lu; Xavier Fettweis; Manchun Li

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-14-3349-2020

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License: CC BY 4.0

Open Access: N/A

Abstract: . Each summer, large volumes of surface meltwater drain off the Greenland ice sheet (GrIS) surface through moulins to the bed, impacting subglacial hydrology and ice flow dynamics. Supraglacial surface routing delays may propagate to englacial and subglacial hydrologic systems, requiring accurate assessment to correctly estimate subglacial effective pressures. We compare hourly supraglacial moulin discharge simulations from three surface meltwater routing models - the synthetic unit hydrograph (SUH), the bare-ice component of surface routing and lake filling (SRLF), and the rescaled width function (RWF) - for four internally drained catchments on the southwestern Greenland ice sheet surface. The routing models are forced identically using surface runoff from the Modèle Atmosphérique Régionale regional climate model (RCM). For each catchment, simulated moulin hydrographs are input to the SHAKTI subglacial hydrologic model to simulate diurnally varying subglacial effective-pressure variations in the vicinity of a single moulin. Overall, all three routing models produce more realistic moulin discharges than simply using RCM runoff outputs without surface routing but produce significant differences in peak moulin discharge and time to peak. In particular, the RWF yields later, smaller peak moulin discharges than the SUH or SRLF due to its representation of slow interfluvial flow between supraglacial meltwater channels, and it can readily accommodate the seasonal evolution of supraglacial stream and river networks. Differences among the three routing models are reflected in a series of simple idealized subglacial hydrology simulations that yield different diurnal effective-pressure amplitudes; however, the supraglacial hydrologic system acts as short-term storage for surface meltwater, and the temporal mean effective pressure is relatively consistent across routing models.

260. Sensitivity of glacier elevation analysis and numerical modeling to CryoSat-2 SIRAL retracking techniques

Authors: Thomas Trantow; Ute C. Herzfeld; Veit Helm; Johan Nilsson

Journal: Computers & Geosciences

Publisher: Elsevier BV

DOI: 10.1016/j.cageo.2020.104610

URL: <https://doi.org/10.1016/j.cageo.2020.104610>

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Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, National Aeronautics and Space Administration

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Abstract: N/A

261. Sensitivity of Greenland ice sheet projections to spatial resolution in higher-order simulations: the Alfred Wegener Institute (AWI) contribution to ISMIP6 Greenland using the Ice-sheet and Sea-level System Model (ISSM)

Authors: Martin Rückamp; Heiko Goelzer; Angelika Humbert

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-14-3309-2020

URL: <https://doi.org/10.5194/tc-14-3309-2020>

Publication Date: 2020-10-02

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References: 75

Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: . Projections of the contribution of the Greenland ice sheet to future sea-level rise include uncertainties primarily due to the imposed climate forcing and the initial state of the ice sheet model. Several state-of-the-art ice flow models are currently being employed on various grid resolutions to estimate future mass changes in the framework of the Ice Sheet Model Intercomparison Project for CMIP6 (ISMIP6). Here we investigate the sensitivity to grid resolution of centennial sea-level contributions from the Greenland ice sheet and study the mechanism at play. We employ the finite-element higher-order Ice-sheet and Sea-level System Model (ISSM) and conduct experiments with four different horizontal resolutions, namely 4, 2, 1 and 0.75 km. We run the simulation based on the ISMIP6 core climate forcing from the MIROC5 global circulation model (GCM) under the high-emission Representative Concentration Pathway (RCP) 8.5 scenario and consider both atmospheric and oceanic forcing in full and separate scenarios. Under the full scenarios, finer simulations unveil up to approximately 5 % more sea-level rise compared to the coarser resolution. The sensitivity depends on the magnitude of outlet glacier retreat, which is implemented as a series of retreat masks following the ISMIP6 protocol. Without imposed retreat under atmosphere-only forcing, the resolution dependency exhibits an opposite behaviour with approximately 5 % more sea-level contribution in the coarser resolution. The sea-level contribution indicates a converging behaviour below a 1 km horizontal resolution. A driving mechanism for differences is the ability to resolve the bedrock topography, which highly controls ice discharge to the ocean. Additionally, thinning and acceleration emerge to propagate further inland in high resolution for many glaciers. A major response mechanism is sliding, with an enhanced feedback on the effective normal pressure at higher resolution leading to a larger increase in sliding speeds under scenarios with outlet glacier retreat.

262. Rate of mass loss from the Greenland Ice Sheet will exceed Holocene values this century

Authors: Jason P. Briner; Joshua K. Cuzzone; Jessica A. Badgeley; Nicolás E. Young; Eric J. Steig; Mathieu Morlighem; Nicole-Jeanne Schlegel; Gregory J. Hakim; Joerg M. Schaefer; Jesse V. Johnson; Alia J. Lesnek; Elizabeth K. Thomas; Estelle Allan; Ole Bennike; Allison A. Cluett; Beata Csatho; Anne de Vernal; Jacob Downs; Eric Larour; Sophie Nowicki

Journal: Nature

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41586-020-2742-6

URL: <https://doi.org/10.1038/s41586-020-2742-6>

Publication Date: 2020-09-30

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Cited by: 72 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

263. Cosmogenic isotope measurements from recently deglaciated bedrock as a new tool to decipher changes in Greenland Ice Sheet size

Authors: Nicolás E. Young; Alia J. Lesnek; Josh K. Cuzzone; Jason P. Briner; Jessica A. Badgeley; Alexandra Balter-Kennedy; Brandon L. Graham; Allison Cluett; Jennifer L. Lamp; Roseanne Schwartz; Thibaut Tuna; Edouard Bard; Marc W. Caffee; Susan R. H. Zimmerman; Joerg M. Schaefer

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/cp-2020-111

URL: <https://doi.org/10.5194/cp-2020-111>

Publication Date: 2020-09-30

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Greenland ice sheet|Deglaciation|Bedrock|Ice-sheet model|Last Glacial Maximum|Ice core

Funding: National Science Foundation

License: CC BY 4.0

Open Access: N/A

Abstract: . During the middle to late Holocene (8.2 ka BP to present), the Greenland Ice Sheet (GrIS) was smaller than its current configuration. Determining the exact dimensions of the Holocene ice-sheet minimum and the duration that the ice margin rested inboard of its current position remains challenging. Contemporary retreat of the GrIS from its historical maximum extent in southwestern Greenland is exposing a landscape that holds clues regarding the configuration and timing of past ice-sheet minima. To quantify the duration of the time the GrIS margin was near its modern extent we develop a new technique on Greenland that utilizes in situ cosmogenic ^{10}Be - ^{14}C - ^{26}Al in bedrock samples that have become ice free only in the last few decades by the retreating ice-sheet margin at Kangiata Nunaata Sermia ($n = 12$ sites; KNS), southwest Greenland. To maximize the utility of this approach, we refine the deglaciation history of the region with stand-alone ^{10}Be measurements ($n = 49$) and traditional ^{14}C ages from sedimentary deposits contained in proglacial-threshold lakes. We combine our reconstructed ice-margin history in the KNS region with additional geologic records from southwestern Greenland and recent model simulations of GrIS change, to constrain the timing of the GrIS minimum in southwest Greenland, the magnitude of Holocene inland GrIS retreat, and explore the regional climate history influencing Holocene ice-sheet behavior. Our ^{10}Be - ^{14}C - ^{26}Al measurements reveal that 1) KNS retreated behind its modern margin just before 10 ka, but likely stabilized near the present GrIS margin for several thousand years before retreating farther inland, and 2) pre-Holocene ^{10}Be detected in several of our sample sites is most easily explained by several thousand years of surface exposure during the Last Interglaciation. Moreover, our new results indicate that the minimum extent of the GrIS likely occurred after ~ 5 ka, and the GrIS margin may have approached its eventual historical maximum extent as early as ~ 2 ka. Recent simulations of GrIS change are able to match the geologic record of ice-sheet change in regions dominated by surface mass balance, but produce a poorer model-data fit in areas influenced by oceanic and dynamic processes. Simulations that achieve the best model-data fit suggest that inland retreat of the ice margin driven by early to middle Holocene warmth may have been

mitigated by increased precipitation. Triple ^{10}Be - ^{14}C - ^{26}Al measurements in recently deglaciated bedrock provide a new tool to help decipher the duration of smaller-than-present ice over multiple timescales. Modern retreat of the GrIS margin in southwest Greenland is revealing a bedrock landscape that was also exposed during the migration of the GrIS margin towards its Holocene minimum extent, but has yet to tap into a landscape that remained ice covered throughout the entire Holocene.

264. Vertical Land Motion From Present-Day Deglaciation in the Wider Arctic

Authors: Carsten Ankjær Ludwigsen (Technical University of Denmark); Shfaqat Abbas Khan (Technical University of Denmark); Ole Baltazar Andersen (Technical University of Denmark); Ben Marzeion (University of Bremen)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020gl088144

URL: <https://doi.org/10.1029/2020gl088144>

Publication Date: 2020-09-29

Volume/Issue: Volume 47, Issue 19

References: 76

Cited by: 15 publications

Language: en

Subjects: N/A

Keywords: vertical land motion, sea level, Arctic, Greenland, coastal sea level (and 1 more)

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: Vertical land motion (VLM) from past and ongoing glacial changes can amplify or mitigate ongoing relative sea level change. We present a high-resolution VLM model for the wider Arctic, that includes both present-day ice loading (PDIL) and glacial isostatic adjustment (GIA). The study shows that the nonlinear elastic uplift from PDIL is significant (0.5-1 mm yr⁻¹) in most of the wider Arctic and exceeds GIA at 15 of 54 Arctic GNSS sites, including sites in nonglaciaded areas of the North Sea region and the east coast of North America. Thereby the sea level change from PDIL (1.85 mm yr⁻¹) is significantly mitigated from VLM caused by PDIL. The combined VLM model was consistent with measured VLM at 85% of the GNSS sites () and outperformed a GIA-only model (). Deviations from GNSS-measured VLM can be attributed to local circumstances causing VLM.

265. Extended enthalpy formulations in the Ice-sheet and Sea-level System Model (ISSM) version 4.17: discontinuous conductivity and anisotropic streamline upwind Petrov-Galerkin (SUPG) method

Authors: Martin Rückamp; Angelika Humbert; Thomas Kleiner; Mathieu Morlighem; Helene Seroussi

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-13-4491-2020

URL: <https://doi.org/10.5194/gmd-13-4491-2020>

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Volume/Issue: Volume 13, Issue 9

References: 48

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: . The thermal state of an ice sheet is an important control on its past and future evolution. Some parts of the ice sheet may be polythermal, leading to discontinuous properties at the cold-temperate transition surface (CTS). These discontinuities require a careful treatment in ice sheet models (ISMs). Additionally, the highly anisotropic geometry of the 3D elements in ice sheet modelling poses a problem for stabilization approaches in advection-dominated problems. Here, we present extended enthalpy formulations within the finite-element Ice-Sheet and Sea-Level System model (ISSM) that show a better performance than earlier implementations. In a first polythermal-slab experiment, we found that the treatment of the discontinuous conductivities at the CTS with a geometric mean produces more accurate results compared to the arithmetic or harmonic mean. This improvement is particularly efficient when applied to coarse vertical resolutions. In a second ice dome experiment, we find that the numerical solution is sensitive to the choice of stabilization parameters in the well-established streamline upwind Petrov-Galerkin (SUPG) method. As standard literature values for the SUPG stabilization parameter do not account for the highly anisotropic geometry of the 3D elements in ice sheet modelling, we propose a novel anisotropic SUPG (ASUPG) formulation. This formulation circumvents the problem of high aspect ratio by treating the horizontal and vertical directions separately in the stabilization coefficients. The ASUPG method provides accurate results for the thermodynamic equation on geometries with very small aspect ratios like ice sheets.

266. The future sea-level contribution of the Greenland ice sheet: a multi-model ensemble study of ISMIP6

Authors: Heiko Goelzer; Sophie Nowicki; Anthony Payne; Eric Larour; Helene Seroussi; William H. Lipscomb; Jonathan Gregory; Ayako Abe-Ouchi; Andrew Shepherd; Erika Simon; Cécile Agosta; Patrick Alexander; Andy Aschwanden; Alice Barthel; Reinhard Calov; Christopher Chambers; Youngmin Choi; Joshua Cuzzone; Christophe Dumas; Tamsin Edwards; Denis Felikson; Xavier Fettweis; Nicholas R. Golledge; Ralf Greve; Angelika Humbert; Philippe Huybrechts; Sebastien Leclerc'h; Victoria Lee; Gunter Leguy; Chris Little; Daniel P. Lowry; Mathieu Morlighem; Isabel Nias; Aurelien Quiquet; Martin Rückamp; Nicole-Jeanne Schlegel; Donald A. Slater; Robin S. Smith; Fiamma Straneo; Lev Tarasov; Roderik van de Wal; Michiel van den Broeke

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-14-3071-2020

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Cited by: 175 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, Japan Society for the Promotion of Science

License: CC BY 4.0

Open Access: N/A

Abstract: . The Greenland ice sheet is one of the largest contributors to global mean sea-level rise today and is expected to continue to lose mass as the Arctic continues to warm. The two predominant mass loss mechanisms are increased surface meltwater run-off and mass loss associated with the retreat of marine-terminating outlet glaciers. In this paper we use a large ensemble of Greenland ice sheet models forced by output from a representative subset of the Coupled Model Intercomparison Project (CMIP5) global climate models to project ice sheet changes and sea-level rise contributions over the 21st century. The simulations are part of the Ice Sheet Model Intercomparison Project for CMIP6 (ISMIP6). We estimate the sea-level contribution together with uncertainties due to future climate forcing, ice sheet model formulations and ocean forcing for the two greenhouse gas concentration scenarios RCP8.5 and RCP2.6. The results indicate that the Greenland ice sheet will continue to lose mass in both scenarios until 2100, with contributions of 90 ± 50 and 32 ± 17 mm to sea-level rise for RCP8.5 and RCP2.6, respectively. The largest mass loss is expected from the south-west of Greenland, which is governed by surface mass balance changes, continuing what is already observed today. Because the contributions are calculated against an unforced control experiment, these numbers do not include any committed mass loss, i.e. mass loss that would occur over the coming century if the climate forcing remained constant. Under RCP8.5 forcing, ice sheet model uncertainty explains an ensemble spread of 40 mm, while climate model uncertainty and ocean forcing uncertainty account for a spread of 36 and 19 mm, respectively. Apart from those formally derived uncertainty ranges, the largest gap in our knowledge is about the physical understanding and implementation of the calving process, i.e. the

interaction of the ice sheet with the ocean.

267. ISMIP6 Antarctica: a multi-model ensemble of the Antarctic ice sheet evolution over the 21st century

Authors: Hélène Seroussi; Sophie Nowicki; Antony J. Payne; Heiko Goelzer; William H. Lipscomb; Ayako Abe-Ouchi; Cécile Agosta; Torsten Albrecht; Xylar Asay-Davis; Alice Barthel; Reinhard Calov; Richard Cullather; Christophe Dumas; Benjamin K. Galton-Fenzi; Rupert Gladstone; Nicholas R. Golledge; Jonathan M. Gregory; Ralf Greve; Tore Hattermann; Matthew J. Hoffman; Angelika Humbert; Philippe Huybrechts; Nicolas C. Jourdain; Thomas Kleiner; Eric Larour; Gunter R. Leguy; Daniel P. Lowry; Christopher M. Little; Mathieu Morlighem; Frank Pattyn; Tyler Pelle; Stephen F. Price; Aurélien Quiquet; Ronja Reese; Nicole-Jeanne Schlegel; Andrew Shepherd; Erika Simon; Robin S. Smith; Fiammetta Straneo; Sainan Sun; Luke D. Trusel; Jonas Van Breedam; Roderik S. W. van de Wal; Ricarda Winkelmann; Chen Zhao; Tong Zhang; Thomas Zwinger

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-14-3033-2020

URL: <https://doi.org/10.5194/tc-14-3033-2020>

Publication Date: 2020-09-17

Volume/Issue: Volume 14, Issue 9

References: 169

Cited by: 248 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration, Office of Science, Research Council of Finland (and 7 more)

License: CC BY 4.0

Open Access: N/A

Abstract: . Ice flow models of the Antarctic ice sheet are commonly used to simulate its future evolution in response to different climate scenarios and assess the mass loss that would contribute to future sea level rise. However, there is currently no consensus on estimates of the future mass balance of the ice sheet, primarily because of differences in the representation of physical processes, forcings employed and initial states of ice sheet models. This study presents results from ice flow model simulations from 13 international groups focusing on the evolution of the Antarctic ice sheet during the period 2015-2100 as part of the Ice Sheet Model Intercomparison for CMIP6 (ISMIP6). They are forced with outputs from a subset of models from the Coupled Model Intercomparison Project Phase 5 (CMIP5), representative of the spread in climate model results. Simulations of the Antarctic ice sheet contribution to sea level rise in response to increased warming during this period varies between -7.8 and 30.0 cm of sea level equivalent (SLE) under Representative Concentration Pathway (RCP) 8.5 scenario forcing. These numbers are relative to a control experiment with constant climate conditions and should therefore be added to the mass loss contribution under climate conditions similar to present-day conditions over the same period. The simulated evolution of the West Antarctic ice sheet varies widely among models, with an overall mass loss, up to 18.0 cm SLE, in response to changes in oceanic conditions. East Antarctica mass change varies between -6.1 and 8.3 cm SLE in the simulations, with a significant increase in surface mass balance

outweighing the increased ice discharge under most RCP 8.5 scenario forcings. The inclusion of ice shelf collapse, here assumed to be caused by large amounts of liquid water ponding at the surface of ice shelves, yields an additional simulated mass loss of 28 mm compared to simulations without ice shelf collapse. The largest sources of uncertainty come from the climate forcing, the ocean-induced melt rates, the calibration of these melt rates based on oceanic conditions taken outside of ice shelf cavities and the ice sheet dynamic response to these oceanic changes. Results under RCP 2.6 scenario based on two CMIP5 climate models show an additional mass loss of 0 and 3 cm of SLE on average compared to simulations done under present-day conditions for the two CMIP5 forcings used and display limited mass gain in East Antarctica.

268. Antarctic ice sheet response to sudden and sustained ice-shelf collapse (ABUMIP)

Authors: Sainan Sun; Frank Pattyn; Erika G. Simon; Torsten Albrecht; Stephen Cornford; Reinhard Calov; Christophe Dumas; Fabien Gillet-Chaulet; Heiko Goelzer; Nicholas R. Golledge; Ralf Greve; Matthew J. Hoffman; Angelika Humbert; Elise Kazmierczak; Thomas Kleiner; Gunter R. Leguy; William H. Lipscomb; Daniel Martin; Mathieu Morlighem; Sophie Nowicki; David Pollard; Stephen Price; Aurélien Quiquet; Hélène Seroussi; Tanja Schlemm; Johannes Sutter; Roderik S. W. van de Wal; Ricarda Winkelmann; Tong Zhang

Journal: Journal of Glaciology

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/jog.2020.67

URL: <https://doi.org/10.1017/jog.2020.67>

Publication Date: 2020-09-14

Volume/Issue: Volume 66, Issue 260

References: 107

Cited by: 87 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: Antarctica's ice shelves modulate the grounded ice flow, and weakening of ice shelves due to climate forcing will decrease their 'buttressing' effect, causing a response in the grounded ice. While the processes governing ice-shelf weakening are complex, uncertainties in the response of the grounded ice sheet are also difficult to assess. The Antarctic BUttressing Model Intercomparison Project (ABUMIP) compares ice-sheet model responses to decrease in buttressing by investigating the 'end-member' scenario of total and sustained loss of ice shelves. Although unrealistic, this scenario enables gauging the sensitivity of an ensemble of 15 ice-sheet models to a total loss of buttressing, hence exhibiting the full potential of marine ice-sheet instability. All models predict that this scenario leads to multi-metre (1-12 m) sea-level rise over 500 years from present day. West Antarctic ice sheet collapse alone leads to a 1.91-5.08 m sea-level rise due to the marine ice-sheet instability. Mass loss rates are a strong function of the sliding/friction law, with plastic laws cause a further destabilization of the Aurora and Wilkes Subglacial Basins, East Antarctica. Improvements to marine ice-sheet models have greatly reduced variability between modelled ice-sheet responses to extreme ice-shelf loss, e.g. compared to the SeaRISE assessments.

269. Glacial trimlines to identify former ice margins and subglacial thermal boundaries: A review and classification scheme for trimline expression

Authors: Camilla M. Rootes; Chris D. Clark

Journal: Earth-Science Reviews

Publisher: Elsevier BV

DOI: 10.1016/j.earscirev.2020.103355

URL: <https://doi.org/10.1016/j.earscirev.2020.103355>

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Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: University of Sheffield

License: N/A

Open Access: N/A

Abstract: N/A

270. A kinematic formalism for tracking ice-ocean mass exchange on the Earth's surface and estimating sea-level change

Authors: Surendra Adhikari; Erik R. Ivins; Eric Larour; Lambert Caron; Helene Seroussi

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-14-2819-2020

URL: <https://doi.org/10.5194/tc-14-2819-2020>

Publication Date: 2020-09-02

Volume/Issue: Volume 14, Issue 9

References: 53

Cited by: 8 publications

Language: en

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Keywords: N/A

Funding: California Institute of Technology

License: CC BY 4.0

Open Access: N/A

Abstract: . Polar ice sheets are important components of the Earth system. As the geometries of land, ocean and ice sheets evolve, they must be consistently captured within the lexicon of geodesy. Understanding the interplay between the processes such as ice-sheet dynamics, solid-Earth deformation, and sea-level adjustment requires both geodetically consistent and mass-conserving descriptions of evolving land and ocean domains, grounded ice sheets and floating ice shelves, and their respective interfaces. Here we present mathematical descriptions of a generic level set that can be used to track both the grounding lines and coastlines, in light of ice-ocean mass exchange and complex feedbacks from the solid Earth and sea level. We next present a unified method to accurately compute the sea-level contribution of evolving ice sheets based on the change in ice thickness, bedrock elevation and mean sea level caused by any geophysical processes. Our formalism can be applied to arbitrary geometries and at all timescales. While it can be used for applications with modeling, observations and the combination of two, it is best suited for Earth system models, comprising ice sheets, solid Earth and sea level, that seek to conserve mass.

271. Accelerated Greenland Ice Sheet Mass Loss Under High Greenhouse Gas Forcing as Simulated by the Coupled CESM2.1-CISM2.1

Authors: Laura Muntjewerf (Delft University of Technology); Raymond Sellevold (Delft University of Technology); Miren Vizcaino (Delft University of Technology); Carolina Ernani da Silva (Delft University of Technology); Michele Petrini (Delft University of Technology); Katherine Thayer-Calder (National Center for Atmospheric Research); Meike D. W. Scherrenberg (Delft University of Technology); Sarah L. Bradley (University of Sheffield); Caroline A. Katsman (Delft University of Technology); Jeremy Fyke (Associated Engineering Group Ltd. Calgary Canada); William H. Lipscomb (National Center for Atmospheric Research); Marcus Lofverstrom (University of Arizona); William J. Sacks (National Center for Atmospheric Research)

Journal: Journal of Advances in Modeling Earth Systems

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019ms002031

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Publication Date: 2020-08-27

Volume/Issue: Volume 12, Issue 10

References: 65

Cited by: 17 publications

Language: en

Subjects: N/A

Keywords: surface mass balance, Greenland ice sheet, anthropogenic climate change, sea level rise

Funding: National Council for Eurasian and East European Research, Nederlandse Organisatie voor Wetenschappelijk Onderzoek

License: CC BY 4.0

Open Access: N/A

Abstract: The Greenland ice sheet (GrIS) is now losing mass at a rate of 0.7 mm of sea level rise (SLR) per year. Here we explore future GrIS evolution and interactions with global and regional climate under high greenhouse gas forcing with the Community Earth System Model version 2.1 (CESM2.1), which includes an interactive ice sheet component (the Community Ice Sheet Model v2.1 [CISM2.1]) and an advanced energy balance-based calculation of surface melt. We run an idealized 350-year scenario in which atmospheric CO₂ concentration increases by 1% annually until reaching four times pre-industrial values at year 140, after which it is held fixed. The global mean temperature increases by 5.2 and 8.5 K by years 131-150 and 331-350, respectively. The projected GrIS contribution to global mean SLR is 107 mm by year 150 and 1,140 mm by year 350. The rate of SLR increases from 2 mm yr⁻¹ at year 150 to almost 7 mm yr⁻¹ by year 350. The accelerated mass loss is caused by rapidly increasing surface melt as the ablation area expands, with associated albedo feedback and increased sensible and latent heat fluxes. This acceleration occurs for a global warming of approximately 4.2 K with respect to pre-industrial and is in part explained by the quasi-parabolic shape of the ice sheet, which favors rapid expansion of the ablation area as it approaches the interior "plateau."

272. Present day Jakobshavn Isbræ (West Greenland) close to the Holocene minimum extent

Authors: Karita Kajanto; Helene Seroussi; Basile de Fleurian; Kerim H. Nisancioglu

Journal: Quaternary Science Reviews

Publisher: Elsevier BV

DOI: 10.1016/j.quascirev.2020.106492

URL: <https://doi.org/10.1016/j.quascirev.2020.106492>

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References: 68

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: European Research Council, European Community's Seventh Framework Program, Norwegian Research Council (and 2 more)

License: N/A

Open Access: N/A

Abstract: N/A

273. The Sensitivity of the Antarctic Ice Sheet to a Changing Climate: Past, Present, and Future

Authors: T. L. Noble (University of Tasmania); E. J. Rohling (University of Southampton, Australian National University, National Oceanography Centre); A. R. A. Aitken (School of Earth Sciences University of Western Australia Perth Western Australia Australia); H. C. Bostock (School of Earth and Environmental Sciences University of Queensland Brisbane Queensland Australia); Z. Chase (University of Tasmania); N. Gomez (McGill University); L. M. Jong (Australian Antarctic Division Department of Agriculture, Water and Environment Kingston Tasmania Australia, University of Tasmania); M. A. King (University of Tasmania); A. N. Mackintosh (Monash University); F. S. McCormack (Monash University); R. M. McKay (Victoria University of Wellington); L. Menviel (University of New South Wales); S. J. Phipps (University of Tasmania); M. E. Weber (University of Bonn); C. J. Fogwill (Geography, Geology and the Environment Keele University Staffordshire UK); B. Gayen (University of Melbourne); N. R. Golledge (Victoria University of Wellington); D. E. Gwyther (University of Tasmania); A. McC. Hogg (Australian National University, ARC Centre of Excellence for Climate System Science Sydney New South Wales Australia); Y. M. Martos (University of Maryland, National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); B. Pena-Molino (University of Tasmania, Ocean and Atmosphere Commonwealth Scientific and Industrial Research Organization Hobart Tasmania Australia); J. Roberts (Australian Antarctic Division Department of Agriculture, Water and Environment Kingston Tasmania Australia, University of Tasmania); T. van de Flierdt (Imperial College London); T. Williams (International Ocean Discovery Program Texas A&M; University College Station TX USA)

Journal: Reviews of Geophysics

Publisher: American Geophysical Union (AGU)

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Publication Date: 2020-08-15

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References: 811

Cited by: 78 publications

Language: en

Subjects: N/A

Keywords: past, processes, sea level, Antarctic Ice Sheet, climate (and 1 more)

Funding: Australian Academy of Science, Australian Research Council, University of Tasmania

License: CC BY 4.0

Open Access: N/A

Abstract: The Antarctic Ice Sheet (AIS) is out of equilibrium with the current anthropogenic-enhanced climate forcing. Paleoenvironmental records and ice sheet models reveal that the AIS has been tightly coupled to the climate system during the past and indicate the potential for accelerated and sustained Antarctic ice mass loss into the future. Modern observations by contrast suggest that the AIS has only just started to respond to climate change in recent decades. The maximum projected sea level contribution from Antarctica to 2100 has increased significantly since the Intergovernmental Panel on Climate Change (IPCC) 5th Assessment Report, although estimates continue to evolve with new observational and theoretical advances. This review brings together recent literature highlighting the

progress made on the known processes and feedbacks that influence the stability of the AIS. Reducing the uncertainty in the magnitude and timing of the future sea level response to AIS change requires a multidisciplinary approach that integrates knowledge of the interactions between the ice sheet, solid Earth, atmosphere, and ocean systems and across time scales of days to millennia. We start by reviewing the processes affecting AIS mass change, from atmospheric and oceanic processes acting on short time scales (days to decades), through to ice processes acting on intermediate time scales (decades to centuries) and the response to solid Earth interactions over longer time scales (decades to millennia). We then review the evidence of AIS changes from the Pliocene to the present and consider the projections of global sea level rise and their consequences. We highlight priority research areas required to improve our understanding of the processes and feedbacks governing AIS change.

274. Studies on the variability of the Greenland Ice Sheet and climate

Authors: Kumiko Goto-Azuma; Tomoyuki Homma; Tomotaka Saruya; Fumio Nakazawa; Yuki Komuro; Naoko Nagatsuka; Motohiro Hirabayashi; Yutaka Kondo; Makoto Koike; Teruo Aoki; Ralf Greve; Jun'ichi Okuno

Journal: Polar Science

Publisher: Elsevier BV

DOI: 10.1016/j.polar.2020.100557

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Subjects: N/A

Keywords: N/A

Funding: Ministry of the Environment, Government of Japan, Ministry of Education, Culture, Sports, Science and Technology

License: N/A

Open Access: N/A

Abstract: N/A

275. Greenland temperature and precipitation over the last 20 000 years using data assimilation

Authors: Jessica A. Badgeley; Eric J. Steig; Gregory J. Hakim; Tyler J. Fudge

Journal: Climate of the Past

Publisher: Copernicus GmbH

DOI: 10.5194/cp-16-1325-2020

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Volume/Issue: Volume 16, Issue 4

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Keywords: N/A

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Open Access: N/A

Abstract: . Reconstructions of past temperature and precipitation are fundamental to modeling the Greenland Ice Sheet and assessing its sensitivity to climate. Paleoclimate information is sourced from proxy records and climate-model simulations; however, the former are spatially incomplete while the latter are sensitive to model dynamics and boundary conditions. Efforts to combine these sources of information to reconstruct spatial patterns of Greenland climate over glacial-interglacial cycles have been limited by assumptions of fixed spatial patterns and a restricted use of proxy data. We avoid these limitations by using paleoclimate data assimilation to create independent reconstructions of mean-annual temperature and precipitation for the last 20 000 years. Our method uses oxygen isotope ratios of ice and accumulation rates from long ice-core records and extends this information to all locations across Greenland using spatial relationships derived from a transient climate-model simulation. Standard evaluation metrics for this method show that our results capture climate at locations without ice-core records. Our results differ from previous work in the reconstructed spatial pattern of temperature change during abrupt climate transitions; this indicates a need for additional proxy data and additional transient climate-model simulations. We investigate the relationship between precipitation and temperature, finding that it is frequency dependent and spatially variable, suggesting that thermodynamic scaling methods commonly used in ice-sheet modeling are overly simplistic. Our results demonstrate that paleoclimate data assimilation is a useful tool for reconstructing the spatial and temporal patterns of past climate on timescales relevant to ice sheets.

276. Results of the third Marine Ice Sheet Model Intercomparison Project (MISMIP+)

Authors: Stephen L. Cornford; Helene Seroussi; Xylar S. Asay-Davis; G. Hilmar Gudmundsson; Rob Arthern; Chris Borstad; Julia Christmann; Thiago Dias dos Santos; Johannes Feldmann; Daniel Goldberg; Matthew J. Hoffman; Angelika Humbert; Thomas Kleiner; Gunter Leguy; William H. Lipscomb; Nacho Merino; Gaël Durand; Mathieu Morlighem; David Pollard; Martin Rückamp; C. Rosie Williams; Hongju Yu

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-14-2283-2020

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Language: en

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Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: . We present the result of the third Marine Ice Sheet Model Intercomparison Project, MISMIP+. MISMIP+ is intended to be a benchmark for ice-flow models which include fast sliding marine ice streams and floating ice shelves and in particular a treatment of viscous stress that is sufficient to model buttressing, where upstream ice flow is restrained by a downstream ice shelf. A set of idealized experiments first tests that models are able to maintain a steady state with the grounding line located on a retrograde slope due to buttressing and then explore scenarios where a reduction in that buttressing causes ice stream acceleration, thinning, and grounding line retreat. The majority of participating models passed the first test and then produced similar responses to the loss of buttressing. We find that the most important distinction between models in this particular type of simulation is in the treatment of sliding at the bed, with other distinctions - notably the difference between the simpler and more complete treatments of englacial stress but also the differences between numerical methods - taking a secondary role.

277. Sea Level Rise in the CESM Large Ensemble: The Role of Individual Climate Forcings and Consequences for the Coming Decades

Authors: John T. Fasullo (University of Colorado Boulder, National Center for Atmospheric Research); Peter R. Gent (National Center for Atmospheric Research); R. Steven Nerem (University of Colorado Boulder)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-19-1001.1

URL: <https://doi.org/10.1175/jcli-d-19-1001.1>

Publication Date: 2020-07-10

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References: 51

Cited by: 5 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, National Science Foundation, National Science Foundation (and 1 more)

License: N/A

Open Access: N/A

Abstract: The emergence of a spatial pattern in the externally forced response (FR) of dynamic sea level (DSL) during the altimeter era has recently been demonstrated using climate models but our understanding of its initial emergence, drivers, and implications for the future is poor. Here the anthropogenic forcings of the DSL pattern are explored using the Community Earth System Model Large Ensemble (CESM-LE) and Single-Forcing Large Ensemble, a newly available set of simulations where values of individual forcing agents remain fixed at 1920 levels, allowing for an estimation of their effects. Statistically significant contributions to the DSL FR are identified for greenhouse gases (GHGs) and industrial aerosols (AERs), with particularly strong contributions resulting from AERs in the mid-twentieth century and GHGs in the late twentieth and twenty-first century. Secondary, but important, contributions are identified for biomass burning aerosols in the equatorial Atlantic Ocean in the mid-twentieth century, and for stratospheric ozone in the Southern Ocean during the late twentieth century. Key to understanding regional DSL patterns are ocean heat content and salinity anomalies, which are driven by surface heat and freshwater fluxes, ocean dynamics, and the spatial structure of seawater thermal expansivity. Potential implications for the interpretation of DSL during the satellite era and the longer records from tide gauges are suggested as a topic for future research.

278. Description and validation of the ice-sheet model Yelmo (version 1.0)

Authors: Alexander Robinson; Jorge Alvarez-Solas; Marisa Montoya; Heiko Goelzer; Ralf Greve; Catherine Ritz

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-13-2805-2020

URL: <https://doi.org/10.5194/gmd-13-2805-2020>

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Cited by: 17 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Japan Society for the Promotion of Science

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Open Access: N/A

Abstract: . We describe the physics and features of the ice-sheet model Yelmo, an open-source project intended for collaborative development. Yelmo is a thermomechanical model, solving for the coupled velocity and temperature solutions of an ice sheet simultaneously. The ice dynamics are currently treated via a "hybrid" approach combining the shallow-ice and shallow-shelf/shelfy-stream approximations, which makes Yelmo an apt choice for studying a wide variety of problems. Yelmo's main innovations lie in its flexible and user-friendly infrastructure, which promotes portability and facilitates long-term development. In particular, all physics subroutines have been designed to be self-contained, so that they can be easily ported from Yelmo to other models, or easily replaced by improved or alternate methods in the future. Furthermore, hard-coded model choices are eschewed, replaced instead with convenient parameter options that allow the model to be adapted easily to different contexts. We show results for different ice-sheet benchmark tests, and we illustrate Yelmo's performance for the Antarctic ice sheet.

279. Evaluation of Iceberg Calving Models Against Observations From Greenland Outlet Glaciers

Authors: Tristan Amaral (University of Idaho); Timothy C. Bartholomaus (University of Idaho); Ellyn M. Enderlin (Boise State University)

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019jf005444

URL: <https://doi.org/10.1029/2019jf005444>

Publication Date: 2020-06-09

Volume/Issue: Volume 125, Issue 6

References: 98

Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: iceberg calving, dynamic discharge, frontal ablation, calving models, ice sheet modeling

Funding: National Science Foundation, National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: Frontal ablation processes at marine-terminating glaciers are challenging to observe and difficult to represent in numerical ice flow models, yet play critical roles in modulating ice sheet mass balance. Current ice sheet models typically rely on simple iceberg calving models to prescribe either terminus positions or iceberg calving rates, but the relative accuracies and uncertainties of these calving models remain largely unconstrained at the ice sheet scale. Here, we evaluate six published iceberg calving models against spatially and temporally diverse observations from 50 marine-terminating outlet glaciers in Greenland. We seek the single model that best reproduces observed conditions across all glaciers, at all observation times, and with low sensitivity to calibration uncertainty. Five of six calving models can produce unbiased estimates of calving position or calving rate at the ice sheet scale. However, time series analysis reveals that, when using a single, optimized model parameter, rate-predicting calving models frequently yield calving rate errors in excess of 10 m d⁻¹. In comparison, terminus position-predicting calving models more accurately track observed changes in terminus position (remaining within ~1 km of the observed grounded terminus position). Overall, our results indicate that the crevasse depth calving model provides the best balance of high accuracy and low sensitivity to imperfect parameter calibration. While the crevasse depth model appears unlikely to capture the true controls on crevasse penetration, numerically, it reproduces observed terminus dynamics with high fidelity and should be considered a leading candidate for use in models of the Greenland Ice Sheet.

280. A two-dimensional, higher-order, enthalpy-based thermomechanical ice flow model for mountain glaciers and its benchmark experiments

Authors: Yuzhe Wang; Tong Zhang; Cunde Xiao; Jiawen Ren; Yanfen Wang

Journal: Computers & Geosciences

Publisher: Elsevier BV

DOI: 10.1016/j.cageo.2020.104526

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Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, National Natural Science Foundation of China

License: N/A

Open Access: N/A

Abstract: N/A

281. Forced Patterns of Sea Level Rise in the Community Earth System Model Large Ensemble From 1920 to 2100

Authors: John T. Fasullo (University of Colorado Boulder, National Center for Atmospheric Research); Peter R. Gent (National Center for Atmospheric Research); R. S. Nerem (University of Colorado Boulder)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019jc016030

URL: <https://doi.org/10.1029/2019jc016030>

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References: 69

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: sea level rise, regional climate change, climate modeling

Funding: National Aeronautics and Space Administration, National Science Foundation, National Center for Atmospheric Research

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Open Access: N/A

Abstract: To provide context for observed sea level rise, the forced response (FR) in dynamic sea level (DSL) during the 20th and 21st centuries is examined in the Community Earth System Model Large Ensemble (LE). After accounting for simulation drift, which in the LE is sizable, the DSL FR is found to be complex, both in space and time. Its evolving character is suggested to arise from both the diversity and transient evolution of climate forcing agents and the slow adjustment timescales of the intermediate and deep oceans. Nonetheless, various intervals of spatially coherent change simulated for the recent past and near future are identified, and their characteristics and associated driving mechanisms are identified and discussed. The mid- to late-20th century DSL FR is characterized by an hemispherically asymmetric pattern of change, with depressed rates of rise in the northern oceans. There is also a dipole of change in the Southern Ocean caused by changes in near-surface zonal winds. Through the late 20th and early 21st centuries, a different pattern of rise emerges with elevated rates in the tropics and depressed rates at high latitudes. Zonal and interbasin variations characterize both intervals and involve the pattern and depth of anomalous ocean heat content storage and spatial contrasts in the expansion coefficient tied mainly to base state temperature, with greater rates of rise in warm regions per unit warming. The relative roles of surface flux and ocean convergence anomalies are examined.

282. Spatial probabilistic calibration of a high-resolution Amundsen Sea Embayment ice sheet model with satellite altimeter data

Authors: Andreas Wernecke; Tamsin L. Edwards; Isabel J. Nias; Philip B. Holden; Neil R. Edwards

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-14-1459-2020

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References: 62

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: . Probabilistic predictions of the sea level contribution from Antarctica often have large uncertainty intervals. Calibration of model simulations with observations can reduce uncertainties and improve confidence in projections, particularly if this exploits as much of the available information as possible (such as spatial characteristics), but the necessary statistical treatment is often challenging and can be computationally prohibitive. Ice sheet models with sufficient spatial resolution to resolve grounding line evolution are also computationally expensive. Here we address these challenges by adopting and comparing dimension-reduced calibration approaches based on a principal component decomposition of the adaptive mesh model BISICLES. The effects model parameters have on these principal components are then gathered in statistical emulators to allow for smooth probability density estimates. With the help of a published perturbed parameter ice sheet model ensemble of the Amundsen Sea Embayment (ASE), we show how the use of principal components in combination with spatially resolved observations can improve probabilistic calibrations. In synthetic model experiments (calibrating the model with altered model results) we can identify the correct basal traction and ice viscosity scaling parameters as well as the bedrock map with spatial calibrations. In comparison a simpler calibration against an aggregated observation, the net sea level contribution, imposes only weaker constraints by allowing a wide range of basal traction and viscosity scaling factors. Uncertainties in sea level rise contribution of 50-year simulations from the current state of the ASE can be reduced with satellite observations of recent ice thickness change by nearly 90 %; median and 90 % confidence intervals are 18.9 [13.9, 24.8] mm SLE (sea level equivalent) for the proposed spatial calibration approach, 16.8 [7.7, 25.6] mm SLE for the net sea level calibration and 23.1 [-8.4, 94.5] mm SLE for the uncalibrated ensemble. The spatial model behaviour is much more consistent with observations if, instead of Bedmap2, a modified bedrock topography is used that most notably removes a topographic rise near the initial grounding line of Pine Island Glacier. The ASE dominates the current Antarctic sea level contribution, but other regions have the potential to become more important on centennial scales. These larger spatial and temporal scales would benefit even more from methods of fast but exhaustive model calibration. Applied to projections of the whole Antarctic ice sheet, our

approach has therefore the potential to efficiently improve our understanding of model behaviour, as well as substantiating and reducing projection uncertainties.

283. Extended enthalpy formulations in the ice flow model ISSM version 4.17: discontinuous conductivity and anisotropic SUPG

Authors: Martin Rückamp; Angelika Humbert; Thomas Kleiner; Mathieu Morlighem; Helene Seroussi

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-2020-78

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Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Classification of discontinuities

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: . The thermal state of an ice sheet is an important control on its past and future evolution. Some parts of the ice sheets may be polythermal, leading to discontinuous properties at the cold-temperate transition surface (CTS). These discontinuities require a careful treatment in ice sheet models (ISMs). Additionally, the highly anisotropic geometry of the 3D elements in ice sheet modelling poses a problem for stabilization approaches in advection dominated problems. Here, we present extended enthalpy formulations within the finite-element Ice Sheet System Model (ISSM) that show a better performance to earlier implementations. In a first polythermal-slab experiment, we found that the treatment of the discontinuous conductivities at the CTS with a geometric mean produce more accurate results compared to the arithmetic or harmonic mean. This improvement is particularly efficient when applied to coarse vertical resolutions. In a second ice dome experiment, we find that the numerical solution is sensitive to the choice of stabilization parameters in the well-established Streamline Upwind Petrov-Galerkin (SUPG) method. As standard literature values for the SUPG stabilization parameter are not accounting for the highly anisotropic geometry of the 3D elements in ice sheet modelling, we propose a novel Anisotropic SUPG (ASUPG) formulation. This formulation circumvents the problem of high aspect-ratio by treating the horizontal and vertical directions separately in the stabilization coefficients. The ASUPG method provides accurate results for the thermodynamic equation on geometries with very small aspect ratios like ice sheets.

284. Aurora Basin, the Weak Underbelly of East Antarctica

Authors: Tyler Pelle (University of California Irvine); Mathieu Morlighem (University of California Irvine); Felicity S. McCormack (University of Tasmania, Monash University)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019gl086821

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Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: Antarctica, balance, mass, Aurora, Totten (and 1 more)

Funding: National Science Foundation, Natural Environment Research Council, Australian Research Council (and 1 more)

License: N/A

Open Access: N/A

Abstract: The East Antarctic Ice Sheet (EAIS) has the potential to raise global sea levels by ~52 m. Here, we model the evolution of select EAIS catchments to 2100 using three basal melt rate parameterizations and force our model with surface mass balance and ocean thermal anomalies from 10 global climate models. While the domain loses mass under low-emission scenarios, it gains ~10-mm sea-level rise equivalent ice volume (SLRe) under high-emission scenarios. The primary region of thinning is within 50 km upstream of Totten Glacier's grounding line. Totten's glacial discharge is modulated by the migration of its grounding line, which is sensitive to brief intrusions of ocean water at temperatures higher than present. Once the grounding line is dislodged, Totten's ice velocity increases by up to 70% of present-day values, resulting in ~6-mm SLRe loss from this sector.

285. Geothermal Heat Flux in Antarctica: Assessing Models and Observations by Bayesian Inversion

Authors: Mareen Lösing; Jörg Ebbing; Wolfgang Szwilius

Journal: Frontiers in Earth Science

Publisher: Frontiers Media SA

DOI: 10.3389/feart.2020.00105

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References: 43

Cited by: 48 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: Deutsche Forschungsgemeinschaft

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

286. Understanding of Contemporary Regional Sea-Level Change and the Implications for the Future

Authors: Benjamin D. Hamlington (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Alex S. Gardner (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Erik Ivins (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Jan T. M. Lenaerts (University of Colorado Boulder); J. T. Reager (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); David S. Trossman (University of Texas at Austin); Edward D. Zaron (Portland State University); Surendra Adhikari (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Anthony Arendt (University of Washington); Andy Aschwanden (University of Alaska Fairbanks); Brian D. Beckley (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, KBR Inc.); David P. S. Bekaert (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Geoffrey Blewitt (University of Nevada); Lambert Caron (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Don P. Chambers (University of South Florida); Hrishikesh A. Chandanpurkar (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Knut Christianson (University of Washington); Beata Csatho (University of Buffalo); Richard I. Cullather (University of Maryland); Robert M. DeConto (University of Massachusetts Amherst); John T. Fasullo (National Center for Atmospheric Research); Thomas Frederikse (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Jeffrey T. Freymueller (Michigan State University); Daniel M. Gilford (Rutgers University); Manuela Girotto (University of California Berkeley); William C. Hammond (University of Nevada); Regine Hock (University of Alaska Fairbanks); Nicholas Holschuh (University of Washington); Robert E. Kopp (University of California Berkeley); Felix Landerer (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Eric Larour (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Mark Merrifield (Scripps Institution of Oceanography / University of California San Diego); Jerry X. Mitrovica (Harvard University); R. Steven Nerem (University of Colorado Boulder); Isabel J. Nias (University of Maryland, University of Colorado Boulder); Veronica Nieves (Institute of Marine Sciences); Sophie Nowicki (University of Colorado Boulder); Kishore Pangaluru (University of California Irvine); Christopher G. Piecuch (Woods Hole Oceanographic Institution); Richard D. Ray (University of Colorado Boulder); David R. Rounce (University of Alaska Fairbanks); Nicole-Jeanne Schlegel (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Hélène Seroussi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Manoochehr Shirzaei (Arizona State University); William V. Sweet (Center for Operational Oceanographic Products and Services NOAA Silver Spring MD USA); Isabella Velicogna (Institute of Marine Sciences); Nadya Vinogradova (Science Mission Directorate NASA Headquarters Washington DC USA); Thomas Wahl (Civil, Environmental and Construction Engineering and National Center for Integrated Coastal Research University of Central Florida Orlando FL USA); David N. Wiese (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Michael J. Willis (University of Colorado Boulder)

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Cited by: 83 publications

Language: en

Subjects: N/A

Keywords: satellite observations, remote sensing, sea level

Funding: National Aeronautics and Space Administration, National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: Global sea level provides an important indicator of the state of the warming climate, but changes in regional sea level are most relevant for coastal communities around the world. With improvements to the sea-level observing system, the knowledge of regional sea-level change has advanced dramatically in recent years. Satellite measurements coupled with in situ observations have allowed for comprehensive study and improved understanding of the diverse set of drivers that lead to variations in sea level in space and time. Despite the advances, gaps in the understanding of contemporary sea-level change remain and inhibit the ability to predict how the relevant processes may lead to future change. These gaps arise in part due to the complexity of the linkages between the drivers of sea-level change. Here we review the individual processes which lead to sea-level change and then describe how they combine and vary regionally. The intent of the paper is to provide an overview of the current state of understanding of the processes that cause regional sea-level change and to identify and discuss limitations and uncertainty in our understanding of these processes. Areas where the lack of understanding or gaps in knowledge inhibit the ability to provide the needed information for comprehensive planning efforts are of particular focus. Finally, a goal of this paper is to highlight the role of the expanded sea-level observation network--particularly as related to satellite observations--in the improved scientific understanding of the contributors to regional sea-level change.

287. Ocean-forced evolution of the Amundsen Sea catchment, West Antarctica, by 2100

Authors: Alanna V. Alevropoulos-Borrill; Isabel J. Nias; Antony J. Payne; Nicholas R. Golledge; Rory J. Bingham

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-14-1245-2020

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References: 78

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: . The response of ice streams in the Amundsen Sea Embayment (ASE) to future climate forcing is highly uncertain. Here we present projections of 21st century response of ASE ice streams to modelled local ocean temperature change using a subset of Coupled Model Intercomparison Project (CMIP5) simulations. We use the BISICLES adaptive mesh refinement (AMR) ice sheet model, with high-resolution grounding line resolving capabilities, to explore grounding line migration in response to projected sub-ice-shelf basal melting. We find a contribution to sea level rise of between 2.0 and 4.5 cm by 2100 under RCP8.5 conditions from the CMIP5 subset, where the mass loss response is linearly related to the mean ocean temperature anomaly. To account for uncertainty associated with model initialization, we perform three further sets of CMIP5-forced experiments using different parameterizations that explore perturbations to the prescription of initial basal melt, the basal traction coefficient and the ice stiffening factor. We find that the response of the ASE to ocean temperature forcing is highly dependent on the parameter fields obtained in the initialization procedure, where the sensitivity of the ASE ice streams to the sub-ice-shelf melt forcing is dependent on the choice of parameter set. Accounting for ice sheet model parameter uncertainty results in a projected range in sea level equivalent contribution from the ASE of between -0.02 and 12.1 cm by the end of the 21st century.

288. Greenland Ice Sheet Contribution to 21st Century Sea Level Rise as Simulated by the Coupled CESM2.1-CISM2.1

Authors: Laura Muntjewerf (Delft University of Technology); Michele Petrini (Delft University of Technology); Miren Vizcaino (Delft University of Technology); Carolina Ernani da Silva (Delft University of Technology); Raymond Sellevold (Delft University of Technology); Meike D. W. Scherrenberg (Delft University of Technology); Katherine Thayer-Calder (National Center for Atmospheric Research); Sarah L. Bradley (University of Sheffield); Jan T. M. Lenaerts (University of Colorado Boulder); William H. Lipscomb (National Center for Atmospheric Research); Marcus Lofverstrom (University of Arizona)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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References: 59

Cited by: 47 publications

Language: en

Subjects: N/A

Keywords: Sea level rise, 21st century, Greenland Ice Sheet

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: The Greenland Ice Sheet (GrIS) mass balance is examined with an Earth system/ice sheet model that interactively couples the GrIS to the broader Earth system. The simulation runs from 1850 to 2100, with historical and SSP5-8.5 forcing. By the mid-21st century, the cumulative GrIS contribution to global mean sea level rise (SLR) is 23 mm. During the second half of the 21st century, the surface mass balance becomes negative in all drainage basins, with an additional SLR contribution of 86 mm. The annual mean GrIS mass loss in the last two decades is 2.7-mm sea level equivalent (SLE) year⁻¹. The increased SLR contribution from the surface mass balance (3.1 mm SLE year⁻¹) is partly offset by reduced ice discharge from thinning and retreat of outlet glaciers. The southern GrIS drainage basins contribute 73% of the mass loss in mid-century but 55% by 2100, as surface runoff increases in the northern basins.

289. SICOPOLIS-AD v1: an open-source adjoint modeling framework for ice sheet simulation enabled by the algorithmic differentiation tool OpenAD

Authors: Liz C. Logan; Sri Hari Krishna Narayanan; Ralf Greve; Patrick Heimbach

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

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Cited by: 4 publications

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Abstract: . We present a new capability of the ice sheet model SICOPOLIS that enables flexible adjoint code generation via source transformation using the open-source algorithmic differentiation (AD) tool OpenAD. The adjoint code enables efficient calculation of the sensitivities of a scalar-valued objective function or quantity of interest (QoI) to a range of important, often spatially varying and uncertain model input variables, including initial and boundary conditions, as well as model parameters. Compared to earlier work on the adjoint code generation of SICOPOLIS, our work makes several important advances: (i) it is embedded within the up-to-date trunk of the SICOPOLIS repository - accounting for 1.5 decades of code development and improvements - and is readily available to the wider community; (ii) the AD tool used, OpenAD, is an open-source tool; (iii) the adjoint code developed is applicable to both Greenland and Antarctica, including grounded ice as well as floating ice shelves, with an extended choice of thermodynamical representations. A number of code refactorization steps were required. They are discussed in detail in an Appendix as they hold lessons for the application of AD to legacy codes at large. As an example application, we examine the sensitivity of the total Antarctic Ice Sheet volume to changes in initial ice thickness, austral summer precipitation, and basal and surface temperatures across the ice sheet. Simulations of Antarctica with floating ice shelves show that over 100 years of simulation the sensitivity of total ice sheet volume to the initial ice thickness and precipitation is almost uniformly positive, while the sensitivities to surface and basal temperature are almost uniformly negative. Sensitivity to austral summer precipitation is largest on floating ice shelves from Queen Maud to Queen Mary Land. The largest sensitivity to initial ice thickness is at outlet glaciers around Antarctica. Comparison between total ice sheet volume sensitivities to surface and basal temperature shows that surface temperature sensitivities are higher broadly across the floating ice shelves, while basal temperature sensitivities are highest at the grounding lines of floating ice shelves and outlet glaciers. A uniformly perturbed region of East Antarctica reveals that, among the four control variables tested here, total ice sheet volume is the most sensitive to variations in austral summer precipitation as formulated in SICOPOLIS. Comparison between adjoint- and finite-difference-derived sensitivities shows good agreement, lending confidence that the AD tool is

producing correct adjoint code. The new modeling infrastructure is freely available at <http://www.sicopolis.net> (last access: 2 April 2020) under the development trunk.

290. Western Greenland ice sheet retreat history reveals elevated precipitation during the Holocene thermal maximum

Authors: Jacob Downs; Jesse Johnson; Jason Briner; Nicolás Young; Alia Lesnek; Josh Cuzzone

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-14-1121-2020

URL: <https://doi.org/10.5194/tc-14-1121-2020>

Publication Date: 2020-03-31

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References: 49

Cited by: 7 publications

Language: en

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Keywords: N/A

Funding: National Science Foundation

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Open Access: N/A

Abstract: . We investigate changing precipitation patterns in the Kangerlussuaq region of western central Greenland during the Holocene thermal maximum (HTM), using a new chronology of ice sheet terminus position through the Holocene and a novel inverse modeling approach based on the unscented transform (UT). The UT is applied to estimate changes in annual precipitation in order to reduce the misfit between modeled and observed terminus positions. We demonstrate the effectiveness of the UT for time-dependent data assimilation, highlighting its low computational cost and trivial parallel implementation. Our results indicate that Holocene warming coincided with elevated precipitation, without which modeled retreat in the Kangerlussuaq region is more rapid than suggested by observations. Less conclusive is whether high temperatures during the HTM were specifically associated with a transient increase in precipitation, as the results depend on the assumed temperature history. Our results highlight the important role that changing precipitation patterns had in controlling ice sheet extent during the Holocene.

291. Bed topography and subglacial landforms in the onset region of the Northeast Greenland Ice Stream

Authors: Steven Franke; Daniela Jansen; Tobias Binder; Nils Dörr; Veit Helm; John Paden; Daniel Steinhage; Olaf Eisen

Journal: Annals of Glaciology

Publisher: Cambridge University Press (CUP)

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Cited by: 24 publications

Language: en

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Abstract: The Northeast Greenland Ice Stream (NEGIS) is an important dynamic component for the total mass balance of the Greenland ice sheet, as it reaches up to the central divide and drains 12% of the ice sheet. The geometric boundary conditions and in particular the nature of the subglacial bed of the NEGIS are essential to understand its ice flow dynamics. We present a record of more than 8000 km of radar survey lines of multi-channel, ultra-wideband radio echo sounding data covering an area of 24 000 km², centered on the drill site for the East Greenland Ice-core Project (EGRIP), in the upper part of the NEGIS catchment. Our data yield a new detailed model of ice-thickness distribution and basal topography in the region. The enhanced resolution of our bed topography model shows features which we interpret to be caused by erosional activity, potentially over several glacial-interglacial cycles. Off-nadir reflections from the ice-bed interface in the center of the ice stream indicate a streamlined bed with elongated subglacial landforms. Our new bed topography model will help to improve the basal boundary conditions of NEGIS prescribed for ice flow models and thus foster an improved understanding of the ice-dynamic setting.

292. Review of Ruckamp et al. 2020

Authors: No authors listed

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/tc-2019-329-rc2

URL: <https://doi.org/10.5194/tc-2019-329-rc2>

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Abstract: N/A

293. Review of Rückamp et al.

Authors: No authors listed

Journal: Unknown Journal

Publisher: Copernicus GmbH

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Cited by: N/A publications

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Keywords: Ampere

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Abstract: N/A

294. Sensitivity of Greenland ice sheet projections to spatial resolution in higher-order simulations: the AWI contribution to ISMIP6-Greenland using ISSM

Authors: Martin Rückamp; Heiko Goelzer; Thomas Kleiner; Angelika Humbert

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/egusphere-egu2020-14565

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Subjects: N/A

Keywords: Forcing (mathematics)|Greenland ice sheet|Ice-sheet model|Antarctic ice sheet

Funding: N/A

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Open Access: N/A

Abstract: <p>Projections of the contribution of the Greenland ice sheet to future sea-level rise include uncertainties primarily due to the imposed climate forcing and the initial state of the ice sheet model. Several state-of-the-art ice flow models are currently being employed on various grid resolutions to estimate future mass changes in the framework of the Ice Sheet Model Intercomparison Project for CMIP6 (ISMIP6). Here we investigate the sensitivity to grid resolution on centennial sea-level contributions from the Greenland ice sheet and study the mechanism at play. To this end, we employ the finite-element higher-order ice flow model ISSM and conduct experiments with four different horizontal resolutions, namely 4, 2, 1 and 0.75 km. We run the simulation based on the ISMIP6 core GCM MIROC5 under the high emission scenario RCP8.5 and consider both atmospheric and oceanic forcing in full and separate scenarios. Under the full scenarios, finer simulations unveil up to 5% more sea-level rise compared to the coarser resolution. The sensitivity depends on the magnitude of outlet glacier retreat, which is implemented as a series of retreat masks following the ISMIP6 protocol. Without imposed retreat under atmosphere-only forcing, the resolution dependency exhibits an opposite behaviour with about 5% more sea-level contribution in the coarser resolution. The sea-level contribution indicates a converging behaviour ≤ 1 km horizontal resolution. A driving mechanism for differences is the ability to resolve the bed topography, which highly controls ice discharge to the ocean. Additionally, thinning and acceleration emerge to propagate further inland in high resolution for many glaciers. A major response mechanism is sliding (despite no climate-induced hydrological feedback is invoked), with an enhanced feedback on the effective normal pressure N at higher resolution leading to a larger increase in sliding speeds under scenarios with outlet glacier retreat.</p>

295. The meltwater feedbacks on ice dynamics, elevation versus lubrication

Authors: Basile de Fleurian; Petra Langebroek; Paul Halas

Journal: Unknown Journal

Publisher: Copernicus GmbH

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Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Meltwater|Elevation (ballistics)|Greenland ice sheet

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Open Access: N/A

Abstract: <p>In recent years, temperatures over the Greenland ice sheet have been rising leading to an increase in surface melt. Projections show that this augmentation of surface melt will continue in the future and spread to higher elevations. As it increases, melt leads to two different feedbacks on the dynamic of the Greenland ice sheet. This augmentation of melt lowers the ice surface and changes its overall geometry hence impacting the ice dynamics through ice deformation. The other feedback comes into play at the base of glaciers. Here, the increase of water availability will impact the distribution of water pressure at the base of glaciers and hence their sliding velocity. The first feedback is relatively well known and relies on our knowledge of the rheology and deformation of ice. The lubrication feedback acting at the bed of glaciers is however highly uncertain on time scales longer than a season. Here we apply the Ice Sheet System Model (ISSM) to a synthetic glacier which geometry is similar to the one of a Greenland ice sheet land terminating glacier. The dynamic contributions from ice deformation and sliding are separated to study their relative evolution. This is permitted by the use of a dynamical subglacial hydrology model that allows to link the basal sliding to the meltwater production through an appropriate friction law. The model is forced through a simple temperature distribution and a Positive Degree Day model which allows to apply a large range of different forcing scenarios. Of particular interest is the evolution of the distribution of the efficient and inefficient component of the subglacial drainage system and their different response to the distribution of melt during the year which directly impact the sliding regime at the base of the glacier.</p>

296. The impact of internal variability in ocean-induced melting on Totten Glacier

Authors: Felicity McCormack; Mathieu Morlighem; David Gwyther; Jason Roberts; Tyler Pelle

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/egusphere-egu2020-4196

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Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Antarctic ice sheet|Future sea level|Glacier mass balance

Funding: N/A

License: N/A

Open Access: N/A

Abstract: <p>The Totten Glacier, located in the Aurora Subglacial Basin of East Antarctica, drains a catchment containing approximately 3.5 m of global sea level rise equivalent ice mass. This glacier has been losing mass over recent decades, and modelling studies indicate that it is the most vulnerable glacier in East Antarctica to warming oceans and atmosphere over the coming century. Satellite altimetry shows high internal variability in ocean-forced melting of the Totten Ice Shelf; however, the extent to which this variability signal impacts the upstream ice sheet dynamics, and therefore its mass balance, is unknown. Here we use the Ice Sheet System Model (ISSM) combined with a plume and basal melting parameterisation called PICOP to investigate the impact of variability in ocean temperature on the evolution of Totten Glacier. We find that the southernmost portion of the Totten Glacier grounding line - from which the majority of the catchment's ice is channeled - is stable within only a limited range of background ocean temperatures close to present-day values. In the stable simulations, the magnitude of the ice mass flux depends on the extent to which the ice shelf is pinned on a bed topography rumple located approximately 10 km downstream of its grounding line, but the period of the mass flux is decadal to multi-decadal in each simulation, irrespective of the magnitude of the variability in ocean forcing. We further find that the impact of variability in ocean melt rates decreases as the mean background ocean temperature increases, suggesting that the mean state may have a relatively more important role in the evolution of the Totten Glacier than variability in ocean forcing. Our results have implications for detection and attribution of climate change and internal climate variability in modeling studies, and may inform fieldwork campaigns mapping bed topography in the Aurora Subglacial Basin.</p>

297. Exceptionally high heat flux needed to sustain the Northeast Greenland Ice Stream

Authors: Silje Smith-Johnsen; Basile de Fleurian; Nicole Schlegel; Helene Seroussi; Kerim Nisancioglu

Journal: The Cryosphere

Publisher: Copernicus GmbH

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Abstract: . The Northeast Greenland Ice Stream (NEGIS) currently drains more than 10 % of the Greenland Ice Sheet area and has recently undergone significant dynamic changes. It is therefore critical to accurately represent this feature when assessing the future contribution of Greenland to sea level rise. At present, NEGIS is reproduced in ice sheet models by inferring basal conditions using observed surface velocities. This approach helps estimate conditions at the base of the ice sheet but cannot be used to estimate the evolution of basal drag in time, so it is not a good representation of the evolution of the ice sheet in future climate warming scenarios. NEGIS is suggested to be initiated by a geothermal heat flux anomaly close to the ice divide, left behind by the movement of Greenland over the Icelandic plume. However, the heat flux underneath the ice sheet is largely unknown, except for a few direct measurements from deep ice core drill sites. Using the Ice Sheet System Model (ISSM), with ice dynamics coupled to a subglacial hydrology model, we investigate the possibility of initiating NEGIS by inserting heat flux anomalies with various locations and intensities. In our model experiment, a minimum heat flux value of 970 mW m⁻² located close to the East Greenland Ice-core Project (EGRIP) is required locally to reproduce the observed NEGIS velocities, giving basal melt rates consistent with previous estimates. The value cannot be attributed to geothermal heat flux alone and we suggest hydrothermal circulation as a potential explanation for the high local heat flux. By including high heat flux and the effect of water on sliding, we successfully reproduce the main characteristics of NEGIS in an ice sheet model without using data assimilation.

298. Modelling thermomechanical ice deformation using an implicit pseudo-transient method (FastICE v1.0) based on graphical processing units (GPUs)

Authors: Ludovic Räss; Aleksandar Licul; Frédéric Herman; Yury Y. Podladchikov; Jenny Suckale

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

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Abstract: . Ice sheets lose the majority of their mass through outlet glaciers or ice streams, corridors of fast ice moving multiple orders of magnitude more rapidly than the surrounding ice. The future stability of these corridors of fast-moving ice depends sensitively on the behaviour of their boundaries, namely shear margins, grounding zones and the basal sliding interface, where the stress field is complex and fundamentally three-dimensional. These boundaries are prone to thermomechanical localisation, which can be captured numerically only with high temporal and spatial resolution. Thus, better understanding the coupled physical processes that govern the response of these boundaries to climate change necessitates a non-linear, full Stokes model that affords high resolution and scales well in three dimensions. This paper's goal is to contribute to the growing toolbox for modelling thermomechanical deformation in ice by leveraging graphical processing unit (GPU) accelerators' parallel scalability. We propose FastICE, a numerical model that relies on pseudo-transient iterations to solve the implicit thermomechanical coupling between ice motion and temperature involving shear heating and a temperature-dependent ice viscosity. FastICE is based on the finite-difference discretisation, and we implement the pseudo-time integration in a matrix-free way. We benchmark the mechanical Stokes solver against the finite-element code Elmer/Ice and report good agreement among the results. We showcase a parallel version of FastICE to run on GPU-accelerated distributed memory machines, reaching a parallel efficiency of 99 %. We show that our model is particularly useful for improving our process-based understanding of flow localisation in the complex transition zones bounding rapidly moving ice.

299. Assimilation of surface observations in a transient marine ice sheet model using an ensemble Kalman filter

Authors: Fabien Gillet-Chaulet

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-14-811-2020

URL: <https://doi.org/10.5194/tc-14-811-2020>

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Abstract: . Marine-based sectors of the Antarctic Ice Sheet are increasingly contributing to sea level rise. The basal conditions exert an important control on the ice dynamics and can be propitious to instabilities in the grounding line position. Because the force balance is non-inertial, most ice flow models are now equipped with time-independent inverse methods to constrain the basal conditions from observed surface velocities. However, transient simulations starting from this initial state usually suffer from inconsistencies and are not able to reproduce observed trends. Here, using a synthetic flow line experiment, we assess the performance of an ensemble Kalman filter for the assimilation of transient observations of surface elevation and velocities in a marine ice sheet model. The model solves the shallow shelf equation for the force balance and the continuity equation for ice thickness evolution. The position of the grounding line is determined by the floatation criterion. The filter analysis estimates both the state of the model, represented by the surface elevation, and the basal conditions, with the simultaneous inversion of the basal friction and topography. The idealised experiment reproduces a marine ice sheet that is in the early stage of an unstable retreat. Using observation frequencies and uncertainties consistent with current observing systems, we find that the filter allows the accurate recovery of both the basal friction and topography after few assimilation cycles with relatively small ensemble sizes. In addition it is found that assimilating the surface observations has a positive impact on constraining the evolution of the grounding line during the assimilation window. Using the initialised state to perform century-scale forecast simulations, we show that grounding line retreat rates are in agreement with the reference; however remaining uncertainties in the basal conditions may lead to significant delays in the initiation of the unstable retreat. These results are encouraging for the application to real glacial systems.

300. Parameter sensitivity analysis of dynamic ice sheet models - numerical computations

Authors: Gong Cheng; Per Lötstedt

Journal: The Cryosphere

Publisher: Copernicus GmbH

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Keywords: N/A

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Open Access: N/A

Abstract: . The friction coefficient and the base topography of a stationary and a dynamic ice sheet are perturbed in two models for the ice: the full Stokes equations and the shallow shelf approximation. The sensitivity to the perturbations of the velocity and the height at the surface is quantified by solving the adjoint equations of the stress and the height equations providing weights for the perturbed data. The adjoint equations are solved numerically and the sensitivity is computed in several examples in two dimensions. A transfer matrix couples the perturbations at the base with the perturbations at the top. Comparisons are made with analytical solutions to simplified problems. The sensitivity to perturbations depends on their wavelengths and the distance to the grounding line. A perturbation in the topography has a direct effect at the ice surface above it, while a change in the friction coefficient is less visible there.

301. Glacial-cycle simulations of the Antarctic Ice Sheet with the Parallel Ice Sheet Model (PISM) - Part 1: Boundary conditions and climatic forcing

Authors: Torsten Albrecht; Ricarda Winkelmann; Anders Levermann

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-14-599-2020

URL: <https://doi.org/10.5194/tc-14-599-2020>

Publication Date: 2020-02-14

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References: 127

Cited by: 44 publications

Language: en

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Open Access: N/A

Abstract: . Simulations of the glacial-interglacial history of the Antarctic Ice Sheet provide insights into dynamic threshold behavior and estimates of the ice sheet's contributions to global sea-level changes for the past, present and future. However, boundary conditions are weakly constrained, in particular at the interface of the ice sheet and the bedrock. Also climatic forcing covering the last glacial cycles is uncertain, as it is based on sparse proxy data. We use the Parallel Ice Sheet Model (PISM) to investigate the dynamic effects of different choices of input data, e.g., for modern basal heat flux or reconstructions of past changes of sea level and surface temperature. As computational resources are limited, glacial-cycle simulations are performed using a comparably coarse model grid of 16 km and various parameterizations, e.g., for basal sliding, iceberg calving, or for past variations in precipitation and ocean temperatures. In this study we evaluate the model's transient sensitivity to corresponding parameter choices and to different boundary conditions over the last two glacial cycles and provide estimates of involved uncertainties. We also discuss isolated and combined effects of climate and sea-level forcing. Hence, this study serves as a "cookbook" for the growing community of PISM users and paleo-ice sheet modelers in general. For each of the different model uncertainties with regard to climatic forcing, ice and Earth dynamics, and basal processes, we select one representative model parameter that captures relevant uncertainties and motivates corresponding parameter ranges that bound the observed ice volume at present. The four selected parameters are systematically varied in a parameter ensemble analysis, which is described in a companion paper.

302. Projecting Antarctica's contribution to future sea level rise from basal ice shelf melt using linear response functions of 16 ice sheet models (LARMIP-2)

Authors: Anders Levermann; Ricarda Winkelmann; Torsten Albrecht; Heiko Goelzer; Nicholas R. Golledge; Ralf Greve; Philippe Huybrechts; Jim Jordan; Gunter Leguy; Daniel Martin; Mathieu Morlighem; Frank Pattyn; David Pollard; Aurelien Quiquet; Christian Rodehacke; Helene Seroussi; Johannes Sutter; Tong Zhang; Jonas Van Breedam; Reinhard Calov; Robert DeConto; Christophe Dumas; Julius Garbe; G. Hilmar Gudmundsson; Matthew J. Hoffman; Angelika Humbert; Thomas Kleiner; William H. Lipscomb; Malte Meinshausen; Esmond Ng; Sophie M. J. Nowicki; Mauro Perego; Stephen F. Price; Fuyuki Saito; Nicole-Jeanne Schlegel; Sainan Sun; Roderik S. W. van de Wal

Journal: Earth System Dynamics

Publisher: Copernicus GmbH

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Publication Date: 2020-02-13

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References: 162

Cited by: 84 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: U.S. Department of Energy, European Research Council, Japan Society for the Promotion of Science (and 1 more)

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Open Access: N/A

Abstract: . The sea level contribution of the Antarctic ice sheet constitutes a large uncertainty in future sea level projections. Here we apply a linear response theory approach to 16 state-of-the-art ice sheet models to estimate the Antarctic ice sheet contribution from basal ice shelf melting within the 21st century. The purpose of this computation is to estimate the uncertainty of Antarctica's future contribution to global sea level rise that arises from large uncertainty in the oceanic forcing and the associated ice shelf melting. Ice shelf melting is considered to be a major if not the largest perturbation of the ice sheet's flow into the ocean. However, by computing only the sea level contribution in response to ice shelf melting, our study is neglecting a number of processes such as surface-mass-balance-related contributions. In assuming linear response theory, we are able to capture complex temporal responses of the ice sheets, but we neglect any self-dampening or self-amplifying processes. This is particularly relevant in situations in which an instability is dominating the ice loss. The results obtained here are thus relevant, in particular wherever the ice loss is dominated by the forcing as opposed to an internal instability, for example in strong ocean warming scenarios. In order to allow for comparison the methodology was chosen to be exactly the same as in an earlier study (Levermann et al., 2014) but with 16 instead of 5 ice sheet models. We include uncertainty in the atmospheric warming response to carbon emissions (full range of CMIP5 climate model sensitivities), uncertainty in the oceanic transport to the Southern Ocean (obtained from the time-delayed and scaled oceanic subsurface warming in CMIP5 models in relation to the global mean surface warming), and the observed range of responses of basal ice shelf melting to oceanic warming outside the ice shelf cavity. This uncertainty in basal ice shelf

melting is then convoluted with the linear response functions of each of the 16 ice sheet models to obtain the ice flow response to the individual global warming path. The model median for the observational period from 1992 to 2017 of the ice loss due to basal ice shelf melting is 10.2 mm, with a likely range between 5.2 and 21.3 mm. For the same period the Antarctic ice sheet lost mass equivalent to 7.4 mm of global sea level rise, with a standard deviation of 3.7 mm (Shepherd et al., 2018) including all processes, especially surface-mass-balance changes. For the unabated warming path, Representative Concentration Pathway 8.5 (RCP8.5), we obtain a median contribution of the Antarctic ice sheet to global mean sea level rise from basal ice shelf melting within the 21st century of 17 cm, with a likely range (66th percentile around the mean) between 9 and 36 cm and a very likely range (90th percentile around the mean) between 6 and 58 cm. For the RCP2.6 warming path, which will keep the global mean temperature below 2 °C of global warming and is thus consistent with the Paris Climate Agreement, the procedure yields a median of 13 cm of global mean sea level contribution. The likely range for the RCP2.6 scenario is between 7 and 24 cm, and the very likely range is between 4 and 37 cm. The structural uncertainties in the method do not allow for an interpretation of any higher uncertainty percentiles. We provide projections for the five Antarctic regions and for each model and each scenario separately. The rate of sea level contribution is highest under the RCP8.5 scenario. The maximum within the 21st century of the median value is 4 cm per decade, with a likely range between 2 and 9 cm per decade and a very likely range between 1 and 14 cm per decade.

303. Investigating the Acceleration of Regional Sea Level Rise During the Satellite Altimeter Era

Authors: Benjamin D. Hamlington (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Thomas Frederikse (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); R. Steven Nerem (University of Colorado Boulder); John T. Fasullo (National Center for Atmospheric Research, Cooperative Institute for Research in Environmental Sciences Boulder CO USA); Surendra Adhikari (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Subjects: N/A

Keywords: acceleration, satellite altimetry, sea level, regional

Funding: National Aeronautics and Space Administration, National Aeronautics and Space Administration

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Abstract: The 25-year record of satellite altimeter-measured sea level has led to improvements in the understanding of sea level change on both regional and global scales. However, the extent to which the pattern of regional sea level rise measured by altimeters is representative of the forced sea level response remains an open question. Internal variability both contributes to regional sea level changes on short time scales and masks the pattern of forced trend and acceleration associated with anthropogenic global warming. Recent studies have demonstrated that the forced trend pattern of regional rise has begun to emerge, although there has been no assessment of a possible associated acceleration. Here, the regional acceleration pattern is estimated from the altimeter sea level record and assessed with regard to the influence of internal variability. While the dominant features in the acceleration pattern can be attributed to internal variability, there is an indication that the forced acceleration pattern may emerge as the record continues to lengthen.

304. The role of subglacial hydrology in ice streams with elevated geothermal heat flux

Authors: Silje Smith-Johnsen; Basile de Fleurian; Kerim H. Nisancioglu

Journal: Journal of Glaciology

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/jog.2020.8

URL: <https://doi.org/10.1017/jog.2020.8>

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References: 51

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

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Open Access: N/A

Abstract: The spatial distribution of geothermal heat flux (GHF) under ice sheets is largely unknown. Nonetheless, it is an important boundary condition in ice-sheet models, and suggested to control part of the complex surface velocity patterns observed in some regions. Here we investigate the effect of including subglacial hydrology when modelling ice streams with elevated GHF. We use an idealised ice stream geometry and a thermomechanical ice flow model coupled to subglacial hydrology in the Ice Sheet System Model (ISSM). Our results show that the dynamic response of the ice stream to elevated GHF is greatly enhanced when including the interactive subglacial hydrology. On the other hand, the impact of GHF on ice temperature is reduced when subglacial hydrology is included. In conclusion, the sensitivity of ice stream dynamics to GHF is likely to be underestimated in studies neglecting subglacial hydrology.

305. Increased Subglacial Sediment Discharge in a Warming Climate: Consideration of Ice Dynamics, Glacial Erosion, and Fluvial Sediment Transport

Authors: Ian Delaney (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Surendra Adhikari (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Publication Date: 2020-02-08

Volume/Issue: Volume 47, Issue 7

References: 77

Cited by: 54 publications

Language: en

Subjects: N/A

Keywords: subglacial erosion, sediment transport, climate change, glacier modeling, alpine sediment dynamics (and 1 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We evaluate changes to subglacial sediment discharge during glacier retreat by considering ice dynamics, bedrock erosion, and sediment transport processes. Coupling these components together within a single framework, we simulate sediment discharge from synthetic alpine glaciers experiencing accelerated glacier melt for 100 years. We find that sediment discharge increases by about 8 times the steady glacier values by the end of the simulation. The enhanced sediment discharge persists through peak water discharge and despite annual bedrock erosion volume decreasing by approximately 30% from the initial value. The greater sediment discharge results from increased melt at the glaciers' higher region, where transport was limited prior to glacier retreat. These findings suggest that large increases in sediment discharge may occur as glaciers retreat. The magnitude of the sediment discharge increase primarily depends on the quantity of sediment stored subglacially.

306. Long-term projections of sea-level rise from ice sheets

Authors: Nicholas R. Golledge (GNS Science Lower Hutt New Zealand, Victoria University of Wellington)

Journal: WIREs Climate Change

Publisher: Wiley

DOI: 10.1002/wcc.634

URL: <https://doi.org/10.1002/wcc.634>

Publication Date: 2020-01-17

Volume/Issue: Volume 11, Issue 2

References: 153

Cited by: 23 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Ministry for Business Innovation and Employment, National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: Under future climate change scenarios it is virtually certain that global mean sea level will continue to rise. But the rate at which this occurs, and the height and time at which it might stabilize, are uncertain. The largest potential contributors to sea level are the Greenland and Antarctic ice sheets, but these may take thousands of years to fully adjust to environmental changes. Modeled projections of how these ice masses will evolve in the future are numerous, but vary both in complexity and projection timescale. Typically, there is greater agreement between models in the present century than over the next millennium. This reflects uncertainty in the physical processes that dominate ice-sheet change and also feedbacks in the ice-atmosphere-ocean system, and how these might lead to nonlinear behavior. Satellite observations help constrain short-term projections of ice-sheet change but these records are still too short to capture the full ice-sheet response. Conversely, geological records can be used to inform long-term ice-sheet simulations but are prone to large uncertainties, meaning that they are often unable to adequately confirm or refute the operation of particular processes. Because of these limitations there is a clear need to more accurately reconstruct sea level changes during periods of the past, to improve the spatial and temporal extent of current ice sheet observations, and to robustly attribute observed changes to driving mechanisms. Improved future projections will require models that capture a more extensive suite of physical processes than are presently incorporated, and which better quantify the associated uncertainties. This article is categorized under: Climate Models and Modeling > Knowledge Generation with Models

307. Sensitivity of Greenland ice sheet projections to spatial resolution in higher-order simulations: the AWI contribution to ISMIP6-Greenland using ISSM

Authors: Martin Rückamp; Heiko Goelzer; Angelika Humbert

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/tc-2019-329

URL: <https://doi.org/10.5194/tc-2019-329>

Publication Date: 2020-01-14

Volume/Issue: N/A

References: N/A

Cited by: 1 publications

Language: N/A

Subjects: N/A

Keywords: Greenland ice sheet|Forcing (mathematics)|Ice-sheet model|Antarctic ice sheet|Future sea level|Glacier mass balance

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: . Projections of the contribution of the Greenland ice sheet to future sea-level rise include uncertainties primarily due to the imposed climate forcing and the initial state of the ice sheet model. Several state-of-the-art ice flow models are currently being employed on various grid resolutions to estimate future mass changes in the framework of the Ice Sheet Model Intercomparison Project for CMIP6 (ISMIP6). Here we investigate the sensitivity to grid resolution on centennial sea-level contributions from the Greenland ice sheet and study the mechanism at play. To this end, we employ the finite-element higher-order ice flow model ISSM and conduct experiments with four different horizontal resolutions, namely 4, 2, 1 and 0.75 km. We run the simulation based on the ISMIP6 core GCM MIROC5 under the high emission scenario RCP8.5 and consider both atmospheric and oceanic forcing in full and separate scenarios. Under the full scenarios, finer simulations unveil up to ~5 % more sea-level rise compared to the coarser resolution. The sensitivity depends on the magnitude of outlet glacier retreat, which is implemented as a series of retreat masks following the ISMIP6 protocol. Without imposed retreat under atmosphere-only forcing, the resolution dependency exhibits an opposite behaviour with about ~ 5 % more sea-level contribution in the coarser resolution. The sea-level contribution indicates a converging behaviour

308. Interpreting englacial layer deformation in the presence of complex ice flow history with synthetic radargrams

Authors: Cooper W. Elsworth; Dustin M. Schroeder; Matthew R. Siegfried

Journal: Annals of Glaciology

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/aog.2019.41

URL: <https://doi.org/10.1017/aog.2019.41>

Publication Date: 2020-01-06

Volume/Issue: Volume 61, Issue 81

References: 69

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: Fast ice flow on the Antarctic continent constitutes much of the mass loss from the ice sheet. However, geophysical methods struggle to constrain ice flow history at depth, or separate the signatures of topography, ice dynamics and basal conditions on layer structure. We develop and demonstrate a methodology to compare layer signatures in multiple airborne radar transects in order to characterize ice flow at depth, or improve coverage of existing radar surveys. We apply this technique to generate synthetic, along-flow radargrams and compare different deformation regimes to observed radargram structure. Specifically, we investigate flow around the central sticky spot of Whillans Ice Stream, West Antarctica. Our study suggests that present-day velocity flowlines are insufficient to characterize flow at depth as expressed in layer geometry, and streaklines provide a better characterization of flow around a basal sticky spot. For Whillans Ice Stream, this suggests that ice flow wraps around the central sticky spot, supported by idealized flow simulations. While tracking isochrone translation and rotation across survey lines is complex, we demonstrate that our approach to combine radargram interpretation and modeling can reveal critical details of past ice flow.

309. Maximum Southwest Greenland Ice Sheet Recession in the Early Holocene

Authors: A. J. Lesnek (University of New Hampshire); J. P. Briner (Department of Geology University at Buffalo Buffalo NY USA); N. E. Young (Columbia University); J. K. Cuzzone (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019gl083164

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Publication Date: 2020-01-04

Volume/Issue: Volume 47, Issue 1

References: 105

Cited by: 26 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation

License: N/A

Open Access: N/A

Abstract: Establishing the timing of maximum Holocene warmth in the Arctic is critical for understanding global climate system response to external forcing. In Greenland, challenges in obtaining climate records that span the full Holocene have hampered efforts to robustly identify when the Holocene Thermal Maximum occurred. Reconstructing land-based ice sheet history can fill this gap because these ice sheet regions respond sensitively to summer temperature. We synthesize new and published ^{10}Be and ^{14}C ages from southwest Greenland to map Greenland ice sheet margin positions from 12 to 7 ka and calculate retreat rates from 12 to 0 ka. We found that maximum Greenland ice sheet recession occurred between ~ 10.4 and 9.1 ka. Our reconstruction suggests that summer air temperatures in southwest Greenland were highest from ~ 10.4 to 9.1 ka, providing support for an early regional Holocene Thermal Maximum. These results can serve as benchmarks for comparison with ice sheet and climate model simulations.

310. Sensitivity of the Northeast Greenland Ice Stream to Geothermal Heat

Authors: S. Smith-Johnsen (University of Bergen, Bjerknes Centre for Climate Research); N.-J. Schlegel (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); B. de Fleurian (University of Bergen, Bjerknes Centre for Climate Research); K. H. Nisancioglu (University of Oslo, University of Bergen, Bjerknes Centre for Climate Research)

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019jf005252

URL: <https://doi.org/10.1029/2019jf005252>

Publication Date: 2019-12-20

Volume/Issue: Volume 125, Issue 1

References: 43

Cited by: 21 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: European Research Council

License: CC BY 4.0

Open Access: N/A

Abstract: Recent observations of ice flow surface velocities have helped improve our understanding of basal processes on Greenland and Antarctica, though these processes still constitute some of the largest uncertainties driving ice flow change today. The Northeast Greenland Ice Stream is driven largely by basal sliding, believed to be related to subglacial hydrology and the availability of heat. Characterization of the uncertainties associated with Northeast Greenland Ice Stream is crucial for constraining Greenland's potential contribution to sea level rise in the upcoming centuries. Here, we expand upon past work using the Ice Sheet System Model to quantify the uncertainties in models of the ice flow in the Northeast Greenland Ice Stream by perturbing the geothermal heat flux. Utilizing a subglacial hydrology model simulating sliding beneath the Greenland Ice Sheet, we investigate the sensitivity of the Northeast Greenland Ice Stream ice flow to various estimates of geothermal heat flux, and implications of basal heat flux uncertainties on modeling the hydrological processes beneath Greenland's major ice stream. We find that the uncertainty due to sliding at the bed is 10 times greater than the uncertainty associated with internal ice viscosity. Geothermal heat flux dictates the size of the area of the subglacial drainage system and its efficiency. The uncertainty of ice discharge from the Northeast Greenland Ice Stream to the ocean due to uncertainties in the geothermal heat flux is estimated at 2.10 Gt/yr. This highlights the urgency in obtaining better constraints on the highly uncertain subglacial hydrology parameters.

311. Sliding Relations for Glacier Slip With Cavities Over Three-Dimensional Beds

Authors: Christian Helanow (Iowa State University); Neal R. Iverson (Iowa State University); Lucas K. Zoet (University of Wisconsin-Madison); Olivier Gagliardini (Institut de Recherche pour le Développement (IRD))

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019gl084924

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Publication Date: 2019-12-17

Volume/Issue: Volume 47, Issue 3

References: 53

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Agence Nationale de la Recherche, National Science Foundation

License: N/A

Open Access: N/A

Abstract: Results of glacier flow models and associated estimates of future sea level rise depend sensitively on the prescribed relation between shear stress and slip velocity at the glacier bed. Using a fully three-dimensional numerical model of ice flow, we compute steady-state sliding relations for where ice slips over a rock bed with three-dimensional, periodic topography. In agreement with studies of two-dimensional beds, water-filled cavities that form down-glacier from bedforms cause basal shear stress to peak at a threshold slip velocity and decrease at higher velocities (i.e., rate-weakening drag). However, the shear stress magnitude and extent of rate-weakening drag depend systematically on lateral topographic variations not considered previously. Moreover, steep up-glacier-facing slopes of bedforms can result in shear stress that increases monotonically over a wide range of slip velocity, helping to stabilize slip. These results highlight the potential variability of sliding relations and their likely sensitivity to the morphological diversity of glacier beds.

312. Deep glacial troughs and stabilizing ridges unveiled beneath the margins of the Antarctic ice sheet

Authors: Mathieu Morlighem; Eric Rignot; Tobias Binder; Donald Blankenship; Reinhard Drews; Graeme Eagles; Olaf Eisen; Fausto Ferraccioli; René Forsberg; Peter Fretwell; Vikram Goel; Jamin S. Greenbaum; Hilmar Gudmundsson; Jinxue Guo; Veit Helm; Coen Hofstede; Ian Howat; Angelika Humbert; Wilfried Jokat; Nanna B. Karlsson; Won Sang Lee; Kenichi Matsuoka; Romain Millan; Jeremie Mouginot; John Paden; Frank Pattyn; Jason Roberts; Sebastian Rosier; Antonia Ruppel; Helene Seroussi; Emma C. Smith; Daniel Steinhage; Bo Sun; Michiel R. van den Broeke; Tas D. van Ommen; Melchior van Wessem; Duncan A. Young

Journal: Nature Geoscience

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s41561-019-0510-8

URL: <https://doi.org/10.1038/s41561-019-0510-8>

Publication Date: 2019-12-12

Volume/Issue: Volume 13, Issue 2

References: 33

Cited by: 536 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration, National Science Foundation, Deutsche Forschungsgemeinschaft (and 2 more)

License: N/A

Open Access: N/A

Abstract: N/A

313. Illustrative Analysis of Probabilistic Sea Level Rise Hazard

Authors: Matthew A. Thomas (Department of Civil, Construction, and Environmental Engineering, Marquette University, Milwaukee, Wisconsin); Ting Lin (Texas Tech University)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-19-0320.1

URL: <https://doi.org/10.1175/jcli-d-19-0320.1>

Publication Date: 2019-12-06

Volume/Issue: Volume 33, Issue 4

References: 29

Cited by: 7 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: Marquette University

License: N/A

Open Access: N/A

Abstract: Sea level rise results from several contributing physical processes, including ocean thermal expansion and glacier and ice sheet mass loss. Future projections of sea level remain highly uncertain due to several sources of aleatory and epistemic uncertainty. Quantifying different sources of sea level rise involves considering possible pathways of future radiative forcing and integrating models of different sea level rise processes. The probabilistic hazard analysis strategy has been proposed for combining sea level rise prediction models and climate forcing scenarios to examine sea level rise prediction uncertainty and the sources of this uncertainty. In this study we carry out an illustrative probabilistic sea level rise hazard analysis using ensembles of sea level rise predictions and emissions scenarios from the literature. This illustrative analysis allows us to estimate the probability that sea level rise will exceed a specified threshold at a given location and time and highlights how sea level rise uncertainty is sensitive to scenario inputs and sea level rise projection modeling choices. Probabilistic hazard is depicted for Earth using sea level rise hazard maps. We also demonstrate how hazard deaggregation can help us quantify the relative contributions of sea level rise sources, prediction models, and climate forcing scenarios to sea level rise hazard. The ice sheet contribution to sea level rise has a large impact on probabilistic projection of sea level rise due to the disagreements between current ice sheet models related to differences in modeling ice sheet instability.

314. Impact of Iceberg Calving on the Retreat of Thwaites Glacier, West Antarctica Over the Next Century With Different Calving Laws and Ocean Thermal Forcing

Authors: Hongju Yu (University of California Irvine); Eric Rignot (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Helene Seroussi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Mathieu Morlighem (University of California Irvine); Youngmin Choi (University of California Irvine)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019gl084066

URL: <https://doi.org/10.1029/2019gl084066>

Publication Date: 2019-12-04

Volume/Issue: Volume 46, Issue 24

References: 52

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: Thwaites Glacier, West Antarctica, has been a major contributor to global sea level rise over the past decades. Prior studies illustrated the critical role of ice shelf melt and iceberg calving based on cliff height in driving the retreat of Thwaites glacier. Here, we simulate its evolution with various calving laws and rates of frontal melt by the ocean in the absence of a buttressing ice shelf. Over the next century, we find that volume losses increase by 15-160% with a von Mises calving law compared to the case where the initial ice shelf is kept and the ice front is fixed at its current position, 10-20% with a buoyancy-driven calving law, and 5-50% with frontal melt caused by ocean thermal forcing. Bed topography exerts the ultimate control on the evolution of Thwaites. In all simulations, once Thwaites Glacier retreats past the western subglacial ridge, the retreat becomes rapidly unstoppable.

315. SELEN4 (SELEN version 4.0): a Fortran program for solving the gravitationally and topographically self-consistent sea-level equation in glacial isostatic adjustment modeling

Authors: Giorgio Spada; Daniele Melini

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-12-5055-2019

URL: <https://doi.org/10.5194/gmd-12-5055-2019>

Publication Date: 2019-12-04

Volume/Issue: Volume 12, Issue 12

References: 95

Cited by: 41 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: . We present SELEN4 (SeaLevel EquationN solver), an open-source program written in Fortran 90 that simulates the glacial isostatic adjustment (GIA) process in response to the melting of the Late Pleistocene ice sheets. Using a pseudo-spectral approach complemented by a spatial discretization on an icosahedron-based spherical geodesic grid, SELEN4 solves a generalized sea-level equation (SLE) for a spherically symmetric Earth with linear viscoelastic rheology, taking the migration of the shorelines and the rotational feedback on sea level into account. The approach is gravitationally and topographically self-consistent, since it considers the gravitational interactions between the solid Earth, the cryosphere, and the oceans, and it accounts for the evolution of the Earth's topography in response to changes in sea level. The SELEN4 program can be employed to study a broad range of geophysical effects of GIA, including past relative sea-level variations induced by the melting of the Late Pleistocene ice sheets, the time evolution of paleogeography and of the ocean function since the Last Glacial Maximum, the history of the Earth's rotational variations, present-day geodetic signals observed by Global Navigation Satellite Systems, and gravity field variations detected by satellite gravity missions like GRACE (the Gravity Recovery and Climate Experiment). The "GIA fingerprints" constitute a standard output of SELEN4. Along with the source code, we provide a supplementary document with a full account of the theory, some numerical results obtained from a standard run, and a user guide. Originally, the SELEN program was conceived by Giorgio Spada (GS) in 2005 as a tool for students eager to learn about GIA, and it has been the first SLE solver made available to the community.

316. Totten Glacier subglacial hydrology determined from geophysics and modeling

Authors: C.F. Dow; F.S. McCormack; D.A. Young; J.S. Greenbaum; J.L. Roberts; D.D. Blankenship

Journal: Earth and Planetary Science Letters

Publisher: Elsevier BV

DOI: 10.1016/j.epsl.2019.115961

URL: <https://doi.org/10.1016/j.epsl.2019.115961>

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Volume/Issue: Volume 531

References: 48

Cited by: 33 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Scientific Committee on Antarctic Research, Natural Sciences and Engineering Research Council of Canada, Canada Research Chairs (and 10 more)

License: N/A

Open Access: N/A

Abstract: N/A

317. Thermal Conductivity of Snow, Firn, and Porous Ice From 3-D Image-Based Computations

Authors: Neige Calonne (Université Grenoble Alpes, Université de Toulouse, Météo-France, CNRS, CNRM, Centre d'Études de la Neige Grenoble France); Lucas Milliancourt (Grenoble INP Institute of Engineering, 3SR Université Grenoble Alpes, CNRS Grenoble France, Institut de Recherche pour le Développement (IRD)); Alexis Burr (Grenoble INP Institute of Engineering, SIMaP Université Grenoble Alpes, CNRS Grenoble France, Institut de Recherche pour le Développement (IRD)); Armelle Philip (Institut de Recherche pour le Développement (IRD)); Christophe L. Martin (Grenoble INP Institute of Engineering, SIMaP Université Grenoble Alpes, CNRS Grenoble France); Frederic Flin (Université Grenoble Alpes, Université de Toulouse, Météo-France, CNRS, CNRM, Centre d'Études de la Neige Grenoble France); Christian Geindreau (Grenoble INP Institute of Engineering, 3SR Université Grenoble Alpes, CNRS Grenoble France)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Volume/Issue: Volume 46, Issue 22

References: 59

Cited by: 40 publications

Language: en

Subjects: N/A

Keywords: ice, snow, firn, conductivity, tomography (and 1 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Estimating thermal conductivity of snow, firn, and porous ice is key for modeling the thermal regime of alpine and polar glaciers. Whereas thermal conductivity of snow was widely investigated, studies on firn and porous ice are very scarce. This study presents the effective thermal conductivity tensor computed from 64 3-D images of microstructures of snow, antarctic firn, and porous ice at -3, -20, and -60 degreesC. We show that, in contrast with snow, conductivity of firn and porous ice correlates linearly with density, is approximately isotropic, and is largely impacted by temperature. We report that performances of commonly used estimates of thermal conductivity vary largely with density. In particular, formulas designed for snow lead to significant underestimations when applied to denser ice structures. We present a new formulation to accurately estimate the thermal conductivity throughout the whole density range, from fresh snow to bubbly ice, and for any temperature conditions encountered in glaciers.

318. Constraining the geothermal heat flux in Greenland at regions of radar-detected basal water

Authors: Soroush Rezvanbehbahani; Leigh A. Stearns; C. J. van der Veen; Gordon K. A. Oswald; Ralf Greve

Journal: Journal of Glaciology

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/jog.2019.79

URL: <https://doi.org/10.1017/jog.2019.79>

Publication Date: 2019-11-04

Volume/Issue: Volume 65, Issue 254

References: 73

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: The spatial distribution of basal water critically impacts the evolution of ice sheets. Current estimates of basal water distribution beneath the Greenland Ice Sheet (GrIS) contain large uncertainties due to poorly constrained boundary conditions, primarily from geothermal heat flux (GHF). The existing GHF models often contradict each other and implementing them in numerical ice-sheet models cannot reproduce the measured temperatures at ice core locations. Here we utilize two datasets of radar-detected basal water in Greenland to constrain the GHF at regions with a thawed bed. Using the three-dimensional ice-sheet model SICOPOLIS, we iteratively adjust the GHF to find the minimum GHF required to reach the bed to the pressure melting point, GHF_{pmp}, at locations of radar-detected basal water. We identify parts of the central-east, south and northwest Greenland with significantly high GHF_{pmp}. Conversely, we find that the majority of low-elevation regions of west Greenland and parts of northeast have very low GHF_{pmp}. We compare the estimated constraints with the available GHF models for Greenland and show that GHF models often do not honor the estimated constraints. Our results highlight the need for community effort to reconcile the discrepancies between radar data, GHF models, and ice core information.

319. Satellite Remote Sensing of the Greenland Ice Sheet Ablation Zone: A Review

Authors: Matthew Cooper (University of California Los Angeles); Laurence Smith (University of California Los Angeles, Brown University)

Journal: Remote Sensing

Publisher: MDPI AG

DOI: 10.3390/rs11202405

URL: <https://doi.org/10.3390/rs11202405>

Publication Date: 2019-10-17

Volume/Issue: Volume 11, Issue 20

References: 364

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration

License: CC BY 4.0

Open Access: N/A

Abstract: The Greenland Ice Sheet is now the largest land ice contributor to global sea level rise, largely driven by increased surface meltwater runoff from the ablation zone, i.e., areas of the ice sheet where annual mass losses exceed gains. This small but critically important area of the ice sheet has expanded in size by ~50% since the early 1960s, and satellite remote sensing is a powerful tool for monitoring the physical processes that influence its surface mass balance. This review synthesizes key remote sensing methods and scientific findings from satellite remote sensing of the Greenland Ice Sheet ablation zone, covering progress in (1) radar altimetry, (2) laser (lidar) altimetry, (3) gravimetry, (4) multispectral optical imagery, and (5) microwave and thermal imagery. Physical characteristics and quantities examined include surface elevation change, gravimetric mass balance, reflectance, albedo, and mapping of surface melt extent and glaciological facies and zones. The review concludes that future progress will benefit most from methods that combine multi-sensor, multi-wavelength, and cross-platform datasets designed to discriminate the widely varying surface processes in the ablation zone. Specific examples include fusing laser altimetry, radar altimetry, and optical stereophotogrammetry to enhance spatial measurement density, cross-validate surface elevation change, and diagnose radar elevation bias; employing dual-frequency radar, microwave scatterometry, or combining radar and laser altimetry to map seasonal snow depth; fusing optical imagery, radar imagery, and microwave scatterometry to discriminate between snow, liquid water, refrozen meltwater, and bare ice near the equilibrium line altitude; combining optical reflectance with laser altimetry to map supraglacial lake, stream, and crevasse bathymetry; and monitoring the inland migration of snowlines, surface melt extent, and supraglacial hydrologic features.

320. Assessing Uncertainty in the Dynamical Ice Response to Ocean Warming in the Amundsen Sea Embayment, West Antarctica

Authors: Isabel J. Nias (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, University of Maryland, University of Bristol); Stephen L. Cornford (Department of Geography Swansea University Swansea UK); Tamsin L. Edwards (King's College London); Noel Gourmelen (University of Edinburgh); Antony J. Payne (University of Bristol)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2019gl084941

URL: <https://doi.org/10.1029/2019gl084941>

Publication Date: 2019-10-15

Volume/Issue: Volume 46, Issue 20

References: 49

Cited by: 30 publications

Language: en

Subjects: N/A

Keywords: Antarctica, calibration, grounding lines, sea-level rise, ice sheet modeling

Funding: European Space Agency, Natural Environment Research Council

License: CC BY 4.0

Open Access: N/A

Abstract: Ice mass loss from the Amundsen Sea Embayment ice streams in West Antarctica is a major source of uncertainty in projections of future sea level rise. Physically based ice flow models rely on a number of parameters that represent unobservable quantities and processes, and accounting for uncertainty in these parameters can lead to a wide range of dynamic responses. Here we perform a Bayesian calibration of a perturbed parameter ensemble, in which we score each ensemble member on its ability to match the magnitude and broad spatial pattern of present-day observations of ice sheet surface elevation change. We apply an idealized melt rate forcing to extend the most likely simulations forward to 2200. We find that diverging grounding line response between ensemble members drives an exaggeration in the upper tail of the distribution of sea level rise by 2200, demonstrating that extreme future outcomes cannot be excluded.

321. Exceptionally High Geothermal Heat Flux Needed to Sustain the Northeast Greenland Ice Stream

Authors: Silje Smith-Johnsen; Basile de Fleurian; Nicole Schlegel; Helene Seroussi; Kerim Nisancioglu

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/tc-2019-212

URL: <https://doi.org/10.5194/tc-2019-212>

Publication Date: 2019-10-01

Volume/Issue: N/A

References: N/A

Cited by: 1 publications

Language: N/A

Subjects: N/A

Keywords: Ice divide|Greenland ice sheet|Ice-sheet model|Geothermal heating

Funding: European Research Council

License: CC BY 4.0

Open Access: N/A

Abstract: . The Northeast Greenland Ice Stream (NEGIS) currently drains more than 10 % of the Greenland Ice Sheet area, and has recently undergone significant dynamic changes. It is therefore critical to accurately represent this feature when assessing the future contribution of Greenland to sea level rise. At present, NEGIS is reproduced in ice sheet models by inferring basal conditions using observed surface velocities. This approach helps estimate conditions at the base of the ice sheet, but cannot be used to estimate the evolution of basal drag in time, so it is not a good representation of the evolution of the ice sheet in future climate warming scenarios. NEGIS is suggested to be initiated by a geothermal heat flux anomaly close to the ice divide, left behind by the movement of Greenland over the Icelandic plume. However, the heat flux underneath the ice sheet is largely unknown, except for a few direct measurements from deep ice core drill sites. Using the Ice Sheet System Model (ISSM), with ice dynamics coupled to a subglacial hydrology model, we investigate the possibility of initiating NEGIS by inserting hotspots with various locations and intensities. We find that a minimum geothermal heat flux value of 970 mW/m² located close to EastGRIP is required locally to reproduce the observed NEGIS velocities, consistent with previous estimates. By including high geothermal heat flux and the effect of water on sliding, we successfully reproduce the main characteristics of NEGIS in an ice sheet model without using data assimilation.

322. Estimating Greenland tidewater glacier retreat driven by submarine melting

Authors: Donald A. Slater; Fiamma Straneo; Denis Felikson; Christopher M. Little; Heiko Goelzer; Xavier Fettweis; James Holte

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-13-2489-2019

URL: <https://doi.org/10.5194/tc-13-2489-2019>

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Abstract: . The effect of the North Atlantic Ocean on the Greenland Ice Sheet through submarine melting of Greenland's tidewater glacier calving fronts is thought to be a key driver of widespread glacier retreat, dynamic mass loss and sea level contribution from the ice sheet. Despite its critical importance, problems of process complexity and scale hinder efforts to represent the influence of submarine melting in ice-sheet-scale models. Here we propose parameterizing tidewater glacier terminus position as a simple linear function of submarine melting, with submarine melting in turn estimated as a function of subglacial discharge and ocean temperature. The relationship is tested, calibrated and validated using datasets of terminus position, subglacial discharge and ocean temperature covering the full ice sheet and surrounding ocean from the period 1960-2018. We demonstrate a statistically significant link between multi-decadal tidewater glacier terminus position change and submarine melting and show that the proposed parameterization has predictive power when considering a population of glaciers. An illustrative 21st century projection is considered, suggesting that tidewater glaciers in Greenland will undergo little further retreat in a low-emission RCP2.6 scenario. In contrast, a high-emission RCP8.5 scenario results in a median retreat of 4.2 km, with a quarter of tidewater glaciers experiencing retreat exceeding 10 km. Our study provides a long-term and ice-sheet-wide assessment of the sensitivity of tidewater glaciers to submarine melting and proposes a practical and empirically validated means of incorporating ocean forcing into models of the Greenland ice sheet.

323. Viscous Effects in the Solid Earth Response to Modern Antarctic Ice Mass Flux: Implications for Geodetic Studies of WAIS Stability in a Warming World

Authors: E. Powell (Harvard University); N. Gomez (McGill University); C. Hay (Department of Earth and Environmental Sciences, Boston College, Chestnut Hill, Massachusetts); K. Latychev (Harvard University, Columbia University); J. X. Mitrovica (Harvard University)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-19-0479.1

URL: <https://doi.org/10.1175/jcli-d-19-0479.1>

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Abstract: The West Antarctic Ice Sheet (WAIS) overlies a thin, variable-thickness lithosphere and a shallow upper-mantle region of laterally varying and, in some regions, very low ($\sim 10^{18}$ Pa s) viscosity. We explore the extent to which viscous effects may affect predictions of present-day geoid and crustal deformation rates resulting from Antarctic ice mass flux over the last quarter century and project these calculations into the next half century, using viscoelastic Earth models of varying complexity. Peak deformation rates at the end of a 25-yr simulation predicted with an elastic model underestimate analogous predictions that are based on a 3D viscoelastic Earth model (with minimum viscosity below West Antarctica of 10^{18} Pa s) by ~ 15 and ~ 3 mm yr $^{-1}$ in the vertical and horizontal directions, respectively, at sites overlying low-viscosity mantle and close to high rates of ice mass flux. The discrepancy in uplift rate can be reduced by adopting 1D Earth models tuned to the regional average viscosity profile beneath West Antarctica. In the case of horizontal crustal rates, adopting 1D regional viscosity models is no more accurate in recovering predictions that are based on 3D viscosity models than calculations that assume a purely elastic Earth. The magnitude and relative contribution of viscous relaxation to crustal deformation rates will likely increase significantly in the next several decades, and the adoption of 3D viscoelastic Earth models in analyses of geodetic datasets [e.g., Global Navigation Satellite System (GNSS); Gravity Recovery and Climate Experiment (GRACE)] will be required to accurately estimate the magnitude of Antarctic modern ice mass flux in the progressively warming world.

324. Quantification of Surface Forcing Requirements for a Greenland Ice Sheet Model Using Uncertainty Analyses

Authors: Nicole-Jeanne Schlegel (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Eric Y. Larour (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Language: en

Subjects: N/A

Keywords: surface mass balance, ice dynamics, modeling, ice sheet

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Abstract: The Greenland Ice Sheet is a substantial reservoir of almost 7 m of sea level equivalent, and on average, climate dictates 60% of its sea level contribution. Changes in ice discharge, driven by perturbations in outlet glacier ice dynamics, constitute the rest. Climate also affects ice discharge, since the flow of interior ice feeding the outlet glaciers evolves in response to surface changes over time. Here, using an ice sheet model and uncertainty quantification, we explore ice flow sensitivity to climate-driven changes in ice surface topography on multidecadal timescales. We find that changes in surface forcing near large outlet glaciers can influence region-wide ice flow. Improvements to climate products should be prioritized in these areas, especially in the Central West and Southeast. Results also suggest that over most of Greenland, surface forcing should be supplied at a spatial resolution of 21 km or finer to accurately simulate ice response to climate change.

325. Eemian Greenland ice sheet simulated with a higher-order model shows strong sensitivity to surface mass balance forcing

Authors: Andreas Plach; Kerim H. Nisancioglu; Petra M. Langebroek; Andreas Born; Sébastien Leclerc'h

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-13-2133-2019

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Keywords: N/A

Funding: European Commission

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Open Access: N/A

Abstract: . The Greenland ice sheet contributes increasingly to global sea level rise. Its history during past warm intervals is a valuable reference for future sea level projections. We present ice sheet simulations for the Eemian interglacial period (~130 000 to 115 000 years ago), a period with warmer-than-present summer climate over Greenland. The evolution of the Eemian Greenland ice sheet is simulated with a 3-D higher-order ice sheet model, forced with a surface mass balance derived from regional climate simulations. Sensitivity experiments with various surface mass balances, basal friction, and ice flow approximations are discussed. The surface mass balance forcing is identified as the controlling factor setting the minimum in Eemian ice volume, emphasizing the importance of a reliable surface mass balance model. Furthermore, the results indicate that the surface mass balance forcing is more important than the representation of ice flow for simulating the large-scale ice sheet evolution. This implies that modeling of the future contribution of the Greenland ice sheet to sea level rise highly depends on an accurate surface mass balance.

326. Low-End Probabilistic Sea-Level Projections

Authors: Gonéri Le Cozannet (BRGM, French Geological Survey, 3 Avenue Claude Guillemin, 45060 Orléans CEDEX, France); Rémi Thiéblemont (BRGM, French Geological Survey, 3 Avenue Claude Guillemin, 45060 Orléans CEDEX, France); Jeremy Rohmer (BRGM, French Geological Survey, 3 Avenue Claude Guillemin, 45060 Orléans CEDEX, France); Déborah Idier (BRGM, French Geological Survey, 3 Avenue Claude Guillemin, 45060 Orléans CEDEX, France); Jean-Charles Manceau (BRGM, French Geological Survey, 3 Avenue Claude Guillemin, 45060 Orléans CEDEX, France); Robin Quique (BRGM, French Geological Survey, 3 Avenue Claude Guillemin, 45060 Orléans CEDEX, France)

Journal: Water

Publisher: MDPI AG

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Language: en

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Keywords: N/A

Funding: ERA4CS, Agence Nationale de la Recherche, Convention on financial support for climate services of the French Ministry for an Ecological and Solidary Transition (MTES)

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Abstract: In the area of sea-level rise, recent research has focused on assessing either likely or high end future sea levels, but less attention has been given to "low-end" sea-level projections, exploring best-case potential sea-level changes and providing the basis for estimating minimum adaptation needs. Here, we provide global and regional probabilistic sea-level projections using conservative projections of glaciers and ice-sheets melting and a selection of models from the Coupled Model Intercomparison Project phase 5 (CMIP5) delivering moderate thermal expansion. Our low-end sea-level projections are higher than previously estimated because they rely on modeling outcomes only, and do not add any expert judgement, aiming essentially at delivering more realistic upper tails. While there are good reasons to believe that our projections are excessively optimistic, they can be used as low-end sea-level projections in order to inform users with low aversion to uncertainty. Our low-end sea-level projection exceeds 0.5 m along most inhabited coasts by 2100 for business as usual greenhouse gas emissions (RCP8.5), which is relevant for adaptation practitioners as long as efficient climate change mitigation policies are not implemented. This means that without efficient climate mitigation, an acceleration of sea-level rise can hardly be avoided during the 21st century.

327. Marine ice sheet instability amplifies and skews uncertainty in projections of future sea-level rise

Authors: Alexander A. Robel (Georgia Institute of Technology); Hélène Seroussi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Gerard H. Roe (University of Washington)

Journal: Proceedings of the National Academy of Sciences

Publisher: Proceedings of the National Academy of Sciences

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Keywords: N/A

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Open Access: N/A

Abstract: Sea-level rise may accelerate significantly if marine ice sheets become unstable. If such instability occurs, there would be considerable uncertainty in future sea-level rise projections due to imperfectly modeled ice sheet processes and unpredictable climate variability. In this study, we use mathematical and computational approaches to identify the ice sheet processes that drive uncertainty in sea-level projections. Using stochastic perturbation theory from statistical physics as a tool, we show mathematically that the marine ice sheet instability greatly amplifies and skews uncertainty in sea-level projections with worst-case scenarios of rapid sea-level rise being more likely than best-case scenarios of slower sea-level rise. We also perform large ensemble simulations with a state-of-the-art ice sheet model of Thwaites Glacier, a marine-terminating glacier in West Antarctica that is thought to be unstable. These ensemble simulations indicate that the uncertainty solely related to internal climate variability can be a large fraction of the total ice loss expected from Thwaites Glacier. We conclude that internal climate variability alone can be responsible for significant uncertainty in projections of sea-level rise and that large ensembles are a necessary tool for quantifying the upper bounds of this uncertainty.

328. A reconstruction of warm-water inflow to Upernavik Isstrøm since 1925 CE and its relation to glacier retreat

Authors: Flor Vermassen; Nanna Andreassen; David J. Wangner; Nicolas Thibault; Marit-Solveig Seidenkrantz; Rebecca Jackson; Sabine Schmidt; Kurt H. Kjær; Camilla S. Andresen

Journal: Climate of the Past

Publisher: Copernicus GmbH

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Abstract: . The mass loss from the Greenland Ice Sheet has increased over the past 2 decades. Marine-terminating glaciers contribute significantly to this mass loss due to increased melting and ice discharge. Periods of rapid retreat of these tidewater glaciers have been linked to the concurrent inflow of warm Atlantic-sourced waters. However, little is known about the variability of these Atlantic-derived waters within the fjords, due to a lack of multi-annual in situ measurements. Thus, to better understand the potential role of ocean warming on glacier retreat, reconstructions that characterize the variability of Atlantic water inflow to the fjords are required. Here, we investigate foraminiferal assemblages in a sediment core from Upernavik Fjord, West Greenland, in which the major ice stream Upernavik Isstrøm terminates. We conclude that the foraminiferal assemblage is predominantly controlled by changes in bottom water composition and provide a reconstruction of Atlantic water inflow to Upernavik Fjord, spanning the period 1925-2012. This reconstruction reveals peak Atlantic water influx during the 1930s and again after 2000, a pattern that is comparable to the Atlantic Multidecadal Oscillation (AMO). The comparison of these results to historical observations of front positions of Upernavik Isstrøm reveals that inflow of warm Atlantic-derived waters likely contributed to high retreat rates in the 1930s and after 2000. However, moderate retreat rates of Upernavik Isstrøm also prevailed in the 1960s and 1970s, showing that glacier retreat continued despite a reduced Atlantic water inflow, albeit at a lower rate. Considering the link between bottom water variability and the AMO in Upernavik Fjord, and the fact that a persistent negative phase of the AMO is expected for the next decade, Atlantic water inflow into the fjord may decrease in the coming decade, potentially minimizing or stabilizing the retreat of Upernavik Isstrøm during this time interval.

329. Thermal Weakening, Convergent Flow, and Vertical Heat Transport in the Northeast Greenland Ice Stream Shear Margins

Authors: N. Holschuh (University of Washington); D. A. Lilien (University of Washington); K. Christianson (University of Washington)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Subjects: N/A

Keywords: shear margin, Northeast Greenland Ice Stream, ice sheet modeling, Greenland, model validation (and 1 more)

Funding: National Aeronautics and Space Administration

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Open Access: N/A

Abstract: Ice streams are bounded by abrupt transitions in speed called shear margins. Some shear margins are fixed by subglacial topography, but others are thought to be self-organizing, evolving by thermal feedback to ice viscosity and basal drag which govern the stress balance of ice sheets. Resistive stresses (and properties governing shear-margin formation) manifest nonuniquely at the surface, motivating the use of subsurface observations to constrain ice sheet models. In this study, we use ice-penetrating radar data to evaluate three 3-D thermomechanical models of the Northeast Greenland Ice Stream, focusing on model reproductions of ice temperature (a primary control on viscosity) and subsurface velocity. Data/model agreement indicates elevated temperatures in the Northeast Greenland Ice Stream margins, with depth-averaged temperatures between 2 degreesC and 6 degreesC warmer in the southeast margin compared to ice in streaming flow, driven by vertical heat transport rather than shear heating. This work highlights complexity in ice divergence across stagnant/streaming transitions.

330. Sensitivity of a calving glacier to ice-ocean interactions under climate change: new insights from a 3-D full-Stokes model

Authors: Joe Todd; Poul Christoffersen; Thomas Zwinger; Peter Råback; Douglas I. Benn

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-13-1681-2019

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Keywords: N/A

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Open Access: N/A

Abstract: . Iceberg calving accounts for between 30 % and 60 % of net mass loss from the Greenland Ice Sheet, which has intensified and is now the single largest contributor to global sea level rise in the cryosphere. Changes to calving rates and the dynamics of calving glaciers represent a significant uncertainty in projections of future sea level rise. A growing body of observational evidence suggests that calving glaciers respond rapidly to regional environmental change, but predictive capacity is limited by the lack of suitable models capable of simulating calving mechanisms realistically. Here, we use a 3-D full-Stokes calving model to investigate the environmental sensitivity of Store Glacier, a large outlet glacier in West Greenland. We focus on two environmental processes: undercutting by submarine melting and buttressing by ice mélange, and our results indicate that Store Glacier is likely to be able to withstand moderate warming perturbations in which the former is increased by 50 % and the latter reduced by 50 %. However, severe perturbation with a doubling of submarine melt rates or a complete loss of ice mélange destabilises the calving front in our model runs. Furthermore, our analysis reveals that stress and fracture patterns at Store's terminus are complex and varied, primarily due to the influence of basal topography. Calving style and environmental sensitivity vary greatly, with propagation of surface crevasses significantly influencing iceberg production in the northern side, whereas basal crevasses dominate in the south. Any future retreat is likely to be initiated in the southern side by a combination of increased submarine melt rates in summer and reduced mélange strength in winter. The lateral variability, as well as the importance of rotational and bending forces at the terminus, underlines the importance of using the 3-D full-Stokes stress solution when modelling Greenland's calving glaciers.

331. Scaling of instability timescales of Antarctic outlet glaciers based on one-dimensional similitude analysis

Authors: Anders Levermann; Johannes Feldmann

Journal: The Cryosphere

Publisher: Copernicus GmbH

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Abstract: . Recent observations and ice-dynamic modeling suggest that a marine ice-sheet instability (MISI) might have been triggered in West Antarctica. The corresponding outlet glaciers, Pine Island Glacier (PIG) and Thwaites Glacier (TG), showed significant retreat during at least the last 2 decades. While other regions in Antarctica have the topographic predisposition for the same kind of instability, it is so far unclear how fast these instabilities would unfold if they were initiated. Here we employ the concept of similitude to estimate the characteristic timescales of several potentially MISI-prone outlet glaciers around the Antarctic coast. Our results suggest that TG and PIG have the fastest response time of all investigated outlets, with TG responding about 1.25 to 2 times as fast as PIG, while other outlets around Antarctica would be up to 10 times slower if destabilized. These results have to be viewed in light of the strong assumptions made in their derivation. These include the absence of ice-shelf buttressing, the one-dimensionality of the approach and the uncertainty of the available data. We argue however that the current topographic situation and the physical conditions of the MISI-prone outlet glaciers carry the information of their respective timescale and that this information can be partially extracted through a similitude analysis.

332. Projecting Antarctica's contribution to future sea level rise from basal ice-shelf melt using linear response functions of 16 ice sheet models (LARMIP-2)

Authors: Anders Levermann; Ricarda Winkelmann; Torsten Albrecht; Heiko Goelzer; Nicholas R. Golledge; Ralf Greve; Philippe Huybrechts; Jim Jordan; Gunter Leguy; Daniel Martin; Mathieu Morlighem; Frank Pattyn; David Pollard; Aurelien Quiquet; Christian Rodehacke; Helene Seroussi; Johannes Sutter; Tong Zhang; Jonas Van Breedam; Robert DeConto; Christophe Dumas; Julius Garbe; G. Hilmar Gudmundsson; Matthew J. Hoffman; Angelika Humbert; Thomas Kleiner; William Lipscomb; Malte Meinshausen; Esmond Ng; Mauro Perego; Stephen F. Price; Fuyuki Saito; Nicole-Jeanne Schlegel; Sainan Sun; Roderik S. W. van de Wal

Journal: Unknown Journal

Publisher: Copernicus GmbH

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Language: N/A

Subjects: N/A

Keywords: Future sea level|Ice-sheet model|Ice-albedo feedback|Antarctic ice sheet|Forcing (mathematics)

Funding: U.S. Department of Energy, European Research Council, Japan Society for the Promotion of Science (and 1 more)

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Open Access: N/A

Abstract: . The sea level contribution of the Antarctic ice sheet constitutes a large uncertainty in future sea level projections. Here we apply a linear response theory approach to 16 state-of-the-art ice sheet models to estimate the Antarctic ice sheet contribution from basal ice shelf melting within the 21st century. The purpose of this computation is to estimate the uncertainty that arises from large uncertainty in the external forcing that future warming may exert onto the ice sheet. While ice shelf melting is considered to be a major if not the largest perturbation of the ice sheet's flow into the ocean, the approach is neglecting a number of processes such as surface mass balance related contributions and mechanisms. In assuming linear response theory, we are able to capture complex temporal responses of the ice sheets, but we neglect any dampening or self-amplifying processes. This is particularly relevant in situations where an instability is dominating the ice loss. Results obtained here are thus relevant in particular wherever the ice loss is dominated by the forcing as opposed to an internal instability, for example in strong warming scenarios. In order to allow for comparison the methodology was chosen to be exactly the same as in an earlier study (Levermann et al., 2014), but with 16 instead of 5 ice sheet models. We include uncertainty in the atmospheric warming response to carbon emissions (full range of CMIP-5 climate model sensitivities), uncertainty in the oceanic transport to the Southern Ocean (obtained from the time-delayed and scaled oceanic subsurface warming in CMIP-5 models in relation to the global mean surface warming) and the observed range of responses of basal ice shelf melting to oceanic warming outside the ice shelf cavity. This uncertainty in basal ice

shelf melting is then convoluted with the linear response functions of each of the 16 ice sheet models to obtain the ice flow response to the individual global warming path. The model median for the observational period from 1992 to 2017 is 9.6 mm with a likely range between 5.2 mm and 20.3 mm compared to the observed sea-level contribution from Antarctica of 7.4 mm with a standard deviation of 3.7 mm (Shepherd et al., 2018). For the so-called business-as-usual warming path, RCP-8.5, we obtain a median contribution of the Antarctic ice sheet to global mean sea-level rise within the 21st century of 17 cm with a likely range (66-percentile around the mean) between 9 cm and 36 cm and a very likely range (90-percentile around the mean) between 6 cm and 59 cm. For the RCP-2.6 warming path which will keep the global mean temperature below two degrees of global warming and is thus consistent with the Paris Climate Agreement yields a median of 13 cm of global mean sea-level contribution. The likely range for the RCP-2.6 scenario is between 7 cm and 25 cm and the very likely range is between 5 cm and 39 cm. The structural uncertainties in the method do not allow an interpretation of any higher uncertainty percentiles. We provide projections for the five Antarctic regions and for each model and each scenario, separately. The rate of sea level contribution is highest under the RCP-8.5 scenario. The maximum within the 21st century of the median value is 4 cm per decade with a likely range between 2 cm/dec and 8 cm/dec and a very likely range between 1 cm/dec and 13 cm/dec.

333. Observing and Modeling Ice Sheet Surface Mass Balance

Authors: Jan T. M. Lenaerts (University of Colorado Boulder); Brooke Medley (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Michiel R. van den Broeke (Utrecht University); Bert Wouters (Delft University of Technology, Utrecht University)

Journal: Reviews of Geophysics

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Keywords: N/A

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Abstract: Surface mass balance (SMB) provides mass input to the surface of the Antarctic and Greenland Ice Sheets and therefore comprises an important control on ice sheet mass balance and resulting contribution to global sea level change. As ice sheet SMB varies highly across multiple scales of space (meters to hundreds of kilometers) and time (hourly to decadal), it is notoriously challenging to observe and represent in models. In addition, SMB consists of multiple components, all of which depend on complex interactions between the atmosphere and the snow/ice surface, large-scale atmospheric circulation and ocean conditions, and ice sheet topography. In this review, we present the state-of-the-art knowledge and recent advances in ice sheet SMB observations and models, highlight current shortcomings, and propose future directions. Novel observational methods allow mapping SMB across larger areas, longer time periods, and/or at very high (subdaily) temporal frequency. As a recent observational breakthrough, cosmic ray counters provide direct estimates of SMB, circumventing the need for accurate snow density observations upon which many other techniques rely. Regional atmospheric climate models have drastically improved their simulation of ice sheet SMB in the last decade, thanks to the inclusion or improved representation of essential processes (e.g., clouds, blowing snow, and snow albedo), and by enhancing horizontal resolution (5-30 km). Future modeling efforts are required in improving Earth system models to match regional atmospheric climate model performance in simulating ice sheet SMB, and in reinforcing the efforts in developing statistical and dynamic downscaling to represent smaller-scale SMB processes.

334. initMIP-Antarctica: an ice sheet model initialization experiment of ISMIP6

Authors: Hélène Seroussi; Sophie Nowicki; Erika Simon; Ayako Abe-Ouchi; Torsten Albrecht; Julien Brondex; Stephen Cornford; Christophe Dumas; Fabien Gillet-Chaulet; Heiko Goelzer; Nicholas R. Golledge; Jonathan M. Gregory; Ralf Greve; Matthew J. Hoffman; Angelika Humbert; Philippe Huybrechts; Thomas Kleiner; Eric Larour; Gunter Leguy; William H. Lipscomb; Daniel Lowry; Matthias Mengel; Mathieu Morlighem; Frank Pattyn; Anthony J. Payne; David Pollard; Stephen F. Price; Aurélien Quiquet; Thomas J. Reerink; Ronja Reese; Christian B. Rodehacke; Nicole-Jeanne Schlegel; Andrew Shepherd; Sainan Sun; Johannes Sutter; Jonas Van Breedam; Roderik S. W. van de Wal; Ricarda Winkelmann; Tong Zhang

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Language: en

Subjects: N/A

Keywords: N/A

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Open Access: N/A

Abstract: . Ice sheet numerical modeling is an important tool to estimate the dynamic contribution of the Antarctic ice sheet to sea level rise over the coming centuries. The influence of initial conditions on ice sheet model simulations, however, is still unclear. To better understand this influence, an initial state intercomparison exercise (initMIP) has been developed to compare, evaluate, and improve initialization procedures and estimate their impact on century-scale simulations. initMIP is the first set of experiments of the Ice Sheet Model Intercomparison Project for CMIP6 (ISMIP6), which is the primary Coupled Model Intercomparison Project Phase 6 (CMIP6) activity focusing on the Greenland and Antarctic ice sheets. Following initMIP-Greenland, initMIP-Antarctica has been designed to explore uncertainties associated with model initialization and spin-up and to evaluate the impact of changes in external forcings. Starting from the state of the Antarctic ice sheet at the end of the initialization procedure, three forward experiments are each run for 100 years: a control run, a run with a surface mass balance anomaly, and a run with a basal melting anomaly beneath floating ice. This study presents the results of initMIP-Antarctica from 25 simulations performed by 16 international modeling groups. The submitted results use different initial conditions and initialization methods, as well as ice flow model parameters and reference external forcings. We find a good agreement among model responses to the surface mass balance anomaly but large variations in responses to the basal melting anomaly. These variations can be attributed to differences in the extent of ice shelves and their upstream tributaries, the numerical treatment of grounding line, and the initial ocean conditions applied, suggesting that ongoing efforts to better represent ice shelves in continental-scale models

should continue.

335. Quality Assessment and Glaciological Applications of Digital Elevation Models Derived from Space-Borne and Aerial Images over Two Tidewater Glaciers of Southern Spitsbergen

Authors: Małgorzata Błaszczyk (University of Silesia in Katowice); Dariusz Ignatiuk (University of Silesia in Katowice); Mariusz Grabiec (University of Silesia in Katowice); Leszek Kolondra (University of Silesia in Katowice); Michał Laska (University of Silesia in Katowice); Leo Decaux (University of Silesia in Katowice); Jacek Jania (University of Silesia in Katowice); Etienne Berthier (Institut de Recherche pour le Développement (IRD)); Bartłomiej Luks (Institute of Geophysics, Polish Academy of Sciences, Księcia Janusza 64, 01-452 Warszawa, Poland); Barbara Barzycka (University of Silesia in Katowice); Mateusz Czapla (University of Silesia in Katowice)

Journal: Remote Sensing

Publisher: MDPI AG

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Open Access: N/A

Abstract: In this study, we assess the accuracy and precision of digital elevation models (DEM) retrieved from aerial photographs taken in 2011 and from Very High Resolution satellite images (WorldView-2 and Pléiades) from the period 2012-2017. Additionally, the accuracy of the freely available Strip product of ArcticDEM was verified. We use the DEMs to characterize geometry changes over Hansbreen and Hornbreen, two tidewater glaciers in southern Spitsbergen, Svalbard. The satellite-based DEMs from WorldView-2 and Pléiades stereo pairs were processed using the Rational Function Model (RFM) without and with one ground control point. The elevation quality of the DEMs over glacierized areas was validated with in situ data: static differential GPS survey of mass balance stakes and GPS kinematic data acquired during ground penetrating radar survey. Results demonstrate the usefulness of the analyzed sources of DEMs for estimation of the total geodetic mass balance of the Svalbard glaciers. DEM accuracy is sufficient to investigate glacier surface elevation changes above 1 m. Strips from the ArcticDEM are generally precise, but some of them showed gross errors and need to be handled with caution. The surface of Hansbreen and Hornbreen has been lowering in recent years. The average annual elevation changes for Hansbreen were more negative in the period 2015-2017 (-2.4 m a⁻¹) than in the period 2011-2015 (-1.7 m a⁻¹). The average annual elevation changes over the studied area of Hornbreen for the period 2012-2017 amounted to -1.6 m a⁻¹. The geodetic mass balance for Hansbreen was more negative than the climatic mass balance estimated using the mass budget method, probably due to underestimation of the ice discharge. From 2011 to

2017, Hansbreen lost on average over 1% of its volume each year. Such a high rate of relative loss illustrates how fast these glaciers are responding to climate change.

336. Evaluating ice-rafted debris as a proxy for glacier calving in Upernavik Isfjord, NW Greenland

Authors: Flor Vermassen (Geological Survey of Denmark and Greenland, University of Copenhagen); David J. Wangner (Geological Survey of Denmark and Greenland, University of Copenhagen); Laurence M. Dyke (Geological Survey of Denmark and Greenland); Sabine Schmidt (CNRS, OASU, EPOC UMR5805 Pessac Cedex France); Amalie E. Cordua (University of Hamburg); Kurt H. Kjær (University of Copenhagen); Konstanze Haubner (Geological Survey of Denmark and Greenland, University of Copenhagen); Camilla S. Andresen (Geological Survey of Denmark and Greenland)

Journal: Journal of Quaternary Science

Publisher: Wiley

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Open Access: N/A

Abstract: Records of ice-rafted debris (IRD) in sediments are commonly used as a proxy for iceberg production and to reconstruct past changes of glacier stability. However, the interpretation of IRD is complex as multiple processes modulate its variability. This study investigates the relationship between IRD variability and glaciological change by measuring IRD records from Upernavik Fjord and comparing these to frontal positions of Upernavik Isstrøm during the past century. Results show that the spatial variability of IRD deposition throughout the fjord is high, indicating that randomness inherent to IRD distorts the calving signal. However, we investigate whether IRD records can be combined to improve the reconstruction, as previously suggested, and show the importance of core site selection and number of cores on this approach. The outer-fjord core compares relatively well to the observed front positions and this is reflected in the composite record: increased IRD deposition in 1937-1946, 1968-1980, and 1996-1999 occurred during periods of faster retreat. Comparison with climatic records shows that the calving episodes in the late '30 s/early '40 s and late '90 s are related to warm ocean and air temperatures, whereas intensified retreat and calving during the '70 s reflects partly an internal glacier response to the fjord geometry.

337. Slowdown in Antarctic mass loss from solid Earth and sea-level feedbacks

Authors: E. Larour (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, University of California Los Angeles); H. Seroussi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); S. Adhikari (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); E. Ivins (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); L. Caron (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); M. Morlighem (University of California Irvine); N. Schlegel (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Science

Publisher: American Association for the Advancement of Science (AAAS)

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Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, National Aeronautics and Space Administration, National Aeronautics and Space Administration (and 15 more)

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Open Access: N/A

Abstract: An uplifting effect The rise in sea level that is occurring from the melting of Antarctica is only going to accelerate as climate warms. However, Larour et al. report that crustal uplift in the Amundsen Sea sector is helping to reduce grounding line retreat, thereby stabilizing the ice sheet and slowing its rate of mass loss (see the Perspective by Steig). This effect will not stop or reverse ice sheet loss, but it could delay the progress of dynamic mass loss of Thwaites Glacier by approximately 20 years. Science , this issue p. eaav7908 ; see also p. 936

338. Uncertainty quantification of the multi-centennial response of the Antarctic ice sheet to climate change

Authors: Kevin Bulthuis; Maarten Arnst; Sainan Sun; Frank Pattyn

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-13-1349-2019

URL: <https://doi.org/10.5194/tc-13-1349-2019>

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Language: en

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Funding: Fonds De La Recherche Scientifique - FNRS

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Open Access: N/A

Abstract: . Ice loss from the Antarctic ice sheet (AIS) is expected to become the major contributor to sea level in the next centuries. Projections of the AIS response to climate change based on numerical ice-sheet models remain challenging due to the complexity of physical processes involved in ice-sheet dynamics, including instability mechanisms that can destabilise marine basins with retrograde slopes. Moreover, uncertainties in ice-sheet models limit the ability to provide accurate sea-level rise projections. Here, we apply probabilistic methods to a hybrid ice-sheet model to investigate the influence of several sources of uncertainty, namely sources of uncertainty in atmospheric forcing, basal sliding, grounding-line flux parameterisation, calving, sub-shelf melting, ice-shelf rheology and bedrock relaxation, on the continental response of the Antarctic ice sheet to climate change over the next millennium. We provide probabilistic projections of sea-level rise and grounding-line retreat, and we carry out stochastic sensitivity analysis to determine the most influential sources of uncertainty. We find that all investigated sources of uncertainty, except bedrock relaxation time, contribute to the uncertainty in the projections. We show that the sensitivity of the projections to uncertainties increases and the contribution of the uncertainty in sub-shelf melting to the uncertainty in the projections becomes more and more dominant as atmospheric and oceanic temperatures rise, with a contribution to the uncertainty in sea-level rise projections that goes from 5 % to 25 % in RCP 2.6 to more than 90 % in RCP 8.5. We show that the significance of the AIS contribution to sea level is controlled by the marine ice-sheet instability (MISI) in marine basins, with the biggest contribution stemming from the more vulnerable West Antarctic ice sheet. We find that, irrespective of parametric uncertainty, the strongly mitigated RCP 2.6 scenario prevents the collapse of the West Antarctic ice sheet, that in both the RCP 4.5 and RCP 6.0 scenarios the occurrence of MISI in marine basins is more sensitive to parametric uncertainty, and that, almost irrespective of parametric uncertainty, RCP 8.5 triggers the collapse of the West Antarctic ice sheet.

339. Evaluation of ice-stream model sensitivities for parameter estimation

Authors: Richard B. Alley; Wenjie Li; Byron R. Parizek; Fuqing Zhang

Journal: Earth and Planetary Science Letters

Publisher: Elsevier BV

DOI: 10.1016/j.epsl.2019.03.035

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Abstract: N/A

340. Brief communication: PICOP, a new ocean melt parameterization under ice shelves combining PICO and a plume model

Authors: Tyler Pelle; Mathieu Morlighem; Johannes H. Bondzio

Journal: The Cryosphere

Publisher: Copernicus GmbH

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Open Access: N/A

Abstract: . Basal melting at the bottom of Antarctic ice shelves is a major control on glacier dynamics, as it modulates the amount of buttressing that floating ice shelves exert onto the ice streams feeding them. Three-dimensional ocean circulation numerical models provide reliable estimates of basal melt rates but remain too computationally expensive for century-scale projections. Ice sheet modelers therefore routinely rely on simplified parameterizations based on either ice shelf depth or more sophisticated box models. However, existing parameterizations do not accurately resolve the complex spatial patterns of sub-shelf melt rates that have been observed over Antarctica's ice shelves, especially in the vicinity of the grounding line, where basal melting is one of the primary drivers of grounding line migration. In this study, we couple the Potsdam Ice-shelf Cavity model (PICO, Reese et al., 2018) to a buoyant plume melt rate parameterization (Lazeroms et al., 2018) to create PICOP, a novel basal melt rate parameterization that is easy to implement in transient ice sheet numerical models and produces a melt rate field that is in excellent agreement with the spatial distribution and magnitude of observations for several ocean basins. We test PICOP on the Amundsen Sea sector of West Antarctica, Totten, and Moscow University ice shelves in East Antarctica and the Filchner-Ronne Ice Shelf and compare the results to PICO. We find that PICOP is able to reproduce inferred high melt rates beneath Pine Island, Thwaites, and Totten glaciers (on the order of 100 m yr⁻¹) and removes the "banding" pattern observed in melt rates produced by PICO over the Filchner-Ronne Ice Shelf. PICOP resolves many of the issues contemporary basal melt rate parameterizations face and is therefore a valuable tool for those looking to make future projections of Antarctic glaciers.

341. Uncovering the Pattern of Forced Sea Level Rise in the Satellite Altimeter Record

Authors: Benjamin D. Hamlington (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); John T. Fasullo (University of Colorado Boulder, National Center for Atmospheric Research); R. Steven Nerem (University of Colorado Boulder); Kwang-Yul Kim (Seoul National University); F. W. Landerer (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Cited by: 29 publications

Language: en

Subjects: N/A

Keywords: satellite altimetry, natural variability, sea level, forced trends

Funding: National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: The continuous and near-global coverage of satellite altimeters has provided an improved understanding of sea level rise on both global and regional scales. With its relatively short record, however, questions have persisted regarding how well satellite-measured trends reflect the trends associated with forced or anthropogenic forcing. Masking the forced sea level rise in the altimeter record is the presence of unaccounted-for, naturally occurring variability. Here, a combined approach using models and observations is used to attribute a portion of the satellite-measured sea level record to natural variability. The agreement between the observations and models makes clear the quality of current models and their ability to realistically capture sea level variability. This paper shows a regional trend pattern where the influence of natural variability has been reduced, providing a clearer picture of how anthropogenic forcing may impact sea level rise around the globe.

342. Response of the flow dynamics of Bowdoin Glacier, northwestern Greenland, to basal lubrication and tidal forcing

Authors: HAKIME SEDDIK; RALF GREVE; DAIKI SAKAKIBARA; SHUN TSUTAKI; MASAHIRO MINOWA; SHIN SUGIYAMA

Journal: Journal of Glaciology

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/jog.2018.106

URL: <https://doi.org/10.1017/jog.2018.106>

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References: 59

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: We use the full-Stokes model Elmer/Ice to investigate the present dynamics of Bowdoin Glacier, a marine-terminating outlet glacier in northwestern Greenland. Short-term speed variations of the glacier were observed, correlating with air temperature and precipitation, and with the semi-diurnal ocean tides. We use a control inverse method to determine the distribution of basal friction. This reveals that most of the glacier area is characterized by near-plug-flow conditions, while some sticky spots are also identified. We then conduct experiments to test the sensitivity of the glacier flow to basal lubrication and tidal forcing at the calving front. Reduction of the basal drag by 10-40% produces speed-ups that agree approximately with the observed range of speed-ups that result from warm weather and precipitation events. In agreement with the observations, tidal forcing and surface speed near the calving front are found to be in anti-phase (high tide corresponds to low speed, and vice versa). However, the amplitude of the semi-diurnal variability is underpredicted by a factor ~ 3 , which is likely related to either inaccuracies in the surface and bedrock topographies or mechanical weakening due to crevassing.

343. LIVVkit 2.1: automated and extensible ice sheet model validation

Authors: Katherine J. Evans; Joseph H. Kennedy; Dan Lu; Mary M. Forrester; Stephen Price; Jeremy Fyke; Andrew R. Bennett; Matthew J. Hoffman; Irina Tezaur; Charles S. Zender; Miren Vizcaíno

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

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Abstract: . A collection of scientific analyses, metrics, and visualizations for robust validation of ice sheet models is presented using the Land Ice Verification and Validation toolkit (LIVVkit), version 2.1, and the LIVVkit Extensions repository (LEX), version 0.1. This software collection targets stand-alone ice sheet or coupled Earth system models, and handles datasets and analyses that require high-performance computing and storage. LIVVkit aims to enable efficient and fully reproducible workflows for postprocessing, analysis, and visualization of observational and model-derived datasets in a shareable format, whereby all data, methodologies, and output are distributed to users for evaluation. Extending from the initial LIVVkit software framework, we demonstrate Greenland ice sheet simulation validation metrics using the coupled Community Earth System Model (CESM) as well as an idealized stand-alone high-resolution Community Ice Sheet Model, version 2 (CISM2), coupled to the Albany/FELIX velocity solver (CISM-Albany or CISM-A). As one example of the capability, LIVVkit analyzes the degree to which models capture the surface mass balance (SMB) and identifies potential sources of bias, using recently available in situ and remotely sensed data as comparison. Related fields within atmosphere and land surface models, e.g., surface temperature, radiation, and cloud cover, are also diagnosed. Applied to the CESM1.0, LIVVkit identifies a positive SMB bias that is focused largely around Greenland's southwest region that is due to insufficient ablation.

344. The impact of model resolution on the simulated Holocene retreat of the southwestern Greenland ice sheet using the Ice Sheet System Model (ISSM)

Authors: Joshua K. Cuzzzone; Nicole-Jeanne Schlegel; Mathieu Morlighem; Eric Larour; Jason P. Briner; Helene Seroussi; Lambert Caron

Journal: The Cryosphere

Publisher: Copernicus GmbH

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Cited by: 27 publications

Language: en

Subjects: N/A

Keywords: Greenland ice sheet|Ice-sheet model

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Open Access: N/A

Abstract: . Geologic archives constraining the variability of the Greenland ice sheet (GrIS) during the Holocene provide targets for ice sheet models to test sensitivities to variations in past climate and model formulation. Even as data-model comparisons are becoming more common, many models simulating the behavior of the GrIS during the past rely on meshes with coarse horizontal resolutions (≥ 10 km). In this study, we explore the impact of model resolution on the simulated nature of retreat across southwestern Greenland during the Holocene. Four simulations are performed using the Ice Sheet System Model (ISSM): three that use a uniform mesh and horizontal mesh resolutions of 20, 10, and 5 km, and one that uses a nonuniform mesh with a resolution ranging from 2 to 15 km. We find that the simulated retreat can vary significantly between models with different horizontal resolutions based on how well the bed topography is resolved. In areas of low topographic relief, the horizontal resolution plays a negligible role in simulated differences in retreat, with each model instead responding similarly to retreat driven by surface mass balance (SMB). Conversely, in areas where the bed topography is complex and high in relief, such as fjords, the lower-resolution models (10 and 20 km) simulate unrealistic retreat that occurs as ice surface lowering intersects bumps in the bed topography that would otherwise be resolved as troughs using the higher-resolution grids. Our results highlight the important role that high-resolution grids play in simulating retreat in areas of complex bed topography, but also suggest that models using nonuniform grids can save computational resources through coarsening the mesh in areas of noncomplex bed topography where the SMB predominantly drives retreat. Additionally, these results emphasize that care must be taken with ice sheet models when tuning model parameters to match reconstructed margins, particularly for lower-resolution models in regions where complex bed topography is poorly resolved.

345. ATAT 1.1, the Automated Timing Accordance Tool for comparing ice-sheet model output with geochronological data

Authors: Jeremy C. Ely; Chris D. Clark; David Small; Richard C. A. Hindmarsh

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-12-933-2019

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Language: en

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Keywords: N/A

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Open Access: N/A

Abstract: . Earth's extant ice sheets are of great societal importance given their ongoing and potential future contributions to sea-level rise. Numerical models of ice sheets are designed to simulate ice-sheet behaviour in response to climate changes but to be improved require validation against observations. The direct observational record of extant ice sheets is limited to a few recent decades, but there is a large and growing body of geochronological evidence spanning millennia constraining the behaviour of palaeo-ice sheets. Hindcasts can be used to improve model formulations and study interactions between ice sheets, the climate system and landscape. However, ice-sheet modelling results have inherent quantitative errors stemming from parameter uncertainty and their internal dynamics, leading many modellers to perform ensemble simulations, while uncertainty in geochronological evidence necessitates expert interpretation. Quantitative tools are essential to examine which members of an ice-sheet model ensemble best fit the constraints provided by geochronological data. We present the Automated Timing Accordance Tool (ATAT version 1.1) used to quantify differences between model results and geochronological data on the timing of ice-sheet advance and/or retreat. To demonstrate its utility, we perform three simplified ice-sheet modelling experiments of the former British-Irish ice sheet. These illustrate how ATAT can be used to quantify model performance, either by using the discrete locations where the data originated together with dating constraints or by comparing model outputs with empirically derived reconstructions that have used these data along with wider expert knowledge. The ATAT code is made available and can be used by ice-sheet modellers to quantify the goodness of fit of hindcasts. ATAT may also be useful for highlighting data inconsistent with glaciological principles or reconstructions that cannot be replicated by an ice-sheet model.

346. On nonlinear strain theory for a viscoelastic material model and its implications for calving of ice shelves

Authors: JULIA CHRISTMANN; RALF MÜLLER; ANGELIKA HUMBERT

Journal: Journal of Glaciology

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/jog.2018.107

URL: <https://doi.org/10.1017/jog.2018.107>

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Volume/Issue: Volume 65, Issue 250

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Cited by: 13 publications

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Subjects: N/A

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Abstract: In the current ice-sheet models calving of ice shelves is based on phenomenological approaches. To obtain physics-based calving criteria, a viscoelastic Maxwell model is required accounting for short-term elastic and long-term viscous deformation. On timescales of months to years between calving events, as well as on long timescales with several subsequent iceberg break-offs, deformations are no longer small and linearized strain measures cannot be used. We present a finite deformation framework of viscoelasticity and extend this model by a nonlinear Glen-type viscosity. A finite element implementation is used to compute stress and strain states in the vicinity of the ice-shelf calving front. Stress and strain maxima of small (linearized strain measure) and finite strain formulations differ by $\sim 5\%$ after 1 and by $\sim 30\%$ after 10 years, respectively. A finite deformation formulation reaches a critical stress or strain faster, thus calving rates will be higher, despite the fact that the exact critical values are not known. Nonlinear viscosity of Glen-type leads to higher stress values. The Maxwell material model formulation for finite deformations presented here can also be applied to other glaciological problems, for example, tidal forcing at grounding lines or closure of englacial and subglacial melt channels.

347. The Open Global Glacier Model (OGGM) v1.1

Authors: Fabien Maussion; Anton Butenko; Nicolas Champollion; Matthias Dusch; Julia Eis; Kévin Fourteau; Philipp Gregor; Alexander H. Jarosch; Johannes Landmann; Felix Oesterle; Beatriz Recinos; Timo Rothenpieler; Anouk Vlug; Christian T. Wild; Ben Marzeion

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References: 72

Cited by: 173 publications

Language: en

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Open Access: N/A

Abstract: . Despite their importance for sea-level rise, seasonal water availability, and as a source of geohazards, mountain glaciers are one of the few remaining subsystems of the global climate system for which no globally applicable, open source, community-driven model exists. Here we present the Open Global Glacier Model (OGGM), developed to provide a modular and open-source numerical model framework for simulating past and future change of any glacier in the world. The modeling chain comprises data downloading tools (glacier outlines, topography, climate, validation data), a preprocessing module, a mass-balance model, a distributed ice thickness estimation model, and an ice-flow model. The monthly mass balance is obtained from gridded climate data and a temperature index melt model. To our knowledge, OGGM is the first global model to explicitly simulate glacier dynamics: the model relies on the shallow-ice approximation to compute the depth-integrated flux of ice along multiple connected flow lines. In this paper, we describe and illustrate each processing step by applying the model to a selection of glaciers before running global simulations under idealized climate forcings. Even without an in-depth calibration, the model shows very realistic behavior. We are able to reproduce earlier estimates of global glacier volume by varying the ice dynamical parameters within a range of plausible values. At the same time, the increased complexity of OGGM compared to other prevalent global glacier models comes at a reasonable computational cost: several dozen glaciers can be simulated on a personal computer, whereas global simulations realized in a supercomputing environment take up to a few hours per century. Thanks to the modular framework, modules of various complexity can be added to the code base, which allows for new kinds of model intercomparison studies in a controlled environment. Future developments will add new physical processes to the model as well as automated calibration tools. Extensions or alternative parameterizations can be easily added by the community thanks to comprehensive documentation. OGGM spans a wide range of applications, from ice-climate interaction studies at millennial timescale to estimates of the contribution of glaciers to past and future sea-level change. It has the potential to become a self-sustained community-driven model for global and regional glacier evolution.

348. Modeling the response of northwest Greenland to enhanced ocean thermal forcing and subglacial discharge

Authors: Mathieu Morlighem; Michael Wood; Hélène Seroussi; Youngmin Choi; Eric Rignot

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-13-723-2019

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Keywords: N/A

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Open Access: N/A

Abstract: . Calving-front dynamics is an important control on Greenland's ice mass balance. Ice front retreat of marine-terminating glaciers may, for example, lead to a loss in resistive stress, which ultimately results in glacier acceleration and thinning. Over the past decade, it has been suggested that such retreats may be triggered by warm and salty Atlantic Water, which is typically found at a depth below 200–300 m. An increase in subglacial water discharge at glacier ice fronts due to enhanced surface runoff may also be responsible for an intensification of undercutting and calving. An increase in ocean thermal forcing or subglacial discharge therefore has the potential to destabilize marine-terminating glaciers along the coast of Greenland. It remains unclear which glaciers are currently stable but may retreat in the future and how far inland and how fast they will retreat. Here, we quantify the sensitivity and vulnerability of marine-terminating glaciers along the northwest coast of Greenland (from 72.5 to 76°N) to ocean forcing and subglacial discharge using the Ice Sheet System Model (ISSM). We rely on a parameterization of undercutting based on ocean thermal forcing and subglacial discharge and use ocean temperature and salinity from high-resolution ECCO2 (Estimating the Circulation and Climate of the Ocean, Phase II) simulations at the fjord mouth to constrain the ocean thermal forcing. The ice flow model includes a calving law based on a tensile von Mises criterion. We find that some glaciers, such as Dietrichson Gletscher or Alison Glacier, are sensitive to small increases in ocean thermal forcing, while others, such as Illullip Sermia or Cornell Gletscher, are remarkably stable, even in a +3 °C ocean warming scenario. Under the most intense experiment, we find that Hayes Gletscher retreats by more than 50 km inland by 2100 into a deep trough, and its velocity increases by a factor of 3 over only 23 years. The model confirms that ice-ocean interactions can trigger extensive and rapid glacier retreat, but the bed controls the rate and magnitude of the retreat. Under current oceanic and atmospheric conditions, we find that this sector of the Greenland ice sheet alone will contribute more than 1 cm to sea level rise and up to 3 cm by 2100 under the most extreme scenario.

349. Assessment of the Greenland ice sheet-atmosphere feedbacks for the next century with a regional atmospheric model coupled to an ice sheet model

Authors: Sébastien Le clec'h; Sylvie Charbit; Aurélien Quiquet; Xavier Fettweis; Christophe Dumas; Masa Kageyama; Coraline Wyard; Catherine Ritz

Journal: The Cryosphere

Publisher: Copernicus GmbH

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Open Access: N/A

Abstract: . In the context of global warming, growing attention is paid to the evolution of the Greenland ice sheet (GrIS) and its contribution to sea-level rise at the centennial timescale. Atmosphere-GrIS interactions, such as the temperature-elevation and the albedo feedbacks, have the potential to modify the surface energy balance and thus to impact the GrIS surface mass balance (SMB). In turn, changes in the geometrical features of the ice sheet may alter both the climate and the ice dynamics governing the ice sheet evolution. However, changes in ice sheet geometry are generally not explicitly accounted for when simulating atmospheric changes over the Greenland ice sheet in the future. To account for ice sheet-climate interactions, we developed the first two-way synchronously coupled model between a regional atmospheric model (MAR) and a 3-D ice sheet model (GRISLI). Using this novel model, we simulate the ice sheet evolution from 2000 to 2150 under a prolonged representative concentration pathway scenario, RCP8.5. Changes in surface elevation and ice sheet extent simulated by GRISLI have a direct impact on the climate simulated by MAR. They are fed to MAR from 2020 onwards, i.e. when changes in SMB produce significant topography changes in GRISLI. We further assess the importance of the atmosphere-ice sheet feedbacks through the comparison of the two-way coupled experiment with two other simulations based on simpler coupling strategies: (i) a one-way coupling with no consideration of any change in ice sheet geometry; (ii) an alternative one-way coupling in which the elevation change feedbacks are parameterized in the ice sheet model (from 2020 onwards) without taking into account the changes in ice sheet topography in the atmospheric model. The two-way coupled experiment simulates an important increase in surface melt below 2000 m of elevation, resulting in an important SMB reduction in 2150 and a shift of the equilibrium line towards elevations as high as 2500 m, despite a slight increase in SMB over the central plateau due to enhanced snowfall. In relation with these SMB changes, modifications of ice sheet geometry favour ice flux convergence towards the margins, with an increase in ice velocities in the GrIS interior due to increased surface slopes and a decrease in ice velocities at the margins due to decreasing ice thickness. This convergence counteracts the SMB signal in these areas. In the two-way coupling, the SMB is also

influenced by changes in fine-scale atmospheric dynamical processes, such as the increase in katabatic winds from central to marginal regions induced by increased surface slopes. Altogether, the GrIS contribution to sea-level rise, inferred from variations in ice volume above floatation, is equal to 20.4 cm in 2150. The comparison between the coupled and the two uncoupled experiments suggests that the effect of the different feedbacks is amplified over time with the most important feedbacks being the SMB-elevation feedbacks. As a result, the experiment with parameterized SMB-elevation feedback provides a sea-level contribution from GrIS in 2150 only 2.5 % lower than the two-way coupled experiment, while the experiment with no feedback is 9.3 % lower. The change in the ablation area in the two-way coupled experiment is much larger than those provided by the two simplest methods, with an underestimation of 11.7 % (14 %) with parameterized feedbacks (no feedback). In addition, we quantify that computing the GrIS contribution to sea-level rise from SMB changes only over a fixed ice sheet mask leads to an overestimation of ice loss of at least 6 % compared to the use of a time-variable ice sheet mask. Finally, our results suggest that ice-loss estimations diverge when using the different coupling strategies, with differences from the two-way method becoming significant at the end of the 21st century. In particular, even if averaged over the whole GrIS the climatic and ice sheet fields are relatively similar; at the local and regional scale there are important differences, highlighting the importance of correctly representing the interactions when interested in basin scale changes.

350. Description and evaluation of the Community Ice Sheet Model (CISM) v2.1

Authors: William H. Lipscomb; Stephen F. Price; Matthew J. Hoffman; Gunter R. Leguy; Andrew R. Bennett; Sarah L. Bradley; Katherine J. Evans; Jeremy G. Fyke; Joseph H. Kennedy; Mauro Perego; Douglas M. Ranken; William J. Sacks; Andrew G. Salinger; Lauren J. Vargo; Patrick H. Worley

Journal: Geoscientific Model Development

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Keywords: N/A

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Abstract: . We describe and evaluate version 2.1 of the Community Ice Sheet Model (CISM). CISM is a parallel, 3-D thermomechanical model, written mainly in Fortran, that solves equations for the momentum balance and the thickness and temperature evolution of ice sheets. CISM's velocity solver incorporates a hierarchy of Stokes flow approximations, including shallow-shelf, depth-integrated higher order, and 3-D higher order. CISM also includes a suite of test cases, links to third-party solver libraries, and parameterizations of physical processes such as basal sliding, iceberg calving, and sub-ice-shelf melting. The model has been verified for standard test problems, including the Ice Sheet Model Intercomparison Project for Higher-Order Models (ISMIP-HOM) experiments, and has participated in the initMIP-Greenland initialization experiment. In multimillennial simulations with modern climate forcing on a 4 km grid, CISM reaches a steady state that is broadly consistent with observed flow patterns of the Greenland ice sheet. CISM has been integrated into version 2.0 of the Community Earth System Model, where it is being used for Greenland simulations under past, present, and future climates. The code is open-source with extensive documentation and remains under active development.

351. Implementation and performance of adaptive mesh refinement in the Ice Sheet System Model (ISSM v4.14)

Authors: Thiago Dias dos Santos; Mathieu Morlighem; H            ; Philippe Remy Bernard Devloo; Jefferson Cardia Sim      

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

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Abstract: . Accurate projections of the evolution of ice sheets in a changing climate require a fine mesh/grid resolution in ice sheet models to correctly capture fundamental physical processes, such as the evolution of the grounding line, the region where grounded ice starts to float. The evolution of the grounding line indeed plays a major role in ice sheet dynamics, as it is a fundamental control on marine ice sheet stability. Numerical modeling of a grounding line requires significant computational resources since the accuracy of its position depends on grid or mesh resolution. A technique that improves accuracy with reduced computational cost is the adaptive mesh refinement (AMR) approach. We present here the implementation of the AMR technique in the finite element Ice Sheet System Model (ISSM) to simulate grounding line dynamics under two different benchmarks: MISMIP3d and MISMIP+. We test different refinement criteria: (a) distance around the grounding line, (b) a posteriori error estimator, the Zienkiewicz-Zhu (ZZ) error estimator, and (c) different combinations of (a) and (b). In both benchmarks, the ZZ error estimator presents high values around the grounding line. In the MISMIP+ setup, this estimator also presents high values in the grounded part of the ice sheet, following the complex shape of the bedrock geometry. The ZZ estimator helps guide the refinement procedure such that AMR performance is improved. Our results show that computational time with AMR depends on the required accuracy, but in all cases, it is significantly shorter than for uniformly refined meshes. We conclude that AMR without an associated error estimator should be avoided, especially for real glaciers that have a complex bed geometry.

352. An Improved Analytical Solution for the Temperature Profile of Ice Sheets

Authors: Soroush Rezvanbehbahani (University of Kansas); C. J. van der Veen (University of Kansas); Leigh A. Stearns (University of Kansas)

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018jf004774

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References: 60

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: temperature, ice sheet, analytical solution

Funding: National Science Foundation

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Open Access: N/A

Abstract: The one-dimensional steady state analytical solution of the energy conservation equation obtained by Robin (1955, <https://doi.org/10.3189/002214355793702028>) is frequently used in glaciology. This solution assumes a linear change in surface velocity from a minimum value equal to minus the mass balance at the surface to zero at the bed. Here we show that this assumption of a linear velocity profile leads to large errors in the calculated temperature profile and especially in basal temperature. By prescribing a nonlinear power function of elevation above the bed for the vertical velocity profile arising from use of the Shallow Ice Approximation, we derive a new analytical solution for temperature. We show that the solution produces temperature profiles identical to numerical temperature solutions with the Shallow Ice Approximation vertical velocity near ice divides. We quantify the importance of strain heating and demonstrate that integrating the strain heating and adding it to the geothermal heat flux at the bed is a reasonable approximation for the interior regions. Our analytical solution does not include horizontal advection components, so we compare our solution with numerical solutions of a two-dimensional advection-diffusion model and assess the applicability and errors of the analytical solution away from the ice divide. We show that several parameters and assumptions impact the spatial extent of applicability of the new solution including surface mass balance rate and surface temperature lapse rate. We delineate regions of Greenland and Antarctica within which the analytical solution at any depth is likely within 2 K of the actual temperatures with horizontal advection.

353. Calving Induced Speedup of Petermann Glacier

Authors: Martin Rückamp (Alfred Wegener Institute); Niklas Neckel (Alfred Wegener Institute); Sophie Berger (Laboratoire de Glaciologie Université libre de Bruxelles Brussels Belgium, Alfred Wegener Institute); Angelika Humbert (University of Bremen, Alfred Wegener Institute); Veit Helm (Alfred Wegener Institute)

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

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References: 47

Cited by: 29 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: This study assesses the response on ice dynamics of Petermann Glacier, a major outlet glacier in northern Greenland, to the 2012 and a possible future calving event. So far Petermann Glacier has been believed to be dynamically stable as another large calving event in 2010 had no significant impact on flow velocity or grounding line retreat. By analyzing a time series of remotely sensed surface velocities, we find an average acceleration of 10% between winter 2011/2012 and winter 2016/2017. This increase in surface velocity is not linear but can be separated into two parts, starting in 2012 and 2016 respectively. By conducting modeling experiments, we show that the first speedup can be directly connected to the 2012 calving event, while the second speedup is not captured. However, on recent remote sensing imagery newly developing fractures are clearly visible ~12 km upstream from the terminus, propagating from the eastern fjord wall to the center of the ice tongue, indicating a possible future calving event. By including these fracture zones as a new terminus position in the modeling domain, we are able to reproduce the second speedup, suggesting that surface velocities remain on the 2016/2017 level after the anticipated calving event. This indicates that, from a dynamical point of view, the terminus region has already detached from the main ice tongue.

354. A confined-unconfined aquifer model for subglacial hydrology and its application to the Northeast Greenland Ice Stream

Authors: Sebastian Beyer; Thomas Kleiner; Vadym Aizinger; Martin Rückamp; Angelika Humbert

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-12-3931-2018

URL: <https://doi.org/10.5194/tc-12-3931-2018>

Publication Date: 2018-12-19

Volume/Issue: Volume 12, Issue 12

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Cited by: 18 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: . Subglacial hydrology plays an important role in ice sheet dynamics as it determines the sliding velocity. It also drives freshwater into the ocean, leading to undercutting of calving fronts by plumes. Modeling subglacial water has been a challenge for decades. Only recently have new approaches been developed such as representing subglacial channels and thin water sheets by separate layers of variable hydraulic conductivity. We extend this concept by modeling a confined-unconfined aquifer system (CUAS) in a single layer of an equivalent porous medium (EPM). The advantage of this formulation is that it prevents unphysical values of pressure at reasonable computational cost. We performed sensitivity tests to investigate the effect of different model parameters. The strongest influence of model parameters was detected in terms of governing the opening and closure of the system. Furthermore, we applied the model to the Northeast Greenland Ice Stream, where an efficient system independent of seasonal input was identified about 500 km downstream from the ice divide. Using the effective pressure from the hydrology model, the Ice Sheet System Model (ISSM) showed considerable improvements in modeled velocities in the coastal region.

355. h-Adaptivity Applied to Ice Sheet Simulation

Authors: Thiago Dias dos Santos; Philippe Remy Bernard Devloo; Jefferson Cardia Simões; Mathieu Morlighem; Hélène Seroussi

Journal: Proceeding Series of the Brazilian Society of Computational and Applied Mathematics

Publisher: SBMAC

DOI: 10.5540/03.2018.006.02.0280

URL: <https://doi.org/10.5540/03.2018.006.02.0280>

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Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Ice-sheet model|Adaptive Mesh Refinement|Antarctic ice sheet|Position (finance)|Line (geometry)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

356. Retreat of Thwaites Glacier, West Antarctica, over the next 100 years using various ice flow models, ice shelf melt scenarios and basal friction laws

Authors: Hongju Yu; Eric Rignot; Helene Seroussi; Mathieu Morlighem

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-12-3861-2018

URL: <https://doi.org/10.5194/tc-12-3861-2018>

Publication Date: 2018-12-11

Volume/Issue: Volume 12, Issue 12

References: 80

Cited by: 39 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration

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Open Access: N/A

Abstract: . Thwaites Glacier (TG), West Antarctica, has experienced rapid, potentially irreversible grounding line retreat and mass loss in response to enhanced ice shelf melting. Results from recent numerical models suggest a large spread in the evolution of the glacier in the coming decades to a century. It is therefore important to investigate how different approximations of the ice stress balance, parameterizations of basal friction and ice shelf melt parameterizations may affect projections. Here, we simulate the evolution of TG using ice sheet models of varying levels of complexity, different basal friction laws and ice shelf melt to quantify their effect on the projections. We find that the grounding line retreat and its sensitivity to ice shelf melt are enhanced when a full-Stokes model is used, a Budd friction is used and ice shelf melt is applied on partially floating elements. Initial conditions also impact the model results. Yet, all simulations suggest a rapid, sustained retreat of the glacier along the same preferred pathway. The fastest retreat rate occurs on the eastern side of the glacier, and the slowest retreat occurs across a subglacial ridge on the western side. All the simulations indicate that TG will undergo an accelerated retreat once the glacier retreats past the western subglacial ridge. Combining all the simulations, we find that the uncertainty of the projections is small in the first 30 years, with a cumulative contribution to sea level rise of 5 mm, similar to the current rate. After 30 years, the contribution to sea level depends on the model configurations, with differences up to 300 % over the next 100 years, ranging from 14 to 42 mm.

357. Comparative simulations of the evolution of the Greenland ice sheet under simplified Paris Agreement scenarios with the models SICOPOLIS and ISSM

Authors: Martin Rückamp; Ralf Greve; Angelika Humbert

Journal: Polar Science

Publisher: Elsevier BV

DOI: 10.1016/j.polar.2018.12.003

URL: <https://doi.org/10.1016/j.polar.2018.12.003>

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Cited by: 32 publications

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Keywords: N/A

Funding: Helmholtz Climate Initiative REKLIM (Regional Climate Change), a joint research project of the Helmholtz Association of German Research Centres, BMBF, Japan Society for the Promotion of Science (JSPS) KAKENHI (and 3 more)

License: N/A

Open Access: N/A

Abstract: N/A

358. The GRISLI ice sheet model (version 2.0): calibration and validation for multi-millennial changes of the Antarctic ice sheet

Authors: Aurélien Quiquet; Christophe Dumas; Catherine Ritz; Vincent Peyaud; Didier M. Roche

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-11-5003-2018

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References: 97

Cited by: 35 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: . In this paper, we present the GRISLI (Grenoble ice sheet and land ice) model in its newest revision (version 2.0). Whilst GRISLI is applicable to any given ice sheet, we focus here on the Antarctic ice sheet because it highlights the importance of grounding line dynamics. Important improvements have been implemented in the model since its original version (Ritz et al., 2001). Notably, GRISLI now includes a basal hydrology model and an explicit flux computation at the grounding line based on the analytical formulations of Schoof (2007) or Tsai et al. (2015). We perform a full calibration of the model based on an ensemble of 300 simulations sampling mechanical parameter space using a Latin hypercube method. Performance of individual members is assessed relative to the deviation from present-day observed Antarctic ice thickness. To assess the ability of the model to simulate grounding line migration, we also present glacial-interglacial ice sheet changes throughout the last 400 kyr using the best ensemble members taking advantage of the capacity of the model to perform multi-millennial long-term integrations. To achieve this goal, we construct a simple climatic perturbation of present-day climate forcing fields based on two climate proxies: atmospheric and oceanic. The model is able to reproduce expected grounding line advances during glacial periods and subsequent retreats during terminations with reasonable glacial-interglacial ice volume changes.

359. Altimeter-era emergence of the patterns of forced sea-level rise in climate models and implications for the future

Authors: John T. Fasullo (National Center for Atmospheric Research); R. Steven Nerem (University of Colorado Boulder)

Journal: Proceedings of the National Academy of Sciences

Publisher: Proceedings of the National Academy of Sciences

DOI: 10.1073/pnas.1813233115

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Cited by: 61 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration, National Science Foundation, U.S. Department of Energy

License: N/A

Open Access: N/A

Abstract: Significance Regional patterns of sea-level rise have been observed from satellites since 1993 and are associated with increased coastal impacts in many regions. It is unknown whether such patterns will be transient, arising from natural climate variations, or persistent, driven by external climate forcing. Here, using climate model ensembles, we demonstrate that forced changes are likely to have contributed significantly to observed altimeter-era patterns of rise and that these patterns may persist for decades to come, with increased intensity as climate change progresses.

360. Comparison of four calving laws to model Greenland outlet glaciers

Authors: Youngmin Choi; Mathieu Morlighem; Michael Wood; Johannes H. Bondzio

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-12-3735-2018

URL: <https://doi.org/10.5194/tc-12-3735-2018>

Publication Date: 2018-11-29

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Keywords: N/A

Funding: National Science Foundation, National Aeronautics and Space Administration

License: CC BY 4.0

Open Access: N/A

Abstract: . Calving is an important mechanism that controls the dynamics of marine terminating glaciers of Greenland. Iceberg calving at the terminus affects the entire stress regime of outlet glaciers, which may lead to further retreat and ice flow acceleration. It is therefore critical to accurately parameterize calving in ice sheet models in order to improve the projections of ice sheet change over the coming decades and reduce the uncertainty in their contribution to sea-level rise. Several calving laws have been proposed, but most of them have been applied only to a specific region and have not been tested on other glaciers, while some others have only been implemented in 1-D flowline or vertical flowband models. Here, we test and compare several calving laws recently proposed in the literature using the Ice Sheet System Model (ISSM). We test these calving laws on nine tidewater glaciers of Greenland. We compare the modeled ice front evolution to the observed retreat from Landsat data collected over the past 10 years, and assess which calving law has better predictive abilities for each glacier. Overall, the von Mises stress calving law is more satisfactory than other laws for simulating observed ice front retreat, but new parameterizations that better capture the different modes of calving should be developed. Although the final positions of ice fronts are different for forecast simulations with different calving laws, our results confirm that ice front retreat highly depends on bed topography, irrespective of the calving law employed. This study also confirms that calving dynamics needs to be 3-D or in plan view in ice sheet models to account for complex bed topography and narrow fjords along the coast of Greenland.

361. Control of Ocean Temperature on Jakobshavn Isbræ's Present and Future Mass Loss

Authors: Johannes H. Bondzio (University of California Irvine); Mathieu Morlighem (University of California Irvine); H  l  ne Seroussi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Michael H. Wood (University of California Irvine); J  r  mie Moug  not (University of California Irvine, Institut des G  osciences de l'Environnement Grenoble France)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018gl079827

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Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: sea level projection, glaciology, ice flow, Jakobshavn Isbræ, statistical approach (and 1 more)

Funding: National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: Large uncertainties in model parameterizations and input data sets make projections of future sea level rise contributions of outlet glaciers challenging. Here we introduce a novel technique for weighing large ensemble model simulations that uses information of key observables. The approach is robust to input errors and yields calibrated means and error estimates of a glacier's mass balance. We apply the technique to Jakobshavn Isbræ, using a model that includes a dynamic calving law, and closely reproduce the observed behavior from 1985 to 2018 by forcing the model with ocean temperatures only. Our calibrated projection suggests that the glacier will continue to retreat and contribute about 5.1 mm to eustatic sea level rise by 2100 under present-day climatic forcing. Our analysis shows that the glacier's future evolution will strongly depend on the ambient oceanic setting.

362. Dynamically coupling full Stokes and shallow shelf approximation for marine ice sheet flow using Elmer/Ice (v8.3)

Authors: Eef C. H. van Dongen; Nina Kirchner; Martin B. van Gijzen; Roderik S. W. van de Wal; Thomas Zwinger; Gong Cheng; Per Lötstedt; Lina von Sydow

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

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References: 47

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: . Ice flow forced by gravity is governed by the full Stokes (FS) equations, which are computationally expensive to solve due to the nonlinearity introduced by the rheology. Therefore, approximations to the FS equations are commonly used, especially when modeling a marine ice sheet (ice sheet, ice shelf, and/or ice stream) for 103 years or longer. The shallow ice approximation (SIA) and shallow shelf approximation (SSA) are commonly used but are accurate only for certain parts of an ice sheet. Here, we report a novel way of iteratively coupling FS and SSA that has been implemented in Elmer/Ice and applied to conceptual marine ice sheets. The FS-SSA coupling appears to be very accurate; the relative error in velocity compared to FS is below 0.5 % for diagnostic runs and below 5 % for prognostic runs. Results for grounding line dynamics obtained with the FS-SSA coupling are similar to those obtained from an FS model in an experiment with a periodical temperature forcing over 3000 years that induces grounding line advance and retreat. The rapid convergence of the FS-SSA coupling shows a large potential for reducing computation time, such that modeling a marine ice sheet for thousands of years should become feasible in the near future. Despite inefficient matrix assembly in the current implementation, computation time is reduced by 32 %, when the coupling is applied to a 3-D ice shelf.

363. Exploration of Antarctic Ice Sheet 100-year contribution to sea level rise and associated model uncertainties using the ISSM framework

Authors: Nicole-Jeanne Schlegel; Helene Seroussi; Michael P. Schodlok; Eric Y. Larour; Carmen Boening; Daniel Limonadi; Michael M. Watkins; Mathieu Morlighem; Michiel R. van den Broeke

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-12-3511-2018

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Language: en

Subjects: N/A

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Funding: N/A

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Open Access: N/A

Abstract: . Estimating the future evolution of the Antarctic Ice Sheet (AIS) is critical for improving future sea level rise (SLR) projections. Numerical ice sheet models are invaluable tools for bounding Antarctic vulnerability; yet, few continental-scale projections of century-scale AIS SLR contribution exist, and those that do vary by up to an order of magnitude. This is partly because model projections of future sea level are inherently uncertain and depend largely on the model's boundary conditions and climate forcing, which themselves are unknown due to the uncertainty in the projections of future anthropogenic emissions and subsequent climate response. Here, we aim to improve the understanding of how uncertainties in model forcing and boundary conditions affect ice sheet model simulations. With use of sampling techniques embedded within the Ice Sheet System Model (ISSM) framework, we assess how uncertainties in snow accumulation, ocean-induced melting, ice viscosity, basal friction, bedrock elevation, and the presence of ice shelves impact continental-scale 100-year model simulations of AIS future sea level contribution. Overall, we find that AIS sea level contribution is strongly affected by grounding line retreat, which is driven by the magnitude of ice shelf basal melt rates and by variations in bedrock topography. In addition, we find that over 1.2 m of AIS global mean sea level contribution over the next century is achievable, but not likely, as it is tenable only in response to unrealistically large melt rates and continental ice shelf collapse. Regionally, we find that under our most extreme 100-year warming experiment generalized for the entire ice sheet, the Amundsen Sea sector is the most significant source of model uncertainty (1032 mm 6sigma spread) and the region with the largest potential for future sea level contribution (297 mm). In contrast, under a more plausible forcing informed regionally by literature and model sensitivity studies, the Ronne basin has a greater potential for local increases in ice shelf basal melt rates. As a result, under this more likely realization, where warm waters reach the continental shelf under the Ronne ice shelf, it is the Ronne basin, particularly the Evans and Rutford ice streams, that are the greatest contributors to potential SLR (161 mm) and to simulation uncertainty (420 mm 6sigma spread).

364. Eemian Greenland ice sheet simulated with a higher-order model shows strong sensitivity to SMB forcing

Authors: Andreas Plach; Kerim H. Nisancioglu; Petra M. Langebroek; Andreas Born

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/tc-2018-225

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Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Eemian|Greenland ice sheet|Future sea level|Forcing (mathematics)|Ice-sheet model|Glacier mass balance

Funding: European Commission

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Open Access: N/A

Abstract: . The Greenland ice sheet (GrIS) contributes increasingly to global sea level rise and its past history is a valuable reference for future sea level projections. We present ice sheet simulations for the Eemian interglacial period (~ 125,000 years ago), the period with the most recent warmer-than-present summer climate over Greenland. The evolution of the Eemian GrIS is simulated with a 3D higher-order ice sheet model forced with surface mass balance (SMB) derived from regional climate simulations. Sensitivity experiments with different SMB, basal friction, and ice flow approximations are discussed. We find that the SMB forcing is the controlling factor setting the Eemian minimum ice volume, emphasizing the importance of a reliable SMB model. Our results suggest that when estimating the contribution from the GrIS to sea level rise during warm periods, such as the Eemian interglacial period, the SMB forcing is more important than the representation of ice flow.

365. Basal control of supraglacial meltwater catchments on the Greenland Ice Sheet

Authors: Josh Crozier; Leif Karlstrom; Kang Yang

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-12-3383-2018

URL: <https://doi.org/10.5194/tc-12-3383-2018>

Publication Date: 2018-10-29

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Cited by: 14 publications

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Open Access: N/A

Abstract: . Ice surface topography controls the routing of surface meltwater generated in the ablation zones of glaciers and ice sheets. Meltwater routing is a direct source of ice mass loss as well as a primary influence on subglacial hydrology and basal sliding of the ice sheet. Although the processes that determine ice sheet topography at the largest scales are known, controls on the topographic features that influence meltwater routing at supraglacial internally drained catchment (IDC) scales (<10s of km) are less well constrained. Here we examine the effects of two processes on ice sheet surface topography: transfer of bed topography to the surface of flowing ice and thermal-fluvial erosion by supraglacial meltwater streams. We implement 2-D basal transfer functions in seven study regions of the western Greenland Ice Sheet ablation zone using recent data sets for bed elevation, ice surface elevation, and ice surface velocities. We find that ~1-10 km scale ice surface features can be explained well by bed topography transfer in regions with different multiyear-averaged ice flow conditions. We use flow-routing algorithms to extract supraglacial stream networks from 2 to 5 m resolution digital elevation models and compare these with synthetic flow networks calculated on ice surfaces predicted by bed topography transfer. Multiple geomorphological metrics calculated for these networks suggest that bed topography can explain general ~1-10 km supraglacial meltwater routing and that thermal-fluvial erosion thus has a lesser role in shaping ice surface topography on these scales. We then use bed topography transfer functions and flow routing to conduct a parameter study predicting how supraglacial IDC configurations and subglacial hydraulic potential would change under varying multiyear-averaged ice flow and basal sliding regimes. Predicted changes to subglacial hydraulic flow pathways directly caused by changing ice surface topography are subtle, but temporal changes in basal sliding or ice thickness have potentially significant influences on IDC spatial distribution. We suggest that changes to IDC size and number density could affect subglacial hydrology primarily by dispersing the englacial-subglacial input of surface meltwater.

366. Modeling the response of Northwest Greenland to enhanced ocean thermal forcing and subglacial discharge

Authors: Mathieu Morlighem; Michael Wood; Hélène Seroussi; Youngmin Choi; Eric Rignot

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/tc-2018-214

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References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Forcing (mathematics)|Glacier ice accumulation|Future sea level|Fjord|Greenland ice sheet|Tidewater glacier cycle|Glacier morphology

Funding: National Aeronautics and Space Administration, Division of Arctic Sciences

License: CC BY 4.0

Open Access: N/A

Abstract: . Calving front dynamics is an important control on Greenland's ice mass balance. Ice front retreat of marine-terminating glaciers may, for example, lead to a loss in resistive stress, which ultimately results in glacier acceleration and thinning. Over the past decade, it has been suggested that such retreats may be triggered by warm and salty Atlantic water, which is typically found at a depth below 200-300 m. An increase in subglacial water discharge at glacier ice fronts due to enhanced surface runoff may also be responsible for an intensification of undercutting and calving. An increase in ocean thermal forcing or subglacial discharge therefore has the potential to destabilize marine terminating glaciers along the coast of Greenland. It remains unclear which glaciers are currently stable but may retreat in the future, and how far inland and how fast they will retreat. Here, we quantify the sensitivity and vulnerability of marine-terminating glaciers along the Northwest coast of Greenland (from 72.5° to 76° N) to the ocean forcing and subglacial discharge using the Ice Sheet System Model (ISSM). We rely on the undercutting parameterization based on ocean thermal forcing and subglacial discharge, and use ocean temperature and salinity from high-resolution ECCO2 (Estimating the Circulation & Climate of the Ocean, Phase II) simulations at the fjords mouth to constrain the ocean thermal forcing. The ice flow model includes a calving law based on a tensile Von Mises criterion. While these parameterizations remain approximations and do not include all the physical processes at play, they have been shown to provide reliable estimates of undercutting and calving rates, respectively, on a number of glaciers along the coast of Greenland. We find that some glaciers, such as Dietrichson Gletscher or Alison Gletscher, are sensitive to small increases in ocean thermal forcing, while others, such as Illullip Sermia or Cornell Gletscher, are remarkably stable and remain stable, even in a 3-degree ocean warming scenario. Under the most intense experiment, we find that Hayes Gletscher retreats by more than 50 km inland into a deep trough and its velocity increases by a factor of 10 over only 15 years. The model confirms that ice-ocean interactions can trigger extensive and rapid glacier retreat, but the bed controls the rate and magnitude of the retreat. Under current oceanic and atmospheric condition, we find that this sector alone will contribute more than 1 cm to sea level, and up

to 3 cm under the most extreme scenario.

367. Retrieval of Englacial Firn Aquifer Thickness From Ice-Penetrating Radar Sounding in Southeastern Greenland

Authors: W. Chu (Department of Geophysics Stanford University Stanford CA USA); D. M. Schroeder (Department of Geophysics Stanford University Stanford CA USA, Department of Electrical Engineering Stanford University Stanford CA USA); M. R. Siegfried (Department of Geophysics Stanford University Stanford CA USA)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018gl079751

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Cited by: 33 publications

Language: en

Subjects: N/A

Keywords: meltwater storage, radar sounding, ice sheet modeling, Greenland Ice Sheet, firn aquifer

Funding: National Aeronautics and Space Administration, National Science Foundation, Stanford University

License: N/A

Open Access: N/A

Abstract: An extensive perennial firn aquifer was previously mapped in Helheim Glacier, Southeast Greenland. However, critical constraints on aquifer thickness have been so far unobtainable without expensive field observations. Here we present a novel method that combines very high frequency airborne deep ice radar and ice sheet modeling to retrieve aquifer thickness and its evolution at Helheim Glacier. Using 2012-2014 radar measurements, we identify three aquifers of 4-25 m thick, with a total area of 1,934 km² and water storage of 2.2 +/- 1.5 Gt, about half the volume of previous estimates. The aquifer system is dynamic, and its thickness varies interannually at a rate similar to changes in surface mass balance. The rapid upstream migration of the saturated aquifer implies that this feature has the potential to increase its storage potential in upper Greenland. Together, a combination of very high- and ultra high frequency radar sounding provides a powerful tool to characterize englacial firn aquifers.

368. Toward Improved Understanding of Changes in Greenland Outlet Glacier Shear Margin Dynamics in a Warming Climate

Authors: Derrick J. Lampkin; Byron Parizek; Eric Y. Larour; Hélène Seroussi; Casey Joseph; John P. Cavanagh

Journal: Frontiers in Earth Science

Publisher: Frontiers Media SA

DOI: 10.3389/feart.2018.00156

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Publication Date: 2018-10-18

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Cited by: 2 publications

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Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration, National Science Foundation

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Open Access: N/A

Abstract: N/A

369. The effect of overshooting 1.5 degreesC global warming on the mass loss of the Greenland ice sheet

Authors: Martin Rückamp; Ulrike Falk; Katja Frieler; Stefan Lange; Angelika Humbert

Journal: Earth System Dynamics

Publisher: Copernicus GmbH

DOI: 10.5194/esd-9-1169-2018

URL: <https://doi.org/10.5194/esd-9-1169-2018>

Publication Date: 2018-10-09

Volume/Issue: Volume 9, Issue 4

References: 59

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: . Sea-level rise associated with changing climate is expected to pose a major challenge for societies. Based on the efforts of COP21 to limit global warming to 2.0 °C or even 1.5 °C by the end of the 21st century (Paris Agreement), we simulate the future contribution of the Greenland ice sheet (GrIS) to sea-level change under the low emission Representative Concentration Pathway (RCP) 2.6 scenario. The IceSheet System Model (ISSM) with higher-order approximation is used and initialized with a hybrid approach of spin-up and data assimilation. For three general circulation models (GCMs: HadGEM2-ES, IPSL-CM5A-LR, MIROC5) the projections are conducted up to 2300 with forcing fields for surface mass balance (SMB) and ice surface temperature (Ts) computed by the surface energy balance model of intermediate complexity (SEMIC). The projected sea-level rise ranges between 21-38 mm by 2100 and 36-85 mm by 2300. According to the three GCMs used, global warming will exceed 1.5 °C early in the 21st century. The RCP2.6 peak and decline scenario is therefore manually adjusted in another set of experiments to suppress the 1.5 °C overshooting effect. These scenarios show a sea-level contribution that is on average about 38 % and 31 % less by 2100 and 2300, respectively. For some experiments, the rate of mass loss in the 23rd century does not exclude a stable ice sheet in the future. This is due to a spatially integrated SMB that remains positive and reaches values similar to the present day in the latter half of the simulation period. Although the mean SMB is reduced in the warmer climate, a future steady-state ice sheet with lower surface elevation and hence volume might be possible. Our results indicate that uncertainties in the projections stem from the underlying GCM climate data used to calculate the surface mass balance. However, the RCP2.6 scenario will lead to significant changes in the GrIS, including elevation changes of up to 100 m. The sea-level contribution estimated in this study may serve as a lower bound for the RCP2.6 scenario, as the currently observed sea-level rise is not reached in any of the experiments; this is attributed to processes (e.g. ocean forcing) not yet represented by the model, but proven to play a major role in GrIS mass loss.

370. Calving glaciers and ice shelves

Authors: Douglas I. Benn (University of St Andrews); Jan A. Åström (CSC IT Center for Science)

Journal: Advances in Physics: X

Publisher: Informa UK Limited

DOI: 10.1080/23746149.2018.1513819

URL: <https://doi.org/10.1080/23746149.2018.1513819>

Publication Date: 2018-10-07

Volume/Issue: Volume 3, Issue 1

References: 116

Cited by: 30 publications

Language: en

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Keywords: N/A

Funding: Natural Environment Research Council

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Open Access: N/A

Abstract: N/A

371. A statistical fracture model for Antarctic ice shelves and glaciers

Authors: Veronika Emetc; Paul Tregoning; Mathieu Morlighem; Chris Borstad; Malcolm Sambridge

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-12-3187-2018

URL: <https://doi.org/10.5194/tc-12-3187-2018>

Publication Date: 2018-10-05

Volume/Issue: Volume 12, Issue 10

References: 80

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: . Antarctica and Greenland hold enough ice to raise sea level by more than 65 m if both ice sheets were to melt completely. Predicting future ice sheet mass balance depends on our ability to model these ice sheets, which is limited by our current understanding of several key physical processes, such as iceberg calving. Large-scale ice flow models either ignore this process or represent it crudely. To model fractured zones, an important component of many calving models, continuum damage mechanics as well as linear fracture mechanics are commonly used. However, these methods have a large number of uncertainties when applied across the entire Antarctic continent because the models were typically tuned to match processes seen on particular ice shelves. Here we present an alternative, statistics-based method to model the most probable zones of the location of fractures and demonstrate our approach on all main ice shelf regions in Antarctica, including the Antarctic Peninsula. We can predict the location of observed fractures with an average success rate of 84 % for grounded ice and 61 % for floating ice and a mean overestimation error rate of 26 % and 20 %, respectively. We found that Antarctic ice shelves can be classified into groups based on the factors that control fracture location.

372. Representation of basal melting at the grounding line in ice flow models

Authors: Hélène Seroussi; Mathieu Morlighem

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-12-3085-2018

URL: <https://doi.org/10.5194/tc-12-3085-2018>

Publication Date: 2018-10-01

Volume/Issue: Volume 12, Issue 10

References: 40

Cited by: 72 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Jet Propulsion Laboratory

License: CC BY 4.0

Open Access: N/A

Abstract: . While a lot of attention has been given to the numerical implementation of grounding lines and basal friction in the grounding zone, little has been done about the impact of the numerical treatment of ocean-induced basal melting in this region. Several strategies are currently being employed in the ice sheet modeling community, and the resulting grounding line dynamics may differ strongly, which ultimately adds significant uncertainty to the projected contribution of marine ice sheets to sea level rise. We investigate here several implementations of basal melt parameterization on partially floating elements in a finite-element framework, based on the Marine Ice Sheet-Ocean Model Intercomparison Project (MISOMIP) setup: (1) melt applied only to entirely floating elements, (2) melt applied over all elements that are crossed by the grounding line, and (3) melt integrated partially over the floating portion of a finite element using two different sub-element integration methods. All methods converge towards the same state when the mesh resolution is fine enough. However, (2) and (3) will systematically overestimate the rate of grounding line retreat in coarser resolutions, while (1) converges faster to the solution in most cases. The differences between sub-element parameterizations are exacerbated for experiments with high melting rates in the vicinity of the grounding line and for a Weertman sliding law. As most real-world simulations use horizontal mesh resolutions of several hundreds of meters at best, and high melt rates are generally present close to the grounding lines, we recommend not using (3) to avoid overestimating the rate of grounding line retreat and to carefully assess the impact of mesh resolution and sub-element melt parameterizations on all simulation results.

373. The influence of a poroelastic till on rapid subglacial flooding and cavity formation

Authors: Duncan R. Hewitt; Gregory P. Chini; Jerome A. Neufeld

Journal: Journal of Fluid Mechanics

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/jfm.2018.624

URL: <https://doi.org/10.1017/jfm.2018.624>

Publication Date: 2018-09-24

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References: 31

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We develop a model of the rapid propagation of water at the contact between elastic glacial ice and a poroelastic subglacial till, motivated by observations of the rapid drainage of supraglacial lakes in Greenland. By treating the ice as an elastic bending beam, the fluid dynamics of contact with the subglacial hydrological network, which is modelled as a saturated poroelastic till, can be examined in detail. The model describes the formation and dynamics of an axisymmetric subglacial cavity, and the spread of pore pressure, in response to injection of fluid. A combination of numerical simulation and asymptotic analysis is used to describe these dynamics for both a rigid and a deformable porous till, and for both laminar and turbulent fluid flow. For constant injection rates and laminar flow, the cavity is isostatic and its spread is controlled by bending of the ice and suction of pore water in the vicinity of the ice-till contact. For a deformable till, this control can be modified: generically, a flexural wave that is initially trapped in advance of the contact point relaxes over time by diffusion of pore pressure ahead of the cavity. While the dynamics are found to be relatively insensitive to the properties of the subglacial till during injection with a constant flux, significant dependence on the till properties is manifest during the subsequent spread of a constant volume. A simple hybrid turbulent-laminar model is presented to account for fast injection rates of water: in this case, self-similar turbulent propagation can initially control the spread of the cavity, but there is a transition to laminar control in the vicinity of the ice-till contact point as the flow slows. Finally, the model results are compared with recent geophysical observations of the rapid drainage of supraglacial lakes in Greenland; the comparison provides qualitative agreement and raises suggestions for future quantitative comparison.

374. MPAS-Albany Land Ice (MALI): a variable-resolution ice sheet model for Earth system modeling using Voronoi grids

Authors: Matthew J. Hoffman; Mauro Perego; Stephen F. Price; William H. Lipscomb; Tong Zhang; Douglas Jacobsen; Irina Tezaur; Andrew G. Salinger; Raymond Tuminaro; Luca Bertagna

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-11-3747-2018

URL: <https://doi.org/10.5194/gmd-11-3747-2018>

Publication Date: 2018-09-18

Volume/Issue: Volume 11, Issue 9

References: 120

Cited by: 58 publications

Language: en

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Funding: Biological and Environmental Research

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Open Access: N/A

Abstract: . We introduce MPAS-Albany Land Ice (MALI) v6.0, a new variable-resolution land ice model that uses unstructured Voronoi grids on a plane or sphere. MALI is built using the Model for Prediction Across Scales (MPAS) framework for developing variable-resolution Earth system model components and the Albany multi-physics code base for the solution of coupled systems of partial differential equations, which itself makes use of Trilinos solver libraries. MALI includes a three-dimensional first-order momentum balance solver (Blatter-Pattyn) by linking to the Albany-LI ice sheet velocity solver and an explicit shallow ice velocity solver. The evolution of ice geometry and tracers is handled through an explicit first-order horizontal advection scheme with vertical remapping. The evolution of ice temperature is treated using operator splitting of vertical diffusion and horizontal advection and can be configured to use either a temperature or enthalpy formulation. MALI includes a mass-conserving subglacial hydrology model that supports distributed and/or channelized drainage and can optionally be coupled to ice dynamics. Options for calving include "eigen calving", which assumes that the calving rate is proportional to extensional strain rates. MALI is evaluated against commonly used exact solutions and community benchmark experiments and shows the expected accuracy. Results for the MISIP3 benchmark experiments with MALI's Blatter-Pattyn solver fall between published results from Stokes and L1L2 models as expected. We use the model to simulate a semi-realistic Antarctic ice sheet problem following the initMIP protocol and using 2 km resolution in marine ice sheet regions. MALI is the glacier component of the Energy Exascale Earth System Model (E3SM) version 1, and we describe current and planned coupling to other E3SM components.

375. What drives 20th century polar motion?

Authors: Surendra Adhikari; Lambert Caron; Bernhard Steinberger; John T. Reager; Kristian K. Kjeldsen; Ben Marzeion; Eric Larour; Erik R. Ivins

Journal: Earth and Planetary Science Letters

Publisher: Elsevier BV

DOI: 10.1016/j.epsl.2018.08.059

URL: <https://doi.org/10.1016/j.epsl.2018.08.059>

Publication Date: 2018-09-13

Volume/Issue: Volume 502

References: 43

Cited by: 45 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: N/A

376. Projections of Future Sea Level Contributions from the Greenland and Antarctic Ice Sheets: Challenges Beyond Dynamical Ice Sheet Modeling

Authors: Unknown Author; Sophie Nowicki; Helene Seroussi

Journal: Oceanography

Publisher: The Oceanography Society

DOI: 10.5670/oceanog.2018.216

URL: <https://doi.org/10.5670/oceanog.2018.216>

Publication Date: 2018-08-16

Volume/Issue: Volume 31, Issue 2

References: 94

Cited by: 22 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

377. Basal friction of Fleming Glacier, Antarctica - Part 1: Sensitivity of inversion to temperature and bedrock uncertainty

Authors: Chen Zhao; Rupert M. Gladstone; Roland C. Warner; Matt A. King; Thomas Zwinger; Mathieu Morlighem

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-12-2637-2018

URL: <https://doi.org/10.5194/tc-12-2637-2018>

Publication Date: 2018-08-15

Volume/Issue: Volume 12, Issue 8

References: 49

Cited by: 22 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: . Many glaciers in the Antarctic Peninsula are now rapidly losing mass. Understanding of the dynamics of these fast-flowing glaciers, and their potential future behaviour, can be improved through ice sheet modelling studies. Inverse methods are commonly used in ice sheet models to infer the spatial distribution of a basal friction coefficient, which has a large effect on the basal velocity and ice deformation. Here we use the full-Stokes Elmer/Ice model to simulate the Wordie Ice Shelf-Fleming Glacier system in the southern Antarctic Peninsula. With an inverse method, we infer the pattern of the basal friction coefficient from surface velocities observed in 2008. We propose a multi-cycle spin-up scheme to reduce the influence of the assumed initial glacial temperature field on the final inversion. This is particularly important for glaciers like the Fleming Glacier, which have areas of strongly temperature-dependent deformational flow in the fast-flowing regions. Sensitivity tests using various bed elevation datasets, ice front positions and boundary conditions demonstrate the importance of high-accuracy ice thickness/bed geometry data and precise location of the ice front boundary.

378. The Scalability of Embedded Structured Grids and Unstructured Grids in Large Scale Ice Sheet Modeling on Distributed Memory Parallel Computers

Authors: Phillip M. Dickens; Christopher Dufour; James Fastook

Journal: 2018 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)

Publisher: IEEE

DOI: 10.1109/ipdpsw.2018.00152

URL: <https://doi.org/10.1109/ipdpsw.2018.00152>

Publication Date: 2018-08-06

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References: 36

Cited by: 1 publications

Language: N/A

Subjects: N/A

Keywords: Distributed memory|Unstructured grid

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

379. SHAKTI: Subglacial Hydrology and Kinetic, Transient Interactions v1.0

Authors: Aleah Sommers; Harihar Rajaram; Mathieu Morlighem

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-11-2955-2018

URL: <https://doi.org/10.5194/gmd-11-2955-2018>

Publication Date: 2018-07-24

Volume/Issue: Volume 11, Issue 7

References: 73

Cited by: 26 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration

License: CC BY 4.0

Open Access: N/A

Abstract: . Subglacial hydrology has a strong influence on glacier and icesheet dynamics, particularly through the dependence of sliding velocity on subglacial water pressure. Significant challenges are involved in modeling subglacial hydrology, as the drainage geometry and flow mechanics are constantly changing, with complex feedbacks that play out between water and ice. A clear tradition has been established in the subglacial hydrology modeling literature of distinguishing between channelized (efficient) and sheetlike (inefficient or distributed) drainage systems or components and using slightly different forms of the governing equations in each subsystem to represent the dominant physics. Specifically, many previous subglacial hydrology models disregard opening by melt in the sheetlike system or redistribute it to adjacent channel elements in order to avoid runaway growth that occurs when it is included in the sheetlike system. We present a new subglacial hydrology model, SHAKTI (Subglacial Hydrology and Kinetic, Transient Interactions), in which a single set of governing equations is used everywhere, including opening by melt in the entire domain. SHAKTI employs a generalized relationship between the subglacial water flux and the hydraulic gradient that allows for the representation of laminar, turbulent, and transitional regimes depending on the local Reynolds number. This formulation allows for the coexistence of these flow regimes in different regions, and the configuration and geometry of the subglacial system evolves naturally to represent sheetlike drainage as well as systematic channelized drainage under appropriate conditions. We present steady and transient example simulations to illustrate the features and capabilities of the model and to examine sensitivity to mesh size and time step size. The model is implemented as part of the Ice Sheet System Model (ISSM).

380. Atmosphere-driven ice sheet mass loss paced by topography: Insights from modelling the south-western Scandinavian Ice Sheet

Authors: Henning Åkesson; Mathieu Morlighem; Kerim H. Nisancioglu; John Inge Svendsen; Jan Mangerud

Journal: Quaternary Science Reviews

Publisher: Elsevier BV

DOI: 10.1016/j.quascirev.2018.07.004

URL: <https://doi.org/10.1016/j.quascirev.2018.07.004>

Publication Date: 2018-07-17

Volume/Issue: Volume 195

References: 139

Cited by: 18 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Research Council of Norway, University of California, Irvine

License: N/A

Open Access: N/A

Abstract: N/A

381. Ocean-Induced Melt Triggers Glacier Retreat in Northwest Greenland

Authors: M. Wood (Department of Earth System Science University of California CA Irvine USA); E. Rignot (Department of Earth System Science University of California CA Irvine USA); I. Fenty (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); D. Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); R. Millan (Department of Earth System Science University of California CA Irvine USA); M. Morlighem (Department of Earth System Science University of California CA Irvine USA); J. Mouginot (Department of Earth System Science University of California CA Irvine USA, University of Grenoble Alpes, Institut de Recherche pour le Développement (IRD)); H. Seroussi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018gl078024

URL: <https://doi.org/10.1029/2018gl078024>

Publication Date: 2018-07-11

Volume/Issue: Volume 45, Issue 16

References: 44

Cited by: 67 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NASA Cryosphere Program, NASA Physical Oceanography program, NASA Ocean Melting Greenland EVS-3 mission (and 1 more)

License: N/A

Open Access: N/A

Abstract: In recent decades, tidewater glaciers in Northwest Greenland contributed significantly to sea level rise but exhibited a complex spatial pattern of retreat. Here we use novel observations of bathymetry and water temperature from NASA's Ocean Melting Greenland mission to quantify the role of warm, salty Atlantic Water in controlling the evolution of 37 glaciers. Modeled ocean-induced undercutting of calving margins compared with ice advection and ice front retreat observed by satellites from 1985 to 2015 indicate that 35 glaciers retreated when cumulative anomalies in ocean-induced undercutting rose above the range of seasonal variability of calving-front positions, while two glaciers standing on shallow sills and colder water did not retreat. Deviations in the observed timing of retreat are explained by residual uncertainties in bathymetry, inefficient mixing of waters in shallow fjords, and the presence of small floating sections. Overall, warmer ocean temperature triggered the retreat, but calving processes dominate ablation (71%).

382. Rising Oceans Guaranteed: Arctic Land Ice Loss and Sea Level Rise

Authors: Twila Moon; Andreas Ahlstrøm; Heiko Goelzer; William Lipscomb; Sophie Nowicki

Journal: Current Climate Change Reports

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s40641-018-0107-0

URL: <https://doi.org/10.1007/s40641-018-0107-0>

Publication Date: 2018-07-10

Volume/Issue: Volume 4, Issue 3

References: 104

Cited by: 34 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Dutch Ministry of Education, Culture and Science, Netherlands Earth System Science Centre, EC Horizon 2020 (and 2 more)

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

383. Improving Greenland Surface Mass Balance Estimates Through the Assimilation of MODIS Albedo: A Case Study Along the K-Transect

Authors: Mahdi Navari (University of Maryland, University of California Los Angeles, National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Steven A. Margulis (University of California Los Angeles); Marco Tedesco (National Aeronautics and Space Administration (NASA) Goddard Institute for Space Studies, National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, Columbia University); Xavier Fettweis (University of Liège); Patrick M. Alexander (National Aeronautics and Space Administration (NASA) Goddard Institute for Space Studies, National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, Columbia University)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018gl078448

URL: <https://doi.org/10.1029/2018gl078448>

Publication Date: 2018-06-19

Volume/Issue: Volume 45, Issue 13

References: 26

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: surface mass balance, data assimilation, Greenland ice sheet, albedo

Funding: National Science Foundation

License: N/A

Open Access: N/A

Abstract: Estimating the Greenland ice sheet surface mass balance (SMB) is an important component of current and future projections of sea level rise. Given the lack of in situ information, imperfect models, and underutilized remote sensing data, it is critical to combine the available observations with a physically based model to better characterize the spatial and temporal variation of the Greenland ice sheet SMB. This work proposes a data assimilation framework that yields SMB estimates that benefit from a state-of-the-art snowpack model (Crocus) and a 16-day albedo product. Comparison of our results against in situ SMB measurements from the Kangerlussuaq transect shows that assimilation of 16-day albedo product reduces the root-mean-square error of the posterior estimates of SMB from 1,240 millimeter water equivalent per year (mmWE/yr) to 230 mmWE/yr and reduces the bias from 1,140 mmWE/yr to -20 mmWE/yr.

384. Crevasses as Indicators of Surge Dynamics in the Bering Bagley Glacier System, Alaska: Numerical Experiments and Comparison to Image Data Analysis

Authors: Thomas Trantow (Department of Electrical, Computer and Energy Engineering University of Colorado CO Boulder USA); Ute C. Herzfeld (Department of Electrical, Computer and Energy Engineering University of Colorado CO Boulder USA, Cooperative Institute for Research in Environmental Sciences Boulder CO USA)

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

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Cited by: 10 publications

Language: en

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Keywords: ice flow, geostatistics, glacier dynamics, surge

Funding: National Science Foundation, U.S. National Science Foundation's (NSF) Geography and Spatial Sciences Program, National Aeronautics and Space Administration's Cryospheric Sciences

License: N/A

Open Access: N/A

Abstract: One of the largest sources of uncertainty in sea level rise prediction is glacial acceleration, of which the surge phenomenon is the least understood type. The surge of the Bering Bagley Glacier System (BBGS), Alaska, in 2011-2013 has provided a rare opportunity to study the surge phenomenon in a large and complex glacier system. A surge results in widespread crevassing throughout the glacier system complicating many traditional techniques used to study glacier dynamics. In this paper, we utilize crevassing as a means to investigate the recent BBGS surge through numerical modeling and geostatistical data analysis. Following the principles of structural glaciology, image-based crevasse characterizations are obtained through geostatistical methods applied to Landsat-7 data, supplemented by airborne field observations. On the modeling side, a 3-D full-Stokes finite element model of the BBGS is developed and applied to investigate ice dynamics and surface structures during the recent surge. A von Mises criterion is adopted to simulate crevassing at the glacier surface, oriented along the axes of maximum principal tensile stress. To facilitate evaluation of model- and data-derived crevasse characteristics, three different comparison methods are introduced. General agreement in the model-data comparisons indicates that the model has the ability to represent the BBGS system during peak acceleration. The crevasse-based approach is also employed to optimize the basal sliding parameter and the von Mises stress threshold in the model. Results further indicate that bed topography is an important constraint in modeling the surge process.

385. Representation of basal melting at the grounding line in ice flow models

Authors: Hélène Seroussi; Mathieu Morlighem

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/tc-2018-117

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Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Line (geometry)|Antarctic ice sheet

Funding: Jet Propulsion Laboratory, Jet Propulsion Laboratory

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Open Access: N/A

Abstract: . While a lot of attention has been given to the numerical implementation of grounding lines and basal friction in the grounding zone, little has been done about the impact of the numerical treatment of ocean-induced basal melting in this region. Several strategies are currently being employed in the ice sheet modeling community, and the resulting grounding line dynamics may differ strongly, which ultimately add significant uncertainty to the projected contribution of marine ice sheets to sea level rise. We investigate here several implementations of basal melt parameterization on partially floating elements in a finite element framework, based on the Marine Ice Sheet-Ocean Model Intercomparison Project (MISOMIP) setup: (1) melt applied only to entirely floating elements, (2) melt applied over the entire elements that are crossed by the grounding line, and (3) melt integrated partially over the floating portion of a finite element using two different sub-element integration methods. All methods converge towards the same state when the mesh resolution is fine enough. However, (2) and (3) will systematically overestimate the rate of grounding line retreat in coarser resolutions, while (1) converges faster to the solution in most cases. The differences between sub-element parameterizations are exacerbated for experiments with large melting rates in the vicinity of the grounding line and for a Weertman sliding law. As most real-world simulations use horizontal mesh resolutions of several hundreds of meters at best, and large melt rates are generally present close to the grounding lines, we recommend using (1) to avoid overestimating the rate of grounding line retreat.

386. Friction at the bed does not control fast glacier flow

Authors: L. A. Stearns (University of Kansas); C. J. van der Veen (University of Kansas)

Journal: Science

Publisher: American Association for the Advancement of Science (AAAS)

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URL: <https://doi.org/10.1126/science.aat2217>

Publication Date: 2018-06-07

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Cited by: 39 publications

Language: en

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Keywords: N/A

Funding: National Science Foundation, National Aeronautics and Space Administration

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Open Access: N/A

Abstract: Sliding at the base Predictions of sea level rise caused by dynamic ice sheet loss rely on a good understanding of what controls how fast the sheets slide over the ground below. The standard approach is to model motion on the basis of an assumed frictional stress between the base of the glacier and a hard underlying bed. Now, however, Stearns and van der Veen show that this method is incorrect. Instead, they suggest that net pressure at the glacier bed controls flow. Science , this issue p. 273

387. Glacial isostatic adjustment modelling: historical perspectives, recent advances, and future directions

Authors: Pippa L. Whitehouse

Journal: Earth Surface Dynamics

Publisher: Copernicus GmbH

DOI: 10.5194/esurf-6-401-2018

URL: <https://doi.org/10.5194/esurf-6-401-2018>

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References: 283

Cited by: 125 publications

Language: en

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Funding: N/A

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Open Access: N/A

Abstract: . Glacial isostatic adjustment (GIA) describes the response of the solid Earth, the gravitational field, and the oceans to the growth and decay of the global ice sheets. A commonly studied component of GIA is "postglacial rebound", which specifically relates to uplift of the land surface following ice melt. GIA is a relatively rapid process, triggering 100 m scale changes in sea level and solid Earth deformation over just a few tens of thousands of years. Indeed, the first-order effects of GIA could already be quantified several hundred years ago without reliance on precise measurement techniques and scientists have been developing a unifying theory for the observations for over 200 years. Progress towards this goal required a number of significant breakthroughs to be made, including the recognition that ice sheets were once more extensive, the solid Earth changes shape over time, and gravity plays a central role in determining the pattern of sea-level change. This article describes the historical development of the field of GIA and provides an overview of the processes involved. Significant recent progress has been made as concepts associated with GIA have begun to be incorporated into parallel fields of research; these advances are discussed, along with the role that GIA is likely to play in addressing outstanding research questions within the field of Earth system modelling.

388. HF/VHF Radar Sounding of Ice from Manned and Unmanned Airborne Platforms

Authors: Emily Arnold (University of Kansas); Fernando Rodriguez-Morales (University of Kansas); John Paden (University of Kansas); Carl Leuschen (University of Kansas); Shawn Keshmiri (University of Kansas); Stephen Yan (University of Alabama); Mark Ewing (University of Kansas); Rick Hale (University of Kansas); Ali Mahmood (University of Kansas); Aaron Blevins (University of Kansas); Akhilesh Mishra (University of Kansas); Teja Karidi (University of Kansas); Bailey Miller (University of Kansas); John Sonntag (AECOM Corporation)

Journal: Geosciences

Publisher: MDPI AG

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References: 68

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation

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Open Access: N/A

Abstract: Ice thickness and bed topography of fast-flowing outlet glaciers are large sources of uncertainty for the current ice sheet models used to predict future contributions to sea-level rise. Due to a lack of coverage and difficulty in sounding and imaging with ice-penetrating radars, these regions remain poorly constrained in models. Increases in off-nadir scattering due to the highly crevassed surfaces, volumetric scattering (due to debris and/or pockets of liquid water), and signal attenuation (due to warmer ice near the bottom) are all impediments in detecting bed-echoes. A set of high-frequency (HF)/very high-frequency (VHF) radars operating at 14 MHz and 30-35 MHz were developed at the University of Kansas to sound temperate ice and outlet glaciers. We have deployed these radars on a small unmanned aircraft system (UAS) and a DHC-6 Twin Otter. For both installations, the system utilized a dipole antenna oriented in the cross-track direction, providing some performance advantages over other temperate ice sounders operating at lower frequencies. In this paper, we describe the platform-sensor systems, field operations, data-processing techniques, and preliminary results. We also compare our results with data from other ice-sounding radars that operate at frequencies both above (Center for Remote Sensing of Ice Sheets (CReSIS) Multichannel Coherent Depth Sounder (MCoRDS)) and below (Jet Propulsion Laboratory (JPL) Warm Ice Sounding Explorer (WISE)) our HF/VHF system. During field campaigns, both unmanned and manned platforms flew closely spaced parallel and repeat flight lines. We examine these data sets to determine image coherency between flight lines and discuss the feasibility of forming 2D synthetic apertures by using such a mission approach.

389. Implementation of higher-order vertical finite elements in ISSM v4.13 for improved ice sheet flow modeling over paleoclimate timescales

Authors: Joshua K. Cuzzzone; Mathieu Morlighem; Eric Larour; Nicole Schlegel; Helene Seroussi

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-11-1683-2018

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Funding: N/A

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Open Access: N/A

Abstract: . Paleoclimate proxies are being used in conjunction with ice sheet modeling experiments to determine how the Greenland ice sheet responded to past changes, particularly during the last deglaciation. Although these comparisons have been a critical component in our understanding of the Greenland ice sheet sensitivity to past warming, they often rely on modeling experiments that favor minimizing computational expense over increased model physics. Over Paleoclimate timescales, simulating the thermal structure of the ice sheet has large implications on the modeled ice viscosity, which can feedback onto the basal sliding and ice flow. To accurately capture the thermal field, models often require a high number of vertical layers. This is not the case for the stress balance computation, however, where a high vertical resolution is not necessary. Consequently, since stress balance and thermal equations are generally performed on the same mesh, more time is spent on the stress balance computation than is otherwise necessary. For these reasons, running a higher-order ice sheet model (e.g., Blatter-Pattyn) over timescales equivalent to the paleoclimate record has not been possible without incurring a large computational expense. To mitigate this issue, we propose a method that can be implemented within ice sheet models, whereby the vertical interpolation along the z axis relies on higher-order polynomials, rather than the traditional linear interpolation. This method is tested within the Ice Sheet System Model (ISSM) using quadratic and cubic finite elements for the vertical interpolation on an idealized case and a realistic Greenland configuration. A transient experiment for the ice thickness evolution of a single-dome ice sheet demonstrates improved accuracy using the higher-order vertical interpolation compared to models using the linear vertical interpolation, despite having fewer degrees of freedom. This method is also shown to improve a model's ability to capture sharp thermal gradients in an ice sheet particularly close to the bed, when compared to models using a linear vertical interpolation. This is corroborated in a thermal steady-state simulation of the Greenland ice sheet using a higher-order model. In general, we find that using a higher-order vertical interpolation decreases the need for a high number of vertical layers, while dramatically reducing model runtime for transient simulations. Results indicate that when using a higher-order vertical interpolation, runtimes for a transient ice sheet relaxation are upwards of 5 to 7 times faster than using a model

which has a linear vertical interpolation, and this thus requires a higher number of vertical layers to achieve a similar result in simulated ice volume, basal temperature, and ice divide thickness. The findings suggest that this method will allow higher-order models to be used in studies investigating ice sheet behavior over paleoclimate timescales at a fraction of the computational cost than would otherwise be needed for a model using a linear vertical interpolation.

390. Complex Basal Thermal Transition Near the Onset of Petermann Glacier, Greenland

Authors: W. Chu (Department of Geophysics, School of Earth, Energy and Environmental Sciences Stanford University Stanford CA USA); D. M. Schroeder (Department of Geophysics, School of Earth, Energy and Environmental Sciences Stanford University Stanford CA USA, Department of Electrical Engineering School of Engineering, Stanford University Stanford CA USA); H. Seroussi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); T. T. Creyts (Columbia University); R. E. Bell (Columbia University)

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

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Cited by: 38 publications

Language: en

Subjects: N/A

Keywords: radar sounding, ice sheet modeling, onset of fast flow, Greenland Ice Sheet, Petermann Glacier (and 1 more)

Funding: National Aeronautics and Space Administration, National Science Foundation, National Science Foundation (and 1 more)

License: N/A

Open Access: N/A

Abstract: The basal thermal regime of ice sheets exerts a strong control on ice-sheet stability and the onset of rapidly streaming flow. However, the nature of this thermal transition where sliding initiates is largely unconstrained by geophysical observations. In the Greenland Ice Sheet, topographic troughs or elevated geothermal heat fluxes typically define the onset of outlet glaciers. In contrast, Petermann Glacier in Northern Greenland does not have any distinct bed troughs or localized geothermal heating associated with its onset, making it an ideal site to investigate the basal thermal state and examine its role in the onset of Petermann Glacier. Here we use radar bed reflectivity and an ice-sheet thermomechanical model to examine the basal thermal regime beneath Petermann Glacier. Our results reveal a complex thermal transition near the onset of Petermann Glacier. As the bed shifts from largely frozen to largely thawed with increasing distances from the ice divide, our results show that this thermal transition happens through alternating bands of frozen and thawed bands. The complex thermal state across the onset region suggests that lateral meltwater injection and local meltwater production determine the location of Petermann Glacier. Given the lack of topographic pinning at the onset location, the upstream margin of Petermann is vulnerable to migrate depending on a combination of advective cooling and meltwater supply from the interior of the ice sheet.

391. Simulating ice thickness and velocity evolution of Upernavik Isstrøm 1849-2012 by forcing prescribed terminus positions in ISSM

Authors: Konstanze Haubner; Jason E. Box; Nicole J. Schlegel; Eric Y. Larour; Mathieu Morlighem; Anne M. Solgaard; Kristian K. Kjeldsen; Signe H. Larsen; Eric Rignot; Todd K. Dupont; Kurt H. Kjær

Journal: The Cryosphere

Publisher: Copernicus GmbH

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Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: . Tidewater glacier velocity and mass balance are known to be highly responsive to terminus position change. Yet it remains challenging for ice flow models to reproduce observed ice margin changes. Here, using the Ice Sheet System Model (Larour et al., 2012), we simulate the ice velocity and thickness changes of Upernavik Isstrøm (north-western Greenland) by prescribing a collection of 27 observed terminus positions spanning 164 years (1849-2012). The simulation shows increased ice velocity during the 1930s, the late 1970s and between 1995 and 2012 when terminus retreat was observed along with negative surface mass balance anomalies. Three distinct mass balance states are evident in the reconstruction: (1849-1932) with near zero mass balance, (1932-1992) with ice mass loss dominated by ice dynamical flow, and (1998-2012), when increased retreat and negative surface mass balance anomalies led to mass loss that was twice that of any earlier period. Over the multi-decadal simulation, mass loss was dominated by thinning and acceleration responsible for 70 % of the total mass loss induced by prescribed change in terminus position. The remaining 30 % of the total ice mass loss resulted directly from prescribed terminus retreat and decreasing surface mass balance. Although the method can not explain the cause of glacier retreat, it enables the reconstruction of ice flow and geometry during 1849-2012. Given annual or seasonal observed terminus front positions, this method could be a useful tool for evaluating simulations investigating the effect of calving laws.

392. Design and results of the ice sheet model initialisation experiments initMIP-Greenland: an ISMIP6 intercomparison

Authors: Heiko Goelzer; Sophie Nowicki; Tamsin Edwards; Matthew Beckley; Ayako Abe-Ouchi; Andy Aschwanden; Reinhard Calov; Olivier Gagliardini; Fabien Gillet-Chaulet; Nicholas R. Golledge; Jonathan Gregory; Ralf Greve; Angelika Humbert; Philippe Huybrechts; Joseph H. Kennedy; Eric Larour; William H. Lipscomb; Sébastien Le clec'h; Victoria Lee; Mathieu Morlighem; Frank Pattyn; Antony J. Payne; Christian Rodehacke; Martin Rückamp; Fuyuki Saito; Nicole Schlegel; Helene Seroussi; Andrew Shepherd; Sainan Sun; Roderik van de Wal; Florian A. Ziemer

Journal: The Cryosphere

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Cited by: 91 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, Office of Science, National Aeronautics and Space Administration (and 3 more)

License: CC BY 4.0

Open Access: N/A

Abstract: . Earlier large-scale Greenland ice sheet sea-level projections (e.g. those run during the ice2sea and SeaRISE initiatives) have shown that ice sheet initial conditions have a large effect on the projections and give rise to important uncertainties. The goal of this initMIP-Greenland intercomparison exercise is to compare, evaluate, and improve the initialisation techniques used in the ice sheet modelling community and to estimate the associated uncertainties in modelled mass changes. initMIP-Greenland is the first in a series of ice sheet model intercomparison activities within ISMIP6 (the Ice Sheet Model Intercomparison Project for CMIP6), which is the primary activity within the Coupled Model Intercomparison Project Phase 6 (CMIP6) focusing on the ice sheets. Two experiments for the large-scale Greenland ice sheet have been designed to allow intercomparison between participating models of (1) the initial present-day state of the ice sheet and (2) the response in two idealised forward experiments. The forward experiments serve to evaluate the initialisation in terms of model drift (forward run without additional forcing) and in response to a large perturbation (prescribed surface mass balance anomaly); they should not be interpreted as sea-level projections. We present and discuss results that highlight the diversity of data sets, boundary conditions, and initialisation techniques used in the community to generate initial states of the Greenland ice sheet. We find good agreement across the ensemble for the dynamic response to surface mass balance changes in areas where the simulated ice sheets overlap but differences arising from the initial size of the ice sheet. The model drift in the control experiment is reduced for models that participated in earlier intercomparison exercises.

393. Implementing an empirical scalar constitutive relation for ice with flow-induced polycrystalline anisotropy in large-scale ice sheet models

Authors: Felicity S. Graham; Mathieu Morlighem; Roland C. Warner; Adam Treverrow

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-12-1047-2018

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References: 78

Cited by: 18 publications

Language: en

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Open Access: N/A

Abstract: . The microstructure of polycrystalline ice evolves under prolonged deformation, leading to anisotropic patterns of crystal orientations. The response of this material to applied stresses is not adequately described by the ice flow relation most commonly used in large-scale ice sheet models - the Glen flow relation. We present a preliminary assessment of the implementation in the Ice Sheet System Model (ISSM) of a computationally efficient, empirical, scalar, constitutive relation which addresses the influence of the dynamically steady-state flow-compatible induced anisotropic crystal orientation patterns that develop when ice is subjected to the same stress regime for a prolonged period - sometimes termed tertiary flow. We call this the ESTAR flow relation. The effect on ice flow dynamics is investigated by comparing idealised simulations using ESTAR and Glen flow relations, where we include in the latter an overall flow enhancement factor. For an idealised embayed ice shelf, the Glen flow relation overestimates velocities by up to 17 % when using an enhancement factor equivalent to the maximum value prescribed in the ESTAR relation. Importantly, no single Glen enhancement factor can accurately capture the spatial variations in flow across the ice shelf generated by the ESTAR flow relation. For flow line studies of idealised grounded flow over varying topography or variable basal friction - both scenarios dominated at depth by bed-parallel shear - the differences between simulated velocities using ESTAR and Glen flow relations depend on the value of the enhancement factor used to calibrate the Glen flow relation. These results demonstrate the importance of describing the deformation of anisotropic ice in a physically realistic manner, and have implications for simulations of ice sheet evolution used to reconstruct paleo-ice sheet extent and predict future ice sheet contributions to sea level.

394. Young Surface of Pluto's Sputnik Planitia Caused by Viscous Relaxation

Authors: Qiang Wei; Yongyun Hu; Yonggang Liu; Douglas N. C. Lin; Jun Yang; Adam P. Showman

Journal: The Astrophysical Journal Letters

Publisher: American Astronomical Society

DOI: 10.3847/2041-8213/aab54f

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References: 48

Cited by: 6 publications

Language: N/A

Subjects: N/A

Keywords: planets and satellites: physical evolution, planets and satellites: composition, planets and satellites: surfaces, Kuiper Belt objects: individual (Pluto), methods: numerical (and 1 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: One of the most prominent features of Pluto observed by the New Horizon mission is the absence of craters on Sputnik Planitia (SP). Vigorous thermal convection could renew the SP surface with sufficient depth at a timescale of $\sim 500,000$ years. Here we present numerical simulations demonstrating that craters can be removed much more quickly across all of SP by viscous relaxation of nitrogen (N_2) ice. The timescale of relaxation is in years if the N_2 layer is 4 km thick and the viscosity is as determined in the lab, and will increase to 104 years if the viscosity is 104 times larger than the measured value. For such high viscosity, the thermal convection will have a timescale of greater than 106 years if it happens at all, so that the relaxation timescale is still more than 2 orders of magnitude shorter. The relaxation timescale decreases with increasing thickness and temperature of the ice layer. The existence of pits on SP can be explained by the surface enhancement of viscosity. Such enhancement does not have significant influence on the relaxation timescale of craters with diameters greater than a few kilometers. Therefore, although convection is required to explain the polygon shapes, it may have a lesser role in the absence of craters on SP. The viscous relaxation mechanism can readily explain the nondetection of both craters and polygon shapes on the southeast SP.

395. Observation-Driven Estimation of the Spatial Variability of 20th Century Sea Level Rise

Authors: B. D. Hamlington (Old Dominion University); A. Burgos (Old Dominion University); P. R. Thompson (Department of Oceanography University of Hawai'i at Manoa Honolulu HI USA); F. W. Landerer (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); C. G. Piecuch (Atmospheric and Environmental Research Inc.); S. Adhikari (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); L. Caron (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); J. T. Reager (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); E. R. Ivins (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Geophysical Research: Oceans

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017jc013486

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References: 38

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: trends, 20th century, sea level

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License: N/A

Open Access: N/A

Abstract: Over the past two decades, sea level measurements made by satellites have given clear indications of both global and regional sea level rise. Numerous studies have sought to leverage the modern satellite record and available historic sea level data provided by tide gauges to estimate past sea level rise, leading to several estimates for the 20th century trend in global mean sea level in the range between 1 and 2 mm/yr. On regional scales, few attempts have been made to estimate trends over the same time period. This is due largely to the inhomogeneity and quality of the tide gauge network through the 20th century, which render commonly used reconstruction techniques inadequate. Here, a new approach is adopted, integrating data from a select set of tide gauges with prior estimates of spatial structure based on historical sea level forcing information from the major contributing processes over the past century. The resulting map of 20th century regional sea level rise is optimized to agree with the tide gauge-measured trends, and provides an indication of the likely contributions of different sources to regional patterns. Of equal importance, this study demonstrates the sensitivities of this regional trend map to current knowledge and uncertainty of the contributing processes.

396. GIA Model Statistics for GRACE Hydrology, Cryosphere, and Ocean Science

Authors: L. Caron (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); E. R. Ivins (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); E. Larour (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); S. Adhikari (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); J. Nilsson (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); G. Blewitt (University of Nevada)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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References: 84

Cited by: 170 publications

Language: en

Subjects: N/A

Keywords: GIA, uncertainty, Bayesian, GRACE, statistics

Funding: NASA Postdoctoral Program and ROSES, Cryosphere Program, NASA Sea-level Change Team (and 1 more)

License: Creative Commons

Open Access: N/A

Abstract: We provide a new analysis of glacial isostatic adjustment (GIA) with the goal of assembling the model uncertainty statistics required for rigorously extracting trends in surface mass from the Gravity Recovery and Climate Experiment (GRACE) mission. Such statistics are essential for deciphering sea level, ocean mass, and hydrological changes because the latter signals can be relatively small (≤ 2 mm/yr water height equivalent) over very large regions, such as major ocean basins and watersheds. With abundant new >7 year continuous measurements of vertical land motion (VLM) reported by Global Positioning System stations on bedrock and new relative sea level records, our new statistical evaluation of GIA uncertainties incorporates Bayesian methodologies. A unique aspect of the method is that both the ice history and 1-D Earth structure vary through a total of 128,000 forward models. We find that best fit models poorly capture the statistical inferences needed to correctly invert for lower mantle viscosity and that GIA uncertainty exceeds the uncertainty ascribed to trends from 14 years of GRACE data in polar regions.

397. Increased West Antarctic and unchanged East Antarctic ice discharge over the last 7 years

Authors: Alex S. Gardner; Geir Moholdt; Ted Scambos; Mark Fahnestock; Stefan Ligtenberg; Michiel van den Broeke; Johan Nilsson

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-12-521-2018

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References: 81

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Abstract: . Ice discharge from large ice sheets plays a direct role in determining rates of sea-level rise. We map present-day Antarctic-wide surface velocities using Landsat 7 and 8 imagery spanning 2013-2015 and compare to earlier estimates derived from synthetic aperture radar, revealing heterogeneous changes in ice flow since ~ 2008. The new mapping provides complete coastal and inland coverage of ice velocity north of 82.4 degrees S with a mean error of < 10 m yr⁻¹, resulting from multiple overlapping image pairs acquired during the daylight period. Using an optimized flux gate, ice discharge from Antarctica is 1929 +/- 40 Gigatons per year (Gt yr⁻¹) in 2015, an increase of 36 +/- 15 Gt yr⁻¹ from the time of the radar mapping. Flow accelerations across the grounding lines of West Antarctica's Amundsen Sea Embayment, Getz Ice Shelf and Marguerite Bay on the western Antarctic Peninsula, account for 88 % of this increase. In contrast, glaciers draining the East Antarctic Ice Sheet have been remarkably constant over the period of observation. Including modeled rates of snow accumulation and basal melt, the Antarctic ice sheet lost ice at an average rate of 183 +/- 94 Gt yr⁻¹ between 2008 and 2015. The modest increase in ice discharge over the past 7 years is contrasted by high rates of ice sheet mass loss and distinct spatial patterns of elevation lowering. The West Antarctic Ice Sheet is experiencing high rates of mass loss and displays distinct patterns of elevation lowering that point to a dynamic imbalance. We find modest increase in ice discharge over the past 7 years, which suggests that the recent pattern of mass loss in Antarctica is part of a longer-term phase of enhanced glacier flow initiated in the decades leading up to the first continent-wide radar mapping of ice flow.

398. A Full-Stokes 3-D Calving Model Applied to a Large Greenlandic Glacier

Authors: Joe Todd (University of St Andrews, University of Cambridge); Poul Christoffersen (University of Cambridge); Thomas Zwinger (CSC-IT Center for Science Espoo Finland); Peter Råback (CSC-IT Center for Science Espoo Finland); Nolwenn Chauché (Access Arctic SARL); Doug Benn (University of St Andrews); Adrian Luckman (Department of Geography, College of Science Swansea University Swansea UK); Johnny Ryan (Centre for Glaciology, Institute of Geography and Earth Sciences Aberystwyth University Aberystwyth UK); Nick Toberg (University of Cambridge); Donald Slater (University of Edinburgh); Alun Hubbard (Centre for Glaciology, Institute of Geography and Earth Sciences Aberystwyth University Aberystwyth UK, Centre for Arctic Gas Hydrate, Environment and Climate, Department of Geology Arctic University of Norway Tromsø Norway)

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017jf004349

URL: <https://doi.org/10.1002/2017jf004349>

Publication Date: 2018-01-30

Volume/Issue: Volume 123, Issue 3

References: 82

Cited by: 60 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Natural Environment Research Council, Norges Forskningsråd, H2020 European Research Council

License: N/A

Open Access: N/A

Abstract: Iceberg calving accounts for around half of all mass loss from both the Greenland and Antarctic ice sheets. The diverse nature of calving and its complex links to both internal dynamics and climate make it challenging to incorporate into models of glaciers and ice sheets. Here we present results from a new open-source 3-D full-Stokes calving model developed in Elmer/Ice. The calving model implements the crevasse depth criterion, which states that calving occurs when surface and basal crevasses penetrate the full thickness of the glacier. The model also implements a new 3-D rediscrretization approach and a time-evolution scheme which allow the calving front to evolve realistically through time. We test the model in an application to Store Glacier, one of the largest outlet glaciers in West Greenland, and find that it realistically simulates the seasonal advance and retreat when two principal environmental forcings are applied. These forcings are (1) submarine melting in distributed and concentrated forms and (2) ice mélange buttressing. We find that ice mélange buttressing is primarily responsible for Store Glacier's seasonal advance and retreat. Distributed submarine melting prevents the glacier from forming a permanent floating tongue, while concentrated plume melting has a disproportionately large and potentially destabilizing effect on the calving front position. Our results also highlight the importance of basal topography, which exerts a strong control on calving, explaining why Store Glacier has remained stable during a period when neighboring glaciers have undergone prolonged interannual retreat.

399. Stabilized equal low-order finite elements in ice sheet modeling - accuracy and robustness

Authors: Christian Helanow; Josefin Ahlkrona

Journal: Computational Geosciences

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10596-017-9713-5

URL: <https://doi.org/10.1007/s10596-017-9713-5>

Publication Date: 2018-01-29

Volume/Issue: Volume 22, Issue 4

References: 61

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Stockholms universitet, Svensk Kärnbränslehantering AB and Posiva Oy, Uppsala Universitet

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

400. Widespread Moulin Formation During Supraglacial Lake Drainages in Greenland

Authors: Matthew J. Hoffman (Los Alamos National Laboratory); Mauro Perego (Center for Computing Research Sandia National Laboratories Albuquerque NM USA); Lauren C. Andrews (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Stephen F. Price (Los Alamos National Laboratory); Thomas A. Neumann (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Jesse V. Johnson (University of Montana); Ginny Catania (University of Texas at Austin); Martin P. Lüthi (Department of Geography University of Zürich Zürich Switzerland)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017gl075659

URL: <https://doi.org/10.1002/2017gl075659>

Publication Date: 2018-01-17

Volume/Issue: Volume 45, Issue 2

References: 79

Cited by: 43 publications

Language: en

Subjects: N/A

Keywords: Greenland, hydrology, fracturing, moulins, ice sheets (and 1 more)

Funding: Los Alamos National Laboratory, Advanced Scientific Computing Research, Biological and Environmental Research (and 3 more)

License: Creative Commons

Open Access: N/A

Abstract: Moulins permit access of surface meltwater to the glacier bed, causing basal lubrication and ice speedup in the ablation zone of western Greenland during summer. Despite the substantial impact of moulins on ice dynamics, the conditions under which they form are poorly understood. We assimilate a time series of ice surface velocity from a network of eleven Global Positioning System receivers into an ice sheet model to estimate ice sheet stresses during winter, spring, and summer in a ~30 x 10 km region. Surface-parallel von Mises stress increases slightly during spring speedup and early summer, sufficient to allow formation of 16% of moulins mapped in the study area. In contrast, 63% of moulins experience stresses over the tensile strength of ice during a short (hours) supraglacial lake drainage event. Lake drainages appear to control moulin density, which is itself a control on subglacial drainage efficiency and summer ice velocities.

401. Seasonal Variability in Regional Ice Flow Due to Meltwater Injection Into the Shear Margins of Jakobshavn Isbræ

Authors: J. P. Cavanagh (University of Maryland); D. J. Lampkin (University of Maryland); T. Moon (University of Bristol)

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016jf004187

URL: <https://doi.org/10.1002/2016jf004187>

Publication Date: 2017-12-11

Volume/Issue: Volume 122, Issue 12

References: 81

Cited by: 11 publications

Language: en

Subjects: N/A

Keywords: crevasses, Arctic glaciology, ice velocity, glacier flow, glacial hydrology

Funding: National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: The impact of meltwater injection into the shear margins of Jakobshavn Isbræ via drainage from water-filled crevasses on ice flow is examined. We use Landsat-8 Operational Land Imager panchromatic, high-resolution imagery to monitor the spatiotemporal variability of seven water-filled crevasse ponds during the summers of 2013 to 2015. The timing of drainage from water-filled crevasses coincides with an increase of 2 to 20% in measured ice velocity beyond Jakobshavn Isbræ shear margins, which we define as extramarginal ice velocity. Some water-filled crevasse groups demonstrate multiple drainage events within a single melt season. Numerical simulations show that hydrologic shear weakening due to water-filled crevasse drainage can accelerate extramarginal flow by as much as ~35% within 10 km of the margins and enhance mass flux through the shear margins by 12%. This work demonstrates a novel mechanism through which surface melt can influence regional ice flow.

402. A JavaScript API for the Ice Sheet System Model (ISSM) 4.11: towards an online interactive model for the cryosphere community

Authors: Eric Larour; Daniel Cheng; Gilberto Perez; Justin Quinn; Mathieu Morlighem; Bao Duong; Lan Nguyen; Kit Petrie; Silva Harounian; Daria Halkides; Wayne Hayes

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-10-4393-2017

URL: <https://doi.org/10.5194/gmd-10-4393-2017>

Publication Date: 2017-12-04

Volume/Issue: Volume 10, Issue 12

References: 25

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Jet Propulsion Laboratory

License: Creative Commons

Open Access: N/A

Abstract: . Earth system models (ESMs) are becoming increasingly complex, requiring extensive knowledge and experience to deploy and use in an efficient manner. They run on high-performance architectures that are significantly different from the everyday environments that scientists use to pre- and post-process results (i.e., MATLAB, Python). This results in models that are hard to use for non-specialists and are increasingly specific in their application. It also makes them relatively inaccessible to the wider science community, not to mention to the general public. Here, we present a new software/model paradigm that attempts to bridge the gap between the science community and the complexity of ESMs by developing a new JavaScript application program interface (API) for the Ice Sheet System Model (ISSM). The aforementioned API allows cryosphere scientists to run ISSM on the client side of a web page within the JavaScript environment. When combined with a web server running ISSM (using a Python API), it enables the serving of ISSM computations in an easy and straightforward way. The deep integration and similarities between all the APIs in ISSM (MATLAB, Python, and now JavaScript) significantly shortens and simplifies the turnaround of state-of-the-art science runs and their use by the larger community. We demonstrate our approach via a new Virtual Earth System Laboratory (VESL) website (<http://vesl.jpl.nasa.gov>, VESL(2017)).

403. Should coastal planners have concern over where land ice is melting?

Authors: Eric Larour (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Erik R. Ivins (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Surendra Adhikari (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Science Advances

Publisher: American Association for the Advancement of Science (AAAS)

DOI: 10.1126/sciadv.1700537

URL: <https://doi.org/10.1126/sciadv.1700537>

Publication Date: 2017-11-16

Volume/Issue: Volume 3, Issue 11

References: 36

Cited by: 35 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: JPL Research, Technology and Development Program, NASA Cryospheric Sciences Program, NASA Modeling and Analysis Program (and 4 more)

License: N/A

Open Access: N/A

Abstract: We show how coastal planners can use sea-level gradient fingerprints to inform mitigation efforts for their local port city.

404. Dynamic Equilibrium Modeling of Snow and Inland Glaciers under the Evolving Climate in Wyoming

Authors: Ryan Johnson (Dept. of Civil and Architectural Engineering, Univ. of Wyoming, 1000 E. University Ave., Laramie, WY 82071.); Noriaki Ohara (Assistant Professor, Dept. of Civil and Architectural Engineering, Univ. of Wyoming, 1000 E. University Ave., Laramie, WY 82071 (corresponding author).)

Journal: Journal of Hydrologic Engineering

Publisher: American Society of Civil Engineers (ASCE)

DOI: 10.1061/(asce)he.1943-5584.0001585

URL: [https://doi.org/10.1061/\(asce\)he.1943-5584.0001585](https://doi.org/10.1061/(asce)he.1943-5584.0001585)

Publication Date: 2017-11-15

Volume/Issue: Volume 23, Issue 1

References: 56

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

405. Recent Progress in Greenland Ice Sheet Modelling

Authors: Heiko Goelzer; Alexander Robinson; Helene Seroussi; Roderik S.W. van de Wal

Journal: Current Climate Change Reports

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s40641-017-0073-y

URL: <https://doi.org/10.1007/s40641-017-0073-y>

Publication Date: 2017-11-13

Volume/Issue: Volume 3, Issue 4

References: 115

Cited by: 44 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Netherlands Earth System Science Centre (NESSC), financially supported by the Dutch Ministry of Education, Culture and Science, Marie Curie Horizon2020 project CONCLIMA, Jet Propulsion Laboratory

License: CC BY 4.0

Open Access: N/A

Abstract: N/A

406. Source-to-source adjoint Algorithmic Differentiation of an ice sheet model written in C

Authors: L. Hascoët (Université Côte d'Azur, INRIA, Sophia-Antipolis, France); M. Morlighem (University of California Irvine)

Journal: Optimization Methods and Software

Publisher: Informa UK Limited

DOI: 10.1080/10556788.2017.1396600

URL: <https://doi.org/10.1080/10556788.2017.1396600>

Publication Date: 2017-11-08

Volume/Issue: Volume 33, Issue 4-6

References: 13

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

407. A usability case study of algorithmic differentiation tools on the ISSM ice sheet model

Authors: Alexander Hück (Scientific Computing, Technische Universität Darmstadt, Darmstadt, Germany); Christian Bischof (Scientific Computing, Technische Universität Darmstadt, Darmstadt, Germany); Max Sagebaum (Scientific Computing, Technische Universität Kaiserslautern, Kaiserslautern, Germany); Nicolas R. Gauger (Scientific Computing, Technische Universität Kaiserslautern, Kaiserslautern, Germany); Benjamin Jurgelucks (Institut für Mathematik, Universität Paderborn, Paderborn, Germany); Eric Larour (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Gilberto Perez (University of California Irvine)

Journal: Optimization Methods and Software

Publisher: Informa UK Limited

DOI: 10.1080/10556788.2017.1396602

URL: <https://doi.org/10.1080/10556788.2017.1396602>

Publication Date: 2017-11-08

Volume/Issue: Volume 33, Issue 4-6

References: 32

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: Solver|Automatic differentiation|Benchmark (surveying)

Funding: German Science Foundation, Deutsche Forschungsgemeinschaft

License: N/A

Open Access: N/A

Abstract: N/A

408. A confined-unconfined aquifer model for subglacial hydrology and its application to the North East Greenland Ice Stream

Authors: Sebastian Beyer; Thomas Kleiner; Vadym Aizinger; Martin Rückamp; Angelika Humbert

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/tc-2017-221

URL: <https://doi.org/10.5194/tc-2017-221>

Publication Date: 2017-11-03

Volume/Issue: N/A

References: N/A

Cited by: 1 publications

Language: N/A

Subjects: N/A

Keywords: Post-glacial rebound

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: . Subglacial hydrology plays an important role in the ice sheet dynamics as it determines the sliding velocity of ice sheets and also drives freshwater into the ocean. Modeling subglacial water has been a challenge for decades, and only recently new approaches have been developed such as representing subglacial channels and thin water sheets by separate layers of variable permeability. We extend this concept by modeling a confined and unconfined aquifer system (CUAS) in a single layer. The advantage of this formulation is that it prevents unphysical values of pressure at reasonable computational cost. We also performed sensitivity tests to investigate the effect of different model parameters. The strongest influence of model parameters was detected in terms governing the opening and closure of channels. Furthermore, we applied the model to the North East Greenland Ice Stream, where an efficient system independent of seasonal input was identified about 500 km downstream from the ice divide. Using the effective pressure from the hydrology model in the Ice Sheet System Model (ISSM) shows considerable improvements of modeled velocities in the coastal region.

409. Modeling the Response of Nioghalvfjordsfjorden and Zachariae Isstrøm Glaciers, Greenland, to Ocean Forcing Over the Next Century

Authors: Y. Choi (University of California Irvine); M. Morlighem (University of California Irvine); E. Rignot (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); J. Mouginot (University of California Irvine); M. Wood (University of California Irvine)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017gl075174

URL: <https://doi.org/10.1002/2017gl075174>

Publication Date: 2017-10-24

Volume/Issue: Volume 44, Issue 21

References: 48

Cited by: 46 publications

Language: en

Subjects: N/A

Keywords: northeast Greenland, glaciology, ocean thermal forcing, calving, inverse modeling (and 1 more)

Funding: National Science Foundation, National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: Recent studies have shown that the Northeast Greenland Ice Sheet region has been undergoing significant acceleration and dynamic thinning since 2010, and these changes are closely related to regional atmospheric and oceanic warming. Here we model the response of Nioghalvfjordsfjorden (79North) and Zachariae Isstrøm (ZI) to ocean forcing to investigate their evolution over the coming decades. Our model suggests that 79North will retreat slowly over the next century, whereas ZI will lose its floating ice tongue completely and retreat rapidly for 70 years. After 70 years, ZI will stabilize 30 km upstream of its current position on a topographic ridge. Frontal melt rates need to reach 6 m/d in the summer to dislodge the glacier from this ridge. ZI will then continue a fast and unstoppable retreat, contributing more than 16.2 mm to global sea level rise by 2100.

410. Developments in Simulating and Parameterizing Interactions Between the Southern Ocean and the Antarctic Ice Sheet

Authors: Xylar S. Asay-Davis; Nicolas C. Jourdain; Yoshihiro Nakayama

Journal: Current Climate Change Reports

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s40641-017-0071-0

URL: <https://doi.org/10.1007/s40641-017-0071-0>

Publication Date: 2017-10-24

Volume/Issue: Volume 3, Issue 4

References: 126

Cited by: 56 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Office of Science, Agence Nationale de la Recherche, Jet Propulsion Laboratory

License: N/A

Open Access: N/A

Abstract: N/A

411. Variations of the Antarctic Ice Sheet in a Coupled Ice Sheet-Earth-Sea Level Model: Sensitivity to Viscoelastic Earth Properties

Authors: David Pollard (Pennsylvania State University); Natalya Gomez (McGill University); Robert M. Deconto (University of Massachusetts Amherst)

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017jf004371

URL: <https://doi.org/10.1002/2017jf004371>

Publication Date: 2017-10-18

Volume/Issue: Volume 122, Issue 11

References: 61

Cited by: 46 publications

Language: en

Subjects: N/A

Keywords: Antarctic ice sheet, ice sheet model, Earth sea level model, viscoelastic Earth profile

Funding: National Science Foundation, National Oceanic and Atmospheric Administration, Natural Sciences and Engineering Research Council of Canada (and 2 more)

License: N/A

Open Access: N/A

Abstract: A coupled ice sheet-solid Earth-sea level model is applied to long-term variations of the Antarctic ice sheet. A set of radially varying viscoelastic profiles in the global Earth model is used to explore feedbacks on ice sheet variability, including one with a very weak upper mantle zone and thin lithosphere representative of West Antarctic regions. Simulations are performed for (1) the deglacial retreat over the last ~20,000 years, (2) the future 5,000 years with greenhouse-gas scenario Representative Concentration Pathway 8.5 (RCP8.5), and (3) the warm Pliocene ~3 Ma. For the deglacial period a large ensemble of 625 simulations is analyzed, with a score computed for each run based on comparisons to geologic and modern data. For each of the five Earth profiles, the top-scoring combinations of the other model parameters in the ensemble are used to perform future and Pliocene simulations. For the last deglacial retreat, the viscoelastic Earth profiles produce relatively small differences in overall ice volume and equivalent sea level. In contrast, profiles with weak upper mantle and thin lithosphere produce strong negative feedback and less ice retreat in the future and Pliocene runs. This is due to the faster pace of ice sheet retreat in these runs, leading to greater lags in the viscous bedrock rebound behind the unloading, which allows for greater influence of the viscosity profiles. However, the differences in grounding-line retreat are located primarily in East Antarctic basins, where a weak upper mantle and thin lithosphere may not be realistic, emphasizing the need for lateral heterogeneity in the Earth model.

412. Error sources in basal yield stress inversions for Jakobshavn Isbræ, Greenland, derived from residual patterns of misfit to observations

Authors: MARIJKE HABERMANN; MARTIN TRUFFER; DAVID MAXWELL

Journal: Journal of Glaciology

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/jog.2017.61

URL: <https://doi.org/10.1017/jog.2017.61>

Publication Date: 2017-10-15

Volume/Issue: Volume 63, Issue 242

References: 41

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: The basal interface of glaciers is generally not directly observable. Geophysical inverse methods are therefore used to infer basal parameters from surface observations. Such methods can also provide information about potential inadequacies of the forward model. Ideally an inverse problem can be regularized so that the differences between modeled and observed surface velocities reflect observational errors. However, deficiencies in the forward model usually result in additional errors. Here we use the spatial pattern of velocity residuals to discuss the main error sources for basal stress inversions for Jakobshavn Isbræ, Greenland. Synthetic tests with prescribed patterns of basal yield stress with varying length scales are then used to investigate different weighting functions for the data-model misfit and for the ability of the inversion to resolve details in basal yield stress. We also test real-data inversions for their sensitivities to prior estimate, forward model parameters, data gaps, and temperature fields. We find that velocity errors are not sufficient to explain the residual patterns of real-data inversions. Conversely, ice-geometry errors and especially simulated errors in model simplifications are capable of reproducing similar error patterns and magnitudes. We suggest that residual patterns can provide useful guidance for forward model improvements.

413. Sensitivity of grounding line dynamics to the choice of the friction law

Authors: JULIEN BRONDEX; OLIVIER GAGLIARDINI; FABIEN GILLET-CHAULET; GAËL DURAND

Journal: Journal of Glaciology

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/jog.2017.51

URL: <https://doi.org/10.1017/jog.2017.51>

Publication Date: 2017-10-04

Volume/Issue: Volume 63, Issue 241

References: 66

Cited by: 83 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: Basal slip accounts for a large part of the flow of ice streams draining ice from Antarctica and Greenland into the ocean. Therefore, an appropriate representation of basal slip in ice flow models is a prerequisite for accurate sea level rise projections. Various friction laws have been proposed to describe basal slip in models. Here, we compare the influence on grounding line (GL) dynamics of four friction laws: the traditional Weertman law and three effective pressure-dependent laws, namely the Schoof, Tsai and Budd laws. It turns out that, even when they are tuned to a common initial reference state, the Weertman, Budd and Schoof laws lead to thoroughly different steady-state positions, although the Schoof and Tsai laws lead to much the same result. In particular, under certain circumstances, it is possible to obtain a steady GL located on a reverse slope area using the Weertman law. Furthermore, the predicted transient evolution of the GL as well as the projected contributions to sea level rise over a 100-year time horizon vary significantly depending on the friction law. We conclude on the importance of choosing an appropriate law for reliable sea level rise projections and emphasise the need for a coupling between ice flow models and physically based subglacial hydrological models.

414. Parallel implementation of a Lagrangian-based model on an adaptive mesh in C++: Application to sea-ice

Authors: Abdoulaye Samaké; Pierre Rampal; Sylvain Bouillon; Einar Ólason

Journal: Journal of Computational Physics

Publisher: Elsevier BV

DOI: 10.1016/j.jcp.2017.08.055

URL: <https://doi.org/10.1016/j.jcp.2017.08.055>

Publication Date: 2017-09-01

Volume/Issue: Volume 350

References: 28

Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Norwegian Research Council, NOTUR

License: N/A

Open Access: N/A

Abstract: N/A

415. Highly temporally resolved response to seasonal surface melt of the Zachariae and 79N outlet glaciers in northeast Greenland

Authors: N. M. Rathmann (Center for Ice and Climate, Niels Bohr Institute University of Copenhagen Copenhagen Denmark); C. S. Hvidberg (Center for Ice and Climate, Niels Bohr Institute University of Copenhagen Copenhagen Denmark); A. M. Solgaard (Geological Survey of Denmark and Greenland); A. Grinsted (Center for Ice and Climate, Niels Bohr Institute University of Copenhagen Copenhagen Denmark); G. H. Gudmundsson (British Antarctic Survey Cambridge United Kingdom); P. L. Langen (Danish Meteorological Institute); K. P. Nielsen (Danish Meteorological Institute); A. Kusk (Technical University of Denmark)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017gl074368

URL: <https://doi.org/10.1002/2017gl074368>

Publication Date: 2017-08-25

Volume/Issue: Volume 44, Issue 19

References: 67

Cited by: 27 publications

Language: en

Subjects: N/A

Keywords: NEGIS, Zachariae, Nioghalvfjærdsfjorden, sea level rise, floating tongue (and 1 more)

Funding: H2020 European Research Council

License: N/A

Open Access: N/A

Abstract: The seasonal response to surface melting of the Northeast Greenland Ice Stream outlets, Zachariae and 79N, is investigated using new highly temporally resolved surface velocity maps for 2016 combined with numerical modeling. The seasonal speedup at 79N of 0.15 km/yr is suggested to be driven by a decrease in effective basal pressure induced by surface melting, whereas for Zachariae its 0.11 km/yr seasonal speedup correlates equally well with the breakup of its large ice mélange. We investigate the influence 76 km long floating tongue at 79N, finding it provides little resistance and that most of it could be lost without impacting the dynamics of the area. Furthermore, we show that reducing the slipperiness along the tongue-wall interfaces produces a velocity change spatially inconsistent with the observed seasonal speedup. Finally, we find that subglacial sticky spots such as bedrock bumps play a negligible role in the large-scale response to a seasonally enhanced basal slipperiness of the region.

416. Observations and modeling of ocean-induced melt beneath Petermann Glacier Ice Shelf in northwestern Greenland

Authors: Cilan Cai (University of California Irvine); Eric Rignot (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Dimitris Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Yoshihiro Nakayama (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017gl073711

URL: <https://doi.org/10.1002/2017gl073711>

Publication Date: 2017-08-17

Volume/Issue: Volume 44, Issue 16

References: 37

Cited by: 33 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: We update observationally based estimates of subaqueous melt, Q_m , beneath Petermann Glacier Ice Shelf (PGIS), Greenland, and model its sensitivity to oceanic thermal forcing, TF, and subglacial runoff, Q_{sg} , using the Massachusetts Institute of Technology general circulation model (MITgcm), in a two-dimensional domain, with 20 m vertical and 40 m horizontal resolution at the grounding line. We adjust the drag coefficient to match the observationally based Q_m . With the inclusion of Q_{sg} , the maximum melt rate ($\dot{m}$) is 2 times larger in summer and 1/3 larger annually than in winter. $\dot{m}$ increases above linear with TF and below linear with Q_{sg} . We estimate that $\dot{m}$ increased by 24% (+8.1 m/yr) beneath PGIS from the 1990s to the 2000s from a 0.21 degreesC warming in ocean temperature and a doubling in Q_{sg} , hence contributing to its thinning. If the PGIS is removed, we estimate that the modeled melt rate near the grounding line will increase 13-16 times.

417. Sea-level response to melting of Antarctic ice shelves on multi-centennial timescales with the fast Elementary Thermomechanical Ice Sheet model (f.ETISh v1.0)

Authors: Frank Pattyn

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-11-1851-2017

URL: <https://doi.org/10.5194/tc-11-1851-2017>

Publication Date: 2017-08-08

Volume/Issue: Volume 11, Issue 4

References: 98

Cited by: 67 publications

Language: en

Subjects: N/A

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License: Creative Commons

Open Access: N/A

Abstract: . The magnitude of the Antarctic ice sheet's contribution to global sea-level rise is dominated by the potential of its marine sectors to become unstable and collapse as a response to ocean (and atmospheric) forcing. This paper presents Antarctic sea-level response to sudden atmospheric and oceanic forcings on multi-centennial timescales with the newly developed fast Elementary Thermomechanical Ice Sheet (f.ETISh) model. The f.ETISh model is a vertically integrated hybrid ice sheet-ice shelf model with vertically integrated thermomechanical coupling, making the model two-dimensional. Its marine boundary is represented by two different flux conditions, coherent with power-law basal sliding and Coulomb basal friction. The model has been compared to existing benchmarks. Modelled Antarctic ice sheet response to forcing is dominated by sub-ice shelf melt and the sensitivity is highly dependent on basal conditions at the grounding line. Coulomb friction in the grounding-line transition zone leads to significantly higher mass loss in both West and East Antarctica on centennial timescales, leading to 1.5 m sea-level rise after 500 years for a limited melt scenario of 10 m a-1 under freely floating ice shelves, up to 6 m for a 50 m a-1 scenario. The higher sensitivity is attributed to higher ice fluxes at the grounding line due to vanishing effective pressure. Removing the ice shelves altogether results in a disintegration of the West Antarctic ice sheet and (partially) marine basins in East Antarctica. After 500 years, this leads to a 5 m and a 16 m sea-level rise for the power-law basal sliding and Coulomb friction conditions at the grounding line, respectively. The latter value agrees with simulations by DeConto and Pollard (2016) over a similar period (but with different forcing and including processes of hydrofracturing and cliff failure). The chosen parametrizations make model results largely independent of spatial resolution so that f.ETISh can potentially be integrated in large-scale Earth system models.

418. Influence of a West Antarctic mantle plume on ice sheet basal conditions

Authors: Helene Seroussi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Erik R. Ivins (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Douglas A. Wiens (Department of Earth and Planetary Sciences Washington University Saint Louis Missouri USA); Johannes Bondzio (University of California Irvine, Alfred Wegener Institute)

Journal: Journal of Geophysical Research: Solid Earth

Publisher: American Geophysical Union (AGU)

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URL: <https://doi.org/10.1002/2017jb014423>

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Volume/Issue: Volume 122, Issue 9

References: 127

Cited by: 61 publications

Language: en

Subjects: N/A

Keywords: basal conditions, ice sheet, Marie Byrd Land, subglacial Lake Whillans, mantle plume

Funding: NASA's Sea Level Change Team, NASA's Cryospheric Science, NASA's Earth Surface and Interior Focus Area

License: N/A

Open Access: N/A

Abstract: The possibility that a deep mantle plume manifests Pliocene and Quaternary volcanism and potential elevated heat flux in West Antarctica has been studied for more than 30 years. Recent seismic images support the plume hypothesis as the cause of Marie Byrd Land (MBL) volcanism and geophysical structure. Mantle plumes may more than double the geothermal heat flux above nominal continental values. A dearth of in situ ice sheet basal data exists that samples the heat flux. Consequently, we examine a realistic distribution of heat flux associated with a possible late Cenozoic mantle plume in West Antarctica and explore its impact on thermal and melt conditions at the ice sheet base. We use a simple analytical mantle plume parameterization to produce geothermal heat flux at the base of the ice sheet. The three-dimensional ice flow model includes an enthalpy framework and full-Stokes stress balance. As both the putative plume location and extent are uncertain, we perform broadly scoped experiments to characterize the impact of the plume on geothermal heat flux and ice sheet basal conditions. The experiments show that mantle plumes have an important local impact on the ice sheet, with basal melting rates reaching several centimeters per year directly above the hotspot. In order to be consistent with observations of basal hydrology in MBL, the upper bound on the plume-derived geothermal heat flux is 150 mW/m². In contrast, the active lake system of the lower part of Whillans Ice Stream suggests a widespread anomalous mantle heat flux, linked to a rift source.

419. Inference of the bottom properties in shallow ice approximation models

Authors: J Monnier; P-E des Bosc

Journal: Inverse Problems

Publisher: IOP Publishing

DOI: 10.1088/1361-6420/aa7b92

URL: <https://doi.org/10.1088/1361-6420/aa7b92>

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References: 44

Cited by: 7 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

420. The mechanisms behind Jakobshavn Isbræ's acceleration and mass loss: A 3-D thermomechanical model study

Authors: Johannes H. Bondzio (University of California Irvine, Alfred Wegener Institute); Mathieu Morlighem (University of California Irvine); Hélène Seroussi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Thomas Kleiner (Alfred Wegener Institute); Martin Rückamp (Alfred Wegener Institute); Jeremie Mouginot (University of California Irvine); Twila Moon (University of Colorado Boulder); Eric Y. Larour (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Angelika Humbert (University of Bremen, Alfred Wegener Institute)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017gl073309

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Publication Date: 2017-06-02

Volume/Issue: Volume 44, Issue 12

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Cited by: 60 publications

Language: en

Subjects: N/A

Keywords: glaciology, Jakobshavn Isbr , calving, modeling

Funding: National Aeronautics and Space Administration, Cryospheric Sciences Program

License: N/A

Open Access: N/A

Abstract: The mechanisms causing widespread flow acceleration of Jakobshavn Isbræ, West Greenland, remain unclear despite an abundance of observations and modeling studies. Here we simulate the glacier's evolution from 1985 to 2016 using a three-dimensional thermomechanical ice flow model. The model captures the timing and 90% of the observed changes by forcing the calving front. Basal drag in the trough is low, and lateral drag balances the ice stream's driving stress. The calving front position is the dominant control on changes of Jakobshavn Isbræ since the ice viscosity in the shear margins instantaneously drops in response to the stress perturbation caused by calving front retreat, which allows for widespread flow acceleration. Gradual shear margin warming contributes 5 to 10% to the total acceleration. Our simulations suggest that the glacier will contribute to eustatic sea level rise at a rate comparable to or higher than at present.

421. Current state and future perspectives on coupled ice-sheet - sea-level modelling

Authors: Bas de Boer; Paolo Stocchi; Pippa L. Whitehouse; Roderik S.W. van de Wal

Journal: Quaternary Science Reviews

Publisher: Elsevier BV

DOI: 10.1016/j.quascirev.2017.05.013

URL: <https://doi.org/10.1016/j.quascirev.2017.05.013>

Publication Date: 2017-05-30

Volume/Issue: Volume 169

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Cited by: 28 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: European Union, NWO Earth and Life Sciences (ALW), NERC Independent Research Fellowship

License: N/A

Open Access: N/A

Abstract: N/A

422. Iceberg calving of Thwaites Glacier, West Antarctica: full-Stokes modeling combined with linear elastic fracture mechanics

Authors: Hongju Yu; Eric Rignot; Mathieu Morlighem; Helene Seroussi

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-11-1283-2017

URL: <https://doi.org/10.5194/tc-11-1283-2017>

Publication Date: 2017-05-30

Volume/Issue: Volume 11, Issue 3

References: 71

Cited by: 32 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Aeronautics and Space Administration

License: Creative Commons

Open Access: N/A

Abstract: . Thwaites Glacier (TG), West Antarctica, has been losing mass and retreating rapidly in the past few decades. Here, we present a study of its calving dynamics combining a two-dimensional flow-band full-Stokes (FS) model of its viscous flow with linear elastic fracture mechanics (LEFM) theory to model crevasse propagation and ice fracturing. We compare the results with those obtained with the higher-order (HO) and the shallow-shelf approximation (SSA) models coupled with LEFM. We find that FS/LEFM produces surface and bottom crevasses that are consistent with the distribution of depth and width of surface and bottom crevasses observed by NASA's Operation IceBridge radar depth sounder and laser altimeter, whereas HO/LEFM and SSA/LEFM do not generate crevasses that are consistent with observations. We attribute the difference to the nonhydrostatic condition of ice near the grounding line, which facilitates crevasse formation and is accounted for by the FS model but not by the HO or SSA models. We find that calving is enhanced when pre-existing surface crevasses are present, when the ice shelf is shortened or when the ice shelf front is undercut. The role of undercutting depends on the timescale of calving events. It is more prominent for glaciers with rapid calving rates than for glaciers with slow calving rates. Glaciers extending into a shorter ice shelf are more vulnerable to calving than glaciers developing a long ice shelf, especially as the ice front retreats close to the grounding line region, which leads to a positive feedback to calving events. We conclude that the FS/LEFM combination yields substantial improvements in capturing the stress field near the grounding line of a glacier for constraining crevasse formation and iceberg calving.

423. Causes of accelerating sea level on the East Coast of North America

Authors: James L. Davis (Columbia University); Nadya T. Vinogradova (Cambridge Climate Institute)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017gl072845

URL: <https://doi.org/10.1002/2017gl072845>

Publication Date: 2017-05-16

Volume/Issue: Volume 44, Issue 10

References: 57

Cited by: 32 publications

Language: en

Subjects: N/A

Keywords: climate impacts, sea level acceleration, sea level change, ECCO, GRACE (and 1 more)

Funding: National Aeronautics and Space Administration

License: Creative Commons

Open Access: N/A

Abstract: The tide-gauge record from the North American East Coast reveals significant accelerations in sea level starting in the late twentieth century. The estimated post-1990 accelerations range from near zero to ~ 0.3 mm yr⁻². We find that the observed sea level acceleration is well modeled using several processes: mass change in Greenland and Antarctica as measured by the Gravity Recovery and Climate Experiment satellites; ocean dynamic and steric variability provided by the GECCO2 ocean synthesis; and the inverted barometer effect. However, to achieve this fit requires estimation of an admittance for the dynamical and steric contribution, possibly due to the coarse resolution of this analysis or to simplifications associated with parameterization of bottom friction in the shallow coastal areas. The acceleration from ice loss alone is equivalent to a regional sea level rise in one century of 0.2 m in the north and 0.75 m in the south of this region.

424. Mass transport waves amplified by intense Greenland melt and detected in solid Earth deformation

Authors: S. Adhikari (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); E. R. Ivins (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); E. Larour (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017gl073478

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Publication Date: 2017-05-16

Volume/Issue: Volume 44, Issue 10

References: 47

Cited by: 15 publications

Language: en

Subjects: N/A

Keywords: geodesy, cryosphere, ice sheets, climate change, solitary waves (and 1 more)

Funding: Cryosphere Program and the Earth Surface and Interior Focus Area

License: Creative Commons

Open Access: N/A

Abstract: The annual cycle and secular trend of Greenland mass loading are well recorded in measurements of solid Earth deformation. Horizontal crustal displacements can potentially track the spatiotemporal detail of mass changes with great fidelity. Our analysis of Greenland crustal motion data reveals that a significant excitation of horizontal amplitudes occurs during the intense melt years. We discover that solitary seasonal waves of substantial mass transport (1.67 ± 0.54 Gt/month) traveled at an average speed of 7.1 km/month through Rink Glacier in 2012. We deduce that intense surface melting enhanced either basal lubrication or softening of shear margins, or both, causing the glacier to thin dynamically in summer. The newly routed upstream subglacial water was likely to be both retarded and inefficient, thus providing a causal mechanism for the prolonged ice transport to continue well into the winter months. As the climate continues to produce increasingly warmer spring and summer, amplified seasonal waves of mass transport may become ever more present with important ramifications for the future sea level rise.

425. Implementing an empirical scalar tertiary anisotropic rheology (ESTAR) into large-scale ice sheet models

Authors: Felicity S. Graham; Mathieu Morlighem; Roland C. Warner; Adam Treverrow

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/tc-2017-54

URL: <https://doi.org/10.5194/tc-2017-54>

Publication Date: 2017-05-05

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . The microstructural evolution that occurs in polycrystalline ice during deformation leads to the development of anisotropic rheological properties that are not adequately described by the most common, isotropic, ice flow relation used in large-scale ice sheet models - the Glen flow relation. We present a preliminary assessment of the implementation in the Ice Sheet System Model (ISSM) of a computationally-efficient, empirical, scalar, tertiary, anisotropic rheology (ESTAR). The effect of this anisotropic rheology on ice flow dynamics is investigated by comparing idealised simulations using ESTAR with those using the isotropic Glen flow relation, where the latter includes a flow enhancement factor. For an idealised embayed ice shelf, the Glen flow relation overestimates velocities by up to 17 % when using an enhancement factor equivalent to the maximum value prescribed by ESTAR. Importantly, no single Glen enhancement factor can accurately capture the spatial variations in flow over the ice shelf. For flow-line studies of idealised grounded flow over a bumpy topography or a sticky base - both scenarios dominated at depth by bed-parallel shear - the differences between simulated velocities using ESTAR and the Glen flow relation vary according to the value of the enhancement factor used to calibrate the Glen flow relation. These results demonstrate the importance of describing the anisotropic rheology of ice in a physically realistic manner, and have implications for simulations of ice sheet evolution used to reconstruct paleo-ice sheet extent and predict future ice sheet contributions to sea level.

426. Two-dimensional prognostic experiments for fast-flowing ice streams from the Academy of Sciences Ice Cap

Authors: Yuri V. Konovalov; Oleg V. Nagornov

Journal: Earth System Dynamics

Publisher: Copernicus GmbH

DOI: 10.5194/esd-8-283-2017

URL: <https://doi.org/10.5194/esd-8-283-2017>

Publication Date: 2017-04-20

Volume/Issue: Volume 8, Issue 2

References: 49

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . Prognostic experiments for fast-flowing ice streams on the southern side of the Academy of Sciences Ice Cap on Komsomolets Island, Severnaya Zemlya archipelago, were undertaken in this study. The experiments were based on inversions of basal friction coefficients using a two-dimensional flow-line thermocoupled model and Tikhonov's regularization method. The modeled ice temperature distributions in the cross sections were obtained using ice surface temperature histories that were inverted previously from borehole temperature profiles derived at the summit of the Academy of Sciences Ice Cap and the elevational gradient of ice surface temperature changes (about 6.5 degreesC km⁻¹). Input data included interferometric synthetic aperture radar (InSAR) ice surface velocities, ice surface elevations, and ice thicknesses obtained from airborne measurements, while the surface mass balance was adopted from previous investigations for the implementation of both the forward and inverse problems. The prognostic experiments revealed that both ice mass and ice stream extent declined for the reference time-independent surface mass balance. Specifically, the grounding line retreated: (a) along the B-B' flow line from ~ 40 to ~ 30 km (the distance from the summit), (b) along the C-C' flow line from ~ 43 to ~ 37 km, and (c) along the D-D' flow line from ~ 41 to ~ 32 km, when considering a time period of 500 years and assuming a time-independent surface mass balance. Ice flow velocities in the ice streams decreased with time and this trend resulted in the overall decline of the outgoing ice flux. Generally, the modeled glacial evolution was in agreement with observations of deglaciation of the Severnaya Zemlya archipelago.

427. Fracture propagation and stability of ice shelves governed by ice shelf heterogeneity

Authors: Chris Borstad (Department of Arctic Geophysics The University Centre in Svalbard Longyearbyen Norway); Daniel McGrath (Colorado State University, U.S. Geological Survey Alaska Science Center Anchorage Alaska USA); Allen Pope (University of Washington, University of Colorado Boulder)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017gl072648

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Publication Date: 2017-04-18

Volume/Issue: Volume 44, Issue 9

References: 47

Cited by: 44 publications

Language: en

Subjects: N/A

Keywords: ice shelf, glaciology, data assimilation, remote sensing, ice sheet modeling (and 1 more)

Funding: U.S. Geological Survey

License: N/A

Open Access: N/A

Abstract: Tabular iceberg calving and ice shelf retreat occurs after full-thickness fractures, known as rifts, propagate across an ice shelf. A quickly evolving rift signals a threat to the stability of Larsen C, the Antarctic Peninsula's largest ice shelf. Here we reveal the influence of ice shelf heterogeneity on the growth of this rift, with implications that challenge existing notions of ice shelf stability. Most of the rift extension has occurred in bursts after overcoming the resistance of suture zones that bind together neighboring glacier inflows. We model the stresses in the ice shelf to determine potential rift trajectories. Calving perturbations to ice flow will likely reach the grounding line. The stability of Larsen C may hinge on a single suture zone that stabilizes numerous upstream rifts. Elevated fracture toughness of suture zones may be the most important property that allows ice shelves to modulate Antarctica's contribution to sea level rise.

428. Instability of the Antarctic Ross Sea Embayment as climate warms

Authors: Terence Hughes (University of Maine); Zihong Zhao (University of Maine); Raymond Hintz (University of Maine); James Fastook (University of Maine)

Journal: Reviews of Geophysics

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016rg000545

URL: <https://doi.org/10.1002/2016rg000545>

Publication Date: 2017-04-14

Volume/Issue: Volume 55, Issue 2

References: 176

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: ice shelves, Byrd Glacier, outlet glaciers, Ross Ice Shelf, West Antarctic Ice Sheet (and 1 more)

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Collapse of the Antarctic Ice Sheet since the Last Glacial Maximum 18,000 years ago is most pronounced in the Ross Sea Embayment, which is partly ice-free during Antarctic summers, thereby breaching the O-ring of ice shelves and sea ice surrounding Antarctica that stabilizes the ice sheet. The O-ring may have vanished during Early Holocene (5000 to 3000 B.C.), Roman (1 to 400 A.D.), and Medieval (900 to 1300 A.D.) warm periods and reappeared during the Little Ice Age (1300 to 1900 A.D.). We postulate further collapse in the embayment during the post-1900 warming may be forestalled because East Antarctic outlet glaciers "nail" the Ross Ice Shelf to the Transantarctic Mountains so it can resist the push from West Antarctic ice streams. Our hypothesis is examined for Byrd Glacier and a static ice shelf using three modeling experiments having plastic, viscous, and viscoplastic solutions as more data and improved modeling became available. Observed crevasse patterns were not reproduced. A new research study is needed to model a dynamic Ross Ice Shelf with all its feeder ice streams, outlet glaciers, and ice calving dynamics in three dimensions over time to fully test our hypothesis. The required model must allow accelerated calving if further warming melts sea ice and disorps the ice shelf. Calving must then successively pull the outlet glacier "nails" so collapse of the marine West Antarctic Ice Sheet proceeds to completion.

429. Continued retreat of Thwaites Glacier, West Antarctica, controlled by bed topography and ocean circulation

Authors: H. Seroussi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Y. Nakayama (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); E. Larour (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); D. Menemenlis (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); M. Morlighem (University of California Irvine); E. Rignot (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); A. Khazendar (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2017gl072910

URL: <https://doi.org/10.1002/2017gl072910>

Publication Date: 2017-04-05

Volume/Issue: Volume 44, Issue 12

References: 60

Cited by: 166 publications

Language: en

Subjects: N/A

Keywords: ice sheet dynamics, Antarctica, coupled model, ocean, Amundsen Sea (and 1 more)

Funding: National Science Foundation Program

License: N/A

Open Access: N/A

Abstract: The Amundsen Sea sector is experiencing the largest mass loss, glacier acceleration, and grounding line retreat in Antarctica. Enhanced intrusion of Circumpolar Deep Water onto the continental shelf has been proposed as the primary forcing mechanism for the retreat. Here we investigate the dynamics and evolution of Thwaites Glacier with a novel, fully coupled, ice-ocean numerical model. We obtain a significantly improved agreement with the observed pattern of glacial retreat using the coupled model. Coupled simulations over the coming decades indicate a continued mass loss at a sustained rate. Uncoupled simulations using a depth-dependent parameterization of sub-ice-shelf melt significantly overestimate the rate of grounding line retreat compared to the coupled model, as the parameterization does not capture the complexity of the ocean circulation associated with the formation of confined cavities during the retreat. Bed topography controls the pattern of grounding line retreat, while oceanic thermal forcing impacts the rate of grounding line retreat. The importance of oceanic forcing increases with time as Thwaites grounding line retreats farther inland.

430. Two-dimensional prognostic experiments for fast-flowing ice streams from the Academy of Sciences Ice Cap

Authors: Y V Konovalov; O V Nagornov

Journal: Journal of Physics: Conference Series

Publisher: IOP Publishing

DOI: 10.1088/1742-6596/788/1/012051

URL: <https://doi.org/10.1088/1742-6596/788/1/012051>

Publication Date: 2017-02-02

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References: 33

Cited by: 1 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

431. Simulating the evolution of Hardangerjøkulen ice cap in southern Norway since the mid-Holocene and its sensitivity to climate change

Authors: Henning Åkesson; Kerim H. Nisancioglu; Rianne H. Giesen; Mathieu Morlighem

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-11-281-2017

URL: <https://doi.org/10.5194/tc-11-281-2017>

Publication Date: 2017-01-27

Volume/Issue: Volume 11, Issue 1

References: 117

Cited by: 24 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: European Research Council, Norges Forskningsråd

License: Creative Commons

Open Access: N/A

Abstract: . Understanding of long-term dynamics of glaciers and ice caps is vital to assess their recent and future changes, yet few long-term reconstructions using ice flow models exist. Here we present simulations of the maritime Hardangerjøkulen ice cap in Norway from the mid-Holocene through the Little Ice Age (LIA) to the present day, using a numerical ice flow model combined with glacier and climate reconstructions. In our simulation, under a linear climate forcing, we find that Hardangerjøkulen grows from ice-free conditions in the mid-Holocene to its maximum extent during the LIA in a nonlinear, spatially asynchronous fashion. During its fastest stage of growth (2300-1300 BP), the ice cap triples its volume in less than 1000 years. The modeled ice cap extent and outlet glacier length changes from the LIA until today agree well with available observations. Volume and area for Hardangerjøkulen and several of its outlet glaciers vary out-of-phase for several centuries during the Holocene. This volume-area disequilibrium varies in time and from one outlet glacier to the next, illustrating that linear relations between ice extent, volume and glacier proxy records, as generally used in paleoclimatic reconstructions, have only limited validity. We also show that the present-day ice cap is highly sensitive to surface mass balance changes and that the effect of the ice cap hypsometry on the mass balance-altitude feedback is essential to this sensitivity. A mass balance shift by +0.5 m w.e. relative to the mass balance from the last decades almost doubles ice volume, while a decrease of 0.2 m w.e. or more induces a strong mass balance-altitude feedback and makes Hardangerjøkulen disappear entirely. Furthermore, once disappeared, an additional +0.1 m w.e. relative to the present mass balance is needed to regrow the ice cap to its present-day extent. We expect that other ice caps with comparable geometry in, for example, Norway, Iceland, Patagonia and peripheral Greenland may behave similarly, making them particularly vulnerable to climate change.

432. A prototype implementation of an embedded simulation system for the study of large scale ice sheets

Authors: Phillip Dickens (University of Maine); Christopher Dufour (University of Maine); James Fastook (University of Maine)

Journal: 2016 Winter Simulation Conference (WSC)

Publisher: IEEE

DOI: 10.1109/wsc.2016.7822225

URL: <https://doi.org/10.1109/wsc.2016.7822225>

Publication Date: 2017-01-20

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References: 17

Cited by: 1 publications

Language: N/A

Subjects: N/A

Keywords: Ice-sheet model

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

433. Semi-brittle rheology and ice dynamics in DynEarthSol3D

Authors: Liz C. Logan; Luc L. Lavier; Eunseo Choi; Eh Tan; Ginny A. Catania

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-11-117-2017

URL: <https://doi.org/10.5194/tc-11-117-2017>

Publication Date: 2017-01-17

Volume/Issue: Volume 11, Issue 1

References: 50

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, King Abdullah University of Science and Technology

License: Creative Commons

Open Access: N/A

Abstract: . We present a semi-brittle rheology and explore its potential for simulating glacier and ice sheet deformation using a numerical model, DynEarthSol3D (DES), in simple, idealized experiments. DES is a finite-element solver for the dynamic and quasi-static simulation of continuous media. The experiments within demonstrate the potential for DES to simulate ice failure and deformation in dynamic regions of glaciers, especially at quickly changing boundaries like glacier termini in contact with the ocean. We explore the effect that different rheological assumptions have on the pattern of flow and failure. We find that the use of a semi-brittle constitutive law is a sufficient material condition to form the characteristic pattern of basal crevasse-aided pinch-and-swell geometry, which is observed globally in floating portions of ice and can often aid in eroding the ice sheet margins in direct contact with oceans.

434. Optimal numerical solvers for transient simulations of ice flow using the Ice Sheet System Model (ISSM versions 4.2.5 and 4.11)

Authors: Feras Habbal; Eric Larour; Mathieu Morlighem; Helene Seroussi; Christopher P. Borstad; Eric Rignot

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-10-155-2017

URL: <https://doi.org/10.5194/gmd-10-155-2017>

Publication Date: 2017-01-10

Volume/Issue: Volume 10, Issue 1

References: 30

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation

License: Creative Commons

Open Access: N/A

Abstract: . Identifying fast and robust numerical solvers is a critical issue that needs to be addressed in order to improve projections of polar ice sheets evolving in a changing climate. This work evaluates the impact of using advanced numerical solvers for transient ice-flow simulations conducted with the JPL-UCI Ice Sheet System Model (ISSM). We identify optimal numerical solvers by testing a broad suite of readily available solvers, ranging from direct sparse solvers to preconditioned iterative methods, on the commonly used Ice Sheet Model Intercomparison Project for Higher-Order ice sheet Models benchmark tests. Three types of analyses are considered: mass transport, horizontal stress balance, and incompressibility. The results of the fastest solvers for each analysis type are ranked based on their scalability across mesh size and basal boundary conditions. We find that the fastest iterative solvers are ~ 1.5-100 times faster than the default direct solver used in ISSM, with speed-ups improving rapidly with increased mesh resolution. We provide a set of recommendations for users in search of efficient solvers to use for transient ice-flow simulations, enabling higher-resolution meshes and faster turnaround time. The end result will be improved transient simulations for short-term, highly resolved forward projections (10-100 year time scale) and also improved long-term paleo-reconstructions using higher-order representations of stresses in the ice. This analysis will also enable a new generation of comprehensive uncertainty quantification assessments of forward sea-level rise projections, which rely heavily on ensemble or sampling approaches that are inherently expensive.

435. Ice Sheet Model Intercomparison Project (ISMIP6) contribution to CMIP6

Authors: Sophie M. J. Nowicki; Anthony Payne; Eric Larour; Helene Seroussi; Heiko Goelzer; William Lipscomb; Jonathan Gregory; Ayako Abe-Ouchi; Andrew Shepherd

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-9-4521-2016

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References: 133

Cited by: 157 publications

Language: en

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License: Creative Commons

Open Access: N/A

Abstract: . Reducing the uncertainty in the past, present, and future contribution of ice sheets to sea-level change requires a coordinated effort between the climate and glaciology communities. The Ice Sheet Model Intercomparison Project for CMIP6 (ISMIP6) is the primary activity within the Coupled Model Intercomparison Project - phase 6 (CMIP6) focusing on the Greenland and Antarctic ice sheets. In this paper, we describe the framework for ISMIP6 and its relationship with other activities within CMIP6. The ISMIP6 experimental design relies on CMIP6 climate models and includes, for the first time within CMIP, coupled ice-sheet-climate models as well as standalone ice-sheet models. To facilitate analysis of the multi-model ensemble and to generate a set of standard climate inputs for standalone ice-sheet models, ISMIP6 defines a protocol for all variables related to ice sheets. ISMIP6 will provide a basis for investigating the feedbacks, impacts, and sea-level changes associated with dynamic ice sheets and for quantifying the uncertainty in ice-sheet-sourced global sea-level change.

436. Contributions of Greenland and Antarctica to Global and Regional Sea Level Change

Authors: Unknown Author; Eric Leuliette; Steven Nerem

Journal: Oceanography

Publisher: The Oceanography Society

DOI: 10.5670/oceanog.2016.107

URL: <https://doi.org/10.5670/oceanog.2016.107>

Publication Date: 2016-12-12

Volume/Issue: Volume 29, Issue 4

References: 30

Cited by: 11 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

437. Extensive winter subglacial water storage beneath the Greenland Ice Sheet

Authors: Winnie Chu (Columbia University); Dustin M. Schroeder (Department of Geophysics, School of Earth, Energy and Environmental Sciences Stanford University Stanford California USA); Helene Seroussi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Timothy T. Creyts (Columbia University); Steven J. Palmer (University of Exeter); Robin E. Bell (Columbia University)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016gl071538

URL: <https://doi.org/10.1002/2016gl071538>

Publication Date: 2016-12-09

Volume/Issue: Volume 43, Issue 24

References: 80

Cited by: 56 publications

Language: en

Subjects: N/A

Keywords: radar sounding, ice sheet modeling, Greenland Ice Sheet, glacier dynamics, subglacial hydrology

Funding: National Aeronautics and Space Administration, National Science Foundation

License: N/A

Open Access: N/A

Abstract: Surface meltwater that reaches the base of the Greenland Ice Sheet exerts a fundamental impact on ice flow, but observations of catchment-wide movement and distribution of subglacial water remain limited. Using radar-sounding data from two seasons, we identify the seasonal distribution of subglacial water in western Greenland. Our analysis provides evidence of widespread subglacial water storage beneath Greenland in the wintertime. The winter storage is located primarily on bedrock ridges with higher bed elevations in excess of 200 m. During the melt season water moves to the subglacial troughs. This inverse relationship with topography indicates that the material properties of the glacier bed strongly influence subglacial drainage development. Both the spatial variations in bed properties and the initial state of the subglacial hydrology system at the start of the melt season lead to differing glacier dynamical responses to surface melting across the Greenland Ice Sheet.

438. Accurate and stable time stepping in ice sheet modeling

Authors: Gong Cheng; Per Lötstedt; Lina von Sydow

Journal: Journal of Computational Physics

Publisher: Elsevier BV

DOI: 10.1016/j.jcp.2016.10.060

URL: <https://doi.org/10.1016/j.jcp.2016.10.060>

Publication Date: 2016-11-02

Volume/Issue: Volume 329

References: 45

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Swedish Research Council Formas, PDC Center for High Performance Computing at KTH

License: N/A

Open Access: N/A

Abstract: N/A

439. An approach to computing discrete adjoints for MPI-parallelized models applied to Ice Sheet System Model 4.11

Authors: Eric Larour; Jean Utke; Anton Bovin; Mathieu Morlighem; Gilberto Perez

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-9-3907-2016

URL: <https://doi.org/10.5194/gmd-9-3907-2016>

Publication Date: 2016-11-01

Volume/Issue: Volume 9, Issue 11

References: 43

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Jet Propulsion Laboratory

License: Creative Commons

Open Access: N/A

Abstract: . Within the framework of sea-level rise projections, there is a strong need for hindcast validation of the evolution of polar ice sheets in a way that tightly matches observational records (from radar, gravity, and altimetry observations mainly). However, the computational requirements for making hindcast reconstructions possible are severe and rely mainly on the evaluation of the adjoint state of transient ice-flow models. Here, we look at the computation of adjoints in the context of the NASA/JPL/UCI Ice Sheet System Model (ISSM), written in C++ and designed for parallel execution with MPI. We present the adaptations required in the way the software is designed and written, but also generic adaptations in the tools facilitating the adjoint computations. We concentrate on the use of operator overloading coupled with the AdjoinableMPI library to achieve the adjoint computation of the ISSM. We present a comprehensive approach to (1) carry out type changing through the ISSM, hence facilitating operator overloading, (2) bind to external solvers such as MUMPS and GSL-LU, and (3) handle MPI-based parallelism to scale the capability. We demonstrate the success of the approach by computing sensitivities of hindcast metrics such as the misfit to observed records of surface altimetry on the northeastern Greenland Ice Stream, or the misfit to observed records of surface velocities on Upernavik Glacier, central West Greenland. We also provide metrics for the scalability of the approach, and the expected performance. This approach has the potential to enable a new generation of hindcast-validated projections that make full use of the wealth of datasets currently being collected, or already collected, in Greenland and Antarctica.

440. A meshfree approach to non-Newtonian free surface ice flow: Application to the Haut Glacier d'Arolla

Authors: Josefin Ahlkrona; Victor Shcherbakov

Journal: Journal of Computational Physics

Publisher: Elsevier BV

DOI: 10.1016/j.jcp.2016.10.045

URL: <https://doi.org/10.1016/j.jcp.2016.10.045>

Publication Date: 2016-10-24

Volume/Issue: Volume 330

References: 44

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Swedish strategic research program eSENCE

License: N/A

Open Access: N/A

Abstract: N/A

441. A Matrix Dependent/Algebraic Multigrid Approach for Extruded Meshes with Applications to Ice Sheet Modeling

Authors: R. Tuminaro; M. Perego; I. Tezaur; A. Salinger; S. Price

Journal: SIAM Journal on Scientific Computing

Publisher: Society for Industrial & Applied Mathematics (SIAM)

DOI: 10.1137/15m1040839

URL: <https://doi.org/10.1137/15m1040839>

Publication Date: 2016-10-06

Volume/Issue: Volume 38, Issue 5

References: 40

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: Algebraic method, Ice sheets, Iterative solvers

Funding: Biological and Environmental Research

License: N/A

Open Access: N/A

Abstract: N/A

442. A modeling study of the effect of runoff variability on the effective pressure beneath Russell Glacier, West Greenland

Authors: Basile de Fleurian (University of California Irvine); Mathieu Morlighem (University of California Irvine); Helene Seroussi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Eric Rignot (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Michiel R. van den Broeke (Utrecht University); Peter Kuipers Munneke (Utrecht University); Jeremie Mouginot (University of California Irvine); Paul C. J. P. Smeets (Utrecht University); Andrew J. Tedstone (University of Edinburgh)

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016jf003842

URL: <https://doi.org/10.1002/2016jf003842>

Publication Date: 2016-09-30

Volume/Issue: Volume 121, Issue 10

References: 47

Cited by: 41 publications

Language: en

Subjects: N/A

Keywords: ice sheet modeling, glacier dynamics, subglacial hydrology, modeling

Funding: National Aeronautics and Space Administration

License: Creative Commons

Open Access: N/A

Abstract: N/A

443. Adaptively constraining radar attenuation and temperature across the Thwaites Glacier catchment using bed echoes

Authors: DUSTIN M. SCHROEDER; HELENE SEROUSSI; WINNIE CHU; DUNCAN A. YOUNG

Journal: Journal of Glaciology

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/jog.2016.100

URL: <https://doi.org/10.1017/jog.2016.100>

Publication Date: 2016-09-21

Volume/Issue: Volume 62, Issue 236

References: 55

Cited by: 46 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: Englacial temperature is a major control on ice rheology and flow. However, it is difficult to measure at the glacier to ice-sheet scale. As a result, ice-sheet models must make assumptions about englacial temperature and rheology, which affect sea level projections. This is problematic if fundamental processes are not captured by models due to a lack of observationally constrained ice temperature values. Although radar sounding data have been exploited to constrain the temperature structure of the Greenland ice sheet using englacial layers, this approach is limited to areas and depths where these layers exist intact. In order to extend empirical radar-based temperature estimation beyond this limitation, we present a new technique for estimating englacial attenuation rates for the entire ice column using adaptive fitting of unfocused radar bed echoes based on the correlation of ice thickness and corrected bed echo power. We apply this technique to an airborne survey of Thwaites Glacier in West Antarctica and compare the results with temperatures and attenuation rates from a numerical ice-sheet model. We find that the estimated attenuation rates reproduce modelled patterns and values across the catchment with the greatest differences near steeply sloping bed topography.

444. Are long tide gauge records in the wrong place to measure global mean sea level rise?

Authors: P. R. Thompson (Department of Oceanography University of Hawai'i at Mānoa Honolulu Hawaii USA); B. D. Hamlington (Old Dominion University); F. W. Landerer (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); S. Adhikari (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016gl070552

URL: <https://doi.org/10.1002/2016gl070552>

Publication Date: 2016-09-19

Volume/Issue: Volume 43, Issue 19

References: 45

Cited by: 40 publications

Language: en

Subjects: N/A

Keywords: global mean sea level, tide gauges

Funding: Climate Program Office

License: N/A

Open Access: N/A

Abstract: Ocean dynamics, land motion, and changes in Earth's gravitational and rotational fields cause local sea level change to deviate from the rate of global mean sea level rise. Here we use observations and simulations of spatial structure in sea level change to estimate the likelihood that these processes cause sea level trends in the longest and highest-quality tide gauge records to be systematically biased relative to the true global mean rate. The analyzed records have an average twentieth century rate of approximately 1.6 mm/yr, but based on the locations of these gauges, we show that the simple average underestimates the twentieth century global mean rate by 0.1 +/- 0.2 mm/yr. Given the distribution of potential sampling biases, we find that <1% probability that observed trends from the longest and highest-quality tide gauge records are consistent with global mean rates less than 1.4 mm/yr.

445. Application of GRACE to the assessment of model-based estimates of monthly Greenland Ice Sheet mass balance (2003-2012)

Authors: Nicole-Jeanne Schlegel; David N. Wiese; Eric Y. Larour; Michael M. Watkins; Jason E. Box; Xavier Fettweis; Michiel R. van den Broeke

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-10-1965-2016

URL: <https://doi.org/10.5194/tc-10-1965-2016>

Publication Date: 2016-09-07

Volume/Issue: Volume 10, Issue 5

References: 119

Cited by: 20 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Jet Propulsion Laboratory

License: Creative Commons

Open Access: N/A

Abstract: . Quantifying the Greenland Ice Sheet's future contribution to sea level rise is a challenging task that requires accurate estimates of ice sheet sensitivity to climate change. Forward ice sheet models are promising tools for estimating future ice sheet behavior, yet confidence is low because evaluation of historical simulations is challenging due to the scarcity of continental-wide data for model evaluation. Recent advancements in processing of Gravity Recovery and Climate Experiment (GRACE) data using Bayesian-constrained mass concentration ("mascon") functions have led to improvements in spatial resolution and noise reduction of monthly global gravity fields. Specifically, the Jet Propulsion Laboratory's JPL RL05M GRACE mascon solution (GRACE_JPL) offers an opportunity for the assessment of model-based estimates of ice sheet mass balance (MB) at ~ 300 km spatial scales. Here, we quantify the differences between Greenland monthly observed MB (GRACE_JPL) and that estimated by state-of-the-art, high-resolution models, with respect to GRACE_JPL and model uncertainties. To simulate the years 2003-2012, we force the Ice Sheet System Model (ISSM) with anomalies from three different surface mass balance (SMB) products derived from regional climate models. Resulting MB is compared against GRACE_JPL within individual mascons. Overall, we find agreement in the northeast and southwest where MB is assumed to be primarily controlled by SMB. In the interior, we find a discrepancy in trend, which we presume to be related to millennial-scale dynamic thickening not considered by our model. In the northwest, seasonal amplitudes agree, but modeled mass trends are muted relative to GRACE_JPL. Here, discrepancies are likely controlled by temporal variability in ice discharge and other related processes not represented by our model simulations, i.e., hydrological processes and ice-ocean interaction. In the southeast, GRACE_JPL exhibits larger seasonal amplitude than predicted by the models while simultaneously having more pronounced trends; thus, discrepancies are likely controlled by a combination of missing processes and errors in both the SMB products and ISSM. At the margins, we find evidence of consistent intra-annual variations in regional MB that deviate distinctively from the SMB annual cycle. Ultimately, these monthly-scale variations, likely associated with hydrology or ice-ocean interaction, contribute to steeper

negative mass trends observed by GRACE_JPL. Thus, models should consider such processes at relatively high (monthly-to-seasonal) temporal resolutions to achieve accurate estimates of Greenland MB.

446. On ISSM and leveraging the Cloud towards faster quantification of the uncertainty in ice-sheet mass balance projections

Authors: E. Larour; N. Schlegel

Journal: Computers & Geosciences

Publisher: Elsevier BV

DOI: 10.1016/j.cageo.2016.08.007

URL: <https://doi.org/10.1016/j.cageo.2016.08.007>

Publication Date: 2016-09-03

Volume/Issue: Volume 96

References: 42

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

447. Shallow ice approximation, second order shallow ice approximation, and full Stokes models: A discussion of their roles in palaeo-ice sheet modelling and development

Authors: N. Kirchner; J. Ahlkrone; E.J. Gowan; P. Lötstedt; J.M. Lea; R. Noormets; L. von Sydow; J.A. Dowdeswell; T. Benham

Journal: Quaternary Science Reviews

Publisher: Elsevier BV

DOI: 10.1016/j.quascirev.2016.01.032

URL: <https://doi.org/10.1016/j.quascirev.2016.01.032>

Publication Date: 2016-08-20

Volume/Issue: Volume 147

References: 73

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Swedish Research Council FORMAS, Svalbard Science Forum's Strategic Grants

License: N/A

Open Access: N/A

Abstract: N/A

448. Similitude of ice dynamics against scaling of geometry and physical parameters

Authors: Johannes Feldmann; Anders Levermann

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-10-1753-2016

URL: <https://doi.org/10.5194/tc-10-1753-2016>

Publication Date: 2016-08-17

Volume/Issue: Volume 10, Issue 4

References: 55

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . The concept of similitude is commonly employed in the fields of fluid dynamics and engineering but rarely used in cryospheric research. Here we apply this method to the problem of ice flow to examine the dynamic similitude of isothermal ice sheets in shallow-shelf approximation against the scaling of their geometry and physical parameters. Carrying out a dimensional analysis of the stress balance we obtain dimensionless numbers that characterize the flow. Requiring that these numbers remain the same under scaling we obtain conditions that relate the geometric scaling factors, the parameters for the ice softness, surface mass balance and basal friction as well as the ice-sheet intrinsic response time to each other. We demonstrate that these scaling laws are the same for both the (two-dimensional) flow-line case and the three-dimensional case. The theoretically predicted ice-sheet scaling behavior agrees with results from numerical simulations that we conduct in flow-line and three-dimensional conceptual setups. We further investigate analytically the implications of geometric scaling of ice sheets for their response time. With this study we provide a framework which, under several assumptions, allows for a fundamental comparison of the ice-dynamic behavior across different scales. It proves to be useful in the design of conceptual numerical model setups and could also be helpful for designing laboratory glacier experiments. The concept might also be applied to real-world systems, e.g., to examine the response times of glaciers, ice streams or ice sheets to climatic perturbations.

449. An ice-sheet-wide framework for englacial attenuation from ice-penetrating radar data

Authors: T. M. Jordan; J. L. Bamber; C. N. Williams; J. D. Paden; M. J. Siegert; P. Huybrechts; O. Gagliardini; F. Gillet-Chaulet

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-10-1547-2016

URL: <https://doi.org/10.5194/tc-10-1547-2016>

Publication Date: 2016-07-20

Volume/Issue: Volume 10, Issue 4

References: 59

Cited by: 21 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Natural Environment Research Council

License: Creative Commons

Open Access: N/A

Abstract: . Radar inference of the bulk properties of glacier beds, most notably identifying basal melting, is, in general, derived from the basal reflection coefficient. On the scale of an ice sheet, unambiguous determination of basal reflection is primarily limited by uncertainty in the englacial attenuation of the radio wave, which is an Arrhenius function of temperature. Existing bed-returned power algorithms for deriving attenuation assume that the attenuation rate is regionally constant, which is not feasible at an ice-sheet-wide scale. Here we introduce a new semi-empirical framework for deriving englacial attenuation, and, to demonstrate its efficacy, we apply it to the Greenland Ice Sheet. A central feature is the use of a prior Arrhenius temperature model to estimate the spatial variation in englacial attenuation as a first guess input for the radar algorithm. We demonstrate regions of solution convergence for two input temperature fields and for independently analysed field campaigns. The coverage achieved is a trade-off with uncertainty and we propose that the algorithm can be "tuned" for discrimination of basal melt (attenuation loss uncertainty ~ 5 dB). This is supported by our physically realistic (~ 20 dB) range for the basal reflection coefficient. Finally, we show that the attenuation solution can be used to predict the temperature bias of thermomechanical ice sheet models and is in agreement with known model temperature biases at the Dye 3 ice core.

450. A Parallel Computational Model for Three-Dimensional, Thermo-Mechanical Stokes Flow Simulations of Glaciers and Ice Sheets

Authors: Wei Leng; Lili Ju; Max Gunzburger; Stephen Price

Journal: Communications in Computational Physics

Publisher: Global Science Press

DOI: 10.4208/cicp.310813.010414a

URL: <https://doi.org/10.4208/cicp.310813.010414a>

Publication Date: 2016-07-15

Volume/Issue: Volume 16, Issue 4

References: 36

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: This paper focuses on the development of an efficient, three-dimensional, thermo-mechanical, nonlinear-Stokes flow computational model for ice sheet simulation. The model is based on the parallel finite element model developed in [14] which features high-order accurate finite element discretizations on variable resolution grids. Here, we add an improved iterative solution method for treating the nonlinearity of the Stokes problem, a new high-order accurate finite element solver for the temperature equation, and a new conservative finite volume solver for handling mass conservation. The result is an accurate and efficient numerical model for thermo-mechanical glacier and ice-sheet simulations. We demonstrate the improved efficiency of the Stokes solver using the ISMIP-HOM Benchmark experiments and a realistic test case for the Greenland ice-sheet. We also apply our model to the EISMINT-II benchmark experiments and demonstrate stable thermo-mechanical ice sheet evolution on both structured and unstructured meshes. Notably, we find no evidence for the "cold spoke" instabilities observed for these same experiments when using finite difference, shallow-ice approximation models on structured grids.

451. A synthesis of the basal thermal state of the Greenland Ice Sheet

Authors: Joseph A. MacGregor (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, University of Texas at Austin); Mark A. Fahnstock (University of Alaska Fairbanks); Ginny A. Catania (University of Texas at Austin); Andy Aschwanden (University of Alaska Fairbanks); Gary D. Clow (University of Colorado Boulder, U.S. Geological Survey Denver Colorado USA); William T. Colgan (Department of Earth and Space Science and Engineering York University Toronto Ontario Canada); S. Prasad Gogineni (University of Kansas); Mathieu Morlighem (University of California Irvine); Sophie M. J. Nowicki (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); John D. Paden (University of Kansas); Stephen F. Price (Los Alamos National Laboratory); Hélène Seroussi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2015jf003803

URL: <https://doi.org/10.1002/2015jf003803>

Publication Date: 2016-06-23

Volume/Issue: Volume 121, Issue 7

References: 100

Cited by: 125 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NSF, NASA, U.S. Geological Survey Climate and Land Use Change Program (and 2 more)

License: N/A

Open Access: N/A

Abstract: The basal thermal state of an ice sheet (frozen or thawed) is an important control upon its evolution, dynamics, and response to external forcings. However, this state can only be observed directly at sparse boreholes or inferred conclusively from the presence of subglacial lakes. Here we synthesize spatially extensive inferences of the basal thermal state of the Greenland Ice Sheet to better constrain this state. Existing inferences include outputs from the eight thermomechanical ice-flow models included in the Sea Level Response to Ice Sheet Evolution (SeaRISE) effort. New remote-sensing inferences of the basal thermal state are derived from Holocene radiostratigraphy, modern surface velocity, and Moderate Resolution Imaging Spectroradiometer (MODIS) imagery. Both thermomechanical modeling and remote inferences generally agree that the Northeast Greenland Ice Stream and large portions of the southwestern ice-drainage systems are thawed at the bed, whereas the bed beneath the central ice divides, particularly their west facing slopes, is frozen. Elsewhere, there is poorer agreement regarding the basal thermal state. Both models and remote inferences rarely represent the borehole-observed basal thermal state accurately near NorthGRIP and DYE-3. This synthesis identifies a large portion of the Greenland Ice Sheet (about one third by area), where additional observations would most improve knowledge of its overall basal thermal state.

452. Greenland Ice Sheet seasonal and spatial mass variability from model simulations and GRACE (2003-2012)

Authors: Patrick M. Alexander; Marco Tedesco; Nicole-Jeanne Schlegel; Scott B. Luthcke; Xavier Fettweis; Eric Larour

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-10-1259-2016

URL: <https://doi.org/10.5194/tc-10-1259-2016>

Publication Date: 2016-06-20

Volume/Issue: Volume 10, Issue 3

References: 62

Cited by: 16 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Office of Polar Programs

License: Creative Commons

Open Access: N/A

Abstract: . Improving the ability of regional climate models (RCMs) and ice sheet models (ISMs) to simulate spatiotemporal variations in the mass of the Greenland Ice Sheet (GrIS) is crucial for prediction of future sea level rise. While several studies have examined recent trends in GrIS mass loss, studies focusing on mass variations at sub-annual and sub-basin-wide scales are still lacking. At these scales, processes responsible for mass change are less well understood and modeled, and could potentially play an important role in future GrIS mass change. Here, we examine spatiotemporal variations in mass over the GrIS derived from the Gravity Recovery and Climate Experiment (GRACE) satellites for the January 2003-December 2012 period using a "mascon" approach, with a nominal spatial resolution of 100 km, and a temporal resolution of 10 days. We compare GRACE-estimated mass variations against those simulated by the Modèle Atmosphérique Régionale (MAR) RCM and the Ice Sheet System Model (ISSM). In order to properly compare spatial and temporal variations in GrIS mass from GRACE with model outputs, we find it necessary to spatially and temporally filter model results to reproduce leakage of mass inherent in the GRACE solution. Both modeled and satellite-derived results point to a decline (of -178.9 ± 4.4 and -239.4 ± 7.7 Gt yr⁻¹ respectively) in GrIS mass over the period examined, but the models appear to underestimate the rate of mass loss, especially in areas below 2000 m in elevation, where the majority of recent GrIS mass loss is occurring. On an ice-sheet-wide scale, the timing of the modeled seasonal cycle of cumulative mass (driven by summer mass loss) agrees with the GRACE-derived seasonal cycle, within limits of uncertainty from the GRACE solution. However, on sub-ice-sheet-wide scales, some areas exhibit significant differences in the timing of peaks in the annual cycle of mass change. At these scales, model biases, or processes not accounted for by models related to ice dynamics or hydrology, may lead to the observed differences. This highlights the need for further evaluation of modeled processes at regional and seasonal scales, and further study of ice sheet processes not accounted for, such as the role of subglacial hydrology in variations in glacial flow.

453. Source Transformation of C++ Codes for Compatibility with Operator Overloading

Authors: Alexander Hück; Jean Utke; Christian Bischof

Journal: Procedia Computer Science

Publisher: Elsevier BV

DOI: 10.1016/j.procs.2016.05.470

URL: <https://doi.org/10.1016/j.procs.2016.05.470>

Publication Date: 2016-06-02

Volume/Issue: Volume 80

References: 31

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

454. Review for "Simulating asymmetric growth and retreat of Hardangerjokulen ice cap in southern Norway since the mid-Holocene", by Akesson and al

Authors: No authors listed

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/tc-2016-62-rc2

URL: <https://doi.org/10.5194/tc-2016-62-rc2>

Publication Date: 2016-05-30

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

455. Report on a successful ADOL-C generated adjoint of the ISSM ice sheet model, with focus on Computer Science issues, in particular linear solvers and MPI

Authors: No authors listed

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-2016-99-rc1

URL: <https://doi.org/10.5194/gmd-2016-99-rc1>

Publication Date: 2016-05-27

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

456. An optimized treatment for algorithmic differentiation of an important glaciological fixed-point problem

Authors: Daniel N. Goldberg; Sri Hari Krishna Narayanan; Laurent Hascoet; Jean Utke

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-9-1891-2016

URL: <https://doi.org/10.5194/gmd-9-1891-2016>

Publication Date: 2016-05-21

Volume/Issue: Volume 9, Issue 5

References: 54

Cited by: 15 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Natural Environment Research Council

License: Creative Commons

Open Access: N/A

Abstract: . We apply an optimized method to the adjoint generation of a time-evolving land ice model through algorithmic differentiation (AD). The optimization involves a special treatment of the fixed-point iteration required to solve the nonlinear stress balance, which differs from a straightforward application of AD software, and leads to smaller memory requirements and in some cases shorter computation times of the adjoint. The optimization is done via implementation of the algorithm of Christianson (1994) for reverse accumulation of fixed-point problems, with the AD tool OpenAD. For test problems, the optimized adjoint is shown to have far lower memory requirements, potentially enabling larger problem sizes on memory-limited machines. In the case of the land ice model, implementation of the algorithm allows further optimization by having the adjoint model solve a sequence of linear systems with identical (as opposed to varying) matrices, greatly improving performance. The methods introduced here will be of value to other efforts applying AD tools to ice models, particularly ones which solve a hybrid shallow ice/shallow shelf approximation to the Stokes equations.

457. Contrasts in the response of adjacent fjords and glaciers to ice-sheet surface melt in West Greenland

Authors: Timothy C. Bartholomaus; Leigh A. Stearns; David A. Sutherland; Emily L. Shroyer; Jonathan D. Nash; Ryan T. Walker; Ginny Catania; Denis Felikson; Dustin Carroll; Mason J. Fried; Brice P. Y. Noël; Michiel R. Van Den Broeke

Journal: Annals of Glaciology

Publisher: International Glaciological Society

DOI: 10.1017/aog.2016.19

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References: 75

Cited by: 47 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: Neighboring tidewater glaciers often exhibit asynchronous dynamic behavior, despite relatively uniform regional atmospheric and oceanic forcings. This variability may be controlled by a combination of local factors, including glacier and fjord geometry, fjord heat content and circulation, and glacier surface melt. In order to characterize and understand contrasts in adjacent tidewater glacier and fjord dynamics, we made coincident ice-ocean-atmosphere observations at high temporal resolution (minutes to weeks) within a 10 000 km² area near Uummannaq, Greenland. Water column velocity, temperature and salinity measurements reveal systematic differences in neighboring fjords that imply contrasting circulation patterns. The observed ocean velocity and hydrography, combined with numerical modeling, suggest that subglacial discharge plays a major role in setting fjord conditions. In addition, satellite remote sensing of seasonal ice flow speed and terminus position reveal both speedup and slow-down in response to melt, as well as differences in calving style among the neighboring glaciers. Glacier force budgets and modeling also point toward subglacial discharge as a key factor in glacier behavior. For the studied region, individual glacier and fjord geometry modulate subglacial discharge, which leads to contrasts in both fjord and glacier dynamics.

458. neXtSIM: a new Lagrangian sea ice model

Authors: Pierre Rampal; Sylvain Bouillon; Einar Ólason; Mathieu Morlighem

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-10-1055-2016

URL: <https://doi.org/10.5194/tc-10-1055-2016>

Publication Date: 2016-05-20

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References: 84

Cited by: 94 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . The Arctic sea ice cover has changed drastically over the last decades. Associated with these changes is a shift in dynamical regime seen by an increase of extreme fracturing events and an acceleration of sea ice drift. The highly non-linear dynamical response of sea ice to external forcing makes modelling these changes and the future evolution of Arctic sea ice a challenge for current models. It is, however, increasingly important that this challenge be better met, both because of the important role of sea ice in the climate system and because of the steady increase of industrial operations in the Arctic. In this paper we present a new dynamical/thermodynamical sea ice model called neXtSIM that is designed to address this challenge. neXtSIM is a continuous and fully Lagrangian model, whose momentum equation is discretised with the finite-element method. In this model, sea ice physics are driven by the combination of two core components: a model for sea ice dynamics built on a mechanical framework using an elasto-brittle rheology, and a model for sea ice thermodynamics providing damage healing for the mechanical framework. The evaluation of the model performance for the Arctic is presented for the period September 2007 to October 2008 and shows that observed multi-scale statistical properties of sea ice drift and deformation are well captured as well as the seasonal cycles of ice volume, area, and extent. These results show that neXtSIM is an appropriate tool for simulating sea ice over a wide range of spatial and temporal scales.

459. Optimal numerical solvers for transient simulations of ice flow using the Ice Sheet System Model (ISSM)

Authors: Feras Habbal; Eric Larour; Eric Rignot; Christopher P. Borstad; Helene Seroussi; Mathieu Morlighem

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-2016-111

URL: <https://doi.org/10.5194/gmd-2016-111>

Publication Date: 2016-05-17

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Solver|Benchmark (surveying)|Ice-sheet model|Transient (computer programming)|Greenland ice sheet

Funding: National Science Foundation

License: Creative Commons

Open Access: N/A

Abstract: . Identifying fast and robust numerical solvers is a critical issue that needs to be addressed in order to improve projections of polar ice sheets evolving in a changing climate. This work evaluates the impact of using sophisticated numerical solvers for transient ice flow simulations using the NASA-JPL/UCI Ice Sheet System Model (ISSM). We identify optimal numerical solvers by testing them on a commonly used ice flow benchmark test, the Ice Sheet Model Intercomparison Project for Higher-Order ice sheet Models (ISMIP-HOM) Experiment F. Three types of analyses are considered: mass transport, horizontal stress balance, and vertical stress balance. A broad suite of solvers is tested, ranging from direct sparse solvers to preconditioned iterative methods. The results of the fastest solvers for each analysis type are ranked based on their scalability across mesh size for each basal sliding conditions specified in Experiment F. We find that the fastest iterative solvers are ~1.5-100 times faster than the default direct solver used in ISSM with speed-ups improving rapidly with increased mesh resolution. We provide a set of recommendations for users in search of efficient solvers to use for transient ice flow simulations, enabling higher-resolution meshes and faster turnaround time. The end result will be improved transient simulations for short-term, highly resolved forward projections (10-100 year time scale) and also improved long-term paleo-reconstructions using higher-order representation of stresses in the ice. This analysis will also enable a new generation of comprehensive uncertainty quantification assessments of forward sea-level rise projections, which rely heavily on ensemble or sampling approaches that are inherently expensive.

460. A simple, physically motivated model of sea-level contributions from the Greenland ice sheet in response to temperature changes

Authors: Alexander M.R. Bakker; Patrick J. Applegate; Klaus Keller

Journal: Environmental Modelling & Software

Publisher: Elsevier BV

DOI: 10.1016/j.envsoft.2016.05.003

URL: <https://doi.org/10.1016/j.envsoft.2016.05.003>

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Volume/Issue: Volume 83

References: 45

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, NSF

License: N/A

Open Access: N/A

Abstract: N/A

461. Adaptive mesh refinement versus subgrid friction interpolation in simulations of Antarctic ice dynamics

Authors: S. L. Cornford; D. F. Martin; V. Lee; A. J. Payne; E. G. Ng

Journal: Annals of Glaciology

Publisher: International Glaciological Society

DOI: 10.1017/aog.2016.13

URL: <https://doi.org/10.1017/aog.2016.13>

Publication Date: 2016-05-13

Volume/Issue: Volume 57, Issue 73

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Cited by: 40 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: At least in conventional hydrostatic ice-sheet models, the numerical error associated with grounding line dynamics can be reduced by modifications to the discretization scheme. These involve altering the integration formulae for the basal traction and/or driving stress close to the grounding line and exhibit lower - if still first-order - error in the MISIMP3d experiments. MISIMP3d may not represent the variety of real ice streams, in that it lacks strong lateral stresses, and imposes a large basal traction at the grounding line. We study resolution sensitivity in the context of extreme forcing simulations of the entire Antarctic ice sheet, using the BISICLES adaptive mesh ice-sheet model with two schemes: the original treatment, and a scheme, which modifies the discretization of the basal traction. The second scheme does indeed improve accuracy - by around a factor of two - for a given mesh spacing, but ~ 1 km resolution is still necessary. For example, in coarser resolution simulations Thwaites Glacier retreats so slowly that other ice streams divert its trunk. In contrast, with ~ 1 km meshes, the same glacier retreats far more quickly and triggers the final phase of West Antarctic collapse a century before any such diversion can take place.

462. An Approach to Computing Discrete Adjoint for MPI-Parallelized Models Applied to the Ice Sheet System Model

Authors: Eric Larour; Jean Utke; Anton Bovin; Mathieu Morlighem; Gilberto Perez

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-2016-99

URL: <https://doi.org/10.5194/gmd-2016-99>

Publication Date: 2016-05-09

Volume/Issue: N/A

References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Hindcast|Greenland ice sheet|Operator (biology)|Ice-sheet model

Funding: Jet Propulsion Laboratory

License: Creative Commons

Open Access: N/A

Abstract: . Within the framework of sea-level rise projections, there is a strong need for hindcast validation of the evolution of polar ice sheets in a way that tightly matches observational records (from radar and altimetry observations mainly). However, the computational requirements for making hindcast reconstructions possible are severe and rely mainly on the evaluation of the adjoint state of transient ice-flow models. Here, we look at the computation of adjoints in the context of the NASA/JPL/UCI Ice Sheet System Model, written in C++ and designed for parallel execution with MPI. We present the adaptations required in the way the software is designed and written but also generic adaptations in the tools facilitating the adjoint computations. We concentrate on the use of operator overloading coupled with the AdjoinableMPI library to achieve the adjoint computation of ISSM. We present a comprehensive approach to 1) carry out type changing through ISSM, hence facilitating operator overloading, 2) bind to external solvers such as MUMPS and GSL-LU and 3) handle MPI-based parallelism to scale the capability. We demonstrate the success of the approach by computing sensitivities of hindcast metrics such as the misfit to observed records of surface altimetry on the North-East Greenland Ice Stream, or the misfit to observed records of surface velocities on Upernavik Glacier, Central West Greenland. We also provide metrics for the scalability of the approach, and the expected performance. This approach has the potential of enabling a new generation of hindcast-validated projections that make full use of the wealth of datasets currently being collected, or already collected in Greenland and Antarctica, such as surface altimetry, surface velocities, and/or gravity measurements.

463. ISSM-SESAR v1.0: mesh-based computation of gravitationally consistent sea-level and geodetic signatures caused by cryosphere and climate driven mass change

Authors: Surendra Adhikari; Erik R. Ivins; Eric Larour

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-9-1087-2016

URL: <https://doi.org/10.5194/gmd-9-1087-2016>

Publication Date: 2016-03-18

Volume/Issue: Volume 9, Issue 3

References: 92

Cited by: 46 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Jet Propulsion Laboratory

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Open Access: N/A

Abstract: . A classical Green's function approach for computing gravitationally consistent sea-level variations associated with mass redistribution on the earth's surface employed in contemporary sea-level models naturally suits the spectral methods for numerical evaluation. The capability of these methods to resolve high wave number features such as small glaciers is limited by the need for large numbers of pixels and high-degree (associated Legendre) series truncation. Incorporating a spectral model into (components of) earth system models that generally operate on a mesh system also requires repetitive forward and inverse transforms. In order to overcome these limitations, we present a method that functions efficiently on an unstructured mesh, thus capturing the physics operating at kilometer scale yet capable of simulating geophysical observables that are inherently of global scale with minimal computational cost. The goal of the current version of this model is to provide high-resolution solid-earth, gravitational, sea-level and rotational responses for earth system models operating in the domain of the earth's outer fluid envelope on timescales less than about 1 century when viscous effects can largely be ignored over most of the globe. The model has numerous important geophysical applications. For example, we compute time-varying computations of global geodetic and sea-level signatures associated with recent ice-sheet changes that are derived from space gravimetry observations. We also demonstrate the capability of our model to simultaneously resolve kilometer-scale sources of the earth's time-varying surface mass transport, derived from high-resolution modeling of polar ice sheets, and predict the corresponding local and global geodetic signatures.

464. Modelled glacier dynamics over the last quarter of a century at Jakobshavn Isbræ

Authors: Ioana S. Muresan; Shfaqat A. Khan; Andy Aschwanden; Constantine Khroulev; Tonie Van Dam; Jonathan Bamber; Michiel R. van den Broeke; Bert Wouters; Peter Kuipers Munneke; Kurt H. Kjær

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-10-597-2016

URL: <https://doi.org/10.5194/tc-10-597-2016>

Publication Date: 2016-03-15

Volume/Issue: Volume 10, Issue 2

References: 80

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . Observations over the past 2 decades show substantial ice loss associated with the speed-up of marine-terminating glaciers in Greenland. Here we use a regional three-dimensional outlet glacier model to simulate the behaviour of Jakobshavn Isbræ (JI) located in western Greenland. Our approach is to model and understand the recent behaviour of JI with a physical process-based model. Using atmospheric forcing and an ocean parametrization we tune our model to reproduce observed frontal changes of JI during 1990-2014. In our simulations, most of the JI retreat during 1990-2014 is driven by the ocean parametrization used and the glacier's subsequent response, which is largely governed by bed geometry. In general, the study shows significant progress in modelling the temporal variability of the flow at JI. Our results suggest that the overall variability in modelled horizontal velocities is a response to variations in terminus position. The model simulates two major accelerations that are consistent with observations of changes in glacier terminus. The first event occurred in 1998 and was triggered by a retreat of the front and moderate thinning of JI prior to 1998. The second event, which started in 2003 and peaked in the summer 2004, was triggered by the final break-up of the floating tongue. This break-up reduced the buttressing at the JI terminus that resulted in further thinning. As the terminus retreated over a reverse bed slope into deeper water, sustained high velocities over the last decade have been observed at JI. Our model provides evidence that the 1998 and 2003 flow accelerations are most likely initiated by the ocean parametrization used but JI's subsequent dynamic response was governed by its own bed geometry. We are unable to reproduce the observed 2010-2012 terminus retreat in our simulations. We attribute this limitation to either inaccuracies in basal topography or to misrepresentations of the climatic forcings that were applied. Nevertheless, the model is able to simulate the previously observed increase in mass loss through 2014.

465. Plastic bed beneath Hofsjökull Ice Cap, central Iceland, and the sensitivity of ice flow to surface meltwater flux

Authors: BRENT MINCHEW; MARK SIMONS; HELGI BJÖRNSSON; FINNUR PÁLSSON; MATHIEU MORLIGHEM; HELENE SEROUSSI; ERIC LAROUE; SCOTT HENSLEY

Journal: Journal of Glaciology

Publisher: Cambridge University Press (CUP)

DOI: 10.1017/jog.2016.26

URL: <https://doi.org/10.1017/jog.2016.26>

Publication Date: 2016-03-11

Volume/Issue: Volume 62, Issue 231

References: 68

Cited by: 46 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: The mechanical properties of glacier beds play a fundamental role in regulating the sensitivity of glaciers to environmental forcing across a wide range of timescales. Glaciers are commonly underlain by deformable till whose mechanical properties and influence on ice flow are not well understood but are critical for reliable projections of future glacier states. Using synoptic-scale observations of glacier motion in different seasons to constrain numerical ice flow models, we study the mechanics of the bed beneath Hofsjökull, a land-terminating ice cap in central Iceland. Our results indicate that the bed deforms plastically and weakens following incipient summertime surface melt. Combining the inferred basal shear traction fields with a Coulomb-plastic bed model, we estimate the spatially distributed effective basal water pressure and show that changes in basal water pressure and glacier accelerations are non-local and non-linear. These results motivate an idealized physical model relating mean basal water pressure and basal slip rate wherein the sensitivity of glacier flow to changes in basal water pressure is inversely related to the ice surface slope.

466. Modelling calving front dynamics using a level-set method: application to Jakobshavn Isbræ, West Greenland

Authors: Johannes H. Bondzio; H            ; Mathieu Morlighem; Thomas Kleiner; Martin R        ; Angelika Humbert; Eric Y. Larour

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-10-497-2016

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Publication Date: 2016-03-03

Volume/Issue: Volume 10, Issue 2

References: 51

Cited by: 54 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . Calving is a major mechanism of ice discharge of the Antarctic and Greenland ice sheets, and a change in calving front position affects the entire stress regime of marine terminating glaciers. The representation of calving front dynamics in a 2-D or 3-D ice sheet model remains non-trivial. Here, we present the theoretical and technical framework for a level-set method, an implicit boundary tracking scheme, which we implement into the Ice Sheet System Model (ISSM). This scheme allows us to study the dynamic response of a drainage basin to user-defined calving rates. We apply the method to Jakobshavn Isbr  , a major marine terminating outlet glacier of the West Greenland Ice Sheet. The model robustly reproduces the high sensitivity of the glacier to calving, and we find that enhanced calving triggers significant acceleration of the ice stream. Upstream acceleration is sustained through a combination of mechanisms. However, both lateral stress and ice influx stabilize the ice stream. This study provides new insights into the ongoing changes occurring at Jakobshavn Isbr   and emphasizes that the incorporation of moving boundaries and dynamic lateral effects, not captured in flow-line models, is key for realistic model projections of sea level rise on centennial timescales.

467. Modeling of Store Gletscher's calving dynamics, West Greenland, in response to ocean thermal forcing

Authors: M. Morlighem (University of California Irvine); J. Bondzio (University of California Irvine, Alfred Wegener Institute); H. Seroussi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); E. Rignot (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); E. Larour (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); A. Humbert (Alfred Wegener Institute); S. Rebuffi (University of California Irvine)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2016gl067695

URL: <https://doi.org/10.1002/2016gl067695>

Publication Date: 2016-02-26

Volume/Issue: Volume 43, Issue 6

References: 34

Cited by: 107 publications

Language: en

Subjects: N/A

Keywords: ice-ocean interactions, glaciology, calving, Store Gletscher, inverse modeling

Funding: National Aeronautics and Space Administration, Cryospheric Sciences Program

License: N/A

Open Access: N/A

Abstract: Glacier-front dynamics is an important control on Greenland's ice mass balance. Warmer ocean waters trigger ice-front retreats of marine-terminating glaciers, and the corresponding loss in resistive stress leads to glacier acceleration and thinning. Here we present an approach to quantify the sensitivity and vulnerability of marine-terminating glaciers to ocean-induced melt. We develop a plan view model of Store Gletscher that includes a level set-based moving boundary capability, a parameterized ocean-induced melt, and a calving law with complete and precise land and fjord topographies to model the response of the glacier to increased melt. We find that the glacier is stabilized by a sill at its terminus. The glacier is dislodged from the sill when ocean-induced melt quadruples, at which point the glacier retreats irreversibly for 27 km into a reverse bed. The model suggests that ice-ocean interactions are the triggering mechanism of glacier retreat, but the bed controls its magnitude.

468. A constitutive framework for predicting weakening and reduced buttressing of ice shelves based on observations of the progressive deterioration of the remnant Larsen B Ice Shelf

Authors: Chris Borstad (Department of Arctic Geophysics University Centre in Svalbard Longyearbyen Norway); Ala Khazendar (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Bernd Scheuchl (University of California Irvine); Mathieu Morlighem (University of California Irvine); Eric Larour (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Eric Rignot (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2015gl067365

URL: <https://doi.org/10.1002/2015gl067365>

Publication Date: 2016-02-12

Volume/Issue: Volume 43, Issue 5

References: 33

Cited by: 37 publications

Language: en

Subjects: N/A

Keywords: glaciology, damage mechanics, ice shelf buttressing

Funding: National Aeronautics and Space Administration

License: N/A

Open Access: N/A

Abstract: The increasing contribution of the Antarctic Ice Sheet to sea level rise is linked to reductions in ice shelf buttressing, driven in large part by basal melting of ice shelves. These ocean-driven buttressing losses are being compounded as ice shelves weaken and fracture. To date, model projections of ice sheet evolution have not accounted for weakening ice shelves. Here we present the first constitutive framework for ice deformation that explicitly includes mechanical weakening, based on observations of the progressive degradation of the remnant Larsen B Ice Shelf from 2000 to 2015. We implement this framework in an ice sheet model and are able to reproduce most of the observed weakening of the ice shelf. In addition to predicting ice shelf weakening and reduced buttressing, this new framework opens the door for improved understanding and predictions of iceberg calving, meltwater routing and hydrofracture, and ice shelf collapse.

469. review of Jordan et al. (2016, TCD)

Authors: Joseph MacGregor

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/tc-2016-8-rc1

URL: <https://doi.org/10.5194/tc-2016-8-rc1>

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Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

470. Shallow ice approximation, second order shallow ice approximation, and full Stokes models: A discussion of their roles in palaeo-ice sheet modelling and development

Authors: N. Kirchner; J. Ahlkrona; E.J. Gowan; P. Lötstedt; J.M. Lea; R. Noormets; L. von Sydow; J.A. Dowdeswell; T. Benham

Journal: Quaternary Science Reviews

Publisher: Elsevier BV

DOI: 10.1016/j.quascirev.2016.01.013

URL: <https://doi.org/10.1016/j.quascirev.2016.01.013>

Publication Date: 2016-02-06

Volume/Issue: Volume 135

References: 73

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Swedish Research Council FORMAS, Svalbard Science Forum's Strategic Grants

License: N/A

Open Access: N/A

Abstract: N/A

471. Complex Greenland outlet glacier flow captured

Authors: Andy Aschwanden; Mark A. Fahnestock; Martin Truffer

Journal: Nature Communications

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/ncomms10524

URL: <https://doi.org/10.1038/ncomms10524>

Publication Date: 2016-02-01

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References: 47

Cited by: 109 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: The Greenland Ice Sheet is losing mass at an accelerating rate due to increased surface melt and flow acceleration in outlet glaciers. Quantifying future dynamic contributions to sea level requires accurate portrayal of outlet glaciers in ice sheet simulations, but to date poor knowledge of subglacial topography and limited model resolution have prevented reproduction of complex spatial patterns of outlet flow. Here we combine a high-resolution ice-sheet model coupled to uniformly applied models of subglacial hydrology and basal sliding, and a new subglacial topography data set to simulate the flow of the Greenland Ice Sheet. Flow patterns of many outlet glaciers are well captured, illustrating fundamental commonalities in outlet glacier flow and highlighting the importance of efforts to map subglacial topography. Success in reproducing present day flow patterns shows the potential for prognostic modelling of ice sheets without the need for spatially varying parameters with uncertain time evolution.

472. Feasibility of improving a priori regional climate model estimates of Greenland ice sheet surface mass loss through assimilation of measured ice surface temperatures

Authors: M. Navari; S. A. Margulis; S. M. Bateni; M. Tedesco; P. Alexander; X. Fettweis

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-10-103-2016

URL: <https://doi.org/10.5194/tc-10-103-2016>

Publication Date: 2016-01-27

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References: 73

Cited by: 10 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Science Foundation

License: Creative Commons

Open Access: N/A

Abstract: . The Greenland ice sheet (GrIS) has been the focus of climate studies due to its considerable impact on sea level rise. Accurate estimates of surface mass fluxes would contribute to understanding the cause of its recent changes and would help to better estimate the past, current and future contribution of the GrIS to sea level rise. Though the estimates of the GrIS surface mass fluxes have improved significantly over the last decade, there is still considerable disparity between the results from different methodologies (e.g., Rae et al., 2012; Vernon et al., 2013). The data assimilation approach can merge information from different methodologies in a consistent way to improve the GrIS surface mass fluxes. In this study, an ensemble batch smoother data assimilation approach was developed to assess the feasibility of generating a reanalysis estimate of the GrIS surface mass fluxes via integrating remotely sensed ice surface temperature measurements with a regional climate model (a priori) estimate. The performance of the proposed methodology for generating an improved posterior estimate was investigated within an observing system simulation experiment (OSSE) framework using synthetically generated ice surface temperature measurements. The results showed that assimilation of ice surface temperature time series were able to overcome uncertainties in near-surface meteorological forcing variables that drive the GrIS surface processes. Our findings show that the proposed methodology is able to generate posterior reanalysis estimates of the surface mass fluxes that are in good agreement with the synthetic true estimates. The results also showed that the proposed data assimilation framework improves the root-mean-square error of the posterior estimates of runoff, sublimation/evaporation, surface condensation, and surface mass loss fluxes by 61, 64, 76, and 62 %, respectively, over the nominal a priori climate model estimates.

473. Sheet, stream, and shelf flow as progressive ice-bed uncoupling: Byrd Glacier, Antarctica and Jakobshavn Isbrae, Greenland

Authors: T. Hughes; A. Sargent; J. Fastook; K. Purdon; J. Li; J.-B. Yan; S. Gogineni

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-10-193-2016

URL: <https://doi.org/10.5194/tc-10-193-2016>

Publication Date: 2016-01-27

Volume/Issue: Volume 10, Issue 1

References: 121

Cited by: 3 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . The first-order control of ice thickness and height above sea level is linked to the decreasing strength of ice-bed coupling along flowlines from an interior ice divide to the calving front of an ice shelf. Uncoupling progresses as a frozen bed progressively thaws for sheet flow, as a thawed bed is progressively drowned for stream flow, and as lateral and/or local grounding vanish for shelf flow. This can reduce ice thicknesses by 90 % and ice elevations by 99 % along flowlines. Original work presented here includes (1) replacing flow and sliding laws for sheet flow with upper and lower yield stresses for creep in cold overlying ice and basal ice sliding over deforming till, respectively, (2) replacing integrating the Navier-Stokes equations for stream flow with geometrical solutions to the force balance, and (3) including resistance to shelf flow caused by lateral confinement in a fjord and local grounding at ice rumpled and ice rises. A comparison is made between our approach and two approaches based on continuum mechanics. Applications are made to Byrd Glacier in Antarctica and Jakobshavn Isbrae in Greenland.

474. SeaRISE experiments revisited: potential sources of spread in multi-model projections of the Greenland ice sheet

Authors: F. Saito; A. Abe-Ouchi; K. Takahashi; H. Blatter

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-10-43-2016

URL: <https://doi.org/10.5194/tc-10-43-2016>

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License: Creative Commons

Open Access: N/A

Abstract: . The present paper revisits the future surface-climate experiments on the Greenland ice sheet proposed by the Sea-level Response to Ice Sheet Evolution (SeaRISE; Bindshadler et al., 2013) study. The projections of the different SeaRISE participants show dispersion, which has not been examined in detail to date. A series of sensitivity experiments are conducted and analyzed using the ice-sheet model for integrated Earth-system studies (IcIES) by replacing one or more formulations of the model parameters with those adopted in other model(s). The results show that large potential sources of the dispersion among the projections of the different SeaRISE participants are differences in the initialization methods and in the surface mass balance methods, and both aspects have almost equal impact on the results. The treatment of ice-sheet margins in the simulation has a secondary impact on the dispersion. We conclude that spinning up the model using fixed topography through the spin-up period while the temperature is allowed to evolve according to the surface temperature history is the preferred representation, at least for the experiment configuration examined in the present paper. A benchmark model experimental setup that most of the numerical models can perform is proposed for future intercomparison projects, in order to evaluate the uncertainties relating to pure ice-sheet model flow characteristics.

475. Application of GRACE to the evaluation of an ice flow model of the Greenland Ice Sheet

Authors: N.-J. Schlegel; D. N. Wiese; E. Y. Larour; M. M. Watkins; J. E. Box; X. Fettweis; M. R van den Broeke

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/tc-2015-224

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References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Greenland ice sheet|Post-glacial rebound|Glacier mass balance|Ice-sheet model

Funding: Jet Propulsion Laboratory

License: Creative Commons

Open Access: N/A

Abstract: . Quantifying the Greenland Ice Sheet's future contribution to sea level rise is a challenging task that requires accurate estimates of ice flow sensitivity to climate change. Forward models of ice flow dynamics are promising tools for estimating future ice sheet behavior, yet confidence is low because evaluation of historical simulations is so challenging due to the scarcity of highly-resolved (spatially and temporally) continental-wide validation data. Recent advancements in processing of Gravity Recovery and Climate Experiment (GRACE) data using Bayesian-constrained mass concentration ("mascon") functions have led to improvements in spatial resolution and noise reduction of estimated monthly global gravity fields. Specifically, the Jet Propulsion Laboratory's JPL RL05M GRACE mascon solution (GRACE-JPL) now offers an opportunity for ice sheet model evaluation within independently resolved 300 km mascons. Here, we investigate how Greenland Ice Sheet mass balance captured through observations - GRACE-JPL - differs from that simulated by the ice flow model - the Ice Sheet System Model (ISSM). For the years 2003-2012, ISSM is forced with regional climate model (RCM) surface mass balance (SMB), and resulting mass balance is directly compared against GRACE-JPL within individual mascons. Overall, we find good agreement in the Northeast, Southwest, and the interior of the ice sheet, where mass balance is primarily controlled by SMB. In the Northwest, seasonal amplitudes match well, but trends in ISSM are muted relative to GRACE-JPL. In the Southeast, GRACE-JPL exhibits larger seasonal amplitude than that predicted by SMB while simultaneously having more pronounced trends. These results indicate that discrepancies in the Northwest are controlled by changes in ice dynamics that are not currently modeled by ISSM, i.e. transient processes driven by ice sheet hydrology and ice-ocean interaction, while discrepancies in the Southeast are controlled by a combination of these missing dynamics and errors in modeled SMB. Along the margins, we find that transient dynamics are responsible for consistent intra-annual variations in regional mass balance that ultimately contribute to the steeper negative mass trends observed by GRACE-JPL. Consequently, ice-ocean interactions and hydrologically-driven processes at relatively high (monthly-to-seasonal) temporal resolutions must be considered for improving upon ice

flow models.

476. Committed retreat of Smith, Pope, and Kohler Glaciers over the next 30 years inferred by transient model calibration

Authors: D. N. Goldberg; P. Heimbach; I. Joughin; B. Smith

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-9-2429-2015

URL: <https://doi.org/10.5194/tc-9-2429-2015>

Publication Date: 2015-12-21

Volume/Issue: Volume 9, Issue 6

References: 57

Cited by: 47 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . A glacial flow model of Smith, Pope and Kohler Glaciers is calibrated by means of control methods against time varying, annually resolved observations of ice height and velocities, covering the period 2002 to 2011. The inversion - termed "transient calibration" - produces an optimal set of time-mean, spatially varying parameters together with a time-evolving state that accounts for the transient nature of observations and the model dynamics. Serving as an optimal initial condition, the estimated state for 2011 is used, with no additional forcing, for predicting grounded ice volume loss and grounding line retreat over the ensuing 30 years. The transiently calibrated model predicts a near-steady loss of grounded ice volume of approximately 21 km³ a⁻¹ over this period, as well as loss of 33 km² a⁻¹ grounded area. We contrast this prediction with one obtained following a commonly used "snapshot" or steady-state inversion, which does not consider time dependence and assumes all observations to be contemporaneous. Transient calibration is shown to achieve a better fit with observations of thinning and grounding line retreat histories, and yields a quantitatively different projection with respect to ice volume loss and ungrounding. Sensitivity studies suggest large near-future levels of unforced, i.e., committed sea level contribution from these ice streams under reasonable assumptions regarding uncertainties of the unknown parameters.

477. Dynamically coupling the non-linear Stokes equations with the shallow ice approximation in glaciology: Description and first applications of the ISCAL method

Authors: Josefin Ahlkrona; Per Lötstedt; Nina Kirchner; Thomas Zwinger

Journal: Journal of Computational Physics

Publisher: Elsevier BV

DOI: 10.1016/j.jcp.2015.12.025

URL: <https://doi.org/10.1016/j.jcp.2015.12.025>

Publication Date: 2015-12-17

Volume/Issue: Volume 308

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Cited by: 19 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Nordic Centre of Excellence

License: N/A

Open Access: N/A

Abstract: N/A

478. Reducing uncertainties in projections of Antarctic ice mass loss

Authors: G. Durand; F. Pattyn

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-9-2043-2015

URL: <https://doi.org/10.5194/tc-9-2043-2015>

Publication Date: 2015-11-09

Volume/Issue: Volume 9, Issue 6

References: 60

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . Climate model projections are often aggregated into multi-model averages of all models participating in an intercomparison project, such as the Coupled Model Intercomparison Project (CMIP). The "multi-model" approach provides a sensitivity test to the models' structural choices and implicitly assumes that multiple models provide additional and more reliable information than a single model, with higher confidence being placed on results that are common to an ensemble. A first initiative of the ice sheet modeling community, SeaRISE, provided such multi-model average projections of polar ice sheets' contribution to sea-level rise. The SeaRISE Antarctic numerical experiments aggregated results from all models devoid of a priori selection, based on the capacity of such models to represent key ice-dynamical processes. Here, using the experimental setup proposed in SeaRISE, we demonstrate that correctly representing grounding line dynamics is essential to infer future Antarctic mass change. We further illustrate the significant impact on the ensemble mean and deviation of adding one model with a known bias in its ability of modeling grounding line dynamics. We show that this biased model can hardly be identified from the ensemble only based on its estimation of volume change, as ad hoc and untrustworthy parametrizations can force any modeled grounding line to retreat. However, tools are available to test parts of the response of marine ice sheet models to perturbations of climatic and/or oceanic origin (MISMIP, MISMIP3d). Based on recent projections of Pine Island Glacier mass loss, we further show that excluding ice sheet models that do not pass the MISMIP benchmarks decreases the mean contribution and standard deviation of the multi-model ensemble projection by an order of magnitude for that particular drainage basin.

479. Collapse of the West Antarctic Ice Sheet after local destabilization of the Amundsen Basin

Authors: Johannes Feldmann (Institute of Physics, Potsdam University, 14476 Potsdam, Germany, Potsdam Institute for Climate Impact Research, 14473 Potsdam, Germany;); Anders Levermann (Institute of Physics, Potsdam University, 14476 Potsdam, Germany, Potsdam Institute for Climate Impact Research, 14473 Potsdam, Germany;)

Journal: Proceedings of the National Academy of Sciences

Publisher: Proceedings of the National Academy of Sciences

DOI: 10.1073/pnas.1512482112

URL: <https://doi.org/10.1073/pnas.1512482112>

Publication Date: 2015-11-03

Volume/Issue: Volume 112, Issue 46

References: 51

Cited by: 124 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Seventh Framework Programme

License: N/A

Open Access: N/A

Abstract: Significance The Antarctic Ice Sheet is losing mass at an accelerating rate, and playing a more important role in terms of global sea-level rise. The Amundsen Sea sector of West Antarctica has most likely been destabilized. Although previous numerical modeling studies examined the short-term future evolution of this region, here we take the next step and simulate the long-term evolution of the whole West Antarctic Ice Sheet. Our results show that if the Amundsen Sea sector is destabilized, then the entire marine ice sheet will discharge into the ocean, causing a global sea-level rise of about 3 m. We thus might be witnessing the beginning of a period of self-sustained ice discharge from West Antarctica that requires long-term global adaptation of coastal protection.

480. Modelling the dynamic response of Jakobshavn Isbræ, West Greenland, to calving rate perturbations

Authors: J. H. Bondzio; H. Seroussi; M. Morlighem; T. Kleiner; M. Rückamp; A. Humbert; E. Larour

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/tcd-9-5485-2015

URL: <https://doi.org/10.5194/tcd-9-5485-2015>

Publication Date: 2015-10-15

Volume/Issue: N/A

References: 44

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Greenland ice sheet|Iceberg|Glacier terminus|Future sea level

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . Calving is a major means of ice discharge of the Antarctic and Greenland Ice Sheets. The breaking off of icebergs changes the ice front configuration of marine terminating glaciers, which affects the stress regime of their upstream areas. Recent observations show the close correlation between the ice front position and the behaviour of many outlet glaciers. However, modelling of a glacier subject to calving poses various challenges. No universal calving rate parametrisation is known, and tracking of a moving ice front and the related boundary conditions in two or three spatial dimensions is non-trivial. Here, we present the theoretical and technical framework for a Level-Set Method, an implicit boundary tracking scheme, which we implemented into the Ice Sheet System Model (ISSM). The scheme allows us to study the dynamic response of a drainage basin to user-defined front ablation rates. We apply the method in a suite of experiments to Jakobshavn Isbræ, a major marine terminating outlet glacier of the western Greenland Ice Sheet. The model robustly reproduces the high sensitivity of the glacier to frontal ablation in form of calving. We find that enhanced calving is able to trigger significant acceleration of the ice stream. Upstream acceleration is sustained through a combination of various feedback mechanisms. However, lateral stress and ice influx into the trough are able to stabilise the ice stream. This study contributes to the present discussion on causes and effects of the continued changes occurring at Jakobshavn Isbræ, and emphasises that the incorporation of seasonal calving and dynamic lateral effects is key for realistic model projections of future global sea level rise on centennial time scales.

481. A new sub-grid surface mass balance and flux model for continental-scale ice sheet modelling: testing and last glacial cycle

Authors: K. Le Morzadec; L. Tarasov; M. Morlighem; H. Seroussi

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-8-3199-2015

URL: <https://doi.org/10.5194/gmd-8-3199-2015>

Publication Date: 2015-10-08

Volume/Issue: Volume 8, Issue 10

References: 39

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: Ice-sheet model

Funding: Natural Sciences and Engineering Research Council of Canada

License: Creative Commons

Open Access: N/A

Abstract: . To investigate ice sheet evolution over the timescale of a glacial cycle, 3-D ice sheet models (ISMs) are typically run at "coarse" grid resolutions (10-50 km) that do not resolve individual mountains. This will introduce to-date unquantified errors in sub-grid (SG) transport, accumulation and ablation for regions of rough topography. In the past, synthetic hypsometric curves, a statistical summary of the topography, have been used in ISMs to describe the variability of these processes. However, there has yet to be detailed uncertainty analysis of this approach. We develop a new flow line SG model for embedding in coarse resolution models. A 1 km resolution digital elevation model was used to compute the local hypsometric curve for each coarse grid (CG) cell and to determine local parameters to represent the hypsometric bins' slopes and widths. The 1-D mass transport for the SG model is computed with the shallow ice approximation. We test this model against simulations from the 3-D Ice Sheet System Model (ISSM) run at 1 km grid resolution. Results show that none of the alternative parameterizations explored were able to adequately capture SG surface mass balance and flux processes. Via glacial cycle ensemble results for North America, we quantify the impact of SG model coupling in an ISM. We show that SG process representation and associated parametric uncertainties, related to the exchange of ice between the SG and CG cells, can have significant (up to 35 m eustatic sea level equivalent for the North American ice complex) impact on modelled ice sheet evolution.

482. Geophysical constraints on the dynamics and retreat of the Barents Sea ice sheet as a paleobenchmark for models of marine ice sheet deglaciation

Authors: Henry Patton (CAGE-Centre for Arctic Gas Hydrate, Environment and Climate, Department of Geology UiT The Arctic University of Norway Tromsø Norway); Karin Andreassen (CAGE-Centre for Arctic Gas Hydrate, Environment and Climate, Department of Geology UiT The Arctic University of Norway Tromsø Norway); Lilja R. Bjarnadóttir (Geological Survey of Norway Trondheim Norway); Julian A. Dowdeswell (University of Cambridge); Monica C. M. Winsborrow (CAGE-Centre for Arctic Gas Hydrate, Environment and Climate, Department of Geology UiT The Arctic University of Norway Tromsø Norway); Riko Noormets (University Centre in Svalbard (UNIS) Longyearbyen Norway); Leonid Polyak (CAGE-Centre for Arctic Gas Hydrate, Environment and Climate, Department of Geology UiT The Arctic University of Norway Tromsø Norway, Byrd Polar Research Center, Ohio State University); Amandine Auriac (Department of Geography Durham University Durham UK); Alun Hubbard (CAGE-Centre for Arctic Gas Hydrate, Environment and Climate, Department of Geology UiT The Arctic University of Norway Tromsø Norway)

Journal: Reviews of Geophysics

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2015rg000495

URL: <https://doi.org/10.1002/2015rg000495>

Publication Date: 2015-10-01

Volume/Issue: Volume 53, Issue 4

References: 356

Cited by: 73 publications

Language: en

Subjects: N/A

Keywords: ice sheet dynamics, Barents Sea ice sheet, ice sheet modeling, Svalbard, marine ice sheet (and 1 more)

Funding: "Glaciations in the Barents Sea area" (GlaciBar), Centres of Excellence

License: N/A

Open Access: N/A

Abstract: Our understanding of processes relating to the retreat of marine-based ice sheets, such as the West Antarctic Ice Sheet and tidewater-terminating glaciers in Greenland today, is still limited. In particular, the role of ice stream instabilities and oceanographic dynamics in driving their collapse are poorly constrained beyond observational timescales. Over numerous glaciations during the Quaternary, a marine-based ice sheet has waxed and waned over the Barents Sea continental shelf, characterized by a number of ice streams that extended to the shelf edge and subsequently collapsed during periods of climate and ocean warming. Increasing availability of offshore and onshore geophysical data over the last decade has significantly enhanced our knowledge of the pattern and timing of retreat of this Barents Sea ice sheet (BSIS), particularly so from its Late Weichselian maximum extent. We present a review of existing geophysical constraints that detail the dynamic evolution of the BSIS through the last glacial cycle, providing numerical modelers and geophysical workers with a benchmark data set with which to tune ice sheet reconstructions and explore ice sheet sensitivities and drivers of dynamic behavior. Although constraining data are generally spatially sporadic across the Barents and Kara Seas, behaviors such as ice sheet thinning, major ice divide migration, asynchronous and rapid flow

switching, and ice stream collapses are all evident. Further investigation into the drivers and mechanisms of such dynamics within this unique paleo-analogue is seen as a key priority for advancing our understanding of marine-based ice sheet deglaciations, both in the deep past and in the short-term future.

483. Reconciling the ICE-6G_C reconstruction of glacial chronology with ice sheet dynamics: The cases of Greenland and Antarctica

Authors: G. R. Stuhne (University of Toronto); W. R. Peltier (University of Toronto)

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2015jf003580

URL: <https://doi.org/10.1002/2015jf003580>

Publication Date: 2015-08-24

Volume/Issue: Volume 120, Issue 9

References: 61

Cited by: 35 publications

Language: en

Subjects: N/A

Keywords: paleoclimate, mass balance, glacial isostatic adjustment, ice sheet dynamics

Funding: Nuclear Waste Management Organization (NWMO), NSERC

License: N/A

Open Access: N/A

Abstract: We describe a theoretical and numerical framework that has been developed to investigate the compatibility of the ICE-6G_C reconstruction of the glaciation histories of the Greenland and Antarctic ice sheets with the latest understanding of ice physics. The ICE-6G_C reconstruction has been produced solely on the basis of the theory of the glacial isostatic adjustment (GIA) process, and it has remained an issue as to whether such reconstructions of the time-dependent thickness variations of grounded continental ice sheets were compatible with physics-based ice mechanical considerations. Our analyses focus on the evolution over the last glacial cycle of these extent ice sheet complexes and demonstrate that the GIA-inferred models are entirely consistent with such considerations when uncertainties in (net) mass balance history are taken fully into account.

484. Dynamics of thermally induced ice streams simulated with a higher-order flow model

Authors: D. J. Brinkerhoff (University of Alaska Fairbanks); J. V. Johnson (University of Montana)

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2015jf003499

URL: <https://doi.org/10.1002/2015jf003499>

Publication Date: 2015-08-10

Volume/Issue: Volume 120, Issue 9

References: 59

Cited by: 21 publications

Language: en

Subjects: N/A

Keywords: instability, modelling, friction, streaming

Funding: NASA

License: N/A

Open Access: N/A

Abstract: We use a new discretization technique to solve the higher-order thermomechanically coupled equations of glacier evolution. We find that under radially symmetric continuum equations, small perturbations in symmetry due to the discretization are sufficient to produce the initiation of nonsymmetric thermomechanical instabilities which we interpret as ice streams, in good agreement with previous studies which have indicated a similar instability. We find that the inclusion of membrane stresses regularizes the size of predicted streams, eliminating the ill-posedness evident in previous investigations of ice stream generation through thermomechanical instability. Ice streams exhibit strongly irregular periodicity which is influenced by neighboring ice streams and the synoptic state of the ice stream. Ice streams are not always the same size but instead appear to follow a temperature-dependent distribution of widths that is robust to grid refinement. The morphology of the predicted ice streams corresponds reasonably well to extant ice streams in physically similar environments.

485. Response of the large-scale subglacial drainage system of Northeast Greenland to surface elevation changes

Authors: N. B. Karlsson; D. Dahl-Jensen

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-9-1465-2015

URL: <https://doi.org/10.5194/tc-9-1465-2015>

Publication Date: 2015-08-06

Volume/Issue: Volume 9, Issue 4

References: 60

Cited by: 18 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: European Research Council

License: Creative Commons

Open Access: N/A

Abstract: . The influence of subglacial water on the dynamics of ice flow has been the object of increasing interest in the past decade. In this study we focus on large-scale, long-term changes in surface elevation over Northeast Greenland and the corresponding changes in subglacial water routeways. Our results show that over timescales ranging from decades to millennia the area may experience redistribution of and fluctuation in subglacial water outflux under the main glacier outlets. The fluctuations in subglacial water routing occur even in the absence of external forcing. Based on these results we conclude that changes in the subglacial water routeways are an intrinsic part of the drainage basin dynamics, where the subglacial system is likely always in a transient state. The results also imply that fluctuations at the margins observed at present might originate from changes several hundred kilometres upstream. Since surface elevation changes may propagate upstream over timescales much longer than the observational period, the cause of the fluctuations may not be present in current observational records.

486. Flow speed within the Antarctic ice sheet and its controls inferred from satellite observations

Authors: Robert J. Arthern (British Antarctic Survey Natural Environment Research Council Cambridge UK); Richard C. A. Hindmarsh (British Antarctic Survey Natural Environment Research Council Cambridge UK); C. Rosie Williams (British Antarctic Survey Natural Environment Research Council Cambridge UK)

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014jf003239

URL: <https://doi.org/10.1002/2014jf003239>

Publication Date: 2015-06-03

Volume/Issue: Volume 120, Issue 7

References: 68

Cited by: 63 publications

Language: en

Subjects: N/A

Keywords: inversion, flow, assimilation, velocity, ice sheet (and 1 more)

Funding: Natural Environment Research Council, European Union Seventh Framework Programme

License: N/A

Open Access: N/A

Abstract: Accurate dynamical models of the Antarctic ice sheet with carefully specified initial conditions and well-calibrated rheological parameters are needed to forecast global sea level. By adapting an inverse method previously used in electric impedance tomography, we infer present-day flow speeds within the ice sheet. This inversion uses satellite observations of surface velocity, snow accumulation rate, and rate of change of surface elevation to estimate the basal drag coefficient and an ice stiffness parameter that influences viscosity. We represent interior ice motion using a vertically integrated approximation to incompressible Stokes flow. This model represents vertical shearing within the ice and membrane stresses caused by horizontal stretching and shearing. Combining observations and model, we recover marked geographical variations in the basal drag coefficient. Relative changes in basal shear stress are smaller. No simple sliding law adequately represents basal shear stress as a function of sliding speed. Low basal shear stress predominates in central East Antarctica, where thick insulating ice allows liquid water at the base to lubricate sliding. Higher shear stress occurs in coastal East Antarctica, where a frozen bed is more likely. Examining Thwaites glacier in more detail shows that the slowest sliding often coincides with elevated basal topography. Differences between our results and a similar adjoint-based inversion suggest that inversion or regularization methods can influence recovered parameters for slow sliding and finer scales; on broader scales we recover a similar pattern of low basal drag underneath major ice streams and extensive regions in East Antarctica that move by basal sliding.

487. Scalable and efficient algorithms for the propagation of uncertainty from data through inference to prediction for large-scale problems, with application to flow of the Antarctic ice sheet

Authors: Tobin Isaac; Noemi Petra; Georg Stadler; Omar Ghattas

Journal: Journal of Computational Physics

Publisher: Elsevier BV

DOI: 10.1016/j.jcp.2015.04.047

URL: <https://doi.org/10.1016/j.jcp.2015.04.047>

Publication Date: 2015-05-07

Volume/Issue: Volume 296

References: 52

Cited by: 112 publications

Language: en

Subjects: N/A

Keywords: Antarctic ice sheet, Low-rank approximation, Adjoint-based Hessian, Nonlinear Stokes equations, Inverse problems (and 6 more)

Funding: Air Force Office of Scientific Research, U.S. Department of Energy, National Science Foundation (and 1 more)

License: N/A

Open Access: N/A

Abstract: N/A

488. A stabilized finite element method for calculating balance velocities in ice sheets

Authors: D. Brinkerhoff; J. Johnson

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-8-1275-2015

URL: <https://doi.org/10.5194/gmd-8-1275-2015>

Publication Date: 2015-05-04

Volume/Issue: Volume 8, Issue 5

References: 26

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . We present a numerical method for calculating vertically averaged velocity fields using a mass conservation approach, commonly known as balance velocities. This allows for an unstructured grid, is not dependent on a heuristic flow routing algorithm, and is both parallelizable and efficient. We apply the method to calculate depth-averaged velocities of the Greenland Ice Sheet, and find that the method produces grid-independent velocity fields for a sufficient parameterization of horizontal plane stresses on flow directions. We show that balance velocity can be used as the forward model for a constrained optimization problem that can be used to fill gaps and smooth strong gradients in InSAR velocity fields.

489. Radar attenuation and temperature within the Greenland Ice Sheet

Authors: Joseph A. MacGregor (University of Texas at Austin); Jilu Li (University of Kansas); John D. Paden (University of Kansas); Ginny A. Catania (University of Texas at Austin); Gary D. Clow (University of Colorado Boulder, U.S. Geological Survey Lakewood Colorado USA); Mark A. Fahnestock (University of Alaska Fairbanks); S. Prasad Gogineni (University of Kansas); Robert E. Grimm (Department of Space Studies Southwest Research Institute Boulder Colorado USA); Mathieu Morlighem (University of California Irvine); Soumyaroop Nandi (University of Kansas); Hélène Seroussi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); David E. Stillman (Department of Space Studies Southwest Research Institute Boulder Colorado USA)

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014jf003418

URL: <https://doi.org/10.1002/2014jf003418>

Publication Date: 2015-04-30

Volume/Issue: Volume 120, Issue 6

References: 90

Cited by: 82 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NSF, NASA

License: N/A

Open Access: N/A

Abstract: N/A

490. Albany/FELIX: a parallel, scalable and robust, finite element, first-order Stokes approximation ice sheet solver built for advanced analysis

Authors: I. K. Tezaur; M. Perego; A. G. Salinger; R. S. Tuminaro; S. F. Price

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-8-1197-2015

URL: <https://doi.org/10.5194/gmd-8-1197-2015>

Publication Date: 2015-04-27

Volume/Issue: Volume 8, Issue 4

References: 74

Cited by: 46 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . This paper describes a new parallel, scalable and robust finite element based solver for the first-order Stokes momentum balance equations for ice flow. The solver, known as Albany/FELIX, is constructed using the component-based approach to building application codes, in which mature, modular libraries developed as a part of the Trilinos project are combined using abstract interfaces and template-based generic programming, resulting in a final code with access to dozens of algorithmic and advanced analysis capabilities. Following an overview of the relevant partial differential equations and boundary conditions, the numerical methods chosen to discretize the ice flow equations are described, along with their implementation. The results of several verification studies of the model accuracy are presented using (1) new test cases for simplified two-dimensional (2-D) versions of the governing equations derived using the method of manufactured solutions, and (2) canonical ice sheet modeling benchmarks. Model accuracy and convergence with respect to mesh resolution are then studied on problems involving a realistic Greenland ice sheet geometry discretized using hexahedral and tetrahedral meshes. Also explored as a part of this study is the effect of vertical mesh resolution on the solution accuracy and solver performance. The robustness and scalability of our solver on these problems is demonstrated. Lastly, we show that good scalability can be achieved by preconditioning the iterative linear solver using a new algebraic multilevel preconditioner, constructed based on the idea of semi-coarsening.

491. The evolving instability of the remnant Larsen B Ice Shelf and its tributary glaciers

Authors: Ala Khazendar; Christopher P. Borstad; Bernd Scheuchl; Eric Rignot; Helene Seroussi

Journal: Earth and Planetary Science Letters

Publisher: Elsevier BV

DOI: 10.1016/j.epsl.2015.03.014

URL: <https://doi.org/10.1016/j.epsl.2015.03.014>

Publication Date: 2015-04-07

Volume/Issue: Volume 419

References: 62

Cited by: 36 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: NASA, NASA, NASA

License: N/A

Open Access: N/A

Abstract: N/A

492. Greenland ice sheet mass balance: a review

Authors: Shfaqat A Khan; Andy Aschwanden; Anders A Bjørk; John Wahr; Kristian K Kjeldsen; Kurt H Kjær

Journal: Reports on Progress in Physics

Publisher: IOP Publishing

DOI: 10.1088/0034-4885/78/4/046801

URL: <https://doi.org/10.1088/0034-4885/78/4/046801>

Publication Date: 2015-03-27

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References: 228

Cited by: 122 publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

493. Enthalpy benchmark experiments for numerical ice sheet models

Authors: T. Kleiner; M. Rückamp; J. H. Bondzio; A. Humbert

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-9-217-2015

URL: <https://doi.org/10.5194/tc-9-217-2015>

Publication Date: 2015-02-06

Volume/Issue: Volume 9, Issue 1

References: 28

Cited by: 24 publications

Language: en

Subjects: N/A

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License: Creative Commons

Open Access: N/A

Abstract: . We present benchmark experiments to test the implementation of enthalpy and the corresponding boundary conditions in numerical ice sheet models. Since we impose several assumptions on the experiment design, analytical solutions can be formulated for the proposed numerical experiments. The first experiment tests the functionality of the boundary condition scheme and the basal melt rate calculation during transient simulations. The second experiment addresses the steady-state enthalpy profile and the resulting position of the cold-temperate transition surface (CTS). For both experiments we assume ice flow in a parallel-sided slab decoupled from the thermal regime. We compare simulation results achieved by three different ice flow-models with these analytical solutions. The models agree well to the analytical solutions, if the change in conductivity between cold and temperate ice is properly considered in the model. In particular, the enthalpy gradient on the cold side of the CTS goes to zero in the limit of vanishing temperate-ice conductivity, as required from the physical jump conditions at the CTS.

494. Simulating the Greenland ice sheet under present-day and palaeo constraints including a new discharge parameterization

Authors: R. Calov; A. Robinson; M. Perrette; A. Ganopolski

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-9-179-2015

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Cited by: 26 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: . In this paper, we propose a new sub-grid scale parameterization for the ice discharge into the ocean through outlet glaciers and inspect the role of different observational and palaeo constraints for the choice of an optimal set of model parameters. This parameterization was introduced into the polythermal ice-sheet model SICOPOLIS, which is coupled to the regional climate model of intermediate complexity REMBO. Using the coupled model, we performed large ensemble simulations over the last two glacial cycles by varying two major parameters: a melt parameter in the surface melt scheme of REMBO and a discharge scaling parameter in our parameterization of ice discharge. Our empirical constraints are the present-day Greenland ice sheet surface elevation, the surface mass balance partition (ratio between total ice discharge and total precipitation) and the Eemian interglacial elevation drop relative to present day in the vicinity of the NEEM ice core. We show that the ice discharge parameterization enables us to simulate both the correct ice-sheet shape and mass balance partition at the same time without explicitly resolving the Greenland outlet glaciers. For model verification, we compare the simulated total and sectoral ice discharge with other estimates. For the model versions that are consistent with the range of observational and palaeo constraints, our simulated Greenland ice sheet contribution to Eemian sea-level rise relative to present-day amounts to 1.4 m on average (in the range of 0.6 and 2.5 m).

495. Thermal boundary conditions on western Greenland: Observational constraints and impacts on the modeled thermomechanical state

Authors: Toby W. Meierbachtol (University of Montana); Joel T. Harper (University of Montana); Jesse V. Johnson (University of Montana); Neil F. Humphrey (Department of Geology and Geophysics University of Wyoming Laramie Wyoming USA); Douglas J. Brinkerhoff (University of Alaska Fairbanks, University of Montana)

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014jf003375

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Publication Date: 2015-01-26

Volume/Issue: Volume 120, Issue 3

References: 52

Cited by: 24 publications

Language: en

Subjects: N/A

Keywords: glaciology, thermodynamics, modeling

Funding: Greenland Analogue Project, NSF

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Open Access: N/A

Abstract: The surface and basal boundary conditions exert an important control on the thermodynamic state of the Greenland Ice Sheet, but their representation in numerical ice sheet models is poorly constrained due to the lack of observations. Here we investigate a land-terminating sector of western Greenland and (1) quantify differences between new observations and commonly used boundary condition data sets and (2) demonstrate the impact of improved boundary conditions on simulated thermodynamics in a higher-order numerical flow model. We constrain near-surface temperature with measurements from two 20 m boreholes in the ablation zone and 10 m firn temperature from the percolation zone. We constrain basal heat flux using in situ measurement in a deep bedrock hole at the study area margin and other existing assessments. To assess boundary condition influences on simulated thermal-mechanical processes, we compare model output to multiple full-thickness temperature profiles collected in the ablation zone. Our observation-constrained basal heat flux is 30 mW m⁻² less than commonly used representations. In contrast, measured near-surface temperatures are warmer than common surface temperature data sets by up to 15 degreesC. Application of lower basal heat flux increases a model cold bias compared to the measured temperature profiles and causes frozen basal conditions across the ablation zone. Temperate basal conditions are reestablished by our warmer surface boundary. Warmer surface ice and firn can introduce several times more energy to the modeled ice mass than what is lost at the bed from reduced basal heat flux, indicating that the thermomechanical state of the ice sheet is highly sensitive to near-surface effects.

496. Experimental determination of a double-valued drag relationship for glacier sliding

Authors: Lucas K. Zoet; Neal R. Iverson

Journal: Journal of Glaciology

Publisher: International Glaciological Society

DOI: 10.3189/2015jog14j174

URL: <https://doi.org/10.3189/2015jog14j174>

Publication Date: 2014-12-16

Volume/Issue: Volume 61, Issue 225

References: 52

Cited by: 28 publications

Language: en

Subjects: N/A

Keywords: Glacier mechanics, Glacier flow, Glacier modelling, Basal ice, Subglacial processes

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The contribution of glaciers to sea-level rise and their effects on landscape evolution depend on the poorly known relationship between sliding speed and drag at the ice/bed interface. Results from experiments with a new rotary laboratory device demonstrate empirically for the first time a double-valued drag relationship like that suggested by some sliding theories: steady drag on a rigid, sinusoidal bed increases, peaks and declines at progressively higher sliding speeds due to growth of cavities in the lee sides of bed undulations. Drag decreases with increased sliding speed if cavities extend beyond the inflection points of up-glacier facing surfaces, so that adverse bed slopes in contact with ice diminish with further cavity growth. These results indicate that shear tractions on glacier beds can potentially decrease due to increases in sliding speed driven by weather or climate variability, promoting even more rapid glacier motion by requiring greater strain rates to produce resistive stresses. Although a double-valued drag relationship has not yet been demonstrated for the complicated geometries of real glacier beds, both its potential major implications and the characteristically convex stoss surfaces of bumps on real glacier beds provide stimulus for exploring the effects of this relationship in ice-sheet models.

497. Inferred basal friction and surface mass balance of the Northeast Greenland Ice Stream using data assimilation of ICESat (Ice Cloud and land Elevation Satellite) surface altimetry and ISSM (Ice Sheet System Model)

Authors: E. Larour; J. Utke; B. Csatho; A. Schenk; H. Seroussi; M. Morlighem; E. Rignot; N. Schlegel; A. Khazendar

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-8-2335-2014

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Publication Date: 2014-12-15

Volume/Issue: Volume 8, Issue 6

References: 108

Cited by: 53 publications

Language: en

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Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . We present a new data assimilation method within the Ice Sheet System Model (ISSM) framework that is capable of assimilating surface altimetry data from missions such as ICESat (Ice Cloud and land Elevation Satellite) into reconstructions of transient ice flow. The new method relies on algorithmic differentiation to compute gradients of objective functions with respect to model forcings. It is applied to the Northeast Greenland Ice Stream, where surface mass balance and basal friction forcings are temporally inverted, resulting in adjusted modeled surface heights that best fit existing altimetry. This new approach allows for a better quantification of basal and surface processes and a better understanding of the physical processes currently missing in transient ice-flow models to better capture the important intra- and interannual variability in surface altimetry. It also demonstrates that large spatial and temporal variability is required in model forcings such as surface mass balance and basal friction, variability that can only be explained by including more complex processes such as snowpack compaction at the surface and basal hydrology at the bottom of the ice sheet. This approach is indeed a first step towards assimilating the wealth of high spatial resolution altimetry data available from EnviSat, ICESat, Operation IceBridge and CryoSat-2, and that which will be available in the near future with the launch of ICESat-2.

498. Ice discharge uncertainties in Northeast Greenland from boundary conditions and climate forcing of an ice flow model

Authors: N.-J. Schlegel (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); E. Larour (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); H. Seroussi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); M. Morlighem (University of California Irvine); J. E. Box (Geological Survey of Denmark and Greenland)

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014jgf003359

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Cited by: 27 publications

Language: en

Subjects: N/A

Keywords: Mass flux, Uncertainty, Ice discharge, Ice flow, Ice stream

Funding: NSF, NASA

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Open Access: N/A

Abstract: N/A

499. Sensitivity of the Weddell Sea sector ice streams to sub-shelf melting and surface accumulation

Authors: A. P. Wright; A. M. Le Brocq; S. L. Cornford; R. G. Bingham; H. F. J. Corr; F. Ferraccioli; T. A. Jordan; A. J. Payne; D. M. Rippin; N. Ross; M. J. Siegert

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-8-2119-2014

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Volume/Issue: Volume 8, Issue 6

References: 48

Cited by: 32 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . A recent ocean modelling study indicates that possible changes in circulation may bring warm deep-ocean water into direct contact with the grounding lines of the Filchner-Ronne ice streams, suggesting the potential for future ice losses from this sector equivalent to ~0.3 m of sea-level rise. Significant advancements have been made in our knowledge of both the basal topography and ice velocity in the Weddell Sea sector, and the ability to accurately model marine ice sheet dynamics, thus enabling an assessment to be made of the relative sensitivities of the diverse collection of ice streams feeding the Filchner-Ronne Ice Shelf. Here we use the BISICLES ice sheet model, which employs adaptive-mesh refinement to resolve grounding line dynamics, to carry out such an assessment. The impact of realistic perturbations to the surface and sub-shelf mass balance forcing fields from our 2000-year "reference" model run indicate that both the Institute and Möller ice streams are highly sensitive to changes in basal melting either near to their respective grounding lines, or in the region of the ice rises within the Filchner-Ronne Ice Shelf. These same perturbations have little impact, however, on the Rutford, Carlson or Foundation ice streams, while the Evans Ice Stream is found to enter a phase of unstable retreat only after melt at its grounding line has increased by 50% of likely present-day values.

500. Hydrostatic grounding line parameterization in ice sheet models

Authors: H. Seroussi; M. Morlighem; E. Larour; E. Rignot; A. Khazendar

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-8-2075-2014

URL: <https://doi.org/10.5194/tc-8-2075-2014>

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Cited by: 86 publications

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Keywords: N/A

Funding: National Aeronautics and Space Administration

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Open Access: N/A

Abstract: . Modeling of grounding line migration is essential to accurately simulate the behavior of marine ice sheets and investigate their stability. Here, we assess the sensitivity of numerical models to the parameterization of the grounding line position. We run the MISMIP3D benchmark experiments using the Ice Sheet System Model (ISSM) and a two-dimensional shelfy-stream approximation (SSA) model with different mesh resolutions and different sub-element parameterizations of grounding line position. Results show that different grounding line parameterizations lead to different steady state grounding line positions as well as different retreat/advance rates. Our simulations explain why some vertically depth-averaged model simulations deviate significantly from the vast majority of simulations based on SSA in the MISMIP3D benchmark. The results reveal that differences between simulations performed with and without sub-element parameterization are as large as those performed with different approximations of the stress balance equations in this configuration. They also demonstrate that the reversibility test is passed at relatively coarse resolution while much finer resolutions are needed to accurately capture the steady-state grounding line position. We conclude that fixed grid SSA models that do not employ such a parameterization should be avoided, as they do not provide accurate estimates of grounding line dynamics, even at high spatial resolution. For models that include sub-element grounding line parameterization, in the MISMIP3D configuration, a mesh resolution finer than 2 km should be employed.

501. Modeling the Evolution of Polar Ice Sheets

Authors: Eric Larour; Nicole Schlegel; Mathieu Morlighem

Journal: Eos, Transactions American Geophysical Union

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014eo450005

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References: N/A

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

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License: Creative Commons

Open Access: N/A

Abstract: The Ice Sheet System Model (ISSM) team, which is funded primarily by NASA (Cryosphere and Modeling, Analysis and Prediction programs), as well as the Jet Propulsion Laboratory, University of California, Irvine, and National Science Foundation's Office of Polar Programs, organized a workshop in collaboration with the Bjerknes Centre for Climate Research (BCCR) at the University of Bergen in Norway, in June 2014.

502. Cold-based debris-covered glaciers: Evaluating their potential as climate archives through studies of ground-penetrating radar and surface morphology

Authors: Sean L. Mackay (Boston University); David R. Marchant (Boston University); Jennifer L. Lamp (Boston University); James W. Head (Brown University)

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014jf003178

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References: 144

Cited by: 32 publications

Language: en

Subjects: N/A

Keywords: Ground-penetrating radar, Climate change, Rock glacier, Antarctica, Debris-covered glacier

Funding: NSF Graduate Research Fellowship, NSF

License: N/A

Open Access: N/A

Abstract: We describe the morphology and internal structure of Mullins and Friedman Glaciers, two cold-based, debris-covered alpine glaciers that occur in neighboring valleys in the McMurdo Dry Valleys, Antarctica. Both glaciers are overlain by a single, dry supraglacial debris layer (8-75 cm thick); each mantling debris layer is marked with near-identical patterns of arcuate ridges and steps, as well as corresponding changes in bulk grain size, meter-scale surface topography, and thermal contraction crack polygons. Results from 24 km of ground-penetrating radar data show that the ice within the uppermost 1-2 km of Mullins and Friedman Glaciers is essentially free of englacial debris (<1% by volume) but thereafter is interspersed with bands of englacial debris (each <3 m thick and dipping up glacier) that intersect the ice surface at all major surface ridges and steps. The similarity in number and pattern of englacial debris bands and corresponding surface ridges and steps across both glaciers, along with model results and observations that call for negligible basal entrainment, is best explained by episodic environmental change at valley headwalls. Our working hypothesis is that layers of englacial debris originate as supraglacial lags that form in ice-accumulation areas during times of reduced net ice accumulation; following renewed net ice accumulation, these lags are subsequently buried by snow and ice, flow englacially, and intersect the ice surface to impart distinctive changes in the texture of supraglacial debris and topographic relief. The implication is that the englacial structure and surface morphology of these cold-based, debris-covered glaciers preserves a consistent record of climate and environmental change.

503. Sensitivity of the dynamics of Pine Island Glacier, West Antarctica, to climate forcing for the next 50 years

Authors: H. Seroussi; M. Morlighem; E. Rignot; J. Mouginot; E. Larour; M. Schodlok; A. Khazendar

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-8-1699-2014

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Publication Date: 2014-09-17

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References: 45

Cited by: 59 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . Pine Island Glacier, a major contributor to sea level rise in West Antarctica, has been undergoing significant changes over the last few decades. Here, we employ a three-dimensional, higher-order model to simulate its evolution over the next 50 yr in response to changes in its surface mass balance, the position of its calving front and ocean-induced ice shelf melting. Simulations show that the largest climatic impact on ice dynamics is the rate of ice shelf melting, which rapidly affects the glacier speed over several hundreds of kilometers upstream of the grounding line. Our simulations show that the speedup observed in the 1990s and 2000s is consistent with an increase in sub-ice-shelf melting. According to our modeling results, even if the grounding line stabilizes for a few decades, we find that the glacier reaction can continue for several decades longer. Furthermore, Pine Island Glacier will continue to change rapidly over the coming decades and remain a major contributor to sea level rise, even if ocean-induced melting is reduced.

504. A model of Greenland ice sheet deglaciation constrained by observations of relative sea level and ice extent

Authors: Benoit S. Lecavalier; Glenn A. Milne; Matthew J.R. Simpson; Leanne Wake; Philippe Huybrechts; Lev Tarasov; Kristian K. Kjeldsen; Svend Funder; Antony J. Long; Sarah Woodroffe; Arthur S. Dyke; Nicolaj K. Larsen

Journal: Quaternary Science Reviews

Publisher: Elsevier BV

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References: 168

Cited by: 188 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Natural Science and Engineering Research Council of Canada, Canada Research Chairs program, Canadian Foundation for Innovation (and 1 more)

License: N/A

Open Access: N/A

Abstract: N/A

505. Role of model initialization for projections of 21st-century Greenland ice sheet mass loss

Authors: G. Aðalgeirsdóttir; A. Aschwanden; C. Khroulev; F. Boberg; R. Mottram; P. Lucas-Picher; J.H. Christensen

Journal: Journal of Glaciology

Publisher: International Glaciological Society

DOI: 10.3189/2014jog13j202

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Volume/Issue: Volume 60, Issue 222

References: 80

Cited by: 24 publications

Language: en

Subjects: N/A

Keywords: Ice and climate, Ice-sheet modeling

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Model simulations of the Greenland ice sheet contribution to 21st-century sea-level rise are performed with a state-of-the-art ice-sheet model (Parallel Ice Sheet Model (PISM)). The climate-forcing fields are obtained from the European Union's Seventh Framework Programme project ice2sea, in which three regional climate models are used to dynamically downscale two scenarios (A1B and E1) from two general circulation models (ECHAM5 and HadCM3). To assess the sensitivity of the projections to the model initial state, four initialization methods are applied. In these experiments, the simulated contribution to sea-level rise by 2100 ranges from an equivalent of 0.2 to 6.8 cm. The largest uncertainties arise from different formulations of the regional climate models (0.8-3.9 cm) and applied scenarios (0.65-1.9 cm), but an important source of uncertainty is the initialization method (0.1-0.8 cm). These model simulations do not account for the recently observed acceleration of ice streams and consequent thinning rates, the changing ice discharge that may result from the spatial and temporal variability of ocean forcing, or the feedback occurring between ice-sheet elevation changes and climate forcing. Thus the results should be considered the lower limit of Greenland ice sheet contributions to sea-level rise, until such processes have been integrated into large-scale ice-sheet models.

506. Optimal initial conditions for coupling ice sheet models to Earth system models

Authors: Mauro Perego (Computational Mathematics Sandia National Laboratories Albuquerque New Mexico USA); Stephen Price (Los Alamos National Laboratory); Georg Stadler (Now at Courant Institute of Mathematical Sciences New York University New York USA, University of Texas at Austin)

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014jf003181

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References: 55

Cited by: 51 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We address complications in the coupling of a dynamic ice sheet model (ISM) and forcing from an Earth system model (ESM), which arise because of the unknown ISM initial conditions. Unless explicitly accounted for during ISM initialization, the ice sheet is far from thermomechanical equilibrium with the surface mass balance forcing from the ESM. Upon coupling to ESM forcing, the result is a shock and unphysical and undesirable transients in ice geometry and other state variables. Under the assumption of thermomechanical equilibrium, we present an approach for finding ISM initial conditions--characterized by optimization of the basal sliding coefficient and basal topography fields--that balance a best fit to surface velocity and basal topography observations against the minimization of unphysical transients when coupling to surface mass balance forcing. A quasi-Newton method is used to solve the resulting large-scale, partial differential equation-constrained optimization problem, where the cost function gradients with respect to the parameter fields are computed using adjoints. After studying properties of our approach on a synthetic test problem, we apply the method toward obtaining optimal initial conditions for a model of the Greenland ice sheet. Our results show that, in the presence of uncertainties in the basal topography, ice thickness should also be treated as an optimization variable. While the focus here is on the coupling between an ISM and ESM-derived surface mass balance, the method is easily extended to include optimal coupling to forcing from an ocean model through submarine melt rates.

507. Representation of sharp rifts and faults mechanics in modeling ice shelf flow dynamics: Application to Brunt/Stancomb-Wills Ice Shelf, Antarctica

Authors: E. Larour (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); A. Khazendar (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); C. P. Borstad (California Institute of Technology); H. Seroussi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); M. Morlighem (University of California Irvine); E. Rignot (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014jf003157

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Volume/Issue: Volume 119, Issue 9

References: 68

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Ice shelves play a major role in buttressing ice sheet flow into the ocean, hence the importance of accurate numerical modeling of their stress regime. Commonly used ice flow models assume a continuous medium and are therefore complicated by the presence of rupture features (crevasses, rifts, and faults) that significantly affect the overall flow patterns. Here we apply contact mechanics and penalty methods to develop a new ice shelf flow model that captures the impact of rifts and faults on the rheology and stress distribution of ice shelves. The model achieves a best fit solution to satellite observations of ice shelf velocities to infer the following: (1) a spatial distribution of contact and friction points along detected faults and rifts, (2) a more realistic spatial pattern of ice shelf rheology, and (3) a better representation of the stress balance in the immediate vicinity of faults and rifts. Thus, applying the model to the Brunt/Stancomb-Wills Ice Shelf, Antarctica, we quantify the state of friction inside faults and the opening rates of rifts and obtain an ice shelf rheology that remains relatively constant everywhere else on the ice shelf. We further demonstrate that better stress representation has widespread application in examining aspects affecting ice shelf structure and dynamics including the extent of ice mélange in rifts and the change in fracture configurations. All are major applications for better insight into the important question of ice shelf stability.

508. Ice-sheet modelling accelerated by graphics cards

Authors: Christian Fredborg Brædstrup; Anders Damsgaard; David Lundbek Egholm

Journal: Computers & Geosciences

Publisher: Elsevier BV

DOI: 10.1016/j.cageo.2014.07.019

URL: <https://doi.org/10.1016/j.cageo.2014.07.019>

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Cited by: 14 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Natur og Univers, Det Frie Forskningsråd

License: N/A

Open Access: N/A

Abstract: N/A

509. Parameterization of basal friction near grounding lines in a one-dimensional ice sheet model

Authors: G. R. Leguy; X. S. Asay-Davis; W. H. Lipscomb

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-8-1239-2014

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Open Access: N/A

Abstract: . Ice sheets and ice shelves are linked by the transition zone, the region where flow dominated by vertical shear stress makes a transition to flow dominated by extensional stress. Adequate resolution of the transition zone is necessary for numerically accurate ice sheet-ice shelf simulations. The required resolution depends on how the basal physics is parameterized. We propose a~new, simple parameterization of the effective pressure near the grounding line, combined with an existing friction law linking effective pressure to basal stress and sliding, in a one-dimensional, fixed-grid, vertically integrated model. This parameterization represents connectivity between the basal hydrological system and the ocean in the transition zone. Our model produces a smooth transition between finite basal friction in the ice sheet and zero basal friction in the ice shelf. In a set of experiments based on the Marine Ice Sheet Model Intercomparison Project (MISMIP), we show that with a smoother basal shear stress, the model yields accurate steady-state results at a fixed-grid resolution of ~1 km.

510. Future Antarctic bed topography and its implications for ice sheet dynamics

Authors: S. Adhikari; E. R. Ivins; E. Larour; H. Seroussi; M. Morlighem; S. Nowicki

Journal: Solid Earth

Publisher: Copernicus GmbH

DOI: 10.5194/se-5-569-2014

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References: 61

Cited by: 31 publications

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Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: . The Antarctic bedrock is evolving as the solid Earth responds to the past and ongoing evolution of the ice sheet. A recently improved ice loading history suggests that the Antarctic Ice Sheet (AIS) has generally been losing its mass since the Last Glacial Maximum. In a sustained warming climate, the AIS is predicted to retreat at a greater pace, primarily via melting beneath the ice shelves. We employ the glacial isostatic adjustment (GIA) capability of the Ice Sheet System Model (ISSM) to combine these past and future ice loadings and provide the new solid Earth computations for the AIS. We find that past loading is relatively less important than future loading for the evolution of the future bed topography. Our computations predict that the West Antarctic Ice Sheet (WAIS) may uplift by a few meters and a few tens of meters at years AD 2100 and 2500, respectively, and that the East Antarctic Ice Sheet is likely to remain unchanged or subside minimally except around the Amery Ice Shelf. The Amundsen Sea Sector in particular is predicted to rise at the greatest rate; one hundred years of ice evolution in this region, for example, predicts that the coastline of Pine Island Bay will approach roughly 45 mm yr⁻¹ in viscoelastic vertical motion. Of particular importance, we systematically demonstrate that the effect of a pervasive and large GIA uplift in the WAIS is generally associated with the flattening of reverse bed slope, reduction of local sea depth, and thus the extension of grounding line (GL) towards the continental shelf. Using the 3-D higher-order ice flow capability of ISSM, such a migration of GL is shown to inhibit the ice flow. This negative feedback between the ice sheet and the solid Earth may promote stability in marine portions of the ice sheet in the future.

511. Constraining past accumulation in the central Pine Island Glacier basin, West Antarctica, using radio-echo sounding

Authors: Nanna B. Karlsson; Robert G. Bingham; David M. Rippin; Richard C.A. Hindmarsh; Hugh F.J. Corr; David G. Vaughan

Journal: Journal of Glaciology

Publisher: International Glaciological Society

DOI: 10.3189/2014jog13j180

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Language: en

Subjects: N/A

Keywords: Radio-echo sounding, Ice-sheet modelling, Ground-penetrating radar, Antarctic glaciology

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License: N/A

Open Access: N/A

Abstract: The potential for future dynamical instability of Pine Island Glacier, West Antarctica, has been addressed in a number of studies, but information on its past remains limited. In this study we use airborne radio-echo sounding (RES) data acquired over Pine Island Glacier to investigate past variations in accumulation pattern. In the dataset a distinctive pattern of layers was identified in the central part of the glacier basin. We use these layers as chronological identifiers in order to construct elevation maps of the internal stratigraphy. The observed internal layer stratigraphy is then compared to calculated stratigraphy from a three-dimensional ice-flow model that has been forced with different accumulation scenarios. The model results indicate that the accumulation pattern is likely to have changed at least twice since the deposition of the deepest identified layer. Additional RES data linked to the Byrd ice core provide an approximate timescale. This timescale suggests that the layers were deposited at the beginning of or during the Holocene period. Thus the widespread changes occurring in the coastal extent of the West Antarctic ice sheet at the end of the last glacial period could have been accompanied by changes in accumulation pattern.

512. Scientific Discourse: Climate Change and Sea-Level Rise

Authors: Biao Chang (Georgia Institute of Technology); Jiabao Guan (Georgia Institute of Technology); Mustafa M. Aral (Georgia Institute of Technology)

Journal: Journal of Hydrologic Engineering

Publisher: American Society of Civil Engineers (ASCE)

DOI: 10.1061/(asce)he.1943-5584.0000860

URL: [https://doi.org/10.1061/\(asce\)he.1943-5584.0000860](https://doi.org/10.1061/(asce)he.1943-5584.0000860)

Publication Date: 2014-06-02

Volume/Issue: Volume 20, Issue 1

References: 83

Cited by: 19 publications

Language: en

Subjects: N/A

Keywords: Empirical model, Sea-level rise, Climate change, Dynamic system model, Physical model

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

513. A system of conservative regridding for ice-atmosphere coupling in a General Circulation Model (GCM)

Authors: E. Fischer; S. Nowicki; M. Kelley; G. A. Schmidt

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-7-883-2014

URL: <https://doi.org/10.5194/gmd-7-883-2014>

Publication Date: 2014-05-19

Volume/Issue: Volume 7, Issue 3

References: 45

Cited by: 17 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . The method of elevation classes, in which the ice surface model is run at multiple elevations within each grid cell, has proven to be a useful way for a low-resolution atmosphere inside a general circulation model (GCM) to produce high-resolution downscaled surface mass balance fields for use in one-way studies coupling atmospheres and ice flow models. Past uses of elevation classes have failed to conserve mass and energy because the transformation used to regrid to the atmosphere was inconsistent with the transformation used to downscale to the ice model. This would cause problems for two-way coupling. A strategy that resolves this conservation issue has been designed and is presented here. The approach identifies three grids between which data must be regridded and five transformations between those grids required by a typical coupled atmosphere-ice flow model. This paper develops a theoretical framework for the problem and shows how each of these transformations may be achieved in a consistent, conservative manner. These transformations are implemented in Glint2, a library used to couple atmosphere models with ice models. Source code and documentation are available for download. Confounding real-world issues are discussed, including the use of projections for ice modeling, how to handle dynamically changing ice geometry, and modifications required for finite element ice models.

514. The trough-system algorithm and its application to spatial modeling of Greenland subglacial topography

Authors: Ute C. Herzfeld; Brian W. McDonald; Bruce F. Wallin; Phillip A. Chen; Helmut Mayer; John Paden; Carlton J. Leuschen

Journal: Annals of Glaciology

Publisher: International Glaciological Society

DOI: 10.3189/2014aog67a001

URL: <https://doi.org/10.3189/2014aog67a001>

Publication Date: 2014-05-15

Volume/Issue: Volume 55, Issue 67

References: 53

Cited by: 2 publications

Language: en

Subjects: N/A

Keywords: Ground-penetrating radar, Arctic glaciology, Glaciological instruments and methods, Radio-echo sounding, Ice-sheet modelling

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Dynamic ice-sheet models are used to assess the contribution of mass loss from the Greenland ice sheet to sea-level rise. Mass transfer from ice sheet to ocean is in a large part through outlet glaciers. Bed topography plays an important role in ice dynamics, since the acceleration from the slow-moving inland ice to an ice stream is in many cases caused by the existence of a subglacial trough or trough system. Problems are that most subglacial troughs are features of a scale not resolved in most ice-sheet models and that radar measurements of subglacial topography do not always reach the bottoms of narrow troughs. The trough-system algorithm introduced here employs mathematical morphology and algebraic topology to correctly represent subscale features in a topographic generalization, so the effects of troughs on ice flow are retained in ice-dynamic models. The algorithm is applied to derive a spatial elevation model of Greenland subglacial topography, integrating recently collected radar measurements (CReSIS data) of the Jakobshavn Isbræ, Helheim, Kangerdlussuaq and Petermann glacier regions. The resultant JakHelKanPet digital elevation model has been applied in dynamic ice-sheet modeling and sea-level-rise assessment.

515. Ice sheets as interactive components of Earth System Models: progress and challenges

Authors: Miren Vizcaino (Delft University of Technology)

Journal: WIREs Climate Change

Publisher: Wiley

DOI: 10.1002/wcc.285

URL: <https://doi.org/10.1002/wcc.285>

Publication Date: 2014-05-12

Volume/Issue: Volume 5, Issue 4

References: 101

Cited by: 35 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Delft University of Technology

License: N/A

Open Access: N/A

Abstract: One of the major impacts of anthropogenic climate change is sea level rise. Reliable estimates of the contribution of ice sheets to future sea level rise are important to policy makers and the civil society. In addition to sea level rise, ice sheet changes can affect the global climate through modified freshwater fluxes in the areas of deep-water convection. Also, ice sheets modify local and large-scale climate through changes in surface albedo and in their own topography. In the past, ice sheets have played a fundamental role in shaping climate and climate transitions. Despite their strong interactions with the climate system, they are not yet standard components of climate models. First attempts have been made in this direction, and it is foreseeable that in several years ice sheets will be included as interactive components of most models. The main challenges for this coupling are related to spatial and temporal resolution, ice sheet initialization, model climate biases, the need for explicit representation of snow/ice surface physics (e.g., albedo evolution, surface melt, refreezing, compaction), and coupling to the ocean component. This article reviews the main processes contributing to the ice sheet mass budget, the suite of ice sheet-climate interactions, and the requirements for modelling them in a coupled system. Focus is given to four major subjects: surface mass balance, ice sheet flow, ocean-ice sheet interaction, and challenges in coupling ice sheet models to climate models. This article is categorized under: Climate Models and Modeling > Model Components

516. Widespread, rapid grounding line retreat of Pine Island, Thwaites, Smith, and Kohler glaciers, West Antarctica, from 1992 to 2011

Authors: E. Rignot (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); J. Mouginot (University of California Irvine); M. Morlighem (University of California Irvine); H. Seroussi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); B. Scheuchl (University of California Irvine)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2014gl060140

URL: <https://doi.org/10.1002/2014gl060140>

Publication Date: 2014-05-12

Volume/Issue: Volume 41, Issue 10

References: 32

Cited by: 619 publications

Language: en

Subjects: N/A

Keywords: mass balance, marine instability, Antarctica, interferometry, glacier dynamics

Funding: N/A

License: N/A

Open Access: N/A

Abstract: We measure the grounding line retreat of glaciers draining the Amundsen Sea sector of West Antarctica using Earth Remote Sensing (ERS-1/2) satellite radar interferometry from 1992 to 2011. Pine Island Glacier retreated 31 km at its center, with most retreat in 2005-2009 when the glacier ungrounded from its ice plain. Thwaites Glacier retreated 14 km along its fast flow core and 1 to 9 km along the sides. Haynes Glacier retreated 10 km along its flanks. Smith/Kohler glaciers retreated the most, 35 km along its ice plain, and its ice shelf pinning points are vanishing. These rapid retreats proceed along regions of retrograde bed elevation mapped at a high spatial resolution using a mass conservation technique that removes residual ambiguities from prior mappings. Upstream of the 2011 grounding line positions, we find no major bed obstacle that would prevent the glaciers from further retreat and draw down the entire basin.

517. An ETKF approach for initial state and parameter estimation in ice sheet modelling

Authors: B. Bonan; M. Nodet; C. Ritz; V. Peyaud

Journal: Nonlinear Processes in Geophysics

Publisher: Copernicus GmbH

DOI: 10.5194/npg-21-569-2014

URL: <https://doi.org/10.5194/npg-21-569-2014>

Publication Date: 2014-04-25

Volume/Issue: Volume 21, Issue 2

References: 49

Cited by: 12 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . Estimating the contribution of Antarctica and Greenland to sea-level rise is a hot topic in glaciology. Good estimates rely on our ability to run a precisely calibrated ice sheet evolution model starting from a reliable initial state. Data assimilation aims to provide an answer to this problem by combining the model equations with observations. In this paper we aim to study a state-of-the-art ensemble Kalman filter (ETKF) to address this problem. This method is implemented and validated in the twin experiments framework for a shallow ice flowline model of ice dynamics. The results are very encouraging, as they show a good convergence of the ETKF (with localisation and inflation), even for small-sized ensembles.

518. Feedbacks between coupled subglacial hydrology and glacier dynamics

Authors: Matthew Hoffman (Los Alamos National Laboratory); Stephen Price (Los Alamos National Laboratory)

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/2013jf002943

URL: <https://doi.org/10.1002/2013jf002943>

Publication Date: 2014-01-20

Volume/Issue: Volume 119, Issue 3

References: 78

Cited by: 75 publications

Language: en

Subjects: N/A

Keywords: coupled modeling, subglacial hydrology, glacier sliding, glacier dynamics, glaciers

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

519. Future Antarctic bed topography and its implications for ice sheet dynamics

Authors: S. Adhikari; E. Ivins; E. Larour; H. Seroussi; M. Morlighem; S. Nowicki

Journal: Solid Earth Discussions

Publisher: Copernicus GmbH

DOI: 10.5194/sed-6-191-2014

URL: <https://doi.org/10.5194/sed-6-191-2014>

Publication Date: 2014-01-17

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Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

520. RIMBAY - a multi-approximation 3D ice-dynamics model for comprehensive applications: model description and examples

Authors: M. Thoma; K. Grosfeld; D. Barbi; J. Determann; S. Goeller; C. Mayer; F. Pattyn

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-7-1-2014

URL: <https://doi.org/10.5194/gmd-7-1-2014>

Publication Date: 2014-01-07

Volume/Issue: Volume 7, Issue 1

References: 67

Cited by: 27 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . Glaciers and ice caps exhibit currently the largest cryospheric contributions to sea level rise. Modelling the dynamics and mass balance of the major ice sheets is therefore an important issue to investigate the current state and the future response of the cryosphere in response to changing environmental conditions, namely global warming. This requires a powerful, easy-to-use, versatile multi-approximation ice dynamics model. Based on the well-known and established ice sheet model of Pattyn (2003) we develop the modular multi-approximation thermomechanic ice model RIMBAY, in which we improve the original version in several aspects like a shallow ice-shallow shelf coupler and a full 3D-grounding-line migration scheme based on Schoof's (2007) heuristic analytical approach. We summarise the full Stokes equations and several approximations implemented within this model and we describe the different numerical discretisations. The results are cross-validated against previous publications dealing with ice modelling, and some additional artificial set-ups demonstrate the robustness of the different solvers and their internal coupling. RIMBAY is designed for an easy adaption to new scientific issues. Hence, we demonstrate in very different set-ups the applicability and functionality of RIMBAY in Earth system science in general and ice modelling in particular.

521. Creep deformation and buttressing capacity of damaged ice shelves: theory and application to Larsen C ice shelf

Authors: C. P. Borstad; E. Rignot; J. Mouginot; M. P. Schodlok

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-7-1931-2013

URL: <https://doi.org/10.5194/tc-7-1931-2013>

Publication Date: 2013-12-18

Volume/Issue: Volume 7, Issue 6

References: 82

Cited by: 84 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . Around the perimeter of Antarctica, much of the ice sheet discharges to the ocean through floating ice shelves. The buttressing provided by ice shelves is critical for modulating the flux of ice into the ocean, and the presently observed thinning of ice shelves is believed to be reducing their buttressing capacity and contributing to the acceleration and thinning of the grounded ice sheet. However, relatively little attention has been paid to the role that fractures play in the ability of ice shelves to sustain and transmit buttressing stresses. Here, we present a new framework for quantifying the role that fractures play in the creep deformation and buttressing capacity of ice shelves. We apply principles of continuum damage mechanics to derive a new analytical relation for the creep of an ice shelf that accounts for the softening influence of fractures on longitudinal deformation using a state damage variable. We use this new analytical relation, combined with a temperature calculation for the ice, to partition an inverse method solution for ice shelf rigidity into independent solutions for softening damage and stabilizing backstress. Using this new approach, field and remote sensing data can be utilized to monitor the structural integrity of ice shelves, their ability to buttress the flow of ice at the grounding line, and thus their indirect contribution to ice sheet mass balance and global sea level. We apply this technique to the Larsen C ice shelf using remote sensing and Operation IceBridge data, finding damage in areas with known crevasses and rifts. Backstress is highest near the grounding line and upstream of ice rises, in agreement with patterns observed on other ice shelves. The ice in contact with the Bawden ice rise is weakened by fractures, and additional damage or thinning in this area could diminish the backstress transmitted upstream. We model the consequences for the ice shelf if it loses contact with this small ice rise, finding that flow speeds would increase by 25% or more over an area the size of the former Larsen B ice shelf. Such a perturbation could potentially destabilize the northern part of Larsen C along pre-existing lines of weakness, highlighting the importance of the feedback between buttressing and fracturing in an ice shelf.

522. Arctic Ocean glacial history

Authors: Martin Jakobsson; Karin Andreassen; Lilja Rún Bjarnadóttir; Dayton Dove; Julian A. Dowdeswell; John H. England; Svend Funder; Kelly Hogan; Ólafur Ingólfsson; Anne Jennings; Nikolaj Krog Larsen; Nina Kirchner; Jon Y. Landvik; Larry Mayer; Naja Mikkelsen; Per Möller; Frank Niessen; Johan Nilsson; Matt O'Regan; Leonid Polyak; Niels Nørgaard-Pedersen; Ruediger Stein

Journal: Quaternary Science Reviews

Publisher: Elsevier BV

DOI: 10.1016/j.quascirev.2013.07.033

URL: <https://doi.org/10.1016/j.quascirev.2013.07.033>

Publication Date: 2013-11-15

Volume/Issue: Volume 92

References: 293

Cited by: 192 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Swedish Research Council, FORMAS, Research Council of Norway (and 3 more)

License: N/A

Open Access: N/A

Abstract: N/A

523. Parameter and state estimation with a time-dependent adjoint marine ice sheet model

Authors: D. N. Goldberg; P. Heimbach

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-7-1659-2013

URL: <https://doi.org/10.5194/tc-7-1659-2013>

Publication Date: 2013-11-06

Volume/Issue: Volume 7, Issue 6

References: 82

Cited by: 63 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . To date, assimilation of observations into large-scale ice models has consisted predominantly of time-independent inversions of surface velocities for basal traction, bed elevation, or ice stiffness, and has relied primarily on analytically derived adjoints of glaciological stress balance models. To overcome limitations of such "snapshot" inversions - i.e., their inability to assimilate time-dependent data for the purpose of constraining transient flow states, or to produce initial states with minimum artificial drift and suitable for time-dependent simulations - we have developed an adjoint of a time-dependent parallel glaciological flow model. The model implements a hybrid shallow shelf-shallow ice stress balance, solves the continuity equation for ice thickness evolution, and can represent the floating, fast-sliding, and frozen bed regimes of a marine ice sheet. The adjoint is generated by a combination of analytic methods and the use of algorithmic differentiation (AD) software. Several experiments are carried out with idealized geometries and synthetic observations, including inversion of time-dependent surface elevations for past thicknesses, and simultaneous retrieval of basal traction and topography from surface data. Flexible generation of the adjoint for a range of independent uncertain variables is exemplified through sensitivity calculations of grounded ice volume to changes in basal melting of floating and basal sliding of grounded ice. The results are encouraging and suggest the feasibility, using real observations, of improved ice sheet state estimation and comprehensive transient sensitivity assessments.

524. Hindcasting to measure ice sheet model sensitivity to initial states

Authors: A. Aschwanden; G. Aðalgeirsdóttir; C. Khroulev

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-7-1083-2013

URL: <https://doi.org/10.5194/tc-7-1083-2013>

Publication Date: 2013-10-22

Volume/Issue: Volume 7, Issue 4

References: 54

Cited by: 70 publications

Language: en

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Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . Validation is a critical component of model development, yet notoriously challenging in ice sheet modeling. Here we evaluate how an ice sheet system model responds to a given forcing. We show that hindcasting, i.e. forcing a model with known or closely estimated inputs for past events to see how well the output matches observations, is a viable method of assessing model performance. By simulating the recent past of Greenland, and comparing to observations of ice thickness, ice discharge, surface speeds, mass loss and surface elevation changes for validation, we find that the short term model response is strongly influenced by the initial state. We show that the thermal and dynamical states (i.e. the distribution of internal energy and momentum) can be misrepresented despite a good agreement with some observations, stressing the importance of using multiple observations. In particular we identify rates of change of spatially dense observations as preferred validation metrics. Hindcasting enables a qualitative assessment of model performance relative to observed rates of change. It thereby reduces the number of admissible initial states more rigorously than validation efforts that do not take advantage of observed rates of change.

525. Dependence of century-scale projections of the Greenland ice sheet on its thermal regime

Authors: H. Seroussi; M. Morlighem; E. Rignot; A. Khazendar; E. Larour; J. Mouginot

Journal: Journal of Glaciology

Publisher: International Glaciological Society

DOI: 10.3189/2013jog13j054

URL: <https://doi.org/10.3189/2013jog13j054>

Publication Date: 2013-10-10

Volume/Issue: Volume 59, Issue 218

References: 64

Cited by: 118 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Observations show that the Greenland ice sheet has been losing mass at an increasing rate over the past few decades, which makes it a major contributor to sea-level rise. Here we use a three-dimensional higher-order ice-flow model, adaptive mesh refinement and inverse methods to accurately reproduce the present-day ice flow of the Greenland ice sheet. We investigate the effect of the ice thermal regime on (1) basal sliding inversion and (2) projections over the next 100 years. We show that steady-state temperatures based on present-day conditions allow a reasonable representation of the thermal regime and that both basal conditions and century-scale projections are weakly sensitive to small changes in the initial temperature field, compared with changes in atmospheric conditions or basal sliding. We conclude that although more englacial temperature measurements should be acquired to validate the models, and a better estimation of geothermal heat flux is needed, it is reasonable to use steady-state temperature profiles for short-term projections, as external forcings remain the main drivers of the changes occurring in Greenland.

526. Capabilities and performance of Elmer/Ice, a new-generation ice sheet model

Authors: O. Gagliardini; T. Zwinger; F. Gillet-Chaulet; G. Durand; L. Favier; B. de Fleurian; R. Greve; M. Malinen; C. Martín; P. Råback; J. Ruokolainen; M. Sacchettini; M. Schäfer; H. Seddik; J. Thies

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-6-1299-2013

URL: <https://doi.org/10.5194/gmd-6-1299-2013>

Publication Date: 2013-08-22

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References: 90

Cited by: 284 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: European Commission

License: Creative Commons

Open Access: N/A

Abstract: . The Fourth IPCC Assessment Report concluded that ice sheet flow models, in their current state, were unable to provide accurate forecast for the increase of polar ice sheet discharge and the associated contribution to sea level rise. Since then, the glaciological community has undertaken a huge effort to develop and improve a new generation of ice flow models, and as a result a significant number of new ice sheet models have emerged. Among them is the parallel finite-element model Elmer/Ice, based on the open-source multi-physics code Elmer. It was one of the first full-Stokes models used to make projections for the evolution of the whole Greenland ice sheet for the coming two centuries. Originally developed to solve local ice flow problems of high mechanical and physical complexity, Elmer/Ice has today reached the maturity to solve larger-scale problems, earning the status of an ice sheet model. Here, we summarise almost 10 yr of development performed by different groups. Elmer/Ice solves the full-Stokes equations, for isotropic but also anisotropic ice rheology, resolves the grounding line dynamics as a contact problem, and contains various basal friction laws. Derived fields, like the age of the ice, the strain rate or stress, can also be computed. Elmer/Ice includes two recently proposed inverse methods to infer badly known parameters. Elmer is a highly parallelised code thanks to recent developments and the implementation of a block preconditioned solver for the Stokes system. In this paper, all these components are presented in detail, as well as the numerical performance of the Stokes solver and developments planned for the future.

527. Why marine ice sheet model predictions may diverge in estimating future sea level rise

Authors: Frank Pattyn (Laboratoire de Glaciologie Université Libre de Bruxelles Brussels Belgium); Gaël Durand (CNRS, LGGE (UMR5183) Grenoble France, Université Grenoble Alpes, LGGE (UMR5183) Grenoble France)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/grl.50824

URL: <https://doi.org/10.1002/grl.50824>

Publication Date: 2013-08-08

Volume/Issue: Volume 40, Issue 16

References: 25

Cited by: 34 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: European Union 7th Framework Programme

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Open Access: N/A

Abstract: Despite major recent efforts, marine ice sheet models aiming at predicting future mass loss from ice sheets still suffer from uncertainties with respect to grounding line migration. A recent model intercomparison provided tools to test how models treat grounding line dynamics in a three-dimensional setting. Here we use these tools to address to what extent differences in mass loss occur according to the approximation to the Stokes equations, describing marine ice sheet flow, used. We find that models that neglect components of vertical shearing in the force budget wrongly estimate ice sheet mass loss by +/-50% over century time scales when compared to models that solve the full Stokes system of equations. Models that only include horizontal stresses also misrepresent velocities and ice shelf geometry, suggesting that interactions between the grounded ice sheet and the ocean will also be modeled incorrectly. Based on these findings, we strongly advise the use of high-order models to compute reliable projections of ice sheet contribution to sea level rise.

528. Inversion of basal friction in Antarctica using exact and incomplete adjoints of a higher-order model

Authors: M. Morlighem (University of California Irvine); H. Seroussi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); E. Larour (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); E. Rignot (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/jgrf.20125

URL: <https://doi.org/10.1002/jgrf.20125>

Publication Date: 2013-07-30

Volume/Issue: Volume 118, Issue 3

References: 35

Cited by: 129 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Basal friction beneath ice sheets remains poorly characterized and yet is a fundamental control on ice mechanics. Here we use a complete map of surface velocity of the Antarctic Ice Sheet to infer the basal friction over the entire continent by combining these observations with a three-dimensional, thermomechanical, higher-order ice sheet numerical model from the Ice Sheet System Model open source software. We demonstrate that inverse methods can be readily applied at the continental scale with appropriate selections of cost function and of scheme of regularization, at a spatial resolution as high as 3 km along the coastline. We compare the convergence of two descent algorithms with the exact and incomplete adjoints to show that the incomplete adjoint is an excellent approximation. The results reveal that the driving stress is almost entirely balanced by the basal shear stress over 80% of the ice sheet. The basal friction coefficient, which relates basal friction to basal velocity, is, however, significantly heterogeneous: it is low on fast moving ice and high near topographic divides. Areas with low values extend far out into the interior, along glacier and ice stream tributaries, almost to the flanks of topographic divides, suggesting that basal sliding is widespread beneath the Antarctic Ice Sheet.

529. Data assimilation and prognostic whole ice sheet modelling with the variationally derived, higher order, open source, and fully parallel ice sheet model VarGlaS

Authors: D. J. Brinkerhoff; J. V. Johnson

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-7-1161-2013

URL: <https://doi.org/10.5194/tc-7-1161-2013>

Publication Date: 2013-07-25

Volume/Issue: Volume 7, Issue 4

References: 62

Cited by: 35 publications

Language: en

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Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: . We introduce a novel, higher order, finite element ice sheet model called VarGlaS (Variational Glacier Simulator), which is built on the finite element framework FEniCS. Contrary to standard procedure in ice sheet modelling, VarGlaS formulates ice sheet motion as the minimization of an energy functional, conferring advantages such as a consistent platform for making numerical approximations, a coherent relationship between motion and heat generation, and implicit boundary treatment. VarGlaS also solves the equations of enthalpy rather than temperature, avoiding the solution of a contact problem. Rather than include a lengthy model spin-up procedure, VarGlaS possesses an automated framework for model inversion. These capabilities are brought to bear on several benchmark problems in ice sheet modelling, as well as a 500 yr simulation of the Greenland ice sheet at high resolution. VarGlaS performs well in benchmarking experiments and, given a constant climate and a 100 yr relaxation period, predicts a mass evolution of the Greenland ice sheet that matches present-day observations of mass loss. VarGlaS predicts a thinning in the interior and thickening of the margins of the ice sheet.

530. Challenges to Understanding the Dynamic Response of Greenland's Marine Terminating Glaciers to Oceanic and Atmospheric Forcing

Authors: Fiammetta Straneo (Woods Hole Oceanographic Institution); Patrick Heimbach (Massachusetts Institute of Technology); Olga Sergienko (Princeton University); Gordon Hamilton (University of Maine); Ginny Catania (University of Texas at Austin); Stephen Griffies (NOAA/Geophysical Fluid Dynamics Laboratory, Princeton, New Jersey); Robert Hallberg (NOAA/Geophysical Fluid Dynamics Laboratory, Princeton, New Jersey); Adrian Jenkins (British Antarctic Survey, Cambridge, United Kingdom); Ian Joughin (University of Washington); Roman Motyka (University of Alaska Fairbanks); W. Tad Pfeffer (University of Colorado Boulder); Stephen F. Price (Los Alamos National Laboratory); Eric Rignot (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Ted Scambos (University of Colorado Boulder); Martin Truffer (University of Alaska Fairbanks); Andreas Vieli (Durham University, Durham, United Kingdom)

Journal: Bulletin of the American Meteorological Society

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Cited by: 122 publications

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License: N/A

Open Access: N/A

Abstract: The recent retreat and speedup of outlet glaciers, as well as enhanced surface melting around the ice sheet margin, have increased Greenland's contribution to sea level rise to 0.6 ± 0.1 mm yr⁻¹ and its discharge of freshwater into the North Atlantic. The widespread, near-synchronous glacier retreat, and its coincidence with a period of oceanic and atmospheric warming, suggests a common climate driver. Evidence points to the marine margins of these glaciers as the region from which changes propagated inland. Yet, the forcings and mechanisms behind these dynamic responses are poorly understood and are either missing or crudely parameterized in climate and ice sheet models. Resulting projected sea level rise contributions from Greenland by 2100 remain highly uncertain. This paper summarizes the current state of knowledge and highlights key physical aspects of Greenland's coupled ice sheet-ocean-atmosphere system. Three research thrusts are identified to yield fundamental insights into ice sheet, ocean, sea ice, and atmosphere interactions, their role in Earth's climate system, and probable trajectories of future changes: 1) focused process studies addressing critical glacier, ocean, atmosphere, and coupled dynamics; 2) sustained observations at key sites; and 3) inclusion of relevant dynamics in Earth system models. Understanding the dynamic response of Greenland's glaciers to climate forcing constitutes both a scientific and technological frontier, given the challenges of obtaining the appropriate measurements from the glaciers' marine termini and the complexity of the

dynamics involved, including the coupling of the ocean, atmosphere, glacier, and sea ice systems. Interdisciplinary and international cooperation are crucial to making progress on this novel and complex problem.

531. Sensitivity of Greenland Ice Sheet Projections to Model Formulations

Authors: H. Goelzer; P. Huybrechts; J.J. Fürst; F.M. Nick; M.L. Andersen; T.L. Edwards; X. Fettweis; A.J. Payne; S. Shannon

Journal: Journal of Glaciology

Publisher: International Glaciological Society

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References: 58

Cited by: 104 publications

Language: en

Subjects: N/A

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Open Access: N/A

Abstract: Physically based projections of the Greenland ice sheet contribution to future sea-level change are subject to uncertainties of the atmospheric and oceanic climatic forcing and to the formulations within the ice flow model itself. Here a higher-order, three-dimensional thermomechanical ice flow model is used, initialized to the present-day geometry. The forcing comes from a high-resolution regional climate model and from a flowline model applied to four individual marine-terminated glaciers, and results are subsequently extended to the entire ice sheet. The experiments span the next 200 years and consider climate scenario SRES A1B. The surface mass-balance (SMB) scheme is taken either from a regional climate model or from a positive-degree-day (PDD) model using temperature and precipitation anomalies from the underlying climate models. Our model results show that outlet glacier dynamics only account for 6-18% of the sea-level contribution after 200 years, confirming earlier findings that stress the dominant effect of SMB changes. Furthermore, interaction between SMB and ice discharge limits the importance of outlet glacier dynamics with increasing atmospheric forcing. Forcing from the regional climate model produces a 14-31 % higher sea-level contribution compared to a PDD model run with the same parameters as for IPCC AR4.

532. An adaptive Newton multigrid method for a model of marine ice sheets

Authors: Guillaume Jovuet; Carsten Gräser

Journal: Journal of Computational Physics

Publisher: Elsevier BV

DOI: 10.1016/j.jcp.2013.06.032

URL: <https://doi.org/10.1016/j.jcp.2013.06.032>

Publication Date: 2013-07-04

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References: 45

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

533. Pine Island Glacier (Antarctica) velocities from Landsat7 images between 2001 and 2011: FFT-based image correlation for images with data gaps

Authors: Roland C. Warner; Jason L. Roberts

Journal: Journal of Glaciology

Publisher: International Glaciological Society

DOI: 10.3189/2013jog12j113

URL: <https://doi.org/10.3189/2013jog12j113>

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Volume/Issue: Volume 59, Issue 215

References: 36

Cited by: 16 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Remote sensing of ice motion by tracking displacement of surface features is a valuable tool in glaciology. Efficient image feature-tracking programs, such as IMCORR, based on fast Fourier transform methods can produce misleading correlations if there are data gaps in either or both of the reference and search images. This is particularly problematic if the data gaps are regular in character, such as for Landsat7 images collected after the failure of the Scan Line Corrector (SLC-off images). We demonstrate that this situation can be alleviated by filling the data gaps with suitably chosen random data. We modified IMCORR to achieve this automatically (source code is included), but generic image-processing software could be used to modify inputs for other correlation packages. We test our method using images of Pine Island Ice Shelf, Antarctica, and document the acceleration of the velocity field for the floating extension of Pine Island Glacier over the decade 2001-11. We also combine our velocities with recent NASA Operation IceBridge ice thickness data from CReSIS to estimate the basal melt rates.

534. Ice-sheet mass balance and climate change

Authors: Edward Hanna; Francisco J. Navarro; Frank Pattyn; Catia M. Domingues; Xavier Fettweis; Erik R. Ivins; Robert J. Nicholls; Catherine Ritz; Ben Smith; Slawek Tulaczyk; Pippa L. Whitehouse; H. Jay Zwally

Journal: Nature

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/nature12238

URL: <https://doi.org/10.1038/nature12238>

Publication Date: 2013-06-04

Volume/Issue: Volume 498, Issue 7452

References: 100

Cited by: 237 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

535. Grounding-line migration in plan-view marine ice-sheet models: results of the ice2sea MISMIP3d intercomparison

Authors: Frank Pattyn; Laura Perichon; Gaël Durand; Lionel Favier; Olivier Gagliardini; Richard C.A. Hindmarsh; Thomas Zwinger; Torsten Albrecht; Stephen Cornford; David Docquier; Johannes J. Fürst; Daniel Goldberg; G. Hilmar Gudmundsson; Angelika Humbert; Moritz Hütten; Philippe Huybrechts; Guillaume Jouvét; Thomas Kleiner; Eric Larour; Daniel Martin; Mathieu Morlighem; Anthony J. Payne; David Pollard; Martin Rückamp; Oleg Rybak; Hélène Seroussi; Malte Thoma; Nina Wilkens

Journal: Journal of Glaciology

Publisher: International Glaciological Society

DOI: 10.3189/2013jog12j129

URL: <https://doi.org/10.3189/2013jog12j129>

Publication Date: 2013-05-09

Volume/Issue: Volume 59, Issue 215

References: 54

Cited by: 185 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Predictions of marine ice-sheet behaviour require models able to simulate grounding-line migration. We present results of an intercomparison experiment for plan-view marine ice-sheet models. Verification is effected by comparison with approximate analytical solutions for flux across the grounding line using simplified geometrical configurations (no lateral variations, no buttressing effects from lateral drag). Perturbation experiments specifying spatial variation in basal sliding parameters permitted the evolution of curved grounding lines, generating buttressing effects. The experiments showed regions of compression and extensional flow across the grounding line, thereby invalidating the boundary layer theory. Steady-state grounding-line positions were found to be dependent on the level of physical model approximation. Resolving grounding lines requires inclusion of membrane stresses, a sufficiently small grid size (<500 m), or subgrid interpolation of the grounding line. The latter still requires nominal grid sizes of <5 km. For larger grid spacings, appropriate parameterizations for ice flux may be imposed at the grounding line, but the short-time transient behaviour is then incorrect and different from models that do not incorporate grounding-line parameterizations. The numerical error associated with predicting grounding-line motion can be reduced significantly below the errors associated with parameter ignorance and uncertainties in future scenarios.

536. Insights into spatial sensitivities of ice mass response to environmental change from the SeaRISE ice sheet modeling project I: Antarctica

Authors: Sophie Nowicki (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Robert A. Bindshadler (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Ayako Abe-Ouchi (University of Tokyo); Andy Aschwanden (University of Alaska Fairbanks); Ed Bueler (University of Alaska Fairbanks); Hyeungu Choi (Sigma Space Corporation); Jim Fastook (University of Maine); Glen Granzow (University of Montana); Ralf Greve (Hokkaido University); Gail Gutowski (University of Texas at Austin); Ute Herzfeld (University of Colorado Boulder); Charles Jackson (University of Texas at Austin); Jesse Johnson (University of Montana); Constantine Khroulev (University of Alaska Fairbanks); Eric Larour (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Anders Levermann (Potsdam Institute for Climate Research Potsdam Germany); William H. Lipscomb (Los Alamos National Laboratory); Maria A. Martin (Potsdam Institute for Climate Research Potsdam Germany); Mathieu Morlighem (University of California Irvine); Byron R. Parizek (Mathematics and Geoscience Penn State DuBois, College Place DuBois Pennsylvania USA); David Pollard (Pennsylvania State University); Stephen F. Price (Los Alamos National Laboratory); Diandong Ren (Curtin University of Technology, Curtin University); Eric Rignot (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Fuyuki Saito (Japan Agency for Marine-Earth Science and Technology Research Institute for Global Change, 3173-25 Showamachi, Kanazawa Yokohama Kanagawa Japan); Tatsuru Sato (Hokkaido University); Hakime Seddik (Hokkaido University); Helene Seroussi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Kunio Takahashi (Japan Agency for Marine-Earth Science and Technology Research Institute for Global Change, 3173-25 Showamachi, Kanazawa Yokohama Kanagawa Japan); Ryan Walker (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, University of Maryland); Wei Li Wang (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center)

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

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Publication Date: 2013-05-07

Volume/Issue: Volume 118, Issue 2

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Language: en

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Keywords: N/A

Funding: Grant-in-Aid for Scientific Research A, U.S. National Science Foundation, Center for Remote Sensing of Ice Sheets (CReSIS) (and 4 more)

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Open Access: N/A

Abstract: Atmospheric, oceanic, and subglacial forcing scenarios from the Sea-level Response to Ice Sheet Evolution (SeaRISE) project are applied to six three-dimensional thermomechanical ice-sheet

models to assess Antarctic ice sheet sensitivity over a 500 year timescale and to inform future modeling and field studies. Results indicate (i) growth with warming, except within low-latitude basins (where inland thickening is outpaced by marginal thinning); (ii) mass loss with enhanced sliding (with basins dominated by high driving stresses affected more than basins with low-surface-slope streaming ice); and (iii) mass loss with enhanced ice shelf melting (with changes in West Antarctica dominating the signal due to its marine setting and extensive ice shelves; cf. minimal impact in the Terre Adelie, George V, Oates, and Victoria Land region of East Antarctica). Ice loss due to dynamic changes associated with enhanced sliding and/or sub-shelf melting exceeds the gain due to increased precipitation. Furthermore, differences in results between and within basins as well as the controlling impact of sub-shelf melting on ice dynamics highlight the need for improved understanding of basal conditions, grounding-zone processes, ocean-ice interactions, and the numerical representation of all three.

537. Implementation and Initial Evaluation of the Glimmer Community Ice Sheet Model in the Community Earth System Model

Authors: William H. Lipscomb (Los Alamos National Laboratory); Jeremy G. Fyke (Los Alamos National Laboratory); Miren Vizcaíno (Utrecht University, University of California Berkeley); William J. Sacks (National Center for Atmospheric Research); Jon Wolfe (National Center for Atmospheric Research); Mariana Vertenstein (National Center for Atmospheric Research); Anthony Craig (National Center for Atmospheric Research); Erik Kluzek (National Center for Atmospheric Research); David M. Lawrence (National Center for Atmospheric Research)

Journal: Journal of Climate

Publisher: American Meteorological Society

DOI: 10.1175/jcli-d-12-00557.1

URL: <https://doi.org/10.1175/jcli-d-12-00557.1>

Publication Date: 2013-05-03

Volume/Issue: Volume 26, Issue 19

References: 59

Cited by: 72 publications

Language: en

Subjects: N/A

Keywords: Coupled models, Ice sheets, Climate models, Arctic

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Abstract: The Glimmer Community Ice Sheet Model (Glimmer-CISM) has been implemented in the Community Earth System Model (CESM). Glimmer-CISM is forced by a surface mass balance (SMB) computed in multiple elevation classes in the CESM land model and downscaled to the ice sheet grid. Ice sheet evolution is governed by the shallow-ice approximation with thermomechanical coupling and basal sliding. This paper describes and evaluates the initial model implementation for the Greenland Ice Sheet (GIS). The ice sheet model was spun up using the SMB from a coupled CESM simulation with preindustrial forcing. The model's sensitivity to three key ice sheet parameters was explored by running an ensemble of 100 GIS simulations to quasi equilibrium and ranking each simulation based on multiple diagnostics. With reasonable parameter choices, the steady-state GIS geometry is broadly consistent with observations. The simulated ice sheet is too thick and extensive, however, in some marginal regions where the SMB is anomalously positive. The top-ranking simulations were continued using surface forcing from CESM simulations of the twentieth century (1850-2005) and twenty-first century (2005-2100, with RCP8.5 forcing). In these simulations the GIS loses mass, with a resulting global-mean sea level rise of 16 mm during 1850-2005 and 60 mm during 2005-2100. This mass loss is caused mainly by increased ablation near the ice sheet margin, offset by reduced ice discharge to the ocean. Projected sea level rise is sensitive to the initial geometry, showing the importance of realistic geometry in the spun-up ice sheet.

538. Insights into spatial sensitivities of ice mass response to environmental change from the SeaRISE ice sheet modeling project II: Greenland

Authors: Sophie Nowicki (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Robert A. Bindshadler (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Ayako Abe-Ouchi (University of Tokyo); Andy Aschwanden (University of Alaska Fairbanks); Ed Bueler (University of Alaska Fairbanks); Hyeungu Choi (Sigma Space Corporation); Jim Fastook (University of Maine); Glen Granzow (University of Montana); Ralf Greve (Hokkaido University); Gail Gutowski (University of Texas at Austin); Ute Herzfeld (University of Colorado Boulder); Charles Jackson (University of Texas at Austin); Jesse Johnson (University of Montana); Constantine Khroulev (University of Alaska Fairbanks); Eric Larour (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Anders Levermann (Physics Institute Potsdam University Potsdam Germany); William H. Lipscomb (Los Alamos National Laboratory); Maria A. Martin (Physics Institute Potsdam University Potsdam Germany); Mathieu Morlighem (University of California Irvine); Byron R. Parizek (Department of Mathematics and Geoscience Penn State DuBois DuBois Pennsylvania USA); David Pollard (Pennsylvania State University); Stephen F. Price (Los Alamos National Laboratory); Diandong Ren (Curtin University of Technology, Curtin University); Eric Rignot (University of California Irvine, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Fuyuki Saito (Research Institute for Global Change Japan Agency for Marine-Earth Science and Technology Yokohama Japan); Tatsuru Sato (Hokkaido University); Hakime Seddik (Hokkaido University); Helene Seroussi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Kunio Takahashi (Research Institute for Global Change Japan Agency for Marine-Earth Science and Technology Yokohama Japan); Ryan Walker (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center, University of Maryland); Wei Li Wang (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center)

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

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Cited by: 81 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Grant-in-Aid for Scientific Research, NASA Cryospheric Sciences Award, U.S. National Science Foundation (and 3 more)

License: N/A

Open Access: N/A

Abstract: The Sea-level Response to Ice Sheet Evolution (SeaRISE) effort explores the sensitivity of the current generation of ice sheet models to external forcing to gain insight into the potential future

contribution to sea level from the Greenland and Antarctic ice sheets. All participating models simulated the ice sheet response to three types of external forcings: a change in oceanic condition, a warmer atmospheric environment, and enhanced basal lubrication. Here an analysis of the spatial response of the Greenland ice sheet is presented, and the impact of model physics and spin-up on the projections is explored. Although the modeled responses are not always homogeneous, consistent spatial trends emerge from the ensemble analysis, indicating distinct vulnerabilities of the Greenland ice sheet. There are clear response patterns associated with each forcing, and a similar mass loss at the full ice sheet scale will result in different mass losses at the regional scale, as well as distinct thickness changes over the ice sheet. All forcings lead to an increased mass loss for the coming centuries, with increased basal lubrication and warmer ocean conditions affecting mainly outlet glaciers, while the impacts of atmospheric forcings affect the whole ice sheet.

539. Resolution of ice streams and outlet glaciers in large-scale simulations of the Greenland ice sheet

Authors: Ralf Greve; Ute C. Herzfeld

Journal: Annals of Glaciology

Publisher: International Glaciological Society

DOI: 10.3189/2013aog63a085

URL: <https://doi.org/10.3189/2013aog63a085>

Publication Date: 2013-04-24

Volume/Issue: Volume 54, Issue 63

References: 52

Cited by: 22 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The dynamic/thermodynamic shallow-ice model SICOPOLIS is applied to the Greenland ice sheet. Paleoclimatic spin-ups from 125 ka BP until today, as well as future-climate experiments 500 years into the future, are carried out with three different grid spacings, namely 20, 10 and 5 km. The scenarios are a subset of those specified by the SeaRISE (Sea-level Response to Ice Sheet Evolution) community effort. The bed topography includes improved troughs for Jakobshavn Isbræ, Helheim, Kangerdlugssuaq and Petermann glaciers, processed by an algorithm that preserves shape, orientation and continuity of the troughs on the 5 km scale. Comparison of simulated and observed present-day surface velocities shows that these ice streams and outlet glaciers are resolved with different accuracies, ranging from poor (20 km grid) to reasonably good (5 km grid). In the future-climate experiments, the simulated absolute ice volumes depend significantly on the resolution, while the sensitivities (ice volumes relative to the constant-climate control run) vary only by a few centimeters of sea-level equivalent.

540. Decadal-scale sensitivity of Northeast Greenland ice flow to errors in surface mass balance using ISSM

Authors: N-J. Schlegel (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); E. Larour (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); H. Seroussi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); M. Morlighem (University of California Irvine); J. E. Box (Ohio State University, Byrd Polar Research Center)

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

DOI: 10.1002/jgrf.20062

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Publication Date: 2013-03-27

Volume/Issue: Volume 118, Issue 2

References: 62

Cited by: 23 publications

Language: en

Subjects: N/A

Keywords: Forcing (mathematics)|Greenland ice sheet|Ice-sheet model|Glacier mass balance

Funding: NSF, NASA

License: N/A

Open Access: N/A

Abstract: The behavior of the Greenland Ice Sheet, which is considered a major contributor to sea level changes, is best understood on century and longer time scales. However, on decadal time scales, its response is less predictable due to the difficulty of modeling surface climate, as well as incomplete understanding of the dynamic processes responsible for ice flow. Therefore, it is imperative to understand how modeling advancements, such as increased spatial resolution or more comprehensive ice flow equations, might improve projections of ice sheet response to climatic trends. Here we examine how a finely resolved climate forcing influences a high-resolution ice stream model that considers longitudinal stresses. We simulate ice flow using a two-dimensional Shelfy-Stream Approximation implemented within the Ice Sheet System Model (ISSM) and use uncertainty quantification tools embedded within the model to calculate the sensitivity of ice flow within the Northeast Greenland Ice Stream to errors in surface mass balance (SMB) forcing. Our results suggest that the model tends to smooth ice velocities even when forced with extreme errors in SMB. Indeed, errors propagate linearly through the model, resulting in discharge uncertainty of 16% or 1.9 Gt/yr. We find that mass flux is most sensitive to local errors but is also affected by errors hundreds of kilometers away; thus, an accurate SMB map of the entire basin is critical for realistic simulation. Furthermore, sensitivity analyses indicate that SMB forcing needs to be provided at a resolution of at least 40 km.

541. Ice-sheet model sensitivities to environmental forcing and their use in projecting future sea level (the SeaRISE project)

Authors: Robert A. Bindshadler; Sophie Nowicki; Ayako Abe-Ouchi; Andy Aschwanden; Hyeungu Choi; Jim Fastook; Glen Granzow; Ralf Greve; Gail Gutowski; Ute Herzfeld; Charles Jackson; Jesse Johnson; Constantine Khroulev; Anders Levermann; William H. Lipscomb; Maria A. Martin; Mathieu Morlighem; Byron R. Parizek; David Pollard; Stephen F. Price; Diandong Ren; Fuyuki Saito; Tatsuru Sato; Hakime Seddik; Helene Seroussi; Kunio Takahashi; Ryan Walker; Wei Li Wang

Journal: Journal of Glaciology

Publisher: International Glaciological Society

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URL: <https://doi.org/10.3189/2013jog12j125>

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References: 103

Cited by: 213 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Ten ice-sheet models are used to study sensitivity of the Greenland and Antarctic ice sheets to prescribed changes of surface mass balance, sub-ice-shelf melting and basal sliding. Results exhibit a large range in projected contributions to sea-level change. In most cases, the ice volume above flotation lost is linearly dependent on the strength of the forcing. Combinations of forcings can be closely approximated by linearly summing the contributions from single forcing experiments, suggesting that nonlinear feedbacks are modest. Our models indicate that Greenland is more sensitive than Antarctica to likely atmospheric changes in temperature and precipitation, while Antarctica is more sensitive to increased ice-shelf basal melting. An experiment approximating the Intergovernmental Panel on Climate Change's RCP8.5 scenario produces additional first-century contributions to sea level of 22.3 and 8.1 cm from Greenland and Antarctica, respectively, with a range among models of 62 and 14 cm, respectively. By 200 years, projections increase to 53.2 and 26.7 cm, respectively, with ranges of 79 and 43 cm. Linear interpolation of the sensitivity results closely approximates these projections, revealing the relative contributions of the individual forcings on the combined volume change and suggesting that total ice-sheet response to complicated forcings over 200 years can be linearized.

542. Achieving Textbook Multigrid Efficiency for Hydrostatic Ice Sheet Flow

Authors: Jed Brown; Barry Smith; Aron Ahmadi

Journal: SIAM Journal on Scientific Computing

Publisher: Society for Industrial & Applied Mathematics (SIAM)

DOI: 10.1137/110834512

URL: <https://doi.org/10.1137/110834512>

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References: 28

Cited by: 24 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

543. Grounding line transient response in marine ice sheet models

Authors: A. S. Drouet; D. Docquier; G. Durand; R. Hindmarsh; F. Pattyn; O. Gagliardini; T. Zwinger

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-7-395-2013

URL: <https://doi.org/10.5194/tc-7-395-2013>

Publication Date: 2013-03-01

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References: 35

Cited by: 29 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: . Marine ice-sheet stability is mostly controlled by the dynamics of the grounding line, i.e. the junction between the grounded ice sheet and the floating ice shelf. Grounding line migration has been investigated within the framework of MISIP (Marine Ice Sheet Model Intercomparison Project), which mainly aimed at investigating steady state solutions. Here we focus on transient behaviour, executing short-term simulations (200 yr) of a steady ice sheet perturbed by the release of the buttressing restraint exerted by the ice shelf on the grounded ice upstream. The transient grounding line behaviour of four different flowline ice-sheet models has been compared. The models differ in the physics implemented (full Stokes and shallow shelf approximation), the numerical approach, as well as the grounding line treatment. Their overall response to the loss of buttressing is found to be broadly consistent in terms of grounding line position, rate of surface elevation change and surface velocity. However, still small differences appear for these latter variables, and they can lead to large discrepancies (> 100%) observed in terms of ice sheet contribution to sea level when cumulated over time. Despite the recent important improvements of marine ice-sheet models in their ability to compute steady state configurations, our results question the capacity of these models to compute short-term reliable sea-level rise projections.

544. Manufactured solutions and the verification of three-dimensional Stokes ice-sheet models

Authors: W. Leng; L. Ju; M. Gunzburger; S. Price

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-7-19-2013

URL: <https://doi.org/10.5194/tc-7-19-2013>

Publication Date: 2013-01-04

Volume/Issue: Volume 7, Issue 1

References: 18

Cited by: 16 publications

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Subjects: N/A

Keywords: N/A

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Open Access: N/A

Abstract: . The manufactured solution technique is used for the verification of computational models in many fields. In this paper, we construct manufactured solutions for the three-dimensional, isothermal, nonlinear Stokes model for flows in glaciers and ice sheets. The solution construction procedure starts with kinematic boundary conditions and is mainly based on the solution of a first-order partial differential equation for the ice velocity that satisfies the incompressibility condition. The manufactured solutions depend on the geometry of the ice sheet, basal sliding parameters, and ice softness. Initial conditions are taken from the periodic geometry of a standard problem of the ISMIP-HOM benchmark tests. The upper surface is altered through the manufactured solution procedure to generate an analytic solution for the time-dependent flow problem. We then use this manufactured solution to verify a parallel, high-order accurate, finite element Stokes ice-sheet model. Simulation results from the computational model show good convergence to the manufactured analytic solution.

545. Greenland ice sheet contribution to sea-level rise from a new-generation ice-sheet model

Authors: F. Gillet-Chaulet; O. Gagliardini; H. Seddik; M. Nodet; G. Durand; C. Ritz; T. Zwinger; R. Greve; D. G. Vaughan

Journal: The Cryosphere

Publisher: Copernicus GmbH

DOI: 10.5194/tc-6-1561-2012

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Publication Date: 2012-12-21

Volume/Issue: Volume 6, Issue 6

References: 46

Cited by: 194 publications

Language: en

Subjects: N/A

Keywords: N/A

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Open Access: N/A

Abstract: . Over the last two decades, the Greenland ice sheet (GrIS) has been losing mass at an increasing rate, enhancing its contribution to sea-level rise (SLR). The recent increases in ice loss appear to be due to changes in both the surface mass balance of the ice sheet and ice discharge (ice flux to the ocean). Rapid ice flow directly affects the discharge, but also alters ice-sheet geometry and so affects climate and surface mass balance. Present-day ice-sheet models only represent rapid ice flow in an approximate fashion and, as a consequence, have never explicitly addressed the role of ice discharge on the total GrIS mass balance, especially at the scale of individual outlet glaciers. Here, we present a new-generation prognostic ice-sheet model which reproduces the current patterns of rapid ice flow. This requires three essential developments: the complete solution of the full system of equations governing ice deformation; a variable resolution unstructured mesh to resolve outlet glaciers and the use of inverse methods to better constrain poorly known parameters using observations. The modelled ice discharge is in good agreement with observations on the continental scale and for individual outlets. From this initial state, we investigate possible bounds for the next century ice-sheet mass loss. We run sensitivity experiments of the GrIS dynamical response to perturbations in climate and basal lubrication, assuming a fixed position of the marine termini. We find that increasing ablation tends to reduce outflow and thus decreases the ice-sheet imbalance. In our experiments, the GrIS initial mass (im)balance is preserved throughout the whole century in the absence of reinforced forcing, allowing us to estimate a lower bound of 75 mm for the GrIS contribution to SLR by 2100. In one experiment, we show that the current increase in the rate of ice loss can be reproduced and maintained throughout the whole century. However, this requires a very unlikely perturbation of basal lubrication. From this result we are able to estimate an upper bound of 140 mm from dynamics only for the GrIS contribution to SLR by 2100.

546. Ice flow sensitivity to geothermal heat flux of Pine Island Glacier, Antarctica

Authors: E. Larour; M. Morlighem; H. Seroussi; J. Schiermeier; E. Rignot

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2012jf002371

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Abstract: Model projections of ice flow in a changing climate are dependent on model inputs such as surface elevation, bedrock position or surface temperatures, among others. Of all these inputs, geothermal heat flux is the one for which uncertainty is greatest. In the area of Pine Island Glacier, Antarctica, available data sets differ by up to a factor of 2.5. Here, we evaluate the impact of such uncertainty on ice flow, using sampling analyses based on the Latin-Hypercube method. First, we quantify the impact of geothermal heat flux errors on ice hardness, a thermal parameter that critically controls the magnitude of ice flow. Second, we quantify the impact of the same errors on mass balance, specifically on the mass flux advecting through thirteen fluxgates distributed across Pine Island Glacier. We contrast our results with similar uncertainties generated by errors in the specification of ice thickness. Model outputs indicate that geothermal heat flux errors yield uncertainties on ice hardness on the order of 5-7%, with maximum uncertainty reaching 15%. Resulting uncertainties in mass balance remain however below 1%. We discuss the uncertainty distribution and its relationship to the amount of heat available at the base of the ice sheet from friction, viscous and geothermal heating. We also show that comparatively, errors in ice thickness contribute more to model uncertainty than errors in geothermal heat flux, especially for fast-flowing ice streams.

547. A damage mechanics assessment of the Larsen B ice shelf prior to collapse: Toward a physically-based calving law

Authors: C. P. Borstad; A. Khazendar; E. Larour; M. Morlighem; E. Rignot; M. P. Schodlok; H. Seroussi

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2012gl053317

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References: 25

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Abstract: Calving is a primary process of mass ablation for glaciers and ice sheets, though it still eludes a general physical law. Here, we propose a calving framework based on continuum damage mechanics coupled with the equations of viscous deformation of glacier ice. We introduce a scalar damage variable that quantifies the loss of load-bearing surface area due to fractures and that feeds back with ice viscosity to represent fracture-induced softening. The calving law is a standard failure criterion for viscous damaging materials and represents a macroscopic brittle instability quantified by a critical or threshold damage. We constrain this threshold using the Ice Sheet System Model (ISSM) by inverting for damage on the Larsen B ice shelf prior to its 2002 collapse. By analyzing the damage distribution in areas that subsequently calved, we conclude that calving occurs after fractures have reduced the load-bearing capacity of the ice by 60 +/- 10%.

548. Spatial variation of englacial radar attenuation: Modeling approach and application to the Vostok flowline

Authors: Joseph A. MacGregor; Kenichi Matsuoka; Edwin D. Waddington; Dale P. Winebrenner; Frank Pattyn

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

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Abstract: Constraining the spatial variation of englacial radar attenuation is critical for accurate inference of the spatial variation of the englacial and basal properties of ice sheets from radar returned power. Here we evaluate attenuation models that account for spatial variations in ice temperature and chemistry and test them along the flowline that passes through the Vostok ice core site, Antarctica. The simplest model, often used but rarely valid, assumes a uniform attenuation rate everywhere along the flowline, so that total attenuation is proportional to ice thickness. The next simplest model uses spatially varying temperatures predicted by an ice-flow model and assumes uniform chemistry. Additional models account for spatially varying chemistry using englacial stratigraphy. We find that the roundtrip attenuation to the bed can easily differ by 10 dB or more between the uniform attenuation-rate model and models that account for variable ice temperature. Such differences are sufficient to confound the delineation of dry and wet beds. Also including spatial variations in chemistry produces smaller differences (<10 dB), but the magnitude of these differences depends on the relative importance of dry and wet deposition of impurities in the past. Accounting for dry-deposited impurities requires ice-flow modeling and results in larger differences from all other models, which assume uniform chemistry or wet deposition only. These results indicate that modeling the spatial variation of attenuation requires a spatially varying temperature model in order to infer bed conditions from bed returned power accurately, and that both ice core data and radar stratigraphy are also strongly desirable.

549. Coupling ice flow models of varying orders of complexity with the Tiling method

Authors: Helene Seroussi; Hachmi Ben Dhia; Mathieu Morlighem; Eric Larour; Eric Rignot; Denis Aubry

Journal: Journal of Glaciology

Publisher: International Glaciological Society

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Language: en

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Keywords: N/A

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Abstract: Ice flow numerical models are essential for predicting the evolution of ice sheets in a warming climate. Recent research emphasizes the need for higher-order and even full-Stokes flow models, instead of the traditional shallow-ice approximation, whose assumptions are not valid in certain critical areas. These higher-order models are, however, computationally intensive and difficult to use at the continental scale. Here we present a new technique, the Tiling method, to couple ice flow models of varying orders of complexity. The goal of the method is to limit the spatial extent of where higherorder models are applied to reduce the computational cost, while maintaining the model precision. We apply this method on synthetic geometries to demonstrate its practical use. We first use a geometry for which all models yield the same results to check the consistency of the method. Then we apply our method to a geometry for which a full-Stokes model is required in the vicinity of the ice front. Our results show that the hybrid models present significant improvements over mono-model approaches and reduce computational times.

550. An assessment of key model parametric uncertainties in projections of Greenland Ice Sheet behavior

Authors: P. J. Applegate; N. Kirchner; E. J. Stone; K. Keller; R. Greve

Journal: The Cryosphere

Publisher: Copernicus GmbH

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Publication Date: 2012-05-30

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Abstract: . Lack of knowledge about the values of ice sheet model input parameters introduces substantial uncertainty into projections of Greenland Ice Sheet contributions to future sea level rise. Computer models of ice sheet behavior provide one of several means of estimating future sea level rise due to mass loss from ice sheets. Such models have many input parameters whose values are not well known. Recent studies have investigated the effects of these parameters on model output, but the range of potential future sea level increases due to model parametric uncertainty has not been characterized. Here, we demonstrate that this range is large, using a 100-member perturbed-physics ensemble with the SICOPOLIS ice sheet model. Each model run is spun up over 125 000 yr using geological forcings and subsequently driven into the future using an asymptotically increasing air temperature anomaly curve. All modeled ice sheets lose mass after 2005 AD. Parameters controlling surface melt dominate the model response to temperature change. After culling the ensemble to include only members that give reasonable ice volumes in 2005 AD, the range of projected sea level rise values in 2100 AD is ~40 % or more of the median. Data on past ice sheet behavior can help reduce this uncertainty, but none of our ensemble members produces a reasonable ice volume change during the mid-Holocene, relative to the present. This problem suggests that the model's exponential relation between temperature and precipitation does not hold during the Holocene, or that the central-Greenland temperature forcing curve used to drive the model is not representative of conditions around the ice margin at this time (among other possibilities). Our simulations also lack certain observed physical processes that may tend to enhance the real ice sheet's response. Regardless, this work has implications for other studies that use ice sheet models to project or hindcast the behavior of the Greenland Ice Sheet.

551. Modeling Ice-Sheet Flow

Authors: Richard B. Alley (Pennsylvania State University); Ian Joughin (University of Washington)

Journal: Science

Publisher: American Association for the Advancement of Science (AAAS)

DOI: 10.1126/science.1220530

URL: <https://doi.org/10.1126/science.1220530>

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Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: Full-stress flow models for ice sheets promise improved projections of sea-level change.

552. Sensitivity of the ice-shelf/ocean system to the sub-ice-shelf cavity shape measured by NASA IceBridge in Pine Island Glacier, West Antarctica

Authors: Michael P. Schodlok; Dimitris Menemenlis; Eric Rignot; Michael Studinger

Journal: Annals of Glaciology

Publisher: International Glaciological Society

DOI: 10.3189/2012aog60a073

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Abstract: Two high-resolution (1 km grid) numerical model simulations of the Amundsen Sea, West Antarctica, are used to study the role of the ocean in the mass loss and grounding line retreat of Pine Island Glacier. The first simulation uses BEDMAP bathymetry under the Pine Island ice shelf, and the second simulation uses NASA IceBridge-derived bathymetry. The IceBridge data reveal the existence of a trough from the ice-shelf edge to the grounding line, enabling warm Circumpolar Deep Water to penetrate to the grounding line, leading to higher melt rates than previously estimated. The mean melt rate for the simulation with NASA IceBridge data is 28 ma⁻¹, much higher than previous model estimates but closer to estimates from remote sensing. Although the mean melt rate is 25% higher than in the simulation with BEDMAP bathymetry, the temporal evolution remains unchanged between the two simulations. This indicates that temporal variability of melting is mostly driven by processes outside the cavity. Spatial melt rate patterns of BEDMAP and IceBridge simulations differ significantly, with the latter in closer agreement with satellite-derived melt rate estimates of ~50ma⁻¹ near the grounding line. Our simulations confirm that knowledge of the cavity shape and its time evolution are essential to accurately capture basal mass loss of Antarctic ice shelves.

553. Sensitivity Analysis of Pine Island Glacier ice flow using ISSM and DAKOTA

Authors: E. Larour; J. Schiermeier; E. Rignot; H. Seroussi; M. Morlighem; J. Paden

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

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Abstract: Assessing output errors of ice flow models is a major challenge that needs to be addressed if we are to increase our confidence level in projections of mass balance in Antarctica and Greenland. Major inputs to ice flow models include geometry (ice thickness and surface elevation), constitutive laws and boundary conditions (geothermal flux, basal drag coefficient, surface temperature). These inputs can be either measured, in which case they carry errors due to instruments, or inferred using inverse methods (such as basal drag which is inverted using InSAR surface velocities) in which case they carry additional errors generated by the inversion process itself. In both cases, these input errors will result in uncertainties that propagate throughout a forward model, and that influence output diagnostics. In order to estimate the resulting error margins on diagnostics such as mass flux, we develop a new framework based on the Design Analysis Kit for Optimization and Terascale Applications (DAKOTA), which we interface to the Ice Sheet System Model (ISSM). We present results on the Pine Island Glacier, West Antarctica, for which we evaluate error margins of mass flux across the whole glacier, given currently known error margins on ice thickness, basal friction and ice hardness. Our results suggest errors in these inputs propagate linearly through the ice flow model, providing a way to 1) calibrate measurement requirements for field campaigns collecting data such as bedrock or surface topography 2) quantify uncertainties in projections of mass balance and 3) assess the sensitivity of model outputs to input parameters. This new error propagation model should help quantify confidence levels that we assign to model projections for the mass balance of Antarctica and Greenland, which will ultimately improve our projections of future sea level rise in a warming climate.

554. Continental scale, high order, high spatial resolution, ice sheet modeling using the Ice Sheet System Model (ISSM)

Authors: E. Larour; H. Seroussi; M. Morlighem; E. Rignot

Journal: Journal of Geophysical Research: Earth Surface

Publisher: American Geophysical Union (AGU)

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Keywords: Ice-sheet model|Ice divide|Antarctic ice sheet

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Abstract: Ice flow models used to project the mass balance of ice sheets in Greenland and Antarctica usually rely on the Shallow Ice Approximation (SIA) and the Shallow-Shelf Approximation (SSA), sometimes combined into so-called "hybrid" models. Such models, while computationally efficient, are based on a simplified set of physical assumptions about the mechanical regime of the ice flow, which does not uniformly apply everywhere on the ice sheet/ice shelf system, especially near grounding lines, where rapid changes are taking place at present. Here, we present a new thermomechanical finite element model of ice flow named ISSM (Ice Sheet System Model) that includes higher-order stresses, high spatial resolution capability and data assimilation techniques to better capture ice dynamics and produce realistic simulations of ice sheet flow at the continental scale. ISSM includes several approximations of the momentum balance equations, ranging from the two-dimensional SSA to the three-dimensional full-Stokes formulation. It also relies on a massively parallelized architecture and state-of-the-art scalable tools. ISSM employs data assimilation techniques, at all levels of approximation of the momentum balance equations, to infer basal drag at the ice-bed interface from satellite radar interferometry-derived observations of ice motion. Following a validation of ISSM with standard benchmarks, we present a demonstration of its capability in the case of the Greenland Ice Sheet. We show ISSM is able to simulate the ice flow of an entire ice sheet realistically at a high spatial resolution, with higher-order physics, thereby providing a pathway for improving projections of ice sheet evolution in a warming climate.

555. A mass conservation approach for mapping glacier ice thickness

Authors: M. Morlighem; E. Rignot; H. Seroussi; E. Larour (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); H. Ben Dhia (Laboratoire MSSMat, UMR 8579, CNRS; École Centrale Paris, Grande Voie des Vignes; Châtenay-Malabry; France); D. Aubry (Laboratoire MSSMat, UMR 8579, CNRS; École Centrale Paris, Grande Voie des Vignes; Châtenay-Malabry; France)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

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Open Access: N/A

Abstract: N/A

556. Ice flux divergence anomalies on 79north Glacier, Greenland

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Abstract: The ice flux divergence of a glacier is an important quantity to examine because it determines the rate of temporal change of its thickness. Here, we combine high-resolution ice surface velocity observations of Nioghalvfjærdsfjorden (79north) Glacier, a major outlet glacier in north Greenland, with a dense grid of ice thickness data collected with an airborne radar sounder in 1998, to examine its ice flux divergence. We detect large variations, up to 100 m/yr, in flux divergence on grounded ice that are incompatible with what we know of the glacier surface mass balance, basal mass balance and thinning rate. We examine the hypothesis that these anomalies are due to the three-dimensional flow of ice around and atop bumps and hollows in basal topography by comparing the flux divergence of three-dimensional numerical models with its surface equivalent. We find that three-dimensional effects have only a small contribution to the observed anomalies. On the other hand, if we degrade the spatial resolution of the data to 10 km the anomalies disappear. Further analysis shows that the source of the anomalies is not the ice velocity data but the interpolation of multiple tracks of ice thickness data onto a regular grid using a scheme (here block kriging) that does not conserve mass or ice flux. This problem is not unique to 79north Glacier but is common to all conventional ice thickness surveys of glaciers and ice sheets; and fundamentally limits the application of ice thickness grids to high-resolution numerical modeling of glacier flow.